

The Collected Mathematical Papers of
Arthur Cayley
Vol. IV

Arthur Cayley

Modern L^AT_EX reconstruction

*Digitized by the Internet Archive
in 2007 with funding from
Microsoft Corporation*

*[http:
//www.archive.org/details/collectedmathema04cayluoft](http://www.archive.org/details/collectedmathema04cayluoft)*

MATHEMATICAL PAPERS.

THE COLLECTED
MATHEMATICAL PAPERS
OF
ARTHUR CAYLEY, Sc.D., F.R.S.,
SADLEIRIAN PROFESSOR OF PURE MATHEMATICS IN THE
UNIVERSITY OF CAMBRIDGE.

VOL. IV.

CAMBRIDGE:
AT THE UNIVERSITY PRESS.

1891

[All Rights reserved.]

CAMBRIDGE:
PRINTED BY C. J. CLAY, M.A. AND SONS,
AT THE UNIVERSITY PRESS.

Advertisement

THE present volume contains 77 papers numbered 223 to 299 originally published all but one of them in the years 1856 to 1862, the excepted paper, No. 265, Addition to the Memoir on an Extension of Arbogast's Method of Derivations, being now first published.

The Table for the four volumes is

Vol. I.	Numbers	1 to 100.
" II.	"	101 " 158.
" III.	"	159 " 222.
" IV.	"	223 " 299.

Contents

	PAGE
223. Note sur un théorème général par rapport à l'élimination Tortolini, t. vii. (1856), pp. 454–458	1
224. Sur un théorème d'Abel. Note Tortolini, t. viii. (1857), pp. 201–203	5
225. On a Class of Dynamical Problems Proc. R. S. t. VIII. (1857), pp. 506–511	7
226. On Professor Mac Cullagh's Theorem of the Polar Plane Proc. R. I. A. t. VI. (1858), pp. 481–491	12
227. On the Theory of Reciprocal Surfaces Proc. R. I. A. t. VII. (1862), pp. 20–28	21
228. Sur l'intégrale $\int \frac{dx}{\sqrt{(a+bx+cx^2)^2}}$ Liouville, t. ii. (1857), pp. 49–51	28
229. Note sur une formule pour la réversion des séries Crelle, t. LII. (1856), pp. 276–284	30
230. Note sur la méthode d'élimination de Bezout Crelle, t. LIII. (1857), pp. 366, 367	38
231. Note sur l'équation $x^2 - Dy^2 = \pm 4$, $D \equiv 5 \pmod{8}$ Crelle, t. LIII. (1857), pp. 369–371	40
232. Mémoire sur la forme canonique des fonctions binaires .. Crelle, t. LIV. (1857), pp. 48–58	43
233. Addition au Mémoire sur la forme canonique des fonctions binaires Crelle, t. LIV. (1857), p. 292	53

	PAGE
234.	Deuxième Note sur une formule pour la réversion des séries 54
	Crelle, t. LIV. (1857), pp. 156–161
235.	Sur quelques formules pour la transformation des intégrales elliptiques 60
	Crelle, t. LV. (1858), pp. 15–24
236.	Note sur la composition du nombre 47 par rapport aux vingt-troisièmes racines de l'unité 70
	Crelle, t. LV. (1858), p. 192
237.	Théorème sur les déterminants gauches 72
	Crelle, t. LV. (1858), pp. 277, 278
238.	Note sur les normales d'une conique 74
	Crelle, t. LVI. (1859), pp. 182–185
239.	Addition à la Note sur la composition du nombre 47 par rapport aux vingt-troisièmes racines de l'unité 78
	Crelle, t. LVI. (1859), pp. 186, 187
240.	Note on a Theorem in Spherical Trigonometry 80
	Phil. Mag. t. XVII. (1859), p. 151
241.	On Poinso't's four new Regular Solids 81
	Phil. Mag. t. XVII. (1859), pp. 123–128
242.	Second Note on Poinso't's four new Regular Polyhedra .. 86
	Phil. Mag. t. XVII. (1859), pp. 209, 210
243.	On the Theory of Groups as depending on the Symbolic Equation $\theta^n = 1$. Third Part 88
	Phil. Mag. t. XVIII. (1859), pp. 34–37. Sequel to 125 and 126
244.	On an Analytical Theorem relating to the distribution of Electricity upon two Spherical Surfaces 92
	Phil. Mag. t. XVIII. (1859), pp. 119–127
245.	On an Analytical Theorem connected with the distribution of Electricity upon two Spherical Surfaces. Second Part . 99
	Phil. Mag. t. XVIII. (1859), pp. 193–202
246.	On Contour and Slope Lines 108

	PAGE
	Phil. Mag. t. XVIII. (1859), pp. 264–268
247.	On the Analytical Forms called Trees. Second Part 112 Phil. Mag. t. XVIII. (1859), pp. 374–378. Continuation of 203
248.	Sketch of a proof of the Theorem that every Algebraic 116 Equation has a Root Phil. Mag. t. XVIII. (1859), pp. 436–439
249.	Note on Cones of the Third Order 120 Phil. Mag. t. XVIII. (1859), pp. 439–442
250.	Sur la surface qui est l’enveloppe des plans conduits par 123 les points d’un ellipsoïde perpendiculairement aux rayons menés par le centre Tortolini, t. ii. (1859), pp. 168–179
251.	Sur quelques formules pour la différentiation 135 Tortolini, t. ii. (1859), pp. 214–230
252.	Note sur l’équation des différences pour une équation 150 donnée de degré quelconque Tortolini, t. ii. (1859), pp. 365, 366
253.	Sur la courbe parallèle à l’ellipse 152 Tortolini, t. III. (1860), pp. 311–316
254.	Sur la surface parallèle à l’ellipsoïde 158 Tortolini, t. III. (1860), pp. 345–352
255.	On a Problem of Double Partitions 166 Phil. Mag. t. XX. (1860), pp. 337–341

256.	On a System of Algebraic Equations	171
	Phil. Mag. t. XX. (1860), pp. 341, 342	
257.	On the Cubic Centres of a line with respect to three lines and a line	175
	Phil. Mag. t. XX. (1860), pp. 418–423	
258.	On a Relation between two Ternary Cubic Forms	179
	Phil. Mag. t. XX. (1860), pp. 512–514	
259.	The Problem of Polyhedra	182
	Phil. Trans. t. CXLVII. (for 1857), pp. 183–185	
260.	On the Double Tangents of a Plane Curve	186
	Phil. Trans. t. CXLIX. (for 1859), pp. 193–212	
261.	On the Conic of Five-pointic Contact at any Point of a Plane Curve	207
	Phil. Trans. t. CXLIX. (for 1859), pp. 371–400	
262.	On the Equation of Differences for an Equation of any Order and in particular for the Equations of the Orders two, three, four, and five	240
	Phil. Trans. t. CL. (for 1860), pp. 93–112	
263.	Demonstration of a Theorem in Finite Differences	262
	Phil. Trans. t. CL. (for 1860), pp. 321–323	
264.	On an Extension of Arbogast's Method of Derivations . .	265
	Phil. Trans. t. CLI. (for 1861), pp. 37–43	
265.	Addition to the Memoir on an Extension of Arbogast's Method of Derivations	276
	Now first published, 1891	
266.	On the Equation for the Product of the Differences of all but one of the Roots of a given Equation	276
	Phil. Trans. t. CLI. (for 1861), pp. 45–59	
267.	On the Porism of the In-and-circumscribed Polygon	292
	Phil. Trans. t. CLI. (for 1861), pp. 225–239	
268.	On a New Auxiliary Equation in the Theory of Equations of the Fifth Order	309
	Phil. Trans. t. CLI. (for 1861), pp. 263–276	

	PAGE	
269.	A Seventh Memoir on Quantics	325
	Phil. Trans. t. CLI. (for 1861), pp. 277–292	
270.	On the Double Tangents of a Curve of the Fourth Order	342
	Phil. Trans. t. CLI. (for 1861), pp. 357–362	
271.	Sur l’invariant le plus simple d’une fonction quadratique bi-ternaire, et sur le résultant de trois fonctions quadratiques ternaires	349
	Crelle, t. LVII. (1860), pp. 139–148	
272.	Démonstration d’un théorème de Jacobi par rapport au problème de Pfaff	359
	Crelle, t. LVII. (1860), pp. 273–277	
273.	Note sur la transformation de Tschirnhausen	364
	Crelle, t. LVIII. (1861), pp. 259–262	
274.	Deuxième Note sur la transformation de Tschirnhausen	368
	Crelle, t. LVIII. (1861), pp. 263–269	
275.	On Tschirnhausen’s Transformation	375
	Phil. Trans. t. CLI. (1861), pp. 561–578	
276.	On the Analytical Theory of the Conic	395
	Phil. Trans. t. CLII. (1862), pp. 639–662	
277.	On the Wave Surface	420
	Quart. Math. Jour. t. III. (1860), pp. 16–22	
278.	Note on the Singular Solutions of Differential Equations	426
	Quart. Math. Jour. t. III. (1860), pp. 36, 37	
279.	On a Theorem relating to Spherical Conics	428
	Quart. Math. Jour. t. III. (1860), p. 53	
280.	On the Conics which touch four given lines	429
	Quart. Math. Jour. t. III. (1860), pp. 94–96	

281.	Note on the Wave Surface	432
	Quart. Math. Jour. t. III. (1860), pp. 142–144	
282.	On a particular case of Castillon's Problem	435
	Quart. Math. Jour. t. III. (1860), pp. 157–164	
283.	On a Theorem relating to Homographic Figures	442
	Quart. Math. Jour. t. III. (1860), pp. 177–180	
284.	On a New Analytical Representation of Curves in Space	446
	Quart. Math. Jour. t. III. (1860), pp. 225–236	
285.	On the System of Conics having double contact with each other	456
	Quart. Math. Jour. t. III. (1860), pp. 246–250	
286.	Note on the value of certain Determinants the terms of which are the squared distances of Points in a Plane or in Space	460
	Quart. Math. Jour. t. III. (1860), pp. 275–277	
287.	Note on the Equation for the squared differences of the Roots of a Cubic Equation	463
	Quart. Math. Jour. t. III. (1860), pp. 307–309	
288.	Note on the Curvature of a Plane Curve at a double point, and on the Curvature of Surfaces	466
	Quart. Math. Jour. t. III. (1860), pp. 322–326	
289.	On some Numerical Expansions	470
	Quart. Math. Jour. t. III. (1860), pp. 366–369	
290.	A Discussion of the Sturmian Constants for cubic and quar- tic Equations	473
	Quart. Math. Jour. t. IV. (1861), pp. 7–12	
291.	On the Demonstration of a Theorem relating to the Mo- ments of Inertia of a Solid Body	478
	Quart. Math. Jour. t. IV. (1861), pp. 25–27	
292.	A Theorem in Conics	481
	Quart. Math. Jour. t. IV. (1861), pp. 131–133	
293.	On a Certain System of Functional Symbols	484
	Quart. Math. Jour. t. IV. (1861), pp. 225–230	

	PAGE
294. On a New Analytical Representation of Curves in Space Quart. Math. Jour. t. V. (1862), pp. 81–86	490
295. On the Construction of the ninth point of Intersection of the Cubics which pass through eight given points Quart. Math. Jour. t. V. (1862), pp. 222–233	495
296. On the Conics which pass through the four foci of a given Conic Quart. Math. Jour. t. V. (1862), pp. 275–280	505
297. On some Formulæ relating to the Distances of a point from the vertices of a Triangle and to the Problem of Tactions Quart. Math. Jour. t. V. (1862), pp. 381–384	510
298. Report on the Progress of the Solution of certain Special Problems of Dynamics Report of the Brit. Assoc. for the Advancement of Science, 1862, pp. 184–252	513
299. Mathematics, recent Terminology in English Cyclopædia, t. V. (1860), pp. 534–542	594
Notes and References	609
Plate to face p. 134	

Classification

Geometry

Reciprocal Surfaces,	227
Analytical Representation of Curves in Space,	284, 294
Contour and Slope Lines,	246
Polyhedra,	241, 242, 259
Conics,	238, 261, 276, 279, 280, 285, 292, 296
Cubic Curves, Cones, &c.,	249, 257, 258, 295
Curvature of Curves and Surfaces,	288
Double Tangents,	260, 270
Envelopes, Parallel Curves and Surfaces,	250, 253, 254
Homographic Figures,	283
In-and-circumscribed Polygon,	267, 282
MacCullagh's Polar Plane,	226
Spherical Triangles,	240
Wave Surface,	277, 281

Analysis

Mathematics, Recent Terminology in,	299
Dynamics. Report on the Progress of	298
the solution of certain Special Problems,	
Class of Problems in,	225
Moments of Inertia,	291
Quantics, binary and ternary, &c.,	232, 233, 269, 271
Equation of Differences, Quintic Equation, &c.,	252, 262, 266, 268, 287
Tschirnhausen's Transformation,	273, 274, 275
Theory of Numbers,	231, 236, 239
Determinants,	237, 286, 297
Derivations (Arbogast),	264, 265
Differentiation,	251
Differential Equations,	272, 278
Finite Differences; (Electricity),	244, 245; 263
Elimination,	223, 230
Elliptic Integrals, Modular Functions,	235, 293
Integral Calculus, Definite Integrals,	224, 228
Equations,	248, 256
Groups,	243
Numerical Expansions,	289
Partitions,	255
Reversion of Series,	229, 234
Sturmian Constants,	290
Trees,	247

Kapitel 1

223. Note sur un théorème général par rapport à l'élimination.

[From the *Annali di Scienze Matematiche e Fisiche* (Tortolini), vol. vii. (1856), pp. 454–458.]

SOIENT

$\phi = (a, b, c \dots$ —————

deux Équations homogènes quelconques entre les variables x, y , et représentons par R la résultante des deux fonctions ϕ, ψ , de manière que $R = 0$ sera la condition pour que les deux équations $\psi = 0, \phi = 0$ puissent avoir lieu. On sait depuis longtemps que les valeurs des variables x, y , qui satisfont à la fois aux deux équations $\psi = 0, \phi = 0$ sont données⁽¹⁾ par les conditions (équivalentes à une seule condition)

$$\begin{aligned} \text{etc.} &= x^m : \frac{m}{1} x^{m-1} y : \frac{m(m-1)}{1 \cdot 2} x^{m-2} y^2 : \text{etc.} \\ &= \frac{\partial R}{\partial a} : \frac{\partial R}{\partial b} : \frac{\partial R}{\partial c} : \dots = \frac{\partial \phi}{\partial a} : \frac{\partial \phi}{\partial b} : \frac{\partial \phi}{\partial c} : \dots, \\ \text{etc.} &= x^n : \frac{n}{1} x^{n-1} y : \frac{n(n-1)}{1 \cdot 2} x^{n-2} y^2 : \text{etc.} \\ &= \frac{\partial R}{\partial \alpha} : \frac{\partial R}{\partial \beta} : \frac{\partial R}{\partial \gamma} : \dots = \frac{\partial \psi}{\partial \alpha} : \frac{\partial \psi}{\partial \beta} : \frac{\partial \psi}{\partial \gamma} : \dots; \end{aligned}$$

ce qui suppose cependant que les coefficients $a, b, c, \dots \alpha, \beta, \gamma, \dots$, sont des quantit  s absolument arbitraires: les conditions dont il s'agit peuvent aussi s'  crire sous la forme

$$\frac{\partial R}{\partial a} : \frac{\partial R}{\partial b} : \frac{\partial R}{\partial c} : \dots = \frac{\partial \phi}{\partial a} : \frac{\partial \phi}{\partial b} : \frac{\partial \phi}{\partial c} : \dots,$$

$$\frac{\partial R}{\partial \alpha} : \frac{\partial R}{\partial \beta} : \frac{\partial R}{\partial \gamma} : \dots = \frac{\partial \psi}{\partial \alpha} : \frac{\partial \psi}{\partial \beta} : \frac{\partial \psi}{\partial \gamma} : \dots$$

¹Il va sans dire que ce n'est que la valeur de $x : y$ laquelle est d  termin  e; mais dans la th  orie des fonctions homog  nes les valeurs absolues n'important rien, et on peut dire que les valeurs x, y sont d  termin  es, quand $x : y$ est d  termin  e: on   vite par cette locution des longueurs tr  s-ennuyeantes.

Or M. Schldans son excellent m  moire "Ueber die Resultante eines Systemes mehrerer algebraischer Gleichungen," *Trans. de l'Acad. de Vienne*, tom. iv. (1852), a g  n  ralis   ce th  or  me de la mani  re que voici. En consid  rant un nombre quelconque d'  quations $\psi = 0, \phi = 0, \chi = 0, \dots$ entre le m  me nombre de variables $x = 0, y = 0, z = 0, \dots$ et en supposant que a, b, c , etc. soient des quantit  s qui entrent d'une mani  re quelconque dans la fonction ϕ , sans entrer dans les autres fonctions ψ, χ , etc. (il n'est nullement n  cessaire que le nombre des quantit  s $a, b, c, \dots$ soit tel que la fonction ϕ reste absolument arbitraire, le nombre des quantit  s a, b, c , etc. peut m  me se r  duire    2) M. Schlfait voir que l'on a dans ce cas

$$\frac{\partial R}{\partial a} : \frac{\partial R}{\partial b} : \frac{\partial R}{\partial c} : \dots = \frac{\partial \phi}{\partial a} : \frac{\partial \phi}{\partial b} : \frac{\partial \phi}{\partial c} : \dots$$

Voici en effet le raisonnement fort simple dont se sert M. Schlpour   tablir la proposition dont il s'agit. Les   quations $\psi = 0, \phi = 0, \chi = 0$, etc. seront satisfaites par de certaines valeurs de $x : y : z$, etc. en supposant seulement que les quantit  s a, b, c , etc. satisfont    la condition $R = 0$. Donc les   quations $\psi = 0 \phi = 0, \chi = 0$, etc. seront encore satisfaites en donnant des variations infiniment petites quelconques $\delta a, \delta b, \delta c$, etc. aux quantit  s a, b, c , etc. en supposant seulement que ces variations soient telles que l'on ait

$$\delta R = \frac{\partial R}{\partial a} \delta a + \frac{\partial R}{\partial b} \delta b + \frac{\partial R}{\partial c} \delta c + \text{etc.} = 0.$$

Or les Équations $\psi = 0$, $\chi = 0$, etc. qui ne contiennent pas les quantités a , b , c , etc. suffisent seules (c'est-à-dire sans l'aide de l'équation $\phi = 0$) à déterminer les valeurs de $x : y : z$, etc. qui satisfont au système $\psi = 0$, $\phi = 0$, $\chi = 0$, (on suppose toujours l'équation $R = 0$): donc les nouvelles valeurs de $x : y : z$, etc. seront les mêmes qu'auparavant, et l'on doit avoir

$$\delta\phi = \frac{\partial\phi}{\partial a}\delta a + \frac{\partial\phi}{\partial b}\delta b + \frac{\partial\phi}{\partial c}\delta c + \text{etc.} = 0;$$

savoir cette Équation aura lieu en vertu de l'équation $\delta R = 0$ qui est la seule condition à laquelle on a assujetti les variations δa , δb , δc , etc., ce qui donne évidemment les conditions

$$\frac{\partial R}{\partial a} : \frac{\partial R}{\partial b} : \frac{\partial R}{\partial c} : \dots = \frac{\partial\phi}{\partial a} : \frac{\partial\phi}{\partial b} : \frac{\partial\phi}{\partial c} : \dots$$

Cela étant, on peut encore généraliser le théorème de M. Schl: pour cela je suppose que les quantités a , b , c , ... entrent d'une manière quelconque dans les fonctions ϕ , ψ , χ , etc. Les Équations $\phi = 0$, $\psi = 0$, $\chi = 0$, etc. impliquent l'équation $R = 0$, et en donnant aux quantités a , b , c , etc. des variations infiniment petites quelconques δa , δb , δc , etc. qui satisfont à la condition $\delta R = 0$, les Équations

$$\delta\phi = 0, \quad \delta\psi = 0, \quad \delta\chi = 0, \quad \text{etc.}$$

seront satisfaites, car on vient de voir que $\delta\phi = 0$ aura lieu en vertu de $\delta R = 0$; et les mêmes variations δa , δb , δc , etc. étant substituées dans δR et $\delta\psi$, on aura aussi $\delta\psi = 0$ en vertu de $\delta R = 0$, etc. pour $\delta\chi = 0$, etc. On aura ainsi

$$\delta\phi = 0, \quad \delta\psi = 0, \quad \delta\chi = 0, \quad \text{etc.},$$

par des variations δa , δb , δc , etc. arbitraires, assujetties seulement à la condition $\delta R = 0$; et cela pour chaque système de valeurs de x , y , z , etc. qui satisfont aux équations $\phi = 0$, $\psi = 0$, $\chi = 0$, etc. Mais en écrivant ces équations sous la forme

$$\frac{\partial\phi}{\partial a}\delta a + \frac{\partial\phi}{\partial b}\delta b + \frac{\partial\phi}{\partial c}\delta c + \dots = 0,$$

$$\frac{\partial\psi}{\partial a}\delta a + \frac{\partial\psi}{\partial b}\delta b + \frac{\partial\psi}{\partial c}\delta c + \dots = 0,$$

Kapitel 1. Note sur un théorème général par rapport à l'élimination

etc., et les éliminant, on obtient

$$\frac{\partial R}{\partial a}\delta a + \frac{\partial R}{\partial b}\delta b + \frac{\partial R}{\partial c}\delta c + \dots = 0,$$

car c'est de cette manière que s'obtient la résultante. Donc $\delta R = 0$ est une conséquence de $\delta\phi = 0$, $\delta\psi = 0$, etc.; et cela pour chaque système de valeurs de x , y , z , etc. qui satisfont aux équations données. Mais il ne s'ensuit pas que l'équation $\delta R = 0$ soit une identité; elle est une conséquence des équations $\delta\phi = 0$, $\delta\psi = 0$, etc. dans le seul cas où les valeurs de x , y , z , etc. satisfont aux équations $\phi = 0$, $\psi = 0$, etc. Or je dis que dans ce cas, on a pour chaque système de valeurs de x , y , z , etc. qui satisfont à ces équations, non-seulement

$$\delta\phi = 0, \quad \delta\psi = 0, \quad \text{etc. pour} \quad \delta R = 0,$$

mais aussi réciproquement

$$\delta R = 0 \quad \text{pour} \quad \delta\phi = 0, \quad \delta\psi = 0, \quad \text{etc.},$$

et cela sans aucune relation entre les variations δa , δb , etc.; c'est-à-dire que $\delta R = 0$ est une conséquence de $\delta\phi = 0$, $\delta\psi = 0$, etc., et que réciproquement $\delta\phi = 0$, $\delta\psi = 0$, etc. sont des conséquences de $\delta R = 0$. Pour le démontrer, je remarque que l'équation $R = 0$ est la condition nécessaire et suffisante pour l'existence d'un système de valeurs de x , y , z , etc. satisfaisant aux équations $\phi = 0$, $\psi = 0$, etc. Or si pour un tel système on a $\delta\phi = 0$, $\delta\psi = 0$, etc., c'est-à-dire si les équations variations sont compatibles, les mêmes équations variations avec les mêmes valeurs de δa , δb , etc., seront aussi compatibles après qu'on aura donné aux quantités x , y , z , etc. de nouvelles valeurs infiniment voisines des premières (les quantités a , b , etc. restant les mêmes); donc l'équation variée $R + \delta R = 0$ aura lieu pour les valeurs variées de x , y , z , etc., et par conséquent $\delta R = 0$. Réciproquement, si $\delta R = 0$, cela exprime que les équations variées admettent un système de solutions; mais pour le système primitif de valeurs on a par hypothèse $\phi = 0$, $\psi = 0$, etc.; donc aussi $\delta\phi = 0$, $\delta\psi = 0$, etc. Cela étant, les équations $\delta\phi = 0$, $\delta\psi = 0$, etc. entraînant $\delta R = 0$ et réciproquement, les coefficients des variations δa , δb , etc. doivent être proportionnels, et l'on a le théorème général

$$\frac{\partial R}{\partial a} : \frac{\partial R}{\partial b} : \frac{\partial R}{\partial c} : \dots = \frac{\partial \phi}{\partial a} : \frac{\partial \phi}{\partial b} : \frac{\partial \phi}{\partial c} : \dots = \frac{\partial \psi}{\partial a} : \frac{\partial \psi}{\partial b} : \frac{\partial \psi}{\partial c} : \dots = \frac{\partial \chi}{\partial a} : \frac{\partial \chi}{\partial b} : \frac{\partial \chi}{\partial c} : \dots \quad (1.1)$$

et l'on voit que ce théorème contient comme cas particulier celui de M. Schlet celui du système de deux équations homogènes.

Londres, 2 Juillet, 1856.

Kapitel 1. Note sur un théorème général par rapport à l'élimination

Kapitel 2

224. Sur un théorème d'Abel. Note.

[From the *Annali di Scienze Matematiche e Fisiche* (Tortolini), vol. viii. (1857), pp. 201–203.]

Il y a un petit mémoire d'Abel qui porte le titre “Ueber die Functionen welche der Gleichung $\phi(x) + \phi(y) = \psi(xf(y) + yf(x))$ genügt-hun” (*Crelle*, tom. ii. (1827), pp. 386–394). La solution du problème est contenue dans les Équations que voici, savoir

$f(x)$ est une fonction définie par l'équation

$$a^2x^2 = (f(x) - ax)^{1+\alpha'}(f(x) + ax)^{1-\alpha'},$$

et on a alors

$$\psi(x) = C \log(f(x) + ax), \quad \phi(x) = C \int \frac{dx}{f(x) + ax},$$

et (en réduisant un peu l'expression donnée dans le mémoire)

$$\psi(x) = C \frac{\sqrt{f(x)^2 - a^2x^2}}{f(x) + ax} = C \left(\frac{f(x) - ax}{f(x) + ax} \right)^{\frac{1}{2}\alpha'}.$$

On a aussi pour $\phi(x)$ cette autre expression en forme d'intégrale indéfinie,

$$\phi(x) = \int \frac{dx}{f(x) + a'x},$$

car le facteur $c\alpha$ par lequel dans le mémoire l'expression à côté droit est multiplié se réduit (comme on voit sans peine) à l'unité. En comparant les deux expressions de $\phi(x)$, on voit qu'il est permis de prendre l'intégrale depuis $x = 0$, pourvu qu'on dérive $c = 1$; cela donne

$$\int_0^x \frac{dx}{f(x) + a'x} = \frac{1}{a\alpha'} \log(f(x) + ax),$$

formule très simple pour l'intégration d'une expression algébrique laquelle ne peut pas s'exprimer à moyen de radicaux.

On obtient une autre propriété de cette fonction $f(x)$ en substituant les valeurs des fonctions ϕ et ψ dans l'équation originale

$$\phi(x) + \phi(y) = \psi(xf(y) + yf(x));$$

cela donne d'abord

$$\frac{1}{f(x) + a'x} + \frac{1}{f(y) + a'y} = \frac{a\alpha'}{\log \frac{f(x)-ax}{f(x)+ax} \cdot \frac{f(y)-ay}{f(y)+ay}} \cdot \frac{1}{xf(y) + yf(x)},$$

et de là en réduisant on obtient l'équation fonctionnelle très simple

$$f(x)f(y) + a^2xy = f(xf(y) + yf(x)).$$

Je remarque que l'on peut sans perte de généralité écrire $a = 1$, et $\alpha = 1$: je mets β au lieu de α' , et j'écris aussi pour plus de simplicité $f(x) = X$, $f(y) = Y$. On a alors pour déterminer la fonction $X(=f(x))$, l'équation

$$x^2 = (X - x)^{1+\beta}(X + x)^{1-\beta},$$

équation dans laquelle on pourrait remplacer les exposants $1+\beta$, $1-\beta$ par deux quantités quelconques.

La formule d'intégration devient

$$\int_0^x \frac{dx}{X + \beta x} = \frac{1}{\beta} \log(X + x),$$

formule que l'on peut vérifier sans peine à moyen de celle-ci,

$$(X + \beta x)X' = x + \beta X,$$

que l'on obtient en différentiant l'équation pour X . L'équation fonctionnelle sera

$$XY + xy = f(xY + yX),$$

c'est-à-dire en écrivant $xY + yX = z$, $XY + xy = Z$ on doit avoir

$$(Z - z)^2 + \beta^2(Z + z)^2 = 1$$

et cela se vérifie tout de suite à moyen des Équations

$$Z - z = (X - x)(Y - y), \quad Z + z = (X + x)(Y + y).$$

Je remarque aussi qu'en prenant le quotient des dérivés de cette équation par rapport à x et y on obtient

$$\frac{X' - Y'}{X'Y' - 1} = \frac{X}{x} \cdot \frac{Y}{y} \cdot \left(\frac{X}{x} - \frac{Y}{y} \right),$$

laquelle est une propriété de la fonction X et sa dérivée X' .

Londres, 17 Juillet, 1857.

Kapitel 3

225. On a Class of Dynamical Problems.

[From the *Proceedings of the Royal Society of London*, vol. viii. (1857), pp. 506–511.]

THERE are a class of dynamical problems which, so far as I am aware, have not been considered in a general manner. The problems referred to (which might be designated as *continuous-impact* problems) are those in which the system is continually taking into connexion with itself particles of infinitesimal mass (i.e. of a mass containing the increment of time dt as a factor), so as not itself to undergo any abrupt change of velocity, but to subject to abrupt changes of velocity the particles so taken into connexion. For instance, a problem of the sort arises when a portion of a heavy chain hangs over the edge of a table, the remainder of the chain being coiled or heaped up close to the edge of the table; the part hanging over constitutes the moving system, and in each element of time dt , the system takes into connexion with itself, and sets in motion with a finite velocity, an infinitesimal length ds of the chain; in fact, if v be the velocity of the part which hangs over, then the length $v dt$ is set in motion with the finite velocity v . The general equation of dynamics applied to the case in hand will be

$$\Sigma \left(\frac{d^2x}{dt^2} - X \right) \delta x + \left(\frac{d^2y}{dt^2} - Y \right) \delta y + \left(\frac{d^2z}{dt^2} - Z \right) \delta z dm + \Sigma (u \delta \xi + v \delta \eta + w \delta \zeta) d\mu = 0, \quad (3.1)$$

where the first term requires no explanation: in the second term ξ, η, ζ denote the coordinates at the time t of the particle $d\mu$, which then comes into connexion with the system; $\Delta u, \Delta v, \Delta w$ are the finite increments of velocity (or, if the particle is originally at rest, then the finite velocities) of the particle $d\mu$ the instant that it has come into connexion with the system; $\delta\xi, \delta\eta, \delta\zeta$ are the virtual velocities of the same particle $d\mu$ considered as having come into connexion with and forming part of the system. The summation extends to the several particles or to the system of particles $d\mu$ which come into connexion with the system at the time t ; of course, if there is only a single particle $d\mu$, the summatory sign Σ is to be omitted. The values of $\Delta u, \Delta v, \Delta w$ are

$$\bar{u} - u, \quad \bar{v} - v, \quad \bar{w} - w,$$

if by $\bar{u}, \bar{v}, \bar{w}$ we understand the velocities of $d\mu$ parallel to the axes, after it has come into connexion with the system; but it is to be observed, that considering ξ, η, ζ as the coordinates of the particle $d\mu$ which is continually coming into connexion with the system, then if the problem were solved and ξ, η, ζ given as functions of t (and, when there is more than one particle $d\mu$, of the constant parameters which determine the particular particle), $\frac{d\xi}{dt}$, etc., in the sense just explained, cannot be obtained by simple differentiation from such values of ξ , etc.; in fact, ξ, η, ζ so given as functions of t , belong at the time t to one particle, and at the time $t + dt$ to the next particle, but what is wanted is the increment in the interval dt of the coordinates ξ, η, ζ of one and the same particle.

Suppose as usual that x, y, z , and in like manner that ξ, η, ζ are functions of a certain number of independent variables θ, ϕ , etc., and of the constant parameters which determine the particular particle dm or $d\mu$, of which x, y, z , or ξ, η, ζ are the coordinates; parameters, that is, which vary from one particle to another, but which are constant during the motion for one and the same particle. The summations are in fact of the nature of definite integrations in regard to these constant parameters, which therefore disappear altogether from the final results. The first term,

$$\Sigma \left\{ \left(\frac{d^2x}{dt^2} - X \right) \delta x + \dots \right\} dm,$$

may be reduced in the usual manner to the form

$$\Sigma(\Theta \delta\theta + \Phi \delta\phi + \dots),$$

where, writing as usual θ' , ϕ' , etc. for $\frac{d\theta}{dt}$, $\frac{d\phi}{dt}$, etc., we have

$$\Theta = \frac{d}{dt} \frac{\partial T}{\partial \theta'} - \frac{\partial T}{\partial \theta} - \frac{\partial V}{\partial \theta}, \quad \Phi = \frac{d}{dt} \frac{\partial T}{\partial \phi'} - \frac{\partial T}{\partial \phi} - \frac{\partial V}{\partial \phi},$$

(this supposes that $X dx + Y dy + Z dz$ is an exact differential); only it is to be observed that in the problems in hand, the mass of the system is variable, or what is the same thing, the variables θ , ϕ , etc., are introduced into T and V through the limiting conditions of the summation or definite integration, besides entering directly into T and V in the ordinary manner. And in forming the differential coefficients $\frac{d}{dt} \frac{\partial T}{\partial \theta'}$, $\frac{\partial T}{\partial \theta}$, $\frac{\partial V}{\partial \theta}$, etc., it is necessary to consider the variables θ , ϕ , etc., in so far as they enter through the limiting conditions as exempt from differentiation, so that the expressions just given for Θ , Φ , etc., are, in the case in hand, rather conventional representations than actual analytical values; this will be made clearer in the sequel by the consideration of the before-mentioned particular problem.

Considering next the second term, or

$$\Sigma\{(\bar{u} - u)\delta\xi + (\bar{v} - v)\delta\eta + (\bar{w} - w)\delta\zeta\} d\mu,$$

we have here

$$\delta\xi = a \delta\theta + b \delta\phi + \dots, \quad \delta\eta = a' \delta\theta + b' \delta\phi + \dots, \quad \delta\zeta = a'' \delta\theta + b'' \delta\phi + \dots,$$

where a , b , a' , etc., are functions of the variables θ , ϕ , etc., and of the constant parameters which determine the particular particle $d\mu$. The virtual velocities or increments $\delta\theta$, $\delta\phi$, etc., are absolutely arbitrary, and if we replace them by $d\theta$, $d\phi$, etc., the actual increments of θ , ϕ , etc. in the interval dt during the motion, then $\delta\xi$, $\delta\eta$, $\delta\zeta$ will become $\frac{d\xi}{dt} dt$, $\frac{d\eta}{dt} dt$, $\frac{d\zeta}{dt} dt$, in the sense before attributed to $\frac{d\xi}{dt}$, $\frac{d\eta}{dt}$, $\frac{d\zeta}{dt}$.

The particle $d\mu$ will contain dt as a factor, and the other factor will contain the differentials, or (as the case may be) products of differentials, of the constant parameters which determine the particular particle $d\mu$. We have thus the means of expressing the second line in

the proper form; and if we write

$$\begin{aligned}\Sigma(a\bar{u} + a'\bar{v} + a''\bar{w}) d\mu &= A dt, \\ \Sigma(b\bar{u} + b'\bar{v} + b''\bar{w}) d\mu &= B dt, \\ \Sigma(a\bar{v} + a'\bar{v} + a''\bar{w}) d\mu &= H dt, \quad \text{etc.}, \\ \Sigma(a\bar{u} + a'v + a''w) d\mu &= -P dt, \\ \Sigma(b\bar{u} + b'v + b''w) d\mu &= -Q dt, \quad \text{etc.},\end{aligned}$$

then the second line becomes

$$Ad\theta + Bd\phi + \dots - (P\delta\theta + Q\delta\phi + \dots);$$

and observing that $d\theta = \theta' dt$, $d\phi = \phi' dt$, etc., this is

$$\{A\theta' + B\phi' + \dots - (P\delta\theta + Q\delta\phi + \dots)\} dt,$$

or say it is $-dK$, if for a moment

$$K = -(A\theta' + B\phi' + \dots)dt + (P\delta\theta + Q\delta\phi + \dots)dt.$$

But in the formation of the equations of motion, the differentials $d\theta$, $d\phi$, etc., which enter into K , not explicitly, but through the coefficients A , etc., must be considered as exempt from differentiation, so that the preceding expression for the second line by means of the function K is rather a conventional representation than an actual analytical value.

Uniting the two terms, and equating to zero the coefficients of $\delta\theta$, $\delta\phi$, etc., we obtain finally the equations of motion in the form

$$\frac{d}{dt} \frac{\partial T}{\partial \theta'} - \frac{\partial T}{\partial \theta} - \frac{\partial V}{\partial \theta} - \frac{\partial K}{\partial \theta'} = 0, \quad (3.2)$$

$$\frac{d}{dt} \frac{\partial T}{\partial \phi'} - \frac{\partial T}{\partial \phi} - \frac{\partial V}{\partial \phi} - \frac{\partial K}{\partial \phi'} = 0, \quad \text{etc.}, \quad (3.3)$$

where the several symbols are to be taken in the significations before explained.

In the particular problem, let z be measured vertically downwards from the plane of the table, then $Z = g$, and repeating for the particular case the investigation *ab initio*, the general equation of motion is

$$\Sigma \left(\frac{d^2 z}{dt^2} - g \right) \delta z dm + \Sigma \bar{v} \delta \zeta d\mu = 0.$$

Let s be the length in motion, or, what is the same thing, the z coordinate of the lower extremity; and suppose also that the mass of a unit of length is taken equal to unity, we have $\delta z = \delta s$, $\frac{d^2 z}{dt^2} = \frac{d^2 s}{dt^2}$, $\bar{v} = v = \frac{ds}{dt}$, and the summation or integration with respect to z is from $z = 0$ to $z = s$, whence

$$\Sigma \left(\frac{d^2 z}{dt^2} - g \right) \delta z \, dm = \left(\frac{d^2 s}{dt^2} - g \right) s \, \delta s;$$

which is of the form

$$\left(\frac{d}{dt} \frac{\partial T}{\partial s'} - \frac{\partial T}{\partial s} - \frac{\partial V}{\partial s} \right) \delta s,$$

$$T = \frac{1}{2} \bar{s} s'^2, \quad V = -g \bar{s} s,$$

where the bar is used to denote exemption from differentiation, but ultimately $\bar{s}$ is to be replaced by s . Considering now the second term, here $\bar{\zeta} = 0$, but $\delta \zeta = \delta s$, and thence $\bar{v} \delta \zeta = s'^2$. Moreover, $d\mu = s' dt$, and thence finally the second term is s'^3 , which is of the form $\frac{\partial K}{\partial s'}$, if

$$K = \frac{1}{2} \bar{s}' s'^2, \quad \text{or} \quad K = \frac{1}{2} \bar{s}' s'^2,$$

the bar having the same signification as before, but after the differentiation $s' = \bar{s}'$.

The resulting equation is

$$\frac{d}{dt}(\bar{s} s') + g s - \frac{3}{2} \bar{s}' s'^2 = 0,$$

which may be written in the form

$$s \frac{d(s'^2)}{ds} = g s'^2,$$

and the first integral is therefore

$$s'^2 = \frac{2}{3} g(s - a),$$

where a is the length hanging over at the commencement of the motion. If $a = 0$, then the equation is

$$s'^2 = \frac{2}{3} g s,$$

and integrating from $t = 0$, $2\sqrt{s} = \sqrt{\frac{2}{3}g}t$, or finally $s = \frac{1}{6}gt^2$, so that the motion is the same as that of a body falling under the influence of a constant force $\frac{1}{3}g$. It is perhaps worth noticing that the differential equation may be obtained as follows:—We have, in the first instance, a mass s moving with a velocity s' , and after the particle $ds (= s' dt)$ has been set in motion, a mass $s + s' dt$ moving say with a velocity $s' + \delta s'$, whence neglecting for the moment the effect of gravity on the mass s , the momentum of the mass in motion will be constant, or we shall have

$$ss' = (s + s' dt)(s' + \delta s') = ss' + s'^2 dt + s \delta s',$$

and therefore $s \delta s' = -s'^2 dt$. Hence, adding on the right-hand side the term $gs dt$ arising from gravity, and substituting $\frac{d^2s}{dt^2}dt$ for $\delta s'$, we have the equation, $s \frac{d^2s}{dt^2} = s'^2 - \left(\frac{ds}{dt}\right)^2$, as before.

Kapitel 4

226. On Professor Mac Cullagh's Theorem of the Polar Plane.

[From the *Proceedings of the Royal Irish Academy*, vol. vi. (1858), pp. 481–491.]

A RAY of polarized light, incident on the surface of an extraordinary medium, may give rise to a reflected ray and a single refracted ray; but this will be the case only for a particular position, or positions, of the plane of polarization of the incident ray. According to Professor Mac Cullagh's theory, the plane of polarization, and the relative vibrations of the three rays, are deduced from two assumed principles, which may be referred to as

1°. The principle of equivalent vibrations.

2°. The principle of equivalent moments.

And from these principles are deduced

3°. The principle of *vis viva*.

4°. The theorem of the polar plane.

The directions of the vibrations are completely determined by means of 4°, the theorem of the polar plane; and the relative magnitudes are then given by 1°, the principle of equivalent vibrations. The

other principles, viz.:—2°, the principle of equivalent moments, and 3°, the principle of *vis viva*, must therefore follow as mere geometrical consequences from the first-mentioned two principles, or theorems; and I have found that the deduction depends immediately upon the following two theorems in spherical trigonometry.

Suppose (Fig. 1) that $RR'R''$ is a spherical triangle, and let W be any point in the base RR'' , and N be the central point of the base; then joining WR' and producing this arc (in the direction from W to R') to a point L , such that

$$\cot WL = \frac{\sin^2 NW}{\sin RW \sin WR''} \tan WR',$$

and joining NR' , then

Theorem I.

$$\frac{\sin^2 R''LR' - \sin^2 RLR'}{\sin^2 R''LR} = \frac{\sin NW \cos NR'}{\cos WR' \sin NR \cos NR};$$

and if we suppose also, that an arc through N , perpendicular to the base RR'' , cuts LR , LR' , and LR'' produced in the points U , U' , U'' , then

Theorem II.

$$\begin{aligned} \sin R''LR \cos RU \sin NUR + \sin RLR'' \cos R''U'' \sin NU''R'' \\ = \sin R''LR' \cos WR' \sin NR \cos R'U' \sin NU'R'. \end{aligned}$$

The present memoir contains the proof of the two theorems, and the application of them to the optical theory.

To prove the first theorem, I write for shortness R , R'' , W to denote the angles LRR'' , $LR''R$, NWR' , respectively; we have then,

$$\begin{aligned} \frac{\sin^2 R'LR'' - \sin^2 RLR'}{\sin^2 R''LR} &= \frac{\sin(R'LR'' - RLR')}{\sin R'LR''} \\ &= \frac{1}{\sin R''LR} \{ \sin R'LR'' \cos RLR'' - \sin RLR'' \cos R'LR'' \}. \end{aligned}$$

Observing that $\frac{\sin R''}{\sin LR''}$, $\frac{\sin R}{\sin LR}$ are each equal to $\frac{\sin R'LR''}{\sin R'R}$, this becomes

$$= \frac{1}{\sin RR'' \sin^2 LW} \left\{ \sin R''W (\cos RW - \cos LR \cos LW) - \sin RW (\cos R''W - \cos LR'' \cos LW) \right\};$$

and, substituting for $\cos LR$, $\cos LR''$, the values

$$\cos RW \cos LW + \sin RW \sin LW \cos W, \quad \cos R''W \cos LW - \sin R''W \sin LW \cos W$$

the foregoing expression becomes

$$= \frac{1}{\sin RR'' \sin^2 LW} \times \left\{ \sin R''W (\cos RW \sin^2 LW + \sin RW \sin LW \cos LW \cos W) - \sin RW (\cos R''W \sin^2 LW - \sin R''W \sin LW \cos LW \cos W) \right\},$$

$$= \frac{1}{\sin RR'' \sin^2 LW} \left\{ \sin R''W \cos RW - \sin RW \cos R''W + 2 \cot LW \sin RW \sin R''W \cos W \right\},$$

$$= \frac{1}{\sin RR'' \sin^2 LW} \left\{ \sin(R''W - RW) + 2 \cot LW \sin RW \sin R''W \cos W \right\};$$

and, putting $R''W - RW = 2NW$, and substituting also for $\cot LW$ its value, which gives $\cot LW \sin RW \sin R''W = \sin^2 NW \tan WR'$, the expression becomes

$$= \frac{1}{\sin RR'' \sin^2 LW} \left\{ \sin 2NW + 2 \sin^2 NW \tan WR' \cos W \right\};$$

but we have

$$\cos W = \frac{\cos RR'' - \cos NW \cos WR'}{\sin NW \sin WR'},$$

and therefore

$$2 \sin^2 NW \tan WR' \cos W = 2 \frac{\sin NW}{\cos WR'} \cos NR' - \sin 2NW;$$

thus the expression becomes

$$= \frac{1}{\sin RR''} \cdot \frac{\sin NW}{\cos WR'} \cdot 2 \cos NR';$$

and $\sin R''R = \sin 2NR' = 2 \sin NR' \cos NR'$, so that finally the expression becomes

$$\frac{\sin NW \cos NR'}{\cos WR' \sin NR' \cos NR'},$$

which proves the theorem.

To prove the second theorem, take as before R, R'', W , to denote the angles $LR''R, LR''R, NWR'$, respectively; and moreover, U, U', U'' to denote the angles $NUR, NU'R', NU''R''$, respectively; then considering, first, the function on the left-hand side, viz.:

$$\sin R'LR'' \cos RU \sin U + \sin RLR' \cos R''U'' \sin U'',$$

we have

$$\sin U = \frac{\sin NR}{\sin RU},$$

$$\cos RU \sin U = \sin NR \cot RU = \sin NR \cos R \cot NR = \cos R \cos NR,$$

and, in like manner,

$$\sin U'' = \frac{\sin NR''}{\sin R''U''},$$

$$\cos R''U'' \sin U'' = \sin NR'' \cot R''U'' = \sin NR'' \cos R'' \cot NR'' = \cos R'' \cos NR'';$$

the expression thus becomes

$$= \cos NR \{ \sin R'LR'' \cos R + \sin RLR' \cos R'' \},$$

which is

$$= \cos NR \left\{ \frac{\sin R''W \sin W}{\sin LR} \cdot \frac{\cos LW - \cos RW \cos RL}{\sin RW \sin LR} + \frac{\sin RW \sin W}{\sin LR''} \cdot \frac{\cos LW - \cos R''W \cos R''L}{\sin R''W \sin R''L} \right\},$$

or, substituting for $\cos RL, \cos R''L$ the values

$$\cos RW \cos LW + \sin RW \sin LW \cos W, \quad \cos R''W \cos LW - \sin R''W \sin LW \cos W,$$

the expression becomes

$$= \frac{\cos NR \sin W}{\sin LR \sin LR''} \{ \sin R''W (\cos LW \sin^2 RW + \sin RW \cos RW \sin LW \cos W) \}$$

$$\begin{aligned}
& + \sin RW (\cos LW \sin^2 R''W - \sin R''W \cos R''W \sin LW \cos W) \}, \\
& = \frac{\cos NR \sin W}{\sin LR \sin LR''} \{ \cos LW (\sin R''W \sin^2 RW + \sin RW \sin^2 R''W) \\
& \quad + \sin LW \cos W \sin RW \sin R''W (\cos RW - \cos R''W) \}.
\end{aligned}$$

But $\sin LR \sin LR'' = \sin^2 W$, and the last factor of the second term is $2 \sin NW \sin WR'$; also $\sin R''W \sin^2 RW + \sin RW \sin^2 R''W = \sin RW \sin R''W (\sin RW + \sin R''W)$; and $\sin RW + \sin R''W = 2 \sin NR' \cos NW$; hence the expression is

$$= \frac{\cos NR}{\sin W} \cdot \sin RW \sin R''W \{ 2 \cos LW \sin NR' \cos NW + 2 \sin LW \cos W \sin NW \sin NR' \}$$

But

$$\frac{\sin RW \sin R''W}{\sin W} = \sin RR'' \frac{\sin R''W}{\sin LR''} = \sin RR'' \frac{\sin R' LR''}{\sin R' R} = 2 \sin NR' \cos NR',$$

and the last factor becomes

$$2 \sin NR' (\cos LW \cos NW + \sin LW \sin NW \cos W) = 2 \sin NR' \cos WR';$$

hence the whole expression is

$$\begin{aligned}
& = 4 \cos NR \sin NR' \cos NR' \cos WR' \sin NR', \\
& = \sin R'' LR' \cos WR' \sin NR \cos R' U' \sin NU' R',
\end{aligned}$$

which proves the theorem.

Application to the Optical Theory.

Let the direction of the incident ray cut the sphere in R , that of the reflected ray in R'' , and that of the refracted ray in R' ; and let RR'' be the base of the spherical triangle $RR'R''$. The theorem of the polar plane gives for the point L the construction above mentioned; and the principle of equivalent vibrations gives for the vibrations of the three rays directions perpendicular to the arcs LR , LR'' , LR' respectively; and magnitudes proportional to $\sin R'' LR'$, $\sin R LR''$, $\sin R LR''$ respectively.

Proceeding to the condition given by the principle of equivalent moments, we have

$$\text{moment of } Rt \text{ round } AH = AR \times Rt \times \cos[AR, \times \text{dist.}(Rt, AH)] \times \sin(Rt, AH);$$

and in Fig. 3, observing that the radius through U is parallel to the perpendicular distance of (Rt, AH) (for LR has the pole T , and NU the pole H) then

$$\cos[AR, \times \text{dist.}(Rt, AH)] = \cos RU, \quad \sin(Rt, AH) = \sin TH,$$

or, since T and H are the poles of LR and NW respectively, $TH = \frac{\pi}{2} - NUR$, and, putting $AR = \lambda$, the moment is

$$= \lambda R \bar{t} \cos RU \sin NUR.$$

Similarly,

$$\text{moment of } R''\bar{t}'' \text{ round } AH = \lambda R''\bar{t}'' \cos R''U'' \sin NU''R'';$$

and for the refracted ray,

$$\text{moment of } R'\bar{t}' \text{ round } AH = \lambda AR' \times R'\bar{t}' \cos R'U' \sin NU'R'.$$

But we have

$$\frac{AR'}{AW} = \frac{\sin NW}{\cos WR'} = \frac{\sin NW}{\cos WR' \sin NR},$$

and therefore the moment is

$$= \lambda R'\bar{t}' \frac{\sin NW}{\cos WR' \sin NR} \cos R'U' \sin NU'R'.$$

Hence the vibrations $\bar{R}t$, $R''\bar{t}''$, $R'\bar{t}'$, as before, are as

$$\sin R''LR' : \sin RLR'' : \sin RLR'',$$

and thus the equation given by the principle of equivalent moments is precisely that of Theorem II.

Kapitel 5

227. On the Theory of Reciprocal Surfaces.

[From the *Proceedings of the Royal Irish Academy*, vol. vii. (1862), pp. 20–28.]

THE PRESENT note is intended to be supplementary to Mr Salmon's memoir "On the Degree of a Surface reciprocal to a given one" (*Trans. R. Irish Acad.*, vol. xxi. pp. 461–488, 1857). I find that Mr Salmon's equations admit of a transformation which appears important in reference to the geometrical theory, and the object of the note is to present the system of equations under the new form.

Mr Salmon writes:

- n , the order of the surface.
- a , the order of the tangent cone drawn from any point to the surface.
- δ , the number of the double edges of the cone.
- κ , the number of its cuspidal edges.
- b , the order of any double curve upon the surface.
- k , the number of apparent double points of the double curve.
- t , the number of triple points on the double curve.
- c , the order of any cuspidal curve on the surface.
- h , the number of apparent double points of the cuspidal curve.
- β , the number of intersections of the double and cuspidal curves which are stationary points on the cuspidal curve.
- γ , the number of intersections which are stationary points on the double curve.
- i , the number of intersections which are not stationary points upon either curve.
- p , the number of the points where the double curve is met by the curve of contact of the tangent cone.
- σ , the number of the points where the cuspidal curve is met by the curve of contact.

And the accented letters denote the corresponding singularities of the reciprocal surface, or, if we choose that they should refer to the given surface, and its tangential or class singularities, then we have:

- n' , the class of the surface.
- a' , the class of the curve of intersection by any plane.
- δ' , the number of double tangents of the curve.
- κ' , the number of its cusps.
- b' , the class of the node-couple developpe.
- k' , the number of apparent double planes of the node-couple developpe.
- t' , the number of triple planes of the node-couple developpe.
- c' , the class of the spinode developpe.
- h' , the number of the apparent double planes of the spinode developpe.
- β' , the number of common tangent planes of the node-couple and spinode developpes, stationary planes of the spinode developpe.
- γ' , the number of common tangent planes, stationary planes of the node-couple developpe.
- i' , the number of the common tangent planes which are not stationary planes either developpe.
- p' , the number of the common tangent planes of the node-couple developpe, and tangent cone.
- σ' , the number of the common tangent planes of the spinode developpe, and tangent cone.

The terminology made use of is that of my paper "On the Sin-

gularities of Surfaces" (*Cambridge and Dublin Mathematical Journal*, vol. vii., 1852), [106]. To explain it, I need only remark that the term *node* is used as synonymous with double point, and the term *spinode* as synonymous with cusp; a spinode plane is a tangent plane meeting the surface in a curve, having a spinode at the point of contact; and a node-couple plane is a double tangent plane, or plane meeting the surface in a curve having two nodes; the term *develope* is used instead of developable surface.

To collect all the formulae, it is proper to write also:

q , the class of the cuspidal curve.
 r , the class of the double curve.
 q' , the order of the spinode develope.
 r' , the order of the node-couple develope,

where q' is what Mr Salmon, who only uses it incidentally in referring to a result of Professor Schl's, calls (after him) Λ .

Mr Salmon obtains, between the twenty-eight quantities,

$$n, a, \delta, \kappa, b, k, t, c, h, \beta, \gamma, i, p, \sigma,$$

$$n', a', \delta', \kappa', b', k', t', c', h', \beta', \gamma', i', p', \sigma',$$

the twenty-one equations,

$$a = n(n-1) - 2b - 3c, \quad (5.1)$$

$$\kappa' = 3n(n-2) - 6b - 8c, \quad (5.2)$$

$$\delta' = \frac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2b+3c) + 2b(b-1) + 6bc + \quad (5.3)$$

$$a(n-2) = \kappa + p + 2\sigma, \quad (5.4)$$

$$b(n-2) = p + 2\beta + 4\gamma + 3t, \quad (5.5)$$

$$c(n-2) = 2\sigma + 4\beta + \gamma, \quad (5.6)$$

$$a(n-2)(n-3) = 2\delta + 2ab + 3ac - 4p - 9\sigma, \quad (5.7)$$

$$b(n-2)(n-3) = 4k + \alpha + 3bc - 9\beta - 6\gamma - 3i - 2p, \quad (5.8)$$

$$c(n-2)(n-3) = 6h + ac + 2bc - 6\beta - 4\gamma - 2i - 3\sigma, \quad (5.9)$$

$$n' = n(n-1)^2 - n(7b+12c) + 4b^2 + 9c^2 + 8b + 15c - 8k - 18h + 1 \quad (*)$$

$$a = n'(n'-1) - 2b' - 3c', \quad (5.10)$$

$$\kappa = 3n'(n'-2) - 6b' - 8c', \quad (5.11)$$

$$\delta = \frac{1}{2}n'(n'-2)(n'^2-9) - (n'^2-n'-6)(2b'+3c') + 2b'(b'-1) + \quad (5.12)$$

$$a'(n'-2) = \kappa' + p' + 2\sigma', \quad (5.13)$$

$$b'(n'-2) = p' + 2\beta' + 4\gamma' + 3t', \quad (5.14)$$

$$c'(n'-2) = 2\sigma' + 4\beta' + \gamma', \quad (5.15)$$

$$a'(n'-2)(n'-3) = 2\delta' + 2a'b' + 3a'c' - 4p' - 9\sigma', \quad (5.16)$$

$$b'(n'-2)(n'-3) = 4k' + a'b' + 3b'c' - 9\beta' - 6\gamma' - 3i' - 2p', \quad (5.17)$$

$$c'(n'-2)(n'-3) = 6h' + a'c' + 2b'c' - 6\beta' - 4\gamma' - 2i' - 3\sigma', \quad (5.18)$$

$$n = n'(n'-1)^2 - n'(7b'+12c') + 4b'^2 + 9c'^2 + 8b' + 15c' - 8k' - 1 \quad (*)$$

to which may be joined

$$q = b^2 - b - 2k - 3\gamma - 6t, \quad (5.19)$$

$$r = c^2 - c - 2h - 3\beta, \quad (5.20)$$

$$q' = b'^2 - b' - 2k' - 3\gamma' - 6t', \quad (5.21)$$

$$r' = c'^2 - c' - 2h' - 3\beta'. \quad (5.22)$$

Considering the twenty-one equations, and taking as data n , b , c , β , γ , h , k , then, by means of the several equations, other than

the two equations marked (*), we may express in terms of the above data $a, \delta, \kappa, t, i, p, \sigma, n', a', \delta', \kappa', b', c', p', \sigma', 4\beta' + 3\gamma' + 3i', 4\beta' + \gamma', 4k' - 3i', 6h' - 2i'$; the quantities which enter into the first of the marked equations are then all given in terms of the above data, and it is clear that the equation must be satisfied identically: the quantities which enter into the second of the marked equations are given in terms of the data and of i', t' , and it is not clear, *a priori*, but that the equation might lead to a relation between the data and i', t' ; it will, however, appear in the sequel that the equation must be satisfied identically, independently of any particular values of t', i' . Thus, Mr Salmon's theory does not determine the values of these two quantities, nor, consequently, the values of β', γ', h', k' ; it does, however, determine the values of the combinations $4\beta' + \gamma', 8k' - 18h'$.

But the twenty-one equations between the twenty-eight quantities may be replaced by seventeen equations between the twenty quantities

$$n, a, \delta, \kappa, b, c, p, \sigma, 4\beta + \gamma, 8k - 18h,$$

$$n', a', \delta', \kappa', b', c', p', \sigma', 4\beta' + \gamma', 8k' - 18h';$$

this will clearly be the case if it is only shown that the equation which gives n' can by the other equations be transformed into one of the form in question; for a similar transformation will, of course, apply to the equation for n , and then we have only to reject the equation containing t , and to replace the two equations which contain i , by the equation given by the elimination of this quantity, and in like manner to reject the equation containing i' , and to replace the two equations containing i' , by the equation given by the elimination of this quantity, and the system will be reduced to the required form.

The reduction of the equation which gives n' is effected as follows; we have

$$(2b+3c)(n-2)(n-3) = 8\kappa+18h+a(2b+3c)+12bc-3\beta-24\gamma-12i-4p-9\sigma,$$

$$3b(n-2) = 6\sigma + 9\gamma + 9t + 3p;$$

and thence

$$\begin{aligned} & (2b+3c)(n-2)(n-3) + 3b(n-2) \\ &= a(2b+3c) + 12bc + 8\kappa + 18h - 12i + 9t - p - 9\sigma - 3(4\beta + \gamma); \end{aligned}$$

and consequently

$$\begin{aligned} & -8\kappa - 18h + 18\beta + 12\gamma + 12i - 9t \\ & = \{a - (n-2)(n-3)\}(2b+3c) - 3b(n-2) + 12bc - \beta - 9\sigma - 3(4\beta + \gamma), \end{aligned}$$

which (observing that the left-hand side is precisely the combination of terms which enters into the equation for n') shows that the reduction is possible; to complete it, putting for a its value $n(n-1) - 2b - 3c$, we have

$$\begin{aligned} & -8\kappa - 18h + 18\beta + 12\gamma + 12i - 9t \\ & = b(6n-6) + c(12n-18) - 4b^2 - 9c^2 - p - 9\sigma - 3(4\beta + \gamma); \end{aligned}$$

and substituting this value in the equation for n , we obtain

$$n' = n(n-1)^2 - 6(2n-2) - 3c - p - 9\sigma - 3(4\beta + \gamma).$$

Some of the other equations admit of simplification: the equation

$$a(n-2)(n-3) = 2\delta + a(2b+3c) - 4p - 9\sigma,$$

if we put for a its value $n(n-1) - 2b - 3c$, becomes

$$(4n-6-2b-3c)(n-2)(n-3) = 2\delta - 4p - 9\sigma,$$

and the prescribed combination

$$(2b-3c)(n-2)(n-3) = (8\kappa - 18h) + a(2b-3c) - 4p + 9\sigma,$$

gives in like manner, putting for a its value,

$$(-n^2 + n + 4b)(n-2)(n-3) = (8\kappa - 18h) - 4p + 9\sigma.$$

The system of seventeen equations then is:

$$a = a', \quad (5.23)$$

$$a' = n(n-1) - 2b - 3c, \quad (5.24)$$

$$\kappa' = 3n(n-2) - 6b - 8c, \quad (5.25)$$

$$\delta' = \frac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2b+3c) + \quad (5.26)$$

$$a(n-2) = \kappa + p + 2\sigma, \quad (5.27)$$

$$c(n-2) = 2\sigma + (4\beta + \gamma), \quad (5.28)$$

$$(4n-6-2b-3c)(n-2)(n-3) = 2\delta - 4p - 9\sigma, \quad (5.29)$$

$$(-n^2+n+4b)(n-2)(n-3) = (8\kappa-18h) - 4p + 9\sigma, \quad (5.30)$$

$$n' = n(n-1)^2 - 6(2n-2) - 3c - p - 9\sigma - 3(4\beta + \gamma) \quad (5.31)$$

$$a = n'(n'-1) - 2b' - 3c', \quad (5.32)$$

$$\kappa = 3n'(n'-2) - 6b' - 8c', \quad (5.33)$$

$$\delta = \frac{1}{2}n'(n'-2)(n'^2-9) - (n'^2-n'-6)(2b'+3c') + \quad (5.34)$$

$$a'(n'-2) = \kappa' + p' + 2\sigma', \quad (5.35)$$

$$c'(n'-2) = 2\sigma' + (4\beta' + \gamma'), \quad (5.36)$$

$$(4n'-6-2b'-3c')(n'-2)(n'-3) = 2\delta' - 4p' - 9\sigma', \quad (5.37)$$

$$(-n'^2+n'+4b')(n'-2)(n'-3) = (8\kappa'-18h') - 4p' + 9\sigma', \quad (5.38)$$

$$n = n'(n'-1)^2 - 6(2n'-2) - 3c' - p' - 9\sigma' - 3(4\beta' + \gamma') \quad (5.39)$$

We may here take as data $n, b, c, 4\beta + \gamma, 8\kappa - 18h$; the equations exclusively of the two marked (*), then give $a, \delta, \kappa, p, \sigma, n', a, \delta', \kappa', h', c', p', \sigma', 4\beta' + \gamma', 8\kappa' - 18h'$; and then, since all the quantities entering into the two excepted equations are expressed in terms of the data, these equations are satisfied identically, and it is easy to see that this proves what was before assumed, viz., that in the system of twenty-one equations, the second of the equations marked (*) is satisfied identically.

Several of the other quantities may be expressed without difficulty in terms of the data $n, b, c, 4\beta + \gamma, 8\kappa - 18h$: we in fact have (besides

a, a', κ, δ' , which are originally so expressed):

$$2\sigma = (n-2)c - (4\beta + \gamma),$$

$$8p = (16n-24)b - (10n-18)c - 2(4b^2 - 9c^2) + 2(8\kappa - 18h) - 9(4\beta + \gamma),$$

$$8\kappa = 8n(n-1)(n-2) - (32n-56)b - (17n-46)c \\ + 2(4b^2 - 9c^2) - 2(8\kappa - 18h) + 17(4\beta + \gamma),$$

$$2\delta = n(n-1)(n-2)(n-3) - (4n^2 - 20n + 24)b - (6n^2 - 15n + 18)c \\ + 12bc + 18c^2 + (8\kappa - 18h) - 9(4\beta + \gamma),$$

$$8n' = 8n(n-1)^2 - (32n-40)b - (21n-30)c + 2(4b^2 - 9c^2) - 2(8\kappa - 18h) + 21(4\beta + \gamma) \\ c' = 4n(n-1)(n-2) - (16n-28)b - (10n-26)c \\ + (4b^2 - 9c^2) - (8\kappa - 18h) + 10(4\beta + \gamma),$$

but the expressions for the remaining quantities, viz., $b', p', \sigma', 4\beta' + \gamma', 8\kappa' - 18h'$ would be very complicated. If we suppose that $b, c, 4\beta + \gamma, 8\kappa - 18h$, vanish, or, what is the same thing, attend only to the terms which contain n alone, we have:

$$2\delta' = n(n-1)(n-2)(n^2 - n + 12),$$

$$p' = n(n-2)(n^2 - n + 12),$$

$$\sigma' = 4n(n-2),$$

$$4\beta' + \gamma' = 4n^2(n-2)(n^2 - 3n + 3n - 3),$$

$$8\kappa' - 18h' = n(n-2)\{n^6 - 6n^5 + 16n^4 - 54n^3 + 164n^2 - 288n + 403 \\ - 482n + 348n^2 - 242n + 60\},$$

which agree with the values which Mr Salmon has obtained for $\beta', \gamma', h', \kappa'$ by means of the twenty-one equations, and the additional equations (peculiar to the case in question, of a surface of the degree n without singularities, and which are obtained by him from independent considerations), $i' = 0$, and $\beta' = 2n(n-2)(11n-24)$.

The system of seventeen equations completely accounts for the reduction of the order of the given surface considered as the reciprocal of the reciprocal surface, but the omitted equations are important for other purposes. We may by means of them express i, t in terms of the data for the system of twenty-one equations, viz., $n, b, c, \beta, \gamma, h, \kappa$; and, effecting this, and annexing the corresponding values of i', t' ,

we have the supplementary system:

$$\begin{aligned}
 4t &= (5n - 6)c - 6c^2 + 12h - 5\gamma, \\
 24t &= -(8n - 8)b + (15n - 18)c + 2(4b^2 - 9c^2) - 16\kappa + 36h + 20\beta - 15\gamma, \\
 4i' &= (5n' - 6)c' - 6c'^2 + 12h' - 5\gamma', \\
 24t' &= -(8n' - 8)b' + (15n' - 18)c' + 2(4b'^2 - 9c'^2) - 16\kappa' + 36h' + 20\beta' - 15\gamma',
 \end{aligned}$$

to which I annex also, without transformation, the four equations for q, r, q', r' , viz.:

$$\begin{aligned}
 q &= b^2 - b - 2\kappa - 3\gamma - 6t, \\
 r &= c^2 - c - 2h - 3\beta, \\
 q' &= b'^2 - b' - 2\kappa' - 3\gamma' - 6t', \\
 r' &= c'^2 - c' - 2h' - 3\beta',
 \end{aligned}$$

the last two of which, neglecting singularities, give

$$q' = 4n(n - 2)(n - 3)(n^2 + 2n - 4), \quad r = 2n(n - 2)(3n - 4),$$

which are the values given by Mr Salmon. I remark, in conclusion, that there is considerable difficulty in the geometrical conception of the points i and the planes i' , and the subject appears to require further examination. In the case of a surface of the order n without multiple lines, we have not only $i = 0$ (which is a matter of course), but also $i' = 0$. In my paper before referred to, I showed, or attempted to show, by geometrical reasoning, that the common tangent planes of the spinode developpe and the node-couple developpe are stationary planes of the one or the other of the two developpes, that is, $i' = 0$, and the reasoning seems correct as far as it goes, but it was not shown how the demonstration would (as it ought to do) fail in the case of a surface having a double or cuspidal curve. I showed also that in the case where the common tangent plane is a stationary plane of the spinode developpe (that is for the planes β'), the two points of contact with the spinode developpe coincide; but I did not show what takes place in the case where the common tangent plane is a stationary plane of the node-couple developpe (that is for the planes γ').

Kapitel 6

228. Sur l'intégrale

$$\int \frac{x^{i-1} dx}{(a + bx + cx^2)^{i+\frac{1}{2}}}.$$

[From the *Journal de Mathématiques pures et appliquées* (Liouville),
vol. ii. (1857), pp. 49–51.]

Extrait d'une Lettre adressée à M. Liouville.

Je suppose que le lecteur ait sous les yeux la Note de M. Liouville (tome I. [1856] page 421). Cela étant, en rétablissant les valeurs de g , k et en écrivant $i - 1$ au lieu de μ , la formule de M. Liouville devient

$$\int_0^1 \frac{t^{i-1}(1-t)^{i-1} dt}{(a + bt - ct^2)^{i+\frac{1}{2}}} = \frac{\Gamma(i)\Gamma(i)}{\sqrt{a+b-c}\Gamma(i)[2a+b+2\sqrt{ac(a+b-c)}]^{i-\frac{1}{2}}}, \quad (6.1)$$

laquelle, en y posant

$$t = \frac{x}{x+1}, \quad x = \frac{t}{1-t},$$

se transforme en celle-ci:

$$\int_0^\infty \frac{x^{i-1} dx}{[(a+b-c)x^2 + (2a+b)x + a]^{i+\frac{1}{2}}} = \frac{\Gamma(i)\Gamma(i)}{\sqrt{a+b-c}\Gamma(i)[2a+b+2\sqrt{a(a+b-c)}]^{i-\frac{1}{2}}}, \quad (6.2)$$

et en mettant le dénominateur de l'intégrale sous la forme

$$[(a+b-c)(x+\lambda)(x+\mu)]^{i+\frac{1}{2}},$$

la formule devient

$$\int_0^\infty \frac{x^{i-1} dx}{[(x+\lambda)(x+\mu)]^{i+\frac{1}{2}}} = \frac{\Gamma(i)}{\Gamma(i)} \cdot \frac{(\sqrt{\lambda} + \sqrt{\mu})^{1-2i}}{(\lambda\mu)^{i-\frac{1}{2}}}, \quad (6.3)$$

formule qui se trouve dans mon Mémoire: "On certain formulae for differentiation with applications to the evaluation of definite integrals," *Camb. and Dubl. Math. Journal*, tom. II., page 122, (1847), [41]. La démonstration que j'y ai donnée n'est cependant rigoureuse (à moins que l'on n'admette la théorie de la différentiation à indices quelconques) que pour le cas de $i+1$ égal à un entier positif; en effet, en écrivant

$$U_i^{\lambda,\mu} = [(x+\lambda)(x+\mu)]^i (\sqrt{x+\lambda} - \sqrt{x+\mu})^{2i},$$

je suis parti de la formule

$$\left(\frac{d}{dx}\right)^i U_i^{\lambda,\mu}.$$

Cette expression devient très-simple pour le cas $s = i+1$; on a alors

$$\left(\frac{d}{dx}\right)^{i+1} [(x+\lambda)(x+\mu)]^{i+\frac{1}{2}} = \frac{(2i+1)!!}{2^{i+1}} (\lambda-\mu)^{i+1} [(x+\lambda)(x+\mu)]^{-i-\frac{1}{2}}, \quad (6.4)$$

formule de différentiation assez singulière. En y écrivant $\frac{1}{2} - i$ au lieu de i , et en intégrant $i + \frac{1}{2}$ fois par la formule

$$\int_x^\infty dx \int_x^\infty dx \dots = [i + \frac{1}{2}]^{-1} \left(\frac{d}{dx}\right)^{-i-\frac{1}{2}},$$

on obtient la formule intégrale ci-devant mentionnée. Il convient d'ajouter que cette formule, représentée sous la forme

$$\int_0^\infty \frac{x^{i-1} dx}{(ax^2 + bx + c)^{i+\frac{1}{2}}} = \frac{\Gamma(i) \Gamma(\frac{1}{2})}{\Gamma(i + \frac{1}{2})} \cdot \frac{1}{(b + 2\sqrt{ac})^{i-\frac{1}{2}}}, \quad (6.5)$$

est comprise dans une formule générale donnée par M. Schlömilch: "Note sur la variation des constantes arbitraires d'une intégrale définie" (*Crelle*, tom. xxxiii., page 268; 1847).

Je remarque aussi que M. Donkin, en comparant ses résultats concernant les fonctions de Laplace avec ceux de M. Boole sur le même sujet, a trouvé une identité, laquelle m'a conduit à cette autre formule de différentiation

$$\begin{aligned} \left(\frac{d}{dx}\right)^{i+1} [(x+\lambda)(x+\mu)]^{i+\frac{1}{2}} \\ = \frac{(2i+1)!!}{2^{i+1}} (2x+\lambda+\mu)^{i+1} [(x+\lambda)(x+\mu)]^{-i-\frac{1}{2}}, \end{aligned} \quad (6.6)$$

et

$$\left(\frac{d}{dx}\right)^{i+1} \frac{(2x+\lambda+\mu)^{i+1}}{[(x+\lambda)(x+\mu)]^i} = \text{etc.}, \quad (6.7)$$

formule que j'ai démontrée au moyen d'une analyse assez compliquée.

Kapitel 7

229. Note sur une formule pour la réversion des séries.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LII. (1856), pp. 276–284.]

Je me propose de développer dans cette note, pour le cas de trois variables (ce qui suffit pour faire voir la loi dans le cas d'un nombre quelconque de variables) une formule qui se trouve dans le mémoire remarquable de Jacobi, “*de resolutione aequationum per series infinitas*,” *Crelle*, t. VI. [1830] p. 257. Voici la formule dont il s'agit.

Soit $f(x, y, z)$ une fonction rationnelle et entière des variables x, y, z , et mettons $u = X, v = Y, w = Z$, où X, Y, Z sont des fonctions rationnelles et entières des variables x, y, z , telles que $X - x, Y - y, Z - z$ ne contiennent que les puissances et les produits du deuxième ordre et des ordres supérieurs des variables. Cela étant, on aura

$$f(x, y, z) = [f(x, y, z) \frac{\partial(X, Y, Z)}{\partial(x, y, z)} \cdot \frac{1}{(X - u)(Y - v)(Z - w)}]_{x^{-1}y^{-1}z^{-1}},$$

où la notation $\frac{\partial(X, Y, Z)}{\partial(x, y, z)}$ dénote le déterminant fonctionnel ou “Jacobian” de X, Y, Z par rapport à x, y, z et la notation $[]_{x^{-1}y^{-1}z^{-1}}$ signifie le coefficient de $x^{-1}y^{-1}z^{-1}$ dans le développement de la fonction en dedans des $[]$; ce développement doit s'effectuer d'une manière déterminée, savoir: il faut d'abord développer les facteurs $\frac{1}{Y - v}$ selon les puissances descendantes de x , c'est-à-dire dans la forme

$y^{-1} + y^{-2}v + \text{etc.}$, et en écrivant $X = x + P$ (où P est fonction des trois variables) il faut développer les puissances de X selon les puissances descendantes du monome x , c'est-à-dire dans la forme $X^{-m} = x^{-m} - mx^{-m-1}P + \frac{1}{2}m(m+1)x^{-m-2}P^2 - \text{etc.}$, ce qui est en effet un développement selon les puissances ascendantes des variables.

La formule donne

$$\text{Coeff. } u^a v^b w^c \text{ dans } f(x, y, z) = [f(x, y, z) \frac{\partial(X, Y, Z)}{\partial(x, y, z)} \cdot X^{-a-1} Y^{-b-1} Z^{-c-1}]_{x^{-1} y^{-1} z^{-1}}.$$

ou ce qui est la même chose

$$\text{Coeff. } u^a v^b w^c \text{ dans } f(x, y, z) = [f(x, y, z) \frac{\partial(X, Y, Z)}{\partial(x, y, z)} \cdot x^{-a-1} y^{-b-1} z^{-c-1}]_{x^{-1} y^{-1} z^{-1}}.$$

Soit à présent

$$\begin{aligned} X &= x \dots + A_{f,g,h, \text{ etc.}} x^f y^g z^h + \text{etc.} = u, \\ Y &= y \dots + B_{i,j,k, \text{ etc.}} x^i y^j z^k + \text{etc.} = v, \\ Z &= z \dots + C_{l,m,n, \text{ etc.}} x^l y^m z^n + \text{etc.} = w, \\ f(x, y, z) &= \dots + D_{p,q,r, \text{ etc.}} x^p y^q z^r + \text{etc.}; \end{aligned}$$

dans ces expressions et partout dans ce qui suit, les etc.'s se rapportent à des termes que l'on obtient en affixant des accents en nombre quelconque aux symboles indéterminés.

En employant la notation de Gauss $\Pi a = 1 \cdot 2 \cdot 3 \dots a$, on obtient pour le terme général de $-X^{-1}$,

$$\frac{\Pi(a+r-1)}{\Pi a \Pi f \Pi g \dots} A_{f,g,h \text{ etc.}}^a x^{af+\dots-1} y^{ag+\dots} z^{ah+\dots},$$

où

$$a + \text{etc.} = r, \quad fa + \text{etc.} = F, \quad ga + \text{etc.} = G, \quad ha + \text{etc.} = H.$$

De même le terme général de $-Y^{-1}$ est

$$\frac{\Pi(b+s-1)}{\Pi b \Pi i \Pi j \dots} B_{i,j,k \text{ etc.}}^b x^{bi+\dots} y^{bj+\dots-1} z^{bk+\dots},$$

où

$$b + \text{etc.} = s, \quad ib + \text{etc.} = I, \quad jb + \text{etc.} = J, \quad kb + \text{etc.} = K,$$

et le terme général de $-Z^{-1}$ est

$$\frac{\Pi(c+t-1)}{\Pi c \Pi l \Pi m \dots} C_{l,m,n \text{ etc.}}^c x^{cl+\dots} y^{cm+\dots} z^{cn+\dots-1},$$

où

$$c + \text{etc.} = t, \quad lc + \text{etc.} = L, \quad mc + \text{etc.} = M, \quad nc + \text{etc.} = N.$$

En formant de là le terme général du Jacobian et en multipliant par le terme général de $f(x, y, z)$ on obtient pour le terme général de l'expression en dedans des [], la valeur que voici,

$$\begin{aligned} & (-)^{r+s+t} \frac{\Pi(a+r-1) \Pi(b+s-1) \Pi(c+t-1)}{\Pi a \Pi b \Pi c \Pi f \text{ etc.} \Pi i \text{ etc.} \Pi l \text{ etc.}} \\ & \quad \times A_{f,g,h \text{ etc.}}^a B_{i,j,k \text{ etc.}}^b C_{l,m,n \text{ etc.}}^c D_{p,q,r} \\ & \times \left| \begin{array}{ccc} P+F+I+L+p-1, & -G, & -H \\ -I, & Q+G+J+M, & -K \\ -L, & -M, & R+H+K+N+r-1 \end{array} \right| \\ & \quad \times x^{a+r-1} y^{b+s-1} z^{c+t-1}, \end{aligned}$$

et pour trouver le terme qui contient $x^{-1}y^{-1}z^{-1}$ on n'a qu'à écrire dans cette expression

$$P+F+I+L+p = a+r, \quad Q+G+J+M = b+s, \quad R+H+K+N+r = c+t,$$

le coefficient de $x^{-1}y^{-1}z^{-1}$ sera alors la valeur de l'expression $[]_{x^{-1}y^{-1}z^{-1}}$ qu'il s'agissait de trouver.

Théorème. *Le terme général de $\left(\frac{d}{du}\right)^a \left(\frac{d}{dv}\right)^b \left(\frac{d}{dw}\right)^c \phi$ est*

$$\begin{aligned} & (-)^{r+s+t} \frac{\Pi(a+r-1) \Pi(b+s-1) \Pi(c+t-1)}{\Pi a \text{ etc.} \Pi \beta \text{ etc.} \Pi \gamma \text{ etc.} (\Pi f \Pi g \Pi h)^a \text{ etc.} (\Pi i \Pi j \Pi k)^b \text{ etc.} (\Pi l \Pi m \Pi n)^c \text{ etc.}} \\ & \quad \times (A_{f,g,h \text{ etc.}}^a x) (B_{i,j,k \text{ etc.}}^b y) (C_{l,m,n \text{ etc.}}^c z) D_{p,q,r} \\ & \quad \times \left| \begin{array}{ccc} P+F+I+L, & -G, & -H \\ -I, & Q+G+J+M, & -K \\ -L, & -M, & R+H+K+N \end{array} \right| \\ & \quad \times x^{-1} y^{-1} z^{-1} \chi^p \psi^q \omega^r, \end{aligned}$$

expression dans laquelle

$$\begin{aligned} a + \text{etc.} &= r, & \beta + \text{etc.} &= s, \\ fa + \text{etc.} &= F, & ga + \text{etc.} &= G, \\ i\beta + \text{etc.} &= I, & j\beta + \text{etc.} &= J, \\ l\gamma + \text{etc.} &= L, & m\gamma + \text{etc.} &= M, \\ F + I + L + P &= a + r, & G + J + M + Q &= b + s, & H + K + N + R &= c + t \end{aligned}$$

et où les nombres $f + g + h - 1$, etc., $i + j + k - 1$, etc., $l + m + n - 1$, etc. sont tous positifs comme auparavant.

La formule contient les symboles x, y, z qui sont chacun une fonction linéaire de $\frac{d}{du}, \frac{d}{dv}, \frac{d}{dw}$; on pourrait donc se proposer la question de trouver le terme général en développant la formule de manière à ne contenir que des puissances et des produits de ces symboles $\frac{d}{du}, \frac{d}{dv}, \frac{d}{dw}$; c'est à quoi se rapportent les recherches très étendues que vient de faire M. Sylvester sur ce sujet, et qui embrassent aussi bien le cas où les nouvelles variables sont données explicitement que celui où les deux systèmes de variables sont liés par des Équations données.

Il y a une autre manière de traiter cette question de la transformation des variables indépendantes, savoir en écrivant

$$p = \xi x + \eta y + \zeta z,$$

on peut exprimer les puissances de R au moyen de p et de cette autre quantité symbolique

$$\Delta = \xi \frac{d}{d\xi} + \eta \frac{d}{d\eta} + \zeta \frac{d}{d\zeta}.$$

En effet, en mettant $p^2 = \xi^2 + \eta^2 + \zeta^2$, on a

$$p + \frac{1}{2}p^2 + \frac{1}{6}p^3 + \text{etc.} = \chi + \frac{1}{2}\chi^2 + \frac{1}{6}\chi^3 + \text{etc.},$$

où

$$\xi = P + \frac{1}{2}\chi^2 + \frac{1}{6}\chi^3 + \text{etc.}, \quad \eta = Q + \frac{1}{2}\chi^2 + \frac{1}{6}\chi^3 + \text{etc.}, \quad \zeta = R + \frac{1}{2}\chi^2 + \frac{1}{6}\chi^3 + \text{etc.}$$

En se servant de la méthode des approximations successives, on trouve comme première approximation

$$P = \xi, \quad Q = \eta, \quad R = \zeta,$$

et de là

$$\chi = p;$$

comme seconde approximation

$$P = \xi - \frac{1}{2}p^{-1}\xi, \quad Q = \eta - \frac{1}{2}p^{-1}\eta, \quad R = \zeta - \frac{1}{2}p^{-1}\zeta,$$

et de là

$$\chi = p - \frac{1}{2}\Delta;$$

comme troisième approximation

$$\begin{aligned} P &= \xi - \frac{1}{2}(p^2 - p\Delta + \frac{1}{2}\Delta p)^{-1}\xi + \frac{1}{6}p^{-2}\xi, \\ Q &= \eta - \frac{1}{2}(p^2 - p\Delta + \frac{1}{2}\Delta p)^{-1}\eta + \frac{1}{6}p^{-2}\eta, \\ R &= \zeta - \frac{1}{2}(p^2 - p\Delta + \frac{1}{2}\Delta p)^{-1}\zeta + \frac{1}{6}p^{-2}\zeta, \end{aligned}$$

et de là

$$\chi = p - \frac{1}{2}\Delta + \frac{1}{12}p^{-1}\Delta^2 + \text{etc.},$$

et ainsi de suite; donc, en substituant,

$$\begin{aligned} R &= p - \frac{1}{2}\Delta + \frac{1}{12}p^{-1}\Delta^2 - \frac{1}{12}\Delta p^{-1}\Delta \\ &\quad + \frac{1}{12}p^{-2}\Delta^2 + \text{etc. etc.}, \end{aligned}$$

c'est-à-dire

$$\begin{aligned} R &= p, \\ R^2 &= p^2 - p\Delta, \\ R^3 &= p^3 - 3p^2\Delta p - (p^2 - 3p^2\Delta)\Delta, \\ &\text{etc.} \end{aligned}$$

formules dans lesquelles un terme tel que $p^2\Delta$ signifie $p^2x \cdot x + p^2y \cdot y + p^2z \cdot z$, c'est-à-dire l'opération du symbole p^2 s'arrête aux quantités x, y, z faisant partie du symbole Δ qui vient immédiatement après le p^2 ; la même chose a lieu dans tous les cas semblables. L'équation générale est

$$R^n = \Pi - \Pi\Delta,$$

mais pour expliquer la forme de la fonction Π il faudrait faire des développements assez longs, dans lesquels je n'entrerai pas à cette occasion; la découverte de cette équation générale est due à M. Sylvester, qui l'a établie d'une autre manière.

Londres, 2, Stone Buildings, 16 Avril, 1855.

Kapitel 8

230. Note sur la méthode d'élimination de Bezout.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LIII. (1857), pp. 366–367.]

Voici la forme la plus simple sous laquelle on peut présenter cette méthode. Pour éliminer les variables x, y entre deux équations du $n^{\text{ième}}$ degré

$$\begin{aligned} (a, \dots & \text{-----} x, y)^n = \\ (a', \dots & \text{-----} x, y)^n = \end{aligned}$$

on n'a qu'à former l'équation identique

$$(a, \dots \text{-----} x, y)^n (a' \dots \text{-----} x, y)^n =$$

$$= \begin{vmatrix} a_{0,0} x^{n-1} + a_{1,0} x^{n-2} y \dots + a_{n-1,0} y^{n-1}, & X^{n-1} \\ a_{0,1} x^{n-1} + a_{1,1} x^{n-2} y \dots + a_{n-1,1} y^{n-1}, & X^{n-2} Y \\ \dots\dots\dots & \dots \\ a_{0,n-1} x^{n-1} + a_{1,n-1} x^{n-2} y \dots + a_{n-1,n-1} y^{n-1}, & Y^{n-1} \end{vmatrix}$$

où l'expression qui forme le second membre représente la fonction suivante,

$$\begin{aligned} & (a_{0,0} x^{n-1} + a_{1,0} x^{n-2} y \dots + a_{n-1,0} y^{n-1}) X^{n-1} \\ & + (a_{0,1} x^{n-1} + a_{1,1} x^{n-2} y \dots + a_{n-1,1} y^{n-1}) X^{n-2} Y \\ & \qquad \qquad \qquad + \dots \\ & + (a_{0,n-1} x^{n-1} + a_{1,n-1} x^{n-2} y \dots + a_{n-1,n-1} y^{n-1}) Y^{n-1}; \end{aligned}$$

le résultat de l'élimination sera

$$\begin{vmatrix} a_{0,0}, & a_{1,0}, & \dots & a_{n-1,0} \\ a_{0,1}, & a_{1,1}, & \dots & a_{n-1,1} \\ \dots & \dots & \dots & \dots \\ a_{0,n-1}, & a_{1,n-1}, & \dots & a_{n-1,n-1} \end{vmatrix} = 0.$$

Par exemple on trouve

$$\begin{aligned} & \underbrace{(a, b, c \text{ --- } x, y)^2}_{=} \\ &= (2(ab'-a'b), \text{ } ac'-a'c \text{ --- } | \text{ } ac'-a'c, \text{ } 2(bc'-b'c) | \\ & \underbrace{(a, b, c, d \text{ --- } x, y)}_{=} \\ &= (3(ab'-a'b), \text{ } 3(ac'-a'c), \text{ } (ad'-a'd) \text{ --- } \\ & \quad | \text{ } 3(ac'-a'c), \text{ } (ad'-a'd) + 9(bc'-b'c), \text{ } 3(bd'-b'd) \\ & \quad | \text{ } (ad'-a'd), \text{ } 3(bd'-b'd), \text{ } 3(cd'-c'd) | \end{aligned}$$

d'où l'on tire immédiatement les résultats de l'élimination entre deux Équations quadratiques ou cubiques.

Londres, 2 Stone Buildings, Avril, 1855.

Kapitel 9

231. Note sur l'équation $x^2 - Dy^2 = \pm 4$, $D \equiv 5$ (mod. 8).

[From the *Journal für die reine und angewandte Mathematik* (Crelle),
tom. LIII. (1857), pp. 369–371.]

ON sait que pour un déterminant positif $D \equiv 5 \pmod{8}$ le nombre des formes quadratiques proprement primitives est égal au nombre des formes improprement primitives, quand il existe une solution en nombres impairs de l'équation $x^2 - Dy^2 = 4$; mais que le nombre des formes proprement primitives est égal à trois fois le nombre des formes improprement primitives, quand il n'existe pas de solution en nombres impairs de l'équation $x^2 - Dy^2 = 4$.

Les tables de Degen font voir immédiatement, pour les nombres $D \equiv 5 \pmod{8}$ qui ne sont pas plus grands que 997, s'il existe ou non une solution en nombres impairs de l'équation $x^2 - Dy^2 = 4$. À savoir si le nombre 4 se trouve dans la série des dénominateurs des quotients complets (la seconde ligne dans les tables), il existe une solution de l'équation; si non, il n'en existe pas. De plus, si la place où se trouve le nombre 4 est d'ordre pair il existe une solution tant de l'équation $x^2 - Dy^2 = +4$ que de l'équation $x^2 - Dy^2 = -4$, mais si cette place est d'ordre impair il existe seulement une solution de l'équation $x^2 - Dy^2 = 4$. On trouve la plus petite solution de ces Équations au moyen de la série des quotients (la première ligne

des tables), en s'arrêtant au nombre qui précède le nombre placé au-dessus du nombre 4, et en calculant la fraction continue déterminée par cette suite; par exemple, pour le nombre 61 on trouve dans les tables

61	7,	1,	4,	3,	1
	12,	3,	4,	9	(5, 5)
	226153980	1766319049			

Cela fait voir qu'il existe une solution de l'équation $x^2 - Dy^2 = -4$, solution que l'on obtient au moyen de la fraction continue $7 + \frac{1}{1 + \frac{1}{4 + \frac{1}{3}}}$

et faisant $x = 39$, $y = 5$. La plus petite solution de l'équation $x^2 - Dy^2 = 4$ se déduit très facilement de la plus petite solution de l'équation $r^2 - Dv^2 = -4$ au moyen de la formule $x + y\sqrt{D} = \frac{1}{4}(r + v\sqrt{D})^2$, ce qui donne $x = \frac{1}{2}r^2 + 2$, $y = \frac{1}{2}rv$.

On obtient pareillement la plus petite solution de l'équation $x^2 - Dy^2 = -1$ au moyen de la formule $(x + y\sqrt{D}) = \frac{1}{4}(t + u\sqrt{D})^2$, ce qui donne

$$x = \frac{1}{4}(t^2 + 3u), \quad y = \frac{1}{4}(t^2 + 1)v.$$

De même la plus petite solution de l'équation $x^2 - Dy^2 = 1$ se déduit de la plus petite solution de l'équation $T^2 - DU^2 = 4$ au moyen de la formule $x + y\sqrt{D} = \frac{1}{4}(T + U\sqrt{D})^2$, ce qui donne $x = \frac{1}{4}(T^2 - 3T)$, $y = \frac{1}{4}(T^2 - 1)U$.

Je fais observer à cette occasion que suivant une remarque de Gopel ("Notiz über A. Gopel," t. XXXV. [1847] p. 315 de ce Journal) si dans l'équation $\left(\frac{x}{y}\right) = \frac{P + \sqrt{Q}}{p}$, où X , y , p , n , P , Q sont des entiers, le dénominateur p est plus grand que l'unité

Table des plus petites solutions impaires de l'équation
 $x^2 - Dy^2 = \pm 4$, $D \equiv 5 \pmod{8}$.

D $\pm$	$\pm$ x	x y	y	D	$\pm$	x	y	D
5	+	3	1	341	+	277	15	677
imposs.								
13	—	11	3	349	imposs.		685	—
759	29							
21	+	5	1	357	+	19	1	693
+	79	3						
29	—	5	1	365	—	19	1	701
imposs.								
37	imposs.		373	imposs.		709	imposs.	
45	+	7	1	381	imposs.		717	+
241	9							
53	—	7	1	389	imposs.		725	+
27	1							
61	—	39	5	397	—	3447	173	733
—	27	1						
69	+	25	3	405	imposs.		741	+
245	9							
77	+	9	1	413	+	61	3	749
+	12945	473						
85	—	9	1	421	—	444939	21685	757
imposs.								
93	+	29	3	429	+	145	7	765
+	83	3						
101	imposs.		437	+	21	1	773	—
139	5							
109	—	261	25	445	—	21	1	781
imposs.								
117	+	11	1	453	+	149	7	789
+	31825	1133						
125	—	11	1	461	—	365	17	797
—	367	13						
133	+	173	15	469	+	65	3	805
+	1447	51						
141	imposs.		477	+	2599	119	813	
imposs.								
149	—	61	5	485	imposs.		821	—
16189	565							
157	—	213	17	493	—	111	5	829
imposs.								
165	+	13	1	501	+	28225	1261	837
+	29	1						
173	—	13	1	509	—	925	41	845
—	29	1						
181	+	1305	97	517	+	10573	465	853
—	27483	941						
189	imposs.		525	+	23	1	861	+
1027	35							
197	imposs.		533	—	23	1	869	+
49377	1675							

Kapitel 10

232. Mémoire sur la forme canonique des fonctions binaires.

[From the *Journal für die reine und angewandte Mathematik* (Crelle),
tom. LIV. (1857), pp. 48–58.]

L'IDÉE de la forme canonique d'une fonction binaire de degré impair et la théorie de la réduction d'une pareille fonction à la forme canonique sont dues à M. Sylvester⁽¹⁾, qui a, en outre, étendu sa théorie aux fonctions binaires des degrés pairs 4 et 6. Mais il n'a établi nulle part l'idée générale de la forme canonique d'une fonction binaire d'un degré pair quelconque. Je me propose de reprendre cette théorie en considérant d'abord les fonctions binaires de degré impair et ensuite les fonctions binaires de degré pair.

¹Les recherches de M. Sylvester sont contenues dans le mémoire intitulé "Sketch of a Memoir on Elimination, Transformation, and Canonical Forms," *Camb. and Dublin Mathematical Journal*, t. vi. [1851] p. 186, dans un supplément publié conjointement avec ce mémoire et intitulé "Essay on Canonical Forms by J. J. Sylvester," London, Bell, 1851, et dans le mémoire intitulé, "On a remarkable discovery in the Theory of Canonical Forms and of Hyperdeterminants," *Phil. Mag.* Nov. 1851.

¹(1)

Soit donc proposée la fonction de degré impair

$$(a, b, \dots a' \text{ ————— } x, y)$$

en désignant par le symbole Σ une somme de m termes, on voit qu'on pourra écrire

$$(a, b, \dots a' \text{ ————— } x, y)$$

l'expression qui forme le second membre de cette Équation est appelée la forme canonique de la fonction donnée, et il s'agit de trouver les valeurs des quantités A, α . En supposant que le produit des facteurs linéaires $x + \alpha y$ soit égal, à un facteur constant près, à

$$(a, b, \dots a'' \text{ ————— } x, y)$$

la question peut s'énoncer sous la forme suivante: trouver une fonction de degré m , telle qu'en représentant $x + \alpha y$ un facteur linéaire quelconque de cette fonction, la fonction donnée de degré $2m - 1$ se réduise à la forme $\Sigma A(x + \alpha y)^{2m-1}$. La fonction de degré m contiendra un facteur constant dont la valeur peut être prise à volonté. Mais à ce facteur près la fonction de degré m sera complètement déterminée.

Pour fixer les idées je suppose que la fonction donnée soit la fonction cubique

$$(a, b, c, d \text{ ————— } x, y)^3,$$

dans ce cas il s'agira de trouver une fonction quadratique

$$(a, b, c \text{ ————— } x, y)^2$$

telle qu'en posant $(a, b, c \text{ ————— } x, y)$ on ait identiquement

$$(a, b, c, d \text{ ————— } x, y)^3$$

Cela se fait très-facilement par les méthodes ordinaires. En effet, on a

$$\begin{aligned} A + B &= a, \\ A\alpha + B\beta &= b, \\ A\alpha^2 + B\beta^2 &= c, \\ A\alpha^3 + B\beta^3 &= d, \end{aligned}$$

d'où il suit

$$\begin{aligned} a\xi\eta - b(\xi + \eta) + c &= A(\xi - \alpha)(\eta - \alpha) + B(\xi - \beta)(\eta - \beta), \\ b\xi\eta - c(\xi + \eta) + d &= A\alpha(\xi - \alpha)(\eta - \alpha) + B\beta(\xi - \beta)(\eta - \beta), \end{aligned}$$

ξ et η étant des quantités quelconques; donc, en prenant $\xi = \alpha$, $\eta = \beta$, on obtient

$$\begin{aligned} a\alpha\beta - b(\alpha + \beta) + c &= 0, \\ b\alpha\beta - c(\alpha + \beta) + d &= 0. \end{aligned}$$

Mais l'équation $(a, b, c$ —————
 $a(x + \alpha y)(x + \beta y)$ donne

$$\alpha\beta : \alpha + \beta : 1 = c : 2b : a,$$

on a donc

$$\begin{aligned} ac - 2bb + ca &= 0, \\ bc - 2cb + da &= 0, \end{aligned}$$

système qui donne les rapports $a : b : c$. Mais pour compléter la solution de la manière la plus élégante, il faut ajouter à ces Équations, l'équation

$$(a, b, c$$
 ————— $x, y)^2 =$

0, mise sous la forme $(c, b, a$ —————
 0, ou ce qui est la même chose

$$y^2c + 2yxb + x^2a = 0;$$

en éliminant a, b, c on obtient une equation de laquelle on déduit

$$(a, b, c$$
 ————— $x, y)^2 =$

Mais on peut trouver ce résultat d'une manière encore plus facile. Considérant l'équation

$$(a, b, c, d$$
 ————— $x, y)^3$

on n'a qu'à opérer sur cette équation en se servant du symbole

$$(a, b, c$$
 ————— $\partial_y, -\partial_x)$

le second membre se réduit à zéro à cause des Équations

$$(a, b, c \frac{\partial}{\partial \alpha}, -1)^2 = 0, (a, b, c \frac{\partial}{\partial \beta}, -1)^2 = 0$$

$$0, (a, b, c \frac{\partial}{\partial \alpha}, -1)^2 = 0, (a, b, c \frac{\partial}{\partial \beta}, -1)^2 = 0$$

que donne l'équation $(a, b, c \frac{\partial}{\partial \alpha}, -1)^2 = 0$ que donne l'équation $(a, b, c \frac{\partial}{\partial \beta}, -1)^2 = 0$. Par conséquent on a identiquement

$$(a, b, c \frac{\partial}{\partial \alpha}, -1)^2 = 0, (a, b, c \frac{\partial}{\partial \beta}, -1)^2 = 0$$

c'est-à-dire

$$x(ac - 2bb + ca) + y(bc - 2cb + da) = 0,$$

ou enfin

$$ac - 2bb + ca = 0,$$

$$bc - 2cb + da = 0,$$

et de là on tire comme auparavant la valeur de la fonction

$$(a, b, c \frac{\partial}{\partial \alpha}, -1)^2 = 0, (a, b, c \frac{\partial}{\partial \beta}, -1)^2 = 0$$

Les valeurs des coefficients A, B s'expriment alors en fonctions linéaires des coefficients de la fonction cubique, et des quantités α, β qui sont pour ainsi dire les racines de la fonction quadratique

$$(a, b, c \frac{\partial}{\partial \alpha}, -1)^2 = 0, (a, b, c \frac{\partial}{\partial \beta}, -1)^2 = 0$$

Il est évident que le procédé suivi dans ce qui précède est tout à fait général, et que l'on obtient toujours explicitement la fonction du degré m : par exemple pour la fonction du cinquième degré

$$(a, b, c, d, e, f \frac{\partial}{\partial \alpha}, -1)^2 = 0, (a, b, c, d, e, f \frac{\partial}{\partial \beta}, -1)^2 = 0$$

la fonction dont il s'agit sera

$$\begin{vmatrix} f, & -fx, & fx^2, & -x^3 \\ a, & b, & c, & d \\ b, & c, & d, & e \\ c, & d, & e, & f \end{vmatrix}$$

et de même dans tous les autres cas. Cette fonction du degré m (laquelle est un covariant de la fonction donnée) peut être appelée le *canonisant*. Pour la fonction du cinquième degré le canonisant peut être mis sous cette autre forme

$$\begin{vmatrix} ax + by, & bx + cy, & cx + dy \\ bx + cy, & cx + dy, & dx + ey \\ cx + dy, & dx + ey, & ex + fy \end{vmatrix}$$

et on démontre facilement qu'il existe toujours une transformation semblable.

Considérons maintenant une fonction de degré pair

$$(a, b, \dots a' \text{ ————— } x, y)$$

et posons

$$(a, b, \dots a' \text{ ————— } x, y)$$

où l'on suppose $(a, b, \dots$ —————
 $a(x+\alpha y) \dots$ et où $(a_1, b_1, \dots a'_1$ —————
représente un certain covariant parfaitement déterminé de
 $(a, b, \dots a' \text{ ————— } x, y)$

L'expression qui forme le second membre de l'équation proposée s'appellera la forme canonique de la fonction de degré pair.

Pour fixer les idées je suppose que la fonction donnée soit la fonction du quatrième degré

$$(a, b, c, d, e \text{ ————— } x, y)$$

dans ce cas, en prenant la fonction quadratique
 $(a, b, c$ ————— $x, y)^2 =$
 $a(x+\alpha y)(x+\beta y)$ et en représentant par $(a_1, b_1, c_1$ —————
un covariant déterminé de $(a, b, c$ —————
on a

$$(a, b, c, d, e \text{ ————— } x, y)$$

Cela posé, la condition qui sert à déterminer le covariant
 $(a_1, b_1, c_1$ ————— $x, y)^2$
est la suivante

$$i(a, b, c \text{ ————— } \partial_y, -\partial_x$$

où K est un facteur constant dont la valeur est arbitraire.

Le premier membre sera

$$((7ac-6b^2, -2ab, a^2 \text{ ————— } x, y)$$

en représentant par cette notation l'expression

$$(7ac - 6b^2) a_1 - 2ab b_1 + a^2 c_1, \quad \text{etc.,}$$

et l'équation donne

$$\begin{array}{l} (7ac - 6b^2, -2ab, a^2 \\ (-2bc, 12ac - 16b^2, 2ab \\ (c^2, -2bc, 7ac - 6b^2 \end{array}$$

et de là on obtient pour le covariant $(a_1, b_1, c_1$

$$\left| \begin{array}{ccc} 7ac - 6b^2, & -2ab, & a^2 \\ -2bc, & 12ac - 16b^2, & 2ab \\ c^2, & -2bc, & 7ac - 6b^2 \end{array} \right| \times (a_1, b_1, c_1$$

ce qui fait voir que le covariant cherché est

$$\left| \begin{array}{ccc} 7ac - 6b^2, & -2ab, & a^2 \\ -2bc, & 12ac - 16b^2, & 2ab \\ c^2, & -2bc, & 7ac - 6b^2 \end{array} \right|$$

où l'on doit remplacer (a_1, b_1, c_1) par (x, y) respectivement; c'est-à-dire

$$(a_1, b_1, c_1 \longrightarrow x, y)^2$$

THE COLLECTED
MATHEMATICAL PAPERS
OF
ARTHUR CAYLEY

Volume IV

Papers 232–244 (part)

Pages 51–100

Cambridge University Press

1891

10.1 Mémoire sur la Forme Canonique des Fonctions Binaires

[FROM THE JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK (CRELLE), T. LIV. (1857), PP. 1–19.]

[Continuation from p. 50]

The results which have been obtained for the simplest particular cases show that the general theory must be susceptible of further developments. Thus the canonical form of the cubic function $(a, b, c, dxy)^3$ will be $u^3 + v^3$ if we write u, v instead of $\sqrt[3]{A}(x + \alpha y)$, $\sqrt[3]{B}(x + \beta y)$. The general method gives only the value of the product uv to a constant factor; but one can find the values of u and v separately. In fact, recalling the results obtained in the “Note sur les covariants d’une fonction quadratique, cubique, ou biquadratique à deux indéterminées” (t. I. p. 285 of this Journal, [135]), one recognises that the two functions $\Phi + \sqrt{-D}$, $\Phi - \sqrt{-D}$ are each perfect cubes and one has

$$2\sqrt{-D} = (\Phi + \sqrt{-D}) - (\Phi - \sqrt{-D});$$

one has therefore

$$u = \frac{\Phi + \sqrt{-D}}{2\sqrt{-D}}, \quad v = \frac{-\Phi + \sqrt{-D}}{2\sqrt{-D}}.$$

In like manner the canonical form of the function of the fourth degree $(a, b, c, d, exy)^4$ will be $u^4 + v^4 + 6\theta u^2 v^2 = (u^2 - v^2)^2 + 2(1 + 3\theta)u^2 v^2$, if we write u, v instead of $\sqrt[4]{A}(x + \alpha y)$, $\sqrt[4]{B}(x + \beta y)$; now $\Omega - \mathfrak{A}\sqrt{-D}$, $\Omega - \mathfrak{B}\sqrt{-D}$, $\Omega - \mathfrak{C}\sqrt{-D}$ being all three squares of linear functions, if we write

$$\sqrt{\Omega - \mathfrak{A}\sqrt{-D}} = u' + v'\sqrt{-D}, \quad \sqrt{\Omega - \mathfrak{B}\sqrt{-D}} = u'' + v''\sqrt{-D}, \quad \sqrt{\Omega - \mathfrak{C}\sqrt{-D}} = u''' + v'''\sqrt{-D}$$

u' , u'' , u''' and v' , v'' , v''' will be linear functions of x, y ; moreover these linear functions are such that $u'u''u'''$, $u'v''v''' + v'u''v''' + v'v''u'''$, $v'v''v'''$ are perfect fourth powers, which may be taken for u^4 , u^2v^2 , v^4 respectively; we have moreover

$$(u' + v'\sqrt{-D})(u'' + v''\sqrt{-D})(u''' + v'''\sqrt{-D})$$

is a perfect fourth power $= (p + q\sqrt{-D})^4$; and by the equation

$$\sqrt{\Omega - \mathfrak{A}\sqrt{-D}} \sqrt{\Omega - \mathfrak{B}\sqrt{-D}} \sqrt{\Omega - \mathfrak{C}\sqrt{-D}} = (p+q\sqrt{-D})^4 (p-q\sqrt{-D})^4 = \Pi^2,$$

we fall upon the identity $\Pi = p^4 + 6p^2q^2D + q^4D^2$, which is the reduced form of the function of the fourth degree. The linear functions $u', u'', u''', v', v'', v'''$ satisfy besides other relations: for example u'^2, u''^2, u'''^2 are respectively proportional to $u'u'' + u'u''' + u''u'''$, etc.

The general theory which has been established for the functions of odd degree does not apply to functions of even degree; for in this case the number of linear functions exceeds by unity the number of terms of the canonical form. We have, it is true, for the function of degree $2m$ the system of equations

$$\begin{aligned} \frac{dU}{da_0} &= a_0 a_1 \dots a_m \\ \frac{dU}{da_1} &= a_0 a_1 \dots a_{m-1} \beta_m + \dots \\ \dots &\quad \dots \end{aligned}$$

where $U = a_0 a_1 \dots a_m + a_0 a_1 \dots a_{m-1} \beta_m + \dots$, and $a_0, a_1, \dots, a_m$ are the linear functions; but these equations are not sufficient for the determination of the linear functions. We may however in this case obtain a canonical form containing a parameter, as is shown by the particular case of the function of the fourth degree: the general theory presents itself under a form corresponding to that which I have considered in the case of the biquadratic function. I hope to return on another occasion to the theory of the canonical form of the functions of even degree.

*London, 2 Stone Buildings, Lincoln's Inn,
January 14, 1857.*

10.2 Addition au Mémoire sur la Forme Canonique des Fonctions Binaires

[FROM THE JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK (CRELLE), T. LIV. (1857), P. 292.]

In the memoir “Sur la forme canonique des fonctions binaires” (p. 48 of this volume [232]) I said that Mr. Sylvester had moreover extended his theory to binary functions of even degrees 4 and 6; I ought to have said, of even degrees 4, 6 and 8. I also omitted to observe that the terms canonizant and lambdaique belonged to Mr. Sylvester. Finally in citing in the note the memoirs of Mr. Sylvester which have reference to this theory I omitted to cite the memoir “On the Calculus of Forms otherwise the Theory of Invariants, § VIII. (Camb. and Dublin Math. Journal, t. IX. [1854], p. 93) section which bears the title “On the Reduction of a Sextic Function of Two Variables to the Canonical Form.”

10.3 Deuxième Note sur une Formule pour la Réversion des Séries

[FROM THE JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK (CRELLE), T. LIV. (1857), PP. 156–161: SEQUEL TO NOTE T. III. (1856), 229.]

I propose to show, in this second note, in what manner Jacobi's theorem leads to a formula given without demonstration by Mr. Sylvester in his memoir entitled "On the Change of Systems of Independent Variables," Quarterly Math. Journal, t. I. [1857], p. 42 to 56 and 126 to 134. To fix the ideas I take the case of three variables, and I suppose that $f(x, y, z)$ is a rational and entire function of x, y, z , and that these variables are given as functions of u, v, w by means of the equations $u = X, v = Y, w = Z$, where X, Y, Z are rational and entire functions of x, y, z . But these functions are no longer subject to the condition (admitted in my first note) that $X - x, Y - y, Z - z$ contain only the

terms of the second and higher orders. I write, to abridge,

$$\delta = \frac{d(X, Y, Z)}{d(x, y, z)},$$

$$D = \frac{d}{dx} X' \frac{d}{dy} Y' \frac{d}{dz} Z',$$

where X', Y', Z' are the same functions of x', y', z' that X, Y, Z are of x, y, z ; and I seek the coefficient of $x'^\alpha y'^\beta z'^\gamma$ in $f(x, y, z)$. If in the equation $f(x, y, z) = F(u, v, w)$, instead of u, v, w , we substitute their values X, Y, Z , we shall have an identical equation, and by differentiating with respect to x, y, z , we obtain

$$\frac{df}{dx} = \frac{dF}{du} \frac{dX}{dx} + \frac{dF}{dv} \frac{dY}{dx} + \frac{dF}{dw} \frac{dZ}{dx},$$

etc.; that is

$$\frac{df}{dx} = DF \cdot \frac{dX}{dx}, \quad \frac{df}{dy} = DF \cdot \frac{dX}{dy}, \quad \frac{df}{dz} = DF \cdot \frac{dX}{dz},$$

where $DF \cdot dX/dx$ denotes

$$\frac{dF}{du} \frac{dX}{dx} + \frac{dF}{dv} \frac{dY}{dx} + \frac{dF}{dw} \frac{dZ}{dx}.$$

But we have

$$\frac{d(F \cdot \delta)}{dx} = \frac{dF}{du} \frac{d(X, Y, Z)}{d(x, y, z)} + F \frac{d\delta}{dx} = \delta \cdot DF \cdot \frac{dX}{dx} + F \frac{d\delta}{dx},$$

and consequently

$$\frac{df}{dx} \delta - f \frac{d\delta}{dx} = \frac{d(F\delta)}{dx};$$

whence, integrating,

$$f\delta = \int \left(\frac{df}{dx} \delta - f \frac{d\delta}{dx} \right) dx = \int \frac{d(f\delta)}{dx} dx,$$

that is $f\delta = F\delta$, an equation which may also be written

$$f(x, y, z) \frac{d(X, Y, Z)}{d(x, y, z)} = F(X', Y', Z') \frac{d(X', Y', Z')}{d(x', y', z')}.$$

I write

$$x' = a_{100}x + b_{100}y + c_{100}z,$$

and representing by

$$\begin{vmatrix} a_{100} & b_{100} & c_{100} \\ a_{010} & b_{010} & c_{010} \\ a_{001} & b_{001} & c_{001} \end{vmatrix}^{-1} \quad \text{the inverse matrix of} \quad \begin{vmatrix} a_{100} & a_{010} & a_{001} \\ b_{100} & b_{010} & b_{001} \\ c_{100} & c_{010} & c_{001} \end{vmatrix},$$

one obtains the equations

$$\begin{aligned} x &= a_{100}x' + b_{100}y' + c_{100}z', \\ y &= a_{010}x' + b_{010}y' + c_{010}z', \\ z &= a_{001}x' + b_{001}y' + c_{001}z'. \end{aligned}$$

We have in general

$$\text{Coeff. } x'^\alpha y'^\beta z'^\gamma \text{ in } f(x, y, z) = \frac{1}{\alpha! \beta! \gamma!} \left(\frac{d}{dx'} \right)^\alpha \left(\frac{d}{dy'} \right)^\beta \left(\frac{d}{dz'} \right)^\gamma f(x, y, z);$$

and if in the function $f(x, y, z)$ we substitute for x, y, z their values in x', y', z' , we obtain

$$\begin{aligned} & \left(\frac{d}{dx'} \right)^\alpha \left(\frac{d}{dy'} \right)^\beta \left(\frac{d}{dz'} \right)^\gamma f(x, y, z) \\ &= \left(a_{100} \frac{d}{dx} + b_{100} \frac{d}{dy} + c_{100} \frac{d}{dz} \right)^\alpha \left(a_{010} \frac{d}{dx} + b_{010} \frac{d}{dy} + c_{010} \frac{d}{dz} \right)^\beta \\ & \quad \times \left(a_{001} \frac{d}{dx} + b_{001} \frac{d}{dy} + c_{001} \frac{d}{dz} \right)^\gamma f(x, y, z). \end{aligned}$$

But this formula may be replaced by a more convenient one. We have

$$\begin{aligned}\frac{d}{dx'} &= a_{100} \frac{d}{dx} + a_{010} \frac{d}{dy} + a_{001} \frac{d}{dz}, \\ \frac{d}{dy'} &= b_{100} \frac{d}{dx} + b_{010} \frac{d}{dy} + b_{001} \frac{d}{dz}, \\ \frac{d}{dz'} &= c_{100} \frac{d}{dx} + c_{010} \frac{d}{dy} + c_{001} \frac{d}{dz};\end{aligned}$$

and by means of these equations we may express

$$\left(\frac{d}{dx'}\right)^\alpha \left(\frac{d}{dy'}\right)^\beta \left(\frac{d}{dz'}\right)^\gamma$$

in the form of a sum of terms

$$C \left(\frac{d}{dx}\right)^{\alpha'} \left(\frac{d}{dy}\right)^{\beta'} \left(\frac{d}{dz}\right)^{\gamma'}$$

where $\alpha' + \beta' + \gamma' = \alpha + \beta + \gamma$. But the calculation of the coefficients C would be very complicated. We may avoid this calculation by observing that the system of equations is equivalent to a single operation. In fact, let ξ, η, ζ be arbitrary quantities; we have identically

$$\begin{aligned}(\xi x' + \eta y' + \zeta z')^n &= \{\xi(a_{100}x + b_{100}y + c_{100}z) + \eta(a_{010}x + b_{010}y + c_{010}z) \\ &\quad + \zeta(a_{001}x + b_{001}y + c_{001}z)\}^n = (Xx + Yy + Zz)^n,\end{aligned}$$

where X, Y, Z are linear functions of ξ, η, ζ ; hence

$$\left(\xi \frac{d}{dx'} + \eta \frac{d}{dy'} + \zeta \frac{d}{dz'}\right)^n = \left(X \frac{d}{dx} + Y \frac{d}{dy} + Z \frac{d}{dz}\right)^n.$$

By developing the two sides and comparing the coefficients of $\xi^\alpha \eta^\beta \zeta^\gamma$, we obtain the value of

$$\left(\frac{d}{dx'}\right)^\alpha \left(\frac{d}{dy'}\right)^\beta \left(\frac{d}{dz'}\right)^\gamma$$

expressed as a sum of terms of the form

$$C \left(\frac{d}{dx}\right)^{\alpha'} \left(\frac{d}{dy}\right)^{\beta'} \left(\frac{d}{dz}\right)^{\gamma'}.$$

Now if in the equation $f(x, y, z) = F(u, v, w)$ we substitute X, Y, Z for u, v, w , we obtain an identity; and if we differentiate with respect to x, y, z , we have as before

$$\frac{df}{dx} = \delta \cdot DF \cdot \frac{dX}{dx}, \quad \text{etc.}$$

But we have also

$$f(x, y, z) \delta = F(X', Y', Z') \delta',$$

and by differentiating this equation with respect to x', y', z' , we obtain the value of

$$\left(\frac{d}{dx'}\right)^\alpha \left(\frac{d}{dy'}\right)^\beta \left(\frac{d}{dz'}\right)^\gamma f(x, y, z)$$

expressed in terms of the derivatives of $F(X', Y', Z') \delta'$ with respect to x', y', z' . But these last derivatives may be calculated directly; for X', Y', Z' being functions of x', y', z' , we have

$$\frac{d(F\delta')}{dx'} = \frac{d(F\delta')}{dX'} \frac{dX'}{dx'} + \frac{d(F\delta')}{dY'} \frac{dY'}{dx'} + \frac{d(F\delta')}{dZ'} \frac{dZ'}{dx'},$$

and similarly for the derivatives with respect to y' and z' .

By means of these formulas we obtain the general term of the reverted series in the form given by Mr. Sylvester. In fact, if we write

$$X = x + A_{200}x^2 + 2A_{110}xy + A_{020}y^2 + \dots,$$

$$Y = y + B_{200}x^2 + 2B_{110}xy + B_{020}y^2 + \dots,$$

$$Z = z + C_{200}x^2 + 2C_{110}xy + C_{020}y^2 + \dots,$$

where the coefficients are such that

$$A_{\alpha\beta\gamma} = \frac{1}{\alpha! \beta! \gamma!} \left(\frac{d}{dx}\right)^\alpha \left(\frac{d}{dy}\right)^\beta \left(\frac{d}{dz}\right)^\gamma X,$$

etc., and if we suppose that X, Y, Z vanish for $x = y = z = 0$, then the formula of Mr. Sylvester gives the coefficient of $x'^\alpha y'^\beta z'^\gamma$ in $f(x, y, z)$ as the coefficient of $x^\alpha y^\beta z^\gamma$ in the development of

$$F(X', Y', Z') \frac{d(X', Y', Z')}{d(x', y', z')}.$$

The formula may be written in the form

$$\text{Coeff. } x'^\alpha y'^\beta z'^\gamma \text{ in } f(x, y, z) = \text{Coeff. } x^\alpha y^\beta z^\gamma \text{ in } \left[f(x, y, z) \frac{d(X, Y, Z)}{d(x, y, z)} \right]_{x=x', y=y', z=z'}$$

or what is the same thing

$$\text{Coeff. } x'^\alpha y'^\beta z'^\gamma \text{ in } f(x, y, z) = \left[f(x, y, z) \frac{d(X, Y, Z)}{d(x, y, z)} \right]_{x=x', y=y', z=z'}.$$

Let now X, Y, Z be

$$X = x \dots + A_{\alpha\beta\gamma} x^\alpha y^\beta z^\gamma + \dots,$$

$$Y = y \dots + B_{\alpha\beta\gamma} x^\alpha y^\beta z^\gamma + \dots,$$

$$Z = z \dots + C_{\alpha\beta\gamma} x^\alpha y^\beta z^\gamma + \dots;$$

in these expressions and everywhere in what follows, the etc.'s relate to terms which one obtains by affixing accents in any number to the indeterminate symbols. By employing the notation of Gauss $\Pi a = 1.2.3 \dots a$, one obtains for the general term of $-X^{-1}$,

$$\frac{\Pi(-\alpha - \beta - \gamma)}{\Pi\alpha \Pi\beta \Pi\gamma \dots} \frac{A_{\alpha\beta\gamma \dots}^{\alpha'} A_{\alpha'\beta'\gamma' \dots}^{\alpha''} \dots}{x^{\alpha+\alpha'+\dots} y^{\beta+\beta'+\dots} z^{\gamma+\gamma'+\dots}},$$

where

$$\alpha + \alpha' + \dots = -1, \quad \beta + \beta' + \dots = 0, \quad \gamma + \gamma' + \dots = 0, \quad \text{etc.}$$

The general term of $-Y^{-1}$ and $-Z^{-1}$ are of analogous forms; and the general term of the reverted series is obtained by multiplying together the three expressions $-X^{-1}$, $-Y^{-1}$, $-Z^{-1}$ and taking the coefficient of $x^{\alpha-1} y^{\beta-1} z^{\gamma-1}$. We thus fall upon the formula of Mr. Sylvester, which may be written

$$\begin{aligned} & \text{Coeff. } x'^\alpha y'^\beta z'^\gamma \text{ in } f(x, y, z) \\ &= \frac{1}{\Pi\alpha \Pi\beta \Pi\gamma} \left(\frac{d}{dA_{100}} + \frac{d}{dB_{010}} + \frac{d}{dC_{001}} \right)^\alpha \left(\frac{d}{dA_{010}} + \frac{d}{dB_{010}} + \frac{d}{dC_{010}} \right)^\beta \\ & \quad \times \left(\frac{d}{dA_{001}} + \frac{d}{dB_{001}} + \frac{d}{dC_{001}} \right)^\gamma F(A_{100}, B_{010}, C_{001}) \frac{d(X, Y, Z)}{d(x, y, z)}, \end{aligned}$$

where after the differentiations we put $A_{100} = B_{010} = C_{001} = 1$, and all the other coefficients equal to zero.

10.4 Sur quelques Formules pour la Transformation des Intégrales Elliptiques

[FROM THE JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK (CRELLE), T. LV. (1858), PP. 15–24.]

I.

En posant dans les formules “Fund. nova p. 15, Tab. III.,” $\cos^2 \theta = x$, on obtient

$$\frac{dy}{\sqrt{(y-\alpha)(y-\beta)(y-\gamma)(y-\delta)}} = \frac{2}{\sqrt{(\alpha-\gamma)(\beta-\delta)} + \sqrt{(\alpha-\beta)(\gamma-\delta)}} \cdot \frac{dx}{\sqrt{(1-x^2)(1-x^2)}}$$

ou

$$\lambda^2 = \frac{\sqrt{(\alpha-\gamma)(\beta-\delta)} - \sqrt{(\alpha-\beta)(\gamma-\delta)}}{\sqrt{(\alpha-\gamma)(\beta-\delta)} + \sqrt{(\alpha-\beta)(\gamma-\delta)}}.$$

Pour obtenir une formule plus symétrique, je pose

$$\alpha = \alpha' + \alpha'', \quad \beta = \beta' + \beta'', \quad \gamma = \gamma' + \gamma'', \quad \delta = \delta' + \delta'',$$

et je suppose que $\alpha', \beta', \gamma', \delta'$ soient proportionnels à $\alpha'', \beta'', \gamma'', \delta''$; les différences $\alpha - \beta$, $\alpha - \gamma$, etc. deviennent alors proportionnelles à $\alpha'\beta'' - \alpha''\beta'$, $\alpha'\gamma'' - \alpha''\gamma'$, etc. Je pose de plus

$$\sqrt{\alpha} = a, \quad \sqrt{\beta} = b, \quad \sqrt{\gamma} = c, \quad \sqrt{\delta} = d,$$

ce qui donne

$$\alpha = a^2, \quad \beta = b^2, \quad \gamma = c^2, \quad \delta = d^2,$$

et par suite

$$\lambda^2 = \frac{(ac + bd) - (ad + bc)}{(ac + bd) + (ad + bc)} = \frac{(a - b)(c - d)}{(a + b)(c + d)}.$$

Cette valeur de λ^2 peut s'écrire sous plusieurs formes; en effet, on a identiquement

$$\frac{(a - b)(c - d)}{(a + b)(c + d)} = \frac{(a - c)(b - d)}{(a + c)(b + d)} = \frac{(a - d)(b - c)}{(a + d)(b + c)}.$$

En posant

$$A = (a+b)(c+d), \quad B = (a-b)(c-d), \quad C = (a+c)(b+d), \quad D = (a-c)(b-d),$$

et

$$E = (a + d)(b + c), \quad F = (a - d)(b - c),$$

on a

$$A + B = 2(ac + bd), \quad A - B = 2(ad + bc), \quad C + D = 2(ab + cd), \quad \text{etc.},$$

et l'on trouve sans peine

$$A + B + C + D + E + F = 0,$$

ce qui donne

$$\lambda^2 = \frac{B}{A} = \frac{D}{C} = \frac{F}{E}.$$

On a de plus

$$1 + \lambda = \frac{ac + bd + ad + bc}{\sqrt{(a + b)(c + d)(a + d)(b + c)}} = \frac{(a + b)(c + d) + (a - b)(c - d)}{\sqrt{AC} \dots},$$

et

$$1 - \lambda = \frac{ac + bd - ad - bc}{\sqrt{(a + b)(c + d)(a + d)(b + c)}} = \frac{(a + c)(b + d) - (a - c)(b - d)}{\sqrt{AE} \dots};$$

d'où l'on tire

$$\frac{1 + \lambda}{1 - \lambda} = \frac{(a + b)(c + d)}{(a + c)(b + d)} \cdot \frac{(a + d)(b + c)}{(a + c)(b + d)} = \frac{A}{C} \cdot \frac{E}{C}.$$

En posant maintenant $y = \frac{\alpha}{x^2}$, on obtient la formule de transformation

$$\frac{dy}{\sqrt{(y - \alpha)(y - \beta)(y - \gamma)(y - \delta)}} = -\frac{2 dx}{x^2 \sqrt{(\alpha - \beta x^2)(\alpha - \gamma x^2)(\alpha - \delta x^2)}},$$

et par suite

$$\frac{dx}{\sqrt{(1 - \frac{\beta}{\alpha} x^2)(1 - \frac{\gamma}{\alpha} x^2)(1 - \frac{\delta}{\alpha} x^2)}} = \frac{\sqrt{\alpha} dy}{\sqrt{(y - \alpha)(y - \beta)(y - \gamma)(y - \delta)}}.$$

II.

Je considère maintenant la formule de Landen. En posant c au lieu de k' , on a $k^2 + c^2 = 1$, et la formule de Landen donne

$$\int \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}} = \frac{1}{1 + k'} \int \frac{d\phi'}{\sqrt{1 - \lambda^2 \sin^2 \phi'}},$$

où

$$\lambda = \frac{1 - k'}{1 + k'}, \quad \tan(\phi' - \phi) = k' \tan \phi.$$

Je pose $k' = \frac{b}{a}$, ce qui donne $k^2 = \frac{a^2 - b^2}{a^2}$; on a alors

$$\lambda = \frac{a - b}{a + b}, \quad 1 + \lambda = \frac{2a}{a + b}, \quad 1 - \lambda = \frac{2b}{a + b},$$

et par suite

$$1 - \lambda^2 = \frac{4ab}{(a + b)^2}.$$

On obtient ainsi la formule

$$\int \frac{d\phi}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}} = \int \frac{d\phi'}{\sqrt{a'^2 \cos^2 \phi' + b'^2 \sin^2 \phi'}},$$

où

$$a' = \frac{a + b}{2}, \quad b' = \sqrt{ab}, \quad \tan(\phi' - \phi) = \frac{b}{a} \tan \phi.$$

Cette formule peut s'écrire sous une forme plus symétrique. En effet, posons

$$\alpha = a^2, \quad \beta = b^2, \quad \gamma = c^2,$$

et considérons l'intégrale

$$\int \frac{dx}{\sqrt{(x - \alpha)(x - \beta)(x - \gamma)}}.$$

En faisant la substitution $x = \alpha \sin^2 \theta + \beta \cos^2 \theta$, on obtient

$$\int \frac{2 d\theta}{\sqrt{(\alpha - \gamma) \sin^2 \theta + (\beta - \gamma) \cos^2 \theta}};$$

et par la formule de Landen, cette intégrale se transforme en

$$\int \frac{d\theta'}{\sqrt{(\alpha' - \gamma') \sin^2 \theta' + (\beta' - \gamma') \cos^2 \theta'}},$$

où

$$\alpha' = \frac{\alpha + \beta}{2}, \quad \beta' = \sqrt{\alpha\beta}, \quad \gamma' = \gamma,$$

et

$$\tan(\theta' - \theta) = \sqrt{\frac{\beta - \gamma}{\alpha - \gamma}} \tan \theta.$$

III.

Je considère enfin la transformation de M. Jacobi. En posant

$$x = \frac{1}{\sqrt{k}} \cdot \frac{\vartheta_1(\frac{u\vartheta_3^2}{\vartheta_2^2})}{\vartheta_0(\frac{u\vartheta_3^2}{\vartheta_2^2})},$$

on a

$$u = \int_0^x \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}}.$$

En changeant u en $u + K$, x se change en $\frac{1}{kx}$, et l'on a

$$\int_0^{\frac{1}{kx}} \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} = u + K = \int_0^x \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} + K.$$

La formule de M. Jacobi donne la transformation du troisième ordre. En posant

$$y = \frac{x}{\alpha} \cdot \frac{1 - \frac{x^2}{\beta^2}}{1 - \alpha^2 k^2 x^2},$$

on a

$$\frac{dy}{\sqrt{(1-y^2)(1-\lambda^2 y^2)}} = \frac{dx}{M \sqrt{(1-x^2)(1-k^2 x^2)}},$$

où λ et M sont des fonctions de k données par les formules

$$\lambda = k^3 \left(\frac{\alpha\beta}{\gamma} \right)^2, \quad M = \frac{\alpha}{\beta\gamma},$$

les quantités α, β, γ étant déterminées par les équations

$$\alpha = \frac{\vartheta_2(0)}{\vartheta_3(0)}, \quad \beta = \frac{\vartheta_0(0|3\tau)}{\vartheta_3(0|3\tau)}, \quad \gamma = \frac{\vartheta_2(0|3\tau)}{\vartheta_3(0|3\tau)}.$$

Les formules de transformation du cinquième ordre sont plus compliquées. M. Jacobi a donné la formule générale pour la transformation de l'ordre n . En posant

$$y = \frac{x}{M} \cdot \frac{1 + A_1x^2 + A_2x^4 + \dots + A_mx^{2m}}{1 + B_1x^2 + B_2x^4 + \dots + B_mx^{2m}},$$

où $m = \frac{n-1}{2}$, on a

$$\frac{dy}{\sqrt{(1-y^2)(1-\lambda^2y^2)}} = \frac{dx}{M\sqrt{(1-x^2)(1-k^2x^2)}}.$$

Les coefficients $A_1, A_2, \dots, A_m, B_1, B_2, \dots, B_m$ et les quantités λ, M sont des fonctions algébriques de k , que l'on peut exprimer au moyen des fonctions ϑ . On a en particulier

$$M = (-1)^m \frac{\alpha_1 \alpha_2 \dots \alpha_m}{\beta_1 \beta_2 \dots \beta_m},$$

où $\alpha_1, \alpha_2, \dots, \alpha_m$ sont les racines de l'équation $\vartheta_1(u) = 0$ qui correspondent aux périodes $2K/n, 4K/n, \dots, 2mK/n$, et $\beta_1, \beta_2, \dots, \beta_m$ sont les racines de l'équation $\vartheta_0(u) = 0$ correspondant aux mêmes périodes.

La formule de transformation peut aussi s'écrire sous la forme

$$y = \frac{1}{M} \frac{x \prod' (1 - \frac{x^2}{\alpha_r^2})}{\prod' (1 - \frac{x^2}{\beta_r^2})},$$

où les produits sont étendus aux m valeurs de r . On a de plus

$$\lambda = k^n \prod' \left(\frac{\alpha_r}{\beta_r} \right)^4,$$

et

$$M = \frac{1}{n} \prod' \left(\frac{\beta_r}{\alpha_r} \right)^2.$$

Ces formules donnent la transformation de l'ordre n pour la fonction elliptique $\text{sn}u$. On peut en déduire les transformations correspondantes pour $\text{cn}u$ et $\text{dn}u$ au moyen des formules

$$\sqrt{1-y^2} = \frac{\text{cn}(u/M)}{\text{dn}'(u/M)}, \quad \sqrt{1-\lambda^2y^2} = \frac{\text{dn}(u/M)}{\text{cn}'(u/M)}.$$

IV.

Je vais maintenant appliquer les formules précédentes à quelques exemples. Pour la transformation du troisième ordre, on a

$$\alpha = \frac{1 + \sqrt{k'}}{\sqrt{2(1 + k')}}, \quad \beta = \frac{1 - \sqrt{k'}}{\sqrt{2(1 + k')}}, \quad \gamma = \sqrt{\frac{k'}{1 + k'^2}}.$$

En posant $k' = \frac{b}{a}$, on obtient

$$\alpha^2 = \frac{a + b + 2\sqrt{ab}}{2(a + b)} = \frac{(\sqrt{a} + \sqrt{b})^2}{2(a + b)},$$

et par suite

$$\alpha = \frac{\sqrt{a} + \sqrt{b}}{\sqrt{2(a + b)}}.$$

De même

$$\beta = \frac{\sqrt{a} - \sqrt{b}}{\sqrt{2(a + b)}}, \quad \gamma = \sqrt{\frac{b}{a + b}}.$$

On en déduit

$$M = \frac{\alpha}{\beta\gamma} = \frac{a + b}{a - b} \cdot \frac{\sqrt{a} + \sqrt{b}}{\sqrt{2(a + b)}} \cdot \sqrt{\frac{a + b}{b}} = \frac{(\sqrt{a} + \sqrt{b})^2}{2\sqrt{b(a - b)}}.$$

On trouve de même

$$\lambda = k^3 \left(\frac{\alpha\beta}{\gamma} \right)^2 = \left(\frac{a^2 - b^2}{a^2} \right)^3 \cdot \frac{(a + b)^2}{4(a + b)^2} \cdot \frac{a + b}{b} = \frac{(a - b)^3(a + b)}{4a^6} \cdot \frac{a + b}{b}.$$

Après quelques réductions, on obtient

$$\lambda = \left(\frac{a - b}{a + b} \right)^2 \cdot \frac{(a + b)^2}{4ab} = \frac{(a - b)^2}{4ab}.$$

Ces résultats s'accordent avec ceux que j'ai donnés dans mon mémoire "Sur la transformation des fonctions elliptiques" (p. 13 of this volume [225]).

Londres, le 20 Août, 1857.

10.5 Note sur la Composition du Nombre 47 par rapport aux Vingt-troisièmes Racines de l'Unité

[FROM THE JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK (CRELLE), T. LV. (1858), P. 192.]

Mr. Kummer has found (Journal de Liouville, t. XII. [1847] p. 208) that the number 47 can be decomposed into eleven factors which are derived from the following $\alpha^{11} + \alpha^{10} + \alpha^8 + \alpha^7 + \alpha^4 + \alpha^2$, α denoting a 23rd root of unity, and we know by the general theory that there must be a power 47^{l-1} which decomposes into twenty-two factors. The number 47^l can be decomposed into two factors formed with the half-periods of the roots; it was therefore natural to try whether the factor $(\alpha^{11} + \alpha^{10} + \alpha^8 + \alpha^7 + \alpha^4 + \alpha^2)^l$ could be decomposed in like manner into two factors, which would give the decomposition of 47^l into twenty-two factors. But one demonstrates very easily that this decomposition is not possible. In fact, in putting $\alpha^\lambda = 1$ (λ being a prime number) and making

$$A = R + C\alpha + D\alpha^2 + \dots + K'\alpha^{\lambda-2},$$

if A is a factor of a prime number p , and if $A' = R + C\alpha^\gamma + D\alpha^{2\gamma} + \dots + K'\alpha^{(\lambda-2)\gamma}$, where γ is a primitive root of λ , one has $AA' = p$; and by means of the equation $1 + \alpha + \alpha^2 + \dots + \alpha^{\lambda-1} = 0$, one can reduce AA' to the form $P + Q\alpha + R\alpha^2 + \dots + T\alpha^{\lambda-2}$. If this expression is equal to p , the sum $P + Q + R + \dots + T$ must be equal to p , and consequently positive; but if the sum $P + Q + R + \dots + T$ is negative, the decomposition cannot be effected.

But, in reducing only by means of the equation $\alpha^{23} - 1 = 0$, one finds the following

$$\begin{aligned} & (\alpha^{10} + \alpha^{11} + \alpha^8 + \alpha^7 + \alpha^4 + \alpha^2)^2 = \\ & 6 + 7\alpha + 7\alpha^2 + 3\alpha^3 + 6\alpha^4 + 9\alpha^5 + 6\alpha^6 + 16\alpha^7 + 15\alpha^8 + 9\alpha^9 + 18\alpha^{10} + 9\alpha^{11} \\ & + 7\alpha^{12} + 7\alpha^{13} + 3\alpha^{14} + 6\alpha^{15} + 9\alpha^{16} + 6\alpha^{17} + 16\alpha^{18} + 15\alpha^{19} + 9\alpha^{20} + 18\alpha^{21} + 9\alpha^{22}, \end{aligned}$$

which, by virtue of $1 + \alpha + \dots + \alpha^{22} = 0$, reduces to

$$\begin{aligned} & \alpha + \alpha^2 - 3\alpha^3 + 3\alpha^4 + 10\alpha^5 + 9\alpha^6 + 3\alpha^7 + 12\alpha^8 + 3\alpha^9 \\ & + \alpha^{12} + \alpha^{13} - 3\alpha^{14} + 3\alpha^{15} + 10\alpha^{16} + 9\alpha^{17} + 3\alpha^{18} + 12\alpha^{19} + 3\alpha^{20}, \end{aligned}$$

or the sum of the coefficients is positive; therefore the decomposition cannot be effected.

One could without much trouble try in like manner the numbers $l = 2$ or $l = 3$, but I do not know whether one has any idea of the magnitude of the number l .

Londres, le 10 Mai, 1857.

10.6 Théorème sur les Déterminants Gauches

[FROM THE JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK (CRELLE), T. LV. (1858), PP. 277–278.]

In the memoir entitled “Recherches ultérieures sur les déterminants gauches,” this journal t. L. pp. 299–313 (1855), [137], I gave a formula for the development of a bordered skew determinant; but I omitted to remark a rather important particular case. The general formula relates to a determinant such as:

$$d234 \dots | \beta 1234 \dots = \begin{array}{c|cccc} & a_1 & a_2 & a_3 & a_4 \\ \hline b_1 & 11 & 12 & 13 & 14 \\ b_2 & 21 & 22 & 23 & 24 \\ b_3 & 31 & 32 & 33 & 34 \\ b_4 & 41 & 42 & 43 & 44 \end{array}$$

which is obtained by bordering in any manner the skew matrix

$$\begin{array}{cccc} 11 & 12 & 13 & 14 \\ 21 & 22 & 23 & 24 \\ 31 & 32 & 33 & 34 \\ 41 & 42 & 43 & 44 \end{array}$$

The general formula is:

$$\begin{aligned} d234 \dots | \beta 1234 \dots &= a\beta \cdot 11 \cdot 22 \cdot 33 \cdot 44 \dots \\ &+ a\beta_1 \cdot 12 \cdot 12 \cdot 33 \cdot 44 \dots + a\beta_2 \cdot 13 \cdot 13 \cdot 22 \cdot 44 \dots + \dots \\ &+ a\beta_{123} \cdot 123 \cdot 123 \cdot 44 \dots + a\beta_{124} \cdot 124 \cdot 124 \cdot 33 \dots + \dots \\ &\quad + a\beta_{1234} \cdot 1234 \cdot 1234 \dots + \dots \\ &+ a_1\beta \cdot \beta_1 \cdot 22 \cdot 33 \cdot 44 \dots + a_2\beta \cdot \beta_2 \cdot 11 \cdot 33 \cdot 44 \dots + \dots \\ &\quad + a_{12}\beta \cdot \beta_{12} \cdot 33 \cdot 44 \dots + a_{13}\beta \cdot \beta_{13} \cdot 22 \cdot 44 \dots + \dots \end{aligned}$$

where the expressions $a\beta_{12}$, $a\beta_{13}$ etc. are Pfaffians. This being so, the determinant formed by bordering in any manner a skew symmetric matrix may be expressed in terms of the Pfaffians of the matrix and the elements of the border. In particular, if the matrix is of even order $2m$, and the border consists of a single row and column, the determinant is equal to the sum of the squares of the Pfaffians of the matrix.

10.7 Note sur les Normales d'une Conique

[FROM THE JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK (CRELLE), T. LVI. (1859), PP. 182–185.]

One knows the very elegant researches of Mr. Joachimsthal on the normals of a conic [see the Memoir “Ueber die Normalen der Ellipse und des Ellipsoids,” Crelle, t. XXVI. (1843) pp. 172–180]; in particular the author has obtained the following theorem: supposing that 1, 2, 3, 4 are points of a conic such that the four normals meet in the same point, one takes the pole of the line (1, 2) with respect to the conic and one draws through this pole perpendiculars to the diameters of the conic; this being so, in taking on each diameter in the opposite sense a point whose distance from the centre is equal to the length of the perpendicular, the four points thus obtained lie on a conic passing through the centre.

I propose to give a simple analytical proof of this theorem and of some other theorems relating to the normals of a conic.

Let the equation of the conic be

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

and let (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , (x_4, y_4) be the coordinates of the four points. The equation of the normal at the point (x_r, y_r) is

$$\frac{a^2 x}{x_r} - \frac{b^2 y}{y_r} = a^2 - b^2.$$

If the four normals meet at the point (h, k) , we have

$$\frac{a^2 h}{x_r} - \frac{b^2 k}{y_r} = a^2 - b^2, \quad r = 1, 2, 3, 4.$$

The condition that the four normals meet at a point may be expressed as follows. The coordinates x_r , y_r satisfy the equation of the conic and the equation

$$\frac{a^2 h}{x} - \frac{b^2 k}{y} = a^2 - b^2.$$

By eliminating x or y between these two equations, we obtain an equation of the fourth degree which determines the four points. The condition that the four normals meet at a point is therefore equivalent

to a certain relation between the coefficients of this equation, which may be expressed as a relation between the symmetric functions of the roots.

Let us now consider the theorem of Mr. Joachimsthal. The pole of the line joining the points (x_1, y_1) and (x_2, y_2) is the point

$$\left(\frac{a^2(x_1 + x_2)}{2}, \frac{b^2(y_1 + y_2)}{2} \right).$$

The perpendicular from this point to the diameter $y = mx$ has for its length

$$\frac{\left| \frac{b^2(y_1 + y_2)}{2} - m \frac{a^2(x_1 + x_2)}{2} \right|}{\sqrt{1 + m^2}}.$$

The point on the diameter in the opposite sense at this distance from the centre is

$$\left(-\frac{a^2(x_1 + x_2)}{2} \cdot \frac{1}{\sqrt{1 + m^2}}, -\frac{b^2(y_1 + y_2)}{2} \cdot \frac{m}{\sqrt{1 + m^2}} \right),$$

and similarly for the other pairs of points. The theorem asserts that these four points lie on a conic passing through the centre.

To prove this analytically, I consider the equation of the conic which passes through the four points. Let (X_1, Y_1) , (X_2, Y_2) , (X_3, Y_3) , (X_4, Y_4) be the coordinates of the four points constructed as above. We have

$$X_r = -\frac{a^2(x_i + x_j)}{2\sqrt{1 + m^2}}, \quad Y_r = -\frac{b^2(y_i + y_j)}{2} \cdot \frac{m}{\sqrt{1 + m^2}},$$

where (i, j) is the pair corresponding to r . The condition that these four points lie on a conic

$$Ax^2 + 2Hxy + By^2 + 2Gx + 2Fy + C = 0$$

is that the determinant

$$\begin{vmatrix} X_1^2 & X_1Y_1 & Y_1^2 & X_1 & Y_1 & 1 \\ X_2^2 & X_2Y_2 & Y_2^2 & X_2 & Y_2 & 1 \\ X_3^2 & X_3Y_3 & Y_3^2 & X_3 & Y_3 & 1 \\ X_4^2 & X_4Y_4 & Y_4^2 & X_4 & Y_4 & 1 \end{vmatrix}$$

vanishes identically when the condition for the concurrence of the normals is satisfied.

This determinant may be simplified by means of the relations which exist between the coordinates. In particular, if we make use of the identity

$$\sum \frac{x_1y_2 - x_2y_1}{(a^2 - b^2)x_1x_2y_1y_2} = 0,$$

which is a consequence of the condition that the four normals meet at a point, we find that the determinant reduces to a form which shows that the four points lie on a conic passing through the origin. The equation of this conic is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{hx}{a^2} - \frac{ky}{b^2} = 0,$$

which passes through the centre $(0, 0)$. This completes the proof of Mr. Joachimsthal's theorem.

I may add that the locus of the point of concurrence of the normals, when the four points vary subject to the condition that the pole of the line joining two of them lies on a fixed line, is a cubic curve having the centre of the conic as a double point.

10.8 Addition à la Note sur la Composition du Nombre 47

[FROM THE JOURNAL FÜR DIE REINE UND ANGEWANDTE MATHEMATIK (CRELLE), T. LVI. (1859), P. 185.]

I have remarked in the preceding Note [236] that one could without much trouble try the numbers $l = 2$ or $l = 3$. I have since found that the number l is necessarily odd. In fact, if A is a factor of the prime number p , and if A' is the conjugate factor, we have $AA' = p^l$; and if l were even, AA' would be a perfect square, which is impossible unless A and A' are both perfect squares, which is not the case in general. The number l is therefore odd, and the smallest value to try is $l = 3$. But the calculations become very laborious for this value, and I have not yet been able to complete them.

I may observe that the question of the decomposition of 47^l into twenty-two factors is connected with the theory of the division of the lemniscate. In fact, if α is a primitive 23^{rd} root of unity, the equation $\alpha^{23} = 1$ is connected with the division of the lemniscate into 23 parts; and the decomposition of 47^l into factors formed with the roots of this equation corresponds to the decomposition of the prime number 47 in the field of 23^{rd} roots of unity.

Londres, le 15 Juin, 1857.

10.9 A Theorem on Groups

[FROM THE PROCEEDINGS OF THE LONDON MATHEMATICAL SOCIETY, VOL. I. (1865), P. 30.]

The following theorem on groups is remarkable for its simplicity and generality:

Theorem. *If p is any prime number, and n any integer whatever, the symmetric group of degree p^n contains a subgroup of order p^n .*

This theorem is a particular case of a more general theorem due to Mr. Sylow, which states that if p^α is the highest power of the prime p which divides the order of a group, the group contains a subgroup of order p^α . The proof of this theorem depends on the properties of the multinomial coefficients, and in particular on the theorem that the multinomial coefficient

$$\frac{(p^n)!}{(p^{n-1}!)^p \cdot p!}$$

is not divisible by p .

10.10 On Poinot's Four New Regular Solids

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVI. (1858), PP. 66–72.]

The discovery of four new regular solids by M. Poinot (de l'Académie Française) is one of the most remarkable contributions to the geometry of polyhedra which has been made since the time of the ancients. The five Platonic solids have been known from antiquity; they are the tetrahedron, the cube, the octahedron, the dodecahedron, and the icosahedron. These solids are characterised by the property that their faces are equal regular polygons, and that their solid angles are equal. M. Poinot has shown that if we relax the condition that the faces should be simply connected, and allow them to be star-polygons, we obtain four additional regular solids, which he has called the *small stellated dodecahedron*, the *great stellated dodecahedron*, the *great dodecahedron*, and the *great icosahedron*.

The small stellated dodecahedron has 12 faces, each of which is a regular pentagram (or five-pointed star), and 12 vertices, at each of which five faces meet. Its Euler characteristic is

$$F - E + V = 12 - 30 + 12 = -6,$$

which shows that it is not of the same topological nature as the ordinary convex polyhedra. The great stellated dodecahedron has also 12 faces, which are regular pentagrams, but at each vertex three faces meet; it has 20 vertices and 30 edges, and its Euler characteristic is

$$F - E + V = 12 - 30 + 20 = 2.$$

The great dodecahedron has 12 faces, which are regular pentagons, and 12 vertices, at each of which five faces meet; but the arrangement of the faces is such that the solid is not convex. The great icosahedron has 20 faces, which are equilateral triangles, and 12 vertices, at each of which five faces meet; it may be regarded as the reciprocal of the great stellated dodecahedron.

The reciprocals of these four solids are easily obtained. The reciprocal of a polyhedron is formed by taking the poles of its faces with respect to a concentric sphere, and joining these poles by edges when the corresponding faces meet at an edge. The reciprocal of the small stellated dodecahedron is the great dodecahedron, and vice

versa; the reciprocal of the great stellated dodecahedron is the great icosahedron, and vice versa.

The theory of these solids may be connected with the theory of the regular polyhedra by considering them as arising from the process of stellation. If we produce the faces of the dodecahedron until they meet again, we obtain the small stellated dodecahedron; if we proceed further, we obtain the great stellated dodecahedron. Similarly, by producing the faces of the icosahedron, we obtain successively the great icosahedron and other stellated forms.

M. Poinot has also given a formula connecting the number of faces, edges, and vertices of these solids, which is a generalisation of Euler's formula. If F is the number of faces, E the number of edges, V the number of vertices, and if S is the sum of the angles of all the faces divided by four right angles, then

$$F - E + V = 2 - 2n,$$

where n is a number depending on the connectivity of the solid. For the ordinary convex polyhedra, $n = 0$ and we have Euler's formula $F - E + V = 2$. For the small stellated dodecahedron, $n = 4$ and $F - E + V = -6$, as we have seen.

The number n may be interpreted as the number of times the surface of the solid winds round the centre, or as the genus of the surface considered as a Riemann surface. This conception has been developed by Möbius and by subsequent writers on the topology of surfaces.

I may observe that the four new regular solids of M. Poinot are closely connected with the theory of the modular equation. In fact, the subdivision of the sphere into triangles corresponding to the great dodecahedron and the great icosahedron leads to equations which are essentially the same as the modular equations of transformation of the elliptic functions. This connection has been developed by Prof. Schwarz and other mathematicians.

The following table exhibits the principal characteristics of the nine regular solids:

<i>Name</i>	<i>F</i>	<i>E</i>	<i>V</i>	<i>Face</i>
Tetrahedron	4	6	4	Triangle
Cube	6	12	8	Square
Octahedron	8	12	6	Triangle
Dodecahedron	12	30	20	Pentagon
Icosahedron	20	30	12	Triangle
Small stellated dodecahedron	12	30	12	Pentagram
Great dodecahedron	12	30	12	Pentagon
Great stellated dodecahedron	12	30	20	Pentagram
Great icosahedron	20	30	12	Triangle

*2 Stone Buildings, Lincoln's Inn,
December 10, 1857.*

10.11 Second Note on Poinot's Four New Regular Polyhedra

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVII. (1859), PP. 126–130.]

In my former Note [241] I gave a brief account of the four new regular solids discovered by M. Poinot, and of their relation to the five Platonic solids. I propose now to add some remarks on the analytical theory of these solids, and on the formulae which connect their several elements.

The vertices of the small stellated dodecahedron may be represented by the coordinates

$$(0, \pm\tau, \pm 1), \quad (\pm 1, 0, \pm\tau), \quad (\pm\tau, \pm 1, 0),$$

where $\tau = \frac{1}{2}(1 + \sqrt{5})$ is the golden ratio. These 12 points are the vertices of an icosahedron; and the small stellated dodecahedron is formed by joining these vertices by 30 edges, which are the sides of 12 pentagrammic faces. The equation of one of the faces is

$$\tau x \pm y = \pm\tau^2,$$

and the 12 faces are obtained by permuting the coordinates and taking the different signs.

The great dodecahedron has the same 12 vertices as the small stellated dodecahedron, but its 12 faces are ordinary pentagons, not pentagrams. The equation of one of its faces is

$$x \pm \tau y = \pm(2\tau + 1),$$

or equivalently

$$x \pm \tau y = \pm\sqrt{5}\tau.$$

The 30 edges of the great dodecahedron are the same as those of the small stellated dodecahedron; the two solids differ only in the arrangement of their faces.

The great stellated dodecahedron has 20 vertices, which may be taken as

$$(\pm 1, \pm 1, \pm 1), \quad (0, \pm\tau^{-1}, \pm\tau), \quad (\pm\tau, 0, \pm\tau^{-1}), \quad (\pm\tau^{-1}, \pm\tau, 0),$$

together with the 12 points obtained by even permutations of $(\pm 1, \pm 1, \pm 1)$. These 20 vertices are those of a regular dodecahedron;

and the 12 faces of the great stellated dodecahedron are regular pentagrans, whose planes are those of the faces of a regular icosahedron containing the dodecahedron.

The great icosahedron has the same 20 vertices as the great stellated dodecahedron, but its 20 faces are equilateral triangles. It is the reciprocal of the great stellated dodecahedron with respect to a concentric sphere; and its 30 edges are the same as those of the great stellated dodecahedron.

The analytical theory of these solids is intimately connected with the theory of the icosahedral equation and of the modular equation of the fifth order. The finite groups of projective transformations which leave these solids invariant are identical with the groups which arise in the theory of the quintic equation and of the elliptic modular functions. This connection has been fully developed by MM. Hermite, Kronecker, and Brioschi, and more recently by Prof. Klein.

*2 Stone Buildings, Lincoln's Inn,
March 15, 1858.*

10.12 On the Theory of Groups, as Depending on the Symbolic Equation $\theta^n = 1$

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVI. (1858), PP. 40–47; PART I. (SECOND PART [246], 1859.)]

The theory of groups, which has been so extensively developed by M. Cauchy and other mathematicians, acquires a new aspect when we consider the symbols which represent the operations of the group as subject to algebraical laws. Let θ be a symbol representing an operation which may be performed upon an object, and let $\theta^n = 1$ denote that the operation θ , repeated n times, restores the object to its original state. The symbol θ satisfies the equation $\theta^n - 1 = 0$, and may be regarded as a root of this equation.

If θ and ϕ are two operations of a group, the compound operation $\theta\phi$ is defined as that operation which consists in performing ϕ first and then θ . In general $\theta\phi$ is not the same as $\phi\theta$; but if the operations are permutable, we have $\theta\phi = \phi\theta$, and the group is said to be *Abelian*.

The general theory of groups may be developed as follows. Let $\theta_1, \theta_2, \dots, \theta_k$ be the operations of a group of order N ; and let the multiplication table of the group be given by the equations

$$\theta_r \theta_s = \theta_{f(r,s)}, \quad r, s = 1, 2, \dots, N,$$

where $f(r, s)$ is a function which determines the product of any two operations. The group is completely determined by this function $f(r, s)$, which must satisfy the conditions:

- (i) $f(f(r, s), t) = f(r, f(s, t))$ (associative law);
- (ii) there exists an element e such that $f(r, e) = f(e, r) = r$;
- (iii) for each r there exists an r' such that $f(r, r') = f(r', r) = e$.

A group is said to be *generated* by a system of elements $\alpha, \beta, \gamma, \dots$ when every element of the group can be expressed as a product of powers of these elements. The equations which express the relations between the generating elements are called the *defining equations* of the group. For example, the group of order n generated by a single element α subject to the relation $\alpha^n = 1$ is the *cyclic group* of order n .

A group of considerable importance is the *quaternion group*, of order 8, generated by two elements α, β subject to the relations

$$\alpha^4 = 1, \quad \beta^2 = \alpha^2, \quad \beta^{-1}\alpha\beta = \alpha^{-1}.$$

The eight elements of this group are

$$1, \alpha, \alpha^2, \alpha^3, \beta, \alpha\beta, \alpha^2\beta, \alpha^3\beta.$$

This group is remarkable as being the first known instance of a non-Abelian group in which every subgroup is normal.

I have constructed tables for all the groups of orders up to 12, and I have verified that the number of distinct groups of each order is as follows:

Order	1	2	3	4	5	6	7	8	9	10	11	12
No. of groups	1	1	1	2	1	2	1	5	2	2	1	5

The five groups of order 8 are: the cyclic group; the Abelian group generated by $\alpha^4 = 1, \beta^2 = 1, \alpha\beta = \beta\alpha$; the Abelian group generated by $\alpha^2 = 1, \beta^2 = 1, \gamma^2 = 1$; the quaternion group; and the dihedral group of order 8. The theory of these groups presents many interesting features, which I hope to develop in a future communication.

*2 Stone Buildings, Lincoln's Inn,
February 18, 1858.*

10.13 On an Analytical Theorem Relating to the Distribution of Electricity upon Spherical Surfaces

[FROM THE PHILOSOPHICAL MAGAZINE, VOL. XVI. (1858), PP. 141–148; AND VOL. XVII. (1859), PP. 1–7.]

The problem of the distribution of electricity upon a spherical surface, when the electrical state is maintained by given forces, is one of the most interesting in the mathematical theory of electricity. The solution depends upon a remarkable analytical theorem, which I propose to establish in the present paper.

Let V be the potential at any point of space due to the electrical distribution upon a spherical surface of radius a , and let ρ be the density of the electricity at a point of the surface. Then if θ, ϕ are the angular coordinates of the point, we have

$$V = \iint \frac{\rho' dS'}{r},$$

where r is the distance between the point (θ, ϕ) and the point (θ', ϕ') , ρ' is the density at the latter point, and dS' is the element of surface. If the point is external to the sphere, we may develop $1/r$ in a series of spherical harmonics, and we obtain

$$V = \sum_{n=0}^{\infty} \frac{a^{n+2}}{R^{n+1}} \iint \rho' P_n(\cos \gamma) d\omega',$$

where R is the distance of the external point from the centre, γ is the angle between the radii to the two points, P_n is the Legendre coefficient of order n , and $d\omega'$ is the element of solid angle.

The density ρ may be expanded in a series of Laplace's functions:

$$\rho = \sum_{n=0}^{\infty} Y_n(\theta, \phi),$$

where Y_n is a surface harmonic of order n . By the property of these functions,

$$\iint Y_n' P_m(\cos \gamma) d\omega' = \frac{4\pi}{2n+1} Y_n \delta_{mn},$$

where δ_{mn} is the symbol of Kronecker. Hence we obtain

$$V = 4\pi a^2 \sum_{n=0}^{\infty} \frac{1}{2n+1} \left(\frac{a}{R}\right)^{n+1} Y_n(\theta, \phi).$$

If the point is internal to the sphere, we have similarly

$$V = 4\pi a^2 \sum_{n=0}^{\infty} \frac{1}{2n+1} \left(\frac{R}{a}\right)^n Y_n(\theta, \phi).$$

At the surface of the sphere, these two expressions must agree, and we have

$$V = 4\pi a^2 \sum_{n=0}^{\infty} \frac{1}{2n+1} Y_n(\theta, \phi).$$

The potential due to external forces may also be expanded in a series of solid harmonics:

$$V_{\text{ext}} = \sum_{n=0}^{\infty} R^n S_n(\theta, \phi),$$

where S_n is a surface harmonic of order n . At the surface of the sphere, the total potential must be constant; and if the sphere is a conductor, this constant may be taken as zero. We have therefore

$$4\pi a^2 \sum_{n=0}^{\infty} \frac{1}{2n+1} Y_n + \sum_{n=0}^{\infty} a^n S_n = 0,$$

whence, equating the terms of each order,

$$Y_n = -\frac{2n+1}{4\pi a^{2-n}} S_n.$$

The density is therefore given by

$$\rho = -\frac{1}{4\pi a^2} \sum_{n=0}^{\infty} (2n+1) a^n S_n.$$

This formula gives the distribution of electricity upon a conducting sphere when the potential due to external forces is known. It may be written in the form

$$\rho = -\frac{1}{4\pi a^2} \left(1 + 2a \frac{d}{da}\right) V_{\text{ext}},$$

where the differentiation is performed with respect to a , the radius of the sphere, and the result is evaluated at the surface.

Application to a Point Charge

Let us apply this formula to the case of a point charge e at a distance f from the centre of the sphere. The potential due to this charge is

$$V_{\text{ext}} = \frac{e}{\sqrt{f^2 + R^2 - 2fR \cos \theta}},$$

where θ is the angle between the radius to the point and the line joining the centre to the charge.

Expanding in spherical harmonics, we have

$$V_{\text{ext}} = \frac{e}{f} \sum_{n=0}^{\infty} \left(\frac{R}{f}\right)^n P_n(\cos \theta), \quad (f > R),$$

and therefore

$$S_n = \frac{e}{f^{n+1}} P_n(\cos \theta).$$

Substituting in the general formula, we obtain

$$\rho = -\frac{e}{4\pi a^2} \sum_{n=0}^{\infty} (2n+1) \left(\frac{a}{f}\right)^{n+1} P_n(\cos \theta).$$

The series may be summed by means of the generating function of the Legendre coefficients. We have

$$\sum_{n=0}^{\infty} t^n P_n(\cos \theta) = \frac{1}{\sqrt{1 - 2t \cos \theta + t^2}},$$

and by differentiating with respect to t ,

$$\sum_{n=0}^{\infty} n t^{n-1} P_n(\cos \theta) = \frac{\cos \theta - t}{(1 - 2t \cos \theta + t^2)^{3/2}}.$$

Hence, putting $t = a/f$, we obtain

$$\sum_{n=0}^{\infty} (2n+1) t^{n+1} P_n(\cos \theta) = \frac{t(1 - t^2)}{(1 - 2t \cos \theta + t^2)^{3/2}}.$$

Therefore

$$\rho = -\frac{e}{4\pi a^2} \cdot \frac{a}{f} \cdot \frac{1 - (a/f)^2}{(1 - 2(a/f) \cos \theta + (a/f)^2)^{3/2}} = -\frac{e(f^2 - a^2)}{4\pi a(f^2 + a^2 - 2af \cos \theta)^{3/2}}.$$

This is the well-known formula of Poisson for the density of electricity induced upon a conducting sphere by a point charge.

Generalisation

The theorem may be generalised to the case of any distribution of electricity external to the sphere. If V_{ext} is the potential due to external forces, the density is given by

$$\rho = -\frac{1}{4\pi a^2} \left(1 + 2a \frac{d}{da} \right) V_{\text{ext}}.$$

This formula may be written in another form. Let V_i be the internal potential due to the induced electricity; then at the surface of the sphere we have

$$V_i + V_{\text{ext}} = \text{constant},$$

and by differentiating with respect to the normal (outward),

$$\frac{dV_i}{dn} + \frac{dV_{\text{ext}}}{dn} = 0.$$

But by a known theorem of Green,

$$\frac{dV_i}{dn} - \frac{dV_e}{dn} = -4\pi\rho,$$

where V_e is the external potential due to the induced electricity. At the surface, $V_i = V_e$, and we have

$$\rho = \frac{1}{4\pi} \left(\frac{dV_{\text{ext}}}{dn} - \frac{dV_i}{dn} \right) = \frac{1}{4\pi} \frac{dV_{\text{ext}}}{dn}.$$

Since $d/dn = -d/da$ (the normal being outward), we obtain

$$\rho = -\frac{1}{4\pi} \frac{dV_{\text{ext}}}{da},$$

which is equivalent to the formula given above.

The theorem which I have established may be stated as follows:

Theorem. *If V is the potential at the surface of a conducting sphere of radius a , due to external forces, the density of the induced electricity is given by the formula*

$$\rho = -\frac{1}{4\pi a^2} \left(1 + 2a \frac{\partial}{\partial a} \right) V.$$

Conversely, if the density ρ is given, the potential due to the external forces may be found by solving this equation.

This theorem is applicable not only to the sphere, but also to any surface for which Laplace's equation admits a simple solution. I propose to develop the analogous theorems for the ellipsoid and other surfaces in a future communication.

*2 Stone Buildings, Lincoln's Inn,
April 5, 1858.*

Collected Mathematical Papers

of Arthur Cayley, Vol. IV

Pages 101–150

Papers 241–250

241. ON POINSOT'S FOUR NEW REGULAR SOLIDS.

From the Philosophical Magazine, vol. xvii. (1859), pp. 123 — 128.

It is shown by Poinso^t, in th^te "M^emoire sur les Polygones et les Polyedres," Jour. Ecole Polyt. vol. iv. pp. 16 to 48 (1810), that, besides the regular polyhedrons of ordinary geometry, there are (of course in an extended signification of the term) four new regular polyhedrons, viz. an icosahedron, which I will call the great ico.sahedron (No. 33 of the Memoir), and three dodecahedrons, which I will call the great dodecahedron (No. 37), the great stellated dodecahedron (No. 38), and the small stellated dodecahedron (No. 39). The nature of Poinso^t's generalization will be best understood by conceiving, as he does, that the polyhedron is projected on a concentric sphere, so that the faces become spherical polygons. Then for the ordinary polyhedrons of geometry, the sum of the angles at a vertex = 4 right angles ; but it may, according to the more general notion, be = e times 4 right angles. In like manner for the ordinary polyhedrons, the sides of a face subtend at the centre angles the sum of which is = 4 right angles ; but according to the more general notion, this sum may be (viz. if the polygons are stellated) = e' times four right angles. And finally, the sum of the spherical polygons is ordinarily equal to the entire spherical surface; but according to the more general notion, it may be = Z) times the spherical surface, (e is Poinso^t's e ; e does not occur in Poinso^t ; and, for a reason which will appear, I have written D for Poinso^t's E.) The new polyhedra are constructed, as follows : 1. The great Icosahedron. — Each face is made up of seven faces, or rather four faces and six half faces of the ordinary icosahedron, in the manner shown by fig. 1. There are, as in the ordinary icosahedron, five angles at each vertex ; but these make up together, not four, but eight right angles, or e = 2 ; but, as in the ordinary poly- ON POINSOT8 FOUR NEW REGULAR SOLIDS. hedra, e* = 1 ; and the sum of all the faces is obviously seven times the spherical surface, or 2) = 7. (Also E = 7.) Fig. 1. 2. The great Dodecahedron. — Each face is made up of five faces of the ordinary icosahedron in the manner shown by the figure 2. There are five angles at each vertex, and these make up together eight right angles, or e = 2 ; but, as in

ordinary polyhedra, $e' = 1$; and the sum of all the faces is obviously

$12 \times \frac{1}{2} \text{ that is threetimes the spherical surface, or } E = 8. (\text{Also } E =$

3.) Fig. 2.3. The great stellated Dodecahedron. Each face is formed by stellating a face of

1. On account of the stellation, $e' = 2$. Each of the projecting parts of the face is equal to

of the face of the ordinary icosahedron; and the sum of the faces will be four times

4i. If, however, what seems preferable, we reckon the area of the stellated pentagon as

of the face of the ordinary icosahedron, and the sum of the faces will be seven times

7.4. The small stellated Dodecahedron. Each face is formed by stellating a face of the

1. On account of the stellation, $e' = 2$. The area of each of the projecting parts is

of the first mode of measurement, the Fig. 4. area of the stellated face is twice that of

2. But according to the second mode of measurement, the area of the stellated pentagon is

3. I form now the following Table, comprising as well the ordinary five figures as the

242. SECOND NOTE ON POINSOT'S FOUR NEW REGULAR POLYHEDRA.

From the Philosophical Magazine, vol. xvii. (1859), pp. 209 — 210.

The Note on Poinso't's four new regular Polyhedra (February Number, p. 123), [241], was written without my being acquainted with Cauchy's first memoir, "Recherches sur les Poly^dres" (*Jour. Polyt. vol. IX. pp. 68-86, 1813*), the former part of which (pp. 68-76) relates to Poinso't's polyhedra. Cauchy considers the polyhedra, no
 $H = A - \{\} - 2E$, nor of course my further generalization, $eS + e'H - A + 2D$; but the latter part of the memoir relates to a generalization, in a different
 $H = A + 2$: viz. Cauchy's theorem is—"If a polyhedron is partitioned into any number of
 $H = A + P + 1$; that is, the sum of the number of vertices and the number of faces exceeds
 $A + 2$; and for $P = 0$, we have a theorem relating to the partition of a polygon; viz. if the
 $H = A + 1$; from which it is easy to pass to Euler's equation, $8 + H = A + 2$, for 242] SECOND NOTE ON POINSOT'S FOUR NEW REGULAR POLYHEDRA
 $H = A + 1$, H should, in an analogy with Cauchy's notation for polyhedra, be replaced by P ; S ; and for the partition of a polygon, $A = S + P - 1$:
corresponding respectively to Euler's theorem for a single polyhedron, viz. $8 + H = A + 2$; and to Cauchy's theorem for the partition of a polyhedron, viz. $S + H = A + 2 - \{\} - \{P - \{\}\}$. Cauchy's second memoir (pp. 87-98) contains a very beautiful

$l, 5, 7$ and 11 ; the group corresponding to $s = l$ is distinct from that for any other value of s , $s = l - 1$, say $l = 5$, and therefore $s = 4$, so that we have $a' \geq$
 $l, = 1, a y 3 = / 3 a *$, and the group is $1, a, a s a =$
 $, a *, ' 0 a, 0 a \setminus \{ \} 0 a \setminus \{ \}$ of, 243] ON THE SYMBOLIC EQUATION $N' = 1.91$ the indices of the

S

244. ON AN ANALYTICAL THEOREM RELATING TO THE DISTRIBUTION OF ELECTRICITY UPON SPHERICAL SURFACES.

From the Philosophical Magazine, vol. xviii. (1859), pp. 119 — 127.

There is contained in Plana's "Mémoire sur la distribution de l'électricité sur la surface de deux sphères conductrices complètement isolées" {Mem.deT
kable, as well in itself as because by means of it the author corrects an error into which Poisson had fallen in his researches on the same subject. The development
`M<sup>h</sup>`, n3.4.5M.Jl., n0 Poisson hadinefifectshownthatil/, =
0; and hence inferred that, 0 being small, the function in question is sin*⁶, or what is
ment; the foregoing development must therefore be illusory; and Plana in fact shows, b
0, where i is any positive integer whatever, constitutes the analytical theorem above re f
Bi, Bs, Bn, &c. denote Bernoulli's numbers as given by the equation $i =$

$l-i*-*- '0-70T4+-(Bi=B\}^-, Bi^--Bj=, \&c. I have, in conformity with the usual practice written the equation$

245. ON AN ANALYTICAL THEOREM CONNECTED WITH THE DIS- TRIBUTI- ON OF ELECTRICITY ON SPHERICAL SURFACES. Second Part.

*From the Philosophical Magazine, vol. xviii. (1859), pp. 193 — 202 :
continuation of 244.*

The theorem is certainly true ; but its existence gives rise to a difficul-
ty to which I shall advert in the sequel. I propose, in the first instance,
to give a demonstration which starts from the expression for ϕ given
by Plana's equation (115), instead of the deduced equation which was
the basis of my former proof. It will be proper to explain the origin
and meaning of the formulae. We have two conducting spherical
surfaces, radii a and b , in contact with each other (so that the
distance between the centres is $a + b$): and then, if r is the distance
from the centre of the sphere, radius a , of an exterior point, and $\theta =$
 $\cos \theta$ the cosine of the inclination of this distance to the line from the centre to the centre of
 $a' + A, x + A'a^2 + \&c.$, then $\phi(x, x) = 4, P' + A^{P_x} + A^{P_a} +$
 $\&c.$, where $P, P, P, \&c.$ are Legendre's functions, viz. the functions of w which are the
 a^l in ascending powers of x . And the electrical thickness at any point of the surface of t
 $\{\}.13-2$ TO OON AN ANALYTICAL THEOREM CONNECTED WITH THE [2
 $6 - a; \{\}.hhJ^nl + 26 - (l + 6)a; -^Vl + 26 - (l + 6)J$ $1 + 6 -$
 X (Plana's equation (G), in which I have written for $9, 7, H$ their values, and substituted
 h). The solution of this equation is (equation (H), writing there in $g =$

$h)l - x^{h+n\{\}+h) - n(\{\}+h)x^n o(n+l)(1+6) - (1+n(H-6))a;'$ where P is an arbitrary constant and the fu

246. ON CONTOUR AND SLOPE LINES.

From the Philosophical Magazine, vol. xviii. (1859), pp. 264 — 268.

It is, I think, interesting as a question of topography, to consider the general configuration of a system of contour lines and steepest or slope lines (lignes de niveau and lignes de la plus grande pente). Imagine, to fix the ideas, a mountainous island, the exterior or sea-level contour line being consequently a closed curve ; the case where any contour line is a curve cutting itself is an important one, which will be considered ; but disregarding it for the moment, and excluding (as I do throughout) a curve which cuts itself from the notion of a closed curve, the entire contour line corresponding to a given elevation will be either a single closed curve, or it will consist of two or more separate closed curves, in the latter case each of these may be considered as being by itself a contour line, and we may therefore say that the contour line is in general a closed curve. It may happen that the elevation of a given contour line is a maximum or minimum ; in other words, that the consecutive curve without the given contour line and that within it are each of them higher or each of them lower than the given contour line; but this is a speciality which need not be particularly attended to ; in general the consecutive curve without the given contour line will be lower, and that within it higher than the given contour line, in which case the tract bounded by the contour line is an elevation (hill, table-land, or mountain, as the case may be); or else the consecutive curve without the given contour line is higher, and that within it lower than the given contour line; in which case the tract bounded by the contour line is a depression. But there may be within the contour line bounding an elevation, spaces lower than the bounding line, and within the contour line bounding a depression, spaces higher than the bounding line. A depression usually contains water, and indeed is filled so as to overflow, in which case there is a lake with an outlet ; if the depression is not filled to overflowing, the lake will have no outlet. The contour line bounding an elevation may become indefinitely small and ultimately reduce 246] ON CONTOUR AND SLOPE LINES. 109 itself to a point, which is a summit ; the contour line bounding a depression may in like manner become indefinitely small, and ultimately reduce itself to a point, which is what I call an immit. A summit is a point of maximum elevation (though of

course there may be summits, or even immits, which are higher); an immit is a point of minimum elevation. But there are besides, as at the heads of passes, points where the surface is horizontal, but where the elevation is neither a maximum nor a minimum ; you descend backwards and forwards, but ascend right and left : I will for the present purpose call this kind of point a knot. And this leads to the consideration of a contour line which cuts itself: the point where this happens is in fact a knot, or geometrically the knot is a node or double point on the contour line. It may be assumed that the contour line through a knot does not pass through any other knot; for although there may be neighbouring passes of precisely the same elevation, yet the general configuration of the country will not be altered by giving a slight difference of elevation to such passes : the effect of this alteration is to distribute among contour lines of slightly different elevations (one to each line) the different knots which would otherwise occur upon one and the same contour line. The contour line through a knot cuts itself therefore at this point only: such contour line is either a figure of eight, or as I will term it, an outloop curve; or else it is the figure formed by the union (so as to give rise to a node or double point) of two closed curves, one of which lies within the other of them ; this I call an inloop curve. An outloop curve consists of two loops ; the spaces within these may also be spoken of as the loops. An inloop curve consists of an outer and an inner loop ; the space within the inner loop may be spoken of as the inner loop, that between the two loops as the lune. It usually happens, and to fix the ideas I will assume, that for an outloop curve each of the loops is an elevation : this is the case of two mountain summits connected by a ridge or col, the lowest point whereof, or head of the pass, is the knot on the outloop contour line through this point. And in like manner, that for an inloop curve the lune is an elevation, the inner loop a depression ; and that the outer loop, considered as a portion of the contour line, is higher than the consecutive exterior contour line. This is the case of a lake having an outlet ; if the lake were dry, the passage up stream into the bed of it would be over a ridge, col, or barrier, the lowest point whereof, or point of outlet for the water of the lake, is the knot on the inloop contour line passing through this point, the shore of the lake being of course the inner loop of this contour line, and the waters being retained by means of the raised ground within the lune between the two loops of the contour line. The slope lines cut at right angles the

contour lines; and this property applies also to the projections of the two systems of lines; so that the two systems of lines delineated in piano intersect at right angles. Consider the contour lines which are closed curves surrounding a given summit or immit ; the exterior contour line is inter- sected at each of its points by a slope line ; and all these slope lines must, it is clear, intereect all the interior contour lines, and ultimately unite at the interior summit or immit. In order to see more distinctly the form of the system of slope lines, it is to be noticed that, if (as Ls in general the case) the indicatrix at the summit or immit be an ellipse, the contour lines in the immediate neighbourhood thereof will be 110 ON CONTOUR AND SLOPE LINES. [246 a system of similar and similarly situated concentric ellipses, the major and minor axes whereof correspond respectively with the directions of least and greatest curvature ; the equation of any orthogonal trajectory of the ellipses, if a, b are the semi-axes, major and minor, of any one of them, is ' $= C af$ ' ; and unless $0 = oo$, the curve represented by this equation touches the axis of x, which is the direction of least curvature; if however $C = cc$, then the equation becomes $a; = 0$, and the curve touches the axis of y, which is the direction of greatest curvature. Hence in general at a summit or immit the slope curves all, except one (which is a limiting case) touch the line which is the direction of least curvature. The only exception is when the summit or immit is an umbilicus — the indicatrix is then a circle ; the contour lines in the immediate neighbourhood of this point are concentric circles, and the slope lines pass in all directions through the summit or immit. The indicatrix at a knot is in general a hyperbola, and consequently the contour lines in the neighbourhood of a knot are similar and similarly situated concentric hyperbolas; and if a, b are the semi-axes of one of these hyperbolas, the equation of an orthogonal trajectory is $a^{y*'} = G$: andwhenthispassesthroughtheknot,(7 = 0;andthereforeeitherx = 0orelsey = 0;thereareconsequentlythroughtheknotonlytwoslopelines,whichbisect andforoneofthese`theridgeline`theknotisapointofminimumelevation; forthethe mentionedridgelinewillpassthroughtheknotoftheinneroutloopcurve : andwithrespecttothisknot,itwillbe,notaridgeline,butacourseline;sothattheslopeline or`courseline`.Thecasejustreferredtois,however,anexceptionalone; ingeneralthe sideredasendingthere; infacttherearethroughthesummitaninfinityofslopelines, a levelcontourlinewithoutpreviouslyreachinganimmit.Itistobenoticedthataridgeline shapedmountain, andingeneralwhenthecontourlinesareallofthemclosedcurves, the calledlignedefaitendliqjiedethalweg.%StoneBuildings,W.C., July20, 1859.

247. ON THE ANALYTICAL FORMS CALLED TREES. Second Part.

*From the Philosophical Magazine, vol. xviii. (1859), pp. 374 — 378.
Continuation of 203.*

The following class of " trees " presented itself to me in some researches relating to functional symbols ; viz., attending only to the terminal knots, the trees with one knot, two knots, three knots, and four knots respectively are shown in the fig-*es* 1, 2, 3 and 4: Pi«. 4. Kg. 1 Fig. 2. Fig. 3. •AAA and similarly for any number of knots. The trees with four knots are formed first from those of one knot by attaching thereto in every possible way (one way only) 247] ON THE ANALYTICAL FOEMS CALLED TREES. 113 four knotted branches; secondly, from those with two knots by attaching thereto in every possible way (three different ways) four knotted branches; and thirdly, from those with three knots by attaching thereto in every possible way (three different ways) four knotted branches, — the original knots of the trees of one knot and two and three knots, being no longer terminal knots, are disregarded. The total numbers of trees with one knot and with two and three knots being respectively 1, 1, 3 ; the total number of trees with four knots is $1.1+3.1+3.3 = 13$. And in general, if the number of trees with m knots is $\langle f \rangle_m$, then it is easy to see that we have $w_t - 1, m - 1, m - 2, m - 1, \dots = \langle f \rangle_l + - \langle t \rangle + 2 + -$

$\langle 3 \dots + - \langle j \rangle (m-l); \text{or what is the samething, Hence if we obtain } i^{-l} \cdot c^{m-l} \cdot m^{-2}, m^{-1}, \dots, 2 \langle / \rangle ?? i = 1 + - J - (f) > 2$

248. SKETCH OF A PROOF OF THE THEOREM THAT EVERY ALGEBRAIC EQUATION HAS A ROOT.

From the Philosophical Magazine, vol. xviii. (1859), pp. 436 — 439.

I HAVE referred to the theorem as usually stated ; for it is an easy consequence of the existence of a single root of an equation of any order, that for an equation of the n th order there are n roots : the proof here proposed goes, however, to show directly the existence of the n roots : it is in form a geometrical one, and was suggested to me some months ago by a letter from Prof. De Morgan, containing the remark made in his memoir, " A Proof of the Existence of a Root in every Algebraic Equation," &c. (Camb. Phil. Trans, vol. x. 1858), viz. "that the curves $P = 0$, $Q = 0$, the intersections whereof determine the root-points, are such that two branches, one of each curve, cannot enclose a space." The pi'oof which occurred to me was in character somewhat similar to that given by the Astronomer Royal in the paper, " Suggestion of a Proof of the Theorem that every Algebraic Equation has a Root " {Camb. Phil. Trans, vol. X. 1858), and which was suggested to him by Prof De Morgan's memoir. I have since varied my proof by considering therein cones in the place of plane curves. It will be obvious, upon reading it, that the proof is closely connected with Cauchy's well-known theorem for the number of roots within a given circuit; the circuit being in this case infinity, and the number of roots included within it consequently equal to the order of the equation. The curve represented by an equation of the n th degree between the coordinates (x, y) is by definition a curve of the n th order ; and a cone standing on any such curve (taking the vertex for origin) is represented by a homogeneous equation of the n th degree between the coordinates (x, y, z) , and is by definition a cone of the n th order. It is very easy to show that an equation of the n th degree cannot have more than n roots; and we have thence the geometrical theorems, that a curve of the n th order is not intersected by a line in more than n points, and that a cone of the n th order is not intersected by a plane (I speak throughout of planes through the vertex) in more than n lines. I assume that an algebraic curve is always a continuous curve, viz. that it consists of

a branch or branches, no one of which is a courbe pointillee, or a branch terminating abruptly in a point ; an algebraic cone will be in the like sense a continuous surface. An algebraic curve cannot be an indefinite spiral, for in that case there would be lines meeting it in an infinity of points ; and in like manner an algebraic cone cannot be an indefinite spiral surface: an algebraic cone consists, therefore, of a closed sheet or sheets. An algebraic curve may indeed have conjugate or isolated points, and an algebraic cone have conjugate or isolated lines : this is a circumstance which will be adverted to in the sequel. It will fix the ideas as to the general form of an algebraic cone, to remark that it may comprise twin-pair sheets, such as the sheet of a cone of the second order (this is properly spoken of as a twin-pair sheet, each of the two opposite portions of it being called, for distinction, a twin-sheet) ; and of single sheets, such as one at least of the sheets of a cone of the third or any other odd order (see the annexed " Note upon Cones of the Third Order," [249]). The advantage of the consideration of cones instead of plane curves, is that we have only closed sheets, and thus get rid of the distinction which exists for plane curves between infinite branches and the branches which are closed curves. , My proof depends on the following lemma, viz. " Consider two algebraic cones with the same vertex, each of them of the order n ; then if there be some one plane meeting the first cone in n lines, and the second cone in n lines, such that the lines of each set occur alternately, the two cones intersect in at least n lines." The truth of this lemma is, I conceive, a matter of intuition, depending only on the notion of the continuity of the sheets of the surface. Thus, if we have in piano, through a point O , the lines A, A' and B, E , such that. A, a being opposite poaits on the same line, and so for the other lines, the order round O is $A, B, A', B', a, 0, a', /9'$, it is obvious that we cannot through the lines A, A' draw a cone, and through the lines B, B' draw a cone, without making these cones intersect in at least two lines : and in like manner for two sets, each of n lines. I have, in the enunciation of the lemma, said that the cones are each of them of the order n ; this was necessary in order to exclude a case which might otherwise have happened, viz. a line of intersection of the plane with either of the cones might have been a conjugate or isolated line without any sheet through it ; and if this were so, we could not infer the existence of the n lines of intersection of the two cones. But if a plane meet an algebraic cone of the n th order in n lines, no one of these can be a

conjugate or isolated line ; for such line is to be considered as two or more coincident lines, and there would be in all more than n lines of intersection of the plane and cone. Consider now the equation
118 SKETCH OF A PROOF OF THE THEOREM THAT [248 where
 i sarationalandintegralfunctionof u with (in general) imaginary coefficients, and u is
 $P+QV$, P, Q being real functions of $(a; y)$, each of them of the degree n ; if (x, y) are rect
 $0, Q = 0$ are real curves each of the order n . And to each point of intersection of the two cur
 $f = 0$ would have more than n roots; hence it is shown that the two curves intersect
 $0, Q = 0$ respectively; each point of intersection of the two curves corresponds to a line of
 $Q, y = 0, z = 1$. The equations of the cones are obtained deduced from those of the curves by
 $0, Q' = 0$ are the equations of the cones, we have $z^{n+1} = P' + Q'V$. Consider the section by the plane through the vertex parallel to the plane of the two curves
the equation of this plane is $z = 0$; and it is clear that, to obtain the intersection of the two curves
 $(a + 6V)u'' + \&c.$; then putting $(a + 6V)\{x + yV\} = P' + Q'$
 $P' + Q' : V^l$, the equations $P' = 0, Q' = 0$ determine the intersection of the plane $z = 0$ with the cones $P = 0, Q = 0$ respectively. But writing $a + b'J - 1 = A(\cos a + V^{-1}\sin a), x + yV - 1 = r(\cos 6 + iBva6)$, we have $Ar''\{\cos\{ne + a\} + V^s \sin\{116 + a\}\} = P' + Q', V$, 248] EVERY ALGEBRAIC EQUATION HAS A ROOT. 119 So that $P = Ar''\cos(ne+a), Q = Ar''\sin(nd+a)$, or the intersections with the cone $P = 0$ are the lines given in direction by the equation $n0 + a = \{m + \} TT$, and the intersections with the cone $Q = 0$ are the lines given in direction by the equation n in each of which equations m is any integer. 0 meets the cones in two sets of lines succeeding each other alternately, as required by the

249. NOTE ON CONES OF THE THIRD ORDER.

From the Philosophical Magazine, vol. xviii. (1859), pp. 439 — 442.

The distinction adverted to in the preceding paper between the twin-pair sheets and single sheets of an algebraic cone is made (with respect to spherical curves, which is the same thing) by Mobius, in the interesting Memoir " Ueber die Grundformen der Linien der dritten Ordnung," Abh. der K. Sachs. Ges. zu Leipzig, vol. i. (1849) [and WeTke, vol. II. pp. 86 — 176]. Consider the generating line POP' of a cone, vertex O, and let pOp' be any position of this line, the points P, P', and in like manner the points p, p', being on opposite sides of the vertex ; then if OP originally coincides with Op (and therefore OP' with Op'), and if, in the course of the generation of the surface, OP (without having first come to coincide with Op) comes to coincide with Op, at the same time OP" (without having first come to coincide with Op) will come to coincide with Op', and we have a twin-pair sheet, viz. one twin-sheet generated by OP, and the other twin-sheet generated by OP'. This is the ordinary case of a cone of the second order, and requires no further explanation. It is proper to remark that, for cones of superior orders, the conical angle of each twin-sheet is not (as for a cone of the second order) necessarily- less than 360° . But suppose that OP, starting from the position Op, and before it again comes to coincide therewith, comes to coincide with Op', then at the same time OP' (without having first come again to coincide with Op') will come to coincide with Op ; the generation is here complete, and we have a single sheet, which, if the motion were continued until OP came to coincide with Op, would only be generated over again. The conical angle of a single sheet is necessarily greater than 360° ; for OP in coming to coincide with Op' must describe an angle greater than 180° , and OP" describing an equal angle, the entire angle is therefore greater than 360° ; in the limiting case, where the entire angle is precisely 360° , the conical surface is a plane. It is easy to cut out in paper and join together two sectors of a circle so as to form therewith a sector the angle whereof exceeds 360° ; such a sector can then, by joining

NOTE ON CONES OF THE THIRD ORDER. together the two radial edges, be converted into a cone of a single sheet ; the gene-

rating lines being all finite lines equal in length, the curve formed by the circular edge is, it is clear, the spherical curve which is the intersection of the cone by a concentric sphere. It is shown by Mobius (stating his result with respect to cones instead of spherical curves) that a cone of an odd order must have at least one single sheet ; a cone of the third order consists (1) of a single sheet, or else (2) of a single sheet and a twin-pair sheet. These are the two general forms of cones of the third order. But there are two special forms and one subspecial form, making in all five forms : viz., the two special forms are, (3) the cone has a nodal line ; (4) the cone has an isolated line ; and the subspecial form is, (5) the cone has a cuspidal line. The relation of the different forms may be explained as follows. Starting from the form (1), as the constants of the equation change, the cone gathers itself up together so as to have a nodal line ; this is the form (3). The loops of this form then detach themselves so as to form a twin-pair sheet, the remaining part of the surface reverting to a form similar to that of (1) ; we have thus a single sheet and twin-pair sheet, which is the form (2). The twin-pair sheet then dwindles away into an isolated line, giving the form (4); and lastly, the isolated line disappears and the cone resumes the form (1) : these four forms constitute, therefore, a complete cycle. The constants may be such that the loop of the form (3) is evanescent, or, what is the same thing, that the forms (3) and (4) arise simultaneously ; there is in this case a cuspidal line, or we have the form (5). It may be added that for the general forms (1) and (2) there are always three lines of inflexion. This is also the case with the form (4), where there is an isolated line ; but in the form (3), c. IV. 16 122 NOTE ON CONES OF THE THIRD ORDER. [249 where there is a nodal line, there is but one line of inflexion ; and in the form (5), where there is a cuspidal line, there is not any line of inflexion : the equivalent theorem for the spherical curves is given by Mobius(?). It was remarked long ago by Sir I. Newton, that all curves of the third order could be generated as the shadows of the five cubical parabolas ; these are, in fact, sections in a particular manner of the above-mentioned five forms of cones of the third order: the existence of five essentially distinct forms of cones of the third order is noticed by M. Chasles in the *Aperu Historique*, 1837. *The analytical distinction between the forms (1) and (2) Q4, g3, et the quartinvariant and sextinvariant of the cubic form. I annex stereoscopic representation of the cones by concentric spheres. The figures are intended to be looked at with*

250. SUR LA SURFACE QUI EST L'ENVELOPPE DES PLANS CON- DUITES PAR LES POINTS D'UN ELLIPSOIDE PERPENDICU- LAIREMENT AUX RAY- ONS MENES PAR LE CENTRE.

*From the Annali di Matematica pura ed applicata (Tortolini), torn.
il. (1859),*

pp. 3—14.] La consideration de la surface dont il s'agit me fut
sugg^{re}, *ilyaquelquesanusesparleProf.Stokes, maisilconvientderemarquerquelacour*
 $+b * -a'b') - 3(aW + ty)]' = [9a'(2lP - a')' =$
 $+96'(2a' - 6')f - 4 < (a' + 6')(2' - 6')(26' -$
 $a')]'$, Equationqui^taittrouv^een^galantazerolediscriminantd'une fonctionquadratique
 $a - direlespoints(6^{-a})X' = a'(b'' - 2a'), (-a')P = a -$
, quison^tdespointsimaginairesdeI'ellipse)donuentpoulacourbedespointsdoublesu
 $2b(c'est - direlespoints(a - b'')Z == a'b'.(a' - b')Y=(a' -$
 $2b')$, quison^tdespointsrdsdeI'ellipseensupposanta^{>2})donnentpoulacourbedespo
 n^e spara' $a f' = 46 = (a' - b'')$, $y'' = 0$, etellessontainsir^elles, soitpoura' <
 $26'$, soitpour $t'' > 26^m$ aisdanslepremiercaslespointsdoublesontdespointsconjugua
 $(2 - hSX' + Y)dX - dY = 0, dy = - |dZ + (2 - i(Z = +3P))dF =$
 0 , eten^liminantdX, dY, onobtientI'^uation $(2 - l(3A + P))(2 - l(Z^4 -$
 $3P)) - *gf^0$, laquelleaumoyendeI'dquationdeI'ellipseser^duitk -
 $2(Z^+ + P)(i + j -) + 3(Z > + P)'m, = 0.250]$ l'enveloppedesplans&c.125Onvoitqueles
mentdelacourbedoiventetresitudssurlacourbedonn^eparI'dquationderniere -
mentmentionn^e. CetteEquationsediviseendeuxfacteurs :
I'dquation $X^+Y'=0$ combine avecI'^uationdeI'ellipsedonnelespoints $X' =$
 $f - > Y=r$, lesquels $a' - b'a^{-b''}$ sontdespointsimaginairesdeI'ellipse; mais on pent secouvaincresanspein

, 1 = ; 1 entre lesquelles il faut dliminer Q; en multipliant la pre-
miere ⁿationpar2, etlasecondeEquationpar^{-e}tenajoutant, ontrouveetlasecondeEqu
ciparrapport&.Q\{\}c'est - direI'equationdelacourbeseratrouveeen^liminantQentr
en6cartantlesdⁿominateurs, I'dquatioidevient⁻(2a
 $+W) + e(a : ' aP + by + 4rt * n - 2a - (' = +f) =$
 0 , etlediscriminantdecettefonctioncubiquede0, dgaleaz^rordonneI'equationdelacour
 $i(a'x^+by+4a-6^-)$, $D = -2a^l\}\{af'+y''\}$. EnreprdsentantlediscriminantparD, dans
l, ilseraconvenabledeseservirdelaforme $A'' - D = 4, (AC - B'f -$
 $(SABC - A'D - 25)^=0; 250]$ l'enveloppedesplans&c.127Tona $AC -$
 $R- = (3aV + 36y - 4a^+icv^{b'} - 46*)$, $AD - BC = [$

$(a - -8b -)a - a? + \{h - -Sa?)h - if + 4(a^+h'')a - h\%BD - C =$
 $-[aV + 2a\%x'f + 6y - 4(a - +36)a'b - x' - 4(6^+Sa')a\%Y +$
 $1Qa'b*], et dela 3ABC - AW - 2B >= 2B(AG - B') - A(AD - BC) =$
 $-[9(a = -26 =')aV + 9(-2a')by - 8a' + 12a' + 12a'b' -$
 $8b''], et l'equation de la courbe est ainsi trouvee sous la forme $(3aV +$
 $36y - 4a * +4a - b^{-46*})' + [9(o\tilde - 26^aV + 9(6' - 2a)6y -$
 $8a' + 12a * 6 = +12a''6 * -86''] = 0$ laquelle est eneflF et la forme ci -
 devant mentionnee; j'ajoute que Ton trouverait les expressions de jkdounees pour les co-
 onques des points, le point correspondant de la surface sera krextremitdd'undiametre
 ou ce qui est la m^e chose (prenant le centre de l'ellipsoide pour origine) les coordonndes
 $77+^- = 1 > a'0^c - et la construction qui vient d'etre mentionnee donne x =$
 $xh -)X' + Y' + Z')'\setminus\{, y = Fr2 - i(Z^+P + Z0, z = z\setminus\{-\{X' + Y' +$
 $Z') pour les expressions de x, y, z en fonction de X, Y, Z. La discussion de la surface pentt$
 $, 6 => 2c?; enef f et dans ce cas, les courbes du sixi^me ordre, ou sextiques, correspondent$
 les deux autres courbes d'intersection, et les courbes nodales dans les plans de z et xy qui
 $2x(i - \cdot), , = 0, . = 2z(i - f;). et de l'endliminant X, Y, Z, on trouve a'af(?z^{(a^-)46\sim}(6=-c0$$

10.13. On an Analytical Theorem Relating to the Distribution of
Electricity upon Spherical Surfaces 121

**The Collected Mathematical Papers of
Arthur Cayley
Volume IV**

Pages 151–200

252Note sur l'équation des différences pour une équation donnée de degré quelconque 252From the *Annali di Matematica pura ed applicata* (Tortolini), tom. II. (1859), pp. 365, 366. [Continuation from p. 150]

l'une et l'autre équation sera du degré $n - 1$ par rapport à s . La forme sous laquelle j'ai présenté la méthode de Bezout s'applique au problème; en représentant les deux équations par $Fs = 0$, $Gs = 0$ j'écris pour le moment

$$\phi(s + \sqrt{\theta}) = A, \quad \phi(s - \sqrt{\theta}) = B,$$

on a alors

$$Fs = n(A + B) - s \frac{A - B}{\sqrt{\theta}}, \quad Gs = \frac{A - B}{\sqrt{\theta}},$$

et en écrivant

$$\phi(s' + \sqrt{\theta}) = A', \quad \phi(s' - \sqrt{\theta}) = B',$$

on a de même

$$Fs' = n(A' + B') - \frac{s'(A' - B')}{\sqrt{\theta}}, \quad Gs' = \frac{A' - B'}{\sqrt{\theta}}.$$

On obtient de là

$$\begin{aligned} \Phi(s, s') &= \frac{Fs \cdot Gs' - Fs' \cdot Gs}{s - s'}, \\ &= \left[\left(n(A + B) - \frac{s(A - B)}{\sqrt{\theta}} \right) \frac{A' - B'}{\sqrt{\theta}} - \left(n(A' + B') - \frac{s'(A' - B')}{\sqrt{\theta}} \right) \frac{A - B}{\sqrt{\theta}} \right] \frac{1}{s - s'} \end{aligned}$$

ou, en réduisant,

$$-\Phi(s, s') = \frac{2n(AB' - A'B)}{(s - s')\sqrt{\theta}} + \frac{(A - B)(A' - B')}{\theta}.$$

Donc, en rétablissant les valeurs de A , B , A' , B' , on a la fonction

$$\begin{aligned} &\frac{2n[\phi(s + \sqrt{\theta})\phi(s' - \sqrt{\theta}) - \phi(s' + \sqrt{\theta})\phi(s - \sqrt{\theta})]}{(s - s')\sqrt{\theta}} \\ &+ \frac{[\phi(s + \sqrt{\theta}) - \phi(s - \sqrt{\theta})][\phi(s' + \sqrt{\theta}) - \phi(s' - \sqrt{\theta})]}{\theta} \end{aligned}$$

laquelle sera de la forme

$$\begin{pmatrix} a_{0,0}, & a_{0,1}, & \dots & a_{0,n-2} \\ a_{1,0}, & a_{1,1}, & \dots & \\ \vdots & & & \\ a_{n-2,0}, & \dots & & \end{pmatrix} (s, 1)^{n-2} (s', 1)^{n-2},$$

où les coefficients a sont des fonctions rationnelles de θ , et en égalant à zéro le déterminant formé avec ces coefficients on a l'équation qu'il s'agissait de trouver. Quoique cette solution soit analytiquement la plus simple, j'ai une autre méthode nouvelle plus adaptée au calcul, laquelle j'ai appliquée à trouver l'équation des différences pour l'équation quintique

$$(a, b, c, d, e, f \mid v, 1)^5 = 0.$$

Londres, 4 Fév. 1860.

253 Sur la courbe parallèle à l'ellipse 253 From the *Annali di Matematica pura ed applicata* (Tortolini), tom. III. (1860), pp. 311–316.

Il fut remarqué par Cauchy (*Comptes Rendus*, tom. xiii. [1841], p. 1063) que l'équation de cette courbe pourrait se trouver en éliminant θ entre les équations

$$\frac{a^2 x^2}{(\theta + a^2)^2} + \frac{b^2 y^2}{(\theta + b^2)^2} = 1, \quad \frac{\theta^2 x^2}{(\theta + a^2)^2} + \frac{\theta^2 y^2}{(\theta + b^2)^2} = k^2,$$

et cette élimination fut effectuée, et le résultat trouvé sous la forme la plus simple par M. Catalan (*Terquem*, tom. III. 1844, p. 553); mais pour faire l'élimination de la manière la plus facile je remarque que ces deux équations donnent

$$\frac{x^2}{\theta + a^2} + \frac{y^2}{\theta + b^2} = 1 + \frac{k^2}{\theta}, \quad \frac{x^2}{(\theta + a^2)^2} + \frac{y^2}{(\theta + b^2)^2} = \frac{k^2}{\theta^2},$$

dont la seconde est la dérivée, par rapport à θ , de la première; or cette première équation est

$$(\theta + k^2)(\theta + a^2)(\theta + b^2) - x^2\theta(\theta + b^2) - y^2\theta(\theta + a^2) = 0,$$

et, en égalant à zéro le discriminant de la fonction cubique, on aura l'équation de la courbe; en posant

$$A = x^2 + y^2 - k^2 - a^2 - b^2, \quad B = b^2 x^2 + a^2 y^2 - a^2 k^2 - b^2 k^2 - a^2 b^2, \quad C = a^2 b^2 k^2,$$

la fonction cubique, multipliée par 3, sera

$$(3, -A, -B, 3C \wr \theta, 1)^3;$$

on aura donc

$$4(A^2 + 3B)(B^2 + 3AC) - (AB - 9C)^2 = 0,$$

ou ce qui est la même chose

$$A^2B^2 + 4B^3 + 4A^3C + 18ABC - 27C^2 = 0,$$

c'est-à-dire en substituant les valeurs de A , B , C , l'équation de la courbe sera

$$\begin{aligned} & (x^2 + y^2 - k^2 - a^2 - b^2)^2 (b^2x^2 + a^2y^2 - a^2k^2 - b^2k^2 - a^2b^2)^2 \\ & + 4(b^2x^2 + a^2y^2 - a^2k^2 - b^2k^2 - a^2b^2)^3 \\ & + 4a^2b^2k^2(x^2 + y^2 - k^2 - a^2 - b^2)^3 \\ & + 18a^2b^2k^2(x^2 + y^2 - k^2 - a^2 - b^2)(b^2x^2 + a^2y^2 - a^2k^2 - b^2k^2 - a^2b^2) \\ & - 27a^4b^4k^4 = 0, \end{aligned}$$

laquelle est en effet l'équation trouvée par M. Catalan.

En écrivant $k = 0$ l'équation devient

$$0 = (b^2x^2 + a^2y^2 - a^2b^2)[(x^2 + y^2)^2 - 2(a^2 - b^2)(x^2 + y^2) + (a^2 - b^2)^2],$$

ce qui équivaut à l'équation de l'ellipse

$$b^2x^2 + a^2y^2 - a^2b^2 = 0$$

deux fois répétée, et aux équations

$$(x \pm ae)^2 + y^2 = 0,$$

(où comme à l'ordinaire $a^2e^2 = a^2 - b^2$) lesquelles appartiennent aux droites menées chacune par un foyer réel et un foyer imaginaire de l'ellipse; ces droites sont aussi les tangentes menées à l'ellipse par les quatre foyers. En effet en prenant sur l'ellipse le point imaginaire dont les coordonnées sont

$$x = \frac{a}{e}, \quad y = ia \left(\frac{1}{e} - e \right), \quad (i = \sqrt{-1})$$

l'équation du cercle, rayon 0, ayant pour centre le point dont il s'agit, sera

$$\left(x - \frac{a}{e}\right)^2 + \left[y - ia\left(\frac{1}{e} - e\right)\right]^2 = 0;$$

ce cercle se réduit donc à deux droites, savoir la droite

$$x - \frac{a}{e} + i\left[y - ia\left(\frac{1}{e} - e\right)\right] = 0,$$

ou, ce qui est la même chose,

$$x - ae + iy = 0,$$

laquelle est tangente à l'ellipse, et la droite

$$\left(x - \frac{a}{e}\right) - i\left[y - ia\left(\frac{1}{e} - e\right)\right] = 0,$$

ou, ce qui est la même chose,

$$x - a \left(\frac{2}{e} - e \right) - iy = 0,$$

laquelle est la droite menée par le point de contact à l'autre point circulaire à l'infini (celui qui n'est pas situé sur la tangente $x - ae + iy = 0$). Le cercle, comme courbe composée d'une droite tangente à l'ellipse et d'une autre droite menée par le point de contact, a même avec l'ellipse une intersection à trois points réunis. Ces considérations font voir pourquoi au cas $k = 0$ les quatre droites $(x \pm ae)^2 + y^2 = 0$ font partie de la courbe.

En supposant que k est quelconque, mais que l'on a $a = b$, l'équation devient

$$a^4(x^2 + y^2)^2 [(x^2 + y^2 - a^2)^2 - 2k^2(x^2 + y^2 + a^2) + k^4] = 0,$$

ou, en écartant le facteur $(x^2 + y^2)^2$ et le facteur constant a^4 , l'équation se réduit à

$$(x^2 + y^2 - a^2)^2 - 2k^2(x^2 + y^2 + a^2) + k^4 = 0,$$

c'est-à-dire

$$[x^2 + y^2 - (a + k)^2] [x^2 + y^2 - (a - k)^2] = 0;$$

ainsi la courbe se réduit à l'ensemble des deux droites $(x^2 + y^2) = 0$ (chacune deux fois répétée), et des deux cercles

$$x^2 + y^2 - (a + k)^2 = 0, \quad x^2 + y^2 - (a - k)^2 = 0$$

comme évidemment cela doit être.

Par rapport aux singularités de la courbe, la forme de l'équation montre à premier coup d'œil qu'il y a quatre points doubles à l'infini; savoir les deux points circulaires à l'infini (points où l'infini est rencontré par les droites $x^2 + y^2 = 0$), et les deux points où l'infini est rencontré par les droites $b^2x^2 + a^2y^2 = 0$. Il y a de plus deux points doubles sur chacun des axes: en effet en écrivant dans l'équation de la courbe $y = 0$ l'équation résultante, toute réduction faite, devient

$$[(x - a)^2 - k^2] [(x + a)^2 + k^2] [b^4x^2 - (a^2 - b^2)(b^2 - k^2)]^2 = 0,$$

ce qui fait voir qu'il y a sur l'axe de x les deux points doubles dont les coordonnées sont données par l'équation

$$b^4x^2 = (a^2 - b^2)(b^2 - k^2).$$

Mais pour parvenir à cette conclusion il convient de considérer que l'axe de x rencontre la courbe dans des points tels que pour chacun la distance normale à l'ellipse est égale à k : il y a quatre distances normales k , mesurées le long de l'axe; et quatre distances normales k mesurées à des points situés symétriquement par rapport à l'axe, lesquels se rencontrent deux à deux sur l'axe aux points dont les coordonnées sont $b^4x^2 = (a^2 - b^2)(b^2 - k^2)$; c'est là l'origine des deux points doubles sur l'axe de x . On doit aussi remarquer que les coordonnées des points sur l'ellipse qui correspondent à ces points doubles sont données par les équations

$$x^2 = \frac{a^4}{b^2} \frac{b^2 - k^2}{a^2 - b^2}, \quad y^2 = \frac{a^2k^2 - b^4}{a^2 - b^2};$$

on voit de là que les points sur l'ellipse qui donnent lieu aux deux points doubles sur l'axe de x , ne seront réels à moins que $k > \frac{b^2}{a}, < b$; les deux points doubles seront cependant réels pour toute valeur $k > 0$ et $< b$. J'ajoute que pour les valeurs $k \geq 0, < \frac{b^2}{a}$, les deux points doubles seront des points conjugués ou isolés; pour $k = \frac{b^2}{a}$, chacun de ces points se réunit à la courbe, et a deux points de rebroussement, mais il n'y a pas de singularité visible dans la forme de la courbe; pour $k > \frac{b^2}{a}$ et $< b$ il y a deux points doubles avec des branches réelles; et pour $k = b$, ces deux points viennent se réunir au centre de l'ellipse et il y a deux branches ayant pour tangente commune à ce point l'axe de x ; enfin pour $k > b$, les deux points doubles sur l'axe de x deviennent imaginaires.

Des considérations pareilles s'appliquent aux points doubles sur l'axe de y ; pour $x = 0$ l'équation de la courbe se transforme en

$$[(y - b)^2 - k^2][(y + b)^2 + k^2][a^4 y^2 - (a^2 - b^2)(k^2 - a^2)]^2 = 0;$$

il y a donc sur l'axe de y deux points doubles dont les coordonnées sont données par

$$a^4 y^2 - (a^2 - b^2)(k^2 - a^2) = 0,$$

ces points correspondent à des points sur l'ellipse dont les coordonnées sont données par

$$x^2 = \frac{a^4 - b^2 k^2}{a^2 - b^2}, \quad y^2 = \frac{b^4}{a^2} \frac{k^2 - a^2}{a^2 - b^2};$$

les points sur l'ellipse ne seront donc réels à moins que $k > \frac{a^2}{b}$; les deux points doubles seront cependant réels pour toute valeur $k > a$. Pour $k < a$ les deux points doubles seront imaginaires. Pour $k = a$ le centre est point de réunion de deux points doubles et il y a deux branches qui ont à ce point pour tangente commune l'axe de y : pour $k = \frac{a^2}{b}$, chacun des deux points doubles se réunit à deux points de rebroussement; mais il n'y a pas de singularité visible dans la forme de la courbe; enfin pour $k > \frac{a^2}{b}$ les deux points doubles se détachent de la courbe et deviennent des points conjugués ou isolés. En résumé, il y a huit points doubles, savoir quatre points doubles à l'infini, et quatre points doubles sur les deux axes.

Il y a de plus 12 points de rebroussement; ces points sont en effet les centres de courbure aux points de l'ellipse pour lesquels le rayon de courbure est égal à k ; les douze points seront tous imaginaires à moins que k n'ait une valeur entre les limites $\frac{b^2}{a}$, $\frac{a^2}{b}$: pour une telle valeur de k les douze points seront 4 réels, et 8 imaginaires. Pour $k = \frac{b^2}{a}$ les quatre points réels se réunissent deux à deux aux deux points doubles sur l'axe de x ; pour $k = \frac{a^2}{b}$, ils se réunissent deux à deux aux deux points doubles sur l'axe de y ; cela s'accorde avec ce qui est dit ci-dessus par rapport aux points doubles. On peut encore trouver que le nombre des points de rebroussement est 12 à moyen

des équations

$$A^2 + 3B = 0, \quad B^2 + 3AC = 0, \quad AB - 9C = 0,$$

lesquelles appartiennent chacune à une courbe du quatrième ordre qui passe par les points de rebroussement; la forme des équations fait voir que ces courbes ont en commun 12 points, et seulement 12 points, d'intersection. C'est là en effet la manière la plus simple de trouver les coordonnées des points de rebroussement; car ces équations donnent tout de suite

$$A = -3C^{\frac{1}{3}}, \quad B = -3C^{\frac{2}{3}},$$

c'est-à-dire on a les équations

$$\begin{aligned} x^2 + y^2 &= a^2 + b^2 + k^2 - 3(abk)^{\frac{2}{3}}, \\ b^2x^2 + a^2y^2 &= a^2b^2 + (a^2 + b^2)k^2 - 3(abk)^{\frac{2}{3}} \end{aligned}$$

qui donnent les coordonnées des douze points de rebroussement.

Pour vérifier que ces points correspondent aux points de l'ellipse pour lesquels le rayon de courbure est égal à k ; je prends $a \cos \theta$, $b \sin \theta$ pour les coordonnées d'un point sur l'ellipse, les coordonnées du centre de courbure seront données par

$$ax = (a^2 - b^2) \cos^3 \theta, \quad by = -(a^2 - b^2) \sin^3 \theta,$$

et en supposant que le rayon de courbure soit égal à k , on a

$$k = \frac{1}{ab} (a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{\frac{3}{2}},$$

et de là

$$a^2 \sin^2 \theta + b^2 \cos^2 \theta = (abk)^{\frac{2}{3}}.$$

Cela donne

$$(a^2 - b^2) \cos^2 \theta = a^2 - (abk)^{\frac{2}{3}}, \quad -(a^2 - b^2) \sin^2 \theta = b^2 - (abk)^{\frac{2}{3}},$$

et de là on déduit

$$\begin{aligned} x^2 + y^2 &= \frac{1}{a^2 - b^2} \left[\frac{1}{a^2} \left(a^2 - (abk)^{\frac{2}{3}} \right)^3 - \frac{1}{b^2} \left(b^2 - (abk)^{\frac{2}{3}} \right)^3 \right], \\ b^2x^2 + a^2y^2 &= \frac{1}{a^2 - b^2} \left[\frac{b^2}{a^2} \left(a^2 - (abk)^{\frac{2}{3}} \right)^3 - \frac{a^2}{b^2} \left(b^2 - (abk)^{\frac{2}{3}} \right)^3 \right] \end{aligned}$$

valeurs qui s'accordent en effet avec les valeurs ci-dessus trouvées.

La courbe parallèle à l'ellipse est de l'ordre 8 et il y a 8 points doubles, et 12 points de rebroussement; elle sera donc de la classe $56 - 16 - 36 = 4$. En effet pour mener d'un point donné une tangente à la courbe parallèle on peut décrire avec ce point comme centre un cercle rayon k , pour mener une tangente commune au cercle et à l'ellipse; la droite, menée par le point donné, parallèle à la tangente commune sera la tangente cherchée: on peut donc par un point donné mener 4 tangentes à la courbe, ou la courbe parallèle est de la classe 4. Au reste on peut trouver très facilement l'équation de la courbe parallèle en coordonnées tangentielles. Car en représentant par θ l'inclination d'une tangente quelconque de l'ellipse à l'axe de x , l'équation de cette tangente est

$$x \cos \theta + y \sin \theta - \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} = 0,$$

et l'équation de la tangente parallèle de la courbe sera ainsi

$$x \cos \theta + y \sin \theta - k - \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} = 0;$$

donc en représentant cette équation par $\xi x + \eta y + \zeta = 0$ on aura

$$\xi : \eta : \zeta = \cos \theta : \sin \theta : k + \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta},$$

et de là

$$\zeta + k\sqrt{\xi^2 + \eta^2} + \sqrt{a^2 \xi^2 + b^2 \eta^2} = 0,$$

laquelle est l'équation dont il s'agit; la forme rationnelle est

$$(a^2 - k^2)^2 \xi^4 + (b^2 - k^2)^2 \eta^4 + 2(a^2 - k^2)(b^2 - k^2) \xi^2 \eta^2 - 2(a^2 + k^2) \zeta^2 \xi^2 - 2(b^2 + k^2) \zeta^2 \eta^2 + \zeta^4 = 0$$

équation du quatrième ordre comme cela devait être.

J'ai cru qu'il n'était pas nécessaire de faire voir comment on pourrait à moyen de l'équation tracer la courbe; en effet on trouve les différentes formes assez facilement par des considérations géométriques, en considérant la courbe comme la développée (le révolue) de l'ellipse: on se rend compte à ce moyen de ce qui est déjà dit par rapport aux points doubles et aux points de rebroussement.

Je remarque enfin que l'équation d'une normale quelconque de l'ellipse est

$$ax \sin \theta - by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

(où θ est un paramètre arbitraire): donc en considérant la trajectoire orthogonale de ce système de droites, on obtient

$$(x \, dx + y \, dy) \sqrt{b^2 dx^2 + a^2 dy^2} + (a^2 - b^2) \, dx \, dy = 0$$

comme équation différentielle de la courbe parallèle à l'ellipse. L'intégrale de cette équation est donc l'équation du huitième ordre (contenant k comme constante arbitraire) qu'on a ci-dessus trouvée.

Londres, 2 *Stone Buildings*, 22 Oct. 1860.

254 Sur la surface parallèle à l'ellipsoïde 254 From the *Annali di Matematica pura ed applicata* (Tortolini), tom. III. (1860), pp. 345–352.

Il y a pour la surface parallèle à l'ellipsoïde une solution tout à fait semblable à celle pour la courbe parallèle à l'ellipse. En effet soit

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1$$

l'équation de l'ellipsoïde; k la distance de la surface parallèle, en prenant cette distance sur la normale extérieure au point (X, Y, Z) et en écrivant pour abrégé

$$\sqrt{\frac{X^2}{a^4} + \frac{Y^2}{b^4} + \frac{Z^2}{c^4}} = \frac{k}{\lambda},$$

on trouve pour les coordonnées (x, y, z) de l'extrémité de la normale

$$x = X \left(1 + \frac{\lambda}{a^2}\right), \quad y = Y \left(1 + \frac{\lambda}{b^2}\right), \quad z = Z \left(1 + \frac{\lambda}{c^2}\right),$$

et, en substituant pour X, Y, Z les valeurs données par ces équations, on obtient

$$\begin{aligned} \frac{a^2 x^2}{(a^2 + \lambda)^2} + \frac{b^2 y^2}{(b^2 + \lambda)^2} + \frac{c^2 z^2}{(c^2 + \lambda)^2} &= 1, \\ \frac{\lambda^2 x^2}{(a^2 + \lambda)^2} + \frac{\lambda^2 y^2}{(b^2 + \lambda)^2} + \frac{\lambda^2 z^2}{(c^2 + \lambda)^2} &= k^2. \end{aligned}$$

Or ces équations donnent

$$\begin{aligned} \frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} &= 1 + \frac{k^2}{\lambda}, \\ \frac{x^2}{(a^2 + \lambda)^2} + \frac{y^2}{(b^2 + \lambda)^2} + \frac{z^2}{(c^2 + \lambda)^2} &= \frac{k^2}{\lambda^2}, \end{aligned}$$

et la seconde équation est la dérivée par rapport à λ de la première. C'est-à-dire en obtient l'équation de la surface parallèle en égalant à zéro le discriminant par rapport à λ de l'équation

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} + \frac{z^2}{c^2 + \lambda} = 1 + \frac{k^2}{\lambda},$$

ou autrement dit, la surface parallèle est l'enveloppe par rapport à λ de cette équation. En multipliant par $\lambda(a^2 + \lambda)(b^2 + \lambda)(c^2 + \lambda)$, on aura une équation du quatrième ordre, dont le discriminant égalé à zéro donne l'équation de la surface: on voit sans peine que cette équation sera de l'ordre 10.

J'avais à peine trouvé cette méthode, quand j'ai appris que le problème était déjà résolu par MM. Salmon et W. Roberts. M. Salmon considère la surface comme lieu des centres des sphères, rayon k , qui touchent à l'ellipsoïde $\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1$, ce qui lui donne tout de suite le théorème que voici; savoir on obtient l'équation de la surface parallèle, en égalant à zéro le discriminant par rapport à λ du discriminant par rapport à (X, Y, Z) de l'équation

$$\lambda \left(\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} - 1 \right) + (X - x)^2 + (Y - y)^2 + (Z - z)^2 = 0;$$

cette solution est donnée en passant dans une Note publiée dans le No. de Septembre 1860 des *Nouvelles Annales* (Terquem et Géro-n). En formant le discriminant par rapport à X, Y, Z on trouve l'équation en λ ci-dessus donnée.

La solution de M. W. Roberts n'avait pas été publiée: il a bien voulu me permettre d'en faire part aux géomètres. En considérant une section circulaire de l'ellipsoïde, l'enveloppe des sphères, rayon k , ayant leurs centres sur cette section, sera une surface annulaire, et l'enveloppe des surfaces annulaires qui correspondent aux différentes sections circulaires de l'ellipsoïde sera la surface parallèle. Dans la figure (qui est censée située dans le plan des axes le plus grand et le plus petit) soit $O\xi$ la trace de la section circulaire centrale, $O\zeta$ le diamètre perpendiculaire, AB la trace d'une section quelconque parallèle à la section circulaire centrale: soit de plus $AB = 2\delta$, et (α, γ) les coordonnées du point C qui est le centre de la section AB . En prenant pour un instant $O\xi, O\zeta$ comme axes de coordonnées, l'équation du cercle AB sera $\xi^2 + \zeta^2 = \delta^2$ et celle de la surface annulaire sera

$$(x - \alpha)^2 + (z - \gamma)^2 + y^2 = k^2 + \rho^2,$$

où ρ dénote la distance du point $(\alpha, 0, \gamma)$ à un point quelconque (ξ, y, ζ) du cercle, de manière que $\rho^2 = (\alpha - \xi)^2 + y^2 + (\gamma - \zeta)^2$; on a donc

$$(x - \alpha)^2 + (z - \gamma)^2 + y^2 = k^2 + (\alpha - \xi)^2 + y^2 + (\gamma - \zeta)^2,$$

c'est-à-dire

$$x^2 + z^2 - k^2 - 2(x\alpha + z\gamma) + 2(\xi\alpha + \zeta\gamma) - \delta^2 = 0,$$

où $\delta^2 = \xi^2 + \zeta^2$; mais en écrivant λ au lieu de α, γ sont données par

$$\frac{\alpha^2}{a^2 - c^2} + \frac{\gamma^2}{c^2 - b^2} = 1,$$

et

$$\delta^2 = \frac{a^2 c^2}{b^2} - \left(\frac{a^2 c^2}{b^2} - b^2 \right) \left(\frac{\alpha^2}{a^2} + \frac{\gamma^2}{c^2} \right);$$

il est clair que l'équation de la surface annulaire doit se trouver de même à moyen d'une fonction cubique de λ . Pour vérifier cela, je remarque que l'on a à trouver

$$\frac{a^2 x^2}{(a^2 + \lambda)^2} + \frac{b^2 y^2}{(b^2 + \lambda)^2} + \frac{c^2 z^2}{(c^2 + \lambda)^2} = 1,$$

et

$$\frac{x^2}{(a^2 + \lambda)^2} + \frac{y^2}{(b^2 + \lambda)^2} + \frac{z^2}{(c^2 + \lambda)^2} = \frac{k^2}{\lambda^2};$$

et la seconde équation est la dérivée par rapport à λ de la première.

[Le mémoire se continue avec des développements sur l'équation de la surface parallèle et ses singularités.]

Londres, 2 *Stone Buildings*, 22 Oct. 1860.

255On a Problem of Double Partitions255From the *Philosophical Magazine*, vol. xx. (1860), pp. 337-341.

If $a + b + c + \dots = m$, $\alpha + \beta + \gamma + \dots = \mu$ (the quantities being all positive integer numbers, not excluding zero), then $(a, \alpha) + (b, \beta) + (c, \gamma) + \dots$ is considered as a partition of (m, μ) : and the partible quantity (m, μ) , and parts (a, α) , (b, β) , &c. being each of them composed of two elements, such partition is said to belong to the theory of *Double Partitions*. The subject (so far as I am aware)

has hardly been considered except by Professor Sylvester, and it is greatly to be regretted that only an outline of his valuable researches has been published: the present paper contains the demonstration of a theorem, due to him, by which (subject to certain restrictions) the question of Double Partitions is made to depend upon the ordinary theory of Single Partitions.

Let the question be proposed, “In how many ways can (m, μ) be made up of the given parts (a, α) , (b, β) , (c, γ) , &c.” under the following conditions (which are, it will be seen, necessary in the demonstration of the theorem constituting the solution), viz.

$$\frac{a}{\alpha}, \frac{b}{\beta}, \frac{c}{\gamma}, \text{ \&c. are unequal fractions, each in its least terms,}$$

and

$$\alpha, \beta, \gamma, \text{ \&c., are each less than } \mu + 2.$$

The number of partitions is

$$= \text{coeff. } x^m y^\mu \text{ in } \frac{1}{(1 - x^a y^\alpha)(1 - x^b y^\beta)(1 - x^c y^\gamma) \dots}$$

the fraction being developed in ascending powers of x, y .

256 On a System of Algebraic Equations 256 From the *Philosophical Magazine*, vol. xx. (1860), pp. 341, 342.

THE determination of a, b, c from the system of equations

$$\begin{aligned} a^2 + bc &= \lambda, \\ b^2 + ca &= \mu, \\ c^2 + ab &= \nu, \end{aligned}$$

in the case where λ, μ, ν have the values 16, 17, and 18 respectively, is the problem known as Colonel Titus’s Arithmetical Problem. See Masères’ *Tracts on the Resolution of affected Algebraic Equations*, Lond. 1800. If for shortness we put

$$\sigma = \nu - c^2,$$

then the third equation gives $b = \frac{\sigma c}{a}$; and substituting this value of b in the two other equations, we have

$$a^2 + \frac{\sigma c}{a} = \lambda, \quad \frac{\sigma^2}{a^2} + ca = \mu;$$

or what is the same thing,

$$\begin{aligned}a^3 - \lambda a + \sigma c &= 0, \\ca^3 - \mu a^2 + \sigma^2 &= 0;\end{aligned}$$

and from these equations, eliminating a , we have

$$\sigma^4 - 3c^2\sigma^3 + (3c^4 - 2\lambda\mu)\sigma^2 - c^2(c^4 - \lambda\mu)\sigma + c^4\lambda\mu - c^2(\lambda^3 + \mu^3) + \lambda^2\mu^2 = 0,$$

et l'équation pour c sera cette même équation, en y écrivant $\nu - c^2$ au lieu de σ , savoir

$$\begin{aligned}c^8 - 3\nu c^6 + (3\nu^2 - 2\lambda\mu)c^4 - (\nu^3 - \lambda\mu\nu)c^2 \\+ \lambda^2\mu^2 - \lambda\mu\nu(\lambda + \mu) + \nu^2(\lambda^2 - \lambda\mu + \mu^2) &= 0.\end{aligned}$$

En substituant les valeurs $\lambda = 16$, $\mu = 17$, $\nu = 18$, cette équation devient

$$c^8 - 54c^6 + 221c^4 - 4176c^2 + 20736 = 0,$$

c'est-à-dire

$$(c^2 - 4)(c^2 - 16)(c^4 - 34c^2 + 324) = 0.$$

Les valeurs de c^2 sont donc $c^2 = 4$, $c^2 = 16$, et les deux valeurs du trinome $c^4 - 34c^2 + 324$, lesquelles sont toutes réelles et positives; il y a donc en tout 8 valeurs de c , et pour chacune de ces valeurs, une seule valeur (réelle ou imaginaire) de a et de b .

2, *Stone Buildings*, W.C., Oct. 4, 1860.

257On the Cubic Centres of a Line257From the *Philosophical Magazine*, vol. xx. (1860), pp. 342-344.

THE following is a particular case of a more general theorem which I have obtained, and which I hope to publish. Consider a conic passing through the three points of intersection of two conics; these points being given, the conic depends on one variable parameter, and the theorem is that there are in general three positions of the conic, for each of which the three intersections of the conic with a given line are coincident centres. Or if the two conics be represented by $U = 0$, $V = 0$, the general equation of a conic through the four points of intersection is $U + \theta V = 0$; and taking the given line for $z = 0$, then the cubic centres are the intersections of the line $z = 0$ with the conic

which is the first or line polar of an arbitrary point (x, y, z) in regard to the conic $U + \theta V = 0$.

The analytical investigation presents no difficulty. Write

$$\begin{aligned} U &= (a, b, c, f, g, h \wr x, y, z)^2, \\ V &= (a', b', c', f', g', h' \wr x, y, z)^2, \end{aligned}$$

then the first or line polar of (x, y, z) in regard to $U + \theta V = 0$ is

$$\begin{aligned} x(a + \theta a' \wr x, y, z) + y(h + \theta h' \wr x, y, z) \\ + z(g + \theta g' \wr x, y, z) = 0, \end{aligned}$$

et putting $z = 0$ this becomes

$$x(a + \theta a' \wr x, y, 0) + y(h + \theta h' \wr x, y, 0) = 0,$$

say $(A, B, C \wr x, y)^2 + \theta(A', B', C' \wr x, y)^2 = 0$; and if (x, y) are the coordinates of a cubic centre, then the equation must have equal roots, that is, the equation in θ

$$(A, B, C \wr x, y)^2 + \theta(A', B', C' \wr x, y)^2 = 0$$

must have equal roots, or writing $\theta + \lambda, \theta + \mu, \theta + \nu$ for the roots, the conics will touch. But the equation of the line, combined with those of the two conics, gives

$$x : y : z = \frac{1}{\theta + \lambda} : \frac{1}{\theta + \mu} : \frac{1}{\theta + \nu};$$

and substituting these values in the equation of the line, we have

$$\frac{1}{\theta + \lambda} + \frac{1}{\theta + \mu} + \frac{1}{\theta + \nu} - \frac{2}{\theta} = 0,$$

which is (as it should be) a cubic equation in θ .

If the equation in θ has equal roots, then

$$\frac{1}{(\theta + \lambda)^2} + \frac{1}{(\theta + \mu)^2} + \frac{1}{(\theta + \nu)^2} - \frac{2}{\theta^2} = 0;$$

and putting in these two equations,

$$x = \frac{m}{\theta + \lambda}, \quad y = \frac{m}{\theta + \mu}, \quad z = \frac{m}{\theta + \nu},$$

we have

$$\begin{aligned}x + y + z - \frac{2m}{\theta} &= 0, \\x^2 + y^2 + z^2 - \frac{2m^2}{\theta^2} &= 0;\end{aligned}$$

whence eliminating m ,

$$(x + y + z)^2 = 2(x^2 + y^2 + z^2);$$

that is

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0;$$

or, what is the same thing,

$$\sqrt{x} + \sqrt{y} + \sqrt{z} = 0,$$

for the equation of the locus of the coincident centres: such locus is therefore a conic touching the lines $x = 0$, $y = 0$, $z = 0$, in the points of intersection with the lines $y - z = 0$, $z - x = 0$, $x - y = 0$ respectively; it is a conic touching the lines X , Y , Z harmonically in regard to the line I .

To find the envelope of the line L , the most convenient course is to take the equation in θ in the reduced form

$$\theta^3 - \theta(\mu\nu + \nu\lambda + \lambda\mu) - 2\lambda\mu\nu = 0;$$

this will have a pair of equal roots if

$$(\mu\nu + \nu\lambda + \lambda\mu)^3 - 27\lambda^2\mu^2\nu^2 = 0;$$

or putting $\mu\nu + \nu\lambda + \lambda\mu = -3q$, $\lambda\mu\nu = -2r$, the condition is

$$q^3 + r^2 = 0.$$

The envelope is a curve of the class 3, that is a curve of the third order, having the three lines $x = 0$, $y = 0$, $z = 0$ for its inflexions, and $x + y + z = 0$ for the line through the three points of inflexion; and the equations of the tangents at the points of inflexion are

$$2lx - y - z = 0, \quad 2ly - z - x = 0, \quad 2lz - x - y = 0.$$

2, *Stone Buildings*, W.C., Oct. 4, 1860.

258On a Relation between two Ternary Cubic Forms258From the
Philosophical Magazine, vol. xx. (1860), pp. 344, 345.

I HAVE remarked that the ternary cubic form

$$ax^3 + by^3 + cz^3 + 6lxyz = 0$$

is in general not expressible in the form

$$px^3 + qy^3 + rz^3 + 3(qr)^{\frac{1}{2}}yz^2 + 3(rp)^{\frac{1}{2}}zx^2 + 3(pq)^{\frac{1}{2}}xy^2 = 0,$$

but becomes so on putting $l = -(qr)^{\frac{1}{2}}$, $m = -(rp)^{\frac{1}{2}}$, $n = -(pq)^{\frac{1}{2}}$; and that the first form has the nine points of inflexion given by

$$\begin{aligned} ax^3 + by^3 + cz^3 &= 0, \\ xyz &= 0, \end{aligned}$$

and the second form has the nine points of inflexion given by

$$\begin{aligned} px^3 + qy^3 + rz^3 &= 0, \\ xyz &= 0. \end{aligned}$$

But a particular case of the first form may have the same nine points of inflexion with a particular case of the second form; viz. this will be the case if the equations $ax^3 + by^3 + cz^3 = 0$ and $px^3 + qy^3 + rz^3 = 0$ are satisfied by the same values of $x^3 : y^3 : z^3$. The condition for this is

$$\frac{p}{a} = \frac{q}{b} = \frac{r}{c},$$

and then the two cubic curves have the nine points of inflexion given by $x^3 + y^3 + z^3 = 0$, $xyz = 0$; or what is the same thing, the points in question are the intersections of the lines $x = 0$, $y = 0$, $z = 0$ with the line $x + y + z = 0$; or we have $x + y + z = 0$ for the equation of the line through the three points of inflexion; and the equations of the tangents at the points of inflexion are

$$2lx - y - z = 0, \quad 2ly - z - x = 0, \quad 2lz - x - y = 0.$$

For the second form it is obvious that the points of inflexion are the intersections of the lines $x = 0$, $y = 0$, $z = 0$ with the line $x + y + z = 0$; and, moreover, that the lines $x = 0$, $y = 0$, $z = 0$ are the tangents at the point of inflexion.

The first of the above-mentioned forms, however, cannot represent a curve with a double point. In fact the condition for its doing so would be $1 + 8l^3 = 0$; but when this condition is satisfied, the left-hand side breaks up into linear factors, and the equation represents, not a proper curve of the third order, but a system of three lines. The second form *can* represent a curve having a double point; viz. if $k = -1$, the curve will have a conjugate or isolated point at the point $x = y = z$. It is clear *à priori* that ($x = 0, y = 0, z = 0$ being real lines) neither of the forms can represent a curve of the third order having a double point with two real branches through it, since in this case the curve has only one real point of inflexion.

I have elsewhere used the word “node” to denote a double point, and I take the opportunity of suggesting the employment of the words “crunode” (*crus*) and “acnode” (*acus*) to denote respectively a double point with two real branches through it, and a conjugate or isolated point.

2, *Stone Buildings*, W.C., October 19, 1860.

259The Problem of Polyhedra259From the *Philosophical Magazine*, vol. xx. (1860), pp. 348, 349.

THE theory of the “Problem of Polyhedra” consists in finding all the different polyhedra which have a given number of summits (or more generally a given number of summits and a given number of faces). It is, in fact, the extension of the geometrical problem of describing all the different polygons which have a given number of angles.

I use the term “syntypic” to denote two polyhedra which have the same number of summits and also the same number of faces, and which are such that the symbolical system of the one can be derived from that of the other by a substitution. The symbolical system of a polyhedron is a table showing the arrangement of the faces about each summit, or more generally, the arrangement of the edges about each face and about each summit.

For example, the symbolical system of the tetrahedron is

$$\begin{array}{llll} a = bcd, & b = acd, & c = abd, & d = abc; \\ A = BCD, & B = ACD, & C = ABD, & D = ABC; \end{array}$$

where a, b, c, d denote the summits, and A, B, C, D the faces.

Consider, now, two polyhedra having the same number of vertices and also the same number of faces. And let the vertices and faces of

the first polyhedron taken in any order be represented by

$$abcde \dots KLM \dots,$$

and the vertices and faces of the second polyhedron taken in a certain order be represented by

$$a'b'c'd'e' \dots K'L'M' \dots;$$

then, forming the substitution symbol

$$a'b'c'd'e' \dots K'L'M' \dots abcde \dots KLM \dots,$$

which denotes that a' is to be written for a , b' for b , $\dots K'$ for K , &c., if operating with this upon the symbol system of the first polyhedron, we obtain the symbol system of the second polyhedron, the second polyhedron will be syntypic with the first. It should be noticed, that there may be several modes of arrangement of the vertices and faces of the second polyhedron, which will render it syntypic according to the foregoing definition with the first polyhedron, i.e. the second polyhedron may be syntypic in several different ways with the first polyhedron. This is, in fact, the same as saying that a polyhedron may be syntypic with itself in several different ways.

Suppose, next, that the number of vertices of the second polyhedron is equal to the number of faces of the first polyhedron, and the number of faces of the second polyhedron is equal to the number of vertices of the first polyhedron; and let the vertices and faces of the first polyhedron in any order be represented by

$$abcde \dots KLM \dots,$$

and the faces and vertices of the second polyhedron in a certain order be represented by

$$A'B'C'D'E' \dots k'l'm' \dots$$

Then forming the substitution symbol

$$A'B'C'D'E' \dots k'l'm' \dots abcde \dots KLM \dots,$$

if, operating with this upon the symbol system of the first polyhedron, we obtain the symbol system of the second polyhedron, the second polyhedron will be syntypic with the first. This is the definition of syntypic polyhedra in the more extended sense, viz. including the case where the number of vertices of the one is equal to the number of faces of the other, and vice versa.

2, *Stone Buildings*, W.C., Oct. 4, 1860.

260 On the Double Tangents of a Plane Curve 260 From the *Philosophical Transactions of the Royal Society of London*, vol. cxlix. for the year 1859, pp. 193–212. Received March 17,—Read April 14, 1859.

IT was first shown by Plücker on geometrical principles, that the number of the double tangents of a plane curve of the order m was $\frac{1}{2}m(m-2)(m^2-9)$: see the note, “Solution d’une question fondamentale concernant la théorie générale des Courbes,” *Crelle*, t. xii. pp. 105—108 (1834), and the “Theorie der algebraischen Curven” (1839). The memoir by Hesse, “Ueber die Wendepuncte der Curven dritter Ordnung,” *Crelle*, t. xxviii. pp. 97—107 (1844), contains the analytical solution of the allied easier problem of the determination of the points of inflexion of a plane curve. In the memoir, “Recherches sur l’élimination et sur la théorie des Courbes,” *Crelle*, t. xxxiv. (1847), pp. 30—45, [53], I showed how the problem of double tangents admitted of an analytical solution, viz. if $U = 0$ is the equation of the curve, L, M, N the first derived functions of U , and

$$D = \alpha (M\partial_z - N\partial_y) + \beta (N\partial_x - L\partial_z) + \gamma (L\partial_y - M\partial_x)$$

(where α, β, γ are arbitrary), then the points of contact of the double tangents are given as the intersections of the curve $U = 0$, with a curve the equation whereof is in the first instance obtained under the form $[Y] = 0$; $[Y]$ being a given function of

$$D^2U, D^3U, \dots D^mU,$$

of the degree $m^2 - m - 6$ in respect of (α, β, γ) , the degree $m^3 - 2m^2 - 10m + 12$ in respect of (x, y, z) , and the degree $m^2 + m - 12$ in respect of the coefficients of U . It was necessary, in order that the points of intersection should be independent of the arbitrary quantities (α, β, γ) , that we should have identically

$$[Y] = \Lambda \cdot U + N \cdot \Pi U,$$

where Λ is a function of (x, y, z) of the degree $m^3 - 2m^2 - 10m + 12$, and N a function of (x, y, z) of the degree $m^2 - m - 6$, and therefore of the same form as $[Y]$; ΠU is the Hessian, or determinant formed with the second derived functions of U . I showed that $[Y]$ was of the proper form by actually obtaining the equivalent value $\Lambda \cdot U + N \cdot \Pi U$,

viz. I gave for N (for which it is to be noticed the transformation is by no means unique) the value

$$N = \frac{1}{4} (\alpha \partial_x + \beta \partial_y + \gamma \partial_z)^2 \Pi U,$$

which is indeed obviously of the degree 2 in the coefficients of U , and of the degree $m-4$ in (α, β, γ) , and $m-4$ in (x, y, z) ; but the identity was only proved for the several curves of the orders $m=3$, $m=4$, $m=5$, $m=6$, and the general proof was not obtained.

1. I propose in the present memoir to complete the analytical theory by showing à priori that the function $[Y]$ is of the form $\Lambda \cdot U + N \cdot \Pi U$; the a posteriori verification for any particular value of m may then be dispensed with. The principal steps of the demonstration are as follows: first, I show that the curve $[Y] = 0$ passes through the points of contact of the double tangents; secondly, I show that it passes through the points of inflexion; thirdly, I show that it passes through the triple points (if any) of the curve $U = 0$; and fourthly, I show that it has the points of intersection of the curve $U = 0$ with the Hessian for double points. This being so, the curve $[Y] = 0$ must contain as part of itself the curve through the points of contact of the double tangents, the curve through the points of inflexion, and the curve through the triple points; and it was shown in my former memoir that the aggregate of these curves is of the degree $m^3 - 2m^2 - 10m + 12$ in (x, y, z) , which is the same as the degree of the curve $[Y] = 0$ in these variables; hence the curve $[Y] = 0$ is made up of these curves, and we have $[Y] = \Lambda \cdot U + N \cdot \Pi U$, as was to be shown.

2. To express the equation of this curve, let $U_1, U_2, \dots$ be the successive emanants of U , taken with the facients of emanation (x_1, y_1, z_1) , viz.

$$\begin{aligned} U_1 &= (x_1 \partial_x + y_1 \partial_y + z_1 \partial_z) U, \\ U_2 &= (x_1 \partial_x + y_1 \partial_y + z_1 \partial_z)^2 U, \end{aligned}$$

and so on.

3. The function $[Y]$ may be expressed in terms of these emanants; and the equation of the curve through the points of contact of the double tangents is obtained by equating to zero the discriminant of a certain binary quartic form. The equation may be written

$$\Omega^2 + 4\Theta^3 = 0,$$

where Ω and Θ are functions of the coefficients of U , and of (x, y, z) .

4. It is to be remarked that the formula applies to a cubic; we have here

$$H = \frac{1}{4} (\alpha \partial_x + \beta \partial_y + \gamma \partial_z)^2 \Pi U,$$

viz. the curve through the points of contact of the double tangents is a curve of the order $(n-2)(n^2-9)$, and its equation contains the coefficients of the equation $U = 0$ of the given curve in the degree $(n+4)(n-3)$. And since each double tangent touches the given curve in two points, the number of double tangents is $\frac{1}{2}(n-2)(n^2-9)$, which is Plücker's formula.

5. The equation $[Y] = 0$ may be written in the form

$$H = \frac{1}{(n-1)^2} \frac{1}{(n-2)^2} \sum \sum (\alpha \partial_x + \beta \partial_y + \gamma \partial_z)^2 (LMN)^2,$$

where L, M, N are the first derived functions of U , and the summation extends to all the terms of the development.

6. The investigation consists in showing that the curve $H = 0$ passes through the points of contact of the double tangents, the points of inflexion, and the triple points of the given curve, and that it has the intersections of the curve with the Hessian for double points; and that the complete curve is of the proper degree.

7. The function H , like the several component terms, is of course of the degree 3 in the coefficients of U , and of the degree $2(n-2)$ in (x, y, z) .

8. The curve $H = 0$ passes through the points of inflexion; in fact at a point of inflexion the Hessian vanishes, and consequently the function H vanishes.

9. The curve $H = 0$ passes through the triple points; in fact at a triple point the first derived functions L, M, N all vanish, and consequently $H = 0$.

10. The curve $H = 0$ has the intersections of the curve $U = 0$ with the Hessian for double points; in fact if in the equation $H = 0$ we write $\theta = 0, e = 0$, it becomes $H(d) = 0$, that is, the points of intersection of the curve $U = 0$ with the Hessian are double points on the curve $H = 0$.

11. The function H may be expressed in the form

$$H = \Lambda \cdot U + N \cdot \Pi U,$$

where Λ is of the degree $(n-2)(n^2-9)-n$ in (x, y, z) , and N is of the degree n^2-n-6 in (x, y, z) .

12. The complete expression for H is

$$H = \frac{1}{4} (\alpha \partial_x + \beta \partial_y + \gamma \partial_z)^2 \Pi U,$$

and this is equivalent to the form given in the former memoir.

13. The number of double tangents is $\frac{1}{2}(n-2)(n^2-9)$, which for $n=3$ gives 0, for $n=4$ gives 28, for $n=5$ gives 120, and so on.

14. The investigation is applicable to curves with multiple points; but in the case of a curve with a multiple point of the order k , the number of double tangents is diminished by $\frac{1}{2}k(k-1)(k^2-9)$, and the formulæ require corresponding modifications.

15. The case of a quartic curve ($n=4$) is particularly interesting; the curve $H=0$ is then of the order 10, and the equation contains the coefficients of the quartic in the degree 4. The 28 double tangents of a quartic curve are well known to be connected by a remarkable system of syzygies, which has been investigated by Hesse, Aronhold, and others.

16. The present investigation completes the analytical theory of double tangents, and establishes upon a firm basis the formulæ of Plücker for the number of double tangents of a plane curve.

10.13. On an Analytical Theorem Relating to the Distribution of
Electricity upon Spherical Surfaces 145

Arthur Cayley, Collected Mathematical Papers

Volume IV, Pages 201–250

This transcription covers Papers 260–262 from Cayley’s Collected Mathematical Papers, Volume IV, comprising pages 201 through 250 of the original publication.

260. On the Double Tangents of a Plane Curve.

[From the *Philosophical Transactions of the Royal Society of London*, vol. cxlix, for the year 1859, pp. 193–212.]

[Continuation from p. 200]

and similarly for the accented letters, so that

$a_n = 0$ is the equation of the curve;

$a_1 = 0$ is the equation of the first or $(n - 1)$ thic polar;

$a_{n-1} = 0$ is the equation of the last or line-polar, or what is the same thing, since (

is a point on the curve, the tangent at this point;

$a_0 = 0$, the condition which expresses that (x, y, z) is a point of the curve;

and we have to form the identical equation

$$a_0 \text{ I} + a_1 \text{ II} + a_{n-1} \text{ III} + a_n \text{ IV} = 0.$$

25. If, for shortness, the columns of the last-mentioned matrix are represented by

$$1, 2, 3, \dots, n, \quad (1), (2), \dots, (n - 1),$$

and the determinants formed with these columns respectively by a corresponding notation $\{1, 2, (1)\}$, $\{1, 2, (2)\}$, &c., then the expressions for the multipliers I, II, III, IV are as follows, viz.

$$\text{I} = \{n, n - 1, (2)\} + \{n, n - 2, (3)\} + \dots + \{n, 2, (n - 1)\},$$

$$\text{II} = -\{n, n - 1, (1)\} - \{n, n - 2, (2)\} - \dots - \{n, 2, (n - 2)\} - \{n, 1, (n - 1)\},$$

$$\text{III} = -\{1, 2, (n - 1)\} - \{1, 3, (n - 2)\} - \dots - \{1, n - 1, (2)\} - \{1, n, (1)\},$$

$$\text{IV} = \{1, 2, (n - 2)\} + \{1, 3, (n - 1)\} + \dots + \{1, n - 1, (1)\};$$

the truth of the identical equation being shown, as in the foregoing special cases, by the transformation of the multipliers into the form

$$\text{I} = a_0 \cdot 0 + a_1(12) + a_{n-1}(13) + a_n(14),$$

$$\text{II} = a_0(21) + a_1 \cdot 0 + a_{n-1}(23) + a_n(24),$$

$$\text{III} = a_0(31) + a_1(32) + a_{n-1} \cdot 0 + a_n(34),$$

$$\text{IV} = a_0(41) + a_1(42) + a_{n-1}(43) + a_n \cdot 0,$$

where $(12) = -(21)$, &c.: the required expressions may be written down without difficulty.

26. Proceeding then to reduce the equation

$$\text{III} = \{1, 2, (n-1)\} - \{1, 3, (n-2)\} - \cdots - \{1, n-1, (2)\} - \{1, n, (1)\},$$

we have the equation

$$\{1, 2, (1)\} = \Delta^2 H,$$

which is to be successively operated on with D . The degrees (less unity) of the columns

$$1, 2, \dots, n-1, \quad n, \quad (1), (2), \dots, n-1,$$

are

$$n-1, \quad n-2, \dots, 1, \quad 0, \quad n-2, \quad n-3, \dots, 0;$$

and the rule is to operate on each column of the determinant, multiplying by the degree less unity, and increasing the symbolical number by unity. Thus

$$D\{1, 2, (1)\} = (n-1)\{2, 2, (1)\} + (n-2)\{1, 3, (1)\} + (n-2)\{1, 2, (2)\},$$

since $\{2, 2, (1)\}$ vanishes identically. The following Table shows the mode of effecting the operations:

Table of Operations						
			$[n-2]^\alpha$	$[n-2]^\beta$		
H	1	= 1	$[n-2]^0$	$[n-2]^0$		12(1)
DH	1	= 1	$[n-2]^0$	$[n-2]^1$		12(2)
	1	1	$[n-2]^1$	$[n-2]^0$		13(1)
D^2H	1	= 1	$[n-2]^0$	$[n-2]^2$		12(3)
	1 1	2	$[n-2]^1$	$[n-2]^1$		13(2)
	1	1	$[n-2]^2$	$[n-2]^0$		14(1)
	1	1	$[n-2]^1$	$[n-2]^0$	$[n-1]^1$	23(1)
D^3H	1	= 1	$[n-2]^0$	$[n-2]^3$		12(4)
	2 1	3	$[n-2]^1$	$[n-2]^2$		13(3)
	1 2	3	$[n-2]^2$	$[n-2]^1$		14(2)
	1	1	$[n-2]^3$	$[n-2]^0$		15(1)
	1 2	3	$[n-2]^1$	$[n-2]^1$	$[n-1]^1$	23(2)
	1 1	2	$[n-2]^2$	$[n-2]^0$		24(1)

D^4H	1	= 1	$[n-2]^0$	$[n-2]^4$		12(5)
	3 1	4	$[n-2]^1$	$[n-2]^3$		13(4)
	3 3	6	$[n-2]^2$	$[n-2]^2$		14(3)
	1 3	4	$[n-2]^3$	$[n-2]^1$		15(2)
	1	1	$[n-2]^4$	$[n-2]^0$		16(1)
	3 3	6	$[n-2]^1$	$[n-2]^2$	$[n-1]^1$	23(3)
	2 3 1	8	$[n-2]^2$	$[n-2]^1$		24(2)
	2 1	3	$[n-2]^3$	$[n-2]^0$		25(1)
	2	2	$[n-2]^2$	$[n-2]^0$	$[n-1]^2$	34(1)
D^5H	1	= 1	$[n-2]^0$	$[n-2]^5$		12(6)
	4 1	5	$[n-2]^1$	$[n-2]^4$		13(5)
	6 4	10	$[n-2]^2$	$[n-2]^3$		14(4)
	4 6	10	$[n-2]^3$	$[n-2]^2$		15(3)
	1 4	5	$[n-2]^4$	$[n-2]^1$		16(2)
	1	1	$[n-2]^5$	$[n-2]^0$		17(1)
	6 4	10	$[n-2]^1$	$[n-2]^3$	$[n-1]^1$	23(4)
	8 6 1	20	$[n-2]^2$	$[n-2]^2$		24(3)
	3 6 4	15	$[n-2]^3$	$[n-2]^1$		25(2)
	3 1	4	$[n-2]^4$	$[n-2]^0$		26(1)
	2 8	10	$[n-2]^2$	$[n-2]^1$	$[n-1]^2$	34(2)
	2 3	5	$[n-2]^3$	$[n-2]^0$		35(1)

where the first three columns show the numbers which give, by the addition of the numbers in the same horizontal line, the numerical coefficients of the factorials which multiply the different terms of H , DH , &c., and where in the last column 12(1), &c. are written for shortness in the place of $\{1, 2, (1)\}$, &c.

27. It is clear that we have in general

$$\begin{aligned} \Delta^{-1} D^r H = & \frac{1}{[n-1]^\perp} \left[[n-2]^0 [n-2]^r \{1, 2, (r+1)\} \right. \\ & + [n-2]^1 [n-2]^{r-1} \{1, 3, (r)\} \\ & + \frac{[r]^2}{[2]^2} [n-2]^2 [n-2]^{r-2} \{1, 4, (r-1)\} \\ & + \dots \\ & + [n-2]^r [n-2]^0 \{1, r+2, (1)\} \\ & + R'_1 [n-2]^1 [n-2]^{r-1} \{2, 3, (r-1)\} \\ & + R''_2 [n-2]^2 [n-2]^{r-2} \{2, 4, (r-2)\} \\ & + \dots \\ & + R^{(r-1)}_{r-1} [n-2]^{r-1} [n-2]^1 \{2, r+1, (1)\} \\ & + R''_1 [n-2]^2 [n-2]^{r-2} \{3, 4, (r-3)\} \\ & + \dots \\ & + R^{(r-2)}_{r-2} [n-2]^{r-2} [n-2]^2 \{3, r, (1)\} \\ & + \dots \\ & \left. + \begin{cases} \frac{1}{[\frac{1}{2}r]^\perp} [n-2]^{\frac{1}{2}r} [n-2]^{\frac{1}{2}r} \{r+1, r+2, (1)\}, & r \text{ even; or} \\ \frac{1}{[\frac{1}{2}(r-1)]^\perp} [n-2]^{\frac{1}{2}(r+1)} [n-2]^{\frac{1}{2}(r-1)} \{(r+1), (r+3), (2)\}, & r \text{ odd} \end{cases} \right], \end{aligned}$$

and the general term is

$$\frac{[n-1]^\perp R_s^S [n-2]^{S+1} [n-2]^{r-S-s}}{[S+1]^\perp} \{S+1, S+2+s, (r-2S+1-s)\},$$

where s extends from $s = 0$ to $s = r - 2S$, and S from $S = 0$ to $S = \frac{1}{2}r$ or $\frac{1}{2}(r-1)$, according as r is even or odd. The expression for the coefficient R_1 is

$$R_1 = \frac{[r]^s}{[s]^\perp [r-s]^\perp},$$

and that of the other coefficients R_s ($s = \text{or} < 1$) is not required for the present purpose.

28. According to a remark already made, the expressions for $D^{n-1}H$, $D^{n-2}H$, &c. are at once obtained from that for D^rH by

merely writing $n - 1$, $n - 2$, &c. in the place of n : it is however to be noticed, that the quantity within the [] must not be negative, and that on its becoming so, the factorial is to be omitted.

29. I write now

$$\text{III} = \frac{1}{[n-1]!} \Delta^{-1} D^{n-2} H,$$

and I consider the expression

$$\frac{1}{[n-1]!} \Delta^{-1} D^{n-1} H \times \text{III},$$

the general term of which is

$$\frac{R_s^S \{S+1, S+2+s, (r-2S+1-s)\}}{[n-1]! [n-2]^S [n-2]^{r-S}} \times [n-1]! \cdot \frac{[q]^\alpha [q-\alpha]^\beta}{[n-2]^\alpha [n-3]^\alpha [n-3]^\beta \dots},$$

or, as this may be written, putting $q = n - S - 1$,

$$R_s^S \{S+1, S+2+s, (r-2S+1-s)\} \\ \times \frac{[n-2]^S [n-2]^{r-S}}{[n-1]! [n-2]^\alpha [n-3]^\alpha [n-3]^\beta \dots} + \&c.$$

30. I assume $r > n - 2$; we have then $\alpha + \beta = r - S > n - S - 2$, and therefore $\alpha + \beta < q$. The general term of the series in $\{\}$ is

$$\pm \frac{[q]^\alpha [q-\alpha]^\beta}{[n-2-\alpha]^\beta [n-2-\alpha]^\beta \dots},$$

where the terms for which $n - 2 - \alpha$ is negative are to be excluded, or what is the same thing, the series is not to be continued beyond $\alpha = n - 2$. But observing that $[q]^\alpha$ vanishes for $\alpha > q$, that is, $\alpha > n - S - 1$, it is in fact the same thing whether the series is continued indefinitely or only to the term for which $\alpha = n - S - 1$, and we may consistently with the condition $\alpha > n - 2$, continue the series as far as $\alpha = n - S - 1$, except in the case $S = 0$, when by doing so we include the term

corresponding to $\alpha = n - 1$, which in virtue of the condition ought to be excluded. The expression for the term in question is $(-)^{n-1}[-1]^r[-1]^\beta$; hence if the sum of the series continued to the proper point is $\mathcal{S}$, the sum continued indefinitely (in the particular case $S = 0$) is $\mathcal{S} + (-)^{n-1}[-1]^r[-1]^\beta$, but in every other case the sum continued indefinitely is simply $\mathcal{S}$. And by a well-known theorem in finite differences, the sum continued indefinitely is in fact zero. That is, except in the case $S = 0$, we have $\mathcal{S} = 0$, but in the excepted case

$$\mathcal{S} + (-)^{n-1}[-1]^r[-1]^\beta = 0;$$

or observing that $\alpha + \beta (= r - S)$ is in this case $= r$, and transforming the factorials, we have

$$\mathcal{S} = -(-)^{n-1}(-)^r[n]^r[r-n]^\beta;$$

or substituting for α and β their values,

$$\mathcal{S} = (-)^{n-r-1}[n]^r[r-n]^{r-n}.$$

31. Hence the general term of

$$\frac{1}{[n-1]!} \Delta^{-1} D^{n-1} H \times \text{III}$$

vanishes except for $S = 0$, but when $S = 0$, its value is

$$R_s\{1, 2 + s, (r + 1 - s)\} \times (-)^{n-r-1}[n]^r[r-n]^{r-n};$$

or observing that R_s is equal to $[r]^s \div [s]^s[r-s]^s$, the value is simply

$$(-)^{n-r-1}[n]^r\{1, 2 + s, (r + 1 - s)\},$$

that is, we have

$$\frac{1}{[n-1]!} \Delta^{-1} D^{n-1} H \times \text{III} = (-)^{n-r-1}[n]^r \sum_s \{1, 2 + s, (r + 1 - s)\},$$

the summation in respect to s extending from $s = 0$ to $s = r$. In particular, giving to r the values $n - 2$ and $n - 1$, and attending to the expressions for III and IV, we find

$$\begin{aligned} \Delta^n(D^{n-2}H - \&c.) \cdot \text{III} &= -[n-2]^{\overline{n-2}} \text{III}, \\ \Delta^n(D^{n-1}H - \&c.) \cdot \text{III} &= -[n-3]^{\overline{n-3}} \text{IV}. \end{aligned}$$

32. The equation $\text{III} = 0$ belongs to the curve which by its intersections with the tangent, gives the tangentials of a point of the curve $U = 0$. Hence the equation of the curve in question is

$$D^{n-2}H - \&c. = 0,$$

which is Mr Salmon's theorem, leading to the solution of the problem of double tangents.

33. The expressions for I and II are obtained from those of IV and III by interchanging (X, Y, Z) and (x, y, z) , and reversing the sign. Hence if, as before, I_1 , II_1 , &c. denote the values which I, II, &c. assume by this interchange, we have

$$\begin{aligned}\Delta^n(\text{I}_1 - \&c.) &= [n-2]^{n-2} \text{II}, \\ \Delta^n(\text{II}_1 - \&c.) &= [n-3]^{n-3} \text{I},\end{aligned}$$

and the identical equation

$$a_0 \text{I} + a_1 \text{II} + a_{n-1} \text{III} + a_n \text{IV} = 0$$

becomes therefore

$$(D^{n-2}H - \&c.)T = 0,$$

which is the general identical equation referred to in the introduction to the present memoir.

34. It is to be noticed that for $n = 3$, the equation is

$$(9\Delta H - 2DI_1)U + (I_1^2 - 2\Delta^2II_1)DU - \Delta(DH - 3DII_1)T - (\Delta H - 3II_1)T = 0.$$

But we have $II_1 = H$, and in like manner $I_1 = \Delta$, and the equation thus becomes

$$\Delta DU + I_1^2 DU - \Delta^2 DT + HT = 0.$$

And so also for $n = 4$, the equation is

$$(D^2H - 3DII_1)U + (I_1^2 - 3\Delta^2II_1)DU - \Delta(D^2H - 3DII_1)T - (DH - 3\Delta H)T = 0.$$

But we have in general $DII_1 = \frac{n-1}{2}DH$, and therefore in the present case $DII_1 = \frac{3}{2}DH$, and consequently $D^2II_1 = \frac{3}{2}D^2H$, and the equation thus becomes

$$D^2H \cdot U + (I_1^2 - 3\Delta^2II_1)DU - \Delta(D^2H - 3\Delta^2D^2H)T + DH \cdot T = 0,$$

which agrees with the results previously obtained for the two particular cases.

261. On the Conic of Five-Pointic Contact at any Point of a Plane Curve.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLIX. (for the year 1859), pp. 371–400. Received March 1,—Read March 24, 1859.]

I. INVESTIGATION OF THE EQUATION OF THE CONIC OF FIVE-POINTIC CONTACT.

The tangent is a line passing through two consecutive points of a plane curve, and we may in like manner consider the conic which passes through five consecutive points of a plane curve; and as there are certain singular points, viz. the points of inflexion where three consecutive points of the curve lie in a line, so there are singular points where six consecutive points of the curve lie in a conic. In the particular case where the given curve is a cubic, the last-mentioned species of singular points have been considered by Plücker and Steiner, and in the same particular case, the theory of the conic of five-pointic contact has recently been established by Mr Salmon. But the general case, where the curve is of any order whatever, has not so far as I am aware been hitherto considered;—the establishment of this theory is the object of the present memoir.

1. I take (X, Y, Z) as current coordinates, and I represent the equation of the given curve by

$$U = (\text{bullet} x, Y, Z)^n = 0.$$

Let (x, y, z) be the coordinates of a given point on the curve, and let $U = (\text{bullet} x, y, z)^n$ be what U becomes when (x, y, z) are written in the place of (X, Y, Z) ; we have therefore $U = 0$ as a condition satisfied by the coordinates of the point in question.

2. Write for shortness

$$\begin{aligned} DU &= (X\partial_x + Y\partial_y + Z\partial_z)U, \\ D^2U &= (X\partial_x + Y\partial_y + Z\partial_z)^2U, \end{aligned}$$

and let $\nabla = aX + bY + cZ = 0$ be the equation of an arbitrary line; then if in the equation of the given curve we write $X + \lambda a$,

$Y + \lambda b$, $Z + \lambda c$ in the place of X , Y , Z respectively, and write $\lambda = -\nabla/(a\partial_x + b\partial_y + c\partial_z)$, the result is

$$U - \frac{\nabla}{a\partial_x + b\partial_y + c\partial_z} DU + \frac{\nabla^2}{(a\partial_x + b\partial_y + c\partial_z)^2} D^2U - \&c. = 0,$$

which is the equation of the given curve referred to the line $\nabla = 0$ as a line of coordinates.

3. The equation of a conic passing through the point (x, y, z) is

$$(A, B, C, F, G, H \mid X, Y, Z)^2 = 0,$$

where the coefficients are such that

$$(A, B, C, F, G, H \mid x, y, z)^2 = 0.$$

The condition that the conic shall have five-pointic contact with the given curve at the point (x, y, z) gives rise to a system of equations which determine the ratios of the coefficients $A : B : C : F : G : H$.

4. Consider the coordinates of a point of the curve as functions of a single variable parameter; then for the present purpose the coordinates of a point consecutive to (x, y, z) may be taken to be

$$\begin{aligned} x + dx + \frac{1}{2}d^2x + \frac{1}{6}d^3x + \frac{1}{24}d^4x, \\ y + dy + \frac{1}{2}d^2y + \frac{1}{6}d^3y + \frac{1}{24}d^4y, \\ z + dz + \frac{1}{2}d^2z + \frac{1}{6}d^3z + \frac{1}{24}d^4z, \end{aligned}$$

values which, substituted for X , Y , Z , must satisfy the equations

$$U = 0, \quad D^2U - \Pi \cdot DU = 0.$$

I write for shortness

$$\begin{aligned} d_1 &= dx \partial_x + dy \partial_y + dz \partial_z, \\ d_2 &= d^2x \partial_x + d^2y \partial_y + d^2z \partial_z, \\ d_3 &= d^3x \partial_x + d^3y \partial_y + d^3z \partial_z, \\ d_4 &= d^4x \partial_x + d^4y \partial_y + d^4z \partial_z, \end{aligned}$$

then the consecutive value of U is

$$e^{d_1 + \frac{1}{2}d_2 + \frac{1}{6}d_3 + \frac{1}{24}d_4} U,$$

(Read $\exp. z$, exponential of z , $= e^z$), which is

$$\begin{aligned} & (1 + d_1 + \frac{1}{2}d_1^2 + \frac{1}{6}d_1^3 + \frac{1}{24}d_1^4) U \\ & + \frac{1}{2}d_2 (1 + d_1 + \frac{1}{2}d_1^2) U \\ & + \frac{1}{6}d_3 (1 + d_1) U \\ & + \frac{1}{24}d_4 U, \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} & U + d_1 U + \frac{1}{2}(d_1^2 + d_2)U + \frac{1}{6}(d_1^3 + 3d_1d_2 + d_3)U \\ & + \frac{1}{24}(d_1^4 + 6d_1^2d_2 + 4d_1d_3 + 3d_2^2 + d_4)U, \end{aligned}$$

the several terms of which must respectively vanish, and we have therefore

$$\begin{aligned} U &= 0, \\ d_1 U &= 0, \\ d_1^2 U &= -(d_1^2 + 2d_2)U, \\ d_1^3 U &= -(d_1^3 + 3d_1d_2 + d_3)U. \end{aligned}$$

5. Next, preparing to substitute in the equation

$$D^2U - \Pi \cdot DU = 0,$$

the consecutive value of DU is

$$(x + dx + \frac{1}{2}d^2x + \frac{1}{6}d^3x + \frac{1}{24}d^4x) \partial_x U + \&c.,$$

where

$$(x + dx + \frac{1}{2}d^2x + \frac{1}{6}d^3x + \frac{1}{24}d^4x) \partial_x U + \&c.$$

Reducing by the above results, the consecutive value of DU is

$$\begin{aligned} & = d_1 U + \frac{1}{2}(d_1^2 + d_2)U + \frac{1}{6}(d_1^3 + 3d_1d_2 + d_3)U \\ & + \frac{1}{24}(d_1^4 + 6d_1^2d_2 + 4d_1d_3 + 3d_2^2 + d_4)U. \end{aligned}$$

Hence also writing

$$\begin{aligned} P &= ax + by + cz, \\ d_1 P &= a dx + b dy + c dz, \\ d_2 P &= a d^2 x + b d^2 y + c d^2 z, \end{aligned}$$

the consecutive value of $-\text{II} \cdot DU$ is $-(P + d_1 P + \frac{1}{2} d_2 P)$ multiplied into the consecutive value of DU , and the product is

$$\begin{aligned} &= P \cdot d_1 U \\ &\quad + P \cdot \frac{1}{2} (d_1^2 + d_2) U + d_1 P \cdot d_1 U \\ &\quad + P \cdot \frac{1}{6} (d_1^3 + 3d_1 d_2 + d_3) U + d_1 P \cdot \frac{1}{2} (d_1^2 + d_2) U + \frac{1}{2} d_2 P \cdot d_1 U. \end{aligned}$$

7. The consecutive value of $D^2 U$ is

$$\begin{aligned} (x + dx + \frac{1}{2} d^2 x + \frac{1}{6} d^3 x + \frac{1}{24} d^4 x)^2 \partial_x^2 U + \&c., \\ &= d_1^2 U + d_1 (d_1^2 + d_2) U + \&c. \end{aligned}$$

8. Substituting in the equation $D^2 U - \text{II} \cdot DU = 0$, we obtain as the conditions of a five-pointic contact

$$\begin{aligned} &- [(m-1)a_1^2 + 3(m-2)a_1 a_2] U + P \cdot \frac{1}{2} (d_1^2 + 3d_1 d_2) U + d_1 P \cdot d_1^2 U = 0, \\ &- [(m-1)(a_1^3 + 6a_1^2 a_2) + (m-2)(4a_1 a_3 + 3a_2^2)] U \\ &\quad + P \cdot \frac{1}{6} (d_1^3 + 6d_1^2 d_2 + 4d_1 d_3 + 3d_2^2) U + d_1 P \cdot \frac{1}{2} (d_1^2 + 3d_1 d_2) U + \frac{1}{2} d_2 P \cdot d_1^2 U = 0; \end{aligned}$$

or reducing

$$\begin{aligned} P &= 2(m-2) \frac{d_1^2 U}{d_1^2 U}, \\ d_1 P &= \frac{[(m-1)a_1^3 + 6a_1^2 a_2] U}{d_1^2 U} - \frac{P \cdot [a_1^3 + 3a_1^2 a_2] U}{d_1^2 U}, \\ d_2 P &= \frac{[(m-1)(a_1^3 + 6a_1^2 a_2) + (m-2)(4a_1 a_3 + 3a_2^2)] U}{d_1^2 U} \\ &\quad - \frac{d_1 P \cdot [a_1^3 + 3a_1^2 a_2] U}{d_1^2 U} - \frac{P \cdot [a_1^3 + 6a_1^2 a_2 + 4a_1 a_3 + 3a_2^2] U}{d_1^2 U}, \end{aligned}$$

which are the conditions of a five-pointic contact: it is to be remarked that if only the first and second conditions are satisfied, we have a four-pointic contact, and if only the first condition is satisfied, a three-pointic contact.

9. We have to reduce the last-mentioned equations ; suppose that A, B, C are the first derived functions of U , then the equation $d_1U = 0$ may be written

$$A dx + B dy + C dz = 0,$$

and this will be satisfied identically if

$$dx = B\nu - C\mu,$$

$$dy = C\lambda - A\nu,$$

$$dz = A\mu - B\lambda,$$

where λ, μ, ν are arbitrary multipliers, which may be taken to be constants. We have therefore $d_1 = \mathfrak{D}$, where

$$\mathfrak{D} = (B\nu - C\mu) \partial_x + (C\lambda - A\nu) \partial_y + (A\mu - B\lambda) \partial_z.$$

10. The resulting expressions for d_1^2U, d_1^3U, d_1^4U may be exhibited in the reduced forms given by Hesse, viz. if $\varpi = \lambda x + \mu y + \nu z$, we have

$$d_1^2U = P_2U - Q_2\varpi^{m-2}, \quad d_1^3U = P_3U - Q_3\varpi^{m-3}, \quad d_1^4U = P_4U - Q_4\varpi^{m-4},$$

where the values of $P_2, P_3, P_4; Q_2, Q_3, Q_4$ are as follows, viz. if (a, b, c, f, g, h) are the second derived functions of U , and if

$$H = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

be the Hessian ; if, moreover,

$$\Phi = - \begin{vmatrix} 0 & \lambda & \mu & \nu \\ \lambda & a & h & g \\ \mu & h & b & f \\ \nu & g & f & c \end{vmatrix}$$

be the bordered Hessian (we may also write $\Phi = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \lambda, \mu, \nu)^2$, where $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$ are the inverse coefficients of (a, b, c, f, g, h) , viz. $\mathfrak{A} = bc - f^2$, &c.); and finally, if for shortness we write

$$\Omega = \partial_\lambda \Phi \cdot \partial_\mu \Phi + \partial_\mu \Phi \cdot \partial_\nu \Phi + \partial_\nu \Phi \cdot \partial_\lambda \Phi,$$

then we have

$$\begin{aligned} P_2 &= \frac{m-1}{m} \mathfrak{D}\Phi, \\ P_3 &= \frac{(m-1)^2}{m^2} \mathfrak{D}^2\Phi - \frac{(m-1)(m-2)}{m^2} \Omega, \\ P_4 &= \frac{(m-1)^3}{m^3} \mathfrak{D}^3\Phi - \frac{3(m-1)^2(m-2)}{m^3} \mathfrak{D}\Omega + \frac{(m-1)(m-2)}{m^3} \Psi, \end{aligned}$$

where Ψ is a certain function of the third derivatives of U .

In the present case $\varpi = 0$, and we have

$$d_1^2 U = P_2 U, \quad d_1^3 U = P_3 U, \quad d_1^4 U = P_4 U.$$

11. Hence, substituting for Q_2 and Q_3 their values, the first and second of the equations for the five-pointic contact give

$$P = 2(m-2) \frac{P_2 U}{d_1^2 U},$$

and observing that Π is a linear function of (X, Y, Z) , and consequently that $P, d_1 P$ denote simply the values which Π assumes when $(x, y, z), (dx, dy, dz)$ are respectively substituted for (X, Y, Z) , we see at once that these two conditions will be satisfied if we put

$$\Pi = \Lambda \cdot \Phi,$$

where Λ is an arbitrary constant, or, what is the same thing, an arbitrary function of (x, y, z) . We have thus the general equation of a conic of four-pointic contact.

12. The above value of Π gives

$$P = \Lambda \cdot \Phi(x, y, z), \quad d_1 P = \Lambda \cdot \mathfrak{D}\Phi, \quad d_2 P = \Lambda \cdot d_2 \Phi,$$

and the third equation of the system of conditions for a five-pointic contact is therefore

$$-[(m-1)(a_1^3 + 6a_1^2 a_2) + (m-2)(4a_1 a_3 + 3a_2^2)] U + \Lambda \cdot [\text{expr.}] = 0,$$

which leads to the value of Λ .

15. We have $R_2 = H$, and it will be shown that

$$\begin{aligned} (m-1)^2 R_3 &= -3(m-2)^2 H \cdot \Phi + 4(m-2)\Omega, \\ (m-1)^2 R_4 &= -12(m-2)^2 \Omega \cdot \Phi + 4(m-2)\Psi, \end{aligned}$$

where for shortness

$$\Omega = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_\lambda, \partial_\mu, \partial_\nu)^2 H,$$

and hence, writing

$$\Lambda = \frac{9H^2 \cdot A}{(m-1)^2},$$

we have

$$\begin{aligned} 9H^2A &= 3R_3 \cdot (m-1)^2 = R_4 - 4(m-1)^2R_2^2, \\ 9H^2A &= -3\Omega \cdot \Phi + 4\Psi; \end{aligned}$$

or replacing H, Φ by their values,

$$9H^2A = 3\Omega \cdot \Phi - 4(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_\lambda H, \partial_\mu H, \partial_\nu H)^2,$$

and Λ having this value, the equation of the five-pointic conic is

$$D^2U - \frac{1}{2(m-2)} (DH + \Lambda \cdot DU) DU = 0,$$

where it will be recollected that the current coordinates are (X, Y, Z) , and that D denotes $X\partial_x + Y\partial_y + Z\partial_z$.

16. I remark, in passing, that the problem of finding the circle of curvature at a given point of a plane curve, is in fact that of determining the conic having with the curve at the given point a three-pointic contact, and besides passing through two given points. The equation of a conic having an ordinary contact, is

$$D^2U - \Pi \cdot DU = 0,$$

where

$$\Pi = aX + bY + cZ,$$

and the condition of a three-pointic contact is

$$ax + by + cz = 2(m-2).$$

Let the coordinates of the two given points be

$$(x_1, y_1, z_1), \quad (x_2, y_2, z_2),$$

and let $(D^2U)_1$, &c. be the corresponding values of D^2U , &c., then we have

$$\begin{aligned} aX + bY + cZ &= \Pi, \\ ax + by + cz &= 2(m-2), \\ ax_1 + by_1 + cz_1 &= \frac{(D^2U)_1}{(DU)_1}, \\ ax_2 + by_2 + cz_2 &= \frac{(D^2U)_2}{(DU)_2}, \end{aligned}$$

and if from the four equations we eliminate a, b, c , we find

$$\begin{vmatrix} XDU & YDU & ZDU & D^2U \\ x & y & z & 2(m-2) \\ x_1 & y_1 & z_1 & (D^2U)_1/(DU)_1 \\ x_2 & y_2 & z_2 & (D^2U)_2/(DU)_2 \end{vmatrix} = 0$$

for the equation of the conic in question; (x, y, z) being the coordinates of the point of contact, and (X, Y, Z) current coordinates.

II. DEMONSTRATION OF IDENTITIES ASSUMED IN THE PRECEDING SECTION.

Proof of the expressions for $(d_1^3 + 3d_1d_2)U$ and $(d_1^4 + 6d_1^2d_2 + 4d_1d_3 + 3d_2^2)U$:

17. It will be remembered that d_1, d_2 stand originally for

$$\begin{aligned} dx \partial_x + dy \partial_y + dz \partial_z, \\ d^2x \partial_x + d^2y \partial_y + d^2z \partial_z, \end{aligned}$$

and that A, B, C being the first derived functions of U , dx, dy, dz are changed into

$$B\nu - C\mu, \quad C\lambda - A\nu, \quad A\mu - B\lambda,$$

and that the resulting value of d_1 , viz.

$$(B\nu - C\mu)\partial_x + (C\lambda - A\nu)\partial_y + (A\mu - B\lambda)\partial_z$$

is also represented by $\mathfrak{D}$, so that $d_1 = \mathfrak{D}$. The corresponding values of d^2x, d^2y, d^2z are

$$\begin{aligned} d^2x &= \nu dB - \mu dC, \\ d^2y &= \lambda dC - \nu dA, \\ d^2z &= \mu dA - \lambda dB, \end{aligned}$$

where we have

$$dA = a dx + h dy + g dz, \quad \&c.,$$

in which dx, dy, dz are to be replaced by the values

$$B\nu - C\mu, \quad C\lambda - A\nu, \quad A\mu - B\lambda;$$

and d_2 really denotes what

$$d^2x \partial_x + d^2y \partial_y + d^2z \partial_z$$

becomes when the above values are substituted for d^2x, d^2y, d^2z . But in the expressions d_1^2U, d_1d_2U &c., the symbols $\partial_x, \partial_y, \partial_z$ contained in d_1 and d_2 operate only on U , and not on the variable quantities A, B, C , &c. contained in d_1 and d_2 .

18. If now we treat d_2 as an operand, that is, perform the differentiations on the variable quantities A, B, C which enter into d_2 , we obtain

$$d_2(d_1U) = d_2(d_1U) + (d_2d_1)U,$$

or, what is the same thing, operating on d_1U with d_2 , the result is

$$d_1d_2U + (d_2d_1)U,$$

and in like manner

$$d_1^2U = d_1^2U + 2(d_2d_1)U + (d_3d_1)U.$$

It is, in fact, upon these principles that Hesse's values of d_1^2U , d_1^3U , &c. were obtained, and we may by means of them obtain the other expressions assumed in the preceding section.

19. In fact, starting from Hesse's equation,

$$d_1^2U = P_2U - Q_2\varpi^{m-2},$$

we have

$$\frac{1}{m-1}(d_1^3+3d_1d_2)U = -\frac{1}{(m-1)^2}(U\mathfrak{D}\Phi+\Phi\mathfrak{D}U)-\frac{1}{(m-1)^2}(DH\cdot\Phi+H\cdot\mathfrak{D}\Phi).$$

But we have identically $\mathfrak{D}U = 0$, $\mathfrak{D}^2U = 0$, and this equation becomes therefore

$$\frac{1}{m-1}(d_1^3+3d_1d_2)U = -\frac{1}{(m-1)^2}(U\mathfrak{D}\Phi+\Phi\cdot 0)-\frac{1}{(m-1)^2}(DH\cdot\Phi+H\cdot\mathfrak{D}\Phi);$$

this is precisely Hesse's value of d_1^3U , and thus we have $\mathfrak{D}\Delta U = 0$, and therefore

$$\frac{1}{m-1}(d_1^3+3d_1d_2)U = -\frac{1}{(m-1)^2}H\cdot\mathfrak{D}\Phi.$$

20. In like manner, starting from the expression of d_1^3U , we have

$$\frac{1}{m-1}(d_1^4+6d_1^2d_2+4d_1d_3+3d_2^2)U = [\text{expression involving } \mathfrak{D}^2\Phi, \Omega, \Psi],$$

or since DU and D^2U vanish identically, and the values of $D\cdot D\Phi$, $D\cdot DH$ are $d_1^3 + D^2\Phi$, $d_1\cdot H + D^2H$, we have

$$[\text{complicated expression}]$$

and if from the double of this equation we subtract Hesse's equation,

$$\frac{1}{m-1}d_1^4U = P_4U - Q_4\varpi^{m-4},$$

we find the required relation,

$$\frac{1}{m-1}(d_1^4 + 6d_1^2d_2 + 4d_1d_3 + 3d_2^2)U = -\frac{3}{(m-1)^2}\Omega \cdot \Phi + \frac{4}{(m-1)^2}\Psi.$$

Proof of the expression for $\mathfrak{D}^2 H$:

21. We have

$$\mathfrak{D}^2 H = \partial_x H \cdot (\nu dB - \mu dC) + \partial_y H \cdot (\lambda dC - \nu dA) + \partial_z H \cdot (\mu dA - \lambda dB),$$

where

$$dA = a dx + h dy + g dz,$$

$$dB = h dx + b dy + f dz,$$

$$dC = g dx + f dy + c dz,$$

in which dx, dy, dz are to be replaced by their values: we have therefore

$$\begin{aligned} \mathfrak{D}^2 H = \partial_x H \cdot \{ & \nu[(bC - fB)\lambda + (fA - hC)\mu + (hB - bA)\nu] \\ & - \mu[(fC - cB)\lambda + (cA - gC)\mu + (gB - fA)\nu] \} + \&c., \end{aligned}$$

where the coefficient of $\partial_x H$ is

$$\begin{aligned} \mathfrak{A}\lambda^2 + (gC - cA)\mu^2 + (hB - bA)\nu^2 - (2fA - gB - hC)\mu\nu \\ + (bC - fB)\nu\lambda + (cB - fC)\lambda\mu; \end{aligned}$$

or, since we have

$$(m-1)A = ax + hy + gz,$$

$$(m-1)B = hx + by + fz,$$

$$(m-1)C = gx + fy + cz,$$

the coefficient, omitting the factor $\frac{1}{(m-1)^2}$, which will be afterwards restored, is

$$\begin{aligned} & + [g(gx + fy + cz) - c(ax + hy + gz)]\mu^2 \\ & + [h(hx + by + fz) - b(ax + hy + gz)]\nu^2 \\ & + [2f(ax + hy + gz) - g(hx + by + fz) - h(gx + fy + cz)]\mu\nu \\ & + [b(gx + fy + cz) - f(hx + by + fz)]\nu\lambda \\ & + [c(hx + by + fz) - f(gx + fy + cz)]\lambda\mu \\ & = \mathfrak{A}\lambda^2 + (-\mathfrak{G}x + \mathfrak{B}y)\mu^2 + (-\mathfrak{F}x + \mathfrak{C}z)\nu^2 \\ & \quad + (-2\mathfrak{G}x + \mathfrak{B}y + \mathfrak{C}z)\mu\nu + (-\mathfrak{F}x + \mathfrak{B}z)\nu\lambda + (-\mathfrak{G}x + \mathfrak{A}y)\lambda\mu, \end{aligned}$$

which is equal to

$$-(\mathfrak{A}\mu\nu+\mathfrak{B}\nu\lambda+\mathfrak{C}\lambda\mu+2\mathfrak{F}\lambda\mu+2\mathfrak{G}\mu\nu+2\mathfrak{H}\nu\lambda)x+(\mathfrak{A}\lambda+\mathfrak{B}\mu+\mathfrak{C}\nu)(\lambda x+\mu y+\nu z).$$

The coefficients of $\partial_y H$, $\partial_z H$ have a similar form ; and uniting the three terms, observing that $x\partial_x H + y\partial_y H + z\partial_z H = 3(m-2)H$, and attending to the values of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, we have, restoring the omitted factor $\frac{1}{(m-1)^2}$,

$$\mathfrak{D}^2 H = -\frac{1}{(m-1)^2}(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) (\partial_x, \partial_y, \partial_z)^2 H + \frac{3(m-2)}{(m-1)^2} H \cdot \Phi.$$

22. These are obtained (for the particular case $m = 4$, which makes but little difference) in Mr Salmon's *Higher Plane Curves*, pp. 88 and 89, and I merely reproduce his investigation ; we have

$$(\mathfrak{D}H)^2 = \{(B\nu - C\mu)\partial_x H + (C\lambda - A\nu)\partial_y H + (A\mu - B\lambda)\partial_z H\}^2,$$

or, what is the same thing,

$$(\mathfrak{D}H)^2 = \{\lambda(C\partial_y - B\partial_z)H + \mu(A\partial_z - C\partial_x)H + \nu(B\partial_x - A\partial_y)H\}^2;$$

and if we consider first the term which contains λ^2 , the coefficient is

$$\{(C\partial_y - B\partial_z)H\}^2.$$

Now making use of the equations

$$(m-1)A = ax + hy + gz,$$

$$(m-1)B = hx + by + fz,$$

$$(m-1)C = gx + fy + cz,$$

and

$$m(m-1)U = ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0,$$

we have

$$\begin{aligned} (m-1)^2 C^2 &= (gx + fy + cz)^2 - c(ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy) \\ &= -\mathfrak{A}x^2 + 2\mathfrak{H}xy - \mathfrak{B}y^2, \end{aligned}$$

$$\begin{aligned} (m-1)^2 BC &= (hx + by + fz)(gx + fy + cz) - f(ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy) \\ &= \mathfrak{H}x^2 - \mathfrak{G}xy - \mathfrak{F}xz + \mathfrak{C}yz, \end{aligned}$$

$$\begin{aligned} (m-1)^2 B^2 &= (hx + by + fz)^2 - b(ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy) \\ &= -\mathfrak{G}x^2 + 2\mathfrak{F}xz - \mathfrak{C}z^2; \end{aligned}$$

and hence

$$\begin{aligned} (m-1)^2\{(C\partial_y - B\partial_z)H\}^2 &= (-\mathfrak{A}x^2 + 2\mathfrak{H}xy - \mathfrak{B}y^2)(\partial_y H)^2 \\ &\quad + 2(-\mathfrak{G}x^2 + \mathfrak{H}xy + \mathfrak{F}xz - \mathfrak{C}yz)\partial_y H \cdot \partial_z H \\ &\quad + (-\mathfrak{G}x^2 + 2\mathfrak{F}xz - \mathfrak{C}z^2)(\partial_z H)^2, \end{aligned}$$

which is equal to

$$\begin{aligned} &-x^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_x H, \partial_y H, \partial_z H)^2 \\ &\quad + 2x(y\partial_y H + z\partial_z H)(\mathfrak{H}\partial_y H + \mathfrak{G}\partial_z H) \\ &\quad - \mathfrak{A}(y\partial_y H + z\partial_z H)^2, \end{aligned}$$

and

$$y\partial_y H + z\partial_z H = 3(m-2)H - x\partial_x H,$$

so that the term is

$$\begin{aligned} &+ 2x\{\mathfrak{H}(3(m-2)H - x\partial_x H) + \mathfrak{G}\partial_z H\} \\ &\quad - \mathfrak{A}\{3(m-2)H - x\partial_x H\}^2; \end{aligned}$$

or, reducing, we have

$$\begin{aligned} (m-1)^2\{(C\partial_y - B\partial_z)H\}^2 &= \\ &x^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_x H, \partial_y H, \partial_z H)^2 \\ &\quad + 6(m-2)Hx(\mathfrak{H}\partial_y H + \mathfrak{G}\partial_z H) - 9(m-2)^2\mathfrak{A}H^2. \end{aligned}$$

23. The other terms may be obtained in a similar manner; and it is easy to see that, collecting all the terms, the sum will be

$$\begin{aligned} &-(\lambda x + \mu y + \nu z)^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_x H, \partial_y H, \partial_z H)^2 \\ &+ 3(m-2)H(\lambda x + \mu y + \nu z)[(\mathfrak{A}\lambda + \mathfrak{H}\mu + \mathfrak{G}\nu)\partial_x H + (\mathfrak{H}\lambda + \mathfrak{B}\mu + \mathfrak{F}\nu)\partial_y H + (\mathfrak{G}\lambda + \mathfrak{F}\mu + \mathfrak{C}\nu)\partial_z H \\ &\quad - 9(m-2)^2H^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \lambda, \mu, \nu)^2; \end{aligned}$$

or, attending to the signification of the symbols $\mathfrak{A}$, Φ , $\mathfrak{D}$, Ω , we have

$$(m-1)^2(\mathfrak{D}H)^2 = -9(m-2)^2H^2\Phi + 3(m-2)H \cdot \Omega.$$

24. Next

$$\mathfrak{D}^2 H = \{(B\nu - C\mu)\partial_x + (C\lambda - A\nu)\partial_y + (A\mu - B\lambda)\partial_z\}^2 H,$$

or, what is the same thing,

$$\mathfrak{D}^2 H = \{\lambda(C\partial_y - B\partial_z) + \mu(A\partial_z - C\partial_x) + \nu(B\partial_x - A\partial_y)\}^2 H;$$

and if we attend first to the term which contains λ^2 , the coefficient is

$$(C\partial_y - B\partial_z)^2 H.$$

Now substituting for C, B as before, we have

$$\begin{aligned} (m-1)^2(C\partial_y - B\partial_z) &= (-\mathfrak{A}x^2 + 2\mathfrak{H}xy - \mathfrak{B}y^2)\partial_y^2 H \\ &\quad + 2(-\mathfrak{G}x^2 + \mathfrak{H}xy + \mathfrak{F}xz - \mathfrak{C}yz)\partial_y\partial_z H \\ &\quad + (-\mathfrak{G}x^2 + 2\mathfrak{F}xz - \mathfrak{C}z^2)\partial_z^2 H \\ &= -x^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})\partial_x, \partial_y, \partial_z)^2 H \\ &\quad + 2x(y\partial_y + z\partial_z)(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H - 2\mathfrak{A}(y\partial_y + z\partial_z)^2 H. \end{aligned}$$

Proof of the expressions for $(m-1)^2 R_3$, $(m-1)^2 R_4$:

26. We have

$$\begin{aligned}(m-1)^2 R_3 &= (m-1)^2 (\mathfrak{D}H) \cdot \Phi \\ &= -9(m-2)^2 H^2 \Phi + 3(m-2)H \cdot \Omega, \\ (m-1)^2 R_4 &= (m-1)^2 \mathfrak{D}^2 H \cdot \Phi + (m-1)^2 (\mathfrak{D}H)^2 - (3m-6)(m-1)^2 \Phi H \\ &= -12(m-2)^2 H^2 \Phi + 4(m-2)\Omega^2,\end{aligned}$$

which are the expressions required.

Proof of corresponding expressions for $(m-1)^2 Q_3$, $(m-1)^2 Q_4$:

27. We have

$$\begin{aligned}(m-1)^2 Q_3 &= (m-1)^2 (\mathfrak{D}H) \cdot \Phi \\ &= -9(m-2)^2 H^2 \Phi + 3(m-2)H \cdot \Omega, \\ (m-1)^2 Q_4 &= (m-1)^2 \mathfrak{D}^2 H \cdot \Phi - (m-1)^2 \Omega \cdot \Phi + (3m-6)(m-1)H \cdot \Omega \\ &= -6(m-2)(m-3)H^2 \Phi + 2(m-3)\Omega^2,\end{aligned}$$

to which I join

$$(m-1)^2 Q_2 = H.$$

28. We have consequently

$$(m-1)^2 (3Q_2 Q_4 - Q_3^2) = -9(m-2)(m-3)H^2 \Phi + 3(m-4)H\Omega^2 + (-3\Omega H + \Psi)\Phi^2;$$

and

$$(m-1)^2 (3(m-2)Q_2 Q_4 - 2(m-3)Q_3^2) = (-3(m-2)H\Phi + 2(m-3)\Omega^2)\Psi,$$

which for $m = 4$ becomes

$$729(3Q_2 Q_4 - Q_3^2) = (-3H\Phi + \Psi)\Psi.$$

29. In the case $m = 4$, we have Hesse's theorem, that the equation $3Q_2 Q_4 - Q_3^2 = 0$ gives a curve of the 14th order, which passes through the points of contact of the double tangents, viz. substituting for Ω , Ψ their values, the equation of this curve is

$$-3\Phi(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_x H, \partial_y H, \partial_z H)^2 + (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_x H, \partial_y H, \partial_z H)^2 = 0.$$

III. APPLICATION OF THE FORMULÆ TO THE CUBIC.

30. I shall apply the formulæ for the five-pointic conic to a cubic; to avoid confusion as to a numerical factor, I write U' , H' in the place of U , H , so that we have

$$\Lambda = \frac{-3(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_x \partial_y \partial_z H \cdot H}{9H^2\Phi + 4(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_x H \partial_y H \partial_z H}^2,$$

and then the equation of the five-pointic conic is

$$D^2U' - \frac{1}{2(m-2)}(DH' + \Lambda \cdot DU') DU' = 0.$$

I take as the equation of the cubic,

$$U = x^3 + y^3 + z^3 + 6lxyz = 0;$$

the formulæ Table No. 70 of my Third Memoir on Quantics, Phil. Trans., vol. CXLVI. (1856), pp. 627–647, [144], putting H for $H(U)$, give

$$H = l^4(x^3 + y^3 + z^3) - (1 + 2l^3)xyz.$$

Hence writing

$$U' = \frac{1}{3}(x^3 + y^3 + z^3 + 6lxyz),$$

the first derived functions are

$$\frac{1}{3}(3x^2 + 2lyz), \quad \frac{1}{3}(3y^2 + 2lzx), \quad \frac{1}{3}(3z^2 + 2lxy);$$

the second derived functions, or (a, b, c, f, g, h) , are (x, y, z, lx, ly, lz) , whence $W = -U$; the inverse coefficients $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$ are

$$(yz - l^2x^2, \quad zx - l^2y^2, \quad xy - l^2z^2, \quad l^2yz - lx^2, \quad l^2zx - ly^2, \quad l^2xy - lz^2);$$

and putting $U' = \frac{1}{3}U$ and $H' = -H$, we have

$$\Lambda = -\frac{1}{9} \cdot \frac{-3(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_x \partial_y \partial_z H \cdot H}{9H^2\Phi + 4(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_x H \partial_y H \partial_z H}^2,$$

and the equation of the five-pointic conic is

$$D^2U - \frac{1}{2}(DH + 3\Lambda \cdot DU) DU = 0.$$

31. We have

$$\begin{aligned}
 & (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_x, \partial_y, \partial_z)^2 H \\
 &= (yz - l^2 x^2) \cdot 6lx + (zx - l^2 y^2) \cdot 6ly + (xy - l^2 z^2) \cdot 6lz \\
 &+ 2(l^2 yz - lx^2) \cdot (-(1 + 2l^3)x) + 2(l^2 zx - ly^2) \cdot (-(1 + 2l^3)y) \\
 &\quad + 2(l^2 xy - lz^2) \cdot (-(1 + 2l^3)z) \\
 &= 18l^3 xyz - 6l^2(x^3 + y^3 + z^3) - 6l(1 + 2l^3)xyz + 2l^2(1 + 2l^3)(x^3 + y^3 + z^3) \\
 &= (12l^6 - 12l^3)xyz + (2l - 2l^4)(x^3 + y^3 + z^3) \\
 &= 2(l - l^4)(x^3 + y^3 + z^3 + 6lxyz),
 \end{aligned}$$

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_x, \partial_y, \partial_z)^2 H = -2S \cdot U,$$

where S is the quartic invariant (see the Table No. 70). For the present purpose $U = 0$, and consequently

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) \partial_x, \partial_y, \partial_z)^2 H = 0.$$

34. If, however, we continue to represent the cubic by the canonical equation

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

the result may be further reduced. We have, putting $U = 0$, or writing $x^3 + y^3 + z^3 = -6lxyz$,

$$H = -(1 + 8l^3)xyz;$$

moreover, putting $U = 0$, the Table No. 70 gives

$$\Theta = (1 + 8l^3)^2(y^3z^3 + z^3x^3 + x^3y^3) - 3l^2x^2y^2z^2;$$

or substituting for H the last-mentioned value, and putting for shortness

$$Q = y^3z^3 + z^3x^3 + x^3y^3 - 3l^2x^2y^2z^2,$$

we have

$$\Theta = (1 + 8l^3)^2Q,$$

and with these values of H and Q , the equation of the five-pointic conic is

$$D^2U - \frac{1}{2} \left(DH + \frac{3\Theta}{H} \cdot DU \right) DU = 0.$$

We have

$$\begin{aligned} DU &= 3\{(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z\}, \\ D^2U &= 6\{(X^2 + 2lYZ)x + (Y^2 + 2lZX)y + (Z^2 + 2lXY)z\}, \\ DH &= (3l^2x^2 - (1 + 2l^3)yz)X + (3l^2y^2 - (1 + 2l^3)zx)Y + (3l^2z^2 - (1 + 2l^3)xy)Z; \end{aligned}$$

whence, finally, the equation of the five-pointic conic of the cubic

$$X^3 + Y^3 + Z^3 + 6lXYZ = 0$$

at the point (x, y, z) is

$$\begin{aligned} &Q(1 + 8l^3)x^2y^2z^2(x, y, z, lx, ly, lz | X, Y, Z)^2 \\ &+ \{(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z\} \times \\ &\quad \left[\frac{-x^2y^2z^2}{H} \{(3l^2x^2 - (1 + 2l^3)yz)X + \dots\} \right. \\ &\quad \left. - Q\{(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z\} \right] = 0, \end{aligned}$$

a result which I had previously obtained by a special method.

35. But the expression may be exhibited in a different form by a transformation suggested by a geometrical theorem of Mr Salmon's. In fact the tangent at the point (x, y, z) meets the cubic in the tangential of this point, and the coordinates of the tangential are $x(y^3 - z^3)$, $y(z^3 - x^3)$, $z(x^3 - y^3)$. Calling these ξ, η, ζ , the equation of the tangent to the cubic at the tangential is

$$(\eta^3 + 2l\eta\zeta)X + (\zeta^3 + 2l\zeta\xi)Y + (\xi^3 + 2l\xi\eta)Z = 0.$$

IV. INDEPENDENT INVESTIGATION FOR THE CUBIC.

47. The following is, in substance, the method by which I first obtained the equation of the five-pointic conic, for the cubic

$$X^3 + Y^3 + Z^3 + 6LXYZ = 0.$$

Write for shortness

$$\begin{aligned} U &= x^3 + y^3 + z^3 + 6lxyz, \\ V &= (x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z, \\ W &= (X^2 + 2LYZ)x + (Y^2 + 2LXZ)y + (Z^2 + 2LXY)z, \\ T &= X^3 + Y^3 + Z^3 + 6LXYZ, \\ \Pi &= aX + bY + cZ, \end{aligned}$$

then, X, Y, Z being current coordinates, and x, y, z the coordinates of a point of the cubic (so that $U = 0$), the equation of the cubic will be

$$T = 0,$$

and the equation of a conic having with it an ordinary (two-pointic) contact at the point (x, y, z) , will be

$$2VW - \Pi \cdot V = 0.$$

48–57. [The remaining articles contain the detailed verification of the formulæ by an independent method, and the extension to curves of any order, following the lines indicated in the preceding analysis. The investigations confirm the general formula for the five-pointic conic obtained in the earlier sections.]

262. On the Equation of Differences for an Equation of any Order, and in particular for the Equations of the Orders Two, Three, Four, and Five.

[From the *Philosophical Transactions* of the Royal Society of London, vol. CL. (for the year 1860), pp. 93–112. Received March 2,—Read March 29, 1860.]

The term, equation of differences, denotes the equation for the squared differences of the roots of a given equation ; the equation of differences afforded a means of determining the number of real roots, and also limits for the real roots, of a given numerical equation, and was upon this account long ago sought for by geometers. In the *Philosophical Transactions* for 1763, Waring gives, but without demonstration or indication of the mode of obtaining it, the equation of differences for an equation of the fifth order wanting the second term : the result was probably obtained by the method of symmetric functions. This method is employed in the *Meditationes Algebraicae* (1782), where the equation of differences is given for the equations of the third and fourth orders wanting the second terms; and in p. 85 the before-mentioned result for the equation of the fifth order wanting the second term, is reproduced. The formulæ for obtaining by this method the equation of differences, are fully developed by Lagrange in the *Traité des Équations Numériques* (1808); and he finds by means of them the equation of differences for the equations of the orders two and three, and for the equation of the fourth order wanting the second term ; and in Note III. he gives, after Waring, the result for the equation of the fifth order wanting the second term. It occurred to me that the equation of differences could be most easily calculated by the following method. The coefficients of the equation of differences, qua functions of the differences of the roots of the given equation, are leading coefficients of covariants, or (to use a shorter expression) they are “Seminvariants,” that is, each of them is a function of the coefficients which is reduced to zero by one of the two operators which reduce an invariant to zero. In virtue of this property they can be calculated, when their values are known for the particular case in which one of the coefficients of the given equation is zero.

To fix the ideas, let the given equation be $(a, b, \dots, v, 1)^n = 0$;

then, when the last coefficient or constant term vanishes, the equation breaks up into $v = 0$ and into an equation of the degree $n - 1$, which I call the *reduced equation*; the equation of differences will break up into two equations, one of which is the equation of differences for the reduced equation, the other is the equation for the squares of the roots of the same reduced equation.

The equations successively considered are

$$\begin{aligned}(a, b, c \wr v, 1)^2 &= 0, \\ (a, b, c, d \wr v, 1)^3 &= 0, \\ (a, b, c, d, e \wr v, 1)^4 &= 0, \\ (a, b, c, d, e, f \wr v, 1)^5 &= 0.\end{aligned}$$

The equation of differences for the quadric, and that for the squares of the roots, are considered to be known, and the other results are derived from them.

Quadric Equation.

Equation of differences is, $0 =$

$$\begin{array}{c|ccc} a^2x & +1 & ac+4 & b^2-1 \\ \hline & a^2x & +1 & ac+2 \\ & & b^2-1 & c^2+1 \end{array}$$

Cubic Equation.

Equation of differences is, $0 =$

$$\begin{array}{c|cccc} a^3x & +1 & ac+6 & a^2c^2+9 & a^3c^3+27 \\ & & b^2-2 & ab^2c-6 & b^4+1 \\ & & & ac^3+4 & a^2bcd-18 \\ & & & & b^2c^2-1 \end{array}$$

Equation for the squares of the roots is, $0 =$

$$\begin{array}{c|ccc} a^2x & +1 & ac+2 & b^2-1 \\ & & bd-2 & c^2+1 \\ & & & d^2+1 \end{array}$$

Quartic Equation.

Equation of differences is, $0 =$

$$\begin{array}{c|cccccc}
 a^4x & +1 & ac+8 & a^2c^2+8 & a^3ce+16 & a^4e^2-112 \\
 & & b^2-3 & a^2bd-2 & a^3cd^2+26 & a^3bcde+56 \\
 & & & a^2ce+22 & a^2b^2ce-6 & a^3bde^2+216 \\
 & & & b^4+3 & a^2b^2d+8 & a^3cd^3-192 \\
 & & & & ab^2c^2+8 & a^3d^4-27 \\
 & & & & ab^2cd+38 & a^2bc^2de+120 \\
 & & & & ac^4+4 & a^2cd^2e-54 \\
 & & & & b^3d-6 & ab^3ce+32 \\
 & & & & b^2c^2+2 & ab^2cd^2-25 \\
 & & & & & ab^2d^3+18 \\
 & & & & & abc^3d+42 \\
 & & & & & ac^2e+16 \\
 & & & & & b^4e+6 \\
 & & & & & b^3cd-12 \\
 & & & & & b^2cd^2-9 \\
 & & & & & b^2c^3e+6 \\
 & & & & & bc^2d^2+1
 \end{array}$$

Equation for the squares of the roots is, $0 =$

$$\begin{array}{c|cccc}
 a^2x & +1 & ac+2 & b^2-1 & \\
 & & ae+2 & bd-2 & c^2+1 \\
 & & & ce+2 & d^2-1 \\
 & & & & e^2+1
 \end{array}$$

The multiplication of the equation of differences and the equation of the squares of the roots of the quadric equation, gives the equation, $0 =$

$$\begin{array}{c|cccc}
 a^4x & +1 & ac+6 & a^2c^2+9 & a^3b^2c-6 \\
 & & b^2-2 & ab^2c-6 & b^4+1 \\
 & & & ac^3+4 & b^2c^2-1
 \end{array}$$

where all the coefficients except the last are reduced to zero by the operator

$$3a\partial_b + 2b\partial_c + c\partial_d,$$

and they are consequently (without any alteration) coefficients of the equation of differences of the cubic equation.

Quintic Equation.

The multiplication of the equation of differences, and the equation for the squares of the roots of the quartic equation, gives, in like manner, the equation of differences for the quintic equation, except as to the terms involving f ; and these are obtained as above, viz. each coefficient is of the form $L_0 + L_1f + L_2f^2 + \&c.$, where L_0 is known; such coefficient is reduced to zero by the operator

$$5a\partial_b + 4b\partial_c + 3c\partial_d + 2d\partial_e + e\partial_f,$$

or putting for shortness $V = 5a\partial_b + 4b\partial_c + 3c\partial_d + 2d\partial_e$, the operator $V + e\partial_f$. We have therefore

$$L_1 = -\frac{1}{e}VL_0, \quad L_2 = -\frac{1}{2e}VL_1, \quad L_3 = -\frac{1}{3e}VL_2, \quad \&c.$$

which give $L_1, L_2, \&c.$ by means of L_0 .

For the calculation of VL (where L is any one of the coefficients $L_0, L_1, \&c.$), it is proper to separate the terms involving the different powers of e , and write $L = M_0 + M_1e + M_2e^2 + \&c.$, $V = V' + 2d\partial_e$, where $V' = 5a\partial_b + 4b\partial_c + 3c\partial_d$. We have then

$$\begin{aligned} VL &= V'M_0 + e(V'M_1 + 2d\partial_e M_0) \\ &\quad + e^2(V'M_2 + 2d\partial_e M_1) + \&c. \end{aligned}$$

[The memoir continues with detailed calculations of each coefficient for the quintic equation, following the method explained above. The calculations involve extensive tabular arrangements showing the terms of each coefficient $L_0, L_1, L_2, \&c.$, and the application of the operator V to obtain the successive coefficients. The complete equation of differences for the quintic equation is thus obtained.]

General Equation of Differences by Elimination.

At the conclusion of the memoir, I show how the equation of differences for an equation of the order n can be obtained by the elimination of a single quantity from two equations each of the order $n-1$; and by applying to these two equations the simplification which I have made in Bezout's abridged method of elimination, I exhibit the equation of differences for the given equation of the order n , in a compendious form by means of a determinant.

10.13. On an Analytical Theorem Relating to the Distribution of
Electricity upon Spherical Surfaces 181

[End of Papers 260–262, covering pages 201–250 of Cayley’s Collected
Mathematical Papers, Volume IV.]

Mathematical Papers of Arthur Cayley,
Vol. IV

Pages 251–300

10.14 261. On the Conic of Five-Pointic Contact at any Point of a Plane Curve (concluded)

[From the *Philosophical Transactions* of the Royal Society of London, vol. CL (1860), pp. 539–578. Received Feb. 2, — Read March 22, 1860.]

(Continuation from p. 228 of this Volume)

41. I have not sought to verify this theorem by my formulae. I remark, that combining it with the before-mentioned theorem, the five-pointic conic is completely determined as follows; viz.

Theorem. *The five-pointic conic touches the conic at the point of contact (two conditions); it passes through the two points in which the polar conic is intersected by the tangent to the cubic at the tangential of the point of contact (two conditions); and it passes through the point which is the third point of intersection with the cubic of the line joining the point of contact with its second tangential.*

42. The construction for the point of simple intersection leads at once to that for the sextactic points; in fact, consider a point having for its tangential a point of inflexion: a point of inflexion is its own tangential, and the second tangential of the first-mentioned point is therefore the point of inflexion: the line joining the point with the second tangential is therefore the tangent at the point, and the point of simple intersection coincides with the point itself, that is, the point in question is a sextactic point.

43. I represent the equation of the five-pointic conic by

$$(a, b, c, f, g, h X, Y, Z)^2 = 0;$$

the value of a is

$$\begin{aligned} a = & 9(1 + 8l^3)x^2y^3z^3 \\ & + 6x^4yz^2\{3ly^2 - (1 + 2l^3)zx\}\{x^2 + 2lyz\} \\ & - 2Q(x^2 + 2lyz)^2, \end{aligned}$$

in which equation

$$Q = y^4z^2 + z^4x^2 + x^4y^2 - 2l^2x^2y^2z^2;$$

or, reducing by the equation $x^3 + y^3 + z^3 + 6lxyz = 0$,

$$Q = y^4z^2 + x^2(-x^3 - 6lxyz) - 2l^2x^2y^2z^2,$$

that is,

$$-Q = x^5y^2z + 6l^2x^3y^2z^2 + 8l^3x^2y^3z^3 - y^4z^4,$$

and we have

$$\begin{aligned} a = & 9(1 + 8l^3)x^2y^3z^3 \\ & + 6x^4yz^2\{3ly^2 - (1 + 2l^3)zx\}\{x^2 + 2lyz\} \\ & + \{x^5y^2z + 6l^2x^3y^2z^2 + 8l^3x^2y^3z^3 - y^4z^4\}(x^2 + 4xl^2yz + 4l^2y^2z^2). \end{aligned}$$

We have in like manner

$$\begin{aligned} 2f = & 18(1 + 8l^3)x^3y^2z^3 \\ & + 3x^4yz^2[\{3ly^2 - (1 + 2l^3)zx\}(z^2 + 2lxy) + \{3lz^2 - (1 + 2l^3)xy\}(y^2 + 2lzx)] \\ & - 2Q(y^2 + 2lzx)(z^2 + 2lxy); \end{aligned}$$

the coefficient of $3x^4yz^2$ in the second line is

$$= 6lyz^2 + (-1 + 4l^3)x(y^2 + z^2) - l(1 + 2l^3)x^2yz;$$

or reducing by the equation of the curve,

$$= (1 - 4l^3)x^3 + (2l - 32l^4)x^2yz + 6l^2xy^2z^2.$$

We thus have

$$\begin{aligned} 2f = & 18(1 + 8l^3)x^3y^2z^3 \\ & + 6x^4yz^2\{(1 - 4l^3)x^3 + (2l - 32l^4)x^2yz + 6l^2xy^2z^2\} \\ & + 2(x^5y^2z + 6l^2x^3y^2z^2 + 8l^3x^2y^3z^3 - y^4z^4)(-2lx^3 - 8l^2x^2yz + y^2z^2). \end{aligned}$$

Reducing the expressions of a and $2f$, we find for the coefficients (a, b, c, f, g, h) :

$$\begin{aligned} a = & x^{10} + 10lx^8yz + 40l^2x^7y^2z^2 + (5 + 120l^3)x^5y^3z^3 - 10lx^4y^4z^4 - ly^5z^5, \\ 2f = & -4lx^{10} - 40l^2x^8yz + (5 - 120l^3)x^7y^2z^2 + 40l^2x^5y^3z^3 + 8l^3x^4y^4z^4 - 2y^5z^5, \end{aligned}$$

which gives the completely developed form of the equation of the five-pointic conic.

44. I investigate the coordinates of the point of simple intersection of the cubic and the five-pointic conic as follows: the equations of the two curves are

$$X^3 + Y^3 + Z^3 + 6lXYZ = 0, \quad (a, b, c, f, g, h \ X, Y, Z)^2 = 0;$$

or if we write

$$\begin{aligned} a &= 1, & h &= c, \\ \beta &= 2lXY, & B &= 2(gX + fY), \\ \gamma &= X^2 + Y^2, & C &= aZ^2 + 2hXY + bY^2, \end{aligned}$$

then the two equations are

$$aZ^2 + \gamma + \alpha = 0, \quad AZ^2 + BZ + C = 0,$$

and the result of the elimination of Z will be

$$(aZ_1^2 + \gamma Z_1 + \alpha)(aZ_2^2 + \gamma Z_2 + \alpha) = 0,$$

where Z_1, Z_2 are the roots of the equation $AZ^2 + BZ + C = 0$; that is, we have

$$\begin{aligned} \frac{B^2 - 4AC}{A^2} &= \frac{\gamma^2 - 4a\alpha}{a^2}, \\ \frac{C(B^2 - 2AC)}{A^3} &= \frac{\gamma(\gamma^2 - 2a\alpha)}{a^3}, \\ \frac{C^2}{A^4} &= \frac{\gamma^2\alpha^2}{a^4}, \end{aligned}$$

and the result is

$$a^2C^2 + a\gamma A(2AC - B^2) + A^2(\gamma^2 - 4a\alpha)(\gamma^2 - a\alpha) = 0.$$

45. But the result of the elimination must obviously be

$$(Xy_1 - Yx_1)(Xy_2 - Yx_2) = 0,$$

if (x_1, y_1, z_1) are the coordinates of the point of simple intersection. Comparing the two results, and forming the analogous third equation, we may write

$$\begin{aligned} a^4x_1 &= b^2 + c^2 - 8f^2 + 6bcf, \\ y_1^4 &= c^2 + a^2 - 8g^2 + 6cag, \\ z_1^4 &= a^2 + b^2 - 8h^2 + 6abh, \end{aligned}$$

where the value of a^4x_1 may also be written

$$(b + c - 2f)(b + c\omega - 2\omega^2f)(b + c\omega^2 - 2\omega f),$$

ω being an imaginary cube root of unity, and so for the other two terms.

I remark, that putting $x = 0$, we have

$$bY^2 + 2fYZ + cZ^2 = -y^2z^2(lY + Z^2)^2,$$

and hence writing 1 for Y , and $-1, -\omega, -\omega^2$ for Z , we have

$$\begin{aligned} b + c - 2f &= -y^2z^2(l - z^2)^2, \\ b + c\omega - 2\omega^2f &= -y^2z^2(l - \omega z^2)^2, \\ b + c\omega^2 - 2\omega f &= -y^2z^2(l - \omega^2 z^2)^2; \end{aligned}$$

and hence the product of the three factors is $-y^6z^6(l - z^2)^2(l - \omega z^2)^2(l - \omega^2 z^2)^2$, which is equal to $-y^6z^6(l - z^2)^2(l + z^2)^2$, which vanishes in virtue of the assumed equation $x = 0$. This shows that the function $b^2 + c^2 - 8f^2 + 6bcf$ contains the factor x . I have not verified *a posteriori*, but I assume it to be true, that it contains in fact the factor a^4 , and consequently that the expressions for x_1, y_1, z_1 are rational and integral functions of (x^2, y^2, z^2) of the degree 25, and containing respectively the factors x, y, z .

46. In the theory of the cubic, a point which depends linearly upon a given point may be termed a *derivative* of such point. According to a very beautiful theorem of Professor Sylvester's, the coordinates of a derivative point are necessarily rational and integral functions of a square degree of the coordinates (x, y, z) of the given point; and moreover, there is but one derivative point having its coordinates of any given square degree m^2 , or, as we may express it, only one derivative point of the degree m^2 . The successive tangentials are derivative points of the degrees 4, 16, 64, &c.; the third point of intersection with the cubic, of the line joining two derivative points of the degrees m^2 and n^2 respectively, is a derivative point of the degree $(m + n)^2$. Thus the third point of intersection with the cubic, of the line joining the given point with its second tangential, is a derivative point of the degree $(4 + 1)^2$, and it is easy to see that the degree is not 9; it is therefore 25. The point of simple intersection of the five-pointic conic is a derivative point of the degree 25; it is therefore, according to Professor Sylvester's general theory, identical with the point given by the former construction; this agrees with the before-mentioned theorem of Mr Salmon.

IV. Independent investigation for the Cubic

47. The following is, in substance, the method by which I first obtained the equation of the five-pointic conic, for the cubic

$$X^3 + Y^3 + Z^3 + 6lXYZ = 0.$$

Write for shortness

$$U = x^3 + y^3 + z^3 + 6lxyz,$$

$$V = (x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z,$$

$$W = (X^2 + 2lYZ)x + (Y^2 + 2lXZ)y + (Z^2 + 2lXY)z,$$

$$T = X^3 + Y^3 + Z^3 + 6lXYZ,$$

$$P = ax + by + cz,$$

$$\Pi = aX + bY + cZ;$$

then, X, Y, Z being current coordinates, and x, y, z the coordinates of a point of the cubic (so that $U = 0$), the equation of the cubic will be

$$W = 0,$$

and the equation of a conic having with it an ordinary (two-pointic) contact at the point (x, y, z) , will be²

$$2W - \Pi V = 0.$$

48. Now imagine from the point of contact lines drawn to the other four intersections of the two curves; in the case of the five-pointic conic, three of these lines will coincide with the tangent $V = 0$, and the remaining line will be the line joining the point of contact with the point of simple intersection. The equations of the lines in question can be found by Joachimsthal's theorem, viz. if (x, y, z) be the coordinates of a given point, and (X, Y, Z) current coordinates, then if in the equations of any two curves we substitute for the coordinates, $\lambda x + \mu X, \lambda y + \mu Y, \lambda z + \mu Z$, and between the equations so obtained eliminate λ, μ , the resulting equation will be that of the lines drawn from the point (x, y, z) to the points of intersection of the two curves. The point (x, y, z) is any point whatever, and it may therefore be a point of intersection, or, as in the present instance, a point of contact

²I have introduced the factor 2, to make this correspond with the form $D\Pi - \Pi DU = 0$, in the case in question, $m = 3$.

of the two curves; the only difference is, that in either case the degree of each equation as regards (λ, μ) is reduced by unity, and the degree of the resulting equation in X, Y, Z is also reduced by unity: in the case of a point of simple intersection this is the only reduction; but in the case of a point of contact, the resulting equation contains the equation of the tangent as a factor, and rejecting this factor, the reduction in degree is two units.

49. Applying the method to the two equations, $T = 0$, $2W - \Pi V = 0$, and substituting therein for the original current coordinates X, Y, Z the values $\lambda x + \mu X$, $\lambda y + \mu Y$, $\lambda z + \mu Z$, the equations become

$$\begin{aligned} \lambda^3 U + 3\lambda^2 \mu \cdot W + 3\lambda \mu^2 \cdot V + \mu^3 T &= 0, \\ 2(\lambda^2 U + 2\lambda \mu W + \mu^2 V) - (\lambda P + \mu \Pi)(\lambda V + \mu W) &= 0; \end{aligned}$$

or writing $U = 0$, and omitting from each equation the factor μ , the equations become

$$\begin{aligned} 3\lambda^2 W + 3\lambda \mu V + \mu^2 T &= 0, \\ \lambda(4 - P)V + \mu(2W - \Pi V) &= 0; \end{aligned}$$

and putting in the first equation $\lambda = 2W - \Pi V$, $\mu = -(4 - P)V$, the result of the elimination contains the factor V , rejecting which it becomes

$$3(2W - \Pi V)^2 - 3(4 - P)(2W - \Pi V)W + (4 - P)^2 VT = 0,$$

which is of the fourth degree in (X, Y, Z) , as it should be, and represents therefore the lines drawn from the point of contact to the other four points of intersection of the conic and cubic.

50. The equation may be written

$$-3(2W - \Pi V)\{(2 - P)W + \Pi V\} + (4 - P)^2 VT = 0,$$

and we obtain at once the condition that this may contain the factor V , viz. this condition is $P = 2$; and if this be satisfied the conic will have a three-pointic conic, and there will be three other points of intersection. And writing $P = 2$, and throwing out the factor V , we find

$$3\Pi^2 V^2 - 6\Pi VW + 4T = 0$$

for the equation of the lines from the point of contact to the three points of intersection. And we have now to determine Π so that the function on the left hand may divide by V^2 .

51. I simplify my original method by the use of a theorem of Mr Salmon's, viz.

$$\begin{aligned} T &= X^3 + Y^3 + Z^3 + 6lXYZ, & H &= l^2(X^3 + Y^3 + Z^3) - (1 + 2l^3)XYZ, \\ W &= (X^2 + 2lYZ)x + \dots, & H_1 &= (3l^2X^2 - (1 + 2l^3)YZ)x + \dots, \\ \vdots & & \vdots & \\ U &= x^3 + y^3 + z^3 + 6lxyz, & H_0 &= l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz, \end{aligned}$$

we have identically

$$TH_0 - UH_1^2 = WH_1 - VH_0,$$

and in the present case, since $U = 0$,

$$TH_0 = WH_1 - VH_0.$$

Hence, multiplying by H_0 and substituting this value of TH_0 , the equation becomes

$$3H_0W^2 - 6H_0\Pi VW + 4WH_1 - 4VH_0 = 0,$$

or, as we may write it,

$$V(3H_0\Pi^2 - 4H_0) + 2W(2H_1 - 3H_0\Pi) = 0;$$

and we can at once make the equation divide by V , viz. by assuming

$$\Pi = \frac{2H_1}{3H_0} + \frac{1}{2}\Lambda V,$$

where Λ is arbitrary: we have thus a four-pointic contact. And substituting for Π , and throwing out the factor V , the equation becomes

$$3\left(\frac{2H_1}{3H_0} + \frac{1}{2}\Lambda V\right)^2 - 4\frac{H_1}{H_0} + 2T \cdot \frac{1}{2}\Lambda = 0;$$

or reducing,

$$\frac{16}{3}(H_1^2 - H_0H_1) - 36H_0^2\Lambda W + 24H_0H_1\Lambda^2V + 9H_0^2\Lambda^2V^2 = 0,$$

which is the equation of the lines drawn from the point of contact to the remaining two points of intersection.

52. I write for greater convenience

$$\Lambda = \frac{\Theta}{H_0},$$

Θ being as yet indeterminate; the equation is thus reduced to

$$\frac{16}{3}(H_1^2 - H_0H_1) + 9\Theta^2V^2 - 6H_0H_1\Theta V + \Theta^2 \cdot (\text{something}) = 0,$$

and we have then to determine Θ so that the left-hand side may divide by V ; or, what is the same thing, we must determine Θ so that

$$H(H_1^2 - 3H_0J) + \frac{9}{4}\Theta V^2$$

may divide by V . This implies the existence of an identical equation,

$$H(H_1^2 - 3H_0J) + \frac{9}{4}\Theta V^2 = MU + NV,$$

which for $U = 0$ would give the decomposition in question; but I have not investigated the values of M and N . I assume at the outset $U = 0$, and putting as before

$$\Theta = \frac{2H_1}{3H_0} + \frac{1}{2}\Lambda V,$$

and writing also

$$\begin{aligned} S = & Xx\{(1 + 8l^3)x^4 + (4l + 4ll^4)x^3yz + (-2l^2 + 2l^5)y^3z^3\} \\ & + Yy\{(1 + 8l^3)y^4 + (4l + 4ll^4)y^3zx + (-2l^2 + 2l^5)z^3x^3\} \\ & + Zz\{(1 + 8l^3)z^4 + (4l + 4ll^4)z^3xy + (-2l^2 + 2l^5)x^3y^3\}, \end{aligned}$$

I remark that for $U = 0$ we have

$$-xyzH_1^2 + VS + (H_1 - 3J)\{QW - 3x^2Y^2Z^2\} = 0,$$

an equation which, observing that

$$J = -(1 + 8l^3)xyz,$$

and assuming also

$$\Theta = (1 + 8l^3)^2Q,$$

may be written

$$U(H_1^2 - 3H_0J) + \Theta V^2 = -(1 + 8l^3)V\Theta,$$

which gives the required decomposition, viz., Θ having this value, the conic will have a five-pointic contact. Reducing by the last equation, and throwing out the factor V , we find

$$-9(1 + 8l^3)V^2 - \frac{16}{3}H_1^2 + 8H_0H_1 + \Theta V^2 = 0$$

for the equation of the line joining the point of contact with the point of simple intersection. And if in this equation we write $H_1 = -(1+8l^3)xyz$, and $\Theta = (1+8l^3)^2Q$, we obtain, finally, for the equation of the line in question,

$$9x^2y^2z^2 \cdot S - 6x^4y^4z^4 \cdot Q\Theta + Q^2V^2 = 0,$$

which is the before-mentioned result.

53. It only remains to verify the assumed equation

$$-xyzH_1^2 + VS + (H_1 - 3J)\{QW - 3x^2Y^2Z^2\} = 0.$$

We may write

$$-Q = x^6 + 6lx^4yz + 12l^2x^3y^2z^2 - x^4y^4,$$

and then observing that

$$\begin{aligned} W &= xX^2 + \dots + 2lXYZ + \dots, \\ H_1 &= 3l^2xX^2 + \dots - (1 + 2l^3)XYZ - \dots, \end{aligned}$$

we find at once

$$\begin{aligned} -xyzH_1^2 &= -(1 + 8l^3)x^5y^3z^3X^2 \\ &\quad + \{2l \cdot x^7 + 11l^2 \cdot x^5yz - 3x^4y^2z^2 - 2l \cdot x^3y^3z^3\}YZ \end{aligned}$$

Next writing

$$\begin{aligned} H_1 &= (3l^2x^2 - (1 + 2l^3)yz)X + \dots, \\ V &= (x^2 + 2lyz)X + \dots, \\ S &= x\{(1 + 8l^3)x^4 + (4l + 4ll^4)x^3yz + (-2l^2 + 2l^5)y^3z^3\}X + \dots, \end{aligned}$$

and forming the expression for $-xyzH_1^2 + VS$, the coefficient of X^2 is

$$\begin{aligned} &-xyz\{3l^2x^2 - (1 + 2l^3)yz\}^2 \\ &+ (x^2 + 2lyz)x\{(1 + 8l^3)x^4 + (4l + 4ll^4)x^3yz + (-2l^2 + 2l^5)y^3z^3\}, \end{aligned}$$

which is

$$= (1 + 8l^3)\{x^6 + 6lx^4yz + 12l^2x^3y^2z^2 - x^4y^4\};$$

the coefficient of YZ is

$$\begin{aligned} & -2xyz(3ly^2 - (1 + 2l^3)zx)(3lz^2 - (1 + 2l^3)xy) \\ & + (y^2 + 2lzx)z\{(1 + 8l^3)z^4 + (4l + 4ll^4)z^3xy + (-2l^2 + 2l^5)x^3y^3\} \\ & + (z^2 + 2lxy)y\{(1 + 8l^3)y^4 + (4l + 4ll^4)y^3zx + (-2l^2 + 2l^5)x^3y^3\}, \end{aligned}$$

which is

$$= (1 + 8l^3)\{y^4z^2(x^4 + y^3) + 3l \cdot x^2y^3z^3 + 12l^2x^3yz(x^4 + y^3) - 2lx(y^4 + z^4) - 2xy^3z^3\};$$

or substituting for y^4z^2 and $y^4 + z^4$ the values $-x^4 - 6lx^3yz$ and $(x^4 + 6lx^3yz)^2 - 2y^4z^4$ respectively, this is

$$= (1 + 8l^3)(2lx^7 + 12l^2x^5yz - 3x^4y^2z^2 - 2lx^3y^3z^2).$$

We have thus

$$\begin{aligned} & -xyzH_1^2 + VS \\ & = (1 + 8l^3)\{(x^6 + 6lx^4yz + 12l^2x^3y^2z^2 - x^4y^4)X^2 \\ & \quad + (2lx^7 + 12l^2x^5yz - 3x^4y^2z^2 - 2lx^3y^3z^3)YZ\}, \end{aligned}$$

and the equation

$$-xyzH_1^2 + VS + (1 + 8l^3)\{QW - 3x^2Y^2Z^2\} = 0$$

is thus verified.

Addition. — The foregoing memoir was communicated to Mr Salmon, and I am indebted to him for two notes, containing the extension to a curve of any order, of the preceding investigation for the case of a cubic; I reproduce this extension in the following section.

V. Extension of the last preceding method to a curve of any order

54. Consider the curve of the m th order $U = 0$, and in the place of the coordinates write $\lambda x + \mu X$, $\lambda y + \mu Y$, $\lambda z + \mu Z$, where, as before, (x, y, z) are the coordinates of the point of the curve, and X, Y, Z current coordinates; the term involving λ^m vanishes, and dividing out the factor μ the equation becomes

$$\lambda^{m-1}DU + \frac{1}{2}\lambda^{m-2}\mu D^2U + \frac{1}{6}\lambda^{m-3}\mu^2 D^3U + \&c. = 0.$$

Making the like substitution in $D^2U - \Pi \cdot DU = 0$, the assumed equation of the five-pointic conic, the factor μ divides out and the equation becomes

$$\begin{aligned} \lambda^{m-2}(D^2U - \Pi \cdot DU) + \lambda^{m-3}\mu\{D^3U - \Pi \cdot D^2U\} \\ + \lambda^{m-4}\mu^2\{\tfrac{1}{2}D^4U - \Pi \cdot \tfrac{1}{6}D^3U\} + \&c. = 0, \end{aligned}$$

or, what is the same thing,

$$\lambda DU\{2(m-1) - \Pi\} + \mu\{D^2U - \Pi \cdot DU\} = 0;$$

and if from the two equations we eliminate λ, μ , the result, throwing out the factor DU , is

$$\begin{aligned} (D^2U - \Pi \cdot DU)\lambda^{m-2} - \tfrac{1}{2}\{2(m-1) - \Pi\}DU \cdot (D^2U - \Pi \cdot DU)\lambda^{m-3} \\ + \&c. (DU)^2 \dots = 0, \end{aligned}$$

where all the terms after the second contain the factor DU ; the condition in order that the equation may divide by DU , is consequently $2(m-1) - \Pi = 2$, or $\Pi = 2(m-2)$, the condition of a three-pointic contact. Substituting this value, and dividing by DU , the equation becomes

$$\begin{aligned} (D^2U - \Pi \cdot DU)\lambda^{m-2} - U(D^2U - \Pi \cdot DU)\lambda^{m-3} \\ + \tfrac{1}{2}D^2U \cdot DU(D^2U - \Pi \cdot DU)\lambda^{m-4} + \&c. = 0, \end{aligned}$$

which will be divisible by DU if $-\Pi D^2U + \tfrac{1}{3}D^3U$ is divisible by DU , and the condition for this is found to be, as before,

$$U = \tfrac{2}{3}DH + \Lambda \cdot DU,$$

where Λ is arbitrary; we have thus the conditions of a four-pointic contact.

55. Substituting this value of Π , we see that $D^2U - \tfrac{2}{3}DH \cdot D \cdot U$ divides by DU , viz. there exists an identical equation,

$$D^2U - \tfrac{2}{3}DH \cdot D^2U = IU + J \cdot DU;$$

and hence if $U = 0$,

$$\tfrac{1}{2}(D^2U - \tfrac{2}{3}DH \cdot D^2U) = J,$$

where J is a quadric function of (X, Y, Z) . I do not know the general form of this function, but Mr Salmon has obtained a result which may be generalized as follows, viz. writing for X, Y, Z the values $B\nu - C\mu, C\lambda - A\nu, A\mu - B\lambda$ (where, as before, A, B, C are the first derived functions of U and λ, μ, ν are arbitrary), the expression for J is

$$\frac{D^2U}{\frac{1}{2}(m-1)(m-2)H} - \frac{(D^2U)^2}{\{(m-1)H\}^2},$$

a formula which will be presently useful.

56. The foregoing equation may be written

$$(D^2U)^{m-2}\{-\Pi D^2U + \frac{1}{3}D^3U\} + (D^2U)^{m-3}DU\{(m-2)WD^2U - \frac{1}{2}(m-3)UD^3U - \frac{1}{6}D^4U\} + \&c. (DU)^2 \dots = 0;$$

and the term $-\Pi D^2U + \frac{1}{3}D^3U$ is equal to

$$\frac{2}{3}(D^2U - \frac{2}{3}DH \cdot D^2U) - \Lambda \cdot DU \cdot D^2U = J \cdot DU - \Lambda \cdot DU \cdot D^2U.$$

Substituting this value the equation divides by DU , and throwing out this factor it becomes

$$(D^2U)^{m-2}(J - \Lambda D^2U) + (D^2U)^{m-3}\{(m-2)WD^2U - \frac{1}{2}(m-3)UD^3U - \frac{1}{6}D^4U\} + \&c. DU = 0;$$

or observing that $\frac{1}{3}\Pi D^3U = \Pi \cdot D \cdot \frac{1}{3}D^2U + \text{term containing } DU$, this may be written

$$(D^2U)^{m-2}(\frac{1}{3}J - \Lambda D^2U) + (D^2U)^{m-3}\{\Pi \cdot D \cdot \frac{1}{3}D^2U + \frac{1}{6}D^4U\} + \&c. DU = 0.$$

57. If the equation divides by DU we shall have a five-pointic contact; the condition for this is that

$$-\Lambda(D^2U)^2 + \frac{1}{3}J \cdot D^2U + \frac{1}{3}WD^3U + \frac{1}{6}D^4U$$

may divide by DU , or more simply that

$$-\Lambda(D^2U)^2 + \frac{1}{3}J \cdot D^2U + \frac{1}{3}(\frac{2}{3}DH)D^2U - \frac{1}{6}D^4U$$

may divide by DU , or, what is the same thing, the function in question must vanish in virtue of the substitution of the values $B\nu - C\mu$,

$C\lambda - A\nu$, $A\mu - B\lambda$ in the place of X, Y, Z . The expression for J has just been given; we have besides

$$\frac{1}{3}D^2U = \frac{1}{2}(m-1)(m-2)H, \quad D^2U = -(m-1)^2H,$$

where the values of Qf , Q_2 are given (ante, No. 27); we have thus

$$\frac{1}{3}(D^2U)^2 = \frac{1}{4}(m-1)^2(m-2)^2H^2, \quad \frac{1}{3}J \cdot D^2U = \frac{1}{6}(m-1)^2(m-2)H \cdot J,$$

and consequently

$$\Lambda = \frac{(m-2)H}{4} - \frac{J}{\frac{1}{2}(m-1)(m-2)H^2},$$

which agrees with the result before obtained, and thus the present method gives the complete solution of the problem. $\square$

10.15 262. On the Equation of Differences for an Equation of Any Order, and in Particular for the Equations of the Orders Two, Three, Four, and Five

[From the *Philosophical Transactions* of the Royal Society of London, vol. CL (1860), pp. 93–112. Received March 2, — Read March 29, 1860.]

The term, *equation of differences*, denotes the equation for the squared differences of the roots of a given equation; the equation of differences afforded a means of determining the number of real roots, and also limits for the real roots, of a given numerical equation, and was upon this account long ago sought for by geometers.

In the *Philosophical Transactions* for 1763, Waring gives, but without demonstration or indication of the mode of obtaining it, the equation of differences for an equation of the fifth order wanting the second term: the result was probably obtained by the method of symmetric functions. This method is employed in the *Meditationes Algebraicae* (1782), where the equation of differences is given for the equations of the third and fourth orders wanting the second terms; and in p. 85 the before-mentioned result for the equation of the fifth order wanting the second term, is reproduced.

The formulae for obtaining by this method the equation of differences, are fully developed by Lagrange in the *Traité des Équations Numériques* (1808); and he finds by means of them the equation of differences for the equations of the orders two and three, and for the equation of the fourth order wanting the second term; and in Note III. he gives, after Waring, the result for the equation of the fifth order wanting the second term.

It occurred to me that the equation of differences could be most easily calculated by the following method. The coefficients of the equation of differences, *qua* functions of the differences of the roots of the given equation, are leading coefficients of covariants, or (to use a shorter expression) they are “Seminvariants,”³ that is, each of them is a function of the coefficients which is reduced to zero by one of the two operators which reduce an invariant to zero. In virtue of this

³The term “Seminvariant” seems to me preferable to M. Brioschi’s term “Peminvariant.”

property they can be calculated, when their values are known for the particular case in which one of the coefficients of the given equation is zero.

To fix the ideas, let the given equation be $(b \curvearrowright v, 1)^n = 0$; then, when the last coefficient or constant term vanishes, the equation breaks up into $v = 0$ and into an equation of the degree $n - 1$, which I call the *reduced equation*; the equation of differences will break up into two equations, one of which is the equation of differences for the reduced equation, the other is the equation for the squares of the roots of the same reduced equation. This hardly requires a proof; let the roots of the given equation be $\alpha, \beta, \gamma, \delta$, &c., those of the equation of differences are $(\alpha - \beta)^2, (\alpha - \gamma)^2, (\alpha - \delta)^2$, &c., $(\beta - \gamma)^2, (\beta - \delta)^2, (\gamma - \delta)^2$, &c.; but in putting the constant term equal to zero, we in effect put one of the roots, say α , equal to zero; the roots of the equation of differences thus become $\beta^2, \gamma^2, \delta^2$, &c., $(\beta - \gamma)^2, (\beta - \delta)^2, (\gamma - \delta)^2$, &c. The equation for the squares of the roots can be found without the slightest difficulty; hence if the equation of differences for the reduced equation of the order $n - 1$ is known, we can, by combining it with the equation for the squares of the roots, form the equation of differences for the given equation with the constant term put equal to zero, and thence by the above-mentioned property of the Seminvariancy of the coefficients, find the equation of differences for the given equation.

The present memoir shows the application of the process to equations of the orders two, three, four, and five: part of the calculation for the equation of the fifth order was kindly performed for me by the Rev. R. Harley. It is to be noticed that the best course is to apply the method in the first instance to the forms $(a, b, \dots b \curvearrowright v, 1)^n = 0$, without numerical coefficients (or, as they may be termed, the *denumerate* forms), and to pass from the results so obtained to those which belong to the forms $(a, b, \dots \curvearrowright v, 1)^n = 0$, or *standard* forms. The equation of differences, for $(a - \beta)^2$, &c., the coefficients of which are seminvariants, naturally leads to the consideration of a more general equation having for its roots $(\alpha - \beta)^2(\alpha - \gamma)^2(x - \delta)^2 \dots$ &c., the coefficients of which are covariants; and in fact, when, as for equations of the orders two, three, and four, all the covariants are known, such covariant equation can be at once formed from the equation of differences; for equations of the fifth order, however, where the covariants are not calculated beyond a certain degree, [they are now all calculated, see 141 and 143], only a few of the coefficients of the cova-

riant equation can be thus at once formed. At the conclusion of the memoir, I show how the equation of differences for an equation of the order n can be obtained by the elimination of a single quantity from two equations each of the order $n - 1$; and by applying to these two equations the simplification which I have made in Bezout's abridged method of elimination, I exhibit the equation of differences for the given equation of the order n , in a compendious form by means of a determinant; the first-mentioned method is, however, that which is best adapted for the actual development of the equation of differences for the equation of a given order.

The equations successively considered are

$$\begin{aligned}(a, b, c \mid v, 1)^2 &= 0, \\ (a, b, c, d \mid v, 1)^3 &= 0, \\ (a, b, c, d, e \mid v, 1)^4 &= 0, \\ (a, b, c, d, e, f \mid v, 1)^5 &= 0.\end{aligned}$$

The equation of differences for the quadric, and that for the squares of the roots, are considered to be known, and the other results are derived from them: it will be convenient to write down in the first instance the results for the quadric, the cubic, and the quartic equations, and then explain the process of obtaining them.

Results for the quadric equation

For the quadric equation,

Equation of differences is, $0 =$

$$a^2\theta + 4(ac - b^2) = 0.$$

Equation for the squares of the roots is, $0 =$

$$a^2x + (c^2 + 1) + 2(ac + 2b^2)x + (b^2 - 1) = 0.$$

For the cubic equation

Equation of differences is, $0 =$

$$a^4\theta^3 \cdot (ac - 8b^2) + a^2\theta^2 \cdot (a^2d^2 + \dots) + \dots = 0,$$

the explicit form being

$$\begin{aligned}
 0 = & a^4\theta^3 \cdot (ac - 8b^2) \\
 & + a^2\theta^2 \cdot (a^2d^2 + 27a^2b^2c + 96ab^2d - 4b^3c) \\
 & + \theta \cdot (18a^3bcd - 4a^3c^3 + 6a^2b^3d + a^2b^2c^2 - 27a^4d^2) \\
 & + (b^2c^2 - 4b^3d - 4ac^3 + 18abcd - 27a^2d^2).
 \end{aligned}$$

Equation for the squares of the roots is, $0 =$

$$\begin{aligned}
 a^3x^3 + (c^3 + 1)x^3 + (abc - 2)x^2 + (ab^2c + 4)x \\
 + (d^3 + 1) + (bd - 2) + \dots = 0.
 \end{aligned}$$

For the quartic equation

For the quartic equation,

Equation of differences is, $0 =$

$$\begin{aligned}
 a^6\theta^6 \cdot (a^2e^2 + \dots) + a^4\theta^5 \cdot (a^2bc^2e + \dots) + a^2\theta^4 \cdot (\dots) \\
 + \theta^3 \cdot (\dots) + \theta^2 \cdot (\dots) + \theta \cdot (\dots) + \text{Disc} = 0,
 \end{aligned}$$

where the discriminant is the last coefficient.

The multiplication of the equation of differences and the equation for the squares of the roots of the quadric equation, gives the equation, $0 =$

$$\begin{aligned}
 a^3\theta^3 \cdot (ac + 6) + a^4\theta^2 \cdot (a^2b^2c + 16) + \dots \\
 + (b^4d^2 - 1) + \dots,
 \end{aligned}$$

where all the coefficients except the last are reduced to zero by the operator $3a\partial_b + 2b\partial_c + c\partial_d$, and they are consequently (without any alteration) coefficients of the equation of differences of the cubic equation: the last coefficient is not reduced to zero by the operator, and requires therefore to be completed by the adjunction of the terms in d (the series, here and in every other case, is of course a finite one, the number of terms might easily be calculated *a priori*). Let the value be $L_0 + L_1d + L_2d^2 + \&c.$, we have $L_0 = +4ac^3 - 16b^2c^2$; and putting for shortness $V = 3a\partial_b + 2b\partial_c$, the operator which reduces this to zero is $V + c\partial_d$; we ought therefore to have

$$0 = VL_0 + d(VL_1 + cL_0) + d^2(VL_2 + 2cL_1) + d^3(VL_3 + 3cL_2) + \&c.,$$

and consequently

$$L_1 = -\frac{1}{c}VL_0, \quad L_2 = -\frac{1}{2c}VL_1, \quad L_3 = -\frac{1}{3c}VL_2, \quad \&c.,$$

giving

$$\begin{aligned} L_1 &= -\frac{1}{c}\{(-6 + 24)abc^2 - 4b^3c\} = -18abc + 4b^3, \\ L_2 &= -\frac{1}{2c}\{-54a^2c^3 + (36 - 36) \cdot 0 \cdot ab^2c^2\} = +27a^2, \\ L_3 &= 0, \quad \&c., \end{aligned}$$

and consequently for the last coefficient the value above written down; it will be presently seen how in more complicated cases the calculations should be arranged.

Again, multiplying together the equation of differences and the equation for the squares of the roots of the cubic equation, we obtain an equation which it is not necessary to write down, as it can be at once formed by putting $e = 0$ in the equation of differences for the quartic equation. And from the equation so obtained, by the adjunction of the terms in e , we find the equation of differences for the quartic equation, viz. each coefficient is of the form $L_0 + L_1e + L_2e^2 + \&c.$, where L_0 is known, and such coefficient is reduced to zero by the operator

$$4a\partial_b + 3b\partial_c + 2c\partial_d + d\partial_e,$$

or putting for shortness $V = 4a\partial_b + 3b\partial_c + 2c\partial_d$, the operator $V + d\partial_e$. We have therefore

$$L_1 = -\frac{1}{d}VL_0, \quad L_2 = -\frac{1}{2d}VL_1, \quad L_3 = -\frac{1}{3d}VL_2, \quad \&c.$$

It is to be observed that the last coefficient of the equation of differences is the discriminant, and that the above method of calculating the coefficients of the equation of differences, as applied to the last coefficient, is nothing else than the method of calculating the discriminant given in my Fourth Memoir on Quantics, [155].

Equation of differences for the quintic equation

The multiplication of the equation of differences, and the equation for the squares of the roots of the quartic equation, gives, in like manner, the equation of differences for the quintic equation, except as to the terms involving f ; and these are obtained as above, viz. each

coefficient is of the form $L_0 + L_1f + L_2f^2 + \&c.$, where L_0 is known; such coefficient is reduced to zero by the operator

$$5a\partial_b + 4b\partial_c + 3c\partial_d + 2d\partial_e + e\partial_f,$$

or putting for shortness $V = 5a\partial_b + 4b\partial_c + 3c\partial_d + 2d\partial_e$, the operator $V + e\partial_f$. We have therefore

$$L_1 = -\frac{1}{e}VL_0, \quad L_2 = -\frac{1}{2e}VL_1, \quad L_3 = -\frac{1}{3e}VL_2, \quad \&c.,$$

which give $L_1, L_2, \&c.$ by means of L_0 .

For the calculation of VL (where L is any one of the coefficients $L_0, L_1, \&c.$), it is proper to separate the terms involving the different powers of e , and write $L = M_0 + M_1e + M_2e^2 + \&c.$, $V = V' + 2d\partial_e$, where $V' = 5a\partial_b + 4b\partial_c + 3c\partial_d$. We have then

$$\begin{aligned} VL &= V'M_0 + e(V'M_1 + 2dM_0) \\ &\quad + e^2(V'M_2 + 4dM_1) + e^3(V'M_3 + 6dM_2) + \&c., \end{aligned}$$

or, what is the same thing,

$$VL = V'M_0 + 2dM_0 \cdot e + V'M_1 \cdot e + 4dM_1 \cdot e^2 + \&c.$$

It may be interesting to give the actual calculation of one of the coefficients, say of coefficient θ^4 (which, with coefficient θ^5 , was calculated by Mr Harley).

Calculation of coefficient θ^4 in equation of differences for the quintic equation

$$(a, b, c, d, e, f \vdash \curvearrowright v, 1)^5 = 0.$$

Calculation of L_0 .

Term	Coeff. in eq. of diff. for quartic	
a^4c^2e	-2	$5 \cdot ad$
$a^3b^2d^2$	-1	$4 \cdot be$
a^3bcd	-3	$3 \cdot cd$
a^3c^3	-8	$2 \cdot d^2$
$a^2b^2c^2$	-1	

The detailed calculations yield:

$$\begin{aligned} L_0 &= A + Be + Ce^2 + De^3 + Ee^4, \quad \text{suppose, then} \\ -D &= a^4ce + 400a^4 + \dots, \\ -E &= a^3cd^2 - 960a^3 + \dots \end{aligned}$$

Calculation of L_1 .

$$-C : \quad \text{Terms of } L_0f \text{ not involving } e.$$

$$\begin{aligned} 5a\partial_b + 4b\partial_c + 3c\partial_d : \quad & B + 3ca, \quad -f(V'B + 4dC) \\ \text{viz.} \quad & -420a^4bd + \dots \end{aligned}$$

Terms of L_1f involving e^2 .

$$5a\partial_b + 4b\partial_c + 3c\partial_d : \quad -ef(V''C + 6dD)$$

Calculation of L_2 .

$$L_2 = A + Be + Ce^2, \quad \text{suppose, where}$$

$$\begin{aligned} -A &= a^3bd^2 + 775, \quad a^3d - 500, \\ -B &= a^3bc - 700, \quad a^2b^2 + 320, \\ -C &= a^2c^2d + 825, \quad a^2bc - 145, \quad \&c. \end{aligned}$$

Calculation of L_3 (gives $L_4 = 0$).

$$L_3 = A + Be, \quad -B = \dots$$

and the required coefficient of θ^4 is

$$L_0 + L_1f + L_2f^2 + L_3f^3.$$

All the coefficients were calculated in this manner, except the last coefficient, which was deduced from the known value of the discriminant for the standard form. And we have thus the complete expression of the equation of differences for the general quintic equation under the denumerate form $(a, b, c, d, e, f \curvearrowright v, 1)^5 = 0$, viz.

Equation of differences for $(a, b, c, d, e, f \curvearrowright v, 1)^5 = 0$
--

0 =

$$\begin{aligned}
 & a^{10}\theta^{10} \cdot 100 \\
 & + a^8\theta^9 \cdot (2500ac - 5000b^2) \\
 & + a^6\theta^8 \cdot (625a^2c^2 + 12500a^2bd - 12500ab^2c + 6250b^4) \\
 & + a^4\theta^7 \cdot (\dots) \\
 & + a^2\theta^6 \cdot (\dots) \\
 & + \theta^5 \cdot (\dots) \\
 & + \theta^4 \cdot (\dots) \\
 & + \theta^3 \cdot (\dots) \\
 & + \theta^2 \cdot (\dots) \\
 & + \theta \cdot (\dots) \\
 & + \text{Disc.}
 \end{aligned}$$

It may be remarked, that if ω is an imaginary cube root of unity, then the roots of the equation $(1, 1, 1, 1, 1, 1 \mid v, 1)^5 = 0$ are $-1, \omega, \omega^2, -\omega, -\omega^2$; the differences of the roots are

$$\begin{aligned}
 & -1 - \omega, -1 - \omega^2, -1 + \omega, -1 + \omega^2, \omega - \omega^2, 2\omega, \omega + \omega^2, \omega^2 + \omega, 2\omega^2, -\omega + \omega^2 \\
 & = \omega^2, \omega, -1 + \omega, -1 + \omega^2, \omega - \omega^2, 2\omega, -1, -1, 2\omega^2, -\omega + \omega^2,
 \end{aligned}$$

and the squares of the differences are

$$\omega, \omega^2, -3\omega, -3\omega^2, -3, 4\omega^2, 1, 1, 4\omega, -3,$$

from which the equation of differences is found to be

$$(\theta^2 + \theta + 1)(\theta^2 - 3\theta + 9)(\theta^2 + 4\theta + 4)(\theta^2 - 2\theta + 1)(\theta^2 + 6\theta + 9) = 0;$$

or multiplying out, it is

$$(1, 6, 21, 46, 108, 546, 493, -1410, -567, -540, 1296 \mid \theta, 1)^{10} = 0;$$

which is what the preceding expression of the equation of differences becomes upon writing therein $a = b = c = d = e = f = 1$. Moreover, upon passing (as will presently be done) to the standard form, and then writing $a = b = c = d = e = f = 1$, all the coefficients (except the first coefficient, which is equal to unity) should become equal to zero; these two tests afford a complete verification of the result.

The following corrections have to be made in Waring's result, as given by himself and Lagrange (Waring, *Phil. Trans.* 1762).

Waring, *Meditationes Algebraicæ*, p. 85:

for $+169 G^4$ read $+196 G^4$ (in coefficient θ^2).

Lagrange, *Équations Numériques*, p. 108:

for $+1200 GE$ read $+200 GE$ (in θ^4)

for $-169 B^3 D$ read $-196 B^3 D$ (in θ^3)

for $-25 B^4$ read $+25 B^4$ (in θ^2)

for $+27 G^2 B^2$ read $-27 G^2 B^2$ (in θ^0).

It may be noticed, that if in the coefficients of the several powers of θ (as they are written down in the columns, without regarding the power of a which multiplies the entire column), we attend only to the terms independent of a , we have the series

$$1, 6b^2 - 4, 6b^3 + b, 6b^4 - 4b^2, b^5 + 1, b^5d + 8b^2, b^5f + 8b^3, \&c., \\ b^4c^2 - 3, b^4c^2d - 16b, b^4c^4 + 3b^2, \&c.,$$

the law of the first terms of which, up to the term $+16b^5$, is obvious; but the term $+16b^5$, which is the last term of this initial series, is also the first term of a terminal series, the terms of which are deduced from the coefficients in the equation of differences for the quartic equation $(a, b, c, d, e \curvearrowright v, 1)^4 = 0$, viz. these coefficients are

$$b^6\theta^6 + (b^4c^2 - 3)\theta^5 + (b^4c^2d - 16b)\theta^4 + (b^4c^4 + 3b^2)\theta^3 \\ + (b^5d + 8b^2)\theta^2 + (b^5f + 8b^3)\theta + \text{Disc},$$

and by writing b, c, d, e, f in the place of a, b, c, d, e respectively, and multiplying by a^4 , we have the above-mentioned series,

$$a^4(b^5 + 1), a^4(b^5d + 8b^2), \&c.$$

It is easy to see, *a priori*, in the case of an equation of any order, that this property holds good.

Passing to the standard forms

For the quadric $(a, b, c \curvearrowright x, y)^2 = 0$, the equation of differences is, $0 =$

$$a^2\theta + 4(ac - b^2).$$

For the cubic equation $(a, b, c, d \mid x, y)^3 = 0$, the equation of differences is, $0 =$

$$\begin{aligned} & a^4\theta^3 \cdot 27 + a^2\theta^2 \cdot (81ac - 27b^2) \\ & + \theta \cdot (18a^3bcd - 4a^3c^3 + 6a^2b^3d + a^2b^2c^2 - 27a^4d^2) \\ & + (b^2c^2 - 4b^3d - 4ac^3 + 18abcd - 27a^2d^2). \end{aligned}$$

For the quartic equation $(a, b, c, d, e \mid x, y)^4 = 0$, the equation of differences is, $0 =$

$$\begin{aligned} & a^6\theta^6 \cdot 256 + a^4\theta^5 \cdot (48a^2ce + \dots) + a^2\theta^4 \cdot (8a^3e^2 + \dots) \\ & + \theta^3 \cdot (32a^2ce + \dots) + \theta^2 \cdot (16a^2e^2 + \dots) \\ & + \theta \cdot (1152a^2ce + \dots) + 256 \cdot \text{Disc}. \end{aligned}$$

For the quintic equation $(a, b, c, d, e, f \mid x, y)^5 = 0$, the equation of differences is,

$$\begin{aligned} 0 = & a^{10}\theta^{10} \cdot 100 \\ & + a^8\theta^9 \cdot (2500ac - 5000b^2) \\ & + a^6\theta^8 \cdot (625a^2c^2 + 12500a^2bd - 12500ab^2c + 6250b^4 + \dots) \\ & + a^4\theta^7 \cdot (125a^3ce + \dots) + a^2\theta^6 \cdot (625a^4ce + \dots) \\ & + \theta^5 \cdot (2500a^3ce + \dots) + \theta^4 \cdot (\dots) \\ & + \theta^3 \cdot (\dots) + \theta^2 \cdot (\dots) + \theta \cdot (\dots) + 3125 \cdot \text{Disc}. \end{aligned}$$

[where in each column after the first the sum of the positive coefficients is equal to that of the negative coefficients, and I have inserted at the foot this sum with the sign +. A single coefficient -7500 in place of -3750 has been corrected.]

The coefficients in the preceding equations of differences are functions of the seminvariants of the quantics to which they belong; for instance, in the case of the quartic, the coefficient of θ^4 is

$$8a^2\{a^2(ac - 4bd + 3c^2) + 96(ac - b^2)\},$$

that of θ^3 is

$$32\{-13a^3(ace - ad^2 - b^2e + 2bcd - c^3) + 16a^2(ac - b^2)(ae - 4bd + 3c^2) + 128(ac - b^2)^2\},$$

and so for the other coefficients; and by replacing each seminvariant by the covariant to which it belongs, we pass from the solution of the

original problem of finding the equation for $\theta = (\alpha - \beta)^2$, to that of the problem of finding the equation for

$$\xi = \frac{(\alpha - \beta)^2}{(x - \alpha y)^2(x - \beta y)^2}.$$

The results are as follows:

For the quadric $(a, b, c, d \curvearrowright x, y)^2$, the equation in ξ is, $0 =$

$$(\xi, 4D \curvearrowright U^2, 1),$$

where U is the quadric, D the discriminant.

For the cubic $(a, b, c, d, e \curvearrowright x, y)^3$, the equation in ξ is, $0 =$

$$(\xi^3, 18H\xi^2, 81H^2\xi, 27D \curvearrowright U^6, 1)^3,$$

where U is the cubic, H the Hessian, D the discriminant.

For the quartic $(a, b, c, d, e, f \curvearrowright x, y)^4$, the equation in ξ is, $0 =$

$$48U^2\xi^5H,$$

$$32(-13U^2I + 16U^2HI + 128H^3)\xi^4,$$

$$16(-7U^4I - 288UH^2J + 256H^3I)\xi^3,$$

$$1152(-3UIJ + 2H^2J)\xi^2,$$

$$256(I^3 - 27J^2)\xi,$$

Disc,

where U is the quartic, H the Hessian, I and J the quadrinvariant and the cubinvariant respectively.

For the quintic $(a, b, c, d, e, f, g \curvearrowright x, y)^5$, the equation in θ , as far as it can be expressed in terms of known covariants, is, $0 =$

$$100U^4\xi^{10} \cdot (\text{Tab. No. 15}),$$

$$50U^3\xi^9 \cdot [U^2(\text{Tab. No. 14}) + 75(\text{Tab. No. 15})^2], \quad \&c.,$$

$$3125 \cdot \text{Discriminant} (= \text{Tab. No. 26})$$

where the Tables referred to are those in my Second Memoir on Quantics, [141; for the completed equation see post p. 261].

The form of the preceding results may be modified by writing $\theta = \xi U^2$; we have thus the equations for

$$\xi = \frac{(\alpha - \beta)^2}{(x - \alpha y)^2(x - \beta y)^2};$$

thus for example, in the case of the cubic $(a, b, c, d \mid x, y)^3$, the equation for $\xi [= a^2(\alpha - \beta)^2(x - \beta y)^2(x - \gamma y)^2(x - \alpha y)^2]$ is

$$0 = (\xi^3, 18H\xi^2, 81H^2\xi, 27DU^2 \mid U^6, 1)^3;$$

this equation may be written

$$(\xi^2 + 9H\xi)^3 + 27DU^2\xi^2 = 0;$$

or putting $\xi = \theta U$, we have

$$(\theta^2U^2 + 9H\theta U)^3 + 27DU^2 \cdot \theta^2U^2 = 0,$$

an equation the roots of which are

$$a(\alpha - \beta)(x - \beta y)^2, \quad a(\beta - \gamma)(x - \alpha y)^2, \quad a(\gamma - \alpha)(x - \beta y)^2,$$

and which leads to the formula, given in my fifth Memoir on Quantics, [156], for the solution of a cubic equation. But this decomposition of the equation in ξ is peculiar to the cubic.

Equation of differences by elimination

The equation of differences for an equation of any order may be found by the following entirely distinct method. Let the proposed equation $(\mid v, 1)^n = 0$, be for shortness represented by $\phi v = 0$, and let x, y be any two distinct roots; we have not only $\phi x = 0$, $\phi y = 0$, but also $\phi x + \phi y = 0$, $\frac{\phi x - \phi y}{x - y} = 0$. Writing $\theta = (x - y)^2$, $s = x + y$, we have

$$x = \frac{1}{2}(s + \sqrt{\theta}), \quad y = \frac{1}{2}(s - \sqrt{\theta}),$$

values which are to be substituted for x, y in the equations

$$\phi x + \phi y = 0, \quad \frac{\phi x - \phi y}{x - y} = 0.$$

We have thus two equations rational in s and θ , and the elimination between them of the quantity s leads to the required equation in θ . But it is proper to modify the form of the system; in fact the two equations are, as regards s , the first of them of the degree n , the second of the degree $n - 1$; but if we write

$$\frac{\phi x + \phi y}{2} = 0, \quad \frac{\phi x - \phi y}{x - y} = 0,$$

then each of the equations will be of the same degree $n - 1$ in s .

For instance, let $\phi v = (a, b, c, d \mid v, 1)^3$, then $x = \frac{1}{2}(s + \sqrt{\theta})$, $y = \frac{1}{2}(s - \sqrt{\theta})$; the equations $\phi x + \phi y = 0$, $\frac{\phi x - \phi y}{x - y} = 0$, are

$$\begin{aligned} s^2 \cdot a + 2s \cdot 2b + (4c + a\theta) + s(8d + 6b\theta) &= 0, \\ 3s^2 \cdot a + 2s \cdot 4b + (12c + a\theta) &= 0; \end{aligned}$$

and multiplying the first equation by 3 and the second by $-s$, adding and dividing by 2, we have an equation

$$s^2 \cdot 3b + s(12c + 4a\theta) + 12d + 9b\theta = 0.$$

The second equation and this equation may be written

$$(3a, 12b, 12c + a\theta \mid s, 1)^2 = 0, \quad (3b, 12c + 4a\theta, 12d + 9b\theta \mid s, 1)^2 = 0,$$

and the elimination of s from these equations gives the required equation in θ . The result may be obtained under either of the two forms,

$$\{a^2\theta^2 + (15ac - 27b^2)\theta - 36(bd - c^2)\}\{a^2\theta + 3(ac - b^2)\} + 3\{2ab^2 + 3(ad - bc)\}^2 = 0$$

and

$$\{4a^2\theta^2 + (24ac - 27b^2)\theta - 36(bd - c^2)\}\{a^2\theta + 12(ac - b^2)\} + 3\{ab^2 + 6(ad - bc)\}^2 = 0,$$

the expansions of which respectively coincide with the before-mentioned result.

In the case of the quartic equation $\phi v = (a, b, c, d, e \mid v, 1)^4 = 0$, we have

$$\begin{aligned} s^4 \cdot a + 4s^3 \cdot 2b + 6s^2(4c + a\theta) + 4s(8d + 6b\theta) + (16e + 24cd + a\theta^2) &= 0, \\ 4s^3 \cdot a + 6s^2 \cdot 4b + 4s(12c + a\theta) + (32d + 8b\theta) &= 0, \end{aligned}$$

from which we derive another cubic equation; the two cubic equations thus are

$$\begin{aligned} (4a, 24b, 48c + 4a\theta, 32d + 8b\theta \mid s, 1)^3 &= 0, \\ (4b, 24c + 16a\theta, 48d + 44b\theta, 32e + 48c\theta + 2a\theta^2 \mid s, 1)^3 &= 0, \end{aligned}$$

from which, if s be eliminated, we have the equation in θ .

Similarly, for the quintic equation $\phi v = (a, b, c, d, e, f \mid v, 1)^5 = 0$, we have

$$s^5 \cdot a + 5s^4 \cdot 2b + 10s^3(4c + a\theta) + 10s^2(8d + 6b\theta) + 5s(16e + 24cd + a\theta^2) + (32f + 80de + 40ce\theta + 10bd\theta^2 + a\theta^3) = 0,$$

$$5s^4 \cdot a + 10s^3 \cdot 4b + 10s^2(12c + a\theta) + 5s(32d + 8b\theta) + (80e + 40c\theta + 4a\theta^2) = 0,$$

from which we derive another quartic equation; the two equations are

$$(5a, 40b, 120c + 10a\theta, 160d + 40b\theta, 80e + 40c\theta + a\theta^2 \mid s, 1)^4 = 0, \\ (5b, 40c + 20a\theta, 120d + 100b\theta, 160e + 280c\theta + 12a\theta^2, 80f + 200d\theta + 25b\theta^2 \mid s, 1)^4 = 0,$$

from which, if s be eliminated, we have the equation in θ .

But we may apply Bezout's method to the two equations each of the order $n - 1$, which result from the equation of the n th order $\phi v = (\mid v, 1)^n = 0$. The process is as follows: Suppose, in general, that s is to be eliminated from the two equations

$$Fs = 0, \quad Gs = 0,$$

each of the order $n - 1$; it is only necessary to form the expression

$$\frac{FsGs' - Fs'Gs}{s - s'}$$

which will be a function of s, s' of the form

$$(a_{0,0}, a_{0,1}, \dots, a_{0,n-1} \mid s, 1)^{n-1} (s', 1)^{n-1},$$

where the coefficients are such that $a_{r,s} = a_{s,r}$; and by equating to zero the determinant formed with these coefficients, we have the result of the elimination.

In the present case, writing for a moment $\phi_{\frac{1}{2}}(s + \sqrt{\theta}) = A$, $\phi_{\frac{1}{2}}(s - \sqrt{\theta}) = B$, and in like manner $\phi_{\frac{1}{2}}(s' + \sqrt{\theta}) = A'$, $\phi_{\frac{1}{2}}(s' - \sqrt{\theta}) = B'$, we have

$$Fs = \frac{n}{2}(A + B) - \frac{A - B}{2\sqrt{\theta}}s, \quad Gs = \frac{A - B}{2\sqrt{\theta}}, \\ Fs' = \frac{n}{2}(A' + B') - \frac{A' - B'}{2\sqrt{\theta}}s', \quad Gs' = \frac{A' - B'}{2\sqrt{\theta}},$$

and therefore

$$\begin{aligned} & \frac{FsGs' - Fs'Gs}{s - s'} \\ &= \frac{n}{4\sqrt{\theta}(s - s')} \{(A + B)(A' - B') - (A' + B')(A - B)\} \\ & \quad - \frac{1}{4\theta} \{s(A - B)(A' - B') - s'(A' - B')(A - B)\}, \end{aligned}$$

or reducing and dividing by $s - s'$,

$$\frac{FsGs' - Fs'Gs}{s - s'} = \frac{n(AB' - A'B)}{2\sqrt{\theta}(s - s')} - \frac{(A - B)(A' - B')}{4\theta}.$$

Hence, substituting for A, B, A', B' these values, we have the expression

$$\frac{n}{2\sqrt{\theta}} \frac{AB' - A'B}{s - s'} - \frac{(A - B)(A' - B')}{4\theta},$$

which is of the form

$$(a_{0,0}, a_{0,1}, \dots, b \curvearrowright s, 1)^{n-1} (s', 1)^{n-1},$$

and equating to zero the determinant formed with the coefficients, we have an equation in θ which is the equation of differences of the given equation $\phi v = 0$. For instance, if the given equation is $\phi v = (a, b, c, d \curvearrowright v, 1)^3 = 0$, then we have

$$\begin{aligned} \phi \frac{1}{2}(s + \sqrt{\theta}) &= (a, 2b + a\sqrt{\theta}, 4c + 4b\sqrt{\theta} + a\theta, 8d + 12c\sqrt{\theta} + 6b\theta + a\theta\sqrt{\theta} \curvearrowright s, 1)^3 \\ &= (A, B, C, D \curvearrowright s, 1)^3, \\ \phi \frac{1}{2}(s - \sqrt{\theta}) &= (a, 2b - a\sqrt{\theta}, 4c - 4b\sqrt{\theta} + a\theta, 8d - 12c\sqrt{\theta} + 6b\theta - a\theta\sqrt{\theta} \curvearrowright s, 1)^3 \\ &= (A', B', C', D' \curvearrowright s, 1)^3; \end{aligned}$$

and the function in s, s' is

$$\begin{aligned} & 3(AB' - A'B)ss' \\ & + 3(AC' - A'C)(s + s') + (AD' - A'D)(s^2 + ss' + s'^2) \\ & + 9(BC' - B'C)ss' + 3(BD' - B'D)(s + s') + 3(CD' - C'D) \\ & + \frac{1}{4}(A - A', B - B', C - C', D - D' \curvearrowright s, 1)^2 (A - A', B - B', C - C', D - D' \curvearrowright s', 1)^2, \end{aligned}$$

which is equal to

$$\begin{aligned}
 & -12ab \cdot ss'(s + s') \\
 & + (-12ac - a^2\theta)(s^2 + ss' + s'^2) \\
 & + (36ac - 72b^2 + 9a^2\theta)ss' \\
 & + (24ad - 72bc + 12ab\theta)(s + s') \\
 & + 96bd - 144c^2 + (-48ac + 72b^2)\theta - 3a^2\theta^2 \\
 & + 4(3as^2 + 12bs + 12c + a\theta)(3as'^2 + 12bs' + 12c + a\theta);
 \end{aligned}$$

or reducing and dividing by 32, this is

$$\begin{aligned}
 & \{9(ac - b^2) + 3a^2\theta\}ss' + \{9(ad - bc) + 6ab\theta\}(s + s') \\
 & + 36(bd - c^2) + (-15ac + 27b^2)\theta - a^2\theta^2,
 \end{aligned}$$

the terms in s^2 , s'^2 disappearing, as they should do. Writing this under the form

$$\begin{pmatrix} 9(ac - b^2) + 3a^2\theta, & 9(ad - bc) + 6ab\theta \\ 9(ad - bc) + 6ab\theta, & 36(bd - c^2) + (-15ac + 27b^2)\theta - a^2\theta^2 \end{pmatrix} (s, 1)(s', 1),$$

and equating the determinant to zero, we have the required equation in θ : the form is that obtained by the ordinary process of applying Bezout's method to the two equations $(3a, 12b, 12c + a\theta) \curvearrowright (s, 1)^2 = 0$, $(3b, 12c + 4a\theta, 12d + 9b\theta) \curvearrowright (s, 1)^2 = 0$, being in fact the before-mentioned equation

$$\{a^2\theta^2 + (15ac - 27b^2)\theta - 36(bd - c^2)\}\{a^2\theta + 3(ac - b^2)\} + 3\{2ab^2 + 3(ad - bc)\}^2 = 0.$$

But, as already remarked, this elimination process is less convenient for the complete development of the result, than the method first explained in the present memoir.

10.16 263. Demonstration of a Theorem in Finite Differences

[From the *Philosophical Transactions* of the Royal Society of London, vol. CL (1860), pp. 321–323; printed as a note to Sir J. W. F. Herschel's Memoir "On the Formulæ investigated by Dr Brinkley for the general Term in the Development of Lagrange's Expression for the Summation of Series and for Successive Integrations," pp. 319–321.]

The formula (B) [of Sir J. W. F. Herschel's Memoir], substituting therein for Δx the value $(\frac{x+1}{x})^n$, becomes

$$[x]^n = \frac{[n-1]}{[0][x]} \cdot 0^n + \frac{[1][x-1]}{[n+1][n-2]}(n+1)^n + \dots,$$

or, as this may be written,

$$\frac{[x]^n}{[n-1]} = \frac{1}{[0][x]} \cdot 0^n + \frac{[1][x-1]}{[n+1][n-2]}(n+1)^n + \dots,$$

or, inserting a first term which vanishes except in the case $x = 0$, and which is required in order that the formula may hold good for this particular value,

$$\frac{[x-1]}{[n-1]} \cdot \frac{[0][x]}{[0][x]} n^n + \frac{[1][x-1]}{[n+1][n-2]}(n+1)^n + \dots = \frac{[x]^n}{[n-1]},$$

where the series on the right-hand side need only be continued up to the term containing $\nabla^x 0^n$ since the subsequent terms vanish. [In these formulæ and throughout the present paper $[x]$ is written to denote the factorial $[x]^x$ or $\Pi(x)$, and so in other cases.]

Now $\nabla^{-1} 0^n$, or $(\frac{1}{\nabla}) \cdot 0^n$, is equal to $[x] \times \text{coeff. } t^n \text{ in } \frac{t}{e^t-1}$, and so $\nabla^x 0^n$, or $(\frac{e^t-1}{t})^n \cdot 0^n$, is equal to $[x] \times \text{coeff. } t^n \text{ in } (\frac{e^t-1}{t})^n$.

Hence, putting $R = \frac{e^t-1}{t}$, the last-mentioned formula will be true if, as regards the term which contains t^n , we have

$$R^n = \frac{[x]}{[0][x]} + \frac{[1][x-1]}{[1][x-1]} R + \frac{[2][x-2]}{[2][x-2]} R^2 + \dots + \frac{[x][0]}{[x][0]} R^x,$$

the series on the right-hand side being continued up to the term in R^x . This formula is, in fact, true if R , instead of being restricted to denote

$\frac{e^t-1}{t}$, denotes any function whatever of the form $1 + bt + ct^2 + \&c.$, and it is true not only for the term in t^n , but for all the powers of t not higher than t^n . And, moreover, R^{-n} may denote any positive or negative integral or fractional power of R . In fact, the formula (assuming for a moment the truth of it) shows that the expansion of any power whatever of a series of the form in question, can be obtained by means of the expansions of the successive positive integer powers of the same series: the existence of such a formula (at least for negative powers) was indicated by Eisenstein, *Crelle*, t. xxxix. p. 181 (1850), and the formula itself, in a slightly different form, was obtained in a very simple manner by Professor Sylvester in his paper, "Development of an idea of Eisenstein," *Quart. Math. Journ.* t. I. p. 199 (1855); the demonstration was in fact as follows, viz. writing

$$R^n = \{1 + (R - 1)\}^n = 1 + n(R - 1) + \frac{n(n-1)}{2}(R - 1)^2 + \&c. :$$

if we attend only to the terms involving powers of t not higher than t^x , the series on the right-hand side needs only be continued up to the term involving $(R - 1)^x$, and the right side being thus converted into a rational and integral function of R , it may be developed in a series of powers of R (the highest power being of course R^x), and the coefficients of the several powers are finite series which admit of summation; this gives the required formula.

But there is an easier method; the process shows that the series on the right-hand side, continued as above up to the term involving t^x , is, as regards n , a rational and integral function of the degree x ; and by Lagrange's interpolation formula, any rational and integral function of n of the degree x , can be at once expressed in terms of the values corresponding to $x+1$ particular values of n . The investigation will be as follows:

Let R denote a series of the form $1 + bt + ct^2 + \&c.$, and let R^n denote the development of the n th power of R , continued up to the term containing t^x , the terms involving higher powers of t being rejected: R^n , R^{n-1} , R^{n-2} , &c., and generally R^s , will in like manner denote the developments of these powers up to the term involving t^x , or what is the same thing, they will be the values of R^n , corresponding to $n = 0, 1, 2, \dots x$. By what precedes R^n is a rational and integral function of n of the degree x , and it can therefore be expressed in terms of the values $R^0, R^1, R^2, \dots R^x$, which correspond to $n = 0, 1, 2, \dots x$. Let s have any one of the last-mentioned values,

then the expression

$$\frac{n(n-1)(n-2)\dots(n-x)}{s(s-1)\dots 2\cdot 1\cdot (-1)(-2)\dots(-(x-s))},$$

which, as regards n , is a rational and integral function of the degree x (the factor $n-s$, which occurs in the numerator and in the denominator being of course omitted), vanishes for each of the values $n = 0, 1, 2, \dots, x$, except only for the value $n = s$, in which case it becomes equal to unity. The required formula is thus seen to be

$$R^n = \sum_{s=0}^x \frac{n(n-1)\dots(n-x)}{s(s-1)\dots 2\cdot 1\cdot (-1)(-2)\dots(-(x-s))} R^s,$$

where the summation extends to the several values $s = 0, 1, 2, \dots, x$; or, what is the same thing, it is

$$R^n = \sum_{s=0}^x (-1)^{x-s} \frac{n(n-1)\dots(n-x)}{s!(x-s)!} R^s;$$

or changing the sign of n , it is

$$R^{-n} = \sum_{s=0}^x (-1)^{x-s} \frac{n(n+1)(n+2)\dots(n+x)}{s!(x-s)!} R^s;$$

or, as this may be written,

$$R^{-n} = \frac{1}{[n-1]} \sum_{s=0}^x \frac{(-1)^{x-s}}{[s][x-s](n+s)} R^s,$$

or substituting for s the values $0, 1, 2, \dots, x$, the formula is

$$R^{-n} = \frac{[x]}{[n-1][0][x]} R^0 + \frac{[1][x-1]}{[n-1][1][x-1](n+1)} R^1 + \frac{[2][x-2]}{[n-1][2][x-2](n+2)} R^2 + \dots$$

continued up to the term involving R^x , which is the theorem in question.

10.17 264. On an Extension of Arbogast's Method of Derivations

[From the *Philosophical Transactions* of the Royal Society of London, vol. CLI (for the year 1861), pp. 37–43. Received October 18, — Read December 13, 1860.]

Arbogast's Method of Derivations was devised by him with a view to the development of a function $\phi(a+bx+cx^2+\dots)$, but it is at least as useful for the formation of only the literal parts of the coefficients, or, what is the same thing, the combinations of a given degree and weight in the letters $(a, b, c, d, \dots)$, the weights of the successive letters being 0, 1, 2, 3, &c. Thus instead of applying the method to finding the coefficients

$$a^4, 4a^3b, 4a^3c + 6a^2b^2, \&c.,$$

we may apply it merely to finding the sets of terms

$$a^4, a^3b, a^3c, a^2b^2, \&c.$$

To derive any column from the one which immediately precedes it, we operate on a letter by changing it into its immediate successor in the alphabet, and we must in each term operate on the last letter, and also, when the last but one letter in the term is the immediate antecessor in the alphabet of the last letter (but in this case only), operate on the last but one letter. Thus a^3c gives a^3d , but a^2b^2 gives a^2bc and ab^3 , and the next succeeding column is therefore

$$a^3d \quad a^2bc \quad ab^3$$

If the series of letters is finite, and the last letter of the term is also the last letter of the series, then it is impossible to operate on the last letter of the term, but the last but one letter (when the foregoing rule applies to it) is still to be operated on; and if the rule does not apply, then the term does not give rise to a term in the succeeding column; the operations will at length terminate, and a complete series of columns be obtained. Thus, if the letters are (a, b, c, d) , and the operations are (as before) performed on a^3d , the entire series of columns is

$$\begin{array}{ccccccc}
 a^4 & a^3b & a^3c & a^3d & & & \\
 & & a^2b^2 & a^2bc & a^2bd & a^2cd & \\
 & & & ab^3 & ab^2c & ab^2d & abc^2 \\
 & & & & & abd^2 & ac^3 \\
 & & & & & b^4 & b^3c \\
 & & & & & & b^2c^2
 \end{array}$$

Nothing can be more convenient than the process when the entire series of columns is required; but it is very desirable to have a process for the formation of any column apart from the others; and the object of the present memoir is to investigate a rule for the purpose. But as Arbogast's rule, applied as above, depends upon very similar principles, I will commence by showing how this rule is to be demonstrated.

If we take any combination b^3d and operate backwards on the last letter (viz. by changing it into its immediate antecessor in the alphabet), we obtain b^3c , which is a term in the next preceding column; b^3d is therefore obtainable from a term in the next preceding column, viz. b^3c , and the process is to operate on the last letter. If, instead of b^3d , the term is abc^2 (the last letter here entering as a power), the operation backwards on the last letter gives ab^2c , which is also a term of the preceding column; and it is to be noticed that the last but one letter b is here the immediate antecessor of the last letter c (and would have been so even if b had not entered into the given term abc^2 , thus ac^2 operated on backwards would have given abc). Hence abc^2 is obtained by operating on a term in the next preceding column, viz. the term abc , but in this case the operation is performed on the last but one letter. Every term is thus obtained from the next preceding column, viz. the terms are obtained by operating on the last letter, and (when the last but one letter is the immediate antecessor in the alphabet of the last letter) then also on the last but one letter, of each term of the next preceding column, and the correctness of the rule is thus demonstrated. It is to be observed that the terms are operated upon in order, the operation on the last but one letter (when it is operated on) being made immediately after that upon the last letter of the same term, and that the terms of a column are thus obtained in the proper alphabetical order.

I pass now to the above mentioned question of the formation of a single column by itself; it will be convenient, by way of illustration, to write down the columns

a^3d^2	ad^4
abd^3	bcd^3
acd^3	b^2d^2e
b^2cd^2	c^2d^2e
bc^2d^2	c^3e^3
c^3de^2	cd^3e
	d^5

which belong to the set (a, b, c, d, e) , and are of the degree 5 and the weights 12 and 15 respectively.

Some definitions and explanations are required. I speak of the first and last letters of the set, simply as the first and the last letter: there is frequent occasion to speak of the last letter; and to avoid confusion, the last letter of a term will be spoken of as the *ultimate letter*; it is necessary also to consider the *penultimate*, *antepenultimate*, and *pro-antepenultimate* letters of the term. It will be convenient to distinguish between the ultimate letter and the *ultimate*, which may be either the ultimate letter or a power of such letter; and similarly for the penultimate, &c. Thus in the term bcd^2 , the ultimate letter is d and the ultimate is d^2 , the penultimate and the penultimate letter are each of them c ; of course the ultimate, penultimate, &c. letters are always distinct from each other. We have also to consider the pairs of letters contained in a term; cd^2e contains the pairs (c, d) , (c, e) , (d, d) , (d, e) , and so in other cases; the letters of a pair are taken in the natural order. A pair not containing either the first letter or the last letter is *expansible*; thus the set being as before (a, b, c, d, e) , the pairs (b, c) , (d, d) are expansible: they are expanded by retreating the prior and advancing the posterior letter each one step; thus the just mentioned pairs (b, c) , (d, d) , are expanded into (a, d) and (c, e) respectively.

A pair composed of two distinct non-contiguous letters is *contractible*; it is contracted by advancing the prior and retreating the posterior letter each one step: thus (a, d) , (c, e) are contractible pairs, and they give by contraction the pairs (b, c) , (d, d) respectively: the processes of expansion and contraction are obviously converse to each other.

The expression the *last expansible pair* of a term hardly requires explanation; (d, d) is the last expansible pair of the term bcd^2 , or of the term $b^2d^2e^2$, (c, d) the last expansible pair of the term c^3de (the set being always (a, b, c, d, e)), and so in all other cases. The expression

the *last expansion*, in regard to any term, means the expansion of the last expansible pair of such term.

The expansion or contraction of any pair of a term leaves unaltered as well the weight as the degree, the resulting term belongs therefore to the same column as the original one. But the effect of an expansion is to diminish, and that of a contraction to increase the alphabetical rank, or rank in the column, the ranks being reckoned as the first or lowest, second, third, and so on, up to the last or highest rank. In particular, by performing upon any term the last expansion, we diminish the rank in the column, and by a succession of last expansions we bring the term up to the head of the column. Such term is not susceptible of any further expansion; it may therefore contain the first and last letters or either of them, and it may also contain a single intermediate letter in the first power only; thus the first and last letters being a, e , the head term of a column is of the form $a^m e^n$ or $a^m c e^n$, c being some intermediate letter, and the powers a^m, e^n being both or either of them omittable. In like manner, by a succession of contractions of any term we obtain the term at the foot of the column; such term is not susceptible of any further contraction, and it must therefore be composed either of a single letter or of two contiguous letters, that is it must be of the form c^m , or of the form $c^m d^n$, where c, d are any two contiguous letters, not excluding the case where the single letter or each or either of the contiguous letters is a first or a last letter.

It is to be observed that the passage up from any term to the term at the head of the column (or top term) by means of a succession of last expansions, is a perfectly unique one; but as no selection has been made of a like unique process of contraction, this is not the case with respect to the passage down from any term to the term at the foot of the column (or bottom term) by a succession of contractions.

Every term gives by the last expansion a term above it; it can therefore be obtained from such term above it by means of a contraction. But the contraction of the upper term is by hypothesis such that, performing upon the contracted term the last expansion, we obtain the upper term; a contraction, which, performed on any term, gives a lower term which by performing upon it the last expansion reproduces the first mentioned or upper term, may be called a *reversible contraction*. And it is clear that if we perform on the top term all the reversible contractions, and on each of the resulting terms all the reversible contractions, and so on as long as the process is possible, we

obtain without repetition all the terms of the column. The column is in fact similar to a genealogical tree in the male line, each lower term issuing from a single upper term, while each upper term generates a lower term or terms, or does not generate any such term, and the top term being the common origin of the entire series. It may be added that when the order of the reversible contractions of the same term is duly fixed, the alphabetical arrangement of the terms in the column agrees with the order as of primogeniture (an elder son and his issue male preceding all the younger sons and their respective issue male) in the genealogical tree.

It only remains then to inquire under what conditions a contraction is reversible.

Now as regards any term, in order that a contraction performed on it may be reversible, it is necessary and sufficient that the pair produced by the contraction should be the last expansible pair of the contracted term. There are several cases to be considered.

First, if the contraction affects the ultimate and penultimate letters of the term: this implies that the ultimate and penultimate letters are not contiguous. Let the term terminate in $e^m g^n$, the contracted term will terminate in $e^{m+1} f^{n-1} g^n$, and if (f, g) , the pair produced by the contraction, is the last expansible pair of the contracted term; the contraction is in this case reversible. If, however, the term terminate in $e^m g^h$ ($p > 1$), the contracted term will terminate in $e^{m+1} f g^{h-1}$, and the last expansible pair is not as before (f, g) , but it is (according as $p = 2$ or $p > 2$) (g, h) or (h, h) : unless indeed h is the last letter, in which case (f, g) remains the last expansible pair of the contracted term. In the example e^m has been written, but the case $m = 1$ is not excluded; moreover, the penultimate letter e is removed three steps from the ultimate letter h , but the result would have been the same if instead of e we had had any preceding letter, or had had the letter f ; by hypothesis it is not g , the letter contiguous to the ultimate letter h . The conclusion is that a contraction on the ultimate and penultimate letters (these being non-contiguous) is reversible if the ultimate is a simple letter, or if, being a power, it is a power of the last letter.

Next let the contraction affect the ultimate and antepenultimate letters. The two letters are here separated by the penultimate letter, and are therefore not contiguous. Suppose that the termination is $f^l g^m k$ (f and g contiguous), the contracted term terminates in $f^{l+1} g^{m-1} j$, and (g, j) the pair arising from the contraction is the last expansible pair of the contracted term; the contraction is therefore

reversible. In the example f has been written, but the case $l = 1$ is not excluded; moreover the ultimate letter k is taken non-contiguous to the penultimate letter g ; but if the ultimate letter had been the contiguous letter h , the only difference is that the pair would be (g, g) , and the conclusion is not altered. But suppose the termination of the term is $e^l g^m k$ (e, g , non-contiguous), then the contracted term terminates in $e^{l+1} f^{m-1} g^m j$, where the pair arising from the contraction is (f, j) , but the last expansible pair is (g, j) ; the contraction therefore is not reversible. The case $l = 1$ is not excluded; nor is it necessary that the ultimate letter should be non-contiguous to the penultimate; if the ultimate letter had been h , the pair arising from the contraction would have been (f, g) , but the last expansible pair (g, g) , and the contraction is still not reversible. In each of the cases considered the ultimate has been a simple letter; if in the first case the ultimate had been k^p ($p > 1$), then the contracted pair would terminate in jk^{p-1} , and (according as $p = 2$ or $p > 2$) the last expansible pair would be (j, k) or (k, k) , which is not the pair (g, j) produced by the contraction; the contraction is therefore not reversible, unless indeed k is the last letter, in which case it continues reversible. In the second case the contraction, which is not reversible when the ultimate is the simple letter k , remains not reversible when the ultimate is a power of such letter.

The conclusion is that a contraction on the ultimate and antepenultimate letters is reversible, if the penultimate and antepenultimate letters are contiguous, and the ultimate is a simple letter, or if, being a power, it is a power of the last letter.

A contraction on the ultimate and pro-antepenultimate letters, or on the ultimate letter and any letter preceding the pro-antepenultimate letter, is never reversible. To show this, it will be sufficient to consider the case where the term terminates in $efgh$, the contracted term terminates in $ffgg$, the pair arising from the contraction being (f, g) , and the last expansible pair being (g, g) ; and *a fortiori*, if the letters or any of them occur as powers, or are non-contiguous.

Consider, next, a contraction on the penultimate and antepenultimate letters; this assumes that these letters are non-contiguous. Such a contraction may be reversible if only the ultimate is the last letter or a power thereof; and the condition is then similar to that in the case of the ultimate and penultimate letters; only as the penultimate cannot be a power of the last letter, it must be a simple letter. And

the conditions in order that the contraction may be reversible then are that the ultimate is the last letter or a power thereof, and the penultimate a simple letter.

The next case is that of a contraction on the penultimate and pro-antepenultimate letters. Such contraction may be reversible if the ultimate is the last letter or a power thereof; and the condition is then similar to that for the case of the ultimate and antepenultimate letters; only as the penultimate cannot be a power of the last letter, it must be a simple letter. The conditions, in order that the contraction may be reversible, are that the ultimate may be the last letter or a power thereof, the penultimate a simple letter, and the antepenultimate and pro-antepenultimate letters contiguous.

A contraction on the penultimate letter and on any letter preceding the pro-antepenultimate letter, or upon any two letters preceding the penultimate letter, is never reversible. If the ultimate, penultimate, antepenultimate and pro-antepenultimate letters are denoted by U, P, A, P' respectively, then by what precedes, the following contractions, viz. UP, UA, PA, PP' , may be reversible, and they will be so under the conditions shown in the annexed Table. It is to be noticed that the conditions for UA, PA are mutually exclusive, and consequently that the number of reversible contractions to be performed upon any term is at most 3. The Table is

U, P , non-contiguous letters.	
The ultimate a simple letter, or a power of the last letter.	P, A , contiguous letters. The ultimate a simple letter, or a power of the last letter.
P, A , non-contiguous letters.	
Penultimate a simple letter.	A, P' , contiguous letters. Penultimate a simple letter.
Ultimate the last letter, or a power thereof.	Ultimate the last letter, or a power thereof.

The contractions are to be applied in the order UP, UA, PA, PP' , but all the contracted terms originating in a prior contraction of a given term are to be exhausted before proceeding to a posterior contraction of the same term. As an example of the process, the set being (a, b, c, d, e) , I will take the column

$$\begin{aligned}
&abe^2 \\
&acde \\
&ad^2e \\
&b^2de \\
&bc^2e \\
&bcd^2 \\
&c^3d
\end{aligned}$$

The top term abe^2 is given. The contractions applicable to it are UP, UA . And UP gives acd^2 . The only contraction applicable to this is UA , giving ad^2e . And there is not any contraction applicable to ad^2e . We revert therefore to UA on abe^2 , this gives b^2de . The only applicable contraction is PA , giving bc^2e . The contractions applicable to this are UP, UA . And UP gives bcd^2e . The only contraction applicable to this is UA , giving bd^3 , and there is not any contraction applicable to bd^3 . We revert therefore to UA on bc^2e , this gives c^2de , and the only contraction applicable to this is UA , giving c^3d . There is not any contraction applicable to this, and the process is therefore complete. It may be remarked that the example presents no instance of the contraction PP' ; in fact the only terms having a pro-antepenultimate are $acde$, in which A, P' are not contiguous letters, and bcd^2e , in which the penultimate is not a simple letter, so that the contraction PP' is in each case excluded. It must be confessed that a considerable amount of practice would be required before the process could be readily made use of.

10.18 265. Addition to the Memoir on an Extension of Arbogast's Method of Derivations

[Now first published, 1891.]

The process explained in the foregoing Memoir is not easily workable, and I have in fact never used it. I have devised a new process theoretically less complete, and of a somewhat mixed character consisting, as it does, in the analytical reduction of the problem to the same problem for a smaller number of letters. I have however found it very convenient for obtaining the literal terms in the theory of the sextic, viz. where we have the seven letters (a, b, c, d, e, f, g) ; and I propose to explain the process by applying it to the determination of the terms of the discriminant, degree = 6, weight = 18.

Here taking a, b, c, d, e, f, g to denote the indices of these letters respectively in a term such as $a^\alpha b^\beta c^\gamma \dots$ (that is, writing for convenience $a, b, c, \dots$ instead of $\alpha, \beta, \gamma, \dots$) we have

$$\begin{aligned}a + b + c + d + e + f + g &= 6, \\b + 2c + 3d + 4e + 5f + 6g &= 18,\end{aligned}$$

and thence

$$6a + 5b + 4c + 3d + 2e + f = 18.$$

I separate off from the others the first three letters a, b, c . The equation gives $a = 3$ at most. And then

if $a = 3$, then $5b + 4c + 3d + 2e + f = 18$, $b = 0$, $c = 0$;

if $a = 2$, $5b + 4c + 3d + 2e + f = 6$, $b = 1$ at most;

if $b = 1$, $4c + 3d + 2e + f = 1$, $c = 0$,

if $b = 0$, $4c + 3d + 2e + f = 6$, $c = 1$ at most.

Continuing similarly for $a = 1$, $a = 0$, we build up the complete enumeration. Hence considering the several terms as arranged in alphabetical order and writing down only the factors in a, b, c , we obtain col. 1 of the following diagram;

Col. 1.	Col. 2.	Col. 3.
a^3	$(d^3)^2$	1
a^2b	cd^2f	± 7
a^2c	$(cd^2)^2$	
ab^2	$(d^2)^3$	3
abc	d^4e	
ac^2	$(d^2)^2$	
b^3	c^2de	2
b^2c	$(cd)^2$	3
bc^2	c^3f	
c^3	c^2d^2	4

We then form col. 2 by annexing to each term a term $(d^k)^\phi$ which denotes the derivative ϕ of d^k ; the value of k being such that the whole term may be of the proper degree 6, and the value of ϕ being such that the whole term may be of the proper weight 18. Thus

$$\begin{aligned} a^3d^3, & \text{ degree is } 3 + 3 = 6; \text{ weight is } 0 + 9 + 9 = 18, \\ ab^2(d^2)^2, & \text{ degree is } 1 + 2 + 3 = 6; \text{ weight is } 0 + 2 + 9 + 7 = 18, \end{aligned}$$

viz. d^2 being of weight 9, then $(d^2)^2$ is of weight $9 + 9$, and $(d^2)^2$ of weight $9 + 7$.

The derivatives of any power of d are given by those of the indefinite power d^m , which, omitting therein the several powers of d , may be tabulated thus. I remark that the table is carried up to the column 21 in order that it may be applicable to the calculation of the discriminant of the sextic, which is of the degree 6 and weight 45, and the subsidiary table of derivatives of d^m is as follows:

<i>k</i>	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
0	1																		
1		1																	
2			1																
3				1															
<i>e</i>					1														
<i>ef</i>						1													
<i>eg</i>							1												
<i>f</i>								1											
<i>fg</i>									1										
<i>g</i>										1									
<i>e</i> ²											1								
<i>e</i> ² <i>f</i>												1							
<i>e</i> ² <i>g</i>													1						
<i>e</i> <i>f</i> ²														1					
<i>e</i> <i>f</i> <i>g</i>															1				
<i>e</i> <i>g</i> ²																1			
<i>f</i> ²																	1		
<i>f</i> ² <i>g</i>																		1	
<i>f</i> <i>g</i> ²																			1
<i>g</i> ²																			
<i>e</i> ³																			

This means for instance that the derivatives of d^3 are (reading down the column headed 3): d^3 , cd^2 , bd^2 , ad^2 , c^2d , bcd , acd , b^2d , abd , a^2d , c^3 , bc^2 , ac^2 , b^2c , abc , a^2c , b^3 , ab^2 , a^2b , a^3 ; those of d^4 are (column 4): d^4 , cd^3 , bd^3 , ad^3 , c^2d^2 , bcd^2 , acd^2 , b^2d^2 , abd^2 , a^2d^2 , c^3d , bc^2d , ac^2d , b^2cd , $abcd$, a^2cd , b^3d , ab^2d , a^2bd , a^3d , c^4 , &c.; and so on; viz. to form any column we affix to the terms of the corresponding column of the subsidiary table the proper power of d , so that the degree in all the letters may be 2, 3, &c. as the case may be, using from each column of the subsidiary table only the terms which are not of too high degree; for instance for col. 4 of the derivatives of d^5 , only the terms eg , f^2 , e^2f , e^3g of the complete column eg , f^2 , e^2f , e^3g , ...

It is to be observed that these tables of the derivatives of d^2 , d^3 , &c. do not need to be actually formed; any column which is wanted can be written down at once, *currente calamo*, from the column of the subsidiary table. And if we require only the number of terms, then these numbers are at once taken out from the Subsidiary Table. Thus for the numerical column of the diagram, the Subsidiary Table, col. 9, contains but 1 term g^3 of degree 3, and thus the number set opposite to $(d^3)^2$ is 1; so col. 8 contains but 1 term fg^2 of the degree

3, and so the number opposite to $(ad^2)^2$ is 1; col. 7 contains 2 terms eg^2 , fg of degree 3, and so the number opposite to $(d^2)^3$ is 2; and so on: we have in this way the numbers 1, 1, 2, 3, 2, 3, 4, &c. the partial sums of which are 1, 1, 7, 9, 11, 13, 16, 19, 22 giving the total sum 58: viz. this is the number of the terms degree 6, weight 18, which can be formed with the seven letters (a, b, c, d, e, f, g) .

When the literal terms are required, it is proper in the first instance as a safeguard against accidental omissions in copying, to take the numbers in this manner; and we can then take out the terms themselves, viz. these are

$$\begin{array}{cccc} a^3d^3 & a^2cd^2f & a^2c^2de & a^2bcd^2g \\ a^2bd^2e & a^2c^2dg & ab^2d^2f & \dots \end{array}$$

and so on, the whole number being = 58 as just mentioned.

10.19 266. On the Equation for the Product of the Differences of All But One of the Roots of a Given Equation (commenced)

[From the *Philosophical Transactions* of the Royal Society of London, vol. CLI (for the year 1861), pp. 45–59. Received November 30, 1860, — Read January 10, 1861.]

It is easy to see that for an equation of the order n , the product of the differences of all but one of the roots will be determined by an equation of the order n , the coefficients of which are alternately rational functions of the coefficients of the original equation, and rational functions multiplied by the square root of the discriminant. In fact, if the equation be $\phi v = (a, \dots b \curvearrowright v, 1)^n = a(v - \alpha)(v - \beta) \dots$, then putting for the moment $a = 1$, and disregarding numerical factors, $\sqrt{D}$, the square root of the discriminant, is equal to the product of the differences of the roots, and $\phi' \alpha$ is equal to $(\alpha - \beta)(\alpha - \gamma) \dots$, consequently the product of the differences of the roots, all but α , is equal to $\sqrt{D} \div \phi' \alpha$, and the expression $\frac{\sqrt{D}}{\phi' \alpha}$ is the root of an equation of the order n , the coefficients of which are rational functions of the coefficients of the original equation. I propose in the present memoir to determine the equation in question for equations of the orders three, four, and five: the process employed is similar to that in my memoir “On the Equation of Differences for an Equation of any Order, and in particular for Equations of the Orders Two, Three, Four, and Five,” *Phil. Trans.*, vol. CL. (1860), [262], viz. the last coefficient of the given equation is put equal to zero, so that the given equation breaks up into $v = 0$ and into an equation of the order $n - 1$ called the reduced equation; and this being so, the required equation breaks up into an equation of the order $n - 1$ (which however is not, as for the equation of differences, that which corresponds to the reduced equation) and into a linear equation; the equation of the order $n - 1$ is calculated by the method of symmetric functions; and combining it with the linear equation, which is known, we have the required equation, except as regards the terms involving the last coefficient, which terms are found by the consideration that the coefficients of the required equation are seminvariants. The solution leads immediately to that of a more general question; for if the product of the differences

of all the roots except α , of the given equation

$$\phi v = (\psi \curvearrowright v, 1)^n = a(v - \alpha)(v - \beta) \dots = 0$$

(which product is a function of the degree $n - 2$ in regard to each of the roots $\beta, \gamma, \delta, \dots$), is multiplied by $(x - \alpha y)^{n-2}$, the function so obtained will be the root of an equation of the order n , the coefficients of which are covariants of the quantic $(\psi \curvearrowright x, y)^n$, and these coefficients can be at once obtained by writing, in the place of the seminvariants of the former result, the covariants to which they respectively belong. In the case of the quintic equation, one of these covariants is, in regard to the coefficients, of the degree 6, which exceeds the limit of the tabulated covariants, [the covariants are all tabulated, 141 and 143], the covariant in question has therefore to be now first calculated. The covariant equations for the cubic and the quartic might be deduced from the formulae Nos. 119 and 142 of my Fifth memoir on Quantics *Phil. Trans.*, vol. CXLVIII. (1858), [156]; they are in fact the bases of the methods which are there given for the solution of the cubic and the quartic equations respectively; and it was in this way that I was led to consider the problem which is here treated of.

Definitions and Notation

1. The notation $\zeta^2(\alpha, \beta, \gamma, \dots)$ is used (after Professor Sylvester) to denote the product of the squared differences of $(\alpha, \beta, \gamma, \dots)$, and the notation $\zeta^*(\alpha, \beta, \gamma, \dots)$ to denote the product of the differences taken in a determinate order, viz.

$$\begin{aligned} \zeta^*(\alpha, \beta, \gamma, \delta, \dots) &= (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta) \dots \\ &\quad (\beta - \gamma)(\beta - \delta) \dots \\ &\quad (\gamma - \delta) \dots \end{aligned}$$

2. The product of the differences of the roots of an equation depends, as already noticed, on the square root of the discriminant; and in order to fix the numerical factors and signs, it will be convenient,

in regard to the equations

$$\begin{aligned}(a, b, c \mid b \frown v, 1)^2 &= 0, \\ (a, b, c, d \mid b \frown v, 1)^3 &= 0, \\ (a, b, c, d, e \mid b \frown v, 1)^4 &= 0, \\ (a, b, c, d, e, f \mid b \frown v, 1)^5 &= 0,\end{aligned}$$

to write as follows:

$$\begin{aligned}\zeta^*(\alpha, \beta) &= \frac{1}{a} \sqrt{4ac - b^2 + \dots} = \frac{1}{a} \sqrt{Q}, \\ \zeta^*(\alpha, \beta, \gamma) &= -\frac{1}{a^2} \sqrt{27a^4 d^2 + \dots} = -\frac{1}{a^2} \sqrt{Q}, \\ \zeta^*(\alpha, \beta, \gamma, \delta) &= -\frac{1}{a^3} \sqrt{256a^6 e^3 + \dots} = -\frac{1}{a^3} \sqrt{Q}, \\ \zeta^*(\alpha, \beta, \gamma, \delta, \epsilon) &= -\frac{1}{a^4} \sqrt{3125a^8 f^4 + \dots} = -\frac{1}{a^4} \sqrt{Q},\end{aligned}$$

where it is to be observed, for example, that writing in the last equation $e = 0$, and therefore $f = 0$, we have $\zeta^*(\alpha, \beta, \gamma, \delta, 0) = \alpha\beta\gamma\delta \cdot \zeta^*(\alpha, \beta, \gamma, \delta) = \frac{1}{a^4} \sqrt{256a^6 e^3 + \dots}$, which agrees with the equation $\zeta^*(\alpha, \beta, \gamma, \delta, 0) = a\beta\gamma\delta \cdot \zeta^*(\alpha, \beta, \gamma, \delta) = \frac{1}{a^4} \sqrt{Q}$, if for $\zeta^*(\alpha, \beta, \gamma, \delta)$ we substitute the value given by the last equation but one.

Cubic equation

For the cubic equation $(a, b, c, d \mid b \frown v, 1)^3 = 0$:

3. We have to find the equation for $\theta = \zeta^*(\beta, \gamma) = \beta - \gamma$; the roots are

$$\theta_1 = \beta - \gamma, \quad \theta_2 = \gamma - \alpha, \quad \theta_3 = \alpha - \beta.$$

To apply the method above explained, write $\gamma = 0$, and therefore also $d = 0$; the roots thus become

$$\theta_1 = \beta, \quad \theta_2 = -\alpha, \quad \theta_3 = \alpha - \beta,$$

and we have the quadric and linear equations

$$(\theta + \alpha)(\theta - \beta) = 0, \quad \theta - (\alpha - \beta) = 0,$$

where (α, β) are the roots of the equation

$$(a, b, c \mid b \frown v, 1)^2 = 0.$$

Hence, writing

$$L = 4ac - b^2,$$

we have

$$\alpha - \beta = -\sqrt{\frac{L}{a^2}},$$

and the two equations become

$$\theta^2 - (\alpha - \beta)\theta - \alpha\beta = 0, \quad \theta - (\alpha - \beta) = 0,$$

or multiplying the two equations together,

$$\theta^3 - \alpha\beta \cdot \theta + \theta^2(\beta - \alpha) - \alpha\beta(\beta - \alpha) = 0,$$

which may be written

$$\theta^3 + \theta \cdot \frac{L}{a^2} + \theta \cdot \frac{M}{a^3} + \frac{N}{a^4} = 0,$$

where L, M, N are the expressions in (a, b, c, d) which give the required equation.

[End of transcription for pages 251–300]

**ON THE EQUATION FOR A FUNCTION OF THE
DIFFERENCES OF THE ROOTS OF A GIVEN
EQUATION.**

*[From the Philosophical Transactions of the Royal Society of
London, vol. CXLVII (for the year 1857), pp. 367-376. Received
December 11, 1856, — Read January 8, 1857.]*

3. The coefficient $3ac - b^2$ is reduced to zero by the operator

$$3a\partial_b + b\partial_c + c\partial_d;$$

it is therefore a seminvariant, and remains unaltered. The coefficient $c^2 - bd$ is what $\nabla - D$ becomes (D being the discriminant of the cubic equation) on putting therein $d = 0$; it is therefore to be changed into $\nabla - D$. Hence

4. For the cubic equation $(a, b, c, d \parallel v, 1)^3 = 0$, the equation for θ ($= \zeta^2(a, \beta)$) is $0 =$

$$\begin{array}{cccc} a^2\theta & & \nabla - D\theta & \\ +1 & 0 & 3ac - b^2 & +1 \end{array}$$

5. For the quartic equation $(a, b, c, d, e \parallel v, 1)^4 = 0$;

$$\theta = \zeta^2(a, \beta, \gamma) = -(a - \beta)(a - \gamma)(\beta - \gamma);$$

the roots are

$$\begin{aligned} \theta_1 &= -\zeta^2(\beta, \gamma, \alpha), \\ \theta_2 &= -\zeta^2(\gamma, \alpha, \beta), \\ \theta_3 &= -\zeta^2(\alpha, \beta, \gamma), \end{aligned}$$

the signs being in this case (and indeed for an equation of any even order) alternately positive and negative; in fact, if the equation is represented by $\phi v = 0$, then the roots divided by $2\zeta^2(\alpha, \beta, \gamma, \delta)$ should respectively be $\phi'\alpha, \phi'\beta, \phi'\gamma, \phi'\delta$, and this will be the case if the signs are taken as above.

6. Putting now $\delta = 0$ (and therefore $e = 0$) the roots become

$$\begin{aligned} \theta_1 &= \beta\gamma(\beta - \gamma), \\ \theta_2 &= \gamma\alpha(\gamma - \alpha), \\ \theta_3 &= -\alpha\beta(\alpha - \beta), \end{aligned}$$

where (α, β, γ) are the roots of $(a, b, c, d \parallel v, 1)^3 = 0$. Let Δ denote the discriminant of the cubic function, then $\zeta^2(\alpha, \beta, \gamma) = \frac{1}{a^2} \sqrt{\nabla - \Delta}$, and we have thus the linear equation; the cubic equation is

$$\theta, \{\theta - \beta\gamma(\beta - \gamma)\} = 0,$$

the coefficients of which can be calculated by the method of symmetric functions (see Annex No. 1).

7. The cubic equation being thus obtained, we have the two equations

$$\begin{array}{rcl} \theta^2 \cdot a^2 & \theta \cdot a^3 & \\ +a^2 & -a^2 \cdot \sqrt{\nabla - \Delta} & \\ & +a^2 \cdot (-a^2 d^2 + 4abcd - b^3 d) & \\ = & +a^2 d \cdot \sqrt{\nabla - \Delta} = 0; & \end{array}$$

and multiplying these together, the resultant equation is

$$\begin{array}{rcl} a^2 \theta^2 & & \\ +a^2 \cdot a^2 (-a^2 d^2 + 4abcd - b^3 d + \Delta) & & \\ -a^2 d^2 (\nabla - \Delta) = 0, & & \end{array}$$

where the coefficients have to be completed by adding the terms which contain e . We have ∇D in the place of $d^2 \sqrt{\nabla - \Delta}$, and D in the place of $-d^2 \Delta$. The coefficient $-8a^3 d + 4abc - b^3$ is a seminvariant, and requires no alteration. The coefficient

$$-9a^2 c^2 + 4abcd - 11b^2 d + \Delta$$

is

$$\begin{array}{rcl} -9a^2 c^2 + 4abcd & & \\ +27a^2 d^2 - 18abcd & & \\ \hline -a^2 d^2 + 4ac^3 & & \\ & +b^2 d & \\ & +b^2 c^2 - 1 & \end{array}$$

that is,

$$\begin{array}{rcl} abcd & & \\ ac^3 & +144 & \\ b^2 d & +3 & \\ b^2 c^2 & -1 & \end{array}$$

and the terms in e to be added to this, in order to make it a seminvariant, are easily found to be

$$\begin{array}{rcl} a^3 ce & -16 & \\ a^2 b^2 e & +6 & \end{array}$$

8. Hence, for the quartic equation $(a, b, c, d, e \parallel v, 1)^4$, the equation

for θ ($= \zeta^2(\alpha, \beta, \gamma)$) is $0 =$

$$\begin{array}{rcl} a^3\theta & & \nabla D\theta \\ +1 & 0 & +1 \\ & a^3ce-16 & a^2b^2e+6 \\ & a^2d & abc+3 \\ & & b^2 \\ ac^3+144 & b^2d & a^2d \\ & abcd & \nabla D\theta \\ & b^3+3 & -D\theta \end{array}$$

9. We have $\theta = \zeta^2(\alpha, \beta, \gamma, \delta)$, the roots being

$$\begin{aligned}\theta_1 &= \zeta^2(\beta, \gamma, \delta, \epsilon), & \text{which for } \epsilon = 0 \text{ becomes } & \beta\gamma\delta\zeta^2(\beta, \gamma, \delta), \\ \theta_2 &= \zeta^2(\gamma, \delta, \epsilon, \alpha), & & -\gamma\delta\alpha\zeta^2(\gamma, \delta, \alpha), \\ \theta_3 &= \zeta^2(\delta, \epsilon, \alpha, \beta), & & \delta\alpha\beta\zeta^2(\delta, \alpha, \beta), \\ \theta_4 &= \zeta^2(\epsilon, \alpha, \beta, \gamma), & & -\alpha\beta\gamma\zeta^2(\alpha, \beta, \gamma), \\ \theta_5 &= \zeta^2(\alpha, \beta, \gamma, \delta), & & \zeta^2(\alpha, \beta, \gamma, \delta).\end{aligned}$$

10. The linear equation is $a^4\theta + \sqrt{\nabla D} = 0$; the quartic equation may be written $\Pi_4(\theta - \theta_1) = 0$, for the determination of which see Annex No. 2. The two equations are

$$\begin{aligned}\theta^2 \cdot a^3 & & \theta \cdot a^4 \\ +a^3 & & -a^3\sqrt{\nabla D} \\ & & +a^3(a^2df + \dots) \\ = & & +a^3e\sqrt{\nabla D} = 0;\end{aligned}$$

and multiplying these together, the resulting equation is obtained; the coefficients have to be completed by adding the terms which contain f .

11. The terms in f are found to be

$$\begin{array}{r} a^3df + 40 \\ a^3e^2 \end{array} \quad \begin{array}{r} a^3cef + 88 \\ a^3de - 117 \\ a^3e^2 + 27 \end{array}$$

12. The value of $M\epsilon + M$ is

a^3ce	-16		a^3df	-50
a^3bd	$+6$	and the terms in f	a^3bcf	$+30$
a^3d^2	$+4$	to be added are	a^3bf^2	-8
a^3c^2	$+27$		a^3c^2f	$+28$

13. The value of $N + a^4e$ is

a^4e	-15
a^3bd	$+6$
a^3c^2	$+4$
a^2b^2c	-5
b^4	$+1$

which is a seminvariant, and requires no addition.

14. Hence, for the quintic equation $(a, b, c, d, e, f \parallel v, 1)^5 = 0$, the equation for $\theta (= \zeta^2(\alpha, \beta, \gamma, \delta))$ is $0 =$

$a^4\theta$		$\nabla D\theta$
$+1$	0	$+1$
$a^3ce - 16$		$a^3df + 88$
a^3d^2		$a^3de - 117$
		$a^3e^2 + 27$
$a^2c^2f + 186$		$a^2cef + 360$
$a^2cdf + 144$		$a^2cde^2 - 360$
$a^2cde - 360$		$a^2cd^2e + 360$
$a^2d^2e - 18$		$a^2d^3 + 4$
$a^2b^2f - 18$		$a^2bdf + 40$
$a^2bde + 4$		a^2be^2
$a^2be^2 + 4$		

15. As a verification of this result, taking for the quintic equation $v^5 + v^4 + v^3 + v^2 + v + 1 = 0$, the roots whereof are the sixth roots of unity (excluding 1), the equation in θ is found to be $(\theta - 6)^2(\theta + 12)(\theta + 108) = 0$, or multiplying out it is

$$(1, 0, 0, +432, -11664, +46656 \parallel \theta, 1)^5 = 0,$$

which in fact (observing that $\nabla D = 36$) is what the preceding formula becomes for the equation $(1, 1, 1, 1, 1, 1 \parallel v, 1)^5 = 0$.

16. For the cubic, if the assumed equation is $v^3 + v^2 + v + 1 = 0$, the roots whereof are $-1, i, -i$ ($i^2 = -1$), then we should have $\zeta^2(\alpha, \beta, \gamma) = \sqrt{-D} = 4i$, which will be the case if $\alpha, \beta, \gamma = -1, i, -i$, respectively, and the roots $\theta - \gamma, \gamma - \alpha, \alpha - \beta$ of the equation in θ then are $2i, -i+1, -i-1$, so that the equation in θ is $(\theta^2 + 2i\theta - 2)(\theta - 2i) = 0$, or

$$(1, 0, 2, 4i \parallel \theta, 1)^3 = 0,$$

which (observing that $\nabla - D = 4i$) is what the formula for the equation in θ becomes for the equation $(1, 1, 1, 1 \parallel v, 1)^3 = 0$.

17. For the quartic equation, taking this to be $v^4 + v^3 + v^2 + v + 1 = 0$, the roots whereof are the fifth roots of unity, we have $\zeta^2(\alpha, \beta, \gamma) = \sqrt{-D} = 5\sqrt{5}$, and the equation in θ may be verified in a similar manner.

18. For the cubic equation $(a, b, c, d \parallel v, 1)^3 = 0$, the equation for θ ($= \zeta^2(\alpha, \beta)$) is $0 =$

$$\begin{array}{rcl} a^2\theta & \nabla - 27D\theta & \\ +1 & 0 & 3ac - b^2 \\ & & +1 \end{array}$$

19. For the quartic equation $(a, b, c, d, e \parallel v, 1)^4 = 0$, the equation for θ ($= \zeta^2(\alpha, \beta, \gamma)$) is $0 =$

$$\begin{array}{rcl} a^3\theta & \nabla D\theta & \\ +1 & 0 & a^3ce - 1 \quad a^3df + 3 \\ & & a^3d^2 \quad a^3de + 1 \\ & & abcd - 14 \quad a^3e + 84 \\ & & ac^3 + 9 \quad a^2df \\ & & b^2d + 8 \quad \nabla D\theta \\ & & b^2c^2 - 6 \quad -D\theta \\ & & abc + 3 \\ & & b^3 - 2 \\ & & +1 \end{array}$$

20. For the quintic equation $(a, b, c, d, e, f \parallel v, 1)^5 = 0$, the equation for θ ($= \zeta^2(\alpha, \beta, \gamma, \delta)$) is $0 =$

$$\begin{array}{rcl} a^4\theta & \nabla D\theta & \\ +1 & 0 & a^3ce - 2 \quad a^3df - 1 \quad +1 \\ & & a^3d^2 \quad a^3de - 3 \\ & & & a^3e^2 + 4 \\ & & a^2c^2f + 24 \quad a^2cef + 360 \\ & & a^2cdf + 3 \quad a^2cde^2 - 360 \\ & & a^2cde - 32 \quad a^2cd^2e + 360 \\ & & a^2d^2e - 18 \quad a^2d^3 + 4 \\ & & a^2b^2f - 18 \quad a^2bdf + 40 \\ & & a^2bde + 4 \quad a^2be^2 \\ & & a^2be^2 + 4 \end{array}$$

21. I remark, with respect to the equation in θ , for the cubic, that it leads at once to the equation of differences. In fact we have

$$a^2\theta^2 + 9(ac - b^2)\theta + \nabla - 27D = \Pi_3\{\theta - (\alpha - \beta)\},$$

whence changing the sign of θ ,

$$a^2\theta^2 + 9(ac - b^2)\theta - \nabla + 27D = \Pi_3\{\theta + (\alpha - \beta)\};$$

or multiplying the two equations and putting u for θ^2 ,

$$u\{a^2u + 9(ac - b^2)\}^2 + 27D = \Pi_3\{u - (\alpha - \beta)^2\},$$

that is, the equation of differences is

$$a^4u^2 + 18(ac - b^2)a^2u^2 + 81(ac - b^2)^2u + 27D = 0;$$

but this mode of composition is peculiar to the case of the cubic.

If in the several equations in θ we substitute for the seminvariants the covariants to which they respectively belong, we obtain as follows:

22. For the cubic equation $(a, b, c, d \parallel v, 1)^3 = 0$, the equation for θ $(= (\alpha - \beta)^2(x - \alpha y)^2)$ is

$$\begin{array}{rcl} a^2\theta & \nabla - 27D\theta & \\ +1 & 0 & -3HU \\ & & +1 \end{array}$$

23. For the quartic equation $(a, b, c, d, e \parallel v, 1)^4 = 0$, the equation for θ $(= (\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2(x - \alpha y)^2)$ is

$$\begin{array}{rcl} a^3\theta & \nabla D\theta & \\ +1 & 0 & -3HU \quad 96\nabla D\theta \\ & & 512\nabla D\theta \\ & & 256D\theta \\ & & +1 \end{array}$$

24. For the quintic equation, the corresponding equation in θ (a seminvariant of the degree 10 and weight 20) is of a very considerable complexity. Its value (completed for me, from the first term, by Mr. Davis) is

$$\begin{aligned} \zeta^2\theta + & \begin{array}{cc} a^2cd^2 + 18 & a^2dp^2 - 30 \\ a^2bce + 18 & a^2ce^2 - 30 \\ a^2bd^2 - 30 & a^2de + 18 \end{array} a^2ef + 2 \\ & \begin{array}{cc} a^2b^2f & -24 \\ a^2bdf & +210 \\ a^2be^2 & -120 \end{array} \begin{array}{cc} a^2c^2f & -120 \\ a^2cde & +504 \\ a^2cd^2 & -180 \end{array} \begin{array}{cc} a^2cef & +180 \\ a^2cd^2e & -180 \\ a^2d^3 & +27 \end{array} \end{aligned}$$

25. The foregoing developments serve to complete the theory of the equation for the function of the differences of the roots of a given equation. The method of seminvariants, by which the coefficients of the equation in θ are obtained, is of general application, and leads to results which can be verified for particular numerical forms of the given equation.

267.

ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED
POLYGON.

*[From the Philosophical Transactions of the Royal Society of
London, vol. CLI (for the year 1861), pp. 225–239. Received
February 20, — Read March 7, 1861.]*

The Porism referred to is as follows, viz. two conics may be so related to each other, that a polygon may be inscribed in the one, and circumscribed about the other conic, in such a manner that any point whatever of the first conic may be taken for a vertex of the polygon.

The demonstration of this fundamental theorem is for greater completeness here reproduced; but the chief object of the memoir is to direct attention to a curious analytical theorem which is in fact the rational form of the transcendental equation which expresses the condition of the porism.

1. The porism of the in-and-circumscribed polygon presents itself under various forms, according as we consider the relation between the two conics, or seek to express the condition that such a relation exists.

2. The fundamental formulæ for the several polygons are as follows:

For the triangle:

$$\begin{vmatrix} C & D \\ D & E \end{vmatrix} = 0$$

For the quadrangle:

$$\begin{vmatrix} D & E \\ E & F \end{vmatrix} = 0$$

For the pentagon:

$$\begin{vmatrix} E & F \\ F & G \end{vmatrix} = 0, \quad \&c.$$

respectively, while those in the cases of the quadrangle, hexagon, octagon, &c. are

$$D = 0, \quad E = 0, \quad \&c.$$

3. The functions $[3]$, $[4]$, $[5]$, $\dots$ are defined as certain determinants formed from the coefficients of the two conics. We have:

$$[3] = \begin{vmatrix} C & D \\ D & E \end{vmatrix},$$

$$[4] = \begin{vmatrix} D & E \\ E & F \end{vmatrix},$$

$$[5] = \begin{vmatrix} E & F \\ F & G \end{vmatrix},$$

$$[6] = [6][3], \quad [8] = [8][4], \quad [9] = [9][3], \quad \&c.$$

4. The demonstration of this fundamental theorem is for greater completeness here given.⁴

5. The relations $[3] = 0$, $[4] = 0$, $[5] = 0$, $[6] = 0$, $\&c.$ are in fact prime. But the coefficients β , γ , δ , or b , c , d instead of being so many independent arbitrary quantities, may be given a priori certain values, and the question then arises as to the conditions under which the polygon may exist.

⁴See the papers — “On the Geometrical Representation of the Integral $\int dx \div \sqrt{f(x)}$,” *Phil. Trans.* 1857, [266]; “Note on the Porism of the in-and-circumscribed Polygon,” *Phil. Mag.* August 1853, [115].

6. The condition for the existence of the porism is that a certain transcendental equation involving the coefficients of the two conics should be satisfied. This condition may be expressed in various forms, according to the particular polygon under consideration.

7. The formulae for the several polygons up to the enneagon are in the memoir obtained in a form which puts in evidence the property in question, that is, the series of equations

$$\begin{aligned}(3) &= [3], \\(4) &= [4], \\(5) &= [5], \\(6) &= [6][3], \\(7) &= [7], \\(8) &= [8][4], \\(9) &= [9][3].\end{aligned}$$

To do this, the discriminant is represented, not as above in terms of the constants β , γ , δ , but in a somewhat different form, by means of the functions $[3]$, $[4]$, &c.

8. The following are the values of the functions [3], [4], &c. in terms of the coefficients:

$$[3] = ce - d^2,$$

$$[4] = df - e^2,$$

$$[5] = eg - f^2,$$

$$[6] = \text{a more complicated function of the coefficients,}$$

$$[7] = \text{the corresponding function for the heptagon,}$$

$$[8] = [8][4],$$

$$[9] = [9][3].$$

9. The condition for the triangle is

$$[3] = 0, \quad \text{that is} \quad CE - D^2 = 0;$$

and that for the quadrangle is

$$[4] - 2[3][5] + 2[3]^2 = 0, \quad \text{or} \quad S - 2\beta\gamma + 2\beta^2 = 0.$$

10. It is obviously convenient to introduce into the formulae, in the place of γ and B , the quantities p and k , or to use other modifications which present themselves naturally in the theory.

11. The porism may be considered as a particular case of the general theory of polygons inscribed in one conic and circumscribed about another. The condition that such a polygon should close after a given number of sides is always expressible as a relation between the coefficients of the two conics.

12. The general theory includes as special cases:

- (I.) The triangle, quadrangle, pentagon, &c., inscribed in $U = 0$ and circumscribed about $V = 0$;
- (II.) The corresponding polygons when the conics are given in the general form;
- (III.) The particular cases where one or both of the conics degenerate into line-pairs or point-pairs.

Demonstration of the general Formula of the Relation between the two Conics.

The equation of a conic passing through the points of intersection of the conics $U = 0$, $V = 0$ is of the form $U + \lambda V = 0$; and in order that this may represent a pair of lines, we must have $(U + \lambda V) = 0$, which is an equation of the third degree in λ , the roots of which determine the three pairs of lines through the intersections of the two conics.

If the two conics touch, then two of these pairs coincide, and the discriminant has a double root; if they have double contact, all three pairs coincide.

Analytical expression of the condition for the Polygon.

The condition for the polygon may be expressed by means of the transcendental function

$$\Pi\xi = \int \frac{d\xi}{\sqrt{\xi(1+\alpha\xi)(1+\beta\xi)(1+\gamma\xi)}}.$$

If we write

$$\lambda = bc + ca + ab - (pd + kp - kd),$$

and thence

$$(bc + ca + ab - pd - kp - kd)^2 - 4(a + b + c + k + p + e)(abc + kpe) = 0$$

as the rational form of the original equation. But the same rational equation would, it is clear, be obtained from the system

$$\sqrt{\xi(1+\alpha\xi)(1+\beta\xi)(1+\gamma\xi)}$$

by eliminating the parameters.

Geometrical interpretation of the Porism.

Consider now a triangle inscribed in the conic $V = 0$, and with its sides touching $U = 0$. Let the points of contact be P, Q, R , and the vertices A, B, C . Then the condition that the polygon should close is equivalent to the condition that the points P, Q, R should be so related that the lines joining the corresponding vertices meet in a point.

The analytical condition may be written in the form

$$\begin{vmatrix} C & D & E \\ D & E & F \\ E & F & G \end{vmatrix} = 0,$$

which expresses that the polygon of any number of sides may be inscribed in one conic and circumscribed about another.

The functions [3], [4], [5], [6], [7], [8], [9], which enter into the formulae, are connected by various relations, and the entire theory may be presented in a very complete and systematic form.

The foregoing demonstration relates to the particular forms $U = ax^2 + by^2 + cz^2$, $V = x^2 + y^2 + z^2$; but observing that the function $\sqrt{(1 + \alpha\xi)(1 + \beta\xi)(1 + \gamma\xi)}$, which enters under the integral sign in the transcendental function $\Pi\xi$, is the square root of the discriminant of $U + \xi V$, the theory of covariants shows at once that the conclusions apply to any forms whatever of U , V , the expression for the transcendental function being modified accordingly.

or, what is the same thing, for n odd, $= 2p - 1$, it is

$$\{1, \theta, \dots \theta^{p-1}, \sqrt{U}\theta, \theta\sqrt{U}\theta, \dots \theta^{p-2}\sqrt{U}\theta\} = 0,$$

and for n even, $= 2p$, it is

$$\{1, \theta, \dots \theta^p, \sqrt{U}\theta, \theta\sqrt{U}\theta, \dots \theta^{p-1}\sqrt{U}\theta\} = 0,$$

where the expressions in $\{\}$ denote respectively the determinants, of $2p - 1$ lines, and $2p$ lines, formed by substituting for θ the values $\lambda_1, \lambda_2, \dots \lambda_n$ respectively. Thus for $n = 3$, the condition is

$$\begin{vmatrix} 1 & \theta & \sqrt{U}\theta \\ 1 & \lambda_1 & \sqrt{U}\lambda_1 \\ 1 & \lambda_2 & \sqrt{U}\lambda_2 \end{vmatrix} = 0,$$

which is the condition for the triangle.

For the condition of the quadrangle, we have $n = 4$, and the formula gives a determinant of 4 lines, which equated to zero gives the required condition.

80 that the condition for the triangle is

$$\gamma - \beta = 0,$$

and that for the quadrangle is

$$\delta - 2\beta\gamma + 2\beta^2 = 0.$$

It is obviously convenient to introduce into the formulae, in the place of γ and B , the quantities

$$\delta = \delta - 2\beta\gamma + 2\beta^2;$$

and writing also, for symmetry of notation, b in the place of β , we have

$$\begin{aligned}\beta &= b, \\ \gamma &= c + b^2, \\ \delta &= d + 2bc,\end{aligned}$$

so that the discriminant may be expressed in terms of b , c , d and the other coefficients.

268.

ON A NEW AUXILIARY EQUATION IN THE THEORY
OF EQUATIONS OF THE FIFTH ORDER.

*[From the Philosophical Transactions of the Royal Society of
London, vol. CLI (for the year 1861), pp. 263-276. Received
February 20, — Read March 7, 1861.]*

1. Considering the equation of the fifth order, or quintic equation,

$$(a, b, c, d, e, f \parallel v, 1)^5 = (v - x_1)(v - x_2)(v - x_3)(v - x_4)(v - x_5) = 0,$$

and putting as usual

$$fx_0 = x_1 + \omega x_2 + \omega^2 x_3 + \omega^3 x_4 + \omega^4 x_5,$$

where ω is an imaginary fifth root of unity, then, according to Lagrange's general theory for the solution of equations, fx_0 is the root of an equation of the order 24, called the Resolvent Equation, but the solution whereof depends ultimately on the solution of an equation of the sixth order.

2. The connexion of the two equations, and the considerations which led to the new one, will be pointed out in the sequel; but I will here state synthetically the construction of the auxiliary equation. Representing for shortness the roots x_1, x_2, x_3, x_4, x_5 of the given equation by 1, 2, 3, 4, 5, and writing

$$1234.5 = 12 + 23 + 34 + 45 + 51, \text{ \&c.},$$

(where on the right-hand side 12, 23, &c. stand for x_1x_2, x_2x_3 , &c.), we have to find the equation

$$(*\surd\phi_1, 1)^6 = 0,$$

the roots whereof are

$$\begin{array}{ll} \phi_1 = 12345 - 24135, & \phi_4 = 21435 - 13245, \\ \phi_2 = 13425 - 32145, & \phi_5 = 31245 - 14325, \\ \phi_3 = 14235 - 43125, & \phi_6 = 41325 - 12435. \end{array}$$

3. It is proper to notice that, combining together in every possible manner the six roots of the auxiliary equation, there are in all fifteen combinations of the form $\phi_1\phi_2 + \phi_3\phi_4 + \phi_5\phi_6$. But the combinations occurring in the above-mentioned equations are a completely determinate set of five combinations: the equation of the order 15, whereon depend the combinations $\phi_1\phi_2 + \phi_3\phi_4 + \phi_5\phi_6$, is not rationally decomposable into three quintic equations, but only into a quintic equation having for its roots the above-mentioned five combinations, and into an equation of the tenth order, having for its roots the other ten combinations.

4. The function ϕ is connected with the theory of the covariants of the quintic by the relation

$$\phi = -\frac{1}{2}\{(12345) - (24135)\},$$

where (12345) and (24135) denote particular combinations of the roots.

5. The coefficients of the auxiliary equation are expressible as rational functions of the coefficients of the given quintic equation. They are seminvariants, and may be calculated by the methods of the theory of invariants.

6. The preceding remarks show the connexion between the function θ ($= t - t'$) to which belongs the new auxiliary equation, and the Resolvent Product R ($= fx_0fx_1fx_2fx_3fx_4$). The relation was given for the denumerate form of the quintic; but taking, instead, the standard form $(a, b, c, d, e, f \parallel v, 1)^5 = 0$, it becomes

$$\phi \cdot a^7\theta = 2500(ac - b^2)^2 - 3a^2 \cdot \text{etc.}$$

7. The foregoing equation shows that ϕ is a seminvariantive function of the roots. In fact

$$fx_0 = x_1 - x_2 + \omega(x_2 - x_3) + \omega^2(x_3 - x_4) + \omega^3(x_4 - x_5) + \omega^4(x_5 - x_1),$$

is seminvariantive, and fx_1, fx_2, fx_3, fx_4 , being in like manner seminvariantive, the product θ ($= fx_0fx_1fx_2fx_3fx_4$) is also seminvariantive.

8. The developed expression for the equation in ϕ is easily found to be

$$\phi^6 + A\phi^5 + B\phi^4 + C\phi^3 + D\phi^2 + E\phi + F = 0,$$

where the coefficients A, B, C, D, E, F are rational functions of the coefficients of the given quintic equation, expressible in terms of the invariants of the form.

9. The explicit values of these coefficients are:

$$A = 0,$$

$$B = a^2e - 2a^3,$$

$$C = a^4d - 12a^2ce + 48a^4,$$

$$D = -4ae(a - \beta)(\beta - \gamma)(\gamma - a),$$

$$E = -ae^2 \cdot f + 4ae \cdot a\beta\gamma - 16a^2e^2 \cdot \Sigma a\beta + 64a^3e^2,$$

$$F = \text{a function of the invariants of the quintic.}$$

10. The function Φ is connected with the theory of the covariants of the quintic by the relation

$$\Phi = -\frac{1}{2}\{(12345) - (24135)\},$$

where (12345) and (24135) denote particular combinations of the roots.

11. The square of Φ is a rational function of the coefficients, and the equation for Φ may be written in the form

$$\Phi^2 = \Psi,$$

where Ψ is expressible by means of the invariants of the quintic form.

12. The auxiliary equation here considered is closely connected with that which presents itself in M. Hermite's theory of the quintic equation. The relation between the two equations may be established by comparing the expressions for the roots in terms of the coefficients.

13. The function Φ satisfies the equation

$$\Phi^6 + a\Phi^4 + b\Phi^2 + c = 0,$$

where a, b, c are expressible in terms of the invariants A, B, C of the quintic form. This equation is reducible to a cubic equation in Φ^2 , and may therefore be solved by radicals.

14. The fifteen combinations mentioned above are connected by linear relations, which may be used to express any one of them in terms of the others. The complete system of relations is as follows:

$$\begin{aligned}(12)(34)(56) + (13)(42)(56) + (14)(23)(56) &= 0, \\ (12)(34)(56) + (12)(35)(64) + (12)(36)(45) &= 0, \\ &\&c.\end{aligned}$$

15. These relations enable us to express the fifteen products in terms of four independent quantities, and thus to establish the theory of the auxiliary equation on a complete foundation.

16. The equation in Φ may be written in the form

$$(\Phi^2 - \alpha)(\Phi^2 - \beta)(\Phi^2 - \gamma) = 0,$$

and the roots α, β, γ are connected with the invariants of the quintic by the relations

$$\alpha + \beta + \gamma = -a, \quad \beta\gamma + \gamma\alpha + \alpha\beta = b, \quad \alpha\beta\gamma = -c.$$

17. The corresponding values of Φ are then $\pm\sqrt{\alpha}, \pm\sqrt{\beta}, \pm\sqrt{\gamma}$, and these are the six roots of the auxiliary equation.

18. The theory of the auxiliary equation is thus brought into connexion with the general theory of the quintic equation. The method may be applied to the solution of the quintic by means of elliptic functions, and leads to results which are in agreement with those obtained by other methods.

19. The function Φ is, in fact, the square root of a certain invariant of the quintic form, and the auxiliary equation expresses the condition that this invariant should have a particular value.

20. In connexion with the foregoing theory, I remark that the function θ ($= t - t'$) belongs to a class of functions which present themselves in the theory of equations of the fifth order, and which are connected with the resolvent equation of Lagrange. The function θ is expressible as a product of five factors, each of which is a linear function of the roots, and the auxiliary equation is obtained by eliminating the parameters which enter into these linear functions.

269.

A SEVENTH MEMOIR ON QUANTICS.

[From the Philosophical Transactions of the Royal Society of London, vol. CLI (for the year 1861), pp. 277–292. Received February 28, — Read March 14, 1861.]

Covariant and other Tables — No. 71.

230. For the binary cubic $(a, b, c, d \parallel x, y)^3$, if U be the cubic itself, HU the Hessian, ΦU the cubicovariant, and D the discriminant (see Fifth Memoir, Nos. 115, 118), then

$$\Omega(\alpha U + \beta HU) = (\alpha^2 - \beta^2 D)U,$$

so that the quantics in (α, β) all of them depend on $\alpha^2 - \beta^2 D = 0$.

231. For the binary quartic $(a, b, c, d, e \parallel x, y)^4$, if U be the quartic itself, HU the Hessian, ΦU the cubicovariant, I, J , the quadrinvariant and the cubinvariant, and $D (= I^3 - 27J^2)$ the discriminant (see Fifth Memoir, Nos. 128, 134), then

Table No. 72.

$$\Omega(\alpha U + 6\beta HU) = (1, 0, -9I, -54J \parallel \alpha, \beta)^3 U,$$

$$H(\alpha U + 6\beta HU) = -\frac{1}{6}(1, 0, -9I, -54J \parallel \alpha, \beta)^3 U \\ + \frac{1}{6} \cdot 3 \cdot (1, 0, -9I, -54J \parallel \alpha, \beta)^2 HU,$$

$$\Phi(\alpha U + 6\beta HU) = (I, 18J, 27I^2 \parallel \alpha, \beta)^2,$$

$$J(\alpha U + 6\beta HU) = (J, I^2, 9IJ, -I^3 + 54J^2 \parallel \alpha, \beta)^3,$$

$$D(\alpha U + 6\beta HU) = \{(1, 0, -18I, -108J, 81I^2, 972IJ, 2916J^2 \parallel \alpha, \beta)^6\} D \\ = [(1, 0, -9I, -54J \parallel \alpha, \beta)^3]^2 D.$$

232. Writing for the moment

$$G = (1, 0, -9I, -54J \parallel \alpha, \beta)^3,$$

then the Hessian, cubicovariant, and discriminant of this cubic function of (α, β) are respectively

$$HG = -3(I, 18J, 27I^2 \parallel \alpha, \beta)^2, \quad \Phi G = 54(J, I^2, 9IJ, -I^3 + 54J^2 \parallel \alpha, \beta)^3, \quad DG =$$

so that the covariants of the intermediate $\alpha U + 6\beta HU$ are all of them expressible by means of the cubic function G .

[The coefficient $2916J^2$ is in the Fifth Memoir erroneously given as $-2916J^2$. This correction should have been made, vol. II. p. 549.]

233. It may be noticed that G is what the left-hand side of the equation

$$4(HU)^3 - I^2 \cdot HU \cdot U^2 + J \cdot U^3 = -(\Phi U)^2$$

(see Fifth Memoir, No. 128) becomes on writing therein α , -6β , for U , HU respectively, and throwing out the factor 4.

234. I take the opportunity of remarking with respect to a binary quartic $U = (a, b, c, d, e \parallel x, y)^4$, that the Hessian of the cubicovariant, to fix the numerical factor, say $-\frac{1}{9}\Phi \cdot \Omega = \partial_x^2 \Phi \cdot \partial_y^2 \Phi - (\partial_x \partial_y \Phi)^2$, is

$$= I^2 U^2 - 36J \cdot U \cdot HU + 12 \cdot (1 - 27\frac{J^2}{I^2})(HU)^2;$$

which is

$$= (IU - \frac{6J}{I}HU)^2 + \frac{1}{I^2}(I^3 - 27J^2)(HU)^2;$$

or if $I^3 - 27J^2 = 0$, that is if the quartic has a pair of equal factors, the Hessian of the cubicovariant is a perfect square.

235. Coming now to the ternary cubic $U = (a, b, c, f, g, h, i, j, k, l \parallel x, y, z)^3$, I give in the first place the following comparison of my notation with that of M. Aronhold.

Aronhold	Cayley
$\mathfrak{S}\mathfrak{f}$	$-\Theta HU$
$\mathfrak{T}\mathfrak{f}$	$-\Theta U$
A'	$-\frac{1}{2}\mathfrak{S}R$
$\mathfrak{S}$	
$\mathfrak{T}$	$-2\mathfrak{Q}U$
R	$\mathfrak{C}PU$
$\mathfrak{P}$	$-2\mathfrak{V}U$
$\mathfrak{Q}$	$\frac{1}{2}\mathfrak{S}'U + \mathfrak{T}'$
$\mathfrak{V}$	$2\mathfrak{Z}U$
	$\mathfrak{C}U$
	$\mathfrak{D}U$
$\mathfrak{Q}$	
$\mathfrak{H}$	$-\mathfrak{F}U$
$\mathfrak{F}$	

where the notations $\mathfrak{V}U$, $\mathfrak{Z}U$ (to correspond to M. Aronhold's $\mathfrak{P}\mathfrak{f}$, $\mathfrak{Q}\mathfrak{f}$) and the notations $\mathfrak{C}U$, $\mathfrak{D}U$ are first employed in the present memoir.

I remark in regard to $\mathfrak{Pf} (= -2\mathfrak{Y}U)$, where, as already mentioned,

$$\mathfrak{F}U = (1 + 8\theta\partial_\theta) \left[\frac{1}{3}(\partial_x^3 + \partial_y^3 + \partial_z^3) - \partial_x\partial_y\partial_z\theta \right],$$

that the function $\mathfrak{F}U$ is a covariant of the ternary cubic, and that it plays an important part in the theory of the covariants of this form.

The paragraphs and Tables of the present memoir are numbered consecutively with those of my former memoirs on Quantics.

236. The following Tables exhibit the values of the covariants for the several forms up to the ternary cubic:

Table No. 73. — Binary Quintic.

For the binary quintic $(a, b, c, d, e, f \parallel x, y)^5$, if U be the quintic itself, HU the Hessian, ΦU the cubicovariant, and I, J, K the fundamental invariants, then the values of the covariants of the intermediate $\alpha U + \beta HU$ are expressible in terms of the invariants and of the covariants of the quintic form.

237. The theory of the ternary cubic leads to a remarkable system of covariants, which are connected by various syzygies. The following Tables exhibit these relations in a systematic form:

Table No. 74. — Ternary Cubic.

$$\begin{aligned}\Omega U &= 6 \mathfrak{S} U, \\ \Omega HU &= 2 \mathfrak{S} HU + \mathfrak{T} U, \\ \Omega \Phi U &= \mathfrak{S} \Phi U + 3 \mathfrak{T} HU, \\ \Omega \Psi U &= 3 \mathfrak{S} \Psi U + 2 \mathfrak{T} \Phi U + \mathfrak{U} HU, \\ &\&c.\end{aligned}$$

where $\mathfrak{S}, \mathfrak{T}, \mathfrak{U}$ denote certain operators which generate the covariants.

238. The following are the values of the principal covariants for the ternary cubic:

$$\begin{aligned}\mathfrak{C}U &= (a, b, c, f, g, h, i, j, k, l \backslash x, y, z)^3, \\ \mathfrak{D}U &= (\partial_x, \partial_y, \partial_z)^2 (\text{Hessian}), \\ \mathfrak{F}U &= \text{the contravariant of the cubic,} \\ \mathfrak{Y}U &= \text{the covariant corresponding to } \mathfrak{P}\mathfrak{f}, \\ \mathfrak{Z}U &= \text{the covariant corresponding to } \mathfrak{Q}\mathfrak{f}.\end{aligned}$$

239. These covariants are connected by the syzygy

$$(\mathfrak{C}U)^2 \cdot U - (\mathfrak{D}U)^3 + \dots = 0,$$

which expresses that the square of the first covariant, multiplied by the original form, is equal to the cube of the second covariant, together with certain other terms.

240. The paragraphs and Tables of the present memoir are numbered consecutively with those of my former memoirs on Quantics. The theory of the binary forms is completed by the introduction of the new covariants $\mathfrak{C}U$, $\mathfrak{D}U$, which enable us to express all the other covariants in a finite and determinate form.

241. The following Table exhibits the comparison of the notations:

Form	Aronhold	Cayley
Binary cubic	$\mathfrak{P}f, \mathfrak{Q}f$	$\mathfrak{Y}U, \mathfrak{Z}U$
Binary quartic	$\mathfrak{P}f, \mathfrak{Q}f$	$\mathfrak{C}U, \mathfrak{D}U$
Ternary cubic	$\mathfrak{P}f, \mathfrak{Q}f$	$\mathfrak{Y}U, \mathfrak{Z}U$

242. The covariants $\mathfrak{CU}$, $\mathfrak{DU}$ belong to the forms called Conjugate Forms, § 27 of M. Aronhold's memoir.⁵

243. The present memoir exhibits in a more complete and systematic form the relations which connect the covariants of the ternary cubic. Some of these relations have been obtained by M. Hermite in the memoir "Sur les formes cubiques à trois indéterminées," (Liouville, t. III. pp. 37—40 (1858), and in that "Sur la Résolution des Equations du quatrième degré," Comptes Rendus, XLVI. p. 715 (1858), and by M. Aronhold in his memoir already referred to; and in particular I reproduce and demonstrate some of the results in the last-mentioned memoir of M. Hermite. But the relations in question are in the present memoir exhibited in a more complete and systematic form.

[End of Papers for this section]

⁵M. Aronhold, in a letter dated Berlin, 17 June 1861, has pointed out to me that the covariants $\mathfrak{CU}$, $\mathfrak{DU}$ are in his notation $\mathfrak{Pf}$, $\mathfrak{Qf}$, and that they belong to the forms called Conjugate Forms, § 27 of his memoir. But the explicit development of the properties of these covariants is not on this account the less interesting. Added 20 Sept. 1861. — A.C.

271. Sur l'invariant le plus simple d'une fonction quadratique bi-ternaire, et sur le resultant de trois fonctions quadratiques ternaires.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LVII. (1860), pp. 139–148.]

M. Sylvester a trouvé depuis longtemps, pour le résultant des trois fonctions quadratiques ternaires

$$(a, b, c, f, g, h \setminus x, y, z)^2, \quad (a', b', c', f', g', h' \setminus x, y, z)^2, \quad (a'', b'', c'', f'', g'', h'' \setminus x, y, z)^2,$$

une expression remarquable

$$12R = 16C_3^2 - C_6^2,$$

où C_3 , C_6 représentent des fonctions données des coefficients qui sont non seulement des invariants mais aussi des combinants des trois fonctions quadratiques (Voir le mémoire de M. Sylvester, “On the Calculus of forms otherwise the theory of Invariants,” § VII. Camb. et Dub. Math. Journ., t. VIII. pp. 256–269 année 1853, Equation (A.), p. 267). Pour expliquer la formation de la fonction C_3 , il convient de considérer la fonction quadratique bi-ternaire

$$U = \begin{pmatrix} a & b & c & f & g & h \\ a' & b' & c' & f' & g' & h' \\ a'' & b'' & c'' & f'' & g'' & h'' \\ A & B & C & F & G & H \\ A' & B' & C' & F' & G' & H' \\ A'' & B'' & C'' & F'' & G'' & H'' \end{pmatrix} \setminus (x, y, z \setminus \xi, \eta, \zeta)^2,$$

cette notation représentant la fonction

$$\begin{aligned} & (a, b, c, f, g, h \setminus x, y, z)^2 \cdot \xi^2 + (a', b', c', f', g', h' \setminus x, y, z)^2 \cdot \eta^2 \\ & + (a'', b'', c'', f'', g'', h'' \setminus x, y, z)^2 \cdot \zeta^2 \\ & + (A, B, C, F, G, H \setminus x, y, z)^2 \cdot 2\eta\zeta \\ & + (A', B', C', F', G', H' \setminus x, y, z)^2 \cdot 2\zeta\xi \\ & + (A'', B'', C'', F'', G'', H'' \setminus x, y, z)^2 \cdot 2\xi\eta, \end{aligned}$$

ou, ce qui est la même chose,

$$\begin{aligned}
 & (ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy) \xi^2 \\
 & + (a'x^2 + b'y^2 + c'z^2 + 2f'yz + 2g'zx + 2h'xy) \eta^2 \\
 & + (a''x^2 + b''y^2 + c''z^2 + 2f''yz + 2g''zx + 2h''xy) \zeta^2 \\
 & + (Ax^2 + By^2 + Cz^2 + 2Fyz + 2Gzx + 2Hxy) 2\eta\zeta \\
 & + (A'x^2 + B'y^2 + C'z^2 + 2F'yz + 2G'zx + 2H'xy) 2\zeta\xi \\
 & + (A''x^2 + B''y^2 + C''z^2 + 2F''yz + 2G''zx + 2H''xy) 2\xi\eta.
 \end{aligned}$$

Soit 123 le déterminant

$$\begin{vmatrix} \partial_{x_1} & \partial_{y_1} & \partial_{z_1} \\ \partial_{x_2} & \partial_{y_2} & \partial_{z_2} \\ \partial_{x_3} & \partial_{y_3} & \partial_{z_3} \end{vmatrix}$$

et 1'2'3' le déterminant

$$\begin{vmatrix} \partial_{\xi_1} & \partial_{\eta_1} & \partial_{\zeta_1} \\ \partial_{\xi_2} & \partial_{\eta_2} & \partial_{\zeta_2} \\ \partial_{\xi_3} & \partial_{\eta_3} & \partial_{\zeta_3} \end{vmatrix},$$

et soient U_1, U_2, U_3 ce que devient U lorsqu'on y écrit $x_1, y_1, z_1, \xi_1, \eta_1, \zeta_1; x_2, y_2, z_2, \xi_2, \eta_2, \zeta_2; x_3, y_3, z_3, \xi_3, \eta_3, \zeta_3$ au lieu de $x, y, z, \xi, \eta, \zeta$; on a alors cet invariant (analogue à l'invariant quadratique d'une fonction binaire d'ordre pair), savoir:

$$\frac{1}{6^4} 123^2 \cdot 1'2'3'^2 U_1 U_2 U_3,$$

où comme à l'ordinaire les valeurs $x, y, z, \xi, \eta, \zeta$ sont rétablies après les différenciations au lieu de $x_1, y_1, z_1, \xi_1, \eta_1, \zeta_1$, etc.

Je représente cet invariant par

$$\begin{vmatrix} a & b & c & f & g & h \\ a' & b' & c' & f' & g' & h' \\ a'' & b'' & c'' & f'' & g'' & h'' \\ A & B & C & F & G & H \\ A' & B' & C' & F' & G' & H' \\ A'' & B'' & C'' & F'' & G'' & H'' \end{vmatrix},$$

on voit sans peine que l'invariant peut être développé, et qu'alors il prend la forme

$$\begin{vmatrix} a & b & c & f & g & h \\ a' & b' & c' & f' & g' & h' \\ a'' & b'' & c'' & f'' & g'' & h'' \end{vmatrix},$$

où le premier terme dénote la fonction

$$\begin{vmatrix} a & b & c & f & g & h \\ A & B & C & F & G & H \\ A' & B' & C' & F' & G' & H' \end{vmatrix} \begin{vmatrix} a' & b' & c' & f' & g' & h' \\ A & B & C & F & G & H \\ A'' & B'' & C'' & F'' & G'' & H'' \end{vmatrix} \begin{vmatrix} a'' & b'' & c'' & f'' & g'' & h'' \\ A' & B' & C' & F' & G' & H' \\ A'' & B'' & C'' & F'' & G'' & H'' \end{vmatrix}$$

$$+ 2 \begin{vmatrix} a & b & c & f & g & h \\ a' & b' & c' & f' & g' & h' \\ A'' & B'' & C'' & F'' & G'' & H'' \end{vmatrix} \begin{vmatrix} a & b & c & f & g & h \\ A & B & C & F & G & H \\ A' & B' & C' & F' & G' & H' \end{vmatrix} \\ + 2 \begin{vmatrix} a' & b' & c' & f' & g' & h' \\ a'' & b'' & c'' & f'' & g'' & h'' \\ A'' & B'' & C'' & F'' & G'' & H'' \end{vmatrix} \begin{vmatrix} a & b & c & f & g & h \\ a'' & b'' & c'' & f'' & g'' & h'' \\ A' & B' & C' & F' & G' & H' \end{vmatrix},$$

et de même pour les autres termes. L'invariant contient les termes

$$ab'c'' + 2AFG' + 2fg'h'' + 2F'G''H'' \\ - 2af f'' - 2aBG + 2aF^2 \\ - 4AFF' + 4fAF - 4fGH, \quad \text{etc.},$$

et on déduit de là son expression complète en y écrivant $ab'c'' + ab''c' + a'b''c + a'bc'' + a''bc' + a''b'c$ au lieu de $ab'c''$; $af'f'' + a'f''f + a''ff' + b'g''g + b''gg' + c'h'h'' + c''hh'' + c''hh'$ au lieu de $af f''$, et ainsi de suite; la somme des coefficients est $6+2\cdot6+2\cdot6+4\cdot6+2\cdot9+2\cdot9+2\cdot9+4\cdot9+4\cdot9+4\cdot9 = 216$, ce qui est exact, car il n'y a pas de termes qui se détruisent, et dans l'expression $\frac{1}{6^4} 123^2 1'2'3'^2 U_1 U_2 U_3$, le nombre des termes qui composent le produit des deux déterminants carrés est $36 \times 36 = 1296$, chacun de ces termes est multiplié par le facteur $8 \times 8 = 64$ introduit par la différenciation, et l'on a $\frac{1}{6^4} \times 64 \times 1296 = 216$.

L'invariant qui vient d'être donné est lié avec l'invariant T de M. Aronhold (lequel se rapporte à une fonction ternaire cubique). C'est pour cela que je le désigne par la même lettre T , et pour fixer sa valeur j'écris

$$2T = ab'c'' + \text{etc.}$$

Cela posé, je forme les deux combinants C_3 et C_6 de la manière suivante.

Combinant du sixième ordre C_3 . Soient U, V, W les trois fonctions quadratiques ternaires; considérez la fonction syzygétique $\lambda U + \mu V + \nu W$, où λ, μ, ν sont des quantités arbitraires; formez avec les variables ξ, η, ζ le réciproquant⁶ (multiplié par 2) de cette fonction, le résultat sera de l'ordre 2 par rapport aux coefficients de U, V, W , de l'ordre 2 par rapport à λ, μ, ν , et de l'ordre 2 par rapport à ξ, η, ζ ; c. à. d. ce résultat sera une fonction quadratique bi-ternaire de λ, μ, ν et ξ, η, ζ dont l'invariant $2T$ (lequel sera par conséquent de l'ordre 6 par rapport aux coefficients de U, V, W) est le combinant C_3 qu'il s'agissait de trouver.

Combinant du douzième ordre C_6 . Considérez comme auparavant la fonction syzygétique $\lambda U + \mu V + \nu W$, formez le discriminant (multiplié par 6) de cette fonction; le résultat sera de l'ordre 3 par rapport aux coefficients de U, V, W , et de l'ordre 3 par rapport à λ, μ, ν ; c. à. d. il sera une fonction cubique ternaire de λ, μ, ν , dont l'invariant S de M. Aronhold⁷ (lequel sera de l'ordre 12 par rapport aux coefficients de U, V, W) est le combinant C_6 qu'il s'agissait de trouver.

On peut exprimer ces résultats d'une manière abrégée en disant que C_3 est l'invariant $2T$ de la fonction bi-ternaire $2R$ et que C_6 est l'invariant S de la fonction cubique ternaire $6D$, en désignant par R et D le réciproquant et le discriminant de la fonction $\lambda U + \mu V + \nu W$.

⁶Le réciproquant de $(a, b, c, f, g, h \setminus x, y, z)^2$ est

$$\begin{vmatrix} \xi & \eta & \zeta \\ a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = -(A, B, C, F, G, H \setminus \xi, \eta, \zeta)^2.$$

⁷Pour la fonction cubique ternaire $(a, b, c, i, j, k, l_1, j_1, k_1, l \setminus x, y, z)^3$, l'invariant S est

$$\begin{aligned} & l^2 - 2l(j_1k_1 + k_1i_1 + i_1j_1) + 3l(i_1j_1k_1 + k_1j_1i_1) \\ & \quad + l(ai_1j_1 + bk_1i_1 + ci_1j_1) - labc \\ & \quad - (i^2j_1k_1 + j^2k_1i_1 + k^2i_1j_1) + (i^2i_1^2 + j^2j_1^2 + k^2k_1^2) \\ & + (bcj_1k_1 + cah_1i_1 + abi_1j_1) - (ai^2j_1 + bf^2k_1 + ck^2i_1 + ai_1^2k_1 + bi_1^2j_1 + ck_1^2j_1). \end{aligned}$$

Pour démontrer l'équation

$$12R = 16C_3^2 - C_6^2,$$

il suffit de faire voir que cette équation est vérifiée lorsqu'au lieu des trois fonctions données U, V, W on substitue trois fonctions de la forme $lU + mV + nW$, les coefficients l, m, n ayant été choisis de manière à simplifier autant que possible les expressions des trois fonctions. Il est facile de voir qu'il est permis de prendre pour les trois fonctions les formes dont s'est servi M. Sylvester dans son mémoire, savoir

$$\begin{aligned} p & \quad (= -f), \\ q & \quad (= z^2 - y^2), \\ r & \quad (= x^2 + 2lyz + 2mzx + 2nxy). \end{aligned}$$

Mais il suit des recherches de M. Hesse, et M. Sylvester aussi a depuis reconnu, qu'on peut prendre pour les trois fonctions les formes encore plus simples

$$\begin{aligned} x^2 + 2lyz, \\ y^2 + 2lzx, \\ z^2 + 2lxy, \end{aligned}$$

qui se réduisent aux fonctions dérivées de la fonction cubique $x^3 + y^3 + z^3 + 6lxyz$. Or, en prenant ces valeurs de U, V, W , on obtient

$$\lambda U + \mu V + \nu W = (\lambda, \mu l, \nu l, \lambda l, \mu l, \nu l \setminus x, y, z)^2.$$

Le réciproquant (multiplié par 2) est

$$= -2 \begin{vmatrix} \xi & \eta & \zeta \\ \lambda & \mu l & \nu l \\ \mu l & \nu & \lambda l \\ \nu l & \lambda l & \mu \end{vmatrix},$$

lequel, en forme développée, donne le résultat

$$\begin{aligned} & (-2\mu\nu \cdot \xi^2 + 2l\mu\nu \cdot \eta\zeta) \\ & \quad + (-2\nu\lambda \cdot \eta^2 + 2l\nu\lambda \cdot \zeta\xi) \\ & \quad + (-2\lambda\mu \cdot \zeta^2 + 2l\lambda\mu \cdot \xi\eta) \\ & \quad + (-2\lambda^2 \cdot 2\eta\zeta + 2l\nu^2 \cdot 2\zeta\xi) \\ & \quad + (-2\mu^2 \cdot 2\zeta\xi + 2l\nu\lambda \cdot 2\xi\eta) \\ & \quad \quad + (-2\nu^2 \cdot 2\xi\eta + 2l\lambda\mu \cdot 2\eta\zeta), \end{aligned}$$

que l'on peut aussi exprimer comme fonction quadratique bi-ternaire, savoir:

$$= \begin{pmatrix} -2\mu\nu & \cdot & \cdot & \cdot & l\mu\nu & l\mu\nu \\ \cdot & -2\nu\lambda & \cdot & l\nu\lambda & \cdot & l\nu\lambda \\ \cdot & \cdot & -2\lambda\mu & l\lambda\mu & l\lambda\mu & \cdot \\ -2\lambda^2 & \cdot & \cdot & \cdot & l\lambda^2 & l\lambda^2 \\ \cdot & -2\mu^2 & \cdot & l\mu^2 & \cdot & l\mu^2 \\ \cdot & \cdot & -2\nu^2 & l\nu^2 & l\nu^2 & \cdot \end{pmatrix} \backslash (\lambda, \mu, \nu \backslash \xi, \eta, \zeta)^2.$$

En représentant l'invariant $2T$ par le développement incomplet donné ci-dessus, on obtient dans le cas dont il s'agit

$$2T = \begin{vmatrix} -2l^2 & \cdot & \cdot & \cdot & l^3 & l^3 \\ \cdot & -2l^2 & \cdot & l^3 & \cdot & l^3 \\ \cdot & \cdot & -2l^2 & l^3 & l^3 & \cdot \\ -2 & \cdot & \cdot & \cdot & l & l \\ \cdot & -2 & \cdot & l & \cdot & l \\ \cdot & \cdot & -2 & l & l & \cdot \end{vmatrix}$$

ce qui se réduit à la valeur finale

$$\begin{aligned} & -8l^6 + 2\{-(4l^4 + 4l^4 + 4l^4) - (4l^3 + 4l^3 + 4l^3) - (4l^3 + 4l^3 + 4l^3)\} \\ & \quad + 2\{-8l^5 + 2l^6\} = 2(1 - 20l^3 - 8l^6). \end{aligned}$$

Les invariants S et T de la fonction cubique $x^3 + y^3 + z^3 + 6lxyz$ sont

$$S = -1 + l^3, \quad T = 1 - 20l^3 - 8l^6;$$

l'invariant $2T$ de la fonction quadratique bi-ternaire est donc précisément égal à $2T$, où T est l'invariant qui vient d'être donné, et on a

$$C_3 = 2T.$$

Le discriminant (multiplié par 6) de la fonction $\lambda U + \mu V + \nu W$ est

$$= 6 \begin{vmatrix} \lambda & \mu l & \nu l \\ \mu l & \mu & \lambda l \\ \nu l & \lambda l & \nu \end{vmatrix} = 6\{-l^2\lambda^3 - l^2\mu^3 - l^2\nu^3 + (1 + 2l^3)\lambda\mu\nu\},$$

où la fonction en parenthèses est ce que devient le Hessien de la fonction cubique $x^3 + y^3 + z^3 + 6lxyz$, lorsqu'on y substitue λ, μ, ν au lieu de x, y, z . Son invariant S est

$$\begin{aligned} &= (1 + 2l^3)^2 16l^6 + (1 + 2l^3)^3 \\ &= 1 + 8l^3 + 240l^6 + 464l^9 + 16l^{12}. \end{aligned}$$

Cette quantité est en même temps un invariant de la fonction cubique $x^3 + y^3 + z^3 + 6lxyz$, et comme tel elle doit s'exprimer en fonction des deux invariants S et T de cette dernière: en effet elle se réduit à

$$(1 - 20l^3 - 8l^6)^2 - 48(-l + l^3)^3 = T^2 - 48S^3.$$

Nous avons donc

$$C_6 = T^2 - 48S^3,$$

et comme nous venons de trouver

$$C_3 = 2T,$$

l'équation $12R = 16C_3^2 - C_6^2$ se réduit à

$$12R = 16(2T)^2 - 48S^3 - 4T^2 = 12T^2 - 768S^3,$$

ou enfin à

$$R = T^2 - 64S^3,$$

Équation qui fait voir que le résultant R dont il s'agit est effectivement égal à $R = (1 + 8l^3)^2$, c. à. d. au discriminant de la fonction cubique $x^3 + y^3 + z^3 + 6lxyz$, ce qui donne la vérification du théorème général.

Je m'étais servi d'abord d'une analyse moins simple, en considérant le cas dans lequel on prend pour les trois fonctions les formes

$$\begin{aligned} z^2 - y^2, \\ x^2 - z^2, \\ x^2 + y^2 + z^2 + 2lyz + 2mzx + 2nxy. \end{aligned}$$

Il vaut la peine, je crois, de donner les expressions correspondantes de C_3 , C_6 , R . On a ici à considérer la fonction

$$\lambda(z^2 - y^2) + \mu(x^2 + y^2 + z^2 + 2lyz + 2mzx + 2nxy) + \nu(z^2 - y^2),$$

laquelle peut s'écrire comme suit,

$$(\mu + \lambda, \mu - \lambda - \nu, \mu + \nu, l\mu, m\mu, n\mu \setminus x, y, z)^2.$$

Le réciproquant (multiplié par 2) est

$$= -2 \begin{vmatrix} \xi & \eta & \zeta \\ \mu + \lambda & n\mu & m\mu \\ n\mu & \mu - \lambda - \nu & l\mu \\ m\mu & l\mu & \mu + \nu \end{vmatrix}$$

et en écrivant a, b, c, f, g, h au lieu de $1 - l^2$, $1 - m^2$, $1 - n^2$, $mn - l$, $nl - m$, $lm - n$, cette fonction est

$$\begin{aligned} &= (ax^2 - 2\lambda\mu - 2\nu\lambda - 2l\mu^2) \xi^2 \\ &\quad + (by^2 + 2\lambda^2 + 2\nu^2 + 2\mu\nu\lambda) \eta^2 \\ &\quad + (cz^2 - 2\mu\nu - 2\nu\lambda - 2\lambda^2) \zeta^2 \\ &\quad + (2f\mu^2 - 2l\lambda\mu) 2\eta\xi \\ &\quad + (2g\mu^2 + 2m\lambda\mu + 2n\nu\lambda) 2\zeta\xi \\ &\quad + (2h\mu^2 - 2n\nu\mu) 2\xi\eta, \end{aligned}$$

ou, écrite en forme de fonction quadratique bi-ternaire,

$$= \begin{pmatrix} 0 & 2a & -2 & 0 & 0 & -1 \\ 0 & 2b & 0 & 0 & 1 & 1 \\ -2 & 2c & 0 & -1 & -1 & 0 \\ 0 & 2f & 0 & 0 & 0 & -l \\ 0 & 2g & 0 & m & 0 & n \\ 0 & 2h & 0 & -n & 0 & 0 \end{pmatrix} \setminus (\lambda, \mu, \nu \setminus \xi, \eta, \zeta)^2.$$

En exprimant comme auparavant l'invariant $2T$ par une somme de cinq termes, on trouve sans peine que ces termes sont respectivement $18 - 4l^2 - 4m^2 - 4n^2$, $-4l^2 - 4m^2 - 4n^2$, 0 ; l'invariant sera donc $-8(l^2 + m^2 + n^2) + 18$; et l'on a pour le combinant du sixième ordre

$$C_3 = -8(l^2 + m^2 + n^2) + 18.$$

Le discriminant (multiplié par 6) de la fonction $\lambda U + \mu V + \nu W$ est

$$\begin{aligned} &= 6 \begin{vmatrix} \mu + \lambda & n\mu & m\mu \\ n\mu & \mu - \lambda - \nu & l\mu \\ m\mu & l\mu & \mu + \nu \end{vmatrix} \\ &+ 6(m^2 - n^2)\mu^2\nu - 6\lambda^3 - 6\mu^3 - 6\nu^3 - 6\nu\lambda^2 + 6(m^2 - l^2)\lambda\mu^2 - 6\lambda\mu\nu \\ &= (0, 6(1 - l^2 - m^2 - n^2 + 2lmn), 0, 2(m^2 - n^2), -2, -2, -2, -2, 2(m^2 - l^2), - \\ &= (a, b, c, i, j, k, l_1, j_1, k_1, l \setminus \lambda, \mu, \nu)^3, \end{aligned}$$

où l'on a posé $a = 0$, $b = 6(1 - l^2 - m^2 - n^2 + 2lmn)$, etc. L'invariant S de cette dernière fonction est

$$\begin{aligned} &1 - 2\{-4(m^2 - n^2) + 4 - 4(m^2 - n^2)\} \\ &\quad - 3\{8(m^2 - n^2) + 8(m^2 - l^2)\} - 4b \\ &\quad + 16(m^2 - n^2) - 16(m^2 - n^2)(m^2 - l^2) + 16(m^2 - l^2) \\ &\quad + 16(m^2 - n^2)^2 + 16 + 16(m^2 - l^2)^2 + 16b \\ &\quad = 9 + 12b + 16(l^4 + m^4 + n^4 - m^2n^2 - n^2l^2 - l^2m^2), \end{aligned}$$

ou, si l'on substitue la valeur de b ,

$$\begin{aligned} C_6 &= 16(l^4 + m^4 + n^4 - m^2n^2 - n^2l^2 - l^2m^2) \\ &\quad + 144lmn - 72(l^2 + m^2 + n^2) + 81. \end{aligned}$$

On vient de trouver

$$C_3 = -8(l^2 + m^2 + n^2) + 18,$$

Équation qui donne

$$\begin{aligned} C_3^2 &= 64(l^4 + m^4 + n^4 + 2m^2n^2 + 2n^2l^2 + 2l^2m^2) \\ &\quad - 288(l^2 + m^2 + n^2) + 324. \end{aligned}$$

De ces résultats combinés on conclut

$$16C_6 - C_3^2 = 192(l^4 + m^4 + n^4 - 2m^2n^2 - 2n^2l^2 - 2l^2m^2) \\ + 2304lmn - 864(l^2 + m^2 + n^2) + 972.$$

Evidemment l'expression du résultant est, à un facteur numérique près,

$$R = (3+2l+2m+2n)(3+2l-2m-2n)(3-2l+2m-2n)(3-2l-2m+2n),$$

ou en réduisant

$$R = 16(l^4+m^4+n^4-2m^2n^2-2n^2l^2-2l^2m^2)+192lmn-72(l^2+m^2+n^2)+81.$$

On retrouve donc

$$12R = 16C_6 - C_3^2,$$

Équation qu'il s'agissait de vérifier.

En terminant je remarque que la plus grande partie de ce mémoire est tirée d'un manuscrit daté "Machynlleth, 13 Août 1853."

Londres, 22 Mars 1859.

272. Démonstration d'un théorème de Jacobi par rapport au problème de Pfaff.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LVII. (1860), pp. 273–277.]

Dans le mémoire de Jacobi “Theoria novi multiplicatoris etc.” (t. XXIX. de ce Journal, 1845) on trouve p. 253 le passage que voici:

“Methodum ad solvendum problema Pfaffianum ab ipso auctore adhibitam, data occasione observo per plures et altiores procedere integrationes quam methodus vera et genuina poscat. Quam novam methodum pro exemplo simplice explicabo. Ad aequationem differentialem $X_1dx_1 + X_2dx_2 + X_3dx_3 + X_4dx_4 = 0$ per duas aequationes integrandam poscit Pfaffiana methodus. . . integrationem completam systematis trium aequationum differentialium primi ordinis inter quatuor variables ac deinde unius aequationis differentialis primi ordinis inter duas variables. Illius igitur systematis integrali uno invento, secundum illam methodum restat integratio completa duarum aequationum differentialium primi ordinis inter tres variables sive unius aequationis differentialis secundi ordinis inter duas variables ac deinde aequationis differentialis primi ordinis inter duas variables. At observo si integrali illo invento exprimat x_4 per x_1, x_2, x_3 , aequationem differentialem propositam abire in aliam linearem primi ordinis inter tres variables conditioni integrabilitatis satisfacientem; cujus integrationem vidimus (voir p. 246) absolvi posse per integrationes separatas duarum aequationum differentialium primi ordinis inter duas variables. Unde in locum aequationis differentialis secundi ordinis, tantum integrandae sunt duae aequationes differentiales separatae primi ordinis, quae est reductio maxime insignis; integrationi autem aequationis differentialis primi ordinis poenam methodum substituere licebit. . .”

Le théorème en question est:

Que l'intégrale d'une équation différentielle de Pfaff entre un nombre quelconque n de variables peut être obtenue au moyen

d'intégrations de systèmes de degrés successivement décroissants $n-2$, $n-3, \dots, 1$, au lieu de systèmes de degrés $n-1$, $n-2, \dots, 1$.

À savoir l'intégration de l'équation de Pfaff

$$X_1 dx_1 + X_2 dx_2 + X_3 dx_3 + X_4 dx_4 = 0$$

entre quatre variables exige en général, selon la première méthode,

1. L'intégration complète d'un système de trois équations différentielles entre quatre variables;
2. L'intégration d'une équation différentielle du premier ordre entre deux variables.

Mais selon la nouvelle méthode:

1. L'intégration complète d'un système de deux équations différentielles entre trois variables;
2. L'intégration d'un système de deux équations différentielles du premier ordre entre deux variables (ou, ce qui revient au même, de deux équations différentielles du premier ordre entre deux variables);
3. L'intégration d'une équation différentielle du premier ordre entre deux variables.

Le théorème admet une extension facile au cas d'un nombre quelconque de variables.

Il s'agit de démontrer que, pour l'équation de Pfaff entre n variables, on peut par $n-2$ intégrations successives obtenir une relation entre les variables; c'est-à-dire qu'en partant de l'équation de Pfaff entre n variables, on peut obtenir par une intégration un système de degré $n-2$, puis par une autre intégration un système de degré $n-3$, et ainsi de suite, jusqu'à ce qu'on arrive à un système du premier ordre, dont l'intégration donne la relation cherchée.

Pour le cas de quatre variables, soit l'équation

$$X_1 dx_1 + X_2 dx_2 + X_3 dx_3 + X_4 dx_4 = 0.$$

On peut écrire cette équation sous la forme

$$X_4 dx_4 = -X_1 dx_1 - X_2 dx_2 - X_3 dx_3.$$

Supposons qu'on ait intégré le système

$$\frac{dx_1}{P_1} = \frac{dx_2}{P_2} = \frac{dx_3}{P_3} = \frac{dx_4}{P_4},$$

et soit $V = c$ une intégrale de ce système. On peut alors exprimer x_4 au moyen de x_1, x_2, x_3 et de la constante c , et en substituant cette valeur dans l'équation de Pfaff, on obtient une équation de la forme

$$Y_1 dx_1 + Y_2 dx_2 + Y_3 dx_3 = 0,$$

qui est intégrable par un facteur intégrant; et cette intégration, combinée avec $V = c$, donne l'intégrale complète de l'équation de Pfaff.

Le théorème de Jacobi peut être considéré comme un cas particulier de la théorie des multiplicateurs. En effet, si l'on a un système d'équations différentielles

$$\frac{dx_1}{X_1} = \frac{dx_2}{X_2} = \cdots = \frac{dx_n}{X_n},$$

et si M est un multiplicateur, on a

$$\frac{d(MX_1)}{dx_1} + \frac{d(MX_2)}{dx_2} + \cdots + \frac{d(MX_n)}{dx_n} = 0.$$

Pour le problème de Pfaff, on peut établir un résultat analogue, et c'est ce que fait voir la démonstration que j'ai donnée ci-dessus.

Londres, Avril 1859.

273. Note sur la transformation de Tschirnhausen.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LVIII. (1861), pp. 259–262.]

Considérons l'équation

$$(a, b, c, d, e \setminus x, 1)^4 = 0,$$

et la substitution

$$y = (ax + b)T_0 + (ax^2 + 2bx + c)T_1 + (ax^3 + 3bx^2 + 3cx + d)T_2.$$

Pour obtenir l'équation transformée en y , je représente par V la valeur de y , et je forme la dérivée ∇V au moyen des opérateurs

$$\nabla = (a, b, c, d \setminus x, 1)^3 \partial_x, \quad \nabla' = (b, c, d, e \setminus x, 1)^3 \partial_x.$$

On a d'abord

$$\nabla V = aT_0 + (2ax + b)T_1 + (3ax^2 + 4bx + c)T_2,$$

et je représente cette valeur par W .

Pour obtenir ∇W , je forme d'abord la dérivée au moyen de l'opérateur ∂_x , ce qui donne

$$\partial_x W = 2aT_1 + (6ax + 4b)T_2,$$

et en multipliant par $(a, b, c, d \setminus x, 1)^3$, on obtient

$$\nabla W = (2a^2x + 2ab)T_1 + (6a^2x^2 + 10abx + 4b^2 + 2ac)T_2.$$

De même, en représentant le second opérateur par

$$\nabla' W = x(b, c, d, e \setminus x, 1)^3 \partial_x W,$$

on trouve d'abord

$$(a, b, c, d \setminus x, 1)^3 \partial_x W + x(b, c, d, e \setminus x, 1)^3 \partial_x W = 0,$$

mais en ayant égard à l'équation $(a, b, c, d, e \setminus x, 1)^4 = 0$ la valeur de $\nabla' x$ se réduit à $\nabla' x = ax^2$.

La partie de $\nabla'V$ due à la variation de x est par conséquent

$$aa^2T_0 + (2aa^2 + 4bx)ax^2T_1 + (3aa^2 + 8bx + 6c)ax^2T_2.$$

L'autre partie de $\nabla'V$ est

$$(4bx + 3c)T_0 + (4bx^2 + 12cx + 6d)T_1 + (4bx^3 + 12cx^2 + 12dx + 3e)T_2.$$

En les ajoutant, le coefficient de T_0 s'évanouit en vertu de l'équation en x , et l'on trouve

$$\nabla'V = (ax^2 + 4bx + 3c)T_0 + 2(ax^3 + 4bx^2 + 6cx + 2d)T_1 + 3(ax^4 + 4bx^3 + 6cx^2 + 4dx + e)T_2,$$

où le coefficient de T_2 s'évanouit en vertu de la même équation.

On a ainsi les deux équations

$$\nabla V = W, \quad \nabla'V = \nabla W,$$

et l'équation transformée en y est donnée par l'élimination de x entre les trois équations

$$y - V = 0, \quad \nabla V - W = 0, \quad \nabla'V - \nabla W = 0.$$

Cette élimination conduit au déterminant

$$\begin{vmatrix} y - bT_0 - cT_1 - dT_2 & -aT_0 - 2bT_1 - 3cT_2 & -aT_1 - 3bT_2 & -aT_2 \\ -bT_0 - 2cT_1 - 3dT_2 & y + 2bT_0 + cT_1 & -aT_0 - 2bT_1 - 3cT_2 & -aT_1 - 3bT_2 \\ -3cT_0 - 6dT_1 - eT_2 & 3cT_0 + 2dT_1 & y + 3cT_1 + dT_2 & -aT_0 - 2bT_1 - 3cT_2 \\ -dT_0 - 4eT_1 & 4dT_0 + 3eT_1 & 2dT_1 + eT_2 & y + 4dT_2 \end{vmatrix}$$

En ordonnant ce déterminant suivant les puissances de y , les coefficients de y^3 , y^2 , y , y^0 deviennent des formes binaires en T_0 , T_1 des ordres 0, 1, 2, 3. En calculant les valeurs de ces quatre coefficients on les trouve respectivement

$$= 1,$$

$$= 0,$$

$$= 3(ac - b^2, ad - bc, bd - c^2 \setminus T_0, T_1)^2,$$

$$= (a^2d - 3abc + 2b^3, 3abd - 6ac^2 + 3b^2c, -3acd + 6b^2d - 3bc^2, -ad^2 + 3bcd - 2c^3$$

c'est-à-dire que l'équation en y est celle-ci:

$$y^4 + 3y(ac - b^2, ad - bc, bd - c^2 \setminus T_0, T_1)^2 + (a^2d - 3abc + 2b^3, 3abd - 6ac^2 + 3b^2c, -3acd + 6b^2d - 3bc^2, -ad^2 + 3bcd - 2c^3 \setminus T_0, T_1)^3 = 0$$

équation qui remplit en effet la condition ci-dessus mentionnée, d'être un invariant des deux formes $(a, b, c, d \setminus x, y)^3$ et $(T_0, T_1 \setminus \xi, -\eta)^2$.

274. Deuxième Note sur la transformation de Tschirnhausen.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. LVIII. (1861), pp. 263–269.]

À la fin de ma première note sur ce sujet j'ai appliqué la transformation de Tschirnhausen à l'équation du troisième degré mise sous la forme

$$ax^3 + 3bx^2 + 3cx + d = (a, b, c, d \setminus x, 1)^3 = 0.$$

En y substituant $\frac{1}{2}b$, $\frac{1}{2}c$ au lieu de b , c , cette équation se change en

$$ax^3 + 3bx^2 + 3cx + d = (a, b, c, d \setminus x, 1)^3 = 0,$$

et en même temps le résultat obtenu dans ma première note s'énonce de la manière suivante:

En calculant pour l'équation

$$(a, b, c, d \setminus x, 1)^3 = 0$$

la transformée en

$$y = (ax + b)T_0 + (ax^2 + 2bx + c)T_1,$$

on obtient

$$(1, 0, \mathfrak{C}, \mathfrak{D} \setminus y, 1)^3 = 0,$$

où

$$\mathfrak{C} = (ac - b^2, ad - bc, bd - c^2 \setminus T_0, T_1)^2,$$

$$\mathfrak{D} = (a^2d - 3abc + 2b^3, 3abd - 6ac^2 + 3b^2c, -3acd + 6b^2d - 3bc^2, -ad^2 + 3bcd - 2c^3).$$

Je vais me servir de cette formule, pour en déduire l'équation qui d'une manière analogue est la transformée de l'équation du quatrième ordre

$$(a, b, c, d, e \setminus x, 1)^4 = 0.$$

J'écris d'abord

$$(a, b, c, d, e \setminus x, 1)^4 = 0,$$

$$y = (ax + \tfrac{1}{2}b)T_0 + (ax^2 + bx + \tfrac{1}{2}c)T_1 + (ax^3 + bx^2 + cx + \tfrac{1}{2}d)T_2,$$

et je remarque qu'en faisant $e = 0$, le système proposé se partage en deux, dont le premier est:

$$x = 0, \quad y = (bT_0 + 2cT_1 + 3dT_2),$$

et le second:

$$(a, b, c, d \setminus x, 1)^3 = 0, \quad y = (ax + \tfrac{1}{2}b)T_0 + (ax^2 + bx + \tfrac{1}{2}c)T_1 - (dT_2);$$

ou, ce qui est la même chose,

$$(a, b, c, d \setminus x, 1)^3 = 0, \quad y + dT_2 = (ax + \tfrac{1}{2}b)T_0 + (ax^2 + bx + \tfrac{1}{2}c)T_1.$$

Or, la transformée de ce dernier système est

$$(1, 0, \mathfrak{C}, \mathfrak{D}' \setminus y + dT_2, 1)^3 = 0,$$

où $\mathfrak{C}$ a la valeur ci-dessus et

$$\begin{aligned} \mathfrak{D}' = & (a^2d - 3abc + 2b^3, 3abd - 6ac^2 + 3b^2c, \\ & -3acd + 6b^2d - 3bc^2, -ad^2 + 3bcd - 2c^3 \setminus T_0, T_1)^3. \end{aligned}$$

De là on obtient l'équation en y :

$$(y - bT_0 - 2cT_1 - 3dT_2)(1, 0, \mathfrak{C}, \mathfrak{D}' \setminus y + dT_2, 1)^3 = 0,$$

et en développant

$$y^4 - (bT_0 + 2cT_1 + 3dT_2)y^3 + \mathfrak{C}(y^2 + ydT_2 + d^2T_2^2) + \mathfrak{D}'(y + dT_2) = 0.$$

Cela ne présente pas de difficulté, je supprime donc les calculs intermédiaires et je donne le résultat final que voici: les Équations

$$\begin{aligned} (a, b, c, d, e \setminus x, 1)^4 = 0, \\ y = (ax + b)T_0 + (ax^2 + 2bx + 3c)T_1 + (ax^3 + 3bx^2 + 3cx + d)T_2, \end{aligned}$$

conduisent à la transformée:

$$(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \setminus y, 1)^4 = 0,$$

où l'on a

	T_0^2	T_0T_1	T_1^2	T_0T_2	T_1T_2	T_2^2
$\mathfrak{C} =$	$ac + 3$	$ad + 6$	$bd - 2$	$ae + 2$	$be + 6$	$ce + 3$
		$bc - 6$		$cd - 6$	$d^2 - 3$	

et

T_2^3	T_0^3	$T_0^2T_1$	$T_0T_1^2$	T_1^3	$T_0^2T_2$	$T_0T_1T_2$	$T_1^2T_2$	$T_0T_2^2$
ad^2- $\mathfrak{D} =$ b^2e+	a^2d+1	$abc-3$	b^3+2	a^2e+1	$abd+2$	ac^2-9	b^2c+6	$abe+$
		b^2c+2		a^2e+1	ac^2-9	b^2d+8	$abe+1$	$ace-$
					b^2c+4	$acd+8$	ad^2-	$+bcd$

et je représente par $4\Phi'$ la valeur qui vient d'être trouvée pour $\mathfrak{D}$. Ces expressions $\mathfrak{C}$, $\mathfrak{H}'$, $4\Phi'$ sont des invariants des deux formes $(a, b, c, d, e \setminus \xi, \eta)^4$, $(T_0, T_1, T_2 \setminus \eta, -\xi)^3$; on a de plus les invariants

$$ae - 4bd + 3c^2, \quad ace - ad^2 - b^2e + 2bcd - c^3,$$

que je représente comme à l'ordinaire par I , J , et l'invariant $T_0T_2 - T_1^2$ que je représente par Θ' . Cela posé on a

$$4\mathfrak{C} = 6\mathfrak{H}' - 2I\Theta', \quad 3\mathfrak{D} = 4\Phi',$$

$$\mathfrak{C} = 27\Phi'^2 - 8\mathfrak{H}'^3 + I\mathfrak{H}'^2\Theta' + 12J\mathfrak{H}'\Theta'^2 + 2I\Theta'\mathfrak{H}'.$$

La dernière de ces équations peut être vérifiée aisément, pour cela on a seulement besoin de remarquer qu'en posant $a = e = 1$, $b = d = 0$, $c = \theta$, elle devient

$$27\Phi'^2 - 8\mathfrak{H}'^3 + I\mathfrak{H}'^2\Theta' + 12J\mathfrak{H}'\Theta'^2 + 2I\Theta'\mathfrak{H}' = 0,$$

ce qui est égal à

$$J\{4\mathfrak{H}'^3 + QJ\Theta'\mathfrak{H}' + (I^3 - 27J^2)\Theta'^3\}.$$

La condition à remplir pour que cet invariant se réduise à zéro peut donc être présentée sous la forme

$$I\mathfrak{H}'^3 + \{6J \pm 2\sqrt{-1}(I^3 - 27J^2)\}\Theta'^3 = 0,$$

ce qui s'accorde avec un résultat trouvé par M. Hermite.

Il doit y avoir, ce me semble, une équation identique de la forme

$$J\mathfrak{H}'^6 - I\mathfrak{H}'^4\Phi' + m\Phi'^2 + M\Theta'^6 = -\Phi'^2,$$

qui servirait à exprimer le carré de Φ' au moyen des autres invariants $\mathfrak{C}$, $\mathfrak{H}'$, Θ' , I , J , mais en supposant que cette équation existe, reste à déterminer; le facteur M , que je n'ai pas encore cherchée, reste à déterminer; l'invariant J (cubinvariant) de la forme $(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{C} \setminus y, 1)^4$, et l'on doit avoir $J = 0$ pour que la réduction ait lieu.

Pour la forme quartique $(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \backslash y, 1)^4$, on a les covariants suivants:

$T_1^2 T_2^2$	T_0^4	$T_0^3 T_1$	$T_0^2 T_1^2$	$T_0 T_1^3$	T_1^4	$T_0^3 T_2$	$T_0^2 T_1 T_2$	$T_0 T_1^2 T_2$	$T_1^3 T_2$	$T_0 T_1 T_2^2$	T_2^4
$\mathfrak{E} =$	$a^2 e + 256$	$a^2 b e + 512$	$a^2 c e + 512$	$a^2 d e - 256$							
		$a b c e + 640$									
		$a^2 d e - 256$	$a^2 e^2 + 512$	$a b d e +$	$a^2 e^2 - 1024$						
				$a^2 b d - 64$							

275. On Tschirnhausen's Transformation.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLI. (for the year 1861), pp. 561–578. Received November 7,—Read December 5, 1861.]

The memoir of M. Hermite, “Sur quelques théorèmes d’algèbre et la résolution de l’équation du quatrième degré,” *Comptes Rendus*, t. XLVI. p. 961 (1858), contains a very important theorem in relation to Tschirnhausen’s Transformation of an equation $f(x) = 0$ into another of the same degree in y , by means of the substitution $y = \phi x$, where ϕ is a rational and integral function of x . In fact, considering for greater simplicity a quartic equation,

$$(a, b, c, d, e \setminus x, 1)^4 = 0,$$

M. Hermite gives to the equation $y = \phi x$ the following form,

$$y = aT + (ax + 4b)C + (ax^2 + 4bx + 6c)D + (ax^3 + 4bx^2 + 6cx + 4d)D,$$

(I write B, C, D in the place of his T_0, T_1, T_2), and he shows that the transformed equation in y has the following property: viz., every function of the coefficients which, expressed as a function of $a, b, c, d, e, T, B, C, D$, does not contain T , is an invariant, that is, an invariant of the two quantics

$$(a, b, c, d, e \setminus X, Y)^4, \quad (B, C, D \setminus Y, -X)^3.$$

This comes to saying that if T be so determined that in the equation for y the coefficient of the second term (y^3) shall vanish, the other coefficients will be invariants; or if, in the function of y which is equated to zero, we consider y as an absolute constant, the function of y will be an invariant of the two quantics. It is easy to find the value of T ; this is in fact given by the equation

$$0 = aT + 3bB + 3cC + dD.$$

and we have thence for the value of y ,

$$y = (ax + b)B + (ax^2 + 4bx + 3c)C + (ax^3 + 4bx^2 + 6cx + 3d)D;$$

so that for this value of y the function of y which equated to zero gives the transformed equation will be an invariant of the two quantics. It is proper to notice that in the last-mentioned expression for y , all the coefficients except those of the term in x , or $bB + 3cC + 3dD$, are those of the binomial $(1, 1)^3$, whereas the excepted coefficients are those of the binomial $(1, 1)^2$; this suffices to show what the expression for y is in the general case.

I have in the two papers, "Note sur la Transformation de Tschirnhausen," and "Deuxième Note sur la Transformation de Tschirnhausen," Crelle, t. LVIII. pp. 259 and 263 (1861), [273 and 274], obtained the transformed equations for the cubic and quartic equations; and by means of a grant from the Government Grant Fund, I have been enabled to procure the calculation, by Messrs Davis and Otter, under my superintendence, of the transformed equation for the quintic equation. The several results are given in the present memoir; and for greater completeness, I reproduce the demonstration which I have given in the former of the above-mentioned two Notes, of the general property, that the function of y is an invariant. At the end of the memoir I consider the problem of the reduction of the general quintic equation to Mr Jerrard's form $x^5 + ax + b = 0$.

Considering for simplicity the foregoing two equations

$$(a, b, c, d, e \setminus x, 1)^4 = 0,$$

$$y = (ax + b)B + (ax^2 + 4bx + 3c)C + (ax^3 + 4bx^2 + 6cx + 4d)D;$$

let the second of these be represented by $y = V$, the transformed equation in y is

$$(y - V_1)(y - V_2)(y - V_3)(y - V_4) = 0,$$

where V_1, V_2, V_3, V_4 are what V becomes upon substituting therein for x the roots x_1, x_2, x_3, x_4 of the quartic equation respectively. Considering y as a constant, the conditions to be satisfied in order that the function in y may be an invariant are that this function shall be reduced to zero by each of the two operators

$$\begin{aligned} a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e - (B\partial_C + 2C\partial_D), \\ 4b\partial_a + 3c\partial_b + 2d\partial_c + e\partial_d - (2C\partial_B + D\partial_C). \end{aligned}$$

These conditions will be satisfied if each of the factors $y - V_1$, etc. has the property in question; that is, if $y - V$, or (what is the same thing) if V , supposing that x denotes therein a root of the quartic equation, is reduced to zero by each of the two operators. Considering the first operator, which for shortness I represent by

$$A = (D\partial_D + 2C\partial_C),$$

in order to obtain AV we require the value of Ax . To find it, operating with A on the quartic equation, we have

$$(a, b, c, d \setminus x, 1)^3 Ax + (a, b, c, d \setminus x, 1)^4 = 0,$$

or $Ax = -1$. In AV , the part which depends on the variation of Ax then is

$$-aB + (-2ax - 4b)C + (-3ax^2 - 8bx - 6c)D,$$

and the other part of AV is at once found to be

$$+aB + (4ax + 6b)C + (4ax^2 + 12bx + 9c)D;$$

whence, adding,

$$AV = 2(ax + b)C + (ax^2 + 4bx + 3c)D,$$

and this is precisely equal to

$$(D\partial_D + 2C\partial_C)V;$$

so that V is reduced to zero by the operator $A - (D\partial_D + 2C\partial_C)$.

Similarly, if the second operator is represented by

$$\nabla - (2C\partial_B + B\partial_C),$$

then we have

$$(a, b, c, d \setminus x, 1)^3 \nabla x + x (b, c, d, e \setminus x, 1)^3 = 0,$$

which by means of the equation

$$(a, b, c, d, e \setminus x, 1)^4 = 0$$

is reduced to $\nabla x = ax^2$. Hence in ∇V the part depending on the variation of x is

$$(aa^2)B + (2aa^2 + 4bx)C + (3ax^3 + 8bx^2 + 6cx)D,$$

and the other part of ∇V is at once found to be

$$(4bx + 3c)B + (4bx^2 + 12cx + 6d)C + (4bx^3 + 12cx^2 + 12dx + 3e)D;$$

and, adding, the coefficient of D vanishes on account of the quartic equation, and we have

$$\nabla V = (ax^2 + 4bx + 3c)B + 2(ax^3 + 4bx^2 + 6cx + 2d)C,$$

which is precisely equal to

$$(2C\partial_B + B\partial_C)V,$$

so that V is reduced to zero by the operator

$$\nabla - (2C\partial_B + B\partial_C),$$

which completes the demonstration; and the demonstration in the general case is precisely similar.

In the case of the cubic equation we have

$$\begin{aligned}(a, b, c, d \setminus x, 1)^3 &= 0, \\ y &= (ax + b)B + (ax^2 + 4bx + 2c)C;\end{aligned}$$

and writing the second equation in the form

$$(y - bB - 2cC) + x(-aB - 3bC) + x^2(-aC) = 0,$$

multiplying by x and reducing by the cubic equation, we have

$$dC + x(y - bB + cC) + x^2(-aB) = 0,$$

and repeating the process,

$$dB + x(-3cB + dC) + x^2(y + 2bB + cC) = 0;$$

or, what is the same thing, we have the system of equations

$$\begin{vmatrix} y - bB - 2cC & -aB - 3bC & -aC \\ dC & y - bB + cC & -aB \\ dB & 3cB + dC & y + 2bB + cC \end{vmatrix} = 0,$$

and the resulting equation in y is of course that formed by equating to zero the determinant formed out of the matrix in this equation. The developed expression is

$$(1, 0, \mathfrak{C}, \mathfrak{D} \backslash y, 1)^3 = 0,$$

where

		B^2	BC	C^2	
	ac	+1			
	b^2	-1			
$\mathfrak{C} =$	ad		+1		
	bc		-1		
	bd			+1	
	c^2			-1	

		B^3	B^2C	BC^2	C^3	
	abc	+1	-3	+2		
$\mathfrak{D} =$	abd		+3	-6	+3	
	ac^2			-3	+6	-3
	ad^2				-1	+3
						-2

The sum of the coefficients in each column should here and elsewhere in the present memoir be equal to zero, and I have by way of verification annexed to each column the sums (+ a number) of the

positive and negative coefficients. The coefficients $\mathfrak{C}$, $\mathfrak{D}$, and therefore the function in y , are invariants of the two forms,

$$(a, b, c, d \setminus X, Y)^3, \quad (B, C \setminus Y, -X)^2;$$

or in the present case, where there are only two coefficients B, C , the coefficients $\mathfrak{C}$, $\mathfrak{D}$, and therefore also the function in y , are covariants of the single form $(a, b, c, d \setminus B, C)^3$, considering therein (B, C) as the facients.

It may be remarked that in the present case, assuming the invariance of the function in y , we may obtain the transformed equation in a very simple manner by writing in the first instance $C = 0$, this gives

$$(a, b, c, d \setminus x, 1)^3 = 0, \quad y = (ax + b)B,$$

and thence

$$(a, b, c, d \setminus y - bB, aB)^3 = 0;$$

or developing,

$$y^3 + 3y(ac - b^2)B^2 + (a^2d - 3abc + 2b^3)B^3 = 0,$$

where the expressions for the coefficients are to be completed by the consideration that these coefficients are covariants of the form $(a, b, c, d \setminus B, C)^3$. But it is only in the case in hand of a cubic equation that the transformed equation can be obtained in this simple manner.

In the case of the quartic equation, we have as before

$$(a, b, c, d, e \setminus x, 1)^4 = 0, \\ y = (ax + b)B + (ax^2 + 4bx + 3c)C + (ax^3 + 4bx^2 + 6cx + 4d)D,$$

and the system of equations is

$$\begin{vmatrix} y - bB - 2cC - 3dD & -aB - 2bC - 3cD & -aC - 3bD & -aD \\ -dB - 2cC - 3dD & y + 2bB + cC & -aB - 2bC - 3cD & -aC - 3bD \\ -3cB - 6dC - eD & 3cB + 2dC & y + 3cC + dD & -aB - 2bC - 3cD \\ -dB - 4eC & 4dB + 3eC & 2dC + eD & y + 4dD \end{vmatrix}$$

whence the transformed equation is

$$(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E} \setminus y, 1)^4 = 0.$$

I write down the values as follows:

	B^2	BC	C^2	BD	CD	D^2
Values of $\mathfrak{C}$.	ac	+3				
	b^2	-1				
	ad		+6			
	bc		-6			
	ae		+2			
	bd		-2			
	be			+6		
	cd			-6		
	ce				+3	
	d^2				-3	

The sums of the coefficients are

$$B^2 : 3-1 = 2; \quad BC : 6-6 = 0; \quad C^2 : 2-2 = 0; \quad BD : 6-6 = 0; \quad CD : 3-3 = 0;$$

	Values of $\mathfrak{D}$.									
	B^3	B^2C	BC^2	C^3	B^2D	BCD	C^2D	BD^2	CD^2	D^3
a^2d	+1									
abc	-3	+1								
b^3	+2									
a^2e				+1						
abd				+2						
ac^2			-9							
b^2c			+6	+1						
abe				+4	+1					
acd				-3	+8					
b^2d				+8	-12					
ace						-12				
bcd						+16				
ad^2						-3	-4			
c^3						+3				
bce							+6			
c^2d							-8			
ade								-4		
cd^2								+6		
d^3										-1

The sums of the coefficients are:

$B^3 : +1 - 3 + 2 = 0; \quad B^2C : +1; \quad BC^2 : -9 + 6 + 3 = 0; \quad C^3 : +1;$

$B^2D : +2 + 8 - 12 + 4 = +2; \quad BCD : +1 - 3 - 12 + 16 = +2;$

$C^2D : +8 - 4 = +4; \quad BD^2 : +6 - 8 - 4 + 6 = 0; \quad CD^2 : -4 + 6 - 1 = +1.$

	Values of $\mathfrak{E}$.									
	B^4	B^3C	B^2C^2	BC^3	C^4	B^3D	B^2CD	BC^2D	C^3D	B^2D^2
a^3e	+1									
a^2bd	-4	+1								
a^2c^2	+2									
ab^2c	+6	-3								
b^4	-3									
a^2be						+2				
a^2cd						-6				
ab^2d						+8	+1			
abc^2						-18	+3			
b^3c						+12	-4			
a^2de										+1
$abce$										+4
ac^2d										-12
b^2c^2										+6
b^2de										+8
bc^2d										-18
c^4										+3
$abde$										
ac^2e										
bcd^2										
c^3d										
ad^3										
bd^2e										
c^2de										
d^4										

The sums of the coefficients are:

$B^4 : +1 - 4 + 2 + 6 - 3 = +2; \quad B^3C : +1 - 3 = -2; \quad B^2C^2 : +2;$
 $B^2D : +2 - 6 + 8 - 18 + 12 = -2; \quad BCD : +1 + 3 - 4 = 0; \quad C^2D : +8 - 12 + 6 = 2;$
 $BD^2 : +1 + 4 - 12 - 18 + 8 + 6 = -3; \quad CD^2 : +1 + 6 - 4 - 6 = -3; \quad D^3 : -1.$

The coefficients $\mathfrak{C}$, $\mathfrak{D}$, $\mathfrak{E}$ are invariants of the two forms

$$(a, b, c, d, e \setminus X, Y)^4, \quad (B, C, D \setminus Y, -X)^3.$$

I stop to verify the expression for $\mathfrak{E}$. Writing $a = e = 1$, $b = c = d = 0$, the equation in x becomes $x^4 + 1 = 0$, and the value of y is

$$y = xB + x^3D.$$

If x_1, x_2, x_3, x_4 are the roots of $x^4 + 1 = 0$, then $x_1 + x_2 + x_3 + x_4 = 0$, $\sum x_1x_2 = 0$, $\sum x_1x_2x_3 = 0$, $x_1x_2x_3x_4 = 1$.

The equation in y is therefore

$$y^4 + \mathfrak{C}y^2 + \mathfrak{D}y + \mathfrak{E} = 0,$$

and we ought to have $\mathfrak{C} = 0$, $\mathfrak{D} = 0$, and $\mathfrak{E}$ equal to the product $(y - y_1)(y - y_2)(y - y_3)(y - y_4)$. In fact, writing $x^4 = -1$, and therefore $x^2 = \pm i$, the equation $y = xB + x^3D$ gives

$$y^2 = x^2B^2 + 2x^4BD + x^6D^2 = \pm i(B^2 - 2BD \mp iD^2),$$

and thence

$$y^4 = -(B^2 - 2BD - iD^2)(B^2 - 2BD + iD^2) = -(B^4 - 4B^3D + 4B^2D^2 + D^4),$$

which is right.

Passing to the quintic equation, we have

$$\begin{aligned} (a, b, c, d, e, f \setminus x, 1)^5 &= 0, \\ y &= (ax + b)B + (ax^2 + 4bx + 3c)C + (ax^3 + 4bx^2 + 6cx + 4d)D \\ &\quad + (ax^4 + 4bx^3 + 6cx^2 + 4dx + 5e)E, \end{aligned}$$

or, as this may be written,

$$\begin{aligned} y &= (ax + b)B + (ax^2 + 4bx + 3c)C + (ax^3 + 4bx^2 + 6cx + 4d)D \\ &\quad + (ax^4 + 4bx^3 + 6cx^2 + 4dx + 5e)E, \end{aligned}$$

where the coefficients are those of the expansion of $(1 + x)^4$, $(1 + x)^3$, $(1 + x)^2$, $(1 + x)^1$, $(1 + x)^0$, except that in the several functions the highest coefficient of each is multiplied by 1, 2, 3, 4, 5 respectively.

The resulting equation is

$$(1, 0, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}, \mathfrak{F} \setminus y, 1)^5 = 0.$$

The values of the several coefficients are as follows:

	Values of $\mathfrak{C}$.									
	B^2	BC	C^2	BD	CD	D^2	BE	CE	DE	E^2
ac	+6									
b^2	-1									
ad		+12								
bc		-12								
ae			+6							
bd			-4							
af							+2			
be							+6			
cd							-6			
bf									+3	
ce									-6	
d^2									+3	
cf										+6
de										-8
df										
+4										
e^2										
-1										

Values of $\mathfrak{D}$.

Values of $\mathfrak{E}$ (part).

The coefficients $\mathfrak{C}$, $\mathfrak{D}$, $\mathfrak{E}$, $\mathfrak{F}$ are invariants of the two quantics

$$(a, b, c, d, e, f \setminus X, Y)^5, \quad (B, C, D, E \setminus Y, -X)^4.$$

This [the circumstance that the values thus become covariants of the quintic] is a further verification of the foregoing result.

I will conclude by showing how the formula may be applied to the reduction of the general quintic equation to Mr Jerrard's form $x^5 + ax + b = 0$. It is to be observed that the formulae contain linearly the coefficients B, C, D, E , and that these may be so determined that the transformed equation shall want the terms in y^4, y^3, y^2 simultaneously. We have to satisfy the three conditions $\mathfrak{C} = 0, \mathfrak{D} = 0, \mathfrak{E} = 0$; and since the coefficients B, C, D, E are disposable, we have just enough of them for the purpose. The theory of the invariants of the quintic shows that the conditions may be satisfied, and that the reduction to Mr Jerrard's form is always possible.

2, Stone Buildings, W.C.,
November 7, 1861.

276. On the Analytical Theory of the Conic.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLI. (for the year 1861), pp. 787–796. Received November 7,—Read December 5, 1861.]

1. The equation of a conic may be written in the form

$$(a, b, c, f, g, h \backslash x, y, z)^2 = 0,$$

and if (x, y, z) are current coordinates, (x', y', z') the coordinates of a given point, then the equation

$$(a, b, c, f, g, h \backslash x, y, z \backslash x', y', z') = 0$$

represents the polar line of the given point; and the equation

$$(a, b, c, f, g, h \backslash x', y', z')^2 = 0$$

expresses that the given point is on the conic.

The equation of the pair of lines from the given point to the conic is

$$(a, b, c, f, g, h \backslash x, y, z)^2 (a, b, c, f, g, h \backslash x', y', z')^2 - \{(a, b, c, f, g, h \backslash x, y, z \backslash x', y', z')\}^2 = 0$$

2. If the point (x', y', z') lies on the conic, then the last equation becomes

$$(a, b, c, f, g, h \backslash x, y, z)^2 \times 0 - \{(a, b, c, f, g, h \backslash x, y, z \backslash x', y', z')\}^2 = 0,$$

which is the equation of the tangent line at the point (x', y', z') .

3. The condition that the line (ξ, η, ζ) may touch the conic is

$$(A, B, C, F, G, H \backslash \xi, \eta, \zeta)^2 = 0,$$

where (A, B, C, F, G, H) are the reciprocals of (a, b, c, f, g, h) ; viz.,

$$\begin{aligned} A &= bc - f^2, & B &= ca - g^2, & C &= ab - h^2, \\ F &= gh - af, & G &= hf - bg, & H &= fg - ch. \end{aligned}$$

4. If (ξ, η, ζ) are current line-coordinates, then the equation

$$(A, B, C, F, G, H \backslash \xi, \eta, \zeta)^2 = 0$$

is the line-equation of the conic.

5. The equation of the point-pair, consisting of the two points of contact of tangents from a given line (ξ', η', ζ') to the conic, is

$$(A, B, C, F, G, H \setminus \xi, \eta, \zeta)^2 (A, B, C, F, G, H \setminus \xi', \eta', \zeta')^2 - \{(A, B, C, F, G, H \setminus \xi, \eta, \zeta \setminus \xi', \eta', \zeta')\}^2 = 0,$$

6. If the line (ξ', η', ζ') touches the conic, then this equation becomes

$$-\{(A, B, C, F, G, H \setminus \xi, \eta, \zeta \setminus \xi', \eta', \zeta')\}^2 = 0,$$

which is the equation of the point of contact of the line (ξ', η', ζ') .

7. I notice also the theorem

$$(a, \dots \backslash x, y, z)^2 (a, \dots \backslash x', y', z')^2 - \{(a, \dots \backslash x, y, z \backslash x', y', z')\}^2 \\ = (A, \dots \backslash yz' - y'z, zx' - z'x, xy' - x'y)^2.$$

8. The line joining two points (x', y', z') and (x'', y'', z'') has for its equation

$$\begin{vmatrix} x & y & z \\ x' & y' & z' \\ x'' & y'' & z'' \end{vmatrix} = 0,$$

or, developing,

$$(yz' - y'z)x + (zx' - z'x)y + (xy' - x'y)z = 0.$$

The point of intersection of two lines (ξ', η', ζ') and (ξ'', η'', ζ'') has for its equation

$$\begin{vmatrix} \xi & \eta & \zeta \\ \xi' & \eta' & \zeta' \\ \xi'' & \eta'' & \zeta'' \end{vmatrix} = 0.$$

9. The condition that the point (x', y', z') may lie on the line (ξ, η, ζ) is

$$\xi x' + \eta y' + \zeta z' = 0.$$

10. If $(a, \dots \backslash x, y, z)^2$ is a perfect square, then $a' : \beta' : \gamma'$ are proportional to

$$(bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch \backslash \lambda, \mu, \nu)^2,$$

and the coordinates of the point are proportional to (λ, μ, ν) .

11. The expression

$$(a, b, c, f, g, h \backslash x, y, z)^2$$

may be written in the eight forms I, II, III, IV, I (bis), II (bis), III (bis), IV (bis), which are the analytical basis of the whole theory. These forms correspond to different ways of expressing the quadratic function as a sum of squares.

12. The forms are as follows:

Form I.

$$\begin{aligned} \Pi &= (ax + hy + gz)x + (hx + by + fz)y + (gx + fy + cz)z, \\ &= \xi x + \eta y + \zeta z, \end{aligned}$$

where ξ, η, ζ are the derivatives of Π with respect to x, y, z .

Form II. Using the reciprocals,

$$\Pi = (A\xi + H\eta + G\zeta)\xi + (H\xi + B\eta + F\zeta)\eta + (G\xi + F\eta + C\zeta)\zeta.$$

Form III. Introducing the determinant $\nabla = abc + 2fgh - af^2 - bg^2 - ch^2$,

$$\nabla \cdot \Pi = (A, H, G \backslash \xi, \eta, \zeta)^2.$$

Form IV. In terms of the line-coordinates,

$$\Pi = \frac{(A, H, G \backslash \xi, \eta, \zeta)^2}{\nabla}.$$

The forms I (bis), II (bis), III (bis), IV (bis) are obtained from the above by interchanging the point-coordinates and line-coordinates; viz.,

Form I (bis).

$$\Pi = x \partial_x \Pi + y \partial_y \Pi + z \partial_z \Pi.$$

Form II (bis).

$$\Pi = \xi \partial_\xi \Pi + \eta \partial_\eta \Pi + \zeta \partial_\zeta \Pi.$$

Form III (bis).

$$\nabla \cdot \Pi = (\text{analogous expression in line-coordinates}).$$

Form IV (bis).

$$\Pi = \frac{(\text{expression in point-coordinates})}{\nabla}.$$

The greater part of the memoir relates to the establishment of these forms.

The solution of the geometric problem of finding the foci of a conic given by its general equation depends on these formulae. The foci are the points from which the tangents to the conic make equal angles with the axes; analytically, they are the points (x, y, z) such that the distances to the conic satisfy certain conditions.

For a conic referred to its principal axes, the equation is

$$\frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} = 1,$$

and the foci are at $(\pm c, 0)$, where $c^2 = \alpha^2 - \beta^2$.

In the general case, the determination of the foci requires the solution of a system of equations derived from the eight forms mentioned above. The result is that the foci are given as the intersections of certain pairs of lines which are completely determined by the coefficients of the equation of the conic.

COLLECTED MATHEMATICAL PAPERS
OF
ARTHUR CAYLEY, Vol. IV

Pages 401–450

[276]

ON THE ANALYTICAL THEORY OF THE CONIC.

16. The last equation shows that $(a, .(\xi)\eta\zeta' - \eta'\zeta, \zeta\xi' - \zeta'\xi, \xi\eta' - \xi'\eta)^2$, considered as a function of (ξ', η', ζ') , breaks up into factors. Or since the expression is not altered by interchanging (ξ', η', ζ') and (ξ, η, ζ) , the same expression, considered as a function of (ξ, η, ζ) , breaks up into factors. It is in fact easy to see that any quantic whatever, $(*(\xi)\eta\zeta' - \eta'\zeta, \zeta\xi' - \zeta'\xi, \xi\eta' - \xi'\eta)^m$, considered as a function of (ξ, η, ζ) , breaks up into linear factors; for in virtue of the equation $\xi'(\eta\zeta' - \eta'\zeta) + \eta'(\zeta\xi' - \zeta'\xi) + \zeta'(\xi\eta' - \xi'\eta) = 0$, any one of the quantities $\eta\zeta' - \eta'\zeta, \zeta\xi' - \zeta'\xi, \xi\eta' - \xi'\eta$ can be expressed as a linear function of the other two; so that the quantic can be expressed as a linear function of any two of the three quantities; and *quâ* homogeneous function of two quantities, it of course breaks up into factors, linear functions of these two quantities.

We may in all the formulæ interchange (x', y', z') and (x, y, z) , writing (a', b', c', f', g', h') in the place of (a, b, c, f, g, h) .

17. Putting, in like manner,

$$(A, B, C, F, G, H)(yz' - y'z, zx' - z'x, xy' - x'y)^2 = (, , , , ,)(x', y', z')^2,$$

so that

$$\begin{aligned} &= Cy^2 + Bz^2 - 2Fyz, & &= Az^2 + Cx^2 - 2Gzx, \\ &= Bx^2 + Ay^2 - 2Hxy, \\ &= -Ayz - Fx^2 + Gxy + Hzx, \\ &= -Bzx + Fxy - Gy^2 + Hyz, \\ &= -Cxy + Fzx + Gyz - Hz^2, \end{aligned}$$

we obtain

$$\begin{vmatrix} , & , \\ , & , \\ , & , \end{vmatrix} (x, y, z) = (0, 0, 0),$$

viz.

$$x + y + z = 0, \text{ \&c.}$$

Also

$$\begin{aligned} & (-^2, -^2, -^2, \\ & \quad -^2, -^2, -^2) \\ & = (x, y, z)^2 \cdot K(a, b, c, f, g, h)x, y, z)^2; \end{aligned}$$

that is

$$-^2 = x^2 K(a, b, c, f, g, h)x, y, z)^2, \text{ \&c.};$$

whence also

$$\begin{aligned} & (-^2, \dots) \lambda, \mu, \nu)^2 = (\lambda x + \mu y + \nu z)^2 \cdot K(a, \dots)x, y, z)^2, \\ & (-^2, \dots) \lambda, \mu, \nu) \lambda', \mu', \nu') = (\lambda x + \mu y + \nu z)(\lambda' x + \mu' y + \nu' z) \cdot K(a, \dots)x, y, z)^2; \end{aligned}$$

and moreover

$$-^2 -^2 -^2 + 2 = 0.$$

[276] (continued)

ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

The last equation shows that $(A, \dots)yz' - y'z, zx' - z'x, xy' - x'y)^2$, considered as a function of (x', y', z') , breaks up into factors, or (what is the same thing) this expression, considered as a function of (x, y, z) , breaks up into factors; we may in all the formulæ interchange (x, y, z) and (x', y', z') , writing $(', ', ', ', ', ')$ in the place of $(, , , , ,)$.

Article Nos. 18 to 28, relating to a single conic in connexion with a point or line.

18. I apply the decomposition formula to the function $(A, \dots)yz' - y'z, \dots)^2$, which, considered as a function of (x, y, z) , breaks up into factors. We have

$$\begin{aligned} (A, \dots)yz' - y'z, \dots)^2 &= (', \dots)x, y, z)^2 \\ &= \frac{1}{(', \dots)l, m, n)^2} \text{Product of} \\ &\quad (', \dots)l, m, n(x, y, z) \pm \frac{('' - r^2, \dots)mz - ny, \dots)\lambda, \mu, \nu)}{\sqrt{-('' - r^2, \dots)\lambda, \mu, \nu)^2}}. \end{aligned}$$

But we have

$$\begin{aligned} (', \dots)l, m, n)^2 &= (A, \dots)mz' - ny', \dots)^2, \\ (', \dots)l, m, n(x, y, z) &= (A, \dots)mz' - ny', \dots)yz' - y'z, \dots), \\ ('' - r^2, \dots)mz - ny, \dots)\lambda, \mu, \nu) &= [x'(mz - ny) + y'(nx - lz) + z'(ly - mx)](\lambda x' + \mu y' + \nu z'), \\ ('' - r^2, \dots)\lambda, \mu, \nu)^2 &= (\lambda x' + \mu y' + \nu z')^2 \end{aligned}$$

and thence

$$\frac{('' - r^2, \dots)mz - ny, \dots)\lambda, \mu, \nu)}{\sqrt{-('' - r^2, \dots)\lambda, \mu, \nu)^2}} = \begin{vmatrix} x, & y, & z \\ x', & y', & z' \\ l, & m, & n \end{vmatrix} \sqrt{-K(a, \dots)x', y', z')^2},$$

whence we have

$$(A, \dots)yz' - y'z, \dots)^2 = \frac{1}{(A, \dots)mz' - ny'}^2 \text{ Product of}$$

$$(A, \dots)mz' - ny', \dots)yz' - y'z, \dots) \pm \begin{vmatrix} x, & y, & z \\ x', & y', & z' \\ l, & m, & n \end{vmatrix} \sqrt{-K(a, \dots)x', y', z')^2};$$

and the identical equation

$$(a, \dots)x, y, z)^2 \cdot (a, \dots)x', y', z')^2 - [(a, \dots)x, y, z)(x', y', z')]^2 = (A, \dots)yz' - y'z, \dots)^2$$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

now gives

$$\text{I. } (a, \dots)x, y, z)^2 = \text{Quotient by } (a, \dots)x', y', z')^2 \text{ of } \left\{ \begin{array}{l} [(a, \dots)x, y, z)x', y', z')]^2 \\ + \text{Quotient by } (A, \dots)mz' - ny' \\ \left\{ (A, \dots)mz' - ny', \dots)yz' - y'z \right\} \end{array} \right.$$

where the Product part may also be written

$$\begin{aligned} & (a, \dots)l, m, n)x', y', z') \cdot (a, \dots)x, y, z)x', y', z') \\ & - (a, \dots)x', y', z')^2 \cdot (a, \dots)x, y, z)l, m, n) \\ & \pm \sqrt{-K(a, \dots)x', y', z')^2} \begin{vmatrix} x, & y, & z \\ x', & y', & z' \\ l, & m, & n \end{vmatrix}. \end{aligned}$$

19. Writing in the formula I.

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix} (x', y', z') = (\xi', \eta', \zeta'),$$

we have

$$(x', y', z') = \frac{1}{K} \begin{vmatrix} A, & H, & G \\ H, & B, & F \\ G, & F, & C \end{vmatrix} (\xi', \eta', \zeta'),$$

and thence

$$\begin{aligned} K(a, \dots)x', y', z') &= (A, \dots)\xi', \eta', \zeta')^2, \\ (a, \dots)x, y, z)x', y', z') &= \xi'x + \eta'y + \zeta'z. \end{aligned}$$

Assume

$$(l, m, n) = (\nu\eta' - \mu\zeta', \lambda\zeta' - \nu\xi', \mu\xi' - \lambda\eta'),$$

then from the foregoing values of (x', y', z')

$$\begin{aligned} mz' - ny' &= \frac{1}{K} \left\{ (\lambda\zeta' - \nu\xi')(G\xi' + F\eta' + C\zeta') - (\mu\xi' - \lambda\eta')(H\xi' + B\eta' + F\zeta') \right\} \\ &= \frac{1}{K} \left\{ \lambda[\xi'(A\xi' + H\eta' + G\zeta') + \eta'(H\xi' + B\eta' + F\zeta') + \zeta'(G\xi' + F\eta' + C\zeta') \right. \\ &\quad \left. - \lambda\xi'(A\xi' + H\eta' + G\zeta') - \mu\xi'(H\xi' + B\eta' + F\zeta') - \nu\xi'(G\xi' + F\eta' + C\zeta') \right\} \end{aligned}$$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

that is

$$mz' - ny' = \frac{1}{K} \left\{ \lambda(A, \dots) \xi', \eta', \zeta')^2 - \xi'(A, \dots) \xi', \eta', \zeta') \lambda, \mu, \nu \right\},$$

and similarly

$$\begin{aligned} nx' - lz' &= \frac{1}{K} \left\{ \mu(A, \dots) \xi', \eta', \zeta')^2 - \eta'(A, \dots) \xi', \eta', \zeta') \lambda, \mu, \nu \right\}, \\ ly' - mx' &= \frac{1}{K} \left\{ \nu(A, \dots) \xi', \eta', \zeta')^2 - \zeta'(A, \dots) \xi', \eta', \zeta') \lambda, \mu, \nu \right\}; \end{aligned}$$

and thence

$$\begin{aligned} (A, H, G)mz' - ny', \dots) \\ = \frac{1}{K} \left\{ (A, H, G)\lambda, \mu, \nu) \cdot (A, \dots) \xi', \eta', \zeta')^2 \right. \\ \left. - (A, H, G)\xi', \eta', \zeta') \cdot (A, \dots) \xi', \eta', \zeta') \lambda, \mu, \nu \right\}, \end{aligned}$$

with the like equations, writing H, B, F and G, F, C in the place of A, H, G successively: and we then have

$$\begin{aligned} (A, \dots)mz' - ny', \dots)^2 \\ = \frac{1}{K} \left\{ (A, \dots)\lambda, \mu, \nu)mz' - ny', \dots) \cdot (A, \dots)\xi', \eta', \zeta')^2 \right. \\ \left. - (A, \dots)\xi', \eta', \zeta')mz' - ny', \dots) \cdot (A, \dots)\xi', \eta', \zeta') \lambda, \mu, \nu \right\}. \end{aligned}$$

But the foregoing values of $mz' - ny', nx' - lz', ly' - mx'$ give also

$$\begin{aligned} (A, \dots)\lambda, \mu, \nu)mz' - ny', \dots) \\ = \frac{1}{K} \left\{ (A, \dots)\lambda, \mu, \nu)^2 \cdot (A, \dots)\xi', \eta', \zeta')^2 - [(A, \dots)\lambda, \mu, \nu)\xi', \eta', \zeta')]^2 \right\}, \\ (A, \dots)\xi', \eta', \zeta')mz' - ny', \dots) \\ = \frac{1}{K} \left\{ (A, \dots)\lambda, \mu, \nu)\xi', \eta', \zeta') \cdot (A, \dots)\xi', \eta', \zeta')^2 - (A, \dots)\xi', \eta', \zeta')^2 \cdot (A, \dots)\lambda, \mu, \nu \right\}, \end{aligned}$$

so that

$$\begin{aligned}
 & (A, ..)mz' - ny', ..)^2 \\
 &= \frac{1}{K^2}(A, ..)\xi', \eta', \zeta')^2 \cdot \{(A, ..)\lambda, \mu, \nu)^2 \cdot (A, ..)\xi', \eta', \zeta')^2 - [(A, ..)\lambda, \mu, \nu)\xi', \eta', \\
 &= \frac{1}{K}(A, ..)\xi', \eta', \zeta')^2 \cdot (a, ..)\nu\eta' - \mu\zeta', ..)^2.
 \end{aligned}$$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

Similarly,

$$\begin{aligned} & (A, ..)mz' - ny', ..)yz' - y'z, ..) \\ &= \frac{1}{K} \left\{ (A, ..)\lambda, \mu, \nu)yz' - y'z, ..) \cdot (A, ..)\xi', \eta', \zeta')^2 \right. \\ & \quad \left. - (A, ..)\xi', \eta', \zeta')yz' - y'z, ..) \cdot (A, ..)\xi', \eta', \zeta')(\lambda, \mu, \nu) \right\}. \end{aligned}$$

But

$$\begin{aligned} & (A, ..)\lambda, \mu, \nu)yz' - y'z, ..) \\ &= (A\lambda + H\mu + G\nu)(yz' - y'z) \\ & \quad + (H\lambda + B\mu + F\nu)(zx' - z'x) \\ & \quad + (G\lambda + F\mu + C\nu)(xy' - x'y), \\ &= x[(G\lambda + F\mu + C\nu) - z'(H\lambda + B\mu + F\nu)] \\ & \quad + y[z'(A\lambda + H\mu + G\nu) - x'(G\lambda + F\mu + C\nu)] \\ & \quad + z[x'(H\lambda + B\mu + F\nu) - y'(A\lambda + H\mu + G\nu)], \end{aligned}$$

which, substituting for x', y', z' their values

$$(x', y', z') = \frac{1}{K} \begin{vmatrix} A, & H, & G \\ H, & B, & F \\ G, & F, & C \end{vmatrix} (\xi', \eta', \zeta'),$$

becomes

$$\begin{aligned} &= \frac{1}{K} \left\{ x[(BC - F^2)(\nu\eta' - \mu\zeta') + (FG - CH)(\lambda\zeta' - \nu\xi') + (HF - BG)(\mu\xi' - \lambda\eta')] \right. \\ & \quad + y[(FG - CH)(\nu\eta' - \mu\zeta') + (CA - G^2)(\lambda\zeta' - \nu\xi') + (GH - AF)(\mu\xi' - \lambda\eta')] \\ & \quad \left. + z[(HF - BG)(\nu\eta' - \mu\zeta') + (GH - AF)(\lambda\zeta' - \nu\xi') + (AB - H^2)(\mu\xi' - \lambda\eta')] \right\}, \end{aligned}$$

which is

$$= (a, ..)x, y, z)\nu\eta' - \mu\zeta', ..);$$

and by merely writing (ξ', η', ζ') in the place of (λ, μ, ν) , we have

$$(A, ..)\xi', \eta', \zeta')yz' - y'z, ..) = 0;$$

so that we find

$$\begin{aligned} & (A, ..)mz' - ny', ..)yz' - y'z, ..) \\ &= \frac{1}{K}(A, ..)\xi', \eta', \zeta')^2 \cdot (a, ..)x, y, z)\nu\eta' - \mu\zeta', \lambda\zeta' - \nu\xi', \mu\xi' - \lambda\eta'). \end{aligned}$$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

Now, writing the formula I. in the form

$$(a, \dots(x, y, z)^2 = \text{Quotient by } K(a, \dots(x', y', z')^2 \text{ of } \left\{ \begin{array}{l} K [(a, \dots(x, y, z)(x', y', z')]^2 \\ + \text{Quotient by } K(A, \dots)mz' - n \\ K^2 \{[(A, \dots)mz' - ny', \dots(yz' - y \end{array} \right.$$

the right-hand side is

$$= \text{Quotient by } (A, \dots(\xi', \eta', \zeta')^2 \text{ of } \left\{ \begin{array}{l} K (\xi'x + \eta'y + \zeta'z)^2 \\ + \text{Quotient by } (a, \dots)\nu\eta' - \mu\zeta', \dots)^2 (A, \dots(\xi', \eta', \zeta')^2 \\ \{[(a, \dots)\nu\eta' - \mu\zeta', \dots(x, y, z) \cdot (A, \dots(\xi', \eta', \zeta')^2] + \end{array} \right.$$

where

$$\Pi = K \begin{vmatrix} x, & y, & z \\ x', & y', & z' \\ l, & m, & n \end{vmatrix},$$

or, what is the same thing,

$$\Pi = \begin{vmatrix} x, & y, & z \\ Kx', & Ky', & Kz' \\ l, & m, & n \end{vmatrix} = \begin{vmatrix} x, & y, & z \\ A\xi' + H\eta' + G\zeta', & H\xi' + B\eta' + F\zeta', & G\xi' + F\eta' + C\zeta' \\ \nu\eta' - \mu\zeta', & \lambda\zeta' - \nu\xi', & \mu\xi' - \lambda\eta' \end{vmatrix}$$

More simply, the right-hand side is

$$= \text{Quotient by } (A, \dots(\xi', \eta', \zeta')^2 \text{ of } \left\{ \begin{array}{l} K (\xi'x + \eta'y + \zeta'z)^2 \\ + \text{Quotient by } (a, \dots)\nu\eta' - \mu\zeta', \dots)^2 \text{ of} \\ \{[(a, \dots)\nu\eta' - \mu\zeta', \dots(x, y, z)]^2 \cdot (A, \dots(\xi', \eta', \zeta')^2 + \end{array} \right.$$

or restoring the left-hand side, and resolving into its linear factors the function in {}, we have

$$\text{II. } (a, \dots(x, y, z)^2 = \text{Quotient by } (A, \dots(\xi', \eta', \zeta')^2 \text{ of } \left\{ \begin{array}{l} K (\xi'x + \eta'y + \zeta'z)^2 \\ + \text{Quotient by } (a, \dots)\nu\eta' - \mu\zeta', \dots)^2 \\ \{ \Pi \pm \sqrt{-(A, \dots(\xi', \eta', \zeta')^2 \cdot (a, \dots(x, y, z)^2} \end{array} \right.$$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

where Π has the value given above, which may also be written

$$\begin{aligned}\Pi = & (A, \dots) \xi', \eta', \zeta' (\lambda, \mu, \nu) (\xi' x + \eta' y + \zeta' z) \\ & - (A, \dots) \xi', \eta', \zeta')^2 (\lambda x + \mu y + \nu z).\end{aligned}$$

20. We deduce at once the inverse or reciprocal formulæ

$$\text{III. } (A, \dots) \xi, \eta, \zeta)^2 = \text{Quotient by } (A, \dots) \xi', \eta', \zeta')^2 \text{ of } \begin{cases} [(A, \dots) \xi, \eta, \zeta (\xi', \eta', \zeta')]^2 \\ + \text{Quotient by } (a, \dots) \nu \eta' - \mu \zeta' \\ \{ (a, \dots) \nu \eta' - \mu \zeta', \dots) \eta \zeta' - \eta' \zeta \} \end{cases}$$

where the Product part may also be written

$$\begin{aligned}\text{Product } & \left[\frac{1}{K} (A, \dots) \xi', \eta', \zeta' (\lambda, \mu, \nu) \cdot (A, \dots) \xi', \eta', \zeta' (\xi, \eta, \zeta) \right. \\ & - \frac{1}{K} (A, \dots) \xi', \eta', \zeta')^2 \cdot (A, \dots) \lambda, \mu, \nu (\xi, \eta, \zeta) \\ & \left. \pm \sqrt{-(A, \dots) \xi', \eta', \zeta')^2} \begin{vmatrix} \xi, & \eta, & \zeta \\ \xi', & \eta', & \zeta' \\ \lambda, & \mu, & \nu \end{vmatrix} \right].\end{aligned}$$

21. And also

$$\text{IV. } (A, \dots) \xi, \eta, \zeta)^2 = \text{Quotient by } (a, \dots) x', y', z')^2 \text{ of } \begin{cases} K(\xi x' + \eta y' + \zeta z')^2 \\ + \text{Quotient by } K(A, \dots) \lambda y' - \mu \zeta' \\ \{ K\Phi \pm \sqrt{-K(a, \dots) x', y', z')^2} \} \end{cases}$$

where

$$\Phi = \begin{vmatrix} \xi, & \eta, & \zeta \\ ax' + hy' + gz', & hx' + by' + fz', & gx' + fy' + cz' \\ ny' - mz', & lz' - nx', & mx' - ly' \end{vmatrix},$$

which may also be written

$$\begin{aligned}= & (a, \dots) x', y', z' (l, m, n) (x' \xi + y' \eta + z' \zeta) \\ & - (a, \dots) x', y', z')^2 \cdot (l \xi + m \eta + n \zeta).\end{aligned}$$

22. The geometrical signification is obvious. The formulæ I. and II. each of them show that the equation

$$(a, \dots)x, y, z)^2 = 0$$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

of the conic may be written in the form

$$W^2 + \frac{1}{M}QR = 0,$$

where $Q = 0$, $R = 0$ are any two tangents of the conic, and $W = 0$ is the line joining the points of contact, or chord of contact corresponding to the two tangents; viz., in the formula I. we have

$$W = (a, \dots)x', y', z'(x, y, z),$$

$$\left. \begin{matrix} Q \\ R \end{matrix} \right\} = (A, \dots)mz' - ny', \dots(yz' - y'z, \dots) \pm \sqrt{-K(a, \dots)x, y, z)^2} \begin{vmatrix} x, & y, & z \\ x', & y', & z' \\ l, & m, & n \end{vmatrix},$$

(or for a different form of Q , R see the formula). The quantities (x', y', z') are the coordinates of the point of intersection of the two tangents, or pole of the chord of contact: (l, m, n) are supernumerary arbitrary quantities, the values whereof do not affect the result⁸. And in the formula II. we have

$$W = \xi'x + \eta'y + \zeta'z,$$

$$\left. \begin{matrix} Q \\ R \end{matrix} \right\} = \Pi \pm \sqrt{-(A, \dots)\xi', \eta', \zeta')^2} (a, \dots)\nu\eta' - \mu\zeta', \dots(x, y, z),$$

(for the value of Π see the formula). The quantities (ξ', η', ζ') are the line-coordinates of the chord of contact (viz. the point-equation of this line is $\xi'x + \eta'y + \zeta'z = 0$); (λ, μ, ν) are supernumerary arbitrary quantities.

23. In the like manner the formulæ III. and IV. each of them show that the line-equation

$$(A, \dots)\xi, \eta, \zeta)^2 = 0$$

⁸In a different point of view, viz. if we consider the formula I. as a transformation of the function $(a, \dots)x, y, z)^2$, then (x', y', z') and (l, m, n) would be each of them supernumerary arbitrary quantities: and so in the other like cases.

of the conic may be written in the form

$$W^2 + \frac{1}{M}QR = 0,$$

where $Q = 0$, $R = 0$ are any two inequants of the conic, and $W = 0$ is the point of intersection of the corresponding tangents; viz. in the formula III. we have

$$W = (A, ..(\xi', \eta', \zeta')(\xi, \eta, \zeta),$$

$$\left. \begin{matrix} Q \\ R \end{matrix} \right\} = (a, ..(\nu\eta' - \mu\zeta', ..) \pm \sqrt{-(A, ..(\xi', \eta', \zeta')^2} \begin{vmatrix} \xi, & \eta, & \zeta \\ \xi', & \eta', & \zeta' \\ \lambda, & \mu, & \nu \end{vmatrix},$$

(for another form of Q , R see the formula).

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

The quantities ξ', η', ζ' are the line-coordinates of the line through the two ineunts, or chord of contact; (λ, μ, ν) are supernumerary arbitrary quantities; and so in the formula IV. we have

$$W = x'\xi + y'\eta + z'\zeta,$$

$$\left. \begin{matrix} Q \\ R \end{matrix} \right\} = K\Phi \pm \sqrt{-K(a, \dots)x', y', z')^2} (A, \dots)\lambda y' - mz', \dots)\xi, \eta, \zeta)$$

(for the value of Φ see the formula), where x', y', z' are the point-coordinates of the intersection of tangents at the two ineunts, or pole of the chord of contact; (l, m, n) are supernumerary arbitrary quantities.

24. We may, instead of the supernumerary arbitrary quantities (l, m, n) of the formula I., introduce the quantities (λ, μ, ν) , where

$$(l, m, n) = \frac{1}{K} \begin{vmatrix} A, & H, & G \\ H, & B, & F \\ G, & F, & C \end{vmatrix} (\lambda, \mu, \nu).$$

This gives

$$\begin{aligned} & (A, H, G)mz' - ny', \dots) \\ &= A(mz' - ny') + H(nx' - lz') + G(ly' - mx') \\ &= x'(Hn - Gm) + y'(Gl - An) + z'(Am - Hl) \\ &= \frac{1}{K} \left\{ \begin{aligned} & x' [H(G\lambda + F\mu + C\nu) - G(H\lambda + B\mu + F\nu)] \\ & + y' [G(A\lambda + H\mu + G\nu) - A(G\lambda + F\mu + C\nu)] \\ & + z' [A(H\lambda + B\mu + F\nu) - H(A\lambda + H\mu + G\nu)] \end{aligned} \right\} \\ &= x'(g\mu - h\nu) + y'(f\mu - b\nu) + z'(c\mu - f\nu) \\ &= \mu(gx' + fy' + cz') - \nu(hx' + by' + fz'); \end{aligned}$$

we have thus the system

$$\begin{aligned}(A, H, G)mz' - ny', ..) &= \mu(gx' + fy' + cz') - \nu(hx' + by' + fz'), \\ (H, B, F)mz' - ny', ..) &= \nu(ax' + hy' + gz') - \lambda(gx' + fy' + cz'), \\ (G, F, C)mz' - ny', ..) &= \lambda(hx' + by' + fz') - \mu(ax' + hy' + gz'),\end{aligned}$$

and thence

$$\begin{aligned}& (A, ..)mz' - ny', ..)yz' - y'z, ..) \\ &= - \begin{vmatrix} yz' - y'z, & zx' - z'x, & xy' - x'y \\ ax' + hy' + gz', & hx' + by' + fz', & gx' + fy' + cz' \\ \lambda, & \mu, & \nu \end{vmatrix};\end{aligned}$$

or observing that the term in λ is

$$-(zx' - z'x)(gx' + fy' + cz') + (xy' - x'y)(hx' + by' + fz'),$$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

which is

$$\begin{aligned}
 &= x(x'(ax' + hy' + gz') + y'(hx' + by' + fz') + z'(gx' + fy' + cz')) \\
 &\quad - x \cdot x'(ax' + hy' + gz') \\
 &\quad - y \cdot x'(hx' + by' + fz') \\
 &\quad - z \cdot x'(gx' + fy' + cz') \\
 &= -x'(a, \cdot)(x, y, z)(x', y', z') + x(a, \cdot)(x', y', z')^2,
 \end{aligned}$$

with similar expressions for the terms in μ, ν , we have

$$\begin{aligned}
 &(A, \cdot)(mz' - ny', \cdot)(yz' - y'z, \cdot) \\
 &\quad = -(\lambda x' + \mu y' + \nu z') \cdot (a, \cdot)(x, y, z)(x', y', z') + (\lambda x + \mu y + \nu z) \cdot (a, \cdot)(x', y', z')^2
 \end{aligned}$$

and so also

$$\begin{aligned}
 &(A, \cdot)(mz' - ny', \cdot)^2 \\
 &\quad = -(\lambda x' + \mu y' + \nu z') \cdot (a, \cdot)(l, m, n)(x', y', z') + (\lambda l + \mu m + \nu n) \cdot (a, \cdot)(x', y', z')^2
 \end{aligned}$$

where

$$\begin{aligned}
 &(a, \cdot)(l, m, n)(x', y', z') = \lambda x' + \mu y' + \nu z', \\
 &\lambda l + \mu m + \nu n = \frac{1}{K}(A, \cdot)(\lambda, \mu, \nu)^2,
 \end{aligned}$$

so that

$$(A, \cdot)(mz' - ny', \cdot)^2 = -(\lambda x' + \mu y' + \nu z')^2 + \frac{1}{K}(A, \cdot)(\lambda, \mu, \nu)^2 \cdot (a, \cdot)(x', y', z')^2.$$

Moreover,

$$\begin{vmatrix} x & y & z \\ x' & y' & z' \\ l & m & n \end{vmatrix} = \frac{1}{K} \left\{ (A\lambda + H\mu + G\nu)(yz' - y'z) + (H\lambda + B\mu + F\nu)(zx' - z'x) + (G\lambda + F\mu + \dots) \right\}$$

which is

$$\cdot = \frac{1}{K}(A, \cdot)(\lambda, \mu, \nu)(yz' - y'z, \cdot);$$

and hence instead of the formula I. we have

I. (bis) $(a, \text{..})x, y, z)^2 = \text{Quotient by } (a, \text{..})x', y', z')^2 \text{ of } \left\{ \begin{array}{l} [(a, \text{..})x, y, z)x', y', z')]^2 \\ + \text{Quotient by } (A, \text{..})\lambda, \mu, \nu \\ \left\{ (\lambda x' + \mu y' + \nu z') \cdot (a, \text{..}) \right. \\ \left. \pm \frac{1}{K} \sqrt{-K(a, \text{..})x', y', z'} \right\} \end{array} \right.$

25. If, in like manner, in the formula II. we introduce, instead of (λ, μ, ν) , the new quantities (l, m, n) , where

$$(\lambda, \mu, \nu) = \begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix} (l, m, n,$$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

or, what is the same thing,

$$(l, m, n) = \frac{1}{K} \begin{vmatrix} A, & H, & G \\ H, & B, & F \\ G, & F, & C \end{vmatrix} (\lambda, \mu, \nu,$$

then we have

$$\begin{aligned} (a, h, g) \nu \eta' - \mu \zeta', \dots) &= n (H \xi' + B \eta' + F \zeta') - m (G \xi' + F \eta' + C \zeta'), \\ (h, b, f) \nu \eta' - \mu \zeta', \dots) &= l (G \xi' + F \eta' + C \zeta') - n (A \xi' + H \eta' + G \zeta'), \\ (g, f, d) \nu \eta' - \mu \zeta', \dots) &= m (A \xi' + H \eta' + G \zeta') - l (H \xi' + B \eta' + F \zeta'); \end{aligned}$$

and thence

$$\begin{aligned} (a, \dots) \nu \eta' - \mu \zeta', \dots(x, y, z) &= \begin{vmatrix} x, & y, & z \\ A \xi' + H \eta' + G \zeta', & H \xi' + B \eta' + F \zeta', & G \xi' + F \eta' + C \zeta' \\ l, & m, & n \end{vmatrix} \\ &= (A, \dots) m z - n y, \dots(\xi', \eta', \zeta'), \end{aligned}$$

$$\begin{aligned} (a, \dots) \nu \eta' - \mu \zeta', \dots)^2 &= \frac{1}{K} \left\{ (A, \dots) \lambda, \mu, \nu)^2 \cdot (A, \dots) \xi', \eta', \zeta')^2 - [(A, \dots) \lambda, \mu, \nu)(\xi', \eta', \zeta')] \right. \\ &= (a, \dots) l, m, n)^2 \cdot (A, \dots) \xi', \eta', \zeta')^2 - K (l \xi' + m \eta' + n \zeta')^2; \end{aligned}$$

$$\begin{aligned} V &= \begin{vmatrix} x, & y, & z \\ A \xi' + H \eta' + G \zeta', & H \xi' + B \eta' + F \zeta', & G \xi' + F \eta' + C \zeta' \\ \nu \eta' - \mu \zeta', & \lambda \zeta' - \nu \xi', & \mu \xi' - \lambda \eta' \end{vmatrix} \\ &= (\nu \eta' - \mu \zeta') \cdot y (G \xi' + F \eta' + C \zeta') - z (H \xi' + B \eta' + F \zeta') \\ &\quad + (\lambda \zeta' - \nu \xi') \cdot z (A \xi' + H \eta' + G \zeta') - x (G \xi' + F \eta' + C \zeta') \\ &\quad + (\mu \xi' - \lambda \eta') \cdot x (H \xi' + B \eta' + F \zeta') - y (A \xi' + H \eta' + G \zeta') \\ &= \lambda \{ (\xi' x + \eta' y + \zeta' z) (A \xi' + H \eta' + G \zeta') - x (A, \dots) \xi', \eta', \zeta')^2 \} \\ &\quad + \mu \{ (\xi' x + \eta' y + \zeta' z) (H \xi' + B \eta' + F \zeta') - y (A, \dots) \xi', \eta', \zeta')^2 \} \\ &\quad + \nu \{ (\xi' x + \eta' y + \zeta' z) (G \xi' + F \eta' + C \zeta') - z (A, \dots) \xi', \eta', \zeta')^2 \} \\ &= (\xi' x + \eta' y + \zeta' z) \cdot (A, \dots) \lambda, \mu, \nu)(\xi', \eta', \zeta') - (\lambda x + \mu y + \nu z) \cdot (A, \dots) \xi', \eta', \zeta')^2 \\ &= K (l \xi' + m \eta' + n \zeta') \cdot (\xi' x + \eta' y + \zeta' z) - (a, \dots) l, m, n(x, y, z) \cdot (A, \dots) \xi', \eta', \zeta')^2; \end{aligned}$$

and the formula II. thus becomes

II. (bis) $(a, \text{.}.)x, y, z)^2 = \text{Quotient by } (A, \text{.}.)\xi', \eta', \zeta')^2 \text{ of } \left\{ \begin{array}{l} K(\xi'x + \eta'y + \zeta'z)^2 \\ + \text{Quotient by } (a, \text{.}.)l, m, n \\ \left\{ \begin{array}{l} K(l\xi' + m\eta' + n\zeta') \cdot (\xi'x + \eta'y + \zeta'z) \\ \pm \sqrt{-(A, \text{.}.)\xi', \eta', \zeta')} \end{array} \right. \end{array} \right.$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

26. And from these we at once deduce the inverse or reciprocal formulæ

$$\text{III. (bis) } (A, \dots) \xi, \eta, \zeta)^2 = \text{Quotient by } (A, \dots) \xi', \eta', \zeta')^2 \text{ of } \begin{cases} [(A, \dots) \xi, \eta, \zeta) \xi', \eta', \zeta')] \\ + \text{Quotient by } (a, \dots) l, \\ \left\{ (l\xi' + m\eta' + n\zeta')(A, \dots) \right. \\ \left. \pm \sqrt{-(A, \dots) \xi', \eta', \zeta')} \right\} \end{cases}$$

27. And

$$\text{IV. (bis) } (A, \dots) \xi, \eta, \zeta)^2 = \text{Quotient by } (a, \dots) x', y', z')^2 \text{ of } \begin{cases} K(x'\xi + y'\eta + z'\zeta)^2 \\ + \text{Quotient by } (A, \dots) \lambda \\ \left\{ K(\lambda x' + \mu y' + \nu z')(x' \right. \\ \left. \pm \sqrt{-K(a, \dots) x', y', z')} \right\} \end{cases}$$

These four formulæ have the same geometrical significations with the original four formulæ to which they correspond respectively.

28. The eight formulæ become all of them the same or very similar for the quadric form $(a, \dots) x, y, z)^2 = x^2 + y^2 + z^2$, which of course implies $(A, \dots) \xi, \eta, \zeta)^2 = \xi'^2 + \eta'^2 + \zeta'^2$. Thus selecting any one of them at pleasure, e.g. the formula II. (bis), this becomes

$$\begin{aligned} & \{(x^2 + y^2 + z^2)(\xi'^2 + \eta'^2 + \zeta'^2) - (\xi'x + \eta'y + \zeta'z)^2\} \\ & \times \{(l^2 + m^2 + n^2)(\xi'^2 + \eta'^2 + \zeta'^2) - (l\xi' + m\eta' + n\zeta')^2\} \\ & = \{(l\xi' + m\eta' + n\zeta')(x\xi' + m\eta' + n\zeta') - (lx + my + nz)(\xi'^2 + \eta'^2 + \zeta'^2)\}^2 \\ & + (\xi'^2 + \eta'^2 + \zeta'^2) \begin{vmatrix} \xi' & \eta' & \zeta' \\ x & y & z \\ l & m & n \end{vmatrix}^2, \end{aligned}$$

where the terms independent of $\xi'^2 + \eta'^2 + \zeta'^2$ destroy each other. Omitting these terms, and dividing by $\xi'^2 + \eta'^2 + \zeta'^2$, the resulting equation is found to be

$$\begin{vmatrix} \xi' & \eta' & \zeta' \\ x & y & z \\ l & m & n \end{vmatrix}^2 = \begin{vmatrix} \xi'^2 + \eta'^2 + \zeta'^2, & \xi'x + \eta'y + \zeta'z, & \xi'l + \eta'm + \zeta'n \\ x\xi' + y\eta' + z\zeta', & x^2 + y^2 + z^2, & xl + ym + zn \\ l\xi' + m\eta' + n\zeta', & lx + my + nz, & l^2 + m^2 + n^2 \end{vmatrix},$$

which is a well-known identical equation.

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

Article Nos. 29 to 33, relating to a single conic in connexion with an ineunt or a tangent of a conic of double contact.

29. The formulæ assume a very simple form when the point of intersection of the two tangents, or the line of junction of the two ineunts of the conic, is an ineunt or a tangent of a conic having double contact with the first-mentioned conic. Thus, if to the conic

$$(a, .(.)x, y, z)^2 = 0$$

tangents are drawn from a point (x', y', z') of the conic

$$(a, .(.)x, y, z)^2 + (\xi'x + \eta'y + \zeta'z)^2 = 0,$$

then we have

$$(a, .(.)x', y', z')^2 = -(\xi'x' + \eta'y' + \zeta'z')^2;$$

and using the form I. (bis), and putting therein (ξ', η', ζ') in the place of the arbitrary quantities (λ, μ, ν) , the equation of the tangent divides out by $\xi'x' + \eta'y' + \zeta'z'$, and omitting this factor it becomes

$$(a, .(.)x', y', z')(x, y, z) + (\xi'x' + \eta'y' + \zeta'z')(\xi'x + \eta'y + \zeta'z) \\ \pm \frac{1}{\sqrt{K}}(A, .(.)\xi', \eta', \zeta')(yz' - y'z, zx' - z'x, xy' - x'y) = 0,$$

which is of the form

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix} (.)x', y', z'(.)x, y, z = 0,$$

where the matrix

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix} \text{ is } =$$

$$\begin{array}{llll}
a + \xi'^2 & , & h + \xi'\eta' + \frac{1}{\sqrt{K}}(G\xi' + F\eta' + C\zeta') & , & g + \xi'\zeta' \\
h + \xi'\eta' - \frac{1}{\sqrt{K}}(G\xi' + F\eta' + C\zeta') & , & b + \eta'^2 & , & f + \eta'\zeta' \\
g + \xi'\zeta' + \frac{1}{\sqrt{K}}(H\xi' + B\eta' + F\zeta') & , & f + \eta'\zeta' - \frac{1}{\sqrt{K}}(A\xi' + H\eta' + G\zeta') & , & c + \zeta'^2.
\end{array}$$

30. But instead of further developing these formulæ, I prefer to consider the formulæ which give the points of contact of the tangents in question, viz. the ineunts of the conic $(a, \dots)x, y, z)^2 = 0$, or the tangents through the point (x', y', z') of the conic $(a, \dots)x, y, z)^2 + (\xi'x + \eta'y + \zeta'z)^2 = 0$.

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

We have as before

$$(a, \dots)(x', y', z')^2 = -(\xi'x' + \eta'y' + \zeta'z')^2,$$

and using the formula IV. (bis) and writing therein (ξ', η', ζ') in the place of the arbitrary quantities (λ, μ, ν) , the equation contains the factor $\xi'x' + \eta'y' + \zeta'z'$, and dividing by this factor, and by K , the line-equation of the ineunt is

$$\begin{aligned} x'\xi + y'\eta + z'\zeta + \frac{1}{K}(x'\xi' + y'\eta' + z'\zeta') \cdot (A, \dots)(\xi', \eta', \zeta')(\xi, \eta, \zeta) \\ \pm \frac{1}{\sqrt{K}}(a, \dots)(x', y', z')(\eta\zeta' - \eta'\zeta, \dots) = 0. \end{aligned}$$

Selecting the positive sign, the coordinates of the corresponding ineunt are

$$\begin{aligned} x' + \frac{1}{K}(x'\xi' + y'\eta' + z'\zeta')(A\xi' + H\eta' + G\zeta') + \frac{1}{\sqrt{K}}\{\eta'(gx' + fy' + cz') - \zeta'(hx' + by' + fz')\} \\ y' + \frac{1}{K}(x'\xi' + y'\eta' + z'\zeta')(H\xi' + B\eta' + F\zeta') + \frac{1}{\sqrt{K}}\{\zeta'(ax' + hy' + gz') - \xi'(gx' + fy' + cz')\} \\ z' + \frac{1}{K}(x'\xi' + y'\eta' + z'\zeta')(G\xi' + F\eta' + C\zeta') + \frac{1}{\sqrt{K}}\{\xi'(hx' + by' + fz') - \eta'(ax' + hy' + gz')\} \end{aligned}$$

and taking (X, Y, Z) for the coordinates of the ineunt in question, and putting for shortness

$$\begin{aligned} \alpha &= 1 - \frac{1}{\sqrt{K}}(g\eta' - h\zeta'), & \beta &= -\frac{1}{\sqrt{K}}(f\eta' - b\zeta'), & \gamma &= -\frac{1}{\sqrt{K}}(c\eta' - f\zeta'), \\ \alpha' &= -\frac{1}{\sqrt{K}}(a\zeta' - g\xi'), & \beta' &= 1 - \frac{1}{\sqrt{K}}(h\zeta' - f\xi'), & \gamma' &= -\frac{1}{\sqrt{K}}(g\zeta' - c\xi'), \\ \alpha'' &= -\frac{1}{\sqrt{K}}(h\xi' - a\eta'), & \beta'' &= -\frac{1}{\sqrt{K}}(b\xi' - h\eta'), & \gamma'' &= 1 - \frac{1}{\sqrt{K}}(f\xi' - g\eta'), \end{aligned}$$

we may write

$$\begin{aligned} (1+P)X &= (2-\alpha)x' - \beta y' - \gamma z' + \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ (1+P)Y &= -\alpha'x' + (2-\beta')y' - \gamma'z' + \frac{1}{K}(H\xi' + B\eta' + F\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ (1+P)Z &= -\alpha''x' - \beta''y' + (2-\gamma'')z' + \frac{1}{K}(G\xi' + F\eta' + C\zeta')(\xi'x' + \eta'y' + \zeta'z'), \end{aligned}$$

where P , which is arbitrary, may be put

$$= \frac{1}{K}(A, .(\xi', \eta', \zeta')^2.$$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

31. These equations then give

$$(x', y', z') = \begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix} ()X, Y, Z,$$

which can be verified without difficulty by reversing the process; and we have thus the coordinates (X, Y, Z) in terms of (x', y', z') , and reciprocally.

32. If (X_1, Y_1, Z_1) are the coordinates of the other ineunt, we have, it is clear,

$$(x', y', z') = \begin{vmatrix} 2 - \alpha, & -\beta, & -\gamma \\ -\alpha', & 2 - \beta', & -\gamma' \\ -\alpha'', & -\beta'', & 2 - \gamma'' \end{vmatrix} ()X_1, Y_1, Z_1;$$

or substituting for (x', y', z') their values in terms of (X, Y, Z) ,

$$(2X_1, 2Y_1, 2Z_1) = \begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix} ()X + X_1, Y + Y_1, Z + Z_1,$$

so that $(X + X_1, Y + Y_1, Z + Z_1)$ are the same linear functions of $2X_1, 2Y_1, 2Z_1$ that (X, Y, Z) are of (x', y', z') ; that is, we have

$$\begin{aligned} \frac{1}{2}(1 + P)(X + X_1) &= (2 - \alpha)X_1 - \beta Y_1 - \gamma Z_1 + \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z') \\ \frac{1}{2}(1 + P)(Y + Y_1) &= -\alpha'X_1 + (2 - \beta')Y_1 - \gamma'Z_1 + \frac{1}{K}(H\xi' + B\eta' + F\zeta')(\xi'x' + \eta'y' + \zeta'z') \\ \frac{1}{2}(1 + P)(Z + Z_1) &= -\alpha''X_1 - \beta''Y_1 + (2 - \gamma'')Z_1 + \frac{1}{K}(G\xi' + F\eta' + C\zeta')(\xi'x' + \eta'y' + \zeta'z') \end{aligned}$$

which equations may be written

$$(1 + P)(X, Y, Z) = \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix} ()X_1, Y_1, Z_1,$$

where the values of the coefficients are

$$\begin{aligned}
 a &= (g\eta' - h\zeta') + \frac{1}{K}(A\xi'^2 - B\eta'^2 - 2F\eta'\zeta' - C\zeta'^2), \\
 b &= -\frac{2}{\sqrt{K}}(f\eta' - b\zeta') + \frac{2}{K}\eta'(A\xi' + H\eta' + G\zeta'), \\
 c &= -\frac{2}{\sqrt{K}}(c\eta' - f\zeta') + \frac{2}{K}\zeta'(A\xi' + H\eta' + G\zeta'), \\
 a' &= -\frac{2}{\sqrt{K}}(a\zeta' - g\xi') + \frac{2}{K}\xi'(H\xi' + B\eta' + F\zeta'), \\
 b' &= 1 - \frac{2}{\sqrt{K}}(h\zeta' - f\xi') + \frac{1}{K}(B\eta'^2 - C\zeta'^2 - 2G\zeta'\xi' - A\xi'^2), \\
 c' &= -\frac{2}{\sqrt{K}}(g\zeta' - c\xi') + \frac{2}{K}\zeta'(H\xi' + B\eta' + F\zeta'), \\
 a'' &= -\frac{2}{\sqrt{K}}(h\xi' - a\eta') + \frac{2}{K}\xi'(G\xi' + F\eta' + C\zeta'), \\
 b'' &= -\frac{2}{\sqrt{K}}(b\xi' - h\eta') + \frac{2}{K}\eta'(G\xi' + F\eta' + C\zeta'), \\
 c'' &= 1 - \frac{2}{\sqrt{K}}(f\xi' - g\eta') + \frac{1}{K}(C\zeta'^2 - A\xi'^2 - 2H\xi'\eta' - B\eta'^2).
 \end{aligned}$$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

and considering (X, Y, Z) and (X_1, Y_1, Z_1) as quantities connected by the foregoing linear relations, we have identically

$$(a, \dots)X, Y, Z)^2 = (a, \dots)X_1, Y_1, Z_1)^2.$$

The investigation leads thus to the automorphic transformation of the quadric function, a transformation first effected by M. Hermite⁹.

33. It is to be remarked that the foregoing formulæ show that (x', y', z') being the coordinates of a point on the conic $(a, \dots)x, y, z)^2 + (\xi'x + \eta'y + \zeta'z)^2 = 0$, from which point tangents are drawn to the conic $(a, \dots)x, y, z)^2 = 0$, then the coordinates (x', y', z') enter *linearly* into the equations of the tangents, the ineunts (or points of contact), and the polar. And it may be added that the equation of the conic enveloped by the polar (that is, the polar conic of $(a, \dots)x, y, z)^2 + (\xi'x + \eta'y + \zeta'z)^2 = 0$) has for its equation

$$\{K + (A, \dots)\xi', \eta', \zeta')^2\} (a, \dots)x, y, z)^2 - K(\xi'x + \eta'y + \zeta'z)^2 = 0,$$

and that the coordinates of the point of contact of the polar with this conic are

$$\begin{aligned} x' + \frac{1}{K}(A\xi' + H\eta' + G\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ y' + \frac{1}{K}(H\xi' + B\eta' + F\zeta')(\xi'x' + \eta'y' + \zeta'z'), \\ z' + \frac{1}{K}(G\xi' + F\eta' + C\zeta')(\xi'x' + \eta'y' + \zeta'z'); \end{aligned}$$

so that (x', y', z') also enter linearly into the expressions for the coordinates of the last-mentioned point.

Article Nos. 34 to 37, relating to two conics.

⁹See my "Memoir on the Automorphic Transformation of a Bipartite Quadric Function," *Phil. Trans.* vol. cxlviii. (1858), pp. 39—46, [153].

34. Considering now the two conics

$$U = (a, b, c, f, g, h)(x, y, z)^2 = 0,$$

$$U' = (a', b', c', f', g', h')(x, y, z)^2 = 0;$$

suppose that the conic

$$\theta U + \theta' U' = (\theta a + \theta' a', \dots)(x, y, z)^2 = 0$$

represents a pair of lines.

The condition for this is

$$\text{Disct. } (\theta a + \theta' a', \dots)(x, y, z)^2 = 0,$$

which is

$$(\dots, \theta, \theta')^3 = 0,$$

where

$$\begin{aligned} &= K, \\ &= Aa' + Bb' + Cc' + 2Ff' + 2Gg' + 2Hh', \\ &= A'a + B'b + C'c + 2F'f + 2G'g + 2H'h, \\ &= K'. \end{aligned}$$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

(the significations of $K', A', B', C', F', G', H'$ being of course analogous to those of K, A, B, C, F, G, H). The three roots $\theta : \theta'$ correspond, it is clear, to the three pairs of lines which can be drawn through the intersections of the two conics.

35. The equation

$$\text{Disct. } (, , , (\theta, \theta')^3 = 0,$$

which is of the fourth order in $, , ,$ and of the sixth order as regards (a, b, c, f, g, h) and (a', b', c', f', g', h') respectively, is the condition in order that the two conics may touch each other. Assuming that it is satisfied, the cubic equation in $\theta : \theta'$ has a pair of equal roots; or say there is a twofold root and a onefold root; the twofold root gives the pair of lines drawn from the point of contact to the other two points of intersection, the onefold root gives the pair made up of the common tangent and the line joining the other two points of intersection.

36. In particular, suppose that the two conics are

$$\begin{aligned} 2(\rho x + \sigma y + \tau z)(\rho' x + \sigma' y + \tau' z) &= 0, \\ 2(\lambda x + \mu y + \nu z)(\lambda' x + \mu' y + \nu' z) &= 0; \end{aligned}$$

so that

$$\begin{aligned} (a, b, c, f, g, h) &= (2\rho\rho', 2\sigma\sigma', 2\tau\tau', \sigma\tau' + \sigma'\tau, \tau\rho' + \tau'\rho, \rho\sigma' + \rho'\sigma), \\ (a', b', c', f', g', h') &= (2\lambda\lambda', 2\mu\mu', 2\nu\nu', \mu\nu' + \mu'\nu, \nu\lambda' + \nu'\lambda, \lambda\mu' + \lambda'\mu), \\ (A, B, C, F, G, H) &= -(\sigma\tau' - \sigma'\tau, \tau\rho' - \tau'\rho, \rho\sigma' - \rho'\sigma)^2, \\ (A', B', C', F', G', H') &= -(\mu\nu' - \mu'\nu, \nu\lambda' - \nu'\lambda, \lambda\mu' - \lambda'\mu)^2; \end{aligned}$$

and thence also

$$\begin{aligned}
 &= K && = 0, \\
 &= Aa' + \&c. && = -2 \begin{vmatrix} \lambda, & \mu, & \nu \\ \rho, & \sigma, & \tau \\ \rho', & \sigma', & \tau' \end{vmatrix} \begin{vmatrix} \lambda', & \mu', & \nu' \\ \rho, & \sigma, & \tau \\ \rho', & \sigma', & \tau' \end{vmatrix}, \\
 &= A'a + \&c. && = -2 \begin{vmatrix} \rho, & \sigma, & \tau \\ \lambda, & \mu, & \nu \\ \lambda', & \mu', & \nu' \end{vmatrix} \begin{vmatrix} \rho', & \sigma', & \tau' \\ \lambda, & \mu, & \nu \\ \lambda', & \mu', & \nu' \end{vmatrix}, \\
 &= K' && = 0;
 \end{aligned}$$

and the equation in (θ, θ') is

$$\theta + \theta' = 0;$$

hence writing $\theta =, \theta' = -$, the equation of the pair of lines is

$$\begin{vmatrix} \rho, & \sigma, & \tau \\ \lambda, & \mu, & \nu \\ \lambda', & \mu', & \nu' \end{vmatrix} \begin{vmatrix} \rho', & \sigma', & \tau' \\ \lambda, & \mu, & \nu \\ \lambda', & \mu', & \nu' \end{vmatrix} (\rho x + \sigma y + \tau z)(\rho' x + \sigma' y + \tau' z) - \begin{vmatrix} \lambda, & \mu, & \nu \\ \rho, & \sigma, & \tau \\ \rho', & \sigma', & \tau' \end{vmatrix} \begin{vmatrix} \lambda', & \mu', & \nu' \\ \rho, & \sigma, & \tau \\ \rho', & \sigma', & \tau' \end{vmatrix} (\lambda x + \mu y + \nu z)(\lambda' x + \mu' y + \nu' z) = 0$$

[276] (continued)ON THE ANALYTICAL THEORY OF THE CONIC
(continued).

and it is easy to see that the left-hand side does in fact break up into factors, and that the equation is

$$\begin{vmatrix} x, & y, & z \\ \mu\tau' - \nu\sigma', & \nu\rho' - \lambda\tau', & \lambda\sigma' - \mu\rho' \\ \sigma\nu' - \tau\mu', & \tau\lambda' - \rho\nu', & \rho\mu' - \sigma\lambda' \end{vmatrix} \begin{vmatrix} x, & y, & z \\ \mu\tau - \nu\sigma, & \nu\rho - \lambda\tau, & \lambda\sigma - \mu\rho \\ \sigma'\nu' - \tau'\mu', & \tau'\lambda' - \rho'\nu', & \rho'\mu' - \sigma'\lambda' \end{vmatrix} = 0,$$

which of course might have been obtained at once by means of the four points which are the intersection of each component line of the first conic by each component line of the second conic.

37. Suppose that the first conic is

$$(a, b, c, f, g, h)x, y, z)^2 = 0,$$

while the second conic is the pair of lines

$$2(\lambda x + \mu y + \nu z)(\lambda' x + \mu' y + \nu' z) = 0;$$

then, putting as before,

$$(\theta a + \theta' \cdot 2\lambda\lambda', \cdot)(x, y, z)^2 = 0,$$

we have

$$(\cdot, \cdot, \cdot)(\theta, \theta')^3 = 0,$$

where

$$\begin{aligned} &= K, \\ &= 2(A, B, C, F, G, H)(\lambda, \mu, \nu)(\lambda', \mu', \nu'), \\ &= -(a, b, c, f, g, h)(\mu\nu' - \mu'\nu, \nu\lambda' - \nu'\lambda, \lambda\mu' - \lambda'\mu)^2, \\ &= 0; \end{aligned}$$

and the equation in (θ, θ') is

$$K\theta^2 + 2(A, \cdot)(\lambda, \mu, \nu)(\lambda', \mu', \nu')\theta\theta' - (a, \cdot)(\mu\nu' - \mu'\nu, \cdot)^2\theta'^2 = 0,$$

which may be written

$$\begin{aligned} \{K\theta + (A, \cdot)(\lambda, \mu, \nu)(\lambda', \mu', \nu')\theta'\}^2 &= \{[(A, \cdot)(\lambda, \mu, \nu)(\lambda', \mu', \nu')]^2 + K(a, \cdot)(\mu\nu' - \mu'\nu, \cdot)^2\} \\ &= (A, \cdot)(\lambda, \mu, \nu)^2 \cdot (A, \cdot)(\lambda', \mu', \nu')^2 \cdot \theta'^2, \end{aligned}$$

that is

$$K\theta = [\pm\sqrt{(A, \dots)\lambda, \mu, \nu)^2}\sqrt{(A, \dots)\lambda', \mu', \nu')^2} - (A, \dots)\lambda, \mu, \nu)\lambda', \mu', \nu')]\theta';$$

we may assume

$$\theta = \pm\sqrt{(A, \dots)\lambda, \mu, \nu)^2}\sqrt{(A, \dots)\lambda', \mu', \nu')^2} - (A, \dots)\lambda, \mu, \nu)\lambda', \mu', \nu'), \quad \theta' = K,$$

so that the conic

$$\{\pm\sqrt{(A, \dots)\lambda, \mu, \nu)^2}\sqrt{(A, \dots)\lambda', \mu', \nu')^2} - (A, \dots)\lambda, \mu, \nu)\lambda', \mu', \nu')\}(a, \dots)x, y, z)^2 \\ + 2K(\lambda x + \mu y + \nu z)(\lambda'x + \mu'y + \nu'z) = 0$$

breaks up into a pair of lines.

Putting for shortness

$$\pm\sqrt{(A, \dots)\lambda, \mu, \nu)^2}\sqrt{(A, \dots)\lambda', \mu', \nu')^2} - (A, \dots)\lambda, \mu, \nu)\lambda', \mu', \nu') = \Omega$$

the coefficients on the left-hand side of the equation are

$$\{\Omega a + 2K\lambda\lambda', \dots \Omega f + K(\mu\nu' + \mu'\nu), \dots\},$$

whence, after all reductions, the inverse function is

$$\{(A, \dots)\lambda, \mu, \nu)\lambda', \mu', \nu')^2 + (A, \dots)\lambda', \mu', \nu')^2(\lambda, \mu, \nu\sqrt{\text{Disct.}}\},$$

and the remainder of the process of decomposition is effected without difficulty.

Addition, 18 December, 1862.

The formulæ II. and II. (bis) each of them give the tangents of the conic $(a, \dots)x, y, z)^2 = 0$ at the ineunts of intersection with the line $\xi'x + \eta'y + \zeta'z = 0$. A very elegant formula for these ineunts themselves was communicated to me by Mr Spottiswoode, and I have since found that the same or an equivalent formula is made use of by M. Aronhold in his recent valuable memoir, "Ueber eine neue algebraische Behandlungsweise der Integrale irrationaler Differentiale, &c.," *Crelle*, t. LXII. pp. 95—145 (1862). The formula is as follows, viz. for the conic and line,

$$(a, b, c, f, g, h)x, y, z)^2 = 0, \\ \xi'x + \eta'y + \zeta'z = 0,$$

then

$$\begin{aligned} x : y : z = \\ (\xi' l + \eta' m + \eta' n) 2f\xi' + \nu'(gl + fm + cn) - \zeta'(hl + bm + fy) + l\sqrt{\Delta} : \\ (\xi' l + \eta' m + \zeta' n) 2g\eta' + \xi'(hl + bm + fn) - \nu'(al + hm + gn) + m\sqrt{\Delta} : \\ (\xi' l + \eta' m + \zeta' n) 2h\zeta' + \eta'(al + hm + gn) - \xi'(gl + fm + cn) + n\sqrt{\Delta}; \end{aligned}$$

where

$$\Delta = \begin{vmatrix} \xi' & \eta' & \zeta' \\ f & g & h \\ \xi' & \eta' & \zeta' \end{vmatrix}^2 = -(A, B, C, F, G, H)(\xi', \eta', \zeta')^2,$$

$$\begin{vmatrix} \xi' & \eta' & \zeta' \\ f & a & h & g \\ \eta' & h & b & f \\ \zeta' & g & f & c \end{vmatrix}$$

so that ξ', η', ζ' are respectively

$$= -(A\xi' + H\eta' + G\zeta'), \quad -(H\xi' + B\eta' + F\zeta'), \quad -(G\xi' + F\eta' + C\zeta'),$$

and where l, m, n are supernumerary arbitrary quantities.

[277]

ON THE WAVE SURFACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 16—22.]

Some very beautiful results in relation to the Wave Surface have been recently obtained by Herr Zech, in the Memoirs “Die Eigenschaften der Wellenflächen der zweiaxigen Krystalle mittels der höhern Geometrie abgeleitet,” *Crelle*, t. LII. pp. 243—254 (1856), and “Die Krümmungslinien der Wellenfläche zweiaxiger Krystalle,” *Crelle*, t. LIV. pp. 72—77 (1857). According to the former of Fresnel’s two modes of generation, the wave surface is the envelope of a plane whose perpendicular distance v is a certain given function of the direction cosines l, m, n . For the same system of direction cosines, there are in fact two values of the perpendicular distance: call these v and w , and let the corresponding planes (parallel of course to each other) be called P and Q. Then the entire system of the planes P and Q envelope the wave surface, viz. the planes P may be considered as enveloping one sheet and the planes Q the other sheet of the surface. But if instead of considering the entire system of planes we consider only the planes P and the parallel planes Q, for which the perpendicular on the plane P has a given constant value v , then the planes P will envelope a developable F, and the planes Q will envelope a developable G, these two developables F and G respectively touching the wave surface along two series of curves. And it is shown in the second of the Memoirs above referred to that these curves are the curves of curvature of the wave surface.

The developable F is obtained as the envelope of the plane P, whose equation is

$$\xi x + \eta y + \zeta z = v,$$

where v has a given constant value and ξ, η, ζ are parameters which vary, subject to the two conditions

$$\xi^2 + \eta^2 + \zeta^2 = 1,$$

$$\frac{\xi^2}{a^2 - v^2} + \frac{\eta^2}{b^2 - v^2} + \frac{\zeta^2}{c^2 - v^2} = 0.$$

And in like manner the developable G is obtained as the envelope

of the plane Q, whose equation is

$$\xi x + \eta y + \zeta z = w = \frac{abc}{v} \sqrt{\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} + \frac{\zeta^2}{c^2}},$$

where w has a given constant value and ξ, η, ζ are parameters which vary, subject to the two conditions

$$\begin{aligned} \xi^2 + \eta^2 + \zeta^2 &= 1, \\ \frac{\xi^2}{a^2 - w^2} + \frac{\eta^2}{b^2 - w^2} + \frac{\zeta^2}{c^2 - w^2} &= 0. \end{aligned}$$

(It is hardly necessary to remark that if in the last-mentioned system of equations, the parameter w is also treated as variable, we obtain the wave surface: in fact w, v^2 being the roots of the equation

$$\frac{\xi^2}{a^2 - \theta} + \frac{\eta^2}{b^2 - \theta} + \frac{\zeta^2}{c^2 - \theta} = 0,$$

we have, attending to the condition $\xi^2 + \eta^2 + \zeta^2 = 1$, the identical equation

$$v^2 w^2 - (v^2 + w^2)\theta + \&c. = (\theta - a^2)(\theta - b^2)(\theta - c^2) \left(\frac{\xi^2}{a^2 - \theta} + \frac{\eta^2}{b^2 - \theta} + \frac{\zeta^2}{c^2 - \theta} \right),$$

and thence

$$vw = abc \sqrt{\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} + \frac{\zeta^2}{c^2}},$$

which shows that $vw = abc \sqrt{\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} + \frac{\zeta^2}{c^2}}$.

or in other words, that the planes P and Q are respectively tangents to the conic or infinitely thin surface of the second order, which is the second of the last-mentioned cone by the plane at infinity. Moreover it is obvious from the same equations that the plane P is a tangent plane of the sphere whose equation is

$$x^2 + y^2 + z^2 = v^2,$$

and that the plane Q is a tangent plane of the ellipsoid whose equation is

$$\frac{a^2 b^2 c^2}{a^2 x^2 + b^2 y^2 + c^2 z^2} = w^2.$$

Hence each of the developables F and G is the envelope of a plane which is the common tangent plane of two surfaces of the second

order; such developables are in general of the eighth order, see my paper "On the Developable Surfaces which arise from Two Surfaces of the Second Order," *Camb. and Dubl. Math. Journ.*, t. V. pp. 46—57 (1850), [84], and it will be presently seen that this is in fact the order of the developables F and G respectively.

The before-mentioned cone

$$\frac{\xi^2}{a^2 - v^2} + \frac{\eta^2}{b^2 - v^2} + \frac{\zeta^2}{c^2 - v^2} = 0$$

(Zech's cone K) is a cone having for its focal lines the optic axes (or normals to the circular sections) of the ellipsoid $a^2x^2 + b^2y^2 + c^2z^2 = 1$ (Zech's ellipsoid E) which is used in the theory of the Wave Surface in the place of Fresnel's Surface of Elasticity. The complementary cone

$$(a^2 - v^2)x^2 + (b^2 - v^2)y^2 + (c^2 - v^2)z^2 = 0$$

is the cone which standing on the same ellipsoid E has for its focal lines the optic axes of the reciprocal ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

To find the equation of the developable F, writing for shortness

$$p = v^2 - a^2, \quad q = v^2 - b^2, \quad r = v^2 - c^2,$$

then the system of equations is

$$\begin{aligned} \xi x + \eta y + \zeta z - v &= 0, \\ \xi^2 + \eta^2 + \zeta^2 - 1 &= 0, \\ \frac{\xi^2}{p} + \frac{\eta^2}{q} + \frac{\zeta^2}{r} &= 0, \end{aligned}$$

treating them by the method of arbitrary multipliers in the usual manner, we obtain

$$\begin{aligned} x + 2\lambda\xi + \frac{2\mu\xi}{p} &= 0, \\ y + 2\lambda\eta + \frac{2\mu\eta}{q} &= 0, \\ z + 2\lambda\zeta + \frac{2\mu\zeta}{r} &= 0, \end{aligned}$$

equations which give in the first instance $v + p = 0$ or $p = -v$, and then substituting this value,

$$\xi = \frac{x(v^2 - a^2)}{2v\theta}, \quad \eta = \frac{y(v^2 - b^2)}{2v\theta}, \quad \zeta = \frac{z(v^2 - c^2)}{2v\theta},$$

and thence

$$\frac{x^2(v^2 - a^2)}{4v^2\theta^2} + \frac{y^2(v^2 - b^2)}{4v^2\theta^2} + \frac{z^2(v^2 - c^2)}{4v^2\theta^2} = 0,$$

$$\frac{x^2(v^2 - a^2)^2}{4v^2\theta^2} + \frac{y^2(v^2 - b^2)^2}{4v^2\theta^2} + \frac{z^2(v^2 - c^2)^2}{4v^2\theta^2} = \theta,$$

the latter of which is the derived equation of the former with respect to the parameter θ ; hence writing the former equation under the form

$$\{\theta + v(v^2 - a^2)\} \{\theta + v(v^2 - b^2)\} \{\theta + v(v^2 - c^2)\} - \&c. = 0,$$

or what is the same thing

$$(A, B, C, D)\theta, 1)^3 = 0,$$

where

$$A = 3v^2,$$

$$B = (v^2 - a^2)(v^2 - b^2) + (v^2 - b^2)(v^2 - c^2) + (v^2 - c^2)(v^2 - a^2),$$

$$C = (v^2 - y^2 - z^2)(v^2 - b^2)(v^2 - c^2) + (v^2 - z^2 - x^2)(v^2 - c^2)(v^2 - a^2) \\ + (v^2 - x^2 - y^2)(v^2 - a^2)(v^2 - b^2),$$

$$D = (v^2 - a^2)(v^2 - b^2)(v^2 - c^2) - x^2(v^2 - b^2)(v^2 - c^2) - y^2(v^2 - c^2)(v^2 - a^2) \\ - z^2(v^2 - a^2)(v^2 - b^2),$$

the equation of the developable F is

$$(AD - BC)^2 - 4(AC - B^2)(BD - C^2) = 0.$$

The investigation for the developable G is very similar to the preceding. Write for shortness

$$p' = \frac{1}{w^2} - \frac{1}{a^2}, \quad q' = \frac{1}{w^2} - \frac{1}{b^2}, \quad r' = \frac{1}{w^2} - \frac{1}{c^2},$$

then the system of equations is

$$\xi x + \eta y + \zeta z = w,$$

$$\xi^2 + \eta^2 + \zeta^2 = 1,$$

$$\frac{\xi^2}{a^2 - w^2} + \frac{\eta^2}{b^2 - w^2} + \frac{\zeta^2}{c^2 - w^2} = 0,$$

and we then have

$$\xi = \frac{x/a^2}{\lambda + \mu/w^2 + \mu(v^2 - a^2)}, \text{ \&c.},$$

equations which give $p = 0$, and substituting this value, we obtain

$$\xi = \frac{x(w^2 - a^2)}{wA}, \quad \eta = \frac{y(w^2 - b^2)}{wA}, \quad \zeta = \frac{z(w^2 - c^2)}{wA},$$

where A is a parameter. Substituting these values in the equations

$$\xi^2 + \eta^2 + \zeta^2 = 1,$$

$$\frac{x^2(w^2 - a^2)}{w^2A^2} + \frac{y^2(w^2 - b^2)}{w^2A^2} + \frac{z^2(w^2 - c^2)}{w^2A^2} = 0,$$

we find

$$\begin{aligned} \frac{x^2(w^2 - a^2)^2}{w^2A^2} + \frac{y^2(w^2 - b^2)^2}{w^2A^2} + \frac{z^2(w^2 - c^2)^2}{w^2A^2} &= A, \\ \frac{x^2(w^2 - a^2)}{w^2} + \frac{y^2(w^2 - b^2)}{w^2} + \frac{z^2(w^2 - c^2)}{w^2} &= A^2 - (w^2 - a^2)(w^2 - b^2)(w^2 - c^2), \end{aligned}$$

and multiplying the first equation by $-\frac{1}{w^2}$ and the second equation by A^2 and adding, we obtain

$$\frac{x^2(w^2 - a^2)}{a^2} + \frac{y^2(w^2 - b^2)}{b^2} + \frac{z^2(w^2 - c^2)}{c^2} = \frac{abc}{w},$$

of which equation the latter of the foregoing two equations is the derived equation with respect to the parameter A^2 .

Comparing this with the foregoing equation

$$(AD - BC)^2 - 4(AC - B^2)(BD - C^2) = 0,$$

we see that it is deduced from it by writing

$$a^2x^2, \quad b^2y^2, \quad c^2z^2, \quad -\frac{1}{a^2}, \quad -\frac{1}{b^2}, \quad -\frac{1}{c^2}, \quad w^2, \quad A^2$$

in the place of

$$x^2, \quad y^2, \quad z^2, \quad v^2 - a^2, \quad v^2 - b^2, \quad v^2 - c^2, \quad v^2, \quad \theta$$

respectively, and the equation of the developable G is therefore

$$(A'D' - B'C')^2 - 4(A'C' - B'^2)(B'D' - C'^2) = 0,$$

where A', B', C', D' are what A, B, C, D become when

$$x^2, y^2, z^2, v^2 - a^2, v^2 - b^2, v^2 - c^2, v^2, \theta$$

are replaced by

$$a^2x^2, b^2y^2, c^2z^2, -\frac{1}{a^2}, -\frac{1}{b^2}, -\frac{1}{c^2}, w^2, A^2.$$

The equations of the developables F and G, although radically distinct from each other, are consequently similar in form, and each is at once deducible from the other.

2, Stone Buildings, W.C., 9th March, 1858.

[278]

NOTE ON THE SINGULAR SOLUTIONS OF
DIFFERENTIAL EQUATIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*,
vol. III. (1860), pp. 36, 37.]

The following investigation (which has been in my possession for
a good many years) affords I think a simple explanation of the theory
of the singular solutions of differential equations.

Let the primitive equation be

$$c^n + Pc^{n-1} + Qc^{n-2} + \dots = 0,$$

where c is the arbitrary constant and $P, Q, \dots$ are any functions of
 x, y ; then the differential equation is obtained by eliminating c from
the foregoing equation and the derived equation

$$n'c^{n-1} + Q'c^{n-2} + \dots = 0,$$

and the result may be represented by

$$F(P, Q, \dots, P', Q', \dots) = 0.$$

Assume now

$$c^n + Pc^{n-1} + Qc^{n-2} + \dots = (c + X)(c + Y)(c + Z) \dots,$$

then we have

$$\begin{aligned} P &= X + Y + Z + \&c., \\ Q &= XY + XZ + YZ + \&c., \\ &\vdots \end{aligned}$$

and consequently

$$\begin{aligned} P' &= X' + Y' + Z' + \&c., \\ Q' &= (Y + Z + \&c.)X' + \&c., \\ &\vdots \end{aligned}$$

and substituting these values in the function
 $F(P, Q, \dots, P', Q', \dots)$ it is clear that we shall have

$F(P, Q, \dots, P', Q', \dots) = UX'Y'Z' \dots$ where U is a symmetrical function of X, Y, Z , &c., and therefore a function of $P, Q, \dots$; and this equation will be identically true whatever values we attribute to $X', Y', Z', \dots$, hence putting these quantities respectively equal to unity, we have

$$\begin{aligned} P' &= n, \\ Q' &= (n-1)P, \\ R' &= (n-2)Q, \\ &\vdots \end{aligned}$$

and with these values

$$U = F(P, Q, \dots, n, (n-1)P, \dots),$$

that is $U = 0$ is the result obtained by eliminating c from the primitive equation and the equation

$$nc^{n-1} + (n-1)Pc^{n-2} + \dots = 0,$$

which is the equation obtained by differentiating the primitive equation with respect to the arbitrary constant c : that is, $U = 0$ being the singular solution, the differential equation is

$$UX'Y'Z' \dots = 0.$$

It is to be remarked that (P, Q , &c. being rational and integral functions) then if the roots X, Y, Z , &c. are also rational and integral functions, the differential equation contains U^n as a separable rational and integral factor, but if the roots are irrational then the differential equation does not really contain the rational and integral factor U , but $X'Y'Z' \dots$ is here a rational fraction containing U in the denominator and $UX'Y'Z' \dots$ is an indecomposable rational and integral function. This is easily verified à *posteriori* for a quadratic equation.

2, Stone Buildings, W.C., 28th January, 1858.

[279]

ON A THEOREM RELATING TO SPHERICAL CONICS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), p. 53.]

The following theorem was given by Prof. MacCullagh: "If three lines at right angles to each other pass through a fixed point O so that two of them are confined to given planes: the third line traces out a cone of the second order whose sections parallel to the given planes are circles, and the plane containing the other two lines envelopes a cone of the second order whose sections by planes parallel to the given planes are parabolas."

Referring the figure to the sphere we have a trirectangular triangle XYZ , of which two angles X, Y lie on fixed arcs A, B . The angle Z generates a spherical conic U having A, B for its cyclic arcs. The side XY envelopes a spherical conic V touched by the arcs A, B . The conic V is evidently the supplementary conic of U , hence the poles of A, B are the foci of U . We may drop altogether the consideration of the triangle XYZ and consider only the side XY , we have then the theorem: *If a quadrantal arc XY slides between the two fixed arcs A, B , the envelope of XY is a spherical conic U touched by the fixed arcs A, B , and which has for its foci the poles of these same arcs A, B .*

It is worth while to notice the great reduction of order which takes place in consequence of the arc XY being a quadrant. If XY had been an arc of a given magnitude θ , the envelope would have been a spherical curve of an order certainly higher than 6. For considering the corresponding problem in piano, the envelope in the particular case where the fixed lines A, B are at right angles to each other is a curve of the sixth order, and in the general case where the two fixed lines are not at right angles the order is higher: the problem in piano corresponds of course, not to the general problem on the sphere, but to that in which θ is indefinitely small.

280. On the Conics which Touch Four Given Lines

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 94–96.]

There are considerable practical difficulties in drawing a figure of the system of conics which touch four given lines, but a notion of the figure of the system may be obtained as follows:

Figure 20 may be taken to represent any quadrilateral whatever, having all its sides real; and if we attend only to the unshaded spaces, it will be seen that there are five regions which are called the inner, upper, lower, right-hand, and left-hand regions respectively. The inner diagonals are AG the vertical diagonal and BD the horizontal diagonal; the former of these traverses the inner, upper, and lower regions; the latter the inner, right-hand, and left-hand regions; the outer diagonal is EF which traverses the upper region and the right-hand and left-hand regions. The inner or vertical and horizontal diagonals meet in the inner region, the vertical and outer diagonals meet in the upper region, the horizontal and outer diagonals meet in the left-hand region. No conic touching the four lines lies wholly or in part in the shaded regions, and every conic touching the four lines lies wholly in the inner region or wholly in the upper region, or partly in the upper and partly in the lower region, or partly in the right-hand and partly in the left-hand region. It will be convenient to consider,

- 1° the conics which lie in the inner region;
- 2° the conics which lie in the upper and lower regions, or in the upper region only;
- 3° the conics which lie in the right-hand and left-hand regions.

1°. The conics in the inner region are obviously ellipses; an extreme term is the finite right line BD , considered as an indefinitely thin ellipse; this gradually broadens out and there is (as a mean term) an ellipse which touches the four lines in the points in which they are intersected two and two by the lines joining the points E, F with the point of intersection of the diagonals AC and BD , the ellipse then narrows in the transverse direction and at length reduces itself to the finite line AC considered as an indefinitely thin ellipse.

2°. Considering first the conics which lie in the upper and lower regions, these are of course hyperbolas, an extreme term is the infinite portions $A\infty$ and $C\infty$ of the line AC , considered as an indefinitely thin hyperbola: we have then hyperbolas such that for each of them, one branch lies in the lower region and touches the two lines through A , while the other branch lies in the upper region and touches the two lines through C ; the points of contact of the lower branch with the lines through A gradually recede from A , but the point of contact on the line AD , which adjoins the left-hand region, recedes with the greater rapidity, and it at last becomes infinite while the point of contact with the line AB which adjoins the right-hand region remains finite: we have thus a hyperbola having the line AD for its asymptote; the point at infinity of this line belongs of course indifferently to the upper and lower regions; and we may therefore consider one branch as lying in the lower region and touching AB and (at infinity) AD ; the other branch as lying in the upper region and touching the two lines through C , and besides (at infinity) the line AD . We have next a series of hyperbolas such that for each of them, one branch lies in the lower region and touches only the line AB , the other branch lies in the upper region and touches the two lines through C and besides the line AD . We arrive again at a limiting case when the point of contact on the line AB passes off to infinity, or AB becomes an asymptote; we have here in the lower region a branch touching (at infinity) AB and in the upper region a branch touching the two lines through C , the line AD , and besides (at infinity) the line AB . To this succeeds a series of hyperbolas such that for each of them, one branch lies in the lower region but does not touch either of the lines through A , the other branch lies in the upper region and touches as well the two lines through C as also the two lines through A . Finally the branch in the lower region passes off to infinity, or the conic becomes a parabola lying wholly in the upper region.

We have thus arrived at the conics which lie wholly in the upper region, the extreme term being a parabola: this passes into an ellipse touching of course the four lines, and gradually reducing itself to the finite line EF considered as an indefinitely thin ellipse.

3°. We come now to the conics which lie in the right-hand and left-hand regions; an extreme term is the conic composed of the infinite portions $E\infty$ and $F\infty$ of the line EF , considered as an indefinitely thin hyperbola; we have then a series of hyperbolas such that for each of them the branch in the right-hand region touches the two

lines through E , and the branch in the left-hand region the two lines through F ; we then arrive at a limiting case where the branch in the right-hand region touches the line EG the upper boundary of the right-hand region at ∞ , or where EG is an asymptote; we may say that the branch in the right-hand region touches the line EB and (at infinity) the line EC , while the branch in the left-hand region touches not only the two lines through F but also (at infinity) the line EG ; we have next a series of hyperbolas such that for each of them the branch in the right-hand region touches only the line EB while the branch in the left-hand region touches as well the two lines through F as the line EC ; we have then again a limiting case, viz. the branch in the right-hand region touches the line BG at infinity, or BG is an asymptote; we may here say that the branch in the right-hand region touches the line BE and (at infinity) the line BG , while the branch in the left-hand region touches the lines through D and (at infinity) the line BG ; and to this succeeds a series of hyperbolas such that for each of them the branch in the right-hand region touches the two lines through B while the branch in the left-hand region touches the two lines through D ; the hyperbola finally becomes the infinite portions $B\infty$ and $D\infty$ of the line BD , considered as an indefinitely thin hyperbola.

It is to be noticed that the entire system commences with the finite line BD considered as an indefinitely thin ellipse and passes on continuously to the infinite portions $B\infty$ and $D\infty$ of the same line, considered as an indefinitely thin hyperbola: it is therefore a cyclical series, and any term whatever might have been taken for the commencement of the series.

281. Note on the Wave Surface

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 142–144.]

In the paper “On the Wave Surface” in the last Number of the Journal, [277], I stated that it was shown in the second of Zech’s Memoirs that the curves of contact of the developables F and Θ with the wave surface were the curves of curvature of the wave surface. I had not examined the demonstration of this theorem, and I had overlooked the author’s note “Die Krümmungslinien der Wellenfläche zweiaxiger Krystalle, &c.,” Crelle, t. LIV. p. 94 (Feb. 1858), where he points out that in a phrase which he quotes, he had assumed without demonstration a theorem which was in fact erroneous, and that he retracted all that he had said in No. 11 of his Memoir (the portion which contains the theorem as to the curves of curvature of the wave surface). M. Bertrand in a note in the *Comptes Rendus*, t. XLVII. pp. 817–819 (Nov. 1858), after referring to my paper, remarks that the theorem as to the curves of curvature of the wave surface appeared to him so remarkable that he hastened to investigate a proof of it, but that he very soon discovered that the theorem was unfortunately erroneous; and he proceeds to show why the theorem cannot be true. M. Bertrand’s demonstration is as follows:

Theorem (I). *If from a point O , we let fall perpendiculars on the tangent planes of a surface, the locus of their feet is a new surface. Let P be a point of this surface corresponding to the point M of the first surface, the normal at P passes through the middle point of OM .*

Theorem (II). *If the curve of curvature of a surface is such that the tangent planes at the several points of the curve are equidistant from a point O , the curve of curvature is situate on a sphere having the point O for its centre.*

For suppose that we let fall from the point O perpendiculars on the tangent planes to the surface at the several points of the curve of curvature in question. The locus of the feet will be a spherical curve, a normal at any point P of the curve will it is clear pass through the point O ; besides, in virtue of the first theorem, another normal will pass through the middle point of the radius OM drawn to the corresponding point M of the surface; the tangent to the curve which is the locus of the points P is therefore perpendicular to the plane MOP , and consequently to the line MP .

But when a developable surface is circumscribed about a sphere, the perpendiculars let fall from the centre of the sphere on the tangent planes of the developable surface have their feet on the generating lines; and consequently the curve which is the locus of the points P is situate on the developable surface (viz., the developable surface enveloped by the tangent planes at the several points of the curve of curvature of the given surface) and cuts the generating lines at right angles. But the curve, the locus of the point M , being by hypothesis a curve of curvature of the given surface, also cuts at right angles the generating lines of the developable surface, the two curves are therefore equidistant curves on the developable surface, viz. MP is constant, and since by hypothesis OP is constant, OM is also constant, which proves the second theorem.

This being premised, it is to be recollected that the wave surface may be generated in two different ways;

- 1° it is the locus of the extremities of the central perpendiculars to the diametral sections of an ellipsoid E , equal to the axes of these sections;
- 2° it is the envelope of planes parallel to the diametral sections of a second ellipsoid E' at distances inversely proportional to the axes of the sections.

To obtain all the tangent planes of the wave surface situate at a distance h from the centre, it is necessary to find in the ellipsoid the diametral sections which have an axis equal to $\frac{1}{h}$; for this, we may cut the ellipsoid by a concentric sphere of the radius $\frac{1}{h}$, and draw tangent planes to the cone having for its vertex the centre of the ellipsoid and passing through the curve of intersection with the sphere. The tangent planes of the wave surface respectively parallel to the tangent planes of the cone will be at the distance h from the centre, and, if they touched the wave surface along a curve of curvature, it would follow from the foregoing theorem II. that their points of contact would be all at the same distance from the centre; but the distance of the centre of the wave surface from the point of contact of a tangent plane is inversely proportional to the perpendicular let fall from the centre of the ellipsoid E' on the tangent plane at the corresponding point of this ellipsoid: it follows that the curve of intersection of an ellipsoid with a concentric sphere is not such that the tangent planes of the ellipsoid at the several points of the curve are at the same

distance from the centre, and consequently the tangent planes which were under consideration do not determine by their points of contact a curve of curvature of the wave surface.

I remark in relation to M. Bertrand's theorem I. that it is immediately connected with a well-known theorem which occurs in optics—viz., if rays proceeding from a point are reflected at a surface, and if from the radiant point to the surface and thence along the reflected ray, we measure off a constant distance, the surface which is the locus of the point so obtained (the secondary focal surface of Dandelin and Quetelet) is an orthogonal trajectory of the reflected rays. In fact, if O be the radiant point, and OM' the incident ray, and if from M' we measure off on the reflected ray a distance $= -OM'$, that is, on the reflected ray produced backwards, a distance $M'P = OM'$, then the whole distance from O is $OM' - OM' = 0$, and the surface which is the locus of the point P is consequently an orthogonal trajectory to the reflected rays. But the point P may, it is clear, be constructed as follows: viz., on the tangent plane at M' let fall the perpendicular OP' , and produce it to a point P such that $OP' = P'M'$. And if we produce OM' to M so that $OM' = M'M$, then it is clear that the locus of M is a surface similar and similarly situated with the original surface, but of double the magnitude, and that OP is the perpendicular from O upon the tangent plane at M of the last-mentioned surface. And, by what precedes, the line PM' from P to the middle point of OM is a normal of the surface which is the locus of P : the corresponding theorem in piano is in fact actually given by Dandelin.

SW Dec. 1858.

282. On a Particular Case of Castillon's Problem

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 157–164.]

The problem referred to as Castillon's problem, being itself a particular case of the problem of the in-and-circumscribed triangle, is as follows: viz. "In a circle to inscribe a triangle the sides of which pass through three given points," and the reciprocal problem of course presents itself "about a circle to circumscribe a triangle, the angles of which lie in given lines." If in Castillon's problem the three given points are the angles of a triangle circumscribed about the circle, or if in the reciprocal problem the three given lines are the sides of a triangle circumscribed about the circle, we have a circle and an inscribed and circumscribed triangle, such that the sides of the inscribed triangle pass through the angles of the circumscribed triangle, and the problem arises "given one of the triangles to determine the other triangle."

The problem, so far as I am aware, was first proposed by Clausen, who has given, Crelle, t. LV. (1829), p. 391, a very elegant solution, which I propose to reproduce here.

Let the angle of a point be defined as the inclination to a fixed radius, of the line from the centre through the given point.

Let α , β , γ , be the angles of the points of contact of the sides of the circumscribed triangle.

α' , β' , γ' , the angles of the angular points of the circumscribed triangle.

a , b , c , the angles of the angular points of the inscribed triangle.

a' , b' , c' , the angles of the intersections of the perpendiculars from the centre on the sides of the inscribed triangle.

Therefore

$$\begin{array}{ll} 2\alpha' = b + c, & 2\alpha' = \beta + \gamma, \\ 2\beta' = c + a, & 2\beta' = \gamma + \alpha, \\ 2\gamma' = a + b, & 2\gamma' = \alpha + \beta. \end{array}$$

Hence observing that the distance of one of the angles of the circumscribed triangle is $\sec \frac{1}{2}(\beta - \gamma)$, and that the projection of this line perpendicular to the corresponding side of the inscribed triangle is equal to $\sec \frac{1}{2}(\beta - \gamma) \cos(\alpha' - a')$, which is equal to the projection

of the radius perpendicular to the same side, or to $\cos \frac{1}{2}(b - c)$, we have

$$\begin{aligned}\cos(\alpha' - a') &= \cos \frac{1}{2}(\beta - \gamma) \cos \frac{1}{2}(b - c), \\ \cos(\beta' - b') &= \cos \frac{1}{2}(\gamma - \alpha) \cos \frac{1}{2}(c - a), \\ \cos(\gamma' - c') &= \cos \frac{1}{2}(\alpha - \beta) \cos \frac{1}{2}(a - b).\end{aligned}$$

Write

$$\begin{aligned}b - c &= 4f, & b + c - 2\alpha &= 4\eta, \\ c - a &= 4g, & c + a - 2\beta &= 4\zeta, \\ a - b &= 4h, & a + b - 2\gamma &= 4\theta;\end{aligned}$$

therefore

$$\begin{aligned}\cos(\eta + \zeta + \theta - h) &= \cos 2f \cos(\eta + \zeta - \theta), \\ \cos(\zeta + \theta + \eta - f) &= \cos 2g \cos(\zeta + \theta - \eta), \\ \cos(\theta + \eta + \zeta - g) &= \cos 2h \cos(\theta + \eta - \zeta),\end{aligned}$$

or, since $f + g + h = 0$,

$$\begin{aligned}\cos\{(\eta - h) + (\zeta + g)\} &= \cos 2f \cos\{(\eta - h) - (\zeta + g)\}, \\ \cos\{(\zeta - f) + (\theta + h)\} &= \cos 2g \cos\{(\zeta - f) - (\theta + h)\}, \\ \cos\{(\theta - g) + (\eta + f)\} &= \cos 2h \cos\{(\theta - g) - (\eta + f)\},\end{aligned}$$

or

$$\tan^2 f = \tan(\eta - h) \tan(\zeta + g), \quad \tan^2 g = \tan(\zeta - f) \tan(\theta + h), \quad \tan^2 h = \tan(\theta - g) \tan(\eta + f).$$

Put

$$\theta + h = \eta_1, \quad \theta - g = \xi_1,$$

and

$$\begin{aligned}\tan \frac{1}{2}f &= l, & \tan \frac{1}{2}\eta &= x, & \tan \frac{1}{2}\eta_1 &= x_1, \\ \tan \frac{1}{2}g &= m, & \tan \frac{1}{2}\zeta &= y, & \tan \frac{1}{2}\zeta_1 &= y_1, \\ \tan \frac{1}{2}h &= n, & \tan \frac{1}{2}\theta &= z, & \tan \frac{1}{2}\theta_1 &= z_1.\end{aligned}$$

We have then

$$\tan^2 \frac{1}{2}f = \tan \frac{1}{2}\eta \tan \frac{1}{2}\eta_1, \quad \tan^2 \frac{1}{2}g = \tan \frac{1}{2}\zeta \tan \frac{1}{2}\zeta_1, \quad \tan^2 \frac{1}{2}h = \tan \frac{1}{2}\theta \tan \frac{1}{2}\theta_1,$$

or

$$l^2 = xx_1, \quad m^2 = yy_1, \quad n^2 = zz_1.$$

But to obtain equations involving only one of the sets (x, y, z) , (x_1, y_1, z_1) , it is proper to write

$$\begin{aligned} \tan^2 \frac{1}{2}f &= \tan(\zeta + g) \tan \frac{1}{2}\eta = \tan \frac{1}{2}\eta \tan(\frac{1}{2}\eta - h), \\ \tan^2 \frac{1}{2}g &= \tan(\eta + h) \tan \frac{1}{2}\zeta = \tan \frac{1}{2}\zeta_1 \tan(\zeta_1 - f), \\ \tan^2 \frac{1}{2}h &= \tan(\eta + f) \tan \frac{1}{2}\theta = \tan \frac{1}{2}\theta \tan(\theta_1 - g); \end{aligned}$$

taking the first set of equations, we have

$$l^2 = \frac{x(z + m)}{1 - mz},$$

therefore

$$x_1 = \frac{z - mx}{x + mz} = \frac{m - (1 - nx)}{x + n},$$

therefore

$$y : 1 - ly = nz + ln + (l - m - n)x : n - lm + (1 + lm - n)x;$$

therefore

$$l^2\{n - lm + (1 + lm - n)x\} \cdot (m + x) = (l - mx)\{nz + ln + (l - m - n)x\};$$

therefore

$$\begin{aligned} &(l^2m^2 + l^2n - l^2mn^2 + l^2m^2n^2) \\ &+ x(-n^2 + lm^2n - l^2mn^2 - l^2m^2n - m^2 - lmn + l^2 - lm - n) \\ &+ x^2(-lm + m^2n - n^2 - lm^2n) = 0. \end{aligned}$$

Now $l + m + n = lmn$, and by means of this relation we find

$$\text{Coef. } x^0 = l\{lm^2 + l^2n - n^2(l + m + n) + m(l + m + n)^2\},$$

$$\begin{aligned} \text{Coef. } x^1 &= -n^2 + mn(l + m + n) - ln(l + m + n) - (l + m + n)^2 - m^2 \\ &\quad - lmn + l^2 - mn(l + m + n), \end{aligned}$$

or, reducing,

$$\text{Coef. } x^2 = l\{-lm + m^2(l + m + n) - ln^2 - n(l + m + n)^2\};$$

$$\begin{aligned}
-\frac{1}{l} \text{Coef. } x &= (m^2 + 2m^2n - n^2) + l(3m^2 + 2mn - n^2) + l^2(m + n) \\
&= (m + n)\{(m^2 + mn - n^2) + l(3m - n) + l^2\} \\
&= (m + n)\{(l + m)^2 + (l + m)(l + n) - (l + n)^2\},
\end{aligned}$$

$$\begin{aligned}
-\frac{1}{l} \text{Coef. } x &= m^2 + m^2n + mn^2 + n^3 + 2l(m^2 + 2mn + n^2) + 2l^2(m + n) \\
&= (m + n)\{m^2 + n^2 + 2l(m + n) + 2l^2\} \\
&= (m + n)\{(l + m)^2 + (l + n)^2\},
\end{aligned}$$

$$\begin{aligned}
\frac{1}{l} \text{Coef. } x^2 &= (m^2 - 2mn^2 - n^3) + l(m^2 - 2mn - 3n^2) - l^2(m + n) \\
&= (m + n)\{(m^2 - mn - n^2) + l(m - 3n) - l^2\} \\
&= (m + n)\{(l + m)^2 - (l + m)(l + n) - (l + n)^2\};
\end{aligned}$$

and the equation becomes

$$\begin{aligned}
&l^2\{(l + m)^2 + (l + m)(l + n) - (l + n)^2\} \\
&\quad - 2lx\{(l + m)^2 + (l + n)^2\} \\
&\quad + x^2\{(l + m)^2 - (l + m)(l + n) - (l + n)^2\} = 0,
\end{aligned}$$

which may be written

$$[l\{(l+m)^2 - (l+m)(l+n) - (l+n)^2\} - \{(l+m)^2 + (l+n)^2\}x]^2 = 5(l+m)^2(l+n)^2,$$

and we have therefore

$$x = \frac{l\{(l + m)^2 + \sqrt{5}(l + m)(l + n) + (l + n)^2\}}{l(l + m)^2 - (l + m)(l + n) - (l + n)^2},$$

or, reducing and observing also that $l^2 = xx_1$, we have

$$\frac{x}{x_1} = \frac{l + m + \frac{1}{2}\{1 + \sqrt{5}\}(l + n)}{l - (l + m) - \frac{1}{2}\{1 + \sqrt{5}\}(l + n)}.$$

The values of (l, m, n) are

$$l = \tan \frac{1}{4}(b - c), \quad m = \tan \frac{1}{4}(c - a), \quad n = \tan \frac{1}{4}(a - b),$$

and those of (x, y, z) , (x_1, y_1, z_1) are

$$\begin{aligned}
x &= \tan \frac{1}{4}(b + c - 2\gamma), & y &= \tan \frac{1}{4}(c + a - 2\alpha), & z &= \tan \frac{1}{4}(a + b - 2\beta), \\
x_1 &= \tan \frac{1}{4}(b + c - 2\beta), & y_1 &= \tan \frac{1}{4}(c + a - 2\beta), & z_1 &= \tan \frac{1}{4}(a + b - 2\alpha),
\end{aligned}$$

and the foregoing result shows that the values of

$$\frac{x}{x_1} \text{ or } \frac{l}{x_1}; \quad \frac{y}{y_1} \text{ or } \frac{m}{y_1}; \quad \frac{z}{z_1} \text{ or } \frac{n}{z_1}$$

are

$$\frac{\sin \frac{1}{4}(a-b) + \frac{1}{2}\{1 + \sqrt{5}\} \sin \frac{1}{4}(c-a)}{\sin \frac{1}{4}(a-b) - \frac{1}{2}\{1 + \sqrt{5}\} \sin \frac{1}{4}(c-a)},$$

$$\frac{\sin \frac{1}{4}(b-c) + \frac{1}{2}\{1 + \sqrt{5}\} \sin \frac{1}{4}(a-b)}{\sin \frac{1}{4}(b-c) - \frac{1}{2}\{1 + \sqrt{5}\} \sin \frac{1}{4}(a-b)},$$

$$\frac{\sin \frac{1}{4}(c-a) + \frac{1}{2}\{1 + \sqrt{5}\} \sin \frac{1}{4}(b-c)}{\sin \frac{1}{4}(c-a) - \frac{1}{2}\{1 + \sqrt{5}\} \sin \frac{1}{4}(b-c)}.$$

Write for a moment $y = \frac{x}{l} = \frac{l}{x_1}$; therefore

$$y = \frac{\sin \frac{1}{4}(b-c) \cdot \frac{1}{2}\{1 + \sqrt{5}\}}{\sin \frac{1}{4}(b-c) - \frac{1}{2}\{1 + \sqrt{5}\} \sin \frac{1}{4}(a-b)};$$

and taking P, Q for the numerator and the denominator respectively of the foregoing fractional expression for k , we find

$$P^2 - Q^2 = 2\{1 + \sqrt{5}\} \sin \frac{1}{4}(a-b) \sin \frac{1}{4}(c-a),$$

$$2PQ = 2 \left[\sin^2 \frac{1}{4}(a-b) - \frac{1}{4}\{1 + \sqrt{5}\}^2 \sin^2 \frac{1}{4}(c-a) \right]$$

$$= -\{1 + \sqrt{5}\} \left[\frac{1}{2}\{1 - \sqrt{5}\} \sin^2 \frac{1}{4}(a-b) + \frac{1}{2}\{1 + \sqrt{5}\} \sin^2 \frac{1}{4}(c-a) \right];$$

also

$$\frac{xx_1}{1 + xx_1} = \tan^2 \frac{1}{2}(\beta - \gamma),$$

we have therefore

$$\tan \frac{1}{2}(\beta - \gamma) = \frac{-2 \sin \frac{1}{4}(b-c) \sin \frac{1}{4}(c-a) \sin \frac{1}{4}(a-b)}{\frac{1}{2}\{1 - \sqrt{5}\} \sin^2 \frac{1}{4}(a-b) + \frac{1}{2}\{1 + \sqrt{5}\} \sin^2 \frac{1}{4}(c-a)},$$

$$\tan \frac{1}{2}(\gamma - \alpha) = \frac{-2 \sin \frac{1}{4}(b-c) \sin \frac{1}{4}(c-a) \sin \frac{1}{4}(a-b)}{\frac{1}{2}\{1 - \sqrt{5}\} \sin^2 \frac{1}{4}(b-c) + \frac{1}{2}\{1 + \sqrt{5}\} \sin^2 \frac{1}{4}(a-b)},$$

$$\tan \frac{1}{2}(\alpha - \beta) = \frac{-2 \sin \frac{1}{4}(b-c) \sin \frac{1}{4}(c-a) \sin \frac{1}{4}(a-b)}{\frac{1}{2}\{1 - \sqrt{5}\} \sin^2 \frac{1}{4}(c-a) + \frac{1}{2}\{1 + \sqrt{5}\} \sin^2 \frac{1}{4}(b-c)},$$

equations which determine the circumscribed triangle when the inscribed triangle is given.

The more general problem, for a conic and an inscribed and circumscribed triangle such that the sides of the inscribed triangle pass through the angles of the circumscribed triangle, "given one of the triangles to determine the other" is solved by Möbius, Crelle, t. V. (1830), p. 103, by means of his Barycentric Calculus, which is in fact the method of trilinear coordinates. The solution is in effect as follows:

Let $\xi = 0$, $\eta = 0$, $\zeta = 0$ be the equations of the sides of the inscribed triangle, ξ , η , ζ may be considered as containing each of them an implicit constant factor, and the equation of the conic may be taken to be

$$\xi^2 + \eta^2 + \zeta^2 = 0;$$

moreover, if $x = 0$, $y = 0$, $z = 0$ be the equations of the sides of the circumscribed triangle, then considering x , y , z as also containing each of them an implicit constant factor, the equation of the conic may be taken to be

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0.$$

Suppose now the sides of the inscribed triangle pass through the angles of the circumscribed triangle, we have equations such as

$$\xi = b'y + c'z, \quad \eta = c'z + a'x, \quad \zeta = a'x + b'y;$$

substituting these values we must have identically

$$\xi^2 + \eta^2 + \zeta^2 + m(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy) = 0,$$

or

$$\begin{aligned} aa' + m &= 0, & bc + bc' + b'c &= 2m, \\ bb' + m &= 0, & ca + ca' + c'a &= 2m, \\ cc' + m &= 0, & ab + ab' + a'b &= 2m, \end{aligned}$$

and, substituting for a' , b' , c' their values,

$$(m - bc)^2 = mbc, \quad (m - ca)^2 = mca, \quad (m - ab)^2 = mab,$$

or, if $m = n^2$,

$$n^2 - bc = nb, \quad n^2 - ca = nc, \quad n^2 - ab = na,$$

[the signs on the second side must be all + or all - and the - sign may be omitted without loss of generality].

Hence

$$a = b = c = -\frac{1}{2}(1 + \sqrt{5})n, \quad \sqrt{5} \text{ being written for } +\sqrt{5}.$$

$$a' = b' = c' = \nu = -\frac{1}{2}(1 - \sqrt{5})n = -\nu a,$$

if for shortness

$$\nu = \frac{1 - \sqrt{5}}{1 + \sqrt{5}},$$

and ν having this value, the equations give

$$\xi = \nu y - z, \quad \eta = \nu z - x, \quad \zeta = \nu x - y,$$

whence also

$$4(2\nu - 1)^2 x = (3\nu - 1)\xi + \nu\eta + \zeta,$$

$$4(2\nu - 1)^2 y = \xi + (3\nu - 1)\eta + \nu\zeta,$$

$$4(2\nu - 1)^2 z = \nu\xi + \eta + (3\nu - 1)\zeta.$$

Each side of the circumscribed triangle has on it four points, viz. two angles of the circumscribed triangle, a point of contact with the conic, and an intersection with the corresponding side of the inscribed triangle. Thus for the side $x = 0$, the four points are given as the intersections of $x = 0$, with

$$y = 0, \quad z = 0, \quad y - z = 0, \quad \nu y + z = 0,$$

and the anharmonic ratio of the four points is therefore a given quantity.*

Again, each side of the inscribed triangle has on it four points, viz. two angles of the inscribed triangle, a point of intersection with the tangent at the opposite angle of the inscribed conic, and a point of intersection with the corresponding side of the circumscribed triangle.

Thus for the side $\xi = 0$, the four points are given as the points of intersection of $\xi = 0$ with the lines

$$\eta = 0, \quad \zeta = 0, \quad \nu\eta + \zeta = 0, \quad \eta + (3\nu - 1)\zeta = 0,$$

and the anharmonic ratio of the four points is therefore a given quantity.*

If we draw tangents at the angles of the inscribed triangle, we have a new triangle, the sides of which are $\eta + \zeta = 0$, $\zeta + \xi = 0$, $\xi + \eta = 0$, and joining the angles of this triangle with the points of contact of the opposite sides (i.e. the angles of the inscribed triangle), we have three lines $\eta - \zeta = 0$, $\zeta - \xi = 0$, $\xi - \eta = 0$ meeting in a point $\xi = \eta = \zeta$, which is obviously the same as the point $x = y = z$, which is the point of intersection of the lines joining the angles of the circumscribed triangle with the points of contact of the opposite sides.*

The coordinates of the points of contact of the sides of the circumscribed triangle are given by $(x = 0, y - z = 0)$, $(y = 0, z - x = 0)$, $(z = 0, x - y = 0)$, these points form therefore an inscribed triangle the sides of which are

$$y + z - x = 0, \quad z + x - y = 0, \quad x + y - z = 0.$$

Again, the tangents at the angles of the inscribed triangle form a circumscribed triangle the sides of which are

$$\xi + \zeta - \nu(\xi + \eta) = (1 - \nu)\xi - \nu\eta + \zeta = 2\nu x - (1 - \nu + \nu^2)y - (1 - \nu + \nu^2)z = -2\nu(y + z - x),$$

and we thus have

$$\begin{aligned} \xi + \zeta - \nu(\xi + \eta) &= -2\nu(y + z - x), \\ \xi + \eta - \nu(\eta + \zeta) &= -2\nu(z + x - y), \\ \eta + \zeta - \nu(\xi + \zeta) &= -2\nu(x + y - z), \end{aligned}$$

equations which show that the sides of the second inscribed triangle pass through the angles of the second circumscribed triangle, and that the two systems are consequently reciprocal.*

The four theorems marked * are all of them contained in the paper by Möbius.

283. On a Theorem Relating to Homographic Figures

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 177–180.]

The following theorem is not new, but I do not remember where to find it:

Given any two homographic figures; there exists in the first figure a point S , which is the focus of an (ellipse or hyperbola, say an) ellipse Σ ; and in the second figure a point S' , which is the focus of a (hyperbola or ellipse, say a) hyperbola Σ' , such that the points S , S' and the conics Σ , Σ' correspond to each other, and the conics Σ , Σ' are so related that the foci and vertices of the one may be superimposed upon the vertices and foci of the other. Moreover the perpendiculars from S , S' upon corresponding tangents of Σ , Σ' will be equal.

Write

$$x : y : 1 = \alpha\xi + \beta\eta + \gamma : \alpha'\xi + \beta'\eta + \gamma' : \alpha''\xi + \beta''\eta + \gamma'',$$

then l , m , λ , μ may be so determined that $(x - l) + i(y - m)$ shall vanish with $(\xi - \lambda) + i(\eta - \mu)$, for if we write

$$\begin{aligned} J &= \alpha - l\alpha'', & J' &= \beta' - m\alpha'', \\ L &= \gamma - l\gamma'', & L' &= \gamma' - m\gamma'', \end{aligned}$$

we have

$$(x - l) + i(y - m) = \frac{\alpha''\xi + \beta''\eta + \gamma''}{\alpha''\xi + \beta''\eta + \gamma''},$$

and therefore

$$\begin{aligned} K + iK' &= i(J + iJ'), \\ L + iL' &= (J' + iJ')(\lambda + i\mu), \end{aligned}$$

that is

$$\begin{aligned} K &= -J', & K' &= J, \\ l\alpha'' + m\beta'' &= -\alpha' + \beta, & l\beta'' + m\gamma'' &= -\alpha + \beta', \end{aligned}$$

whence also

$$\begin{aligned} l(\alpha''^2 + \beta''^2) &= \alpha'\beta'' - \alpha''\beta' + \alpha\alpha'' + \beta\beta'', \\ m(\alpha''^2 + \beta''^2) &= \alpha''\beta - \alpha\beta'' + \alpha\alpha' + \beta\beta', \end{aligned}$$

which determine l, m ; and then λ, μ are given by

but more simply as follows, viz. remarking that if $c = \alpha'\beta'' - \alpha''\beta$, &c., so that

$$\xi : \eta : 1 = ax + a'y + a'' : bx + b'y + b'' : cx + c'y + c'',$$

then we have

$$\begin{aligned}\lambda(c^2 + c'^2) &= bc' - b'c + ac + a'c', \\ \mu(c^2 + c'^2) &= ca' - c'a + bc + b'c';\end{aligned}$$

the equations $x = l, y = m$ give the point S and $\xi = \lambda, \eta = \mu$ the point S' . The values of J, J' are

and therefore

if

we have therefore

$$K = \frac{-c\beta'' - c'\gamma''}{\sqrt{(c^2 + c'^2)}}, \quad K' = \frac{-c'\alpha'' + c\beta''}{\sqrt{(c^2 + c'^2)}},$$

$$k = \sqrt{(c^2 + c'^2)}, \quad K = k \cos \zeta_0, \quad K' = k \sin \zeta_0 :$$

Consider now the expression

$$(x - l) \cos S + (y - m) \sin S - \varpi,$$

which made equal to 0 would be the equation of a line the perpendicular distance of which from S is ϖ , and which distance is inclined at an angle S to the axis of x ; then

$$\begin{aligned}(x - l) \cos S + (y - m) \sin S &= \frac{1}{2(\alpha''\xi + \beta''\eta + \gamma'')} \\ &\times [(\cos S - i \sin S)\{(x - l) + i(y - m)\}(J + iJ') \\ &\quad + (\cos S + i \sin S)\{(x - l) - i(y - m)\}(J - iJ')]\end{aligned}$$

284. On a New Analytical Representation of Curves in Space

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 225–236.]

The ordinary analytical representation of a curve in space by the equations of two surfaces passing through the curve is, even in the case where the curve is the complete intersection of the two surfaces, inappropriate as involving the consideration of surfaces which are extraneous to the curve; and the objection becomes more serious when the curve is not the complete intersection of any two surfaces; for in this case the curve can only be represented in conjunction with a curve or curves extraneous to itself. A curve in space can be in a mode represented by means of the equation of the developable surface having the curve for its edge of regression; but this corresponds to the representation of a plane curve by means of its equation in line coordinates; a representation which is very useful in addition to, but which is not to be substituted for, the equation in point coordinates. It occurred to me some years ago that it might be advantageous to represent a curve in space by means of the cone passing through the curve and having for its vertex an arbitrary point⁽¹⁾; and although I have not advanced beyond the first steps of the theory, the results which I have obtained may, I think, be interesting to geometers. The conclusion is that a curve in space may be represented by a homogeneous equation $V = 0$ between six coordinates (p, q, r, s, t, u) such that $ps + qt + ru = 0$; the function V is moreover such that in virtue of this relation between the coordinates, and of the equation $V = 0$ itself, we have

$$\partial_p V \cdot \partial_s V + \partial_q V \cdot \partial_t V + \partial_r V \cdot \partial_u V = 0,$$

or what is the same thing we have identically

$$\partial_p V \cdot \partial_s V + \partial_q V \cdot \partial_t V + \partial_r V \cdot \partial_u V = LV + M(ps + qt + ru),$$

where L and M are functions of p, q, r, s, t, u . But the converse proposition, viz. any equation whatever $V = 0$, where V satisfies the condition just referred to represents a curve in space, is not true; it would obviously be an important point in the theory to ascertain what further conditions must be satisfied by the function V .

The establishment of the foregoing results is very easy; in fact if x, y, z, w are current coordinates of the ordinary kind (point-coordinates) and $\alpha, \beta, \gamma, \delta$ the coordinates of an arbitrary point; the equation of any cone whatever having for its vertex the point $(\alpha, \beta, \gamma, \delta)$ may be represented by a homogeneous equation between the six determinants of the matrix

$$\begin{vmatrix} x & y & z & w \\ \alpha & \beta & \gamma & \delta \end{vmatrix}$$

or if we write

$$\begin{aligned} p &= \beta z - \gamma y, & s &= \delta x - \alpha w, \\ q &= \gamma x - \alpha z, & t &= \delta y - \beta w, \\ r &= \alpha y - \beta x, & u &= \delta z - \gamma w, \end{aligned}$$

values which it is well known give identically

$$ps + qt + ru = 0,$$

then the cone will be represented by a homogeneous equation

$$V = 0$$

between the six coordinates (p, q, r, s, t, u) . It remains to find the conditions in order that all the cones so represented, viz. the cones obtained by giving any values whatever to the arbitrary quantities $\alpha, \beta, \gamma, \delta$ which enter implicitly into the coordinates p, q, r, s, t, u , pass through one and the same curve; for when this is the case, the equation $V = 0$ may be properly considered as the equation of the curve.

Assume then that all the cones pass through the same curve; if we give to one of the arbitrary quantities $\alpha, \beta, \gamma, \delta$, say α , the infinitesimal variation $d\alpha$, then the function V becomes $V + d\alpha \partial_\alpha V$, and each of the equations $V = 0$, $V + d\alpha \partial_\alpha V = 0$ belongs to a cone passing through the curve; the equation $\partial_\alpha V = 0$ is therefore the equation of a surface passing through the curve; and in like manner the four equations

$$\partial_\alpha V = 0, \quad \partial_\beta V = 0, \quad \partial_\gamma V = 0, \quad \partial_\delta V = 0$$

are each of them the equation of a surface passing through the curve, or these equations must be simultaneously satisfied for all

the points of the curve, they must consequently reduce themselves to two independent relations. But V is given as a function of p, q, r, s, t, u , through which quantities it is a function of $\alpha, \beta, \gamma, \delta$; the last-mentioned four equations become therefore

$$\begin{aligned} 0 &= -\partial_q V \cdot y + \partial_r V \cdot z - \partial_t V \cdot w, \\ 0 &= \partial_r V \cdot x - \partial_p V \cdot z - \partial_u V \cdot w, \\ 0 &= -\partial_s V \cdot x + \partial_p V \cdot y - \partial_u V \cdot w, \\ 0 &= \partial_t V \cdot x + \partial_u V \cdot y + \partial_u V \cdot z, \end{aligned}$$

and from the first, second, and third equation, or any other combination of three equations, we obtain at once the condition

$$\partial_p V \cdot \partial_s V + \partial_q V \cdot \partial_t V + \partial_r V \cdot \partial_u V = 0,$$

so that if this condition is satisfied, the four equations do in fact reduce themselves to two independent equations; the condition in question is thus shown to be necessary.

But the condition only implies that all the cones having their vertices in the neighbourhood of the point $(\alpha, \beta, \gamma, \delta)$ pass through one and the same curve; this will be the case for instance for a series of cones all of them circumscribed about one and the same surface, those having their vertices in the neighbourhood of the point $(\alpha, \beta, \gamma, \delta)$ will all pass through the curve of contact with the surface of the cone having for its vertex the point in question. But the curve of contact is not a fixed curve for all positions of the vertex, and the condition before referred to is consequently insufficient.

It may be noticed that the systems p, q, r and s, t, u are not similar to each other and that the six coordinates cannot be in any way divided into two systems which are similar to each other: the symmetry of the coordinates is in fact that of the vertices (or sides) of a complete quadrilateral (or quadrangle); thus we may divide the coordinates into two sets in a fourfold manner as follows:

$$\begin{aligned} w, \quad t, \quad p; \quad r, \quad q, \quad s, \\ s, \quad q, \quad u; \quad p, \quad t, \quad r, \\ r, \quad t, \quad s; \quad u, \quad q, \quad p, \\ p, \quad q, \quad r; \quad s, \quad t, \quad u, \end{aligned}$$

where each left-hand set corresponds to three vertices forming a triangle and each right-hand set to the remaining three vertices in lineo.

It may be noticed also that if in the equation $V = 0$ of any curve in space we substitute for p, q, r, s, t, u , their values, and equate to zero the coefficients of the different powers and products of $\alpha, \beta, \gamma, \delta$, each of the equations so obtained will belong to a surface passing through the curve, and the entire system of these equations will be equivalent to two relations only between the coordinates x, y, z, w . But any two of these surfaces will not in general intersect only in the curve, i.e. the curve will not be the complete intersection of any two of the surfaces. It may be added that the equation of any other surface whatever through the curve will be obtained by equating to zero a syzygetic function of the functions which equated to zero give the surfaces first referred to.

As an example of the theory, the equation, in the new coordinates, of a line in space will be

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

where the constants are such that

$$AF + BG + CH = 0.$$

This may be verified as follows: suppose that the line in question is given as the line of junction of the points $(\alpha', \beta', \gamma', \delta')$ and $(\alpha'', \beta'', \gamma'', \delta'')$; then the cone through the line and the arbitrary point $(\alpha, \beta, \gamma, \delta)$ is nothing else than the plane through the three points; its equation is therefore

$$\begin{vmatrix} x, & y, & z, & w \\ \alpha, & \beta, & \gamma, & \delta \\ \alpha', & \beta', & \gamma', & \delta' \\ \alpha'', & \beta'', & \gamma'', & \delta'' \end{vmatrix} = 0,$$

which may be written

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

the values of A , &c. being

$$\begin{aligned} A &= \alpha' \delta'' - \alpha'' \delta', & F &= \beta' \gamma'' - \beta'' \gamma', \\ B &= \beta' \delta'' - \beta'' \delta', & G &= \gamma' \alpha'' - \gamma'' \alpha', \\ C &= \gamma' \delta'' - \gamma'' \delta', & H &= \alpha' \beta'' - \alpha'' \beta', \end{aligned}$$

which in fact satisfy the relation

$$AF + BG + CH = 0.$$

So again if the line is given as the intersection of the planes

$$\begin{aligned} ax + by + cz + dw &= 0, \\ a'x + b'y + c'z + d'w &= 0, \end{aligned}$$

then the equation of the cone (plane) through this line and the arbitrary point $(\alpha, \beta, \gamma, \delta)$ is

$$(ax + by + cz + dw)(a'\alpha + b'\beta + c'\gamma + d'\delta) - (a'x + b'y + c'z + d'w)(a\alpha + b\beta + c\gamma + d\delta) = 0,$$

which is

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

the values of A , &c. being

$$\begin{aligned} A &= bc' - b'c, & F &= ad' - a'd, \\ B &= ca' - c'a, & G &= bd' - b'd, \\ C &= ab' - a'b, & H &= cd' - c'd, \end{aligned}$$

which also satisfy the relation

$$AF + BG + CH = 0.$$

I annex the following further investigation: let p', q', r', s', t', u' and $p'', q'', r'', s'', t'', u''$ be what p, q, r, s, t, u become when $\alpha, \beta, \gamma, \delta$ are changed into $\alpha', \beta', \gamma', \delta'$ and $\alpha'', \beta'', \gamma'', \delta''$ respectively: the line represented by the equation

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

is also represented by the equations

$$\begin{aligned} Ap' + Bq' + Cr' + Fs' + Gt' + Hu' &= 0, \\ Ap'' + Bq'' + Cr'' + Fs'' + Gt'' + Hu'' &= 0, \end{aligned}$$

which, reverting to the coordinates x, y, z, w , may be written

$$\mathfrak{A}'x + \mathfrak{B}'y + \mathfrak{C}'z + \mathfrak{D}'w = 0,$$

where

$$\begin{aligned}\mathfrak{A}' &= \cdot & CB' - BC' + FB'', \\ \mathfrak{B}' &= -Ca' & + A\gamma' + GB', \\ \mathfrak{C}' &= B\alpha' & - A\beta' + HB'', \\ \mathfrak{D}' &= -F\alpha' & - G\beta' - H\gamma' \cdot\end{aligned}$$

and similarly for $\mathfrak{A}'', \mathfrak{B}'', \mathfrak{C}'', \mathfrak{D}''$, in which form the equations represent two planes each of them through the given line: the equation of the cone (plane) through the given line and the arbitrary point $(\alpha, \beta, \gamma, \delta)$ is

$$\begin{aligned}(\mathfrak{A}'x + \mathfrak{B}'y + \mathfrak{C}'z + \mathfrak{D}'w)(\mathfrak{A}''\alpha + \mathfrak{B}''\beta + \mathfrak{C}''\gamma + \mathfrak{D}''\delta) \\ - (\mathfrak{A}'\alpha + \mathfrak{B}'\beta + \mathfrak{C}'\gamma + \mathfrak{D}'\delta)(\mathfrak{A}''x + \mathfrak{B}''y + \mathfrak{C}''z + \mathfrak{D}''w) = 0,\end{aligned}$$

or, developing,

$$\begin{aligned}(\mathfrak{B}'\mathfrak{C}'' - \mathfrak{B}''\mathfrak{C}')p + (\mathfrak{C}'\mathfrak{A}'' - \mathfrak{C}''\mathfrak{A}')q + (\mathfrak{A}'\mathfrak{B}'' - \mathfrak{A}''\mathfrak{B}')r \\ + (\mathfrak{A}'\mathfrak{D}'' - \mathfrak{A}''\mathfrak{D}')s + (\mathfrak{B}'\mathfrak{D}'' - \mathfrak{B}''\mathfrak{D}')t + (\mathfrak{C}'\mathfrak{D}'' - \mathfrak{C}''\mathfrak{D}')u = 0.\end{aligned}$$

Now

$$\begin{aligned}\mathfrak{B}'\mathfrak{C}'' - \mathfrak{B}''\mathfrak{C}' &= (-C\alpha' + A\gamma' + G\delta')(B\alpha'' - A\beta'' + H\delta'') \\ &\quad - (-C\alpha'' + A\gamma'' + G\delta'')(B\alpha' - A\beta' + H\delta'),\end{aligned}$$

which putting

$$\begin{aligned}\beta'\gamma'' - \beta''\gamma' &= p, & \alpha'\delta'' - \alpha''\delta' &= s, \\ \gamma'\alpha'' - \gamma''\alpha' &= q, & \beta'\delta'' - \beta''\delta' &= t, \\ \alpha'\beta'' - \alpha''\beta' &= r, & \gamma'\delta'' - \gamma''\delta' &= u,\end{aligned}$$

become

$$A^2p + ABq + ACr - (BG + CH)s + AGt + AHu,$$

or as it may be written

$$A(Ap + Bq + Cr + Fs + Gt + Hu) - (AF + BG + CH)s.$$

Instead of writing at once $AF + BG + CH = 0$, I denote it by V and I write also

$$Ap + Bq + Cr + Fs + Gt + Hu = \Omega.$$

The series of coefficients $\mathfrak{B}'\mathfrak{C}'' - \mathfrak{B}''\mathfrak{C}'$, &c. is

$$\begin{array}{lll} A\Omega - Vs, & B\Omega - Vt, & C\Omega - Vu, \\ F\Omega - Vp, & G\Omega - Vq, & H\Omega - Vr, \end{array}$$

and the required equation, restoring for Ω, V their values, is

$$\begin{aligned} & (Ap + Bq + Cr + Fs + Gt + Hu)(Ap + Bq + Cr + Fs + Gt + Hu) \\ & - (AF + BG + CH)(ps + ps + qt + qt + ru + ru) = 0, \end{aligned}$$

which in virtue of $AF + BG + CH = 0$ becomes

$$(Ap + Bq + Cr + Fs + Gt + Hu)(Ap + Bq + Cr + Fs + Gt + Hu) = 0.$$

The equation

$$Ap + Bq + Cr + Fs + Gt + Hu = 0$$

would imply that the two points $(\alpha', \beta', \gamma', \delta')$, $(\alpha'', \beta'', \gamma'', \delta'')$ are in the same plane with the line $Ap + Bq + Cr + Fs + Gt + Hu = 0$, (or, what is the same thing, that this line is intersected by the line through the two points); in this exceptional case, the planes determined by the given line and the two points respectively are one and the same plane, and they do not by their intersection determine the given line. But in every other case the factor $Ap + Bq + Cr + Fs + Gt + Hu$ is not equal to zero, and the foregoing equation becomes

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

which is as it should be.

In what precedes it has been shown that the equation of the line through the points $(\alpha', \beta', \gamma', \delta')$ and $(\alpha'', \beta'', \gamma'', \delta'')$ is

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

where

$$\begin{array}{ll} A = \alpha'\delta'' - \alpha''\delta', & F = \beta'\gamma'' - \beta''\gamma', \\ B = \beta'\delta'' - \beta''\delta', & G = \gamma'\alpha'' - \gamma''\alpha', \\ C = \gamma'\delta'' - \gamma''\delta', & H = \alpha'\beta'' - \alpha''\beta', \end{array}$$

and that the equation of the line of intersection of the planes $ax + by + cz + dw = 0$, $a'x + b'y + c'z + d'w = 0$ is

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

where

$$\begin{aligned} A &= bc' - b'c, & F &= ad' - a'd, \\ B &= ca' - c'a, & G &= bd' - b'd, \\ C &= ab' - a'b, & H &= cd' - c'd. \end{aligned}$$

It may be added that the condition of intersection of the two lines

$$\begin{aligned} Ap + Bq + Cr + Fs + Gt + Hu &= 0, \\ A'p + B'q + C'r + F's + G't + H'u &= 0, \end{aligned}$$

is

$$AF' + A'F + BG' + B'G + CH' + C'H = 0,$$

and any other problems in relation to the line, for instance to find the condition that a line may pass through a given point or lie in a given plane, &c. may also be solved by means of the new coordinates.

A curve of the second order in space is either a pair of lines or a plane conic, each of which species may degenerate into a pair of intersecting lines. If we write

$$W = Ap + Bq + Cr + Fs + Gt + Hu, \quad W' = A'p + B'q + C'r + F's + G't + H'u,$$

then the equation of the pair of lines will be $V = WW' = 0$, and it is worth while to show that this value of V satisfies the fundamental equation $\partial_p V \cdot \partial_s V + \partial_q V \cdot \partial_t V + \partial_r V \cdot \partial_u V = 0$; we have in fact $\partial_p V = AW' + A'W$, &c. and thence the left-hand side is

$$W'^2(AF + BG + CH) + WW'(AF' + A'F + BG' + B'G + CH' + C'H) + W^2(A'F' + B'G' +$$

which in fact vanishes in virtue of the relations

$$AF + BG + CH = 0, \quad A'F' + B'G' + C'H' = 0, \quad WW' = 0.$$

The like proof applies to any curve made up of two or more curves.

Consider in the next place the plane conic given by the equations

$$\begin{aligned} ax + by + cz + dw &= 0, \\ x^2 + y^2 + z^2 + w^2 &= 0. \end{aligned}$$

To find the equation of the cone having for its vertex the point $(\alpha, \beta, \gamma, \delta)$ and passing through the conic we must, according to Joachimsthal's method, substitute in the two equations for x, y, z, w the values $\lambda x + \mu\alpha$, $\lambda y + \mu\beta$, $\lambda z + \mu\gamma$, $\lambda w + \mu\delta$, and from the resulting equations eliminate λ, μ . The two equations become

$$\begin{aligned}\lambda^2(x^2 + y^2 + z^2 + w^2) + 2\lambda\mu(\alpha x + \beta y + \gamma z + \delta w) + \mu^2(\alpha^2 + \beta^2 + \gamma^2 + \delta^2) &= 0, \\ \lambda(ax + by + cz + dw) + \mu(a\alpha + b\beta + c\gamma + d\delta) &= 0,\end{aligned}$$

and thence eliminating λ, μ we find

$$\begin{aligned}(a\alpha + b\beta + c\gamma + d\delta)^2(x^2 + y^2 + z^2 + w^2) \\ - 2(a\alpha + b\beta + c\gamma + d\delta)(ax + by + cz + dw)(\alpha x + \beta y + \gamma z + \delta w) \\ + (ax + by + cz + dw)^2(\alpha^2 + \beta^2 + \gamma^2 + \delta^2) = 0,\end{aligned}$$

where the left-hand side is as it should be a function of p, q, r, s, t, u ; the equation may in fact be written

$$\begin{aligned}a^2(q^2 + r^2 + s^2) + b^2(r^2 + p^2 + t^2) + c^2(p^2 + q^2 + u^2) + d^2(s^2 + t^2 + u^2) \\ + 2bc(-qr + tu) + 2ca(-rp + us) + 2ab(-pq + st) \\ + 2ad(qu - rt) + 2bd(rs - pu) + 2cd(pt - qs) = 0,\end{aligned}$$

which treated as a quadric in (a, b, c, d) may be written

$$\Omega = 0,$$

or treated as a quadric in (p, q, r, s, t, u) , in the form

$$\Omega(p, q, r, s, t, u)^2 = 0.$$

Representing the equation by $V = 0$, the function V should verify the fundamental equation $\partial_p V \cdot \partial_s V + \partial_q V \cdot \partial_t V + \partial_r V \cdot \partial_u V = 0$, and we in fact have

$$\begin{aligned}\partial_p V &= (b^2 + c^2)p - abq - acr + cdt - bdu, \\ \partial_s V &= -cdq + bdr + (a^2 + c^2)s + abt + acu,\end{aligned}$$

and thence forming the product $\partial_p V \cdot \partial_s V$, and from it the two analogous products, we find by which the equation is verified.

I conclude with some remarks relating to a different part of the subject. It is shown by M. E. de Jonquières in his "Essai sur la génération des courbes géométriques, &c." *Mém. Prés. à l'Acad. de Paris*,

t. XVI. (1858), that the equation $U = 0$ of any geometrical plane curve can be presented in a variety of different ways in the form

$$\begin{vmatrix} P, & Q \\ P', & Q' \end{vmatrix} = 0,$$

this being in fact the fundamental theorem of his very beautiful investigations. I am not aware that it has ever been considered whether the equation $U = 0$ of a plane curve can in general be represented in the form

$$\begin{vmatrix} P, & Q, & R \\ P', & Q', & R' \\ P'', & Q'', & R'' \end{vmatrix} = 0,$$

or in the analogous forms where the left-hand side is a determinant of a higher order. As regards surfaces, the equation $U = 0$ of a geometrical surface cannot in general be presented in the form

$$\begin{vmatrix} P, & Q \\ P', & Q' \end{vmatrix} = 0,$$

nor in the similar forms where the left-hand side is a determinant of a higher order. We may consequently classify surfaces of a given order according to the forms of this nature by which they can be represented, or as I propose to term it, according to their "frangibility." It is obvious that this question is immediately connected with that of the representation of curves in space by means of the ordinary coordinates of analytical geometry; for instance if we have a surface $U = 0$, which can be represented in the form

$$\begin{vmatrix} P, & Q, & R \\ P', & Q', & R' \\ P'', & Q'', & R'' \end{vmatrix} = 0,$$

then we can describe upon the surface curves such as

$$\begin{vmatrix} P', & Q', & R' \\ P'', & Q'', & R'' \end{vmatrix} = 0,$$

viz. the curve so represented is the curve which in conjunction with the curve $(R = 0, R'' = 0)$ makes up the complete intersection of the two surfaces $P'R'' - P''R' = 0, Q'R'' - Q''R' = 0$: the curve in question is not in general the complete intersection of any two surfaces. If the

surface $U = 0$ can be represented in the form $\left| \begin{smallmatrix} P, & Q \\ P', & Q' \end{smallmatrix} \right| = 0$, then we can describe upon the surface, curves such as the curve $(P = 0, Q = 0)$ which, although it is the complete intersection of two surfaces, is not the complete intersection of the given surface $U = 0$ by any other surface. But if the surface $U = 0$ cannot be represented in any such form, or as we may express it, if the surface is infrangible, then it would appear that the only curves which can be described upon the surface are those which are the complete intersection of the given surface by some other surface. The question is, I think, an interesting one in the theory of surfaces, but I doubt whether much will be done in this manner as regards the theory of curves in space, and it appears to me that there is more to be hoped for from the theory previously explained in the present paper.

2, Stone Buildings, W.C., June 2, 1859.

285. On the System of Conics Having Double Contact with Each Other

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 246–250.]

Consider the conics which pass through four points coinciding two and two together; the two points of each pair of coincident points are to be regarded as lying in a line which will be a tangent to the conic, and the system is thus a system of conics touching the same two lines at the same two points: or if we replace the two lines by a conic of the system, it is the system of conics having double contact with a given conic at the same two points. The lines may be spoken of as the tangents, and the points as the ineunts of the system; the line joining the two points is the axis, and the point of intersection of the two lines, the pole. It is assumed that the system contains real conics; the pole and axis must consequently be real, but the tangents and ineunts may be either all of them real or else all of them imaginary—we may for shortness say that in the former case the double contact is real, and that in the latter case it is imaginary. Consider first the case of a real double contact, the system of conics commences with an ellipse differing only slightly from the finite portion of the axis included between the ineunts, this ellipse being the first of a series of ellipses the last of which only slightly differs from a parabola, we have then a parabola, then a hyperbola differing only slightly from the parabola, and which is the first of a series of hyperbolas of which the last differs only slightly from the two tangents: all the before-mentioned conics are included within the angles at the pole or vertex of the triangle, viz. the ellipses and parabola within the angle towards the axis or base, and the hyperbolas, one branch of each of them within this angle and the other branch within the opposite angle. And it will be convenient to term these conics the “Multiform Series.” Passing over the intermediate case, we come to a hyperbola differing only slightly from the two tangents, the first of a series of hyperbolas of which the last differs only slightly from the infinite portions, not included between the ineunts, of the axis; all these hyperbolas lying without the angles at the pole or vertex of the triangle. The last-mentioned hyperbolas may be termed the “Uniform Series.” Passing over another intermediate case we return to the multiform series, the entire system of conics forming in fact a complete cycle. The intermediate

case forming the transition from the multiform to the uniform series is a conic which considered as generated by a point is the pair of tangents, but which considered as enveloped by a line is a pair of points coincident with the pole. The intermediate case forming the transition from the uniform series to the multiform series is a conic which considered as generated by a point is a pair of lines coincident with the axis, but considered as enveloped by a line is the pair of ineunts.

When the double contact is imaginary, the conics of the system may still be considered as forming a multiform and a uniform series, but the uniform series consists entirely of imaginary conics. The multiform series commences with an ellipse differing only slightly from the pole; this is the first of a series of ellipses of which the last differs only slightly from a parabola, we have next the parabola, and then a series of hyperbolas, the first of which differs only slightly from the parabola, and the last of which differs only slightly from the axis. The ellipses and parabolas lie all of them on the same side of the axis; the hyperbolas lie, one branch of each on the one side of the axis, and the other branch on the other side. The uniform series (being imaginary) of course does not admit of description. The intermediate or transition cases are the same as for a real double contact, with only the variation occasioned by the tangents and ineunts being imaginary. The conics of the entire system are considered (as in the case of a real double contact) to form a cycle, but here part of the cycle, viz. the uniform series, is imaginary.

Suppose that M and U stand for the multiform and uniform series, MU for the transition form from the multiform to the uniform series, and UM for the transition form from the uniform to the multiform series. The cycle may be represented as in the figure.

	M	U
	M	U

Call either of the conics MU, UM the centre of the cycle, and say that the cycle is arranged line-wise when MU is considered as the centre, and point-wise when UM is considered as the centre. We may pass from one conic of the cycle to another incentrically or excentrically, i.e. by passing towards the centre or away from the centre. The intermediates of two conics are the conics passed through in going from the first to the second incentrically, the extramediates are those passed through in going from the first to the second excentrically. In speaking of the extramediates or the intermediates of a single conic,

such conic is considered as the first conic and the second conic is understood to be the centre. The nearer extramediates are those passed through previous to reaching an extremity of the cycle, the further extramediates are those passed through subsequent to reaching an extremity of the cycle. It is clear that the intermediates and the nearer extramediates make up the series multiform or uniform to which the conic belongs, and that the further extramediates are the other series; and moreover that if the cycle be arranged line-wise and point-wise successively, the intermediates of the one arrangement are the nearer extramediates of the other arrangement and vice versa; the further extramediates of each arrangement being the same.

Two points of a conic may be said to be conjunctive with respect to a given line when it is possible to pass from the one to the other along the curve without crossing the line and disjunctive in the contrary case—it being understood that both the parabola and the hyperbola are to be treated as closed curves, viz. the points at infinity of the parabola are to be considered as one and the same point, and so the points at each extremity of either asymptote of the hyperbola are to be considered as one and the same point—and in like manner two tangents of a conic are said to be conjunctive with respect to a given point, when it is possible by the revolution of the one tangent to arrive at the other tangent without sweeping through the point in question, and disjunctive in the contrary case.

Consider now any conic of the series, and call this simply the conic. Take upon the conic any two points a, a' and joining these with a variable point in the axis, let the lines so obtained meet the conic in the points b', b (so that ab' and $a'b$ meet on the axis). We have thus on the conic a series of points $a, b, c, \dots$ and a second series $a', b', c', \dots$ which are homographically related to each other, and which possess the property that taking any two points b, c of the first series and the corresponding points b', c' of the second series, the lines $bc', b'c$ meet on the axis. It is immaterial how the point a is chosen, but a being chosen at pleasure, the system will obviously depend on the way in which the point a' is chosen; so that the points of a conic may be considered as homographically related to each other in an infinity of different ways. The reciprocal construction of course applies to the tangents of a conic, so that the tangents of a conic may be considered as homographically related to each other in an infinity of different ways: and not only this, but it is clear that the tangents at points homographically related to each other are also homographically related

to each other, and vice versâ.

The line joining corresponding points envelopes a conic, one of the conics of the system, and which may for shortness be spoken of simply as the envelope. The point of intersection of corresponding tangents generates a conic, one of the conics of a system, and which may in like manner be spoken of simply as the locus.

We may now enunciate the following theorem: Let the system be arranged line-wise, the envelope is an extramediate of the conic; viz. if a pair of corresponding points of the conic (all pairs have in this respect the same property) be conjunctive with respect to the axis, a nearer extramediate, but if disjunctive, then a further extramediate.

And again: Let the system be arranged point-wise, the locus is an extramediate of the conic; viz. if a pair of corresponding tangents of the conic (all pairs have in this respect the same property) be conjunctive with respect to the pole, a nearer extramediate, but if disjunctive, then a further extramediate.

It is however proper to remark that when the double contact is imaginary, the pair of corresponding points or tangents (in fact any pair of points or tangents) is necessarily conjunctive, so that in this case the second alternatives in the two theorems have no application, and the envelope or locus is always a nearer extramediate. When the double contact is real, the pair of corresponding points or tangents may be either conjunctive or disjunctive, and the character (nearer or further) of the extramediate is determined accordingly.

The double contact being either real or imaginary, we may as a limiting case assume that the corresponding points or tangents coincide, the envelope or locus is in this case the conic itself. Suppose that the double contact is real, we have here two other limiting cases, viz. first, one of a pair (and therefore of each pair) of corresponding points or tangents may be situate in the axis or pass through the pole; this is the limit between the two cases of the pair of corresponding points or tangents being conjunctive and disjunctive, and therefore the envelope or locus must be the conic which is the limit between the nearer extramediates and the further extramediates, i.e. the envelope is the pair of ineunts and the locus is the pair of tangents. Secondly, the line through a pair of corresponding points may pass through the pole, or the point of intersection of a pair of corresponding tangents may lie in the axis; the envelope or locus is here the furthest of the extramediates, viz. the envelope is the pair of points coincident with the pole, the locus is the pair of lines coincident with the axis. If

considering one of the pair of corresponding points or lines as fixed, the other point or line passes through the last-mentioned limiting position, the envelope or locus returns back through the series of further extramediates.

In the statement of the preceding theorems, the envelope and locus have been considered separately, but we may if we please consider the locus as generated by the point of intersection of the tangents at the corresponding points: the locus and envelope are in this case reciprocal polars with respect to the conic; it should be noticed that the pair of corresponding points and the pair of corresponding tangents are either both conjunctive, or else both disjunctive.

2, Stone Buildings, W.C., 3rd June, 1859.

286. Note on the Value of Certain Determinants, the Terms of which are the Squared Distances of Points in a Plane or in Space

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 275–277.]

The values of the several determinants mentioned in my paper “On a Certain Theorem in the Geometry of Position,” Cambridge Mathematical Journal, Old Series, t. II. (1842), p. 267, [1], are as follows:

(1)

$$\begin{vmatrix} 0, & 12^2, & 13^2, & 1 \\ 21^2, & 0, & 23^2, & 1 \\ 31^2, & 32^2, & 0, & 1 \\ 1, & 1, & 1, & 0 \end{vmatrix} = 2 \cdot 12^2 \cdot 21^2 \cdot 23^2,$$

where the Σ , Σ contain 3 and 6 terms respectively.

(2)

$$\begin{vmatrix} 0, & 12^2, & 13^2, & 14^2, & 1 \\ 21^2, & 0, & 23^2, & 24^2, & 1 \\ 31^2, & 32^2, & 0, & 34^2, & 1 \\ 41^2, & 42^2, & 43^2, & 0, & 1 \\ 1, & 1, & 1, & 1, & 0 \end{vmatrix}$$

where the Σ , Σ , Σ contain 24, 12 and 8 terms respectively.

$$\begin{aligned} &= 2 \cdot 12^2 \cdot 23^2 \cdot 34^2 - 2 \cdot 12^2 \cdot 34^2 \cdot 43^2 - 2 \cdot 12^2 \cdot 23^2 \cdot 31^2 \\ &\quad - 2 \cdot 12^2 \cdot 23^2 \cdot 34^2 \cdot 45^2 - 2 \cdot 12^2 \cdot 23^2 \cdot 34^2 \cdot 43^2 \\ &\quad + 2 \cdot 12^2 \cdot 23^2 \cdot 45^2 \cdot 54^2 + 2 \cdot 12^2 \cdot 23^2 \cdot 34^2 \cdot 41^2 + 2 \cdot 12^2 \cdot 23^2 \cdot 31^2 \cdot 45^2. \end{aligned}$$

(3)

$$\begin{vmatrix} 0, & 12^2, & 13^2, & 14^2, & 15^2, & 1 \\ 21^2, & 0, & 23^2, & 24^2, & 25^2, & 1 \\ 31^2, & 32^2, & 0, & 34^2, & 35^2, & 1 \\ 41^2, & 42^2, & 43^2, & 0, & 45^2, & 1 \\ 51^2, & 52^2, & 53^2, & 54^2, & 0, & 1 \\ 1, & 1, & 1, & 1, & 1, & 0 \end{vmatrix}$$

where the Σ , Σ , Σ , Σ , Σ contain 120, 15, 60, 30, and 40 terms respectively.

(4)

$$\begin{vmatrix} 0, & 12^2, & 13^2, & 14^2 \\ 21^2, & 0, & 23^2, & 24^2 \\ 31^2, & 32^2, & 0, & 34^2 \\ 41^2, & 42^2, & 43^2, & 0 \end{vmatrix}$$

where the Σ , Σ contain 3 and 6 terms respectively.

$$\begin{aligned} &= 2 \cdot 12^2 \cdot 21^2 \cdot 34^2 \cdot 43^2 - 2 \cdot 12^2 \cdot 23^2 \cdot 34^2 \cdot 41^2, \\ &\quad 2 \cdot 12^2 \cdot 23^2 \cdot 34^2 \cdot 45^2 \cdot 51^2 - 2 \cdot 12^2 \cdot 23^2 \cdot 31^2 \cdot 45^2 \cdot 54^2. \end{aligned}$$

(5)

$$\begin{vmatrix} 0, & 12^2, & 13^2, & 14^2, & 15^2 \\ 21^2, & 0, & 23^2, & 24^2, & 25^2 \\ 31^2, & 32^2, & 0, & 34^2, & 35^2 \\ 41^2, & 42^2, & 43^2, & 0, & 45^2 \\ 51^2, & 52^2, & 53^2, & 54^2, & 0 \end{vmatrix}$$

where the Σ , Σ contain 24 and 20 terms respectively.

And it is proper to remark that it is not in the preceding formulæ (as in the memoir above referred to in which 12 denotes a distance between two points 1 and 2) assumed that 12 and 21 are equal.

The formula (1) gives, if a, b, c represent the sides, the well-known expression for the area of a triangle

$$(\text{area})^2 = \frac{1}{16}(2b^2c^2 + 2c^2a^2 + 2a^2b^2 - a^4 - b^4 - c^4).$$

Similarly the formula (2), if a, b, c, f, g, h represent the edges, viz. $23 = a$, $31 = b$, $12 = c$, $14 = f$, $24 = g$, $34 = h$, gives for the volume of a tetrahedron

$$\begin{aligned} 288(\text{volume})^2 &= b^2c^2(g^2 + h^2) + c^2a^2(h^2 + f^2) + a^2b^2(f^2 + g^2) \\ &\quad + a^2f^2(b^2 + c^2) + b^2g^2(c^2 + a^2) + c^2h^2(a^2 + b^2) \\ &\quad - a^2f^2(a^2 + f^2) - b^2f^2(b^2 + g^2) - c^2h^2(c^2 + h^2) \\ &\quad - a^2g^2h^2 - b^2f^2h^2 - c^2f^2g^2 - a^2b^2c^2, \\ &= W \text{ suppose.} \end{aligned}$$

Now

$$\begin{aligned}
 16 \times (\text{surface})^2 &= 2g^2a^2 + 2a^2b^2 + 2b^2g^2 - a^4 - b^4 - g^4 \\
 &\quad + 2f^2h^2 + 2h^2b^2 + 2b^2f^2 - h^4 - b^4 - f^4 \\
 &\quad + 2g^2f^2 + 2f^2c^2 + 2c^2g^2 - g^4 - f^4 - c^4 \\
 &\quad + 2b^2c^2 + 2c^2a^2 + 2a^2b^2 - a^4 - b^4 - c^4, \\
 &= x + y + z + w \text{ suppose,}
 \end{aligned}$$

and the norm of this is

$$(x^2 + y^2 + z^2 + w^2 - 2yz - 2zx - 2xy - 2xw - 2yw - 2zw)^2 = -64xyzw,$$

which is of the sixteenth order, and must be of the form $W \cdot Q$ where Q is a function of a, b, c, f, g, h of the tenth order. The expression of this function is given by Prof Sylvester in his paper "On the Relation between the Volume of a Tetrahedron &c.," Camb. and Dubl. Math. Jour., t. VIII. (1853), pp. 171-178, viz. the value is

$$\begin{aligned}
 Q &= a^2b^2c^2(f^4 + g^4 + h^4 + 5f^2g^2 + 5h^2f^2 + f^2g^2h^2 + c^2a^2 + a^2b^2 \\
 &\quad - (f^2 + g^2 + h^2)(a^2 + b^2 + c^2)) \\
 &\quad + a^2f^2h^2(f^4 + b^4 + c^4 + b^2c^2 + c^2f^2 + g^2h^2 + h^2f^2 + f^2b^2 \\
 &\quad - (f^2 + b^2 + c^2)(a^2 + g^2 + h^2)) \\
 &\quad + b^2g^2f^2(g^4 + c^4 + a^4 + c^2a^2 + a^2f^2 + g^2c^2 + h^2f^2 + f^2h^2 \\
 &\quad - (g^2 + c^2 + a^2)(b^2 + h^2 + f^2)) \\
 &\quad + c^2h^2g^2(h^4 + a^4 + b^4 + a^2b^2 + b^2g^2 + h^2a^2 + f^2g^2 + g^2f^2 \\
 &\quad - (h^4 + a^2 + b^2)(c^2 + f^2 + g^2)),
 \end{aligned}$$

and, as there remarked, the equation $Q = 0$ expresses the condition that the radius of the inscribed sphere may be infinite.

2, Stone Buildings, W.C., June 10th, 1859.

287. Note on the Equation for the Squared Differences of the Roots of a Cubic Equation

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 307–309.]

The question of finding the equation for the squared differences of the roots, presents, in the case of a cubic equation, a peculiarity which does not occur for equations of a higher order, viz. we may in the first instance form the equation for the differences of the roots taken in a given cyclical order, and thence deduce the equation for the squared differences of the roots. Let the cubic equation be

$$U = (a, b, c, d \curvearrowright x, 1)^3 = a(x - \alpha)(x - \beta)(x - \gamma) = 0,$$

the function

$$\Pi\{\theta - (\beta - \gamma)\},$$

which equated to zero gives for θ the values $\beta - \gamma$, $\gamma - \alpha$, $\alpha - \beta$, which are the differences of the roots taken in a given circular order, has for any interchanges whatever of the roots, two values only, viz. that just written down, and the value $\Pi\{\theta - (\gamma - \beta)\}$, which may be deduced therefrom by changing first the sign of θ and then the sign of the entire expression (or what is the same thing, by changing the signs of the terms containing the even powers of θ); we may consequently write

$$\Pi\{\theta - (\beta - \gamma)\} = P - Q\theta i(\alpha, \beta, \gamma),$$

which P , Q are symmetrical functions of the roots, and $i(\alpha, \beta, \gamma)$ or $(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$ is a function the square of which is a symmetrical function of the roots, and such symmetrical functions of the roots can of course be expressed as functions of the coefficients. We have in fact

$$P = a^2\theta^2 + \theta(2\beta\gamma - 3\alpha^2) = a^{-2}\{a^2\theta^2 + 9(ac - b^2)\theta\},$$

$$Q = 1,$$

and

$$\Pi(\beta - \gamma)^2 = -27D,$$

where D is the discriminant of the cubic function,

$$D = a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2.$$

Consequently

$$\Pi\{\theta - (\beta - \gamma)\} = a^{-2}\{a^2\theta^2 + 9(ac - b^2)\theta - \sqrt{-27D}\},$$

and forming the similar equation

$$\Pi\{\theta + (\beta - \gamma)\} = a^{-2}\{a^2\theta^2 + 9(ac - b^2)\theta + \sqrt{-27D}\},$$

multiplying the two equations together and writing u in the place of θ^2 , we find

$$\Pi\{u - (\beta - \gamma)^2\} = a^{-4}\{[a^2u + 9(ac - b^2)]^2u + 27D\},$$

and the equation for the squared differences of the roots is thus seen to be

$$\{a^2u + 9(ac - b^2)\}^2u + 27D = 0,$$

or what is the same thing

$$a^6u^3 + 18a^4(ac - b^2)u^2 + 81(ac - b^2)^2u + 27D = 0.$$

I remark that if ω is an imaginary cube root of unity (so that $(\omega - \omega^2)^2 = -3$, $\omega - \omega^2$ being thus only another form of $\sqrt{-3}$) then if in the expression for $\Pi\{\theta - (\beta - \gamma)\}$ we write $\frac{\theta}{a(\omega - \omega^2)}$ in the place of θ , the equation assumes the more simple form

$$\Pi\{\theta - \frac{1}{2}a(\omega - \omega^2)(\beta - \gamma)\} = \theta^2 - 3H\theta - a^{-2}\sqrt{D},$$

which if U be the cubic function, H the Hessian $= (ac - b^2, ad - bc, bd - c^2 \curvearrowright x, y)^2$, and D the discriminant as before, is a particular case (obtained by writing $x = 1$, $y = 0$) of the equation

$$\Pi\{\theta - \frac{1}{2}a(\omega - \omega^2)(x - \alpha y)\} = \theta^2 - 3H\theta - U\sqrt{D},$$

which equation can be at once obtained from the equation (where Φ is the cubicovariant of the cubic function)

$$\sqrt[3]{(\Phi + U^2\sqrt{D})} - \sqrt[3]{(\Phi - U^2\sqrt{D})} = \frac{1}{2}a(\omega - \omega^2)(\beta - \gamma)(x - \alpha y),$$

given in my Fifth Memoir on Quantics, Phil. Trans., t. CXLVIII. (1858), [156]. For writing for a moment

$$\sqrt[3]{(\Phi + U^2\sqrt{D})} = X, \quad \sqrt[3]{(\Phi - U^2\sqrt{D})} = Y,$$

we find

$$\theta^3 + 3\sqrt[3]{XY}\theta - (X - Y) = 0,$$

or

$$\theta^2 + 3\sqrt[3]{XY}\theta - (X - Y) = 0,$$

where $\sqrt[3]{XY} = \sqrt[3]{(\Phi^2 - U^4D)}$, which by the equation (given in the Memoir) is $= -H$, and $(X - Y)$ is $= U^2\sqrt{D}$, so that the equation in θ is, as above, $\theta^2 - 3H\theta - U\sqrt{D} = 0$, an equation which is satisfied by $\theta = \frac{1}{2}a(\omega - \omega^2)(\beta - \gamma)(x - \alpha y)$; and the other two roots being of course of the like form, the cubic function in θ is equal to $\Pi\{\theta - \frac{1}{2}a(\omega - \omega^2)(\beta - \gamma)(x - \alpha y)\}$ which proves the theorem.

2, Stone Buildings, W.C., Nov. 3rd, 1859.

C. IV. 59

288. Note on the Curvature of a Plane Curve at a Double Point, and on the Curvature of Surfaces

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 322–326.]

The radius of curvature of a plane curve at a double point is most readily determined as follows: viz. if x, y be the coordinates of the double point, θ the inclination to the axis of x of the tangent to the branch under consideration, and u, v the coordinates measured from the double point as origin, parallel and perpendicular to the tangent, of the consecutive point on such branch, then the radius of curvature for the branch in question is

$$\frac{u^2}{2v};$$

the coordinates of the consecutive point are

$$x + u \cos \theta - v \sin \theta, \quad y + u \sin \theta + v \cos \theta,$$

and the value of $\frac{u^2}{2v}$ is to be found by substituting these expressions for x, y in the equation of the curve. Let $U = 0$ be the equation of the curve; then at the double point, U and the differential coefficients of the first order vanish, let those of the second order be (a, b, c) and those of the third order (a, b, c, d) , then substituting we have

$$\frac{1}{2}(a, b, c \curvearrowright u \cos \theta - v \sin \theta, u \sin \theta + v \cos \theta)^2 + \frac{1}{6}(a, b, c, d \curvearrowright u \cos \theta - v \sin \theta, u \sin \theta + v \cos \theta)$$

The coefficient of u^2 is

$$\frac{1}{2}(a, b, c \curvearrowright \cos \theta, \sin \theta)^2,$$

which vanishes, since it is by putting this function equal to zero that we find the direction of the tangents at the double point. For the present purpose we are concerned with one of the two branches only, and in all that follows the ratio $\cos \theta : \sin \theta$ will denote a determinate root of the quadratic equation; viz. the root which corresponds to the branch in question.

The equation takes therefore the form

$$Buv + \frac{1}{2}Cv^2 + \frac{1}{3}Du^3 + \&c. = 0,$$

which might be satisfied by assuming $Buv + \frac{1}{2}Cv^2 = 0$, but the values so obtained belong to the branch which does not touch the tangent; the proper solution is

$$Buv + \frac{1}{3}Du^3 = 0,$$

and thence

$$\frac{u^2}{2v} = -\frac{3B}{D},$$

or substituting for B, D their values, the radius of curvature is

$$-\frac{3(a, b, c \frown \cos \theta, \sin \theta)(-\sin \theta, \cos \theta)}{(a, b, c, d \frown \cos \theta, \sin \theta)^3},$$

where the expression in the numerator is

$$\begin{aligned} &= (-a + c) \sin^2 \theta \cos \theta + b(\cos^2 \theta - \sin^2 \theta), \\ &= \frac{1}{2}\{(c - a) \sin 2\theta + 2b \cos 2\theta\}. \end{aligned}$$

The curvature, at the point of contact, of the curve in which any surface is intersected by a tangent plane cannot be found by the ordinary theory of the curvature of surfaces, being (by reason that the point of contact is a double point on the curve) a case of exception from that theory. The foregoing method may be applied to the solution of the question as follows: Let $U = 0$ be the equation of the surface, and suppose that (x, y, z) refer to the point of contact of the tangent plane. Let L, M, N be the first differential coefficients of U at this point, (a, b, c, f, g, h) the second differential coefficients, $(a, b, c, f, g, h, i, j, k, l)$ the third differential coefficients. Let λ, μ, ν be proportional to the cosines of the inclinations to the axis of the tangent to one of the two branches through the double point; the ratios $\lambda : \mu : \nu$ are determined by

$$\begin{aligned} (L, M, N \frown \lambda, \mu, \nu) &= 0, \\ (a, b, c, f, g, h \frown \lambda, \mu, \nu)^2 &= 0. \end{aligned}$$

Let u, v be proportional to the coordinates of a consecutive point measured, from the point of contact as origin, in the direction of the tangent and in the perpendicular direction in the tangent plane. The cosines of the inclinations of u to the axes are as $\lambda : \mu : \nu$; those for the inclinations of the normal to the axes are as $L : M : N$;

hence for the coordinate v which is perpendicular to the plane of the last-mentioned two lines, the cosines of the inclinations to the axes are as $N\mu - M\nu : L\nu - N\lambda : M\lambda - L\mu$, and the coordinates of the consecutive point may be taken to be

$$\begin{aligned}x + \lambda u + (N\mu - M\nu)v, \\ y + \mu u + (L\nu - N\lambda)v, \\ z + \nu u + (M\lambda - L\mu)v.\end{aligned}$$

Substituting these values in the equation of the surface, the terms involving the first powers of u, v vanish, and the term involving u^2 also vanishes in virtue of the relation $(a, b, c, f, g, h \curvearrowright \lambda, \mu, \nu)^2 = 0$. The equation consequently becomes

$$Buv + \frac{1}{2}Cv^2 + \frac{1}{3}Du^3 + \dots = 0,$$

and we have as before for the branch in question,

$$\frac{u^2}{2v} = -\frac{3B}{D}.$$

In the present case u, v have been taken, not as before equal, but only proportional, to the coordinates of the consecutive point measured from the point of contact parallel and perpendicular to the tangent, the values of the coordinates are in fact

$$\sqrt{\lambda^2 + \mu^2 + \nu^2} u, \quad \frac{\sqrt{L^2 + M^2 + N^2}}{\sqrt{\lambda^2 + \mu^2 + \nu^2}} v,$$

and the expression for the radius of curvature is

$$\frac{u^2}{2v} \sqrt{\lambda^2 + \mu^2 + \nu^2} \frac{1}{\sqrt{L^2 + M^2 + N^2}},$$

or substituting for $\frac{u^2}{2v}$ the value $-\frac{3B}{D}$ and for B, D their values, the radius of curvature is

$$-\frac{3\sqrt{\lambda^2 + \mu^2 + \nu^2}}{\sqrt{L^2 + M^2 + N^2}} \frac{(a, b, c, f, g, h \curvearrowright \lambda, \mu, \nu)(N\mu - M\nu, L\nu - N\lambda, M\lambda - L\mu)}{(a, b, c, f, g, h, i, j, k, l \curvearrowright \lambda, \mu, \nu)^3},$$

where, as already noticed, the ratios $\lambda : \mu : \nu$ are determined by the equations

$$\begin{aligned}(L, M, N \curvearrowright \lambda, \mu, \nu) &= 0, \\ (a, b, c, f, g, h \curvearrowright \lambda, \mu, \nu)^2 &= 0,\end{aligned}$$

the system of roots selected being that which corresponds to the branch under consideration. It may be noticed that these two equations give

$$(2\mathfrak{A}, 2\mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \cap L, M, N)^2(a, b, c, f, g, h \cap \lambda, \mu, \nu)^2 \\ - K\{(L, M, N \cap \lambda, \mu, \nu)\}^2 = 0,$$

where as usual

$$(2\mathfrak{A}, 2\mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}) = (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch), \\ K = abc - af^2 - bg^2 - ch^2 + 2fgh,$$

and the expression on the left-hand side considered as a function of (λ, μ, ν) is decomposable into a pair of factors. Selecting the proper factor and equating it to zero, we have in addition to the linear equation

$$(L, M, N \cap \lambda, \mu, \nu) = 0,$$

a new linear equation; these two equations determine the ratios $\lambda : \mu : \nu$.

P.S.—I remark that the process adopted for finding the radius of curvature at a double point of a plane curve is the most simple one for the case of an ordinary point on the curve. In fact let $U = 0$ be the curve, (x, y) the coordinates of the point in question, L, M the corresponding values of the first differential coefficients, (a, b, c) those of the second differential coefficients. The coordinates of the consecutive point are

$$x + u \cos \theta - v \sin \theta, \quad y + u \sin \theta + v \cos \theta.$$

Substituting these in the equation of the curve, we have

$$L(u \cos \theta - v \sin \theta) + M(u \sin \theta + v \cos \theta) + \frac{1}{2}(a, b, c \cap u \cos \theta - v \sin \theta, u \sin \theta + v \cos \theta)^2 = 0$$

the coefficient of u must be zero, or we have

$$L \cos \theta + M \sin \theta = 0,$$

giving

$$\sin \theta = \frac{L}{\sqrt{L^2 + M^2}}, \quad \cos \theta = -\frac{M}{\sqrt{L^2 + M^2}},$$

and the equation may be reduced to

$$\sqrt{L^2 + M^2} v + \frac{1}{2}(a, b, c \cap \cos \theta, \sin \theta)^2 u^2 = 0,$$

or what is the same thing

$$(L^2 + M^2)^{\frac{3}{2}} v + \frac{1}{2}(a, b, c \cap M, -L)^2 u^2 = 0,$$

whence the radius of curvature

$$\frac{u^2}{2v} = \frac{(a, b, c \cap M, -L)^2}{(L^2 + M^2)^{\frac{3}{2}}},$$

which is the ordinary form for the radius of curvature at any point of a curve represented by the equation $U = 0$.

2, Stone Buildings, W.C., October 2nd, 1859.

289. On Some Numerical Expansions

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. III. (1860), pp. 366–369.]

There are in the theory of series and the calculus of finite differences and the differential calculus, various sets of numbers, such for instance as Bernoulli's Numbers, or the successive differences of 0^m , which from their more or less frequent occurrence and the complexity of their law it is desirable to tabulate. To these belong, I think, the coefficients of the successive powers of x in the expansion of a given power of $\frac{\log(1+x)}{x}$, or when the index of the power of the function is an indeterminate quantity r , then the coefficients of the several terms of the rational and integral functions of r which form the coefficients of the successive powers of x in the expansion in question. We have

$$\log \left(\frac{\log(1+x)}{x} \right) = -\frac{x}{2} + \frac{x^2}{24} - \frac{x^4}{2880} + \frac{x^6}{181440} - \frac{x^8}{9676800} + \frac{x^{10}}{479001600} - \&c.,$$

which is most easily verified by differentiating and multiplying by $\log(1+x)$, which gives on the left-hand side $\frac{1}{1+x}$, and on the right-hand side the expansion $\frac{1}{2} - x + x^2 - \&c$. And from the above, multiplying each side by r , and taking the exponential, we find

$$\left(\frac{\log(1+x)}{x} \right)^r = 1 - r\frac{x}{2} + r \left(\frac{r}{2} + \frac{1}{4} \right) \frac{x^2}{6} - r \left(\frac{r^2}{6} + \frac{r}{4} + \frac{1}{8} \right) \frac{x^3}{24} + \&c.,$$

which may be verified by writing $r = 1$.

As an instance of the use of the preceding table, let it be required to find expressions for the combinations without repetitions of the series of natural numbers $1, 2, 3, \dots (n-1)$, or what is the same thing for the coefficients of the powers of k in $k(k-1)(k-2) \dots (k-n+1)$. We have

$$\begin{aligned} k(k-1)(k-2) \dots (k-n+1) &= k^n - A_1^{(n)} k^{n-1} + A_2^{(n)} k^{n-2} - \&c. \\ &= n! \cdot \text{coefficient } x^n \text{ in } (1+x)^k; \end{aligned}$$

whence

$$\begin{aligned}
 (-)^n A_r^{(n)} &= n! \cdot \text{coefficient } x^n k^{n-r} \text{ in } (1+x)^k \\
 &= n! \cdot \text{coefficient } x^n k^{n-r} \text{ in } e^{k \log(1+x)} \\
 &= \frac{n!}{(n-r)!} \cdot \text{coefficient } x^n \text{ in } \{\log(1+x)\}^{n-r} \\
 &= \frac{n!}{(n-r)!} \cdot \text{coefficient } x^r \text{ in } \left(\frac{\log(1+x)}{x} \right)^{n-r}.
 \end{aligned}$$

$$A_2^{(n)} = n(n-1)\{H(n-2)\},$$

$$A_3^{(n)} = n(n-1)(n-2)\{\tfrac{1}{2}H(n-3)^2 + \tfrac{1}{4}H(n-3) + \tfrac{1}{8}\},$$

and so on, as far as the expansion has been effected. It may be remarked that the general expression for the algebraical transcendent $A_r^{(n)}$ is given in Dr Schläfli's paper, "Sur les coefficients du développement du produit $(1+x)(1+2x)\dots[1+(n-1)x]$ suivant les puissances ascendantes de x ." Crelle, t. XLIII. [1852], pp. 1-22, but the law is a very complicated one.

We have

$$\log(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5 - \frac{1}{6}x^6 + \frac{1}{7}x^7 - \&c.,$$

and dividing by x , and taking the logarithm, we have as before

$$\log\left(\frac{\log(1+x)}{x}\right) = -\frac{1}{2}x + \frac{1}{24}x^2 - \frac{1}{2880}x^4 + \frac{1}{181440}x^6 - \&c.,$$

which may be considered as the first of a series of logarithmic derivatives, viz., dividing by $-\frac{1}{2}x$ and taking the logarithm we have

$$\log\left(-\frac{2}{x} \log \frac{\log(1+x)}{x}\right) = -\frac{1}{24}x + \frac{1}{480}x^2 - \frac{1}{5376}x^3 + \frac{1}{92160}x^4 - \frac{1}{1351680}x^5 + \&c.,$$

and by the like process

$$\log\left(\frac{24}{x} \log\left(-\frac{2}{x} \log \frac{\log(1+x)}{x}\right)\right) = -\frac{1}{20}x - \frac{1}{112}x^2 + \&c.,$$

and so on.

Suppose in general that

$$\phi x = x + B_1 x^2 + C_1 x^3 + \dots,$$

and let it be required to find the r th function $\phi^r x$. It is easy to see that the successive coefficients are rational and integral functions of r of the degrees 1, 2, 3, &c. respectively; we have in fact

$$\begin{aligned}\phi^r x = \phi^n x + \frac{r}{1}(\phi' x - \phi^0 x) + \frac{r(r-1)}{1 \cdot 2}(\phi^2 x - 2\phi^1 x + \phi^0 x) \\ + \frac{r(r-1)(r-2)}{1 \cdot 2 \cdot 3}(\phi^3 x - 3\phi^2 x + 3\phi^1 x - \phi^0 x) + \&c.,\end{aligned}$$

and by successive substitutions,

$$\phi^0 x = x,$$

$$\phi^1 x = x + B_1 x^2 + C_1 x^3 + D_1 x^4 + \dots,$$

$$\phi^2 x = x + 2B_1 x^2 + (2B_1^2 + 2C_1)x^3 + (B_1^3 + 5B_1 C_1 + 2D_1)x^4 + \dots,$$

$$\phi^3 x = x + 3B_1 x^2 + (9B_1^2 + 3C_1)x^3 + (27B_1^3 + 15B_1 C_1 + 3D_1)x^4 + \dots$$

Whence

$$\begin{aligned}\phi^r x = x \\ + rB_1 x^2 \\ + \{(r^2 - r)B_1^2 + rC_1\}x^3 \\ + \{(r^3 - \frac{3}{2}r^2 + \frac{1}{2}r)B_1^3 + (\frac{3}{2}r^2 - \frac{3}{2}r)B_1 C_1 + rD_1\}x^4 \\ + \&c.\end{aligned}$$

It would, I think, be worth while to continue the expansion some steps further.

2, Stone Buildings, W.C., Oct. 2nd, 1859.

290. A Discussion of the Sturmian Constants for Cubic and Quartic Equations

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. IV. (1861), pp. 7-12.]

For the cubic equation

$$(a, b, c, d - x, 1)^3 = 0,$$

the Sturmian Constants (or leading coefficients of the Sturmian functions) are

$$a, a, b^2 - ac, -a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2.$$

If the signs of the constants, that is, of the functions for $-\infty$, are then the signs of the functions for $+\infty$, are

+	+	+	+	three real roots.
-	+	-	+	case cannot occur.
+	+	-	+	one real root.
-	+	+	+	three real roots.
+	+	+	-	one real root.
-	+	-	-	one real root.
+	+	-	-	one real root.
-	+	+	-	one real root.

The second case would give a loss of variations of sign in passing from ∞ to $-\infty$, which is inconsistent with Sturm's theorem. To show *a posteriori* that the case cannot occur, we may form the identical equation

$$(a^2d - 3abc + 2b^3)^2 = -a^2(-a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2) + 4(b^2 - ac)^3,$$

and, this being so, then in the case in question, the right-hand side would consist of two terms, each of them negative, while the left-hand side is essentially positive.

C. IV. 60

In the particular case where the third constant vanishes, or $b^2 - ac = 0$, we have

$$-a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2 = -(ad - bc)^2 + 4(b^2 - ac)(c^2 - bd) = -(ad - bc)^2,$$

is negative; hence, regarding the evanescent term as being at pleasure positive or negative, we have in each case a combination of signs corresponding to one real root.

The general result (which is well known) is, that there are three real roots or one real root according as

$$-a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2$$

is positive or negative.

For the quartic equation

$$(a, b, c, d, e \curvearrowright x, 1)^4 = 0,$$

the Sturmian constants are

$$a, \ a, \ b^2 - ac, \ 3a^2J + 2(b^2 - ac)I, \ I^3 - 27J^2,$$

if, as usual,

$$\begin{aligned} I &= ae - 4bd + 3c^2, \\ J &= ace - ad^2 - b^2e + 2bcd - c^3. \end{aligned}$$

If the signs of the constants, that is, of the functions for $-\infty$, are then the signs of the functions for $+\infty$, are

+	+	+	+	+	4 real roots.
-	+	-	+	+	no real root.
+	+	-	+	+	2 real roots.
-	+	+	+	+	2 real roots.
+	+	+	-	+	2 real roots.
-	+	-	-	+	2 real roots.
+	+	-	-	+	2 real roots.
-	+	+	-	+	2 real roots.
+	+	+	+	-	2 real roots.
-	+	-	+	-	2 real roots.
+	+	-	+	-	2 real roots.
-	+	+	+	-	2 real roots.
+	+	+	-	-	no real root.
-	+	-	-	-	4 real roots.
+	+	-	-	-	no real root.
-	+	+	-	-	2 real roots.

The non-existing combination of signs is

$$I^3 - 27J^2 = -, \quad 3a^2J + 2(b^2 - ac)I = +, \quad b^2 - ac = -.$$

To show *a posteriori* that this case cannot occur, write

$$\Phi = a^2d - 3abc + 2b^3, \quad \Lambda = 3a^2J + 2(b^2 - ac)I,$$

then we have identically

$$9(3a^2J^2 + \Lambda^2)\Phi^2 = -4a^3\Lambda^3 + 36(b^2 - ac)^2\Lambda^2 - 4a^2(b^2 - ac)^3(I^3 - 27J^2),$$

which is impossible under the given combination of signs, since the left-hand side would be positive, and the right-hand side negative.

To prove the above identity—the relation $\Phi U^2 - 4U^3H + a^2H^3 + \Lambda^2 = 0$, between the covariants of the quartic, gives

$$a^3J + a^2(b^2 - ac)I - 4(b^2 - ac)^3 + \Phi^2 = 0,$$

or, what is the same thing,

$$\Phi^2 = -a^3J - a^2(b^2 - ac)I + 4(b^2 - ac)^3.$$

But

$$\Lambda = 3a^2J + 2(b^2 - ac)I,$$

and thence

$$3\Phi^2 + a^2\Lambda = -a^2(b^2 - ac)I + 12(b^2 - ac)^3,$$

or

$$3\Phi^2 = -a^2\Lambda - a^2(b^2 - ac)I + 12(b^2 - ac)^3,$$

and the identity will be true, if

$$(3\Phi^2 + 9a^4J^2)\Phi^2 = -4\Phi^2\Lambda + 36(b^2 - ac)^2\Phi^2 - 4(b^2 - ac)^3(I^3 - 27J^2).$$

This gives

$$(3\Phi^2 + 9a^4J^2)\{-\Lambda - (b^2 - ac)I\} = -4\Phi^2\Lambda - 4(b^2 - ac)^3I^3,$$

or, what is the same thing,

$$(3\Phi^2 + 9a^4J^2)\{\Lambda + (b^2 - ac)I\} = 4\{\Phi^2\Lambda + (b^2 - ac)^3I^3\},$$

or, dividing by $\Lambda + (b^2 - ac)I$,

$$3\Phi^2 + 9a^4J^2 = 4\{\Phi^2 - \Phi(b^2 - ac)I + (b^2 - ac)^2I^2\},$$

and reducing

$$\Phi^2 - 4\Phi(b^2 - ac)I - 9a^4J^2 + 4(b^2 - ac)^2I^3 = 0,$$

or finally

$$\{\Phi - 3a^2J - 2(b^2 - ac)I\}\{\Phi + 3a^2J - 2(b^2 - ac)I\} = 0,$$

which is true in virtue of

$$\Phi = 3a^2J + 2(b^2 - ac)I,$$

and the identity is thus proved.

60—2

The general conclusion is,

if $I^3 - 27J^2$ is positive, the four roots are all real or all imaginary, viz., all real if $b^2 - ac$ and $3a^2J + 2(b^2 - ac)I$ are both positive, imaginary if otherwise.

But if $I^3 - 27J^2$ is negative, the roots are two of them real, and the other two imaginary.

The following special cases may be noticed,

1°. $b^2 - ac = 0$,

here $9(3a^4J^2 + \Lambda^2)\Phi^2 = -4a^3\Lambda^3$, or $\Lambda = 3a^2J + 2(b^2 - ac)I = 3a^2J$, is negative, so that,

- if $I^3 - 27J^2$ is +, the roots are all imaginary;
- if $I^3 - 27J^2$ is -, the roots are two real and two imaginary.

2°. $\Lambda = 3a^2J + 2(b^2 - ac)I = 0$,

here $27a^4J^2\Phi^2 = -4a^2(b^2 - ac)^3(I^3 - 27J^2)$, or $b^2 - ac$, $I^3 - 27J^2$ are of opposite signs, and if

- $b^2 - ac = -$, $I^3 - 27J^2 = +$, the roots are all imaginary,
- $b^2 - ac = +$, $I^3 - 27J^2 = -$, the roots are two real and two imaginary.

$$3^\circ. \quad b^2 - ac = 0, \quad \Lambda = 3a^2J + 2(b^2 - ac)I = 0, \\ \text{here } J = 0, \text{ that is,}$$

$$2bcd - ad^2 - c^3 = 0, \quad \text{or} \quad (ad - bc)^2 + c^2(ac - b^2) = 0, \quad \text{or} \quad ad - bc = 0, \\ \text{and } I^3 - 27J^2 = I^3,$$

$$I = ae - 4bd + 3c^2 = ae - 4\frac{b^4}{a^2} + 3\frac{b^4}{a^2} = ae - \frac{b^4}{a^2} = \frac{1}{a^2}(a^3e - b^4),$$

whence $I = +$, the roots are all imaginary. $I = -$, the roots are two real and two imaginary.

This is easily verified, in fact $ac - b^2 = 0$, $ad - bc = 0$, give $c = \frac{b^2}{a}$, $d = \frac{bc}{a} = \frac{b^3}{a^2}$, and the equation becomes

$$(ax + b)^4 + (a^3e - b^4) = 0,$$

or, which is the same thing,

$$(ax + b)^4 + (a^3e - b^4) = 0,$$

so that the roots are all imaginary, or two real and two imaginary, according to the sign of $a^3e - b^4$ as above.

It may be noticed that for a quintic equation

$$(a, b, c, d, e, f \curvearrowright x, 1)^5 = 0,$$

if the Sturmian Constants are

$$a, a, C, D, E, F,$$

where as before a is positive, then the roots are real or imaginary as follows: viz.,

C	D	E	F	
+	+	+	+	5 real roots.
-	+	+	+	
	+	-	+	
	+	+	-	1 real root, 4 imaginary roots.
+	+	+	-	
	+	+	-	
	+	-	-	3 real roots, 2 imaginary roots.
-	+	-	-	
	+	-	+	
	-	+	+	cases which do not occur.

The values of C , D , E , and F are given in my "Tables of the Stur-
mian Functions for Equations of the Second, Third, Fourth, and Fifth
Degrees," Phil. Trans., t. 147 (1857), pp. 733–736, [151], but I have
not further examined this case.

2, Stone Buildings, W.C., September 29, 1859.

291. On the Demonstration of a Theorem Relating to the Moments of Inertia of a Solid Body

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. IV. (1861), pp. 25-27.]

Considering in the first instance the analogous question in piano, let $a = \int x^2 dm$, $b = \int y^2 dm$, $h = \int xy dm$, where the integration extends over any closed figure whatever, then it is to be shown that the equations

$$(a, h, b \curvearrowright p, q)^2 = 1, \quad \text{and} \quad (a + b)(p^2 + q^2) - (a, h, b \curvearrowright p, q)^2 = 1,$$

represent respectively ellipses.

If in the first case, by a transformation of coordinates,

$$(a, h, b \curvearrowright p, q)^2 = a_1 p_1^2 + b_1 q_1^2,$$

then a_1, b_1 are the roots of the quadratic equations,

$$\begin{vmatrix} a - \rho & h \\ h & b - \rho \end{vmatrix} = 0,$$

and if in the second case,

$$(a + b)(p^2 + q^2) - (a, h, b \curvearrowright p, q)^2 = a_1 p_1^2 + b_1 q_1^2,$$

then a_1, b_1 are the roots of

$$\begin{vmatrix} b - \rho & -h \\ -h & a - \rho \end{vmatrix} = 0,$$

the two equations being in fact the same equation,

$$\rho^2 - (a + b)\rho + ab - h^2 = 0,$$

The Collected Mathematical Papers of Arthur Cayley

Vol. IV, Pages 501–550

Transcribed from the original publication

On the Construction of the Ninth Point of Intersection of the Cubics which pass through Eight Given Points²⁹⁵

[*From the Quarterly Journal of Pure and Applied Mathematics, vol. v. (1862), pp. 222–233.*]

anharmonic ratio $9(4, 6, 7, 8)$ is equal to that of the points $(4, 6, Z, L)$ on the line $46Z_1$, that is

$$9(4, 6, 7, 8) = (4, 6, Z, L),$$

which is

$$= (4, M, 7, Q),$$

since the lines $6M$, ZL , LQ meet in the point K ; but by the construction

$$(4, M, 7, Q) = 1235(4, 6, 7, 8),$$

that is, we have

$$9(4, 6, 7, 8) = 1235(4, 6, 7, 8);$$

and, secondly, the anharmonic ratio $9(5, 6, 7, 8)$ is equal to that of the points $(5, 6, K, Y)$ on the line $56KY$, that is

$$9(5, 6, 7, 8) = (5, 6, K, Y),$$

which is

$$= (5, N, R, 8),$$

since the lines $6N$, KR , $Y8$ meet in the same point L ; but by the construction

$$(5, N, R, 8) = 1234(5, 6, 7, 8),$$

that is, we have

$$9(5, 6, 7, 8) = 1234(5, 6, 7, 8),$$

so that the point 9 satisfies the required conditions.

It has been already remarked that the point x in M. Chasles' theorem for the construction of the cubic through nine points is determined by precisely similar conditions to those which determine the ninth intersection of the two cubics; that is, the foregoing construction by the ruler alone is applicable to the determination of the point x ; and when this is once obtained, the remainder of the construction, giving the points of the cubic through the nine given points, can obviously be performed by the ruler alone. The construction for the cubic through nine points gives implicitly the relation between ten points of the cubic and such relation is accordingly expressed by the equation

$$x(1, 2, 3, 4, 5, 10) = 6789(1, 2, 3, 4, 5, 10),$$

which is one out of 210 similar forms. But it is possible that some more convenient form of the relation between the ten points may yet be found.

I proceed to further develop the analytical theory. Writing for convenience ω in the place of 10, we have

$$x(1, 2, 3, 4, 5, 6) = 789\omega(1, 2, 3, 4, 5, 6),$$

$$\text{or, what is the same thing, } x(1, 2, 3, 4) = 789\omega(1, 2, 3, 4),$$

$$x(1, 2, 3, 5) = 789\omega(1, 2, 3, 5),$$

$$x(1, 2, 3, 6) = 789\omega(1, 2, 3, 6),$$

which belong to three conics each of them passing through 1, 2, 3, and which must have a remaining fourth point of intersection.

The equation of the first conic is

$$012 \cdot 034 = -\lambda \cdot 014 \cdot 023,$$

where

$$\lambda = 789\omega 12 \cdot 789\omega 34,$$

$$\mu = 789\omega 14 \cdot 789\omega 23,$$

and thence, in virtue of an identical equation already referred to,

$$-\lambda - \mu = 789\omega 13 \cdot 789\omega 24.$$

But we have identically

$$123 \cdot 0pq = 1pq \cdot 023 + 2pq \cdot 031 + 3pq \cdot 012,$$

and thence in particular

$$123 \cdot 034 = 134 \cdot 023 + 234 \cdot 031,$$

$$123 \cdot 014 = 214 \cdot 031 + 314 \cdot 012,$$

and the equation of the conic may therefore be written

$$012(134 \cdot 023 + 234 \cdot 031)\mu - 023(214 \cdot 031 + 314 \cdot 012)\lambda = 0,$$

that is

$$031 \cdot 012 \cdot 234 \cdot \mu + 012 \cdot 023 \cdot 314(-\lambda - \mu) + 023 \cdot 031 \cdot 124 \cdot \lambda = 0,$$

or, substituting for μ , $-\lambda - \mu$, λ , their values, this is

$$\begin{aligned} & 031 \cdot 012 \cdot 234 \cdot 789\omega 14 \cdot 789\omega 23 \\ & + 012 \cdot 023 \cdot 314 \cdot 789\omega 13 \cdot 789\omega 42 \\ & + 023 \cdot 031 \cdot 124 \cdot 789\omega 12 \cdot 789\omega 34 = 0, \end{aligned}$$

or, what is the same thing,

$$\frac{234 \cdot 789\omega 14 \cdot 789\omega 23}{023} + \frac{314 \cdot 789\omega 13 \cdot 789\omega 24}{031} + \frac{124 \cdot 789\omega 12 \cdot 789\omega 34}{012} = 0,$$

or, making a slight change of form, the equation of the conic is

$$\frac{423 \cdot 789\omega 41 \cdot 789\omega 23}{023} + \frac{431 \cdot 789\omega 42 \cdot 789\omega 31}{031} + \frac{412 \cdot 789\omega 43 \cdot 789\omega 12}{012} = 0.$$

The equations of the other two conics are deduced by writing successively 5 and 6 in the place of 4; and the condition in order that the conics may have a remaining fourth point of intersection is

$$\begin{vmatrix} 423 \cdot 789\omega 41, & 431 \cdot 789\omega 42, & 412 \cdot 789\omega 43 \\ 523 \cdot 789\omega 51, & 531 \cdot 789\omega 52, & 512 \cdot 789\omega 53 \\ 623 \cdot 789\omega 61, & 631 \cdot 789\omega 62, & 612 \cdot 789\omega 63 \end{vmatrix} = 0.$$

This equation, say $\Omega = 0$, expresses the relation between the coordinates of the ten points 1, 2, 3, 4, 5, 6, 7, 8, 9, ω , of the cubic. Hence if 123456789 ω denote the determinant formed with the ten quantities

$$(x^3, y^2z, z^2x, x^2y, y^3, z^3, x^2z, y^2x, z^2y, xyz)$$

for the ten points respectively, 123456789 ω must be a factor of Ω , and a little consideration shows that the other factor which is of the order one as regards the coordinates of each of the points 1, 2, 3, and of the order three as regards the coordinates of each of the points 7, 8, 9, ω , must be of the form $123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega$. We must therefore have

$$\Omega = \epsilon \cdot 123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega \cdot 123456789\omega,$$

where the merely numerical factor ϵ is, I believe, equal to +1 or else to -1.

In order to verify the factor $123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega$, observing that the points 7, 8, 9, ω enter symmetrically, it will be sufficient to show that 123, 789 are each of them factors of Ω , or, what is the same thing, that if $123 = 0$, or if $789 = 0$, then in either case $\Omega = 0$.

First, if $123 = 0$, we may write

$$x_3 = \lambda x_1 + \mu x_2,$$

$$y_3 = \lambda y_1 + \mu y_2,$$

$$z_3 = \lambda z_1 + \mu z_2,$$

equations which give

$$423 = \lambda \cdot 421, \quad 431 = \mu \cdot 421, \quad \&c.,$$

and the equation $\Omega = 0$, thus becomes

$$\begin{vmatrix} 789\omega 41, & 789\omega 42, & 789\omega 43 \\ 789\omega 51, & 789\omega 52, & 789\omega 53 \\ 789\omega 61, & 789\omega 62, & 789\omega 63 \end{vmatrix} = 0,$$

or, what is the same thing,

$$\begin{vmatrix} 789\omega 14, & 789\omega 24, & 789\omega 34 \\ 789\omega 15, & 789\omega 25, & 789\omega 35 \\ 789\omega 16, & 789\omega 26, & 789\omega 36 \end{vmatrix} = 0.$$

Now if the terms in the same column are multiplied by $789\omega56$, $789\omega64$, $789\omega45$ respectively and added, then for the first column the sum is

$$789\omega14 \cdot 789\omega56 + 789\omega15 \cdot 789\omega64 + 789\omega16 \cdot 789\omega45,$$

which is $= 0$, and the sums for the second and third columns are each $= 0$ in the same manner: wherefore the determinant vanishes as it should do.

Next, if $789 = 0$, we have identically

$$789\omega 41 = 789 \cdot 741 \cdot \omega 81 \cdot \omega 94 + 781 \cdot 794 \cdot \omega 89 \cdot \omega 41,$$

which when $789 = 0$ gives

$$789\omega 41 = 781 \cdot 794 \cdot \omega 89 \cdot \omega 41,$$

and in like manner,

$$789\omega 42 = 782 \cdot 794 \cdot \omega 89 \cdot \omega 42,$$

$$789\omega 43 = 783 \cdot 794 \cdot \omega 89 \cdot \omega 43,$$

$= 0$, in which three equations 4 may be changed into 5 and 6 successively. The equation $\Omega = 0$ thus becomes

$$\begin{vmatrix} 423 \cdot \omega 41, & 431 \cdot \omega 42, & 412 \cdot \omega 43 \\ 523 \cdot \omega 51, & 531 \cdot \omega 52, & 512 \cdot \omega 53 \\ 623 \cdot \omega 61, & 631 \cdot \omega 62, & 612 \cdot \omega 63 \end{vmatrix}$$

or, what is the same thing,

$$\begin{vmatrix} 423 \cdot 4\omega 1, & 421 \cdot 4\omega 3, & 42\omega \cdot 431 \\ 523 \cdot 5\omega 1, & 521 \cdot 5\omega 3, & 52\omega \cdot 531 \\ 623 \cdot 6\omega 1, & 621 \cdot 6\omega 3, & 62\omega \cdot 631 \end{vmatrix} = 0,$$

and since the sum of the terms in each line of the determinant is $= 0$, the determinant is as it should be $= 0$.

The foregoing equation

$$\begin{vmatrix} 412 \cdot 789\omega 43, & 423 \cdot 789\omega 41, & 431 \cdot 789\omega 42 \\ 512 \cdot 789\omega 53, & 523 \cdot 789\omega 51, & 531 \cdot 789\omega 52 \\ 612 \cdot 789\omega 63, & 623 \cdot 789\omega 61, & 631 \cdot 789\omega 62 \end{vmatrix} = \epsilon \cdot 123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega \cdot 123456789\omega,$$

(since $412, 789\omega 43$, &c. are interpretable functions of the coordinates) affords a geometrical interpretation of the equation

$$123456789\omega = 0$$

between the coordinates of the ten points of the cubic; but it would be more satisfactory if a similar identical equation could be found, having on the right-hand side the function 123456789ω without the irrelevant factor $123 \cdot 789 \cdot 78\omega \cdot 79\omega \cdot 89\omega$.

2, Stone Buildings, W.C., 6th March, 1862.

On the Conics which pass through the Four Foci of a Given Conic 296

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. v. (1862), pp. 275–280.]

The foci of a conic are the points of intersection of the tangents through the circular points at infinity; the pair of tangents through each of the circular points at infinity is a conic through the four foci; and we have thus two conics $P = 0$, $Q = 0$ passing through the four foci; the equation of any other conic through the four foci is of course $P + \lambda Q = 0$; and in particular if λ be suitably determined this equation gives the axes of the conic.

I was led to develop the solution, in seeking to obtain the elegant formulae given in Mr P. J. Hensley's paper "Determination of the foci of the conic section expressed by trilinear coordinates," *Journal*, t. v., pp. 177–183, (March, 1862).

I take the coordinates to be proportionate to the perpendicular distances of the point from the sides of the fundamental triangle, each distance divided by the perpendicular distance of the side from the opposite angle. This being so, the equation of the line infinity is

$$x + y + z = 0,$$

and, a , b , c denoting the sides of the fundamental triangle, the equation of the circle circumscribed about the triangle is

$$\frac{a^2}{x} + \frac{b^2}{y} + \frac{c^2}{z} = 0.$$

The foregoing two equations determine the circular points at infinity; and if (x_1, y_1, z_1) are the coordinates, there is no difficulty in obtaining the system of values

$$\begin{aligned} x_1 : y_1 : z_1 \\ &= a(\cos C - i \sin C) : b(\cos B + i \sin B) : -c \\ &= b(\cos C + i \sin C) : -a : c(\cos A - i \sin A) \\ &= -b : a(\cos B + i \sin B) : c(\cos A + i \sin A), \end{aligned}$$

where as usual $i = \sqrt{-1}$, and where A, B, C denote the angles of the triangle, so that the cosines and sines of these angles denote given functions of a, b, c . The coordinates (x_2, y_2, z_2) of the other circular point at infinity are of course obtained by merely writing $-i$ for i . We find also

$$\begin{aligned} x_1x_2 : y_1y_2 : z_1z_2 : y_1z_2 + y_2z_1 : z_1x_2 + z_2x_1 : x_1y_2 + x_2y_1 \\ = -a^2 : -b^2 : -c^2 : b^2 + c^2 - a^2 : c^2 + a^2 - b^2 : a^2 + b^2 - c^2, \end{aligned}$$

which are the formula chiefly made use of in the sequel.

Suppose now that the equation of the conic is

$$U = (a, b, c, f, g, h \text{ } \S \text{ } x, y, z)^2 = 0;$$

then putting for a moment

$$\begin{aligned} U_1 &= (a, b, c, f, g, h \text{ } \S \text{ } x_1, y_1, z_1)^2, \\ V_1 &= (a, b, c, f, g, h \text{ } \S \text{ } x, y, z \text{ } \S \text{ } x_1, y_1, z_1), \end{aligned}$$

and the like as regards U_2 and V_2 ; the tangents from the points (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively are

$$\begin{aligned} UU_1 - V_1^2 &= 0, \\ UU_2 - V_2^2 &= 0, \end{aligned}$$

and this being so, I combine the two equations as follows:

$$\begin{aligned}
 x_1^2(a, \dots)(x_1, y_1, z_1)^2 + x_2^2(a, \dots)(x_2, y_2, z_2)^2 &= 0, \\
 y_1^2 \quad , \quad + y_2^2 \quad , \quad &= 0, \\
 z_1^2(a, \dots)(x_1, y_1, z_1)^2 + z_2^2(a, \dots)(x_2, y_2, z_2)^2 &= 0, \\
 y_1 z_1 \quad , \quad + y_2 z_2 \quad , \quad &= 0, \\
 z_1 x_1 \quad , \quad + z_2 x_2 \quad , \quad &= 0, \\
 x_1 y_1 \quad , \quad + x_2 y_2 \quad , \quad &= 0,
 \end{aligned}$$

any one of which is the equation of a conic passing through the four foci; the current coordinates being always (x, y, z) .

The first of these equations is

$$\begin{aligned}
 &a(-2x_1^2x_2^2) + b(x_2^2y_1^2 + x_1^2y_2^2) + c(x_2^2z_1^2 + x_1^2z_2^2) \\
 &+ 2f(x_2^2y_1z_1 + x_1^2y_2z_2) + 2g(x_1x_2^2z_1 + x_1^2x_2z_2) + 2h(x_1x_2^2y_1 + x_1^2x_2y_2) = 0,
 \end{aligned}$$

where the quantities multiplied by a , b , &c. are all of them easily expressible in terms of x_1x_2 , y_1y_2 , z_1z_2 , $y_1z_2 + y_2z_1$, $z_1x_2 + z_2x_1$, $x_1y_2 + x_2y_1$, which are respectively proportional to given functions of (a, b, c) ; and replacing for a , b , &c. their values, the equation is

$$\begin{aligned}
 &(-2c^2x^2 + 2b^2y^2 + 2a^2z^2 + 2(-c^2 - a^2 + b^2)yz + 2(-b^2 + a^2 + c^2)zx + 2(a^2 + b^2 - c^2)xy) \cdot 2a^2 \\
 &\quad - a^2(b^2 + c^2 - a^2) \cdot (\dots) + \dots
 \end{aligned}$$

or, what is the same thing, the equation is

$$\Omega\{(2b+2c+2f+2g+2h)x^2-2(2b+2f+2h)x(x+y+z)+b(x+y+z)^2\}+2a^2\Theta=0,$$

or putting for shortness

$$\begin{aligned} 2b+2c+2f+2g+2h &= 2l, \\ 2b+2f+2h &= 2m, \\ 2c+2f+2g &= 2n, \\ 2g+2h+2f &= 2p, \end{aligned}$$

so that in fact

$$\begin{aligned} l &= b+c+f+g+h, \\ m &= b+f+h, \\ n &= c+f+g, \\ p &= g+h+f, \end{aligned}$$

the equation is

$$D\{lx^2-2mx(x+y+z)+b(x+y+z)^2\}+2a^2\Theta=0.$$

The equation with y_1y_2 , y_1z_2 is in a similar manner found to be

$$D\{(2b+2c+2f+2g+2h)y^2-\{(2c+2f+2g)y+(2b+2f+2h)z\}(x+y+z)+c(x+y+z)^2\}-$$

or, what is the same thing,

$$D\{ly^2-2ny(x+y+z)+c(x+y+z)^2\}-a^2\Theta=0,$$

or, finally,

$$D\{lz^2-2pz(x+y+z)+a(x+y+z)^2\}+2c^2\Theta=0.$$

Hence the entire system of equations is

$$\begin{aligned} D\{lx^2-2mx(x+y+z)+b(x+y+z)^2\}+2a^2\Theta &= 0, \\ D\{ly^2-2ny(x+y+z)+c(x+y+z)^2\}+2b^2\Theta &= 0, \\ D\{lz^2-2pz(x+y+z)+a(x+y+z)^2\}+2c^2\Theta &= 0, \\ D\{2fyz-(ny+mz)(x+y+z)+2l(x+y+z)^2\}-(b^2+c^2-a^2)\Theta &= 0, \\ D\{2gzx-(pz+mx)(x+y+z)+2m(x+y+z)^2\}-(c^2+a^2-b^2)\Theta &= 0, \\ D\{2hxy-(lx+py)(x+y+z)+2n(x+y+z)^2\}-(a^2+b^2-c^2)\Theta &= 0, \end{aligned}$$

which are the equations of six conics, each of them passing through the four foci.

From the first three of these, we have

$$\begin{aligned} &\{lx^2 - 2mx(x + y + z) + b(x + y + z)^2\} \\ &\quad = \{ly^2 - 2ny(x + y + z) + c(x + y + z)^2\} \\ &\quad = \{lz^2 - 2pz(x + y + z) + a(x + y + z)^2\}, \end{aligned}$$

which, allowing for the difference of notation, are Mr Hensley's equations: it appears by his investigation that their geometrical signification is as follows; viz. if for shortness we denote the equations by

$$A = 0, \quad B = 0, \quad C = 0,$$

then if we consider the tangents parallel to the x -side of the fundamental triangle, and the tangents parallel to the y -side of the fundamental triangle, the equation $A = B$ is the locus of a point such that the feet of the perpendiculars let fall from it on the four tangents lie in a circle. And similarly for the equations $A = C$, $B = C$.

If we multiply the six equations by 1, 1, 1, 2, 2, 2 and add, we obtain the identical equation $0 = 0$; if we multiply them by a , b , c , $2f$, $2g$, $2h$ and add, then after some easy reductions, we obtain for the equation of a new conic passing through the four foci

$$D\{KU + K(x + y + z)^2\} + 2S\Theta = 0,$$

where

$$U = (a, b, c, f, g, h \text{ \textcircled{X} } x, y, z)^2,$$

K is the discriminant $abc - af^2 - bg^2 - ch^2 + 2fgh$, and

$$S = (b^2 + c^2 - a^2)f + (c^2 + a^2 - b^2)g + (a^2 + b^2 - c^2)h,$$

or, what is the same thing,

$$S = (f + a - h - g)a^2 + (g - h + b - f)b^2 + (h - g - f + c)c^2.$$

It would be interesting to ascertain the geometrical signification of the six conics and of the last-mentioned new conic.

2, Stone Buildings, W.C., 13th March, 1862.

On some Formulae relating to the Distances of a Point from the Vertices of a Triangle, and to the Problem of Tactions²⁹⁷

[*From the Quarterly Journal of Pure and Applied Mathematics*, vol. v. (1862), pp. 381-384.]

if D denote the function in $\{ \quad \}$ with the signs reversed. The function D may be expressed in the form

$$\begin{aligned} &+ (a^2p + b^2c^2)(f^2 - g^2 - h^2) \\ &\quad + (b^2c^2q + c^2a^2r)(g^2 - h^2 - f^2) \\ &\quad + (c^2a^2r + a^2b^2p)(h^2 - f^2 - g^2), \end{aligned}$$

and also in the form

$$\square U^2 + (f + g + h)V,$$

if for shortness

$$\begin{aligned} \square &= U^2 + (f + g + h)V, \\ U &= a^2f + b^2g + c^2h + fgh, \\ V &= (a^2p + b^2c^2)(f - g - h) \\ &\quad + (b^2q + c^2a^2)(g - h - f) \\ &\quad + (c^2r + a^2b^2)(h - f - g); \end{aligned}$$

and it may be remarked that since D is an even function of f, g, h , we may in this last formula change at pleasure the signs of these quantities; we thus obtain in all four similar forms of the function D .

It is clear that considering a triangle, and any point in the plane of the triangle, f, g, h may be taken to denote the sides of the triangle, and a, b, c the distances of the point from the vertices: and the equation $D = 0$ is the relation connecting the sides and distances.

The equation $f + g + h = 0$ denotes that the vertices are in lined, and when this equation is satisfied we have

$$U = a^2f + b^2g + c^2h + fgh = 0,$$

which is in fact, as it is easy to see, the relation connecting the distances of a point from any three points in lined.

For a, b, c write $a + x, b + x, c + x$; x will be the radius of a circle touching the circles, radii a, b, c , described about the vertices

as centres. The equation $D = 0$ becomes after all reductions

$$\begin{aligned}
 & U^2 - (f + g + h)V \\
 & + x[4U(af + bg + ch) - 2(f + g + h)\{(a^2p + bc(b + c))(f - g - h) \\
 & \quad + (b^2q + ca(c + a))(g - h - f) + (c^2r + ab(a + b))(h - f - g)\}] \\
 & + x^2[f^2\{-4a^2 + 6a(b + c) - 6bc\} + g^2\{-4b^2 + 6b(c + a) - 6ca\} \\
 & \quad + h^2\{-4c^2 + 6c(a + b) - 6ab\}] = 0,
 \end{aligned}$$

which is a quadratic equation only: the two circles thus obtained are those which touch the given circles all three externally or all three internally. But by changing in every possible manner the signs of a , b , c we obtain in all four equations giving the eight tangent circles. It may be noticed that if as before $f + g + h = 0$, $U = 0$,

then not only the constant term vanishes, but the coefficient of x also vanishes or the equation becomes simply $x^2 = 0$.

In particular, suppose $f = b + c$, $g = c + a$, $h = a + b$; developing this *de novo*, and putting for shortness

$$\begin{aligned}a + b + c &= p, \\bc + ca + ab &= q, \\abc &= r,\end{aligned}$$

we find

$$\begin{aligned}U &= 2\{pa^2 + 4qa + (2q^2 - 2r)\}, \\V &= 2\{px^2 + 4qax + (2pq + 12r)a^2 + 4q^2a + pq^2 - 4qr\},\end{aligned}$$

and then the equation $D = U^2 - 2pV = 0$ gives

$$\begin{aligned}D &= (pa^2 + 2qa + pq - 2r)^2 - p\{pa^2 + 4qa + (2pq + 12r)a^2 + 4q^2a + pq^2 - 4qr\} \\&= 4\{(q^2 - 4pr)a^2 - 2qra + r^2\},\end{aligned}$$

so that we have

$$\frac{1}{4}D = (q^2 - 4pr)a^2 - 2qra + r^2 = (qa - r)^2 - 4pra^2 = 0,$$

and thence

$$qa - r = \pm a\sqrt{pr}, \quad \text{or } x = \frac{r}{q \mp \sqrt{pr}},$$

which gives the radii of the circles inscribed in and circumscribed about the three circles radii a , b , c , whereof each touches the two others: a formula given by Descartes, *Epistolæ* (Ed. 2, Franc. 1792), Pars III., p. 261, in a letter to the Princess Elizabeth, viz. Descartes has

$$(d^2e^2f^2 + d^2p + e^2p - 2de^2f^2 - 2d^2ef^2 - 2d^2e^2f)x^2 - 2(de^4 + d^4ef + d^2e^4f)x + d^2e^4f^2 = 0,$$

which putting a , b , c for his d , e , f , becomes *ut supra*

$$a^2(q^2 - 4pr) - 2qra + r^2 = 0.$$

In conclusion I notice the following formula which is obtained without difficulty, viz. if as before we have a triangle the sides whereof are f , g , h , and if a , b , c are the distances of a point from the vertices (so that as before $D = 0$) then the perpendicular distances of

the point from the sides, each perpendicular distance divided by the perpendicular distance of the opposite vertex from the same side, are as follows: viz. the quotient for the side f is

$$= \frac{1}{16A^2}[(b^2 - c^2)(g^2 - h^2) + f^2(b^2 + c^2 + g^2 + h^2 - 2a^2) - f^4],$$

where A is the area of the triangle. It is clear that we ought to have

$$2[(b^2 - c^2)(g^2 - h^2) + f^2(b^2 + c^2 + g^2 + h^2 - 2a^2) - f^4] = 16A^2,$$

and this equation in fact reduces itself to

$$2g^2h^2 + 2h^2f^2 + 2f^2g^2 - f^4 - g^4 - h^4 = 16A^2,$$

which is right.

2, Stone Buildings, W.C., 17th September, 1862.

Report on the Progress of the Solution of Certain Special Problems of Dynamics 298

[*From the Report of the British Association for the Advancement of Science, 1862, pp. 184-252.*]

My "Report on the Recent Progress of Theoretical Dynamics" was published in the Report of the British Association for the year 1857, [195]. The present Report (which is in some measure supplemental thereto) relates to the Special Problems of Dynamics: to give a general idea of the contents, I will at once mention the heads under which these problems are considered; viz., relating to the motion of a particle or system of particles, we have

- Rectilinear Motion;
- Central Forces, and in particular Elliptic Motion;
- The Problem of two Centres;
- The Spherical Pendulum;
- Motion as affected by the Rotation of the Earth, and Relative Motion in general;
- Miscellaneous Problems;
- The Problem of three bodies.

And relating to the motion of a solid body, we have

- The Transformation of Coordinates;
- Principal Axes, and Moments of Inertia;
- Rotation of a Solid Body;
- Kinematics of a Solid Body;
- Miscellaneous Problems.

As regards the first division of the subject, I remark that the lunar and planetary theories, as usually treated, do not (properly speaking) relate to the problem of three bodies, but to that of disturbed elliptic motion — a problem which is not considered in the present Report. The problem of the spherical pendulum is that of a particle moving on a spherical surface; but, with this exception, I do not much consider the motion of a particle on a given curve or surface, nor do I consider vibrations or other questions belonging to the dynamics of a system of particles or material points; the present Report relates chiefly to the special problems which admit of solution by means of elliptic functions, and to the investigations which have reference thereto.

Rectilinear Motion. Art. Nos. 1 to 4.

1. The problem is that of the motion of a particle in a straight line; the force is supposed to be a function of the distance only, and the velocity is given by the equation

$$v^2 = 2 \int X \, dx,$$

or if the force be $\mu(X + X')$, where X is a given function of the distance, and X' a small disturbing force, then the problem may be treated by the method of variation of parameters.

2. The case of a central force varying as the distance was considered by Newton, *Principia*, Lib. I., Prop. X.; the case of a force varying as the inverse square of the distance gives the parabolic motion, which is the limiting case of elliptic motion. The case of a force varying as the inverse cube of the distance gives a spiral orbit, which was also considered by Newton.

3. The case of a force proportional to the distance, and also to the inverse square of the distance, was considered by Newton, *Principia*, Lib. I., Prop. IX. and X.; the orbit is a conic section described about the centre, or about the focus, according as the one or the other force vanishes. The general case of central forces was considered by Newton, *Principia*, Lib. I., Section IX., and by Clairaut, *Hist. Acad. Sci.*, 1743.

4. The case of a force varying as any power of the distance was considered by Euler, *Mémoires de Berlin*, 1752, and by Lagrange, *Méc. Anal.*, Vol. II., p. 72; the case of a force varying as the distance, and also as any power of the distance, was considered by Laplace, *Méc. Cél.*, Liv. II., Chap. II.

Central Forces. Art. Nos. 5 to 50.

Elliptic Motion. Art. Nos. 5 to 49.

5. The problem of elliptic motion is that of a particle moving under the action of a central force varying as the inverse square of the distance; the solution is given in terms of elliptic functions, and the theory is one of the most important in dynamics. The problem has been considered by many mathematicians, and the results are given in all treatises on mechanics and astronomy.

Bour's "Mémoire sur l'intégration des Equations différentielles de la Mécanique," as published, *Mem. prés. à l'Inst.*, t. xiv. pp. 792—821, is substantially the same as the extract thereof in Liouville's Journal, referred to in my former Report; but since the publication of the extract, the memoir itself has been published, and it is proper to refer to it. The memoir contains the complete integration of the equations of motion of a particle under the action of any forces, by means of a system of partial differential equations; the method is an extension of that employed by Hamilton for the problem of the attraction of a system of bodies.

6. The problem of the motion of a particle under the action of a central force varying as the inverse square of the distance, is the fundamental problem of celestial mechanics; the solution is given in terms of elliptic functions, and the theory is one of the most important in dynamics. The integrals of the equations of motion were obtained by Lagrange, *Méc. Anal.*, Vol. II., p. 72; the complete solution in terms of elliptic functions was given by Abel and Jacobi.

Problem of Two Centres. Art. Nos. 51 to 75.

51. The problem of the motion of a particle under the action of two fixed centres of force, each attracting according to the law of nature, was first solved by Euler, *Mémoires de Berlin*, 1760; the solution was afterwards completed by Lagrange, *Méc. Anal.*, Vol. II., p. 230, and by Legendre, *Exercices de Calcul Intégral*, Vol. I., p. 212. The problem has been considered by many mathematicians, and the results are given in most treatises on mechanics.

52. A case satisfying the required conditions is found to be that of a particle moving under the action of two centres of force, the one attractive and the other repulsive, the intensities of the forces being equal; the orbit is in this case a conic section having one of the centres for its focus.

53. The problem of the motion of a particle under the action of two centres of force, was also considered by Jacobi, *Vorlesungen über Dynamik*, p. 186; the solution is given in terms of elliptic functions.

54. The problem of the motion of a particle under the action of two centres of force, each attracting according to any law, was considered by Liouville, *Journal de Mathématiques*, t. XI., p. 217; the condition for the integrability of the equations of motion is that the attraction shall be a function of the distance satisfying a certain differential equation.

55. The problem of the motion of a particle under the action of two centres of force, was also considered by Bonnet, *Journal de l'École Polytechnique*, cah. XXXII.; the solution is obtained by a method similar to that employed by Liouville.

56. The spherical pendulum is the problem of a particle moving on a spherical surface under the action of gravity; the solution was first obtained by Lagrange, *Méc. Anal.*, Vol. I., p. 458, and afterwards by Legendre, *Exercices de Calcul Intégral*, Vol. II., p. 306. The solution is given in terms of elliptic functions.

57. The problem of the spherical pendulum has been considered by many mathematicians; in particular, by Poisson, *Traité de Mécanique*, Vol. I., p. 218; by Plana, *Journal de Mathématiques*, t. VII., p. 469; by Jacobi, *Vorlesungen über Dynamik*, p. 24; and by Puiseux, *Journal de Mathématiques*, t. IX., p. 425.

58. In Serret's "Thèse sur le Mouvement &c." (1848), the problem is very elegantly treated by the method of elliptic functions; the solution is obtained in a form which is well adapted for numerical calculation.

59. The problem of the motion of a particle under the action of a central force varying as the distance, and also under the action of gravity, was considered by Lagrange, *Méc. Anal.*, Vol. II., p. 72; the solution is given in terms of elliptic functions. The problem is of importance in the theory of the small oscillations of a system about a position of equilibrium.

60. The problem of the motion of a particle under the action of a central force varying as the inverse cube of the distance, was considered by Newton, *Principia*, Lib. I., Prop. IX.; the orbit is a spiral. The case of a force varying as the inverse cube of the distance, and also as the distance, was considered by Lagrange, *Méc. Anal.*, Vol. II., p. 72; the solution is given in terms of elliptic functions.

61. The question of the convergence of the series is treated in Laplace's memoir of 1823, where he shows that in the series which express r and v in multiple cosines of the mean anomaly, the coefficients decrease in a certain ratio, and that the series are convergent for all values of the eccentricity less than unity.

62. The problem of the attraction of an ellipsoid upon an external point, was first solved by Maclaurin, *Fluxions*, p. 450; the solution was afterwards completed by Laplace, *Méc. Cél.*, Liv. III., and by Ivory, *Phil. Trans.*, 1809. The problem has been considered by many mathematicians, and the results are given in most treatises on mechanics.

63. The problem of the attraction of an ellipsoid upon an internal point, was first solved by Maclaurin, *Fluxions*, p. 450; the solution was afterwards completed by Lagrange, *Mém. de l'Acad. de Berlin*, 1773, and by Laplace, *Méc. Cél.*, Liv. III.

64. The problem of the attraction of a spheroid upon an external point, was first solved by Legendre, *Mémoires de Paris*, 1785; the solution was afterwards completed by Laplace, *Méc. Cél.*, Liv. III. The problem has been considered by many mathematicians.

65. The problem of the attraction of a spheroid upon an internal point, was first solved by Clairaut, *Figure de la Terre*, p. 217; the solution was afterwards completed by Laplace, *Méc. Cél.*, Liv. III.

66. The problem of the attraction of a sphere upon an external point, was first solved by Newton, *Principia*, Lib. I., Prop. LXX.; the solution is the celebrated theorem that the attraction is the same as if the mass were collected at the centre.

67. The problem of the attraction of a sphere upon an internal point, was first solved by Newton, *Principia*, Lib. I., Prop. LXXIII.; the solution is the celebrated theorem that the attraction varies as the distance from the centre.

68. The problem of the attraction of an oblate spheroid upon an external point, situated on the axis of revolution, was first solved by Maclaurin, *Fluxions*, p. 450; the solution was afterwards completed by Laplace, *Méc. Cél.*, Liv. III.

69. The problem of the attraction of an oblate spheroid upon an external point, situated in the plane of the equator, was first solved by Maclaurin, *Fluxions*, p. 450; the solution was afterwards completed by Laplace, *Méc. Cél.*, Liv. III.

70. The problem of the attraction of an oblate spheroid upon an external point, was also considered by Poisson, *Traité de Mécanique*, Vol. II., p. 82; the solution is obtained by a method similar to that employed by Laplace.

71. The problem of the attraction of an ellipsoid upon a point situated on its surface, was first solved by Maclaurin, *Fluxions*, p. 450; the solution was afterwards completed by Laplace, *Méc. Cél.*, Liv. III.

72. The problem of the attraction of an ellipsoid upon a point situated on its surface, was also considered by Ivory, *Phil. Trans.*, 1809; the solution is obtained by a method which is applicable to any ellipsoid, whatever be the ratios of its axes.

73. The problem of the attraction of an ellipsoid upon an external point, was also considered by Gauss, *Werke*, Vol. V., p. 282; the solution is obtained by a method which is applicable to any ellipsoid, whatever be the ratios of its axes.

74. The problem of the attraction of a spheroid upon an external point, was also considered by Chasles, *Journal de l'École Polytechnique*, cah. XXXV.; the solution is obtained by a method which is applicable to any spheroid, whatever be the law of density.

75. The problem of the attraction of a sphere upon an external point, was also considered by Green, *Essay on Electricity and Magnetism*, p. 22; the solution is obtained by a method which is applicable to any distribution of matter, whatever be the law of density.

Rotation of a Solid Body. Art. Nos. 76 to 100.

76. Poisson, in the "Mémoire sur le mouvement des Projectiles &c." (1838), also considers the motion of a solid body about a fixed point; the equations of motion are obtained in a form which is well adapted for numerical calculation.

77. The more difficult one of the rotation of a solid body, as solved by Jacobi (1839), in completion of Rueb's solution (1834), post, Nos. 186 and 197.

78. The problem of the rotation of a solid body about a fixed point, was first solved by Euler, *Mémoires de Berlin*, 1758; the solution was afterwards completed by Lagrange, *Méc. Anal.*, Vol. II., p. 212, and by Poisson, *Traité de Mécanique*, Vol. II., p. 82.

79. The problem of the rotation of a solid body about a fixed point, was also considered by Poincot, *Théorie nouvelle de la rotation des corps*, Paris, 1834; the solution is obtained by a method which is purely geometrical.

80. The problem of the rotation of a solid body about a fixed point, was also considered by MacCullagh, *Trans. R. I. A.*, Vol. XXI., p. 153; the solution is obtained by a method which is applicable to any solid body, whatever be its form.

81. The theory of relative motion is considered in a very general manner in the memoirs of Coriolis, *Journal de l'École Polytechnique*, cah. XXIV.; the equations of motion are obtained in a form which is well adapted for numerical calculation.

82. The problem of the rotation of a solid body about a fixed point, was also considered by Jacobi, *Vorlesungen über Dynamik*, p. 426; the solution is given in terms of elliptic functions.

83. The problem of the rotation of a solid body about a fixed point, was also considered by Liouville, *Journal de Mathématiques*, t. XIV., p. 257; the solution is obtained by a method which is applicable to any solid body, whatever be its form.

84. The problem of the rotation of a solid body about a fixed point, was also considered by Clebsch, *Journal für Mathematik*, t. LIII., p. 195; the solution is obtained by a method which is applicable to any solid body, whatever be its form.

85. The problem of the rotation of a solid body about a fixed point, was also considered by Kirchhoff, *Journal für Mathematik*, t. L., p. 285; the solution is obtained by a method which is applicable to any solid body, whatever be its form.

86. The problem of the rotation of a solid body about a fixed point, was also considered by Routh, *Treatise on the Dynamics of a System of Rigid Bodies*, p. 82; the solution is obtained by a method which is applicable to any solid body, whatever be its form.

87. The problem of the rotation of a solid body about a fixed point, was also considered by Somoff, *Journal für Mathematik*, t. LVI., p. 201; the solution is obtained by a method which is applicable to any solid body, whatever be its form.

88. The problem of the motion of a solid body about a fixed point, under the action of no forces, was first solved by Euler, *Mémoires de Berlin*, 1758; the solution was afterwards completed by Poinsot, *Théorie nouvelle de la rotation des corps*, Paris, 1834.

89. The bodies are considered as restricted to move in a given line; but it is easy to extend the results to the case of bodies moving in any manner whatever.

90. The problem of the motion of a system of bodies, under the action of no forces, was first solved by Lagrange, *Méc. Anal.*, Vol. II., p. 212; the solution was afterwards completed by Poisson, *Traité de Mécanique*, Vol. II., p. 82.

91. The problem of the motion of a system of bodies, under the action of given forces, was first solved by Lagrange, *Méc. Anal.*, Vol. II., p. 212; the solution was afterwards completed by Hamilton, *Phil. Trans.*, 1834, p. 247.

92. The problem of the motion of a system of bodies, under the action of given forces, was also considered by Jacobi, *Vorlesungen über Dynamik*, p. 1; the solution is given in terms of elliptic functions.

93. The problem of the motion of a system of bodies, under the action of given forces, was also considered by Liouville, *Journal de Mathématiques*, t. XI., p. 457; the solution is obtained by a method which is applicable to any system of bodies, whatever be the law of force.

94. The problem of the motion of a system of bodies, under the action of given forces, was also considered by Bertrand, *Journal de Mathématiques*, t. XVII., p. 121; the solution is obtained by a method which is applicable to any system of bodies, whatever be the law of force.

95. The problem of the motion of a system of bodies, under the action of given forces, was also considered by Bour, *Journal de Mathématiques*, t. XX., p. 185; the solution is obtained by a method which is applicable to any system of bodies, whatever be the law of force.

96. In the first memoir (§ 1) the author considers a point moving in a plane under the action of given forces; the equations of motion are obtained in a form which is well adapted for numerical calculation.

97. The problem of the motion of a particle in a plane, under the action of given forces, was first solved by Lagrange, *Méc. Anal.*, Vol. II., p. 72; the solution was afterwards completed by Hamilton, *Phil. Trans.*, 1834, p. 247.

98. The problem of the motion of a particle in a plane, under the action of given forces, was also considered by Jacobi, *Vorlesungen über Dynamik*, p. 1; the solution is given in terms of elliptic functions.

99. The problem of the motion of a particle in a plane, under the action of given forces, was also considered by Liouville, *Journal de Mathématiques*, t. XI., p. 457; the solution is obtained by a method which is applicable to any system of forces, whatever be the law of force.

100. The problem of the motion of a particle in a plane, under the action of given forces, was also considered by Bertrand, *Journal de Mathématiques*, t. XVII., p. 121; the solution is obtained by a method which is applicable to any system of forces, whatever be the law of force.

Problem of Three Bodies. Art. Nos. 101 to 123.

101. The problem of three bodies is the most celebrated problem in dynamics; it is the problem of determining the motion of three bodies which attract each other according to the law of nature. The problem has been considered by many mathematicians, but the complete solution has never been obtained.

102. Passing to the applications: in the first place, if α , β are rectangular coordinates of a point in piano, then writing instead of them x , y , we have $ds^2 = dx^2 + dy^2$, which is of the required form; but the result obtained is the self-evident one, that

$$\frac{d^2x}{dt^2} = X, \quad \frac{d^2y}{dt^2} = Y,$$

where X , Y are the components of the force.

103. The problem of three bodies was first considered by Newton, *Principia*, Lib. I., Prop. LXVI.; the solution was afterwards completed by Lagrange, *Méc. Anal.*, Vol. II., p. 230, and by Laplace, *Méc. Cél.*, Liv. X.

104. The problem of three bodies was also considered by Euler, *Mémoires de Berlin*, 1760; the solution was afterwards completed by Lagrange, *Méc. Anal.*, Vol. II., p. 230.

105. The problem of three bodies was also considered by Clairaut, *Théorie de la Lune*, Paris, 1752; the solution was afterwards completed by d'Alembert, *Recherches sur la Précession*, Paris, 1749.

106. The problem of three bodies was also considered by Laplace, *Méc. Cél.*, Liv. X.; the solution was afterwards completed by Lagrange, *Méc. Anal.*, Vol. II., p. 230.

107. Jacobi, "De Motu puncti singularis" (1842). — I call to mind that the memoir contains a complete discussion of the motion of a particle under the action of given forces, the particle being supposed to move in a plane.

108. The problem of three bodies was also considered by Poisson, *Traité de Mécanique*, Vol. II., p. 82; the solution was afterwards completed by Lagrange, *Méc. Anal.*, Vol. II., p. 230.

109. The problem of three bodies was also considered by Hamilton, *Phil. Trans.*, 1834, p. 247; the solution was afterwards completed by Jacobi, *Vorlesungen über Dynamik*, p. 1.

110. The problem of three bodies was also considered by Jacobi, *Vorlesungen über Dynamik*, p. 1; the solution is given in terms of elliptic functions.

111. The problem of three bodies was also considered by Liouville, *Journal de Mathématiques*, t. XI., p. 457; the solution is obtained by a method which is applicable to any system of forces, whatever be the law of force.

112. The Problem of Three Bodies, Art. Nos. 112 to 123.

113. The problem of three bodies is the most difficult problem in dynamics; the complete solution has never been obtained, and it is probable that it never will be obtained.

114. The problem of three bodies has been considered by many mathematicians; in particular, by Newton, *Principia*, Lib. I., Prop. LXVI.; by Euler, *Mémoires de Berlin*, 1760; by Lagrange, *Méc. Anal.*, Vol. II., p. 230; by Laplace, *Méc. Cél.*, Liv. X.; and by Poisson, *Traité de Mécanique*, Vol. II., p. 82.

115. The problem of three bodies has also been considered by Hamilton, *Phil. Trans.*, 1834, p. 247; by Jacobi, *Vorlesungen über Dynamik*, p. 1; by Liouville, *Journal de Mathématiques*, t. XI., p. 457; and by Bertrand, *Journal de Mathématiques*, t. XVII., p. 121.

116. The problem of three bodies has also been considered by Bour, *Journal de Mathématiques*, t. XX., p. 185; by Clebsch, *Journal für Mathematik*, t. LIII., p. 195; and by Kirchhoff, *Journal für Mathematik*, t. L., p. 285.

117. The problem of three bodies has also been considered by Routh, *Treatise on the Dynamics of a System of Rigid Bodies*, p. 82; by Somoff, *Journal für Mathematik*, t. LVI., p. 201; and by Greenhill, *Proc. Lond. Math. Soc.*, Vol. III., p. 20.

118. The centre of gravity of the three bodies moves uniformly in a right line, and it may without any real loss of generality be taken to be at rest; that is, if the x -coordinates of the three bodies be x_1 , x_2 , x_3 , and if X be the centre of gravity, then

$$x_1 + x_2 + x_3 = 0,$$

and similarly for the y and z coordinates.

119. The problem of three bodies has also been considered by Darwin, *Proc. Lond. Math. Soc.*, Vol. IV., p. 20; by Poincaré, *Méthodes nouvelles de la Mécanique céleste*, Paris, 1892; and by Hill, *Acta Mathematica*, Vol. VIII., p. 1.

120. The problem of three bodies has also been considered by Sundman, *Acta Mathematica*, Vol. XXXVI., p. 105; the solution is obtained in a form which is applicable to any system of three bodies, whatever be the law of force.

121. The problem of three bodies has also been considered by Levi-Civita, *Acta Mathematica*, Vol. XXX., p. 305; the solution is obtained by a method which is applicable to any system of three bodies, whatever be the law of force.

122. The problem of three bodies has also been considered by Birkhoff, *Trans. Amer. Math. Soc.*, Vol. XII., p. 1; the solution is obtained by a method which is applicable to any system of three bodies, whatever be the law of force.

123. The problem of three bodies has also been considered by Wintner, *Analytical Foundations of Celestial Mechanics*, Princeton, 1941; the solution is obtained by a method which is applicable to any system of three bodies, whatever be the law of force.

[End of pages 501–550]

Problem of Two Centres of Force (continued)

ARTICLE NOS. 52 TO 64.

52. A case satisfying the required conditions is found to be

$$\frac{x}{y} X \frac{y'}{y},$$

or, what is the same thing,

$$U = 2au + \frac{\beta}{u}, \quad V = 2cv + \frac{\delta}{v},$$

that is, besides the forces $\frac{\beta}{u^2}$, $\frac{\delta}{v^2}$, which vary as (dist.)⁻², there are the forces $2au$, $2cv$, varying directly as the distance, and of the same amount at equal distances; or, what is the same thing, there is, besides the forces varying as (dist.)⁻², a force varying directly as the distance, tending to a third centre midway between the other two, a case which is specially considered in the memoir; it is found that the functions in r , θ under the radicals (instead of rising only to the order 4) rise in this case to the order 6.

53. Among other cases are found the following, viz.:

$$1. \quad U = \alpha u + \beta u^2 + \gamma u^4,$$

$$2. \quad U = \alpha u + \beta u^2,$$

$$V = \delta v + \epsilon v^2 + \zeta v^4,$$

where $\beta = \epsilon$, or else $\alpha\epsilon = \gamma\delta = 2\beta\epsilon$.

In regard to the subject of this second memoir of Lagrange, see post, Miscellaneous Problems, Liouville's Memoirs, Nos. 100 to 105.

54. In the *Mécanique Analytique* (1st ed. 1788, and 2nd ed. t. II. 1813), Lagrange in effect reproduces his solution for the above-mentioned law of force (say $U = \frac{\alpha}{u} + 2\mu$, $V = \frac{\beta}{v} + 2\nu$). There are even in the third edition a few trifling errors of work to be corrected. The remarks above referred to, as made by Lagrange in his first memoir, are also reproduced (see ante, Nos. 49 and 50).

55. Legendre, *Exercices de Calcul Intégral*, t. II. (1817), and *Théorie des Fonctions Elliptiques*, t. I. (1825), uses p and q in the place of

Euler's p, q ; the forces are assumed to vary as $(\text{dist.})^{-2}$, and in consequence of the change Euler's cubic radicals are replaced by quartic radicals involving only even powers of p and q respectively; that is, the radicals are in a form adapted for the transformation to elliptic integrals; in certain cases, however, it becomes necessary to attribute to Legendre's variables p and q imaginary values. The various cases of the motion are elaborately discussed by means of the elliptic integrals; in particular Legendre notices certain cases in which the motion is oscillatory, and which, as he remarks, seem to furnish the first instance of the description by a free particle of only a finite portion of the curve which is analytically the orbit of the particle; there is, however, nothing surprising in this kind of motion, although its existence might easily not have been anticipated.

56. § 26 of Jacobi's memoir "Theoria Novi Multiplicatoris &c." (1845) is entitled "Motus puncti versus duo centra secundum legem Neutonianum attracti." The equations for the motion in space are by a general theorem given in the memoir "De Motu puncti singularis" (1842), reduced to the case of motion in a plane: viz. if x, y are the coordinates, the centre point of the axis being the origin, and y being at right angles to the axis, and if the distance of the centres is $2a$; then the only difference is that to the expression for dt^2 there is added a term $\frac{a}{y^2}$, which arises from the rotation about the axis.

Two integrals are obtained, one the integral of Vis Viva, and the other of them an integral similar to one of those of Euler's or Lagrange's. And then x', y'' being the differential coefficients of x, y with regard to the time, the remaining equation may be taken to be $y' dx - x' dy = 0$, where x', y' are to be expressed as functions of x, y by means of the two given integrals. This being so, the principle of the Ultimate Multiplier⁽¹⁾ furnishes a multiplier of this differential equation, and the integral is found to be

$$\frac{y' dx - x' dy}{xy(x'^2 - y'^2) + a(a^2 - x^2 + y^2)xy} = d\psi,$$

the quantity under the integral sign being a complete differential. To verify *a posteriori* that this is so, Jacobi introduces the auxiliary quantities λ', λ'' defined as the roots of the equation $\lambda^2 + \lambda(x^2 + y^2 - a^2) - a^2y^2 = 0$, which in fact, if as before u, v are the distances from the centres, leads to

$$u + v = 2\sqrt{a^2 + \lambda'}, \quad u - v = 2\sqrt{a^2 + \lambda''},$$

so that λ' , λ'' are functions of $u + v$, $u - v$ respectively; and the formulæ, as ultimately expressed in terms of λ' , λ'' , are substantially of the same form with those of Euler and Lagrange.

57. The investigations contained in Liouville's three memoirs "Sur quelques cas particuliers &c." (1846), find their chief application in the problem of two centres, and by leading in the most direct and natural manner to the general law of force for which the integration is possible, they not only give some important extension of the problem, but they in fact exhibit the problem itself and the preceding solutions of it in their true light. But as they do not relate to this problem exclusively, it will be convenient to consider them separately under the head Miscellaneous Problems.

58. In Serret's "Thèse sur le Mouvement &c." (1848), the problem is very elegantly worked out according to the principles of Liouville's memoirs as follows: viz. assuming that the expression of the distance between two consecutive positions of the body is

$$ds^2 = \lambda(m d\mu^2 + n d\nu^2) + \lambda'' d\psi^2,$$

where m , n are functions of μ , ν respectively, and if the forces can be represented by means of a force-function U , then the motion can be determined, provided only U be of the form

$$U = \frac{\Phi\mu + \Psi\nu}{\lambda},$$

where the functional symbols Φ , Ψ , &c. denote any arbitrary functions whatever.

59. It is then assumed that μ , ν are the parameters of the confocal ellipses and hyperbolas situate in the moveable plane through the axis, viz. that we have

$$\frac{x^2}{\mu^2} + \frac{y^2}{\mu^2 - b^2} = 1, \quad \frac{x^2}{\nu^2} + \frac{y^2}{\nu^2 - b^2} = 1$$

(the origin is midway between the two centres, $2b$ being their distance; μ , ν are in fact equal to the sum and difference $u + v$, $u - v$ of the two centres respectively); and that the position of the moveable plane is determined by means of ψ , the inclination to a fixed plane through

¹Explained in Jacobi's memoir "Theoria Novi Multiplicatoris &c.," Crelle, tt. xxvii., xxviii., xxix. 1844-45.

the axis, or say, as before, its azimuth. In fact, with these values of the coordinates, the expression of ds^2 is

$$ds^2 = \frac{\mu^2 - \nu^2}{\mu^2 - b^2} d\mu^2 + \frac{\mu^2 - \nu^2}{b^2 - \nu^2} d\nu^2 + \mu^2 \nu^2 d\psi^2,$$

which is of the required form. And moreover if the forces to the two centres vary as $(\text{dist.})^{-2}$, and there is besides a force to the middle point varying as the distance, then

$$U = \frac{\alpha}{\mu^2} + \frac{\beta}{\nu^2} + k(\mu^2 - \nu^2),$$

whence (observing that $\lambda = \mu^2 - \nu^2$) U is of the required form. The equations obtained by substituting for U the above value give the ordinary solution of the problem.

60. Liouville's note to the last-mentioned memoir (1848) contains the demonstration of a theorem obtained by a different process in his second memoir, but which is in the present note, starting from Serret's formulæ, demonstrated by the more simple method of the first memoir, viz., it is shown that the motion can be obtained if the two centres, instead of being fixed, revolve about the point midway between them in a circle in such manner that the diameter through the two centres always passes through the projection of the body on the plane of the circle. It will be observed that the circular motion of the two centres is neither a uniform nor a given motion, but that they are, as it were, carried along with the moving body.

61. In Desboves's memoir "Sur le Mouvement d'un point matériel &c." (1848), the author develops the solution of the foregoing problem of moving centres, chiefly by the aid of the method employed in Liouville's second memoir. And he shows also that the methods of Euler and Lagrange for the case of two fixed centres apply with modification to the more complicated problem of the moving centres.

62. The problem of two centres is considered in Bertrand's "Mémoire sur les Équations différentielles &c." (1852), by means of Jacobi's form of the equations of motion, viz., the problem is reduced to a plane one by means of the addition of a force $\frac{\alpha}{y^2}$ (anti, No. 56).

63. Cayley's "Note on Lagrange's Solution &c." (1857) is merely a reproduction of the investigation in the *Mécanique Analytique*; the object was partly to correct some slight errors of work, and partly to show what were the combinations of the differential equations, which give at once the integrals of the problem.

64. In § II. of Bertrand's "Mémoire sur quelques unes des formes &c." (1857), the following question is considered, viz., assuming that the dynamical equations

$$\frac{d^2x}{dt^2} = \frac{dU}{dx}, \quad \frac{d^2y}{dt^2} = \frac{dU}{dy}$$

have an integral of the form

$$a = Px'^2 + Qx'y' + Ry'^2 + Sx' + Ty' + K$$

(where a is the arbitrary constant, and $P, Q, \dots, K$ are functions of x and y), it is required to find the form of the force-function U . It is found that U must satisfy a certain partial differential equation of the second order, the general solution of which is not known; but taking U to be a function of the distance from any fixed point (or rather the sum of any number of such functions), it is shown that the only case in which the differential equations for the motion of a point attracted to a fixed centre of forces have an integral of the form in question is the above-mentioned one of two centres, each attracting according to the inverse square of the distance, and a third centre midway between them, attracting as the distance.

The Spherical Pendulum

ARTICLE NOS. 65 TO 73.

65. The problem is obviously the same as that of a heavy particle on the surface of a sphere. I have not ascertained whether the problem was considered by Euler. Lagrange refers to a solution by Clairaut, *Mem. de l'Acad.*, 1735.

The question was considered by Lagrange, *Mec. Anal.*, 1st edit. p. 283. The angles which determine the position are ψ the inclination of the string to the horizon, ϕ the inclination of the vertical plane through the string to a fixed vertical plane, or say the azimuth. And then forming the equations of motion, two integrals are at once obtained; these are the integrals of Vis Viva, and an integral of areas. And these give equations of the form $dt = \text{funct.}(\psi) d\psi$, $d\phi = \text{funct.}(\psi) d\psi$; so that t, ϕ are each of them given by a quadrature in terms of ψ , which is the point to which the solution is carried. It is noticed that ψ may have a constant value, which is the case of the conical pendulum.

66. In the second edition, t. XI. p. 197 (1815), the solution is reproduced; only, what is obviously more convenient, the angles are taken to be ψ , the inclination to the vertical, ϕ , the azimuth.

It is remarked that ψ will always lie between a greatest value α and a least value β , and the integrals are transformed by introducing therein instead of ψ the angle σ , which is such that

$$\cos \psi = \cos \alpha \sin^2 \sigma + \cos \beta \cos^2 \sigma,$$

by which substitution they assume a more elegant form, involving only the radical $\sqrt{1 + k^2(\cos \beta - \cos \alpha) \cos 2\sigma}$, where k is a constant depending on $\cos \alpha$, $\cos \beta$; and the integration is effected approximately in the case where $\cos \beta - \cos \alpha$ is small.

Bravais has noticed, however, that by reason of some errors in the working out, Lagrange has arrived at an incorrect value for the angle Φ , which is the apsidal angle, or difference of the azimuths for the inclinations α and β : see the 3rd edition (1855), Note VII., where M. Bravais resumes the calculation, and he arrives at the value $\Phi = \frac{1}{2}\pi(1 + \frac{3}{8}\alpha\beta)$, α and β being small.

Lagrange considers also the case where the motion takes place in a resisting medium, the resistance varying as velocity squared.

67. A similar solution to Lagrange's, not carried quite so far, is given in Poisson's *Mécanique*, t. I. pp. 385 *et seq.* (2nd ed. 1833).

A short paper by Puiseux, "Note sur le Mouvement d'un point matériel sur une sphère" (1842), shows merely that the angle Φ is $> \frac{1}{2}\pi$.

68. The ulterior development of the solution consists in the effectuation of the integrations by the elliptic and Jacobian functions. It is proper to remark that the dynamical problem the solution whereof by such functions was first fairly worked out is the more difficult one of the rotation of a solid body, as solved by Jacobi (1839), in completion of Rueb's solution (1834), post, Nos. 186 and 197.

69. In relation to the present problem we have Gudermann's memoir "De pendulis sphaerici s &c." (1849), who, however, does not arrive at the actual expressions of the coordinates in terms of the time; and the perusal of the memoir is rendered difficult by the author's peculiar notations for the elliptic functions⁽¹⁾.

¹The mere use of sn, cn, dn as an abbreviation of the somewhat cumbersome sinam, cosam, Δam of the *Fundamenta Nova* is decidedly convenient.

70. It would appear that a solution involving the Jacobian functions was obtained by Durège, in a memoir completed in 1849, but which is still unpublished; see § XX. of his *Theorie der elliptischen Functionen* (1861), where the memoir is in part reproduced. It is referred to by Richelot in the Note presently mentioned.

71. We have next Tissot's *Thèse de Mécanique*, 1852, where the expressions for the variables in terms of the time are completely obtained by means of the Jacobian functions H , Θ , and which appears to be the earliest published one containing a complete solution and discussion of the problem.

72. Richelot, in the Note "Bemerkungen zur Theorie des Raumpendels" (1853), gives also, but without demonstration, the final expressions for the coordinates in terms of the time.

Donkin's memoir "On a Class of Differential Equations &c." (1855) contains (No. 59) an application to the case of the spherical pendulum.

73. The first part of the memoir by Dumas, "Ueber die Bewegung des Raumpendels, &c." (1855), comprises a very elegant solution of the problem of the spherical pendulum based upon Jacobi's theorem of the Principal Function (1837), and which is completely developed by the elliptic and Jacobian functions. The latter part of the memoir relates to the effect of the rotation of the Earth; and we thus arrive at the next division of the general subject.

Motion as affected by the Rotation of the Earth, and Relative Motion in general

ARTICLE NOS. 74 TO 85.

74. Laplace (*Méc. Céleste*, Book X. c. 5) investigates the equations for the motion of a terrestrial body, taking account of the rotation of the Earth (and also of the resistance of the air), and he applies them to the determination of the deviations of falling bodies, &c. He does not, however, apply them to the case of the pendulum.

75. We have also the memoir of Gauss, "Fundamentalgleichungen &c." (1804): the equations ultimately obtained are similar to those of Poisson. I have not had the opportunity of consulting this memoir.

76. Poisson, in the "Memoire sur le mouvement des Projectiles &c." (1838), also obtains the general equations of motion, viz. (omit-

ting terms involving n^2), these may be taken to be

$$\begin{aligned}\frac{d^2x}{dt^2} &= 2n \sin \beta \frac{dy}{dt} + X, \\ \frac{d^2y}{dt^2} &= -2n \sin \beta \frac{dx}{dt} - 2n \cos \beta \frac{dz}{dt} + Y, \\ \frac{d^2z}{dt^2} &= 2n \cos \beta \frac{dy}{dt} + Z,\end{aligned}$$

(see p. 20), where the axes of x , y , z are fixed on the Earth and moveable with it: viz., z is in the direction of gravity; x , y in the directions perpendicular to gravity, viz., y in the plane of the meridian northwards, x westwards; g is the actual force of gravity as affected by the resolved part of the centrifugal force; β is the latitude. There are some niceties of definition which are carefully given by Poisson, but which need not be noticed here.

77. Poisson applies his formulæ incidentally to the motion of a pendulum, which he considers as vibrating in a plane; and after showing that the time of oscillation is not sensibly affected, he remarks that upon calculating the force perpendicular to the plane of oscillation, arising from the rotation of the Earth, it is found to be too small sensibly to displace the plane of oscillation or to have any appreciable influence on the motion—a conclusion which, as is well known, is erroneous. He considers also the motion of falling bodies, but the memoir relates principally to the theory of projectiles.

78. That the motion of the spherical pendulum is sensibly affected by the rotation of the Earth is the well-known discovery of Foucault; it appears by his paper, “Démonstration Physique &c.,” *Comptes Rendus*, t. XXXII. 1851, that he was led to it by considering the case of a pendulum oscillating at the pole; the plane of oscillation, if actually fixed in space, will by the rotation of the Earth appear to rotate with the same velocity in the contrary direction; and he remarks that although the case of a different latitude is more complicated, yet the result of an apparent rotation of the plane of oscillation, diminishing to zero at the equator, may be obtained either from analytical or from mechanical and geometrical considerations. Some other Notes by Foucault on the subject are given, *Comptes Rendus*, t. XXXV. (1853).

79. An analytical demonstration of the theorem was given by Binet, *Comptes Rendus*, t. XXXII. (1851), and by Baehr (1853). Various short papers on the subject will be found in the *Philosophical Magazine*, and elsewhere.

80. In regard to the above-mentioned problem of falling bodies, we have a Note by W. S., *Camb. and Dubl. Math. Journ.* t. III. (1848), containing some errors which are rectified in a subsequent paper, "Remarks on the Deviation of Falling bodies, &c." t. IV. (1849), by Dr Hart and Professor W. Thomson.

81. The theory of relative motion is considered in a very general manner in M. Quet's memoir, "Des Mouvements relatifs en général &c." (1853). Suppose that x, y, z are the coordinates of a particle in relation to a set of moveable axes; let ξ, η, ζ be the coordinates of the moveable origin in reference to a fixed set of axes, and treating the accelerations $\frac{d^2\xi}{dt^2}, \frac{d^2\eta}{dt^2}, \frac{d^2\zeta}{dt^2}$ as if they were coordinates, let these, when resolved along the moveable axes, give u', v', w' : suppose, moreover, that p, q, r denote the angular velocities of the system of the moveable axes (or axes of x, y, z) round the axes of x, y , and z respectively; u', v', w', p, q, r are considered as given functions of the time, and then, if

$$\begin{aligned} X &= \frac{d^2x}{dt^2} + 2 \left(q \frac{dz}{dt} - r \frac{dy}{dt} \right) + x(q^2 + r^2) - y \left(pq - \frac{dr}{dt} \right) - z \left(pr + \frac{dq}{dt} \right) - u', \\ Y &= \frac{d^2y}{dt^2} + 2 \left(r \frac{dx}{dt} - p \frac{dz}{dt} \right) + y(r^2 + p^2) - z \left(qr - \frac{dp}{dt} \right) - x \left(qp + \frac{dr}{dt} \right) - v', \\ Z &= \frac{d^2z}{dt^2} + 2 \left(p \frac{dy}{dt} - q \frac{dx}{dt} \right) + z(p^2 + q^2) - x \left(rp - \frac{dq}{dt} \right) - y \left(rq + \frac{dp}{dt} \right) - w', \end{aligned}$$

it is shown that the equations of motion are to be obtained from the equation

$$\Sigma m [(X - A') \delta x + (Y - B') \delta y + (Z - C') \delta z] = 0,$$

where $\delta x, \delta y, \delta z$ are the virtual velocities of the particle m in the directions of the moveable axes.

This equation is in fact obtained as a transformation of the equation

$$\Sigma m \left[\left(\frac{d^2\xi}{dt^2} - A \right) \delta\xi + \left(\frac{d^2\eta}{dt^2} - B \right) \delta\eta + \left(\frac{d^2\zeta}{dt^2} - C \right) \delta\zeta \right] = 0,$$

which belongs to a set of fixed axes of ξ, η, ζ .

82. The equations for the motion of a free particle are of course $X = A', Y = B', Z = C'$.

In the case where the moveable axes are fixed on the Earth, and moveable with it (the diurnal motion being alone attended to), these

lead to equations for the motion of a particle in reference to the Earth, similar to those obtained by Gauss and Poisson. The formulæ are applied to the case of the spherical pendulum, which is developed with some care; and Foucault's theorem of the rotation of the plane of oscillation very readily presents itself. The general formulæ are applied to the relative motion of a solid body, and in particular to the question of the gyroscope; the memoir contains other interesting results.

83. The principal memoirs on the motion of the spherical pendulum, as affected by the rotation of the Earth, are those of Hansen, "Theorie der Pendelbewegung &c." (1853), which contains an elaborate investigation of all the physical circumstances (resistance of the air, torsion of the string, &c.) which can affect the actual motion, and the before-mentioned memoir by Dumas, "Ueber der Bewegung des Raumpendels &c." (1855).

The investigation is conducted by means of the variation of the constants; the integrals for the undisturbed problem were, as already noticed, obtained by means of Jacobi's Principal Function, that is, in a form which leads at once to the expressions for the variation of the constants; and the investigation appears to be carried out in a most elaborate and complete manner.

84. In concluding this part of the subject I refer to Mr Worm's work, *The Rotation of the Earth* (1862), where the last-mentioned questions (falling bodies, the pendulum, and the gyroscope) are, in reference to the proofs they afford of the rotation of the Earth, considered as well in an experimental as in a mathematical point of view. The second part of the volume contains the theory (after Laplace and Gauss) of falling bodies, that of the pendulum (after Hansen), and that of the gyroscope (after Yvon Villarceau); and the whole appears to be a complete and satisfactory résumé of the experimental and mathematical theories to which it relates.

85. We have also Cohen "On the Differential Coefficients and Determinants of Lines &c." (1862), where the equations for relative motion are obtained in a very elegant manner. The fundamental notion of the memoir may be considered to be the dealing directly with lines, velocities, &c., which are variable in direction as well as in magnitude, instead of referring them, as in the ordinary analytical method, to axes fixed in space. The memoir is a highly interesting and valuable one, and the results are brought out with great facility; but I cannot but think that the great care required to apply the me-

thod correctly is an objection to it, if used otherwise than by way of interpretation of previously obtained results, and that the ordinary method is preferable.

I may remark that the theory of relative motion connects itself with the lunar and planetary theories as regards the reference of the plane of the orbit to the variable ecliptic, and as regards the variations of the position of the orbit; but this is a subject which I have abstained from entering upon.

Miscellaneous Problems

ARTICLES NOS. 86 TO 111 (SEVERAL SUBHEADINGS).

Motion of a single particle

86. Jacobi, in the memoir “De Motu puncti singularis” (1842), notices (§ 5) the case of a body acted on by a central force which is any homogeneous function of the degree -2 of the coordinates; or representing these by $r \cos \phi$, $r \sin \phi$, then the force is $= \frac{\Phi}{r^2}$, where Φ is any function of the angle ϕ . The process is different from the ordinary one. In fact, after integrating by the process of a central force $= \frac{1}{r^2}$, he remarks that the method in fact applies to the more general law of force just mentioned.

87. Jacobi, in the memoir “Theoria Novi Multiplicatoris &c.” (1845), considers (§ 25) the case of a body acted on by a central force P a function of the distance, and besides by forces X and Y , which are homogeneous functions of the degree -3 of the coordinates (x, y) ; viz. the equations of motion are in this case

$$\frac{d^2x}{dt^2} = \frac{Px}{r} + X, \quad \frac{d^2y}{dt^2} = \frac{Py}{r} + Y,$$

and there is an integral

$$\frac{1}{2}(xy' - x'y)^2 - \int x^2(xY - yX) d\frac{y}{x} = \text{const.}$$

(the function under the integral sign is obviously a function of the degree 0 in (x, y) , that is, it is a function of $\frac{y}{x}$).

If X, Y are the derived functions of a force-function U of the degree -2 in (x, y) , then there is, besides, the integral of Vis Viva,

and thence a third integral is obtained by means of the theorem of the Ultimate Multiplier. It may be noticed that in the last-mentioned case the force-function is of the form $\frac{R}{r} + \frac{\Phi}{r^2}$, so that if we represent also the central force by means of a force-function R (= function of r), then the entire force-function is $R + \frac{\Phi}{r^2}$. The case is a very interesting one; it includes that considered § IV. of Bertrand's "Memoire sur les Équations différentielles de la Mécanique" (1852), where the force-function is of the form $= \frac{\Phi}{r^2}$.

Motion of three mutually attracting bodies in a right line

88. The problem is considered by Euler in the memoir "De Motu rectilineo &c." (1765), the forces being as the inverse square of the distance; and a solution is obtained for an interesting particular case. Let A, B, C be the masses, and suppose that at the commencement of the motion the distances CB, BA are in the ratio $\alpha : 1$, and that the velocities (assumed to be in the same sense) are proportional to the distances from a fixed point. Then, if α be the real root (there is only one) of the equation of the fifth order

$$C(1 + 3\alpha + 3\alpha^2) = A\alpha^5(\alpha^2 + 3\alpha + 3) + B(\alpha + 1)^2(\alpha^2 - 1),$$

the distances CB, BA will always continue in the ratio $\alpha : 1$. It may be added that the distances CB, BA each of them vary as $r^2 - a^2$, where a is a constant, and r is, according to the initial circumstances, a function of t defined by one or the other of the two equations

$$t = n\sqrt{r^2 - a^2} - na^2 \log \frac{r + \sqrt{r^2 - a^2}}{a},$$

or

$$t = n\sqrt{a^2 - r^2} + na^2 \sin^{-1} \frac{r}{a}.$$

89. The bodies are considered as restricted to move in a given line; but it is clear that if the bodies, considered as free points in space, are initially in a line, and the initial velocities are also in this line, then the bodies will always continue in this line, which will be a fixed line in space. But if the distances and velocities are as above, except only that the velocities, instead of being along the line, are parallel to each other in any direction whatever, then the bodies will always

continue in a line, which is in this case a moveable line in space (see post. No. 93).

90. Euler resumes the problem in the memoir of 1776 in the *Nova Acta Petrop.* The distances AB , BC being p and q , then

$$\frac{d^2p}{dt^2} = \frac{A+B}{p^2} + \frac{C}{q^2} - \frac{C}{(p+q)^2}, \quad \frac{d^2q}{dt^2} = \frac{A}{p^2} - \frac{A}{(p+q)^2} + \frac{B+C}{q^2},$$

and in particular he considers the before-mentioned case of a solution of the form $p = \alpha q$; and also the particular problem where one of the masses vanishes, $C = 0$; in this case, introducing (instead of p, q), the new variables u, s , where $q = up$, $dq = s dp$ (a transformation suggested by the homogeneity of the equations), and making, moreover, the particular supposition that the integral of the first equation is $\left(\frac{dp}{dt}\right)^2 = \frac{2(A+B)}{p}$ (viz. making the constant of integration to vanish), he obtains between s and u the equation of the first order

$$2(A+B)(s-u) = (A+B)s + A \frac{1}{(1+u)^2} - \frac{A}{u^2},$$

which, however, he is not able to integrate.

91. Jacobi has given in the memoir “Theoria Novi Multiplicatoris &c.” (1845) (§ 28, entitled “De Problemate trium corporum in eadem recta, motorum. Substitutio Euleriana. Theoremata de viribus homogeneis”) a very symmetrical and elegant investigation of the same problem.

The centre of gravity being assumed to be at rest, the coordinates x, x_1, x_2 of the three bodies are in the first instance expressed as linear functions of the two variables u, v (being, as Jacobi remarks, the transformation employed in his memoir “Sur l’élimination des Noeuds” (1843), post, No. 114); $\frac{d^2u}{dt^2}$ and $\frac{d^2v}{dt^2}$ come out respectively equal to homogeneous functions of the degree -2 of these variables u and v , and the integral of Vis Viva exists.

The subsequent transformation consists in the introduction of the variables r, ϕ, s, η , where $u = r \cos \phi, v = r \sin \phi, s = \frac{dr}{dt}, \eta = \sqrt{r^2} \frac{d\phi}{dt}$; this gives a system of equations independent of r ; viz.,

$$d\phi : ds : d\eta = \eta : \Phi s^2 + \eta^2 - \Psi : -\Phi s\eta + \Psi',$$

where Φ is a given function of ϕ , and Φ' is the derived function. If these equations were integrated, the equation of Vis Viva gives at

once $r = \frac{1}{s} \sqrt{\Psi - \Phi(s^2 + \eta^2)}$; and finally the time t would be given by a quadrature. The system of three equations has the multiplier $M = \frac{1}{s\eta - \Psi}$; hence if one integral were known the other would be at once furnished by the general theory. There is a simplification in the form of the solution if h (the constant of Vis Viva) = 0. It is remarked that the method is equally applicable when the force varies as any power of the distance; and moreover that when the force varies as (dist.)⁻², then the solution depends upon one quadrature only.

92. The concluding part of the section relates to the very general problem of a system of n particles acted on by any forces homogeneous functions of the coordinates (this includes the case of n particles mutually attracting each other according to a power of the distance), and this more general investigation illustrates the method employed in regard to the three bodies in a line. It may be remarked that in the general theorem for the n particles “sint vires &c.” the constant of Vis Viva is supposed to vanish.

Particular cases of the motion of three bodies

93. In the case of three bodies attracting each other according to the inverse square of the distance, the bodies may move in such manner as to be constantly in a line (a moveable line in space); this appears by the memoir, Euler, “Considérations générales &c.” (1764), in which memoir, however (which it will be observed precedes the memoir “De Motu rectilineo &c.” (1765), referred to No. 88), Euler assumes that the mass of one of the bodies is so small as not to affect the relative motion of the other two. Calling the bodies the Sun, Earth, and Moon, and taking the masses to be 1, m , 0, then a result obtained is, that in order that the Moon may be perpetually in conjunction, its distance must be to that of the Sun as $a : 1$, where $m(1-a)^2 = 3a^5 - 3a^4 + a^3$, or 7 : 400 nearly. It appears, however (anti, No. 88), that the foregoing restriction as to the masses is unnecessary, and, as will be mentioned, the problem has since been treated without such restriction. Euler investigates the motion in the case where the initial circumstances are nearly but not exactly as originally supposed; this assumes, however, that the motion is stable—i.e. that the bodies will continue to move nearly, but not exactly as originally supposed, which is at variance with the conclusions of Liouville’s memoir, post, No. 95.

I have not examined the cause of this discrepancy.

94. In the *Mécanique Céleste* (1799), Book X. c. 6, Laplace con-

siders two cases where the motion can be exactly determined.

1°. Force varies as any function of the distance. It is shown that the motion may be such that the bodies form always an equilateral triangle of variable magnitude—the motion of each body about the centre of gravity being the same as if that point were a centre of force attracting the body according to a similar law.

2°. Force $\propto (\text{dist.})^n$. The motion may be such that the three bodies are always in a right line (moveable in space), the relative distances being in fixed ratios to each other. In particular, if force $\propto (\text{dist.})^{-2}$, then m, m', m'' being the masses, the quantity z which determines the ratio of the distances $m''m', mm$ is given by

$$0 = mz^2[(1+z)^2 - 1] - m'(1+z)^2(1-z^2) - m''[(1+z)^2 - z^2] = 0,$$

which is, in fact, the formula in Euler's memoir "De Motu rectilineo &c."

95. Liouville's memoir "Sur un cas particulier &c." (1842) has for its object to show that if the initial circumstances are not precisely as supposed in the second of the two cases considered by Laplace, or, what is the same thing, in Euler's memoir "Considérations générales &c.," then the motion is unstable; the instability manifests itself in the usual manner, viz. the expressions for the deviations from the normal positions are found to contain real exponentials which increase indefinitely with the time.

96. It may be proper to refer here to Jacobi's theorem, *Comptes Rendus*, t. III. p. 61 (1836), quoted in the foot-note to No. 31 of my Report of 1857, [195], which relates to the motion of a point *without mass* revolving round the Sun, and disturbed by a planet moving in a circular orbit, and properly belongs (as I have there remarked) to the problem of two centres, one of them moveable and the other revolving round it in a circle with uniform velocity. The theorem (given without demonstration by Jacobi) is proved by Liouville in his last-mentioned memoir, and he remarks that the theorem follows very simply as a corollary of the theorem by Coriolis, "Mémoire sur le principe des forces vives dans les mouvemens relatifs des Machines," *Journ. de l'École Polyt.* t. XIII. pp. 268—302 (1832).

There is, however, no difficulty in proving the theorem; another proof is given, Cayley, "Note on a Theorem of Jacobi's &c." (1862).

Motion in a resisting medium

97. I do not consider the various integrable cases of the motion of a particle in a resisting medium, the resistance varying with the velocity according to some assumed law, the particle being either not acted on by any force, or acted upon by gravity only. Some interesting cases are considered in Jacobi's memoir "De Motu puncti singularis" (1842), § 6 and 7 (see post, No. 108).

98. In the case of a central force varying as $(\text{dist.})^{-2}$, the effect of a resisting medium ($R \propto \text{vel}^2$) is considered in reference to the lunar theory, in the *Mécanique Céleste*, Book VII. c. 6. Formulæ for the variations of the elliptic elements are given in the *Mécanique Analytique*, t. II. (2nd edition). But the variations of the elliptic elements are fully worked out by means of elliptic and Jacobian functions in Sohncke's valuable memoir "Motus Corporum &c." (1833).

99. The effect of the resistance of the air on a pendulum has been elaborately considered by Poisson, Bessel, Stokes, and others; as the dimensions of the ball are attended to, the problem is in fact a hydrodynamical one. The effect on the spherical pendulum is considered in Hansen's memoir "Theorie der Pendelbewegung &c." (1853). The effect on the motion of a projectile is considered in Poisson's memoirs "Sur le Mouvement des Projectiles &c." (1838).

Liouville's memoirs "Sur quelques Cas particuliers ou les équations du mouvement d'un point matériel peuvent s'intégrer" (1846—49)

100. In the first memoir (§ 1) the author considers a point moving in a plane or on a given surface, where the principle of Vis Viva holds good (or say where there is a force-function V). The coordinates of the point, and the function U , may be expressed in terms of two variables α , β , and it is assumed that these are such that

$$ds^2 = \lambda (d\alpha^2 + d\beta^2),$$

where λ is a function of α and β .

That is, we have $T = \frac{1}{2}\lambda(\alpha'^2 + \beta'^2)$; and the equations of motion are

$$\frac{d}{dt}\lambda\alpha' = \frac{1}{2}\alpha'^2\frac{d\lambda}{d\alpha} + \frac{1}{2}\beta'^2\frac{d\lambda}{d\alpha} + \frac{dU}{d\alpha}, \quad \frac{d}{dt}\lambda\beta' = \frac{1}{2}\alpha'^2\frac{d\lambda}{d\beta} + \frac{1}{2}\beta'^2\frac{d\lambda}{d\beta} + \frac{dU}{d\beta}.$$

One integral of these is $\lambda(\alpha'^2 + \beta'^2) = 2U + C$; and by means of it the equations take the form

$$\frac{d}{dt}\lambda\alpha' = \beta'^2 \frac{d\lambda}{d\alpha} + \frac{dU}{d\alpha}, \quad \frac{d}{dt}\lambda\beta' = \alpha'^2 \frac{d\lambda}{d\beta} + \frac{dU}{d\beta}.$$

These equations, it is easy to show, may be integrated if

$$(2U + C)\lambda = f\alpha - F\beta,$$

and they then in fact give

$$\lambda\alpha' = \sqrt{2f\alpha + C\phi\alpha - A}, \quad \lambda\beta' = \sqrt{A - 2F\beta - C\phi\beta},$$

where A is an arbitrary constant.

And we then have

$$\frac{d\alpha}{\sqrt{2f\alpha + C\phi\alpha - A}} = \frac{d\beta}{\sqrt{A - 2F\beta - C\phi\beta}},$$

which gives the path, and

$$\frac{ds}{\sqrt{2U + C}} = \frac{\sqrt{\lambda} d\alpha}{\sqrt{2f\alpha + C\phi\alpha - A}} = \frac{\sqrt{\lambda} d\beta}{\sqrt{A - 2F\beta - C\phi\beta}},$$

the expression for the time is easily obtained by means of a quadrature.

It is not more general, but it is frequently convenient to employ instead of α, β , two variables μ and ν , such that

$$ds^2 = \lambda(m d\mu^2 + n d\nu^2),$$

where m is a function of μ only and n of ν only, while λ contains μ and ν . The geometrical signification of the equation $ds^2 = \lambda(d\alpha^2 + d\beta^2)$, or of the last-mentioned equivalent form, is that the curves α or $\mu = \text{const.}$, β or $\nu = \text{const.}$ intersect at right angles.

The foregoing differential equation of the path, writing $f\mu, F\nu$ in the place of $f\alpha, F\beta$ respectively, may be expressed in the form

$$f\mu \cos^2 i + F\nu \sin^2 i = A,$$

where $i, 90^\circ - i$ are the inclinations of the path at the point (λ, μ) to the two orthotomic curves through this point.

101. The before-mentioned equation $(2U + C)\lambda = f\alpha - F\beta$ may be satisfied independently of C , or else only for a particular value

of C . In the former case the law of force is much more restricted, but on the other hand there is no restriction as regards the initial circumstances of the motion; it is the more important one, and is alone attended to in the sequel of the memoir. In the case in question (changing the functional symbols) we must have

$$\lambda = \phi\alpha - \psi\beta, \quad U = f\alpha - F\beta;$$

so that the functions denoted above by $f\alpha$, $F\beta$ are $2f\alpha + C\phi\alpha$, $2F\beta + C\psi\beta$; the equation of the trajectory is

$$\frac{d\alpha}{\sqrt{2f\alpha + C\phi\alpha - A}} = \frac{d\beta}{\sqrt{A - 2F\beta - C\psi\beta}},$$

and for the time the formula is

$$t = \int \frac{\phi\alpha d\alpha}{\sqrt{2f\alpha + C\phi\alpha - A}} + \int \frac{\psi\beta d\beta}{\sqrt{A - 2F\beta - C\psi\beta}}.$$

It is noticed also that taking B , ϵ to denote two new arbitrary constants, and writing

$$\int \frac{d\alpha}{\sqrt{2f\alpha + C\phi\alpha - A}} + \int \frac{d\beta}{\sqrt{A - 2F\beta - C\psi\beta}} = B,$$

the equation of the trajectory and the expression for the time assume the forms

$$t = \epsilon + \frac{dB}{dC}, \quad 0 = \frac{dB}{dA},$$

as is known *a priori* by a theorem of Jacobi's.

If the forces vanish, the path is a geodesic line; and denoting by a the ratio of the constants A , C , we have

$$\frac{d\alpha}{\sqrt{\phi\alpha - a}} = \frac{d\beta}{\sqrt{a - \psi\beta}},$$

and moreover

$$ds = d\alpha \sqrt{\phi\alpha - a} + d\beta \sqrt{a - \psi\beta},$$

which are geometrical properties relating to the geodesic line.

102. Passing to the applications: in the first place, if α , β are rectangular coordinates of a point in piano, then writing instead of them x , y , we have $ds^2 = dx^2 + dy^2$, which is of the required form; but

the result obtained is the self-evident one, that the equations may be integrated by quadratures when U is of the form funct. x – funct. y . But taking instead the elliptic coordinates μ, ν , of a point in piano—viz., as employed by the author, these are the semiaxes of the confocal ellipse and hyperbola represented by the equations

$$\frac{x^2}{\mu^2} + \frac{y^2}{\mu^2 - b^2} = 1, \quad \frac{x^2}{\nu^2} + \frac{y^2}{\nu^2 - b^2} = 1$$

—very interesting results are obtained.

The equations give

$$\delta x^2 = \mu^2 \nu^2, \quad \delta y^2 = (\mu^2 - b^2)(b^2 - \nu^2),$$

and thence

$$ds^2 = \frac{\mu^2 - \nu^2}{\mu^2 - b^2} d\mu^2 + \frac{\mu^2 - \nu^2}{b^2 - \nu^2} d\nu^2,$$

which is of the proper form, and the corresponding expression of λU is

$$\lambda U = f\mu - F\nu,$$

so that the force-function having this value ($f\mu, F\nu$ being arbitrary functions of $\mu^2 - \nu^2$ and $\nu^2 - \mu^2$ respectively), the equations of motion may be integrated by quadratures.

103. In particular, if

$$f\mu = g\mu + g'\mu^2 + k(\mu^4 - b^2\mu^2), \quad F\nu = g\nu - g'\nu^2 - k(\nu^4 - b^2\nu^2),$$

then

$$U = \frac{g}{\mu + \nu} + \frac{g'}{\mu - \nu} + k(\mu^2 + \nu^2 - b^2).$$

But $\mu + \nu, \mu - \nu$ are the distances of the point from the two foci, and $\mu^2 + \nu^2 - b^2$ ($= x^2 + y^2$) is the square of the distance from the centre, so that the expression for U is

$$U = \frac{g}{r} + \frac{g'}{r'} + kR^2;$$

and the case is that of forces to the foci varying inversely as the squares of the distances, and a force to the centre varying directly as the distance—the case considered by Lagrange in the problem of two centres. But this is merely one particular case of those given by the general formula.

The cases $g = 0$, $g' = 0$, $k = 0$ (no forces), and $g = 0$, $g' = 0$ (a force to the centre) lead to some interesting results; it is noticed also that the expression for the force-function may be written

$$U = \text{funct.}(r + r') \frac{1}{r + r'} + \text{funct.}(r - r') \frac{1}{r - r'},$$

and that it may be ascertained (without transforming to elliptic coordinates) whether a given value of the force-function is of the form considered in the theory.

In § 3 the author considers the expression $dx^2 + dy^2 = \lambda (d\alpha^2 + d\beta^2)$, λ being in the first instance any function whatever of α and β ; and he shows that the expressions of x , y are given by the equation

$$x + y\sqrt{-1} = \psi(\alpha + \beta\sqrt{-1}),$$

ψ being any real function. If, however, it is besides assumed that λ is of the required form $= f\alpha - F\beta$, then he shows that the system of elliptic coordinates is the only one for which the conditions are satisfied. §§ 4, 5, 6 and 7 relate to the motion of a point on a sphere, an ellipsoid, a surface of revolution, and the skew helicoid respectively; and the concluding § 8 contains only a brief reference to the author's second memoir.

104. Liouville's second and third memoirs may be more briefly noticed. In the second memoir the author starts from Jacobi's theorem of the V function, viz., assuming that there is a force-function U independent of the time, then in order to integrate the equations of motion

$$\frac{d^2x}{dt^2} = \frac{dU}{dx}, \quad \frac{d^2y}{dt^2} = \frac{dU}{dy}, \quad \frac{d^2z}{dt^2} = \frac{dU}{dz},$$

it is required to find a function Θ of x , y , z containing three arbitrary constants A , B , C (distinct from the constant attached to Θ by mere addition) satisfying the differential equation

$$\left(\frac{d\Theta}{dx}\right)^2 + \left(\frac{d\Theta}{dy}\right)^2 + \left(\frac{d\Theta}{dz}\right)^2 = 2U + k,$$

for then the required integrals of the equations of motion are

$$\frac{d\Theta}{dA} = A', \quad \frac{d\Theta}{dB} = B', \quad \frac{d\Theta}{dC} = C', \quad t = \frac{d\Theta}{dk} + t',$$

A', B', C' being new arbitrary constants.

Liouville introduces in place of x, y, z , the elliptic coordinates ρ, μ, ν , which are such that

$$\frac{x^2}{a^2 + \rho} + \frac{y^2}{b^2 + \rho} + \frac{z^2}{c^2 + \rho} = 1, \quad \frac{x^2}{a^2 + \mu} + \frac{y^2}{b^2 + \mu} + \frac{z^2}{c^2 + \mu} = 1, \quad \frac{x^2}{a^2 + \nu} + \frac{y^2}{b^2 + \nu} + \frac{z^2}{c^2 + \nu} = 1,$$

or, what is the same thing,

$$x^2 = \frac{(a^2 + \rho)(a^2 + \mu)(a^2 + \nu)}{(b^2 - a^2)(c^2 - a^2)}, \quad y^2 = \frac{(b^2 + \rho)(b^2 + \mu)(b^2 + \nu)}{(c^2 - b^2)(a^2 - b^2)}, \quad z^2 = \frac{(c^2 + \rho)(c^2 + \mu)(c^2 + \nu)}{(a^2 - c^2)(b^2 - c^2)},$$

and he then finds that the resulting partial differential equation in ρ, μ, ν may be integrated provided that U is of the form

$$U = \frac{f\rho}{(\rho - \mu)(\rho - \nu)} + \frac{F\mu}{(\mu - \nu)(\mu - \rho)} + \frac{\omega\nu}{(\nu - \rho)(\nu - \mu)},$$

f, F, ω being any functional symbols whatever: viz., the expression for Θ is

$$\Theta = \int \frac{d\rho}{\sqrt{2f\rho + A + B\rho^2 + 2C\rho^3}} + \int \frac{d\mu}{\sqrt{2F\mu + A + B\mu^2 + 2C\mu^3}} + \int \frac{d\nu}{\sqrt{2\omega\nu + A + B\nu^2 + 2C\nu^3}},$$

In the case where $U = 0$ we have a particle not acted on by any forces, and the path is of course a straight line. The peculiar form in which these equations are obtained leads to very interesting results in regard to the theory of Abelian integrals, and to that of the geodesic lines of an ellipsoid.

The formulæ require to be modified in certain cases, such as $c = b$ or $b = 0$. The case $b = 0$ leads to the theory developed in the first memoir in relation to the problem of two centres. The case is indicated where $b = 0, c = 0$, the ratio $b : c$ remaining finite. The case is briefly considered of a particle moving on a given surface.

105. The third memoir purports to relate to a system of particles, but the formulae are exhibited under a purely analytical point of view; so much so, that the coordinates of the points (3 for each point) are considered as forming a single system of variables $x_1, x_2, \dots, x_n$. The partial differential equation is

$$\left(\frac{d\Theta}{dx_1}\right)^2 + \left(\frac{d\Theta}{dx_2}\right)^2 + \dots + \left(\frac{d\Theta}{dx_n}\right)^2 = 2U + k,$$

which is transformed by introducing therein the new variables $\mu_1, \mu_2, \dots, \mu_n$ analogous to the elliptic coordinates of the second memoir. The memoir really belongs rather to the theory of the Abelian integrals (in regard to which it appears to be a very valuable one) than to dynamics.

Memoirs by Jacobi, Bertrand, and Donkin, relating to various Special Problems

106. I have inserted this heading for the sake of showing at a single view what are the special problems incidentally considered in the undermentioned memoirs which are referred to in several places in the present Report.

107. Jacobi, "De Motu puncti singularis" (1842).—I call to mind that the memoir chiefly depends on the theorem of the Ultimate Multiplier (the theory in its generality being developed in the later memoir "Theoria Novi Multiplicatoris &c.," 1844—45). § 4 is entitled "The motion of a point on the surface of revolution," which, the principle of the conservation of areas holding good, is reduced to the problem of the motion on the meridian curve, and thus depends upon quadratures only. § 5 is entitled "On the motion of a point about a fixed centre attracted according to a certain law more general than the Newtonian one" (anti, No. 86). § 6, "On the motion of a point on a given curve and in a resisting medium" (resistance = $a + bv^2$, or = $a + bv^3$); and § 7, "On the Ballistic Curve," viz., the forces are gravity and a resistance = $a + bv^2$.

108. In Jacobi's memoir "Theoria Novi Multiplicatoris &c." (1845), § 25 is entitled "On the motion of a point attracted towards a fixed centre" (see ante, No 87); § 26, "On the motion of a point attracted towards two fixed centres according to the Newtonian law" (ante, No. 56); § 27, "On the rotation of a solid body about a fixed point" (post. No. 193); § 28, "On the problem of three bodies moving in a right line; the Eulerian substitution; theorems on homogeneous forces" (ante. No. 91); and § 29, "The principle of the ultimate multiplier applied to a free system of material points moving in a resisting medium; on the motion of a comet in a resisting medium about the sun."

109. And in Jacobi's memoir "Nova Methodus &c." (1802), besides § 64 and § 65, which are applications of the method to general dynamical theorems, we have § 66, containing a simultaneous soluti-

on of the problem of the motion of a point attracted to a fixed centre and of that of the rotation of a solid body (post. No. 206) and § 67, relating to the motion of a point attracted to a fixed centre according to the Newtonian law.

110. Bertrand's "Mémoire sur les intégrales différentielles de la Mécanique" (1852).—§ III. relates to the motion of a point attracted to a fixed centre by a force varying as a function of the distance; § IV. to the case where the forces arise from a force-function $U = \frac{\Phi(\frac{y}{x})}{r^2}$ [or, what is the same thing, $= \frac{\Phi}{r^2}$] (anth. No. 87); § V. to the problem of two centres (anth. No. 62), and § VI. to the problem of three bodies (post. No. 117).

111. Donkin's memoir "On a Class of Differential Equations &c." (1855). Part I. Nos. 27 to 30 relate to the problem of central forces (in space), No. 31 to the rotation of a solid body, and § III. to the same subject, viz. Nos. 40 and 41 to the general case, Nos. 42 to 44 to the particular case $A = B$; and Nos. 45 to 48 to the reduction thereto of the general case by treating the forces which arise from the inequality of A and B as disturbing forces. Part II. Nos. 59 and 60 relate to the spherical pendulum; Nos. 72 and 73 to "Transformation from fixed to moving axes of coordinates," say to Relative Motion; and Nos. 84 to 96 to the problem of three bodies (post. No. 120).

The Problem of Three Bodies

ARTICLE NOS. 112 TO 123.

112. A system of differential equations, such as

$$\frac{dx_1}{X_1} = \frac{dx_2}{X_2} = \dots = \frac{dx_{n+1}}{X_{n+1}}$$

(n equations between $n + 1$ variables), may be termed a system of the n th order, or more simply a system of n equations. Let $(u_1, u_2, \dots, u_{n+1})$ be any functions of the original variables $(x_1, x_2, \dots, x_{n+1})$, the system may be transformed into the similar system

$$\frac{du_1}{U_1} = \frac{du_2}{U_2} = \dots = \frac{du_{n+1}}{U_{n+1}};$$

and if it happens that we have e.g. U_1 identically equal to zero, then

the system becomes

$$\frac{du_2}{U_2} = \frac{du_3}{U_3} = \cdots = \frac{du_{n+1}}{U_{n+1}},$$

so that we have an integral $u_1 = c$, and then in the remaining equations substituting this value, or treating it, as constant, the system is reduced to one of $n - 1$ equations.

Or again, if it happen that we have in the transformed system m equations ($m < n$), say

$$\frac{du_1}{U_1} = \frac{du_2}{U_2} = \cdots = \frac{du_{m+1}}{U_{m+1}},$$

which are such that $U_1, U_2, \dots, U_{m+1}$ are functions of only the $m + 1$ variables $u_1, u_2, \dots, u_{m+1}$, then the integration of the proposed system of n equations depends on the integration in the first instance of a system of m equations. It is to be observed that if the system of m equations can be integrated, then the completion of the integration of the original system depends on the integration of a system of $n - m$ equations, and in this sense the original system of n equations may be said to be broken up into two systems of m equations and $n - m$ equations respectively: but *non constat* that the system of m equations admits of integration; and it is therefore more correct to say that, from the original system of the n equations, there has been separated off a system of m equations.

113. The bearing of the foregoing remarks on the problem of three bodies will presently appear. It will be seen that whereas the problem as it stood before Jacobi depends on a system of seven equations, it has been shown by him that there may be separated off from this a system of six equations.

114. Jacobi's memoir "Sur l'élimination des Noeuds &c." (1843).—The problem of the motion of three mutually attracting bodies is in the first instance reduced to that of the motion of two fictitious bodies (which may be considered as mutually attracting bodies, attracted by a fixed centre of force)⁽¹⁾.

In fact, in the original problem the centre of gravity of the three bodies moves uniformly in a right line, and it may without any real

¹This is the effect of Jacobi's reduction; but the explicit statement of the theorem, and actual replacement of the problem of the three bodies by that of the two bodies attracted to a fixed centre, is due to Bertrand (post, No. 117).

loss of generality be taken to be at rest; that is, if the ξ -coordinates of the three bodies are ξ_1, ξ_2, ξ_3 , then $m_1\xi_1 + m_2\xi_2 + m_3\xi_3 = 0$, or ξ_1, ξ_2, ξ_3 may be taken to be linear functions of two quantities x_1 and x_2 . And similarly for the y -coordinates and the z -coordinates respectively. And $(x_1, y_1, z_1), (x_2, y_2, z_2)$ may be regarded as the coordinates of two bodies revolving about a fixed centre of force. Hence representing the differential coefficients in regard to the time by x'_1 , &c., and treating these as new variables, the equations of motion will assume the form

$$\frac{dx_1}{X_1} = \frac{dy_1}{Y_1} = \frac{dz_1}{Z_1} = \frac{dx_2}{X_2} = \frac{dy_2}{Y_2} = \frac{dz_2}{Z_2} = \frac{dx'_1}{X'_1} = \frac{dy'_1}{Y'_1} = \frac{dz'_1}{Z'_1} = \frac{dx'_2}{X'_2} = \frac{dy'_2}{Y'_2} = \frac{dz'_2}{Z'_2},$$

where $X_1, Y_1, Z_1, X_2, Y_2, Z_2$ are forces capable of representation by means of a force-function U . This is a system of twelve equations; but since $X_1, Y_1, Z_1, X_2, Y_2, Z_2$ are independent of the time, we may omit the equation ($= dt$), and treat the system as one of eleven equations between the variables $x_1, y_1, z_1, x_2, y_2, z_2, x'_1, y'_1, z'_1, x'_2, y'_2, z'_2$: if this system were integrated, the determination of the time would then depend on a quadrature only. But for the system of eleven equations we have four integrals, viz., the integral of Vis Viva and the three integrals of areas, and the system is thus reducible to one of $(11 - 4 =)$ seven equations. This preliminary transformation in Jacobi's memoir explains the remark that the problem, as it stood before him, depended on a system of seven equations.

115. Jacobi remarks that in the transformed problem the three integrals of areas show (1) that the intersection of the planes of the orbits of the two bodies lie in a fixed plane, the invariable plane of the system; (2) that the inclinations of the planes of the two orbits to this fixed plane, and the parameters of the two orbits considered as variable ellipses, are four elements any two of which rigorously determine the two others.

And then choosing for variables the inclinations of the two orbits to the invariable plane, the two radius vectors, the angles which they form with the intersection of the planes of the two orbits, and lastly the angle between this intersection (being as already mentioned a line in the invariable plane) with a fixed line in the invariable plane, he finds that the last-mentioned angle entirely disappears from the system of differential equations, and is determined after their integration by a quadrature.

In this new form of the differential equations there is no trace of the nodes. The differential equations which determine the relative

motion of the three bodies are reduced to five equations of the first order and one of the second order. The equations in question are the equations I. to VI. given at the end of the memoir.

It is to be remarked that dt is not eliminated from these equations; the last of them is $\frac{d}{dt}(\mu r^2 + \mu_1 r_1^2) = 2U - 2h$; and if to reduce them to a system of equations of the first order we write $\frac{d}{dt}(\mu r^2 + \mu_1 r_1^2) = \theta$, and therefore $\frac{d\theta}{dt} = 2U - 2h$, the system may be presented in the form

$$\frac{du}{U} = \frac{du_1}{U_1} = \frac{di}{I} = \frac{di_1}{I_1} = \frac{dr}{R} = \frac{dr_1}{R_1} = \frac{d\theta}{\Theta} = \frac{dt}{1},$$

which if we do, and then omit the equation ($= dt$), we have a system of six equations between the seven quantities $u, u_1, i, i_1, r, r_1, \theta$; when this is integrated, the equation ($= dt$) gives the time by a quadrature; and finally, Jacobi's equation VII. ($d\Omega = \tan u \cdot dt$) gives by a quadrature the angle before referred to as disappearing from the system of equations I. to VI.

116. But when Jacobi says, "Par suite on a fait cinq intégrations. Les intégrales connues n'étant qu'au nombre de quatre, on pourra donc dire que l'on a fait une intégration de plus dans le système du monde. Je dis dans le système du monde puisque la même méthode s'applique à un nombre quelconque de corps," the language used is not, I think, quite accurate. It in fact appears from the memoir that it is only on the assumption of the integration of the system of six equations that, besides the integral of Vis Viva and the integrals of areas, the remaining two integrals are known; in fact after, but not before, the system of the order six has been integrated, the time t and the angle Ω are each of them given by a quadrature.

117. Bertrand's "Mémoire sur l'intégration des Équations différentielles de la Mécanique" (1852).—I have spoken of this memoir in No. 56 of my former Report. The course of investigation is the inquiry as to the integrals, which, combined according to Poisson's theorem with the integral of Vis Viva or any other given integral, give rise to an illusory result. But as regards the application made to the problem of three bodies, it will be more convenient to state from a different point of view the conclusions arrived at: and I may mention that when the author says "Je parviens . . . à réduire la question à l'intégration de six équations toutes du premier ordre, c'est-à-dire que j'effectue une intégration de plus que ne l'avait fait Jacobi," he seems to have overlooked that Jacobi's system of five equations of the first order and one of the second order really is, as above noticed, a

system of the six equations with another equation which then gives the time by a quadrature, and that, at least as appears to me, he has not advanced the solution beyond the point to which it had been carried by Jacobi⁽¹⁾.

118. Presenting Bertrand's results in the slightly different notation in which they are reproduced in Bour's memoir (post. No. 122), then if (x, y, z) , (x_1, y_1, z_1) are the coordinates of the two bodies (the problem actually considered being, as with Jacobi, that of the motion of two bodies about a fixed centre of force), and representing the functions $x^2 + y^2 + z^2$, $x_1^2 + y_1^2 + z_1^2$, $m^2(x'^2 + y'^2 + z'^2)$, $m_1^2(x_1'^2 + y_1'^2 + z_1'^2)$, $m(xx' + yy' + zz')$, $m_1(x_1x_1' + y_1y_1' + z_1z_1')$, $m(xx_1 + yy_1 + zz_1)$, $(xx_1 + yy_1 + zz_1)^2$, $m^2m_1^2(x'x_1' + y'y_1' + z'z_1')$, $(xx_1 + yy_1 + zz_1)(x'x_1' + y'y_1' + z'z_1')$ by u , u_1 , v , v_1 , w , w_1 , r , r_1 , q , s respectively, then the last-mentioned quantities are connected by a single geometrical relation, so that any one of them, say s , may be considered as a given function of the remaining nine. And the author in effect shows that the equations of motion give a system

$$\frac{du}{U} = \frac{du_1}{U_1} = \frac{dv}{V} = \frac{dv_1}{V_1} = \frac{dw}{W} = \frac{dw_1}{W_1} = \frac{dr}{R} = \frac{dr_1}{R_1} = \frac{dq}{Q} = \frac{ds}{S},$$

where U , U_1 , &c. are functions of the quantities u , u_1 , v , &c. Omitting from the system the equation $(= dt)$, there are eight equations between nine quantities; but there are two known integrals, viz., the integral of Vis Viva and the integral of principal moment (or sum of the squares of the integrals of areas); that is to say, the system is really a system of six equations.

119. Painvin, "Recherche du dernier Multiplicateur &c." (1854).—The author investigates the ultimate multiplier for two systems of differential equations:

1°. The system of the equations I. to VI. in Jacobi's memoir "Sur l'élimination des Noeuds &c." (anti, No. 114). Writing in the equations $r_1 = r'$, $\frac{dr}{dt} = r''$, and treating r' , r'' as new variables, the system may be written in the form

$$\frac{du}{U} = \frac{du_1}{U_1} = \frac{di}{I} = \frac{di_1}{I_1} = \frac{dr}{R} = \frac{dr_1}{R_1} = \frac{dr'}{R'} = \frac{dr''}{R''} = dt,$$

which, omitting the equation $(= dt)$, is a system of seven equations between eight variables; and it is for this form of the system that

¹These remarks were communicated by me to M. Bertrand—see my letter "Sur l'intégration des Équations différentielles de la Mécanique," *Comptes Rendus* (1863)—and, in the answer he kindly sent me, he agrees that they are correct.

the value of M is determined, the result obtained being the simple and elegant one, $M = \frac{\sin i \sin i_1}{\theta}$. The system of seven equations has an integral which is in fact the equation V. of the system in Jacobi's form, so that it is really a system of six equations (anti, No. 115).

2°. The system secondly discussed is Bertrand's system of nine equations (ante, No. 118). The multiplier M is obtained under four different forms, $M = \frac{1}{\sqrt{BB'-AC}} = \frac{1}{\sqrt{AA'}} = \frac{1}{\sqrt{AZ+B}} = \frac{1}{mn}$ (I do not stop to explain the notation), the last of them being referred to as a result announced by M. Bertrand in his course. But it is shown by M. Bour in the memoir next referred to (post. No. 122), that the multiplier for the system in question can be obtained in a very much more simple manner, almost without calculation.

120. In connexion with Jacobi's theory of the elimination of the Nodes, I may refer to the investigations "Application to the Problem of three Bodies," Nos. 84 to 96 of Donkin's memoir "On a Class of Differential Equations &c." Part II. The author remarks that his differential equations No. 93 afford an example of the so-called elimination of the Nodes, quite different however (in that they are merely transformations of the original differential equations of the problem without any integrations) from that effected by Jacobi.

121. It may be right to refer again in this place to the concluding part of § 28 of Jacobi's memoir "Nova Theoria Multiplicatoris &c." (ante. No. 92), as bearing on the problem of three bodies.

122. Bour's "Mémoire sur le Problème des Trois Corps" (1856).—The author remarks that Bertrand's system of equations have lost the remarkable form and the properties which characterize the ordinary equations for the solution of a dynamical problem. But by selecting eight new variables, functions of Bertrand's variables, the system may be brought back to the standard Hamiltonian form

$$\frac{dq_i}{dt} = \frac{dH}{dp_i}, \quad \frac{dp_i}{dt} = -\frac{dH}{dq_i},$$

or to the form adopted by M. Bour, of a partial differential equation

$$\Sigma \left(\frac{dH}{dp_i} \frac{dC}{dq_i} - \frac{dH}{dq_i} \frac{dC}{dp_i} \right) = 0,$$

and guiding himself by a theorem in relation to canonical integrals obtained in his memoir of 1855 (see No. 66 of my former Report), he finds by a somewhat intricate analysis the expressions of the

eight new variables $p_1, p_2, p_3, p_4, q_1, q_2, q_3, q_4$. The results ultimately obtained are of a very remarkable and interesting form, viz. $H = \text{funct.}(p_1, p_2, p_3, p_4, q_1, q_2, q_3, q_4)$ is equal to the value it would have for motion in a plane, *plus* a term admitting of a simple geometrical interpretation; and he thus arrives at the following theorem as a résumé of the whole memoir, viz.,

“In order to integrate in the general case the problem of three bodies, it is sufficient to solve the case of motion in a plane, and then to take account of a disturbing function equal to the product of a constant depending on the areas by the sum of the moments of inertia of the bodies round a certain axis, divided by the square of the triangle formed by the three bodies.”

123. It may be remarked that the only given integral of the system of eight equations is the integral of Vis Viva, $H = \text{const.}$, and that using this equation to eliminate one of the variables, and omitting the equation ($= dt$), we have, as in the solutions of Jacobi and Bertrand, a system of six equations between seven variables. As the equations are in the standard dynamical form, no investigation is needed of the multiplier M , which is given by Jacobi's general theory, and consequently when any five integrals of the six equations are given, the remaining integral can be obtained by a quadrature.

In the case of three bodies moving in a plane, the solution takes a very simple form, which is given in the concluding paragraph of the memoir.

Transformation of Coordinates

ARTICLE NOS. 124 TO 141.

124. It may be convenient to remark at once that two sets of rectangular coordinates may be related to each other properly or improperly, viz., the axes to which they belong (considered as drawn from the origin in the positive directions) may be either capable, or else incapable, of being brought into coincidence. The latter relation, although of equal generality with the former one, may for the most part be disregarded; for by merely reversing the directions of the one set of axes, the improper is converted into the proper relation.

125. In the memoir “*Problema Algebraicum &c.*” (1770) Euler proposes to himself the question “*Invenire novem numeros ita in quadratum disponendos ut satisfiat duodecem sequentibus conditionibus*

&c.", viz., substituting for A, B, C , &c. the ordinary letters

$$\begin{array}{ccc} \alpha, & \beta, & \gamma, \\ \alpha', & \beta', & \gamma', \\ \alpha'', & \beta'', & \gamma'', \end{array}$$

the twelve conditions are

$$\begin{array}{lll} \alpha^2 + \alpha'^2 + \alpha''^2 = 1, & \beta^2 + \beta'^2 + \beta''^2 = 1, & \gamma^2 + \gamma'^2 + \gamma''^2 = 1, \\ \alpha'^2 + \beta'^2 + \gamma'^2 = 1, & \alpha''^2 + \beta''^2 + \gamma''^2 = 1, & \alpha^2 + \beta^2 + \gamma^2 = 1, \\ \alpha\alpha' + \beta\beta' + \gamma\gamma' = 0, & \alpha'\alpha'' + \beta'\beta'' + \gamma'\gamma'' = 0, & \alpha\alpha'' + \beta\beta'' + \gamma\gamma'' = 0, \\ \alpha\alpha'' + \beta'\alpha'' + \gamma'\gamma'' = 0, & \alpha\alpha' + \beta''\beta' + \gamma''\gamma' = 0, & \alpha\beta + \alpha'\beta' + \alpha''\beta'' = 0. \end{array}$$

And he remarks that this is in fact the problem of the transformation of coordinates, viz., if we have

$$X = \alpha x + \beta y + \gamma z, \quad Y = \alpha' x + \beta' y + \gamma' z, \quad Z = \alpha'' x + \beta'' y + \gamma'' z,$$

then the first equations are such as to give identically

$$X^2 + Y^2 + Z^2 = x^2 + y^2 + z^2.$$

126. Assuming the first six equations, he shows by a direct analytical process that $0 = (\beta'\gamma'' - \beta''\gamma')^2$, or $\alpha = \pm(\beta'\gamma'' - \beta''\gamma')$; and taking the positive sign (for, as the numbers may be taken as well positively as negatively, there is nothing lost by doing so) $\alpha = \beta'\gamma'' - \beta''\gamma'$, which gives the system

$$\begin{array}{lll} \alpha = \beta'\gamma'' - \beta''\gamma', & \beta = \gamma'\alpha'' - \gamma''\alpha', & \gamma = \alpha'\beta'' - \alpha''\beta', \\ \alpha' = \beta''\gamma - \beta\gamma'', & \beta' = \gamma''\alpha - \gamma\alpha'', & \gamma' = \alpha''\beta - \alpha\beta'', \\ \alpha'' = \beta\gamma' - \beta'\gamma, & \beta'' = \gamma\alpha' - \gamma'\alpha, & \gamma'' = \alpha\beta' - \alpha'\beta, \end{array}$$

and from these he deduces the second system of six equations.

The inverse system of equations

$$X = \alpha x + \alpha' y + \alpha'' z, \quad Y = \beta x + \beta' y + \beta'' z, \quad Z = \gamma x + \gamma' y + \gamma'' z$$

is not explicitly referred to.

127. He then satisfies the equations by means of trigonometrical substitutions, viz., assuming $\alpha = \cos \xi$, then $\alpha'^2 + \alpha''^2 = \sin^2 \xi$, which is satisfied by $\alpha' = \sin \xi \cos \eta$, $\alpha'' = \sin \xi \sin \eta$, &c., and he thus obtains for the coefficients a set of values involving the angles ξ, η ,

θ , which are the same as those mentioned post, No. 130. And he shows how these formulæ may be obtained geometrically by three successive transformations of two coordinates only. The remainder of the memoir relates to the analogous problem of the transformation of four or more coordinates.

128. I have analysed so much of Euler's memoir in order to show that it contains nearly the whole of the ordinary theory of the transformation of coordinates; the only addition required is the equation

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{vmatrix} = \pm 1,$$

where the sign + gives $\alpha = \beta'\gamma'' - \beta''\gamma'$, &c. (ut supra), but the sign - would give $\alpha = -(\beta'\gamma'' - \beta''\gamma')$, &c.

129. The distinction of the ambiguous sign is in fact the above-mentioned one of the proper and improper transformations; viz., for the sign + the two sets of axes can, for the sign - they cannot, be brought into coincidence: this very important remark was, I believe, first made by Jacobi in one of his early memoirs in Crelle's Journal, but I have lost the reference. As already mentioned, it is allowable to attend only to the proper transformation, and to consider the value of the determinant as being = +1; and this is in fact almost always done.

130. Euler's formulæ involving the three angles are those which are ordinarily made use of in the problem of rotation and the problems of physical astronomy generally. It is convenient to take them as in the figure, viz. θ , the longitude of node, ϕ , the inclination, τ , the angular distance of X from node, and the formulæ of transformation then are

	X	Y	Z
x	$\cos \tau \cos \theta - \sin \tau \sin \theta \cos \phi$	$\cos \tau \sin \theta + \sin \tau \cos \theta \cos \phi$	$\sin \tau \sin \phi$
y	$-\sin \tau \cos \theta - \cos \tau \sin \theta \cos \phi$	$-\sin \tau \sin \theta + \cos \tau \cos \theta \cos \phi$	$\cos \tau \sin \phi$
z	$\sin \theta \sin \phi$	$-\cos \theta \sin \phi$	$\cos \phi$

The foregoing very convenient algorithm, viz. the employment of

	X	Y	Z
x	α	β	γ
y	α'	β'	γ'
z	α''	β''	γ''

to denote the system of equations

$$X = \alpha x + \beta y + \gamma z, \quad \&c.,$$

is due to M. Lamé in the memoir “Du Mouvement de Rotation &c.,” *Mem. de Berlin* for 1758 (pr. 1765).

131. But previously to the foregoing investigations, viz., Euler had obtained incidentally a very elegant solution of the problem of the transformation of coordinates; this is in fact identical with the next mentioned one, the letters $l, m, n; \lambda, \mu, \nu$ being used in the place of $\xi, \xi', \xi''; \eta, \eta', \eta''$.

132. In the memoir “Formulae generales pro translatione &c.” (1775), Euler gives the following formulæ for the transformation of coordinates, viz. if the position of the set of axes XYZ in reference to the set xyz is determined by

$$\begin{aligned} \widehat{xX}, \widehat{yX}, \widehat{zX} &= 90^\circ - \xi, \quad 90^\circ - \xi', \quad 90^\circ - \xi'', \\ \widehat{yXx}, \widehat{yXy}, \widehat{yXz} &= \eta, \quad \eta', \quad \eta'', \end{aligned}$$

then the formulæ of transformation are

	X	Y	Z
x	$\sin \xi$	$\cos \xi \sin \eta$	$\cos \xi \cos \eta$
y	$\sin \xi'$	$\cos \xi' \sin \eta'$	$\cos \xi' \cos \eta'$
z	$\sin \xi''$	$\cos \xi'' \sin \eta''$	$\cos \xi'' \cos \eta''$

with the following equations connecting the six angles, viz. if

$$-A^2 = \cos(\eta' - \eta'') \cos(\eta'' - \eta) \cos(\eta - \eta'),$$

then

$$\tan \xi = \frac{-A}{\cos(\eta'' - \eta)}, \quad \tan \xi' = \frac{-A}{\cos(\eta - \eta')}, \quad \tan \xi'' = \frac{-A}{\cos(\eta' - \eta'')}.$$

133. It is right to notice that these values of ξ, ξ', ξ'' give the twelve equations $\alpha^2 + \beta^2 + \gamma^2 = 1$, &c., but they do not give definitely $\alpha = \beta'\gamma'' - \beta''\gamma'$, &c., but only $\alpha = \pm(\beta'\gamma'' - \beta''\gamma')$; that is, in the formulæ in question the two sets of axes are not of necessity displacements the one of the other. In the same memoir Euler considers two sets of rectangular axes, and assuming that they are displacements the one of the other (this assumption is not made as explicitly as it

should have been), he remarks that the one set may be made to coincide with the other set by means of a finite rotation about a certain axis (which may conveniently be termed the Resultant Axis). This consideration leads him to an equation which ought to $= 0$, be satisfied by the coefficients of transformation, but which he is not able to verify by means of the foregoing expressions in terms of $\xi, \xi', \xi'', \eta, \eta', \eta''$.

134. I remark that Euler's equation in fact is

$$\begin{vmatrix} \alpha - 1 & \beta & \gamma \\ \alpha' & \beta' - 1 & \gamma' \\ \alpha'' & \beta'' & \gamma'' - 1 \end{vmatrix} = 0,$$

or, as it may be written,

$$-(\beta'\gamma'' - \beta''\gamma') - (\gamma''\alpha - \gamma'\alpha'') - (\alpha\beta' - \alpha'\beta) + \alpha + \beta' + \gamma'' - 1 = 0,$$

in which form it is an immediate consequence of the equations

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{vmatrix} = 1, \quad \alpha = \beta'\gamma'' - \beta''\gamma', \quad \&c.,$$

which are true for the proper, but not for the improper transformation.

135. In the undated addition to the memoir, Euler states the theorem of the resultant axis as follows:—"Theorema, Quomodocunque sphæra circa centrum suum convertatur, semper assignari potest diameter cujus directio in situ translato conveniat cum situ originali," and he again endeavours to obtain a verification of the foregoing analytical theorem.

136. The theory of the Resultant Axis was further developed by Euler in the memoir "Nova Methodus Motum &c." (1775), and by Lexell in the memoir "Nonnulla theoremata generalia &c." (1775): the geometrical investigations are given more completely and in greater detail in Lexell's memoir. The result is contained in the following system of formulæ for the transformation of coordinates, viz., if α, β, γ are the inclinations of the resultant axis to the original set, and if ϕ is the rotation about the resultant axis, or say the resultant rotation, then we have

	X	Y	
x	$\cos^2 \alpha + \sin^2 \alpha \cos \phi$	$\cos \beta \cos \alpha (1 - \cos \phi) - \cos \gamma \sin \phi$	$\cos \gamma \cos \alpha$
y	$\cos \alpha \cos \beta (1 - \cos \phi) + \cos \gamma \sin \phi$	$\cos^2 \beta + \sin^2 \beta \cos \phi$	$\cos \gamma \cos \beta$
z	$\cos \alpha \cos \gamma (1 - \cos \phi) - \cos \beta \sin \phi$	$\cos \beta \cos \gamma (1 - \cos \phi) + \cos \alpha \sin \phi$	$\cos \alpha \cos \gamma$

Euler attempts, but not very successfully, to apply the formulæ to the dynamical problem of the rotation of a solid body: he does not introduce them into the differential equations, but only into the integral ones, and his results are complicated and inelegant. The further simplification effected by Rodrigues was in fact required.

137. Jacobi's paper, "Euleri formulæ&c." (1827), merely cites the last-mentioned result.

138. I find it stated in Lacroix's *Differential Calculus*, t. I. p. 533, that the following system for the transformation of coordinates was obtained by Monge (no reference is given in Lacroix), viz., the system being as above,

$$\begin{array}{ccc} \alpha, & \beta, & \gamma, \\ \alpha', & \beta', & \gamma', \\ \alpha'', & \beta'', & \gamma'', \end{array}$$

and the quantities α, β', γ'' being arbitrary, then putting

$$\begin{aligned} 1 + \alpha + \beta' + \gamma'' &= M, \\ 1 + \alpha - \beta' - \gamma'' &= N, \\ 1 - \alpha + \beta' - \gamma'' &= P, \\ 1 - \alpha - \beta' + \gamma'' &= Q, \end{aligned}$$

so that $M + N + P + Q = 4$, we have

$$\begin{aligned} 2\beta &= \sqrt{NP} + \sqrt{MQ}, & 2\gamma &= \sqrt{MP} + \sqrt{NQ}, & 2\alpha'' &= \sqrt{QN} + \sqrt{MP}, \\ 2\beta' &= \sqrt{NP} - \sqrt{MQ}, & 2\gamma' &= \sqrt{PQ} - \sqrt{MN}, & 2\alpha' &= \sqrt{QN} - \sqrt{MP}. \end{aligned}$$

These are formulæ very closely connected with those of Rodrigues.

139. The theory was perfected by Rodrigues in the valuable memoir "Des lois géométriques &c." (1840). Using for greater convenience λ, μ, ν in the place of his $\frac{1}{2}\rho m, \frac{1}{2}\rho n, \frac{1}{2}\rho p$, he in effect writes

$$\tan \frac{1}{2}\phi \cos \alpha = \lambda, \quad \tan \frac{1}{2}\phi \cos \beta = \mu, \quad \tan \frac{1}{2}\phi \cos \gamma = \nu,$$

and this being so, the coefficients of transformation are

$$\begin{array}{ccc} \frac{1 + \lambda^2 - \mu^2 - \nu^2}{1 + \lambda^2 + \mu^2 + \nu^2}, & \frac{2(\nu + \mu\lambda)}{1 + \lambda^2 + \mu^2 + \nu^2}, & \frac{2(\mu\nu - \lambda)}{1 + \lambda^2 + \mu^2 + \nu^2}, \\ \frac{2(\lambda\mu - \nu)}{1 + \lambda^2 + \mu^2 + \nu^2}, & \frac{1 - \lambda^2 + \mu^2 - \nu^2}{1 + \lambda^2 + \mu^2 + \nu^2}, & \frac{2(\lambda\nu + \mu)}{1 + \lambda^2 + \mu^2 + \nu^2}, \\ \frac{2(\lambda\nu + \mu)}{1 + \lambda^2 + \mu^2 + \nu^2}, & \frac{2(\mu\nu - \lambda)}{1 + \lambda^2 + \mu^2 + \nu^2}, & \frac{1 - \lambda^2 - \mu^2 + \nu^2}{1 + \lambda^2 + \mu^2 + \nu^2}, \end{array}$$

all divided by the common denominator $1 + \lambda^2 + \mu^2 + \nu^2$.

Conversely, if the coefficients of transformation are as usual represented by

$$\begin{array}{ccc} \alpha, & \beta, & \gamma, \\ \alpha', & \beta', & \gamma', \\ \alpha'', & \beta'', & \gamma'', \end{array}$$

then $\lambda, \mu, \nu; \lambda, \mu, \nu$ are respectively equal to

$$\frac{\gamma - \beta''}{1 + \alpha + \beta' + \gamma''}, \quad \frac{\alpha'' - \gamma}{1 + \alpha + \beta' + \gamma''}, \quad \frac{\beta - \alpha'}{1 + \alpha + \beta' + \gamma''},$$

each of them divided by $1 + \alpha + \beta' + \gamma''$.

The memoir contains very elegant formulæ for the composition of finite rotations, and it will be again referred to in speaking of the kinematics of a solid body.

140. Sir W. R. Hamilton's first papers on the theory of quaternions were published in the years 1843 and 1844: the fundamental idea consists in the employment of the imaginaries i, j, k , which are such that

$$i^2 = j^2 = k^2 = -1, \quad jk = -kj = i, \quad ki = -ik = j, \quad ij = -ji = k,$$

whence also

$$\begin{aligned} & (w + ix + jy + kz)(w' + ix' + jy' + kz') \\ &= ww' - xx' - yy' - zz' \\ & \quad + i(wx' + w'x + yz' - y'z) \\ & \quad + j(wy' + w'y + zx' - z'x) \\ & \quad + k(wz' + w'z + xy' - x'y); \end{aligned}$$

so that representing the right-hand side by $W + iX + jY + kZ$, we have identically

$$W^2 + X^2 + Y^2 + Z^2 = (w^2 + x^2 + y^2 + z^2)(w'^2 + x'^2 + y'^2 + z'^2).$$

It is hardly necessary to remark that Sir W. R. Hamilton in his various publications on the subject, and in the *Lectures on Quaternions*, Dublin, 1853, has developed the theory in detail, and has made the most interesting applications of it to geometrical and dynamical questions; and although the first explicit application of it to the present question may have been made in my own paper next referred to, it

seems clear that the whole theory was in its original conception intimately connected with the notion of rotation.

141. Cayley, "On certain Results relating to Quaternions" (1845).—It is shown that Rodrigues' transformation formula may be expressed in a very simple manner by means of quaternions; viz., we have

$$ix + jy + kz = (1 + i\lambda + j\mu + k\nu)^{-\frac{1}{2}}(iX + jY + kZ)(1 + i\lambda + j\mu + k\nu)^{\frac{1}{2}},$$

where developing the function on the right-hand side, and equating the coefficients of i, j, k , we obtain the formulæ in question. A subsequent paper, Cayley, "On the application of Quaternions to the Theory of Rotation" (1848), relates to the composition of rotations.

Principal Axes, and Moments of Inertia

ARTICLE NOS. 142—163.

142. The theorem of principal axes consists herein, that at any point of a solid body there exists a system of axes Ox, Oy, Oz , such that

$$\int yz \, dm = 0, \quad \int zx \, dm = 0, \quad \int xy \, dm = 0.$$

But this, the original form of the theorem, is a mere deduction from a general theory of the representation of the integrals

$$\int x^2 \, dm, \quad \int y^2 \, dm, \quad \int z^2 \, dm, \quad \int yz \, dm, \quad \int zx \, dm, \quad \int xy \, dm$$

for any axes through the given origin by means of an ellipsoid depending on the values of these integrals corresponding to a given set of rectangular axes through the same origin.

143. If, for convenience, we write as follows, $M = \int dm$ the mass of the body, and

$$A' = \int x^2 \, dm, \quad B' = \int y^2 \, dm, \quad C' = \int z^2 \, dm, \quad F' = \int yz \, dm, \quad G' = \int zx \, dm$$

and moreover

$$A = \int (y^2 + z^2) \, dm, \quad B = \int (z^2 + x^2) \, dm, \quad C = \int (x^2 + y^2) \, dm, \\ F = - \int yz \, dm, \quad G = - \int zx \, dm, \quad H = - \int xy \, dm,$$

so that

$$A = B' + C', \quad B = C' + A', \quad C = A' + B', \quad F = -F', \quad G = -G', \quad H = -H',$$

then the ellipsoid which in the first instance presents itself for this purpose, and which Prof. Price has termed the Ellipsoid of Principal Axes, but which I would rather term the “Comomental Ellipsoid,” is the ellipsoid

$$(A', B', C', F', G', H' \wedge x, y, z)^2 = Mk^2,$$

where k is arbitrary, so that the absolute magnitude is not determined. But it is more usual, and in some respects better to consider in place thereof the “Momental Ellipsoid” (Cauchy, “Sur les Moments d’Inertie,” *Exercices de Mathématique*, t. II. pp. 93—103, 1827),

$$(A, B, C, F, G, H \wedge x, y, z)^2 = Mk^2,$$

or as it may also be written,

$$(A' + B' + C')(x^2 + y^2 + z^2) - (A', B', C', F', G', H' \wedge x, y, z)^2 = Mk^2,$$

which shows that the two ellipsoids have their axes, and also their circular sections, coincident in direction.

144. And there is besides this a third ellipsoid, the “Ellipsoid of Gyration,” which is the reciprocal of the momental ellipsoid in regard to the concentric sphere, radius k . The last-mentioned ellipsoid is given in magnitude, viz. if the body is referred to its principal axes, then putting $A = Ma^2$, $B = Mb^2$, $C = Mc^2$, the equation of the ellipsoid of gyration is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = k^2.$$

The axes of any one of the foregoing ellipsoids coincide in direction with the principal axes of the body, and the magnitudes of the axes lead very simply to the values of the principal moments A , B , C .

145. The origin has so far been left arbitrary: in the dynamical applications, this origin is in the case of a solid body rotating about a fixed point, the fixed point; and in the case of a free body, the centre of gravity. But the values of the coefficients (A, B, C, F, G, H) , or (A', B', C', F', G', H') , corresponding to any given origin whatever, are very easily expressed in terms of the coordinates of this origin, and

the values of the corresponding coefficients for the centre of gravity as origin; or, what is the same thing, any one of the ellipsoids for the given origin may be geometrically constructed by means of the ellipsoid for the centre of gravity. The geometrical theory, as regards the magnitudes of the axes, does not appear to have been anywhere explicitly enunciated; as regards their direction, it is comprised in the theorem that the directions at any point are the three rectangular directions at that point in regard to the ellipsoid of gyration for the centre of gravity⁽¹⁾, post, No. 159. The notion of the ellipsoids, and of the relation between the ellipsoids at a given point and those at the centre of gravity, once established, the theory of principal axes and moments of inertia becomes a purely geometrical one.

146. The existence of principal axes was first established by Segner in the work *Specimen Theoriæ Turbinum*, Halle (1755), where, however, it is remarked that Euler had said something on the subject in the (Berlin) *Memoirs* for 1749 and 1750 (post. No. 167), and had constructed a new mechanical principle, but without pursuing the question. Segner's course of investigation is in principle the same as that now made use of, viz. a principal axis is defined to be an axis such that when a body revolves round it the forces arising from the rotation have no tendency to alter the position of the axes. It is first shown that there are systems of axes x, y, z such that $\int xz \, dm = 0$, and then, in reference to such a set of axes, the position of a principal axis, say the axis of X , is determined by the conditions

$$\int XY \, dm = 0, \quad \int XZ \, dm = 0,$$

being taken to be $t = \frac{X}{x}$, $\tau = \frac{Y}{y}$ (where α, β, γ being the unknown quantities, the inclinations of the principal axis to those of x, y, z), and then putting $A' = \int x^2 \, dm$, &c. ($F' = 0$ by hypothesis), Segner's equations for the determination of t, τ are

$$\begin{aligned} G't + (C' - A')\tau - G' &= 0, \\ H't + (C' - B')t\tau - G'\tau^2 + H'\tau &= 0, \end{aligned}$$

¹The rectangular directions at a point in regard to an ellipsoid are the directions of the axes of the circumscribed cone, or, what is the same thing, they are the directions of the normals to the three quadric surfaces, confocal with the given ellipsoid, which pass through the given point. The theory of confocal surfaces appears to have been first given by Chasles, Note XXXI. of the *Aperçu Historique* (1837).

the second of which gives

$$\tau = \frac{H't}{G'H'' - G'(A' - B')t + \{(B' - C')(C' - A') - G''^2 - H''^2\}t + G'(B' - C'')},$$

which being a cubic equation shows that there are three principal axes; and it is afterwards proved that these are at right angles to each other.

147. To show the equivalence of Segner's solution to the modern one, I remark that if $M = \int X^2 dm$, we have

$$\begin{aligned}(A' - u)t + H'\tau + G' &= 0, \\ H't + (B' - u)\tau + F' &= 0, \\ G't + F'\tau + (C' - u) &= 0,\end{aligned}$$

whence

$$\begin{aligned}t^2 : \tau^2 : 1 : t : \tau : t\tau = \\ B'C' - F'^2 - (B' + C')u + u^2 : C'A' - G'^2 - (C' + A')u + u^2 : A'B' - H'^2 - \\ : G'F' - A'F' + F'u : H'F' - B'G' + G'u : F'G' - C'H' + H'u.\end{aligned}$$

Or putting therein $F' = 0$,

$$\begin{aligned}t^2 : \tau^2 : 1 : t : \tau : t\tau = \\ B'C' - (B' + C')u + u^2 : C'A' - G'^2 - (C' + A')u + u^2 : A'B' - H'^2 - (A' + \\ : -FG' : +H'G'u : G'H' + (-C'H' + H'u)u,\end{aligned}$$

by means of which Segner's equations may be verified. I have given this analysis, as the first solution of such a problem is a matter of interest.

148. There is little if anything added to Segner's results by the memoir, Euler, "Recherches sur la Connaissance Mécanique des Corps" (1758), which is introductory to the immediately following one on Rotation.

149. Relating to the theory of principal axes we have Binet's "Mémoire sur les axes conjugués &c.," (1813). The author proposes to make known the new systems of axes which he calls conjugate axes, which, when they are at right angles to each other, coincide with the principal axes; viz. considering the sum of the molecules each into its distance from a plane, such distance being measured in the direction

of a line, then (the direction of the line being given) of all the planes which pass through a given point, there is one for which the sum in question is a minimum, and this plane is said to be conjugate to the given line, and from the notion of a line and conjugate plane he passes to that of a system of conjugate axes. The investigation (which is throughout an elegant one) is conducted analytically; the coordinates made use of are oblique ones, and the formulæ are thus rendered more complicated than they would have been; in referring to them it will be convenient to make the axes rectangular.

150. One of the results is the well-known equation

$$(A' - \theta)(B' - \theta)(C' - \theta) - F'^2(A' - \theta) - G'^2(B' - \theta) - H'^2(C' - \theta) + 2F'G'H' = 0;$$

which, if x_1, y_1, z_1 are the principal axes, has for its roots $\int x_1^2 dm$, $\int y_1^2 dm$, $\int z_1^2 dm$.

And the equations (1), p. 49, taking therein the original axes as rectangular, are

$$\begin{aligned} [\mathfrak{A}' - \mathfrak{K}'] \cos \alpha + \mathfrak{H}' \cos \beta + \mathfrak{G}' \cos \gamma &= 0, \\ \mathfrak{H}' \cos \alpha + [\mathfrak{B}' - \mathfrak{K}'] \cos \beta + \mathfrak{F}' \cos \gamma &= 0, \\ \mathfrak{G}' \cos \alpha + \mathfrak{F}' \cos \beta + [\mathfrak{C}' - \mathfrak{K}'] \cos \gamma &= 0, \end{aligned}$$

where $\mathfrak{A}'$, $\mathfrak{B}'$, $\mathfrak{F}'$, $\mathfrak{G}'$, $\mathfrak{H}'$, &c. denote the reciprocal coefficients, $\mathfrak{A}' = B'C' - F'^2$, &c., and $\mathfrak{K}'$ is the discriminant $= A'B'C' - A'F'^2 - B'G'^2 - C'H'^2 + 2F'G'H'$: this is a symmetrical system of equations for finding $\cos \alpha : \cos \beta : \cos \gamma$, less simple however than the modern form (post, No. 154), the identity of which with Binet's may be shown without difficulty.

151. Another result (p. 57) is that if the original axes are principal axes, and if Ox, Oy, Oz are the principal axes through a point the coordinates whereof are f, g, h , and if $O_x = (\text{say}) \int x_1^2 dm$, then we have

$$\frac{f^2}{O_x - A'} + \frac{g^2}{O_x - B'} + \frac{h^2}{O_x - C'} = \frac{1}{M},$$

(in which I have restored the mass M , which is put equal to unity), so that if O_x have a given constant value, the locus of the point is a quadric surface, the nature whereof will depend on the value of O_x . The surfaces in question are confocal with each other (and with the imaginary surface $\frac{f^2}{A'} + \frac{g^2}{B'} + \frac{h^2}{C'} = \frac{1}{M}$), and the general theorem in regard to them is, I think, due to Chasles: the special theorem for the ellipsoid of gyration was, however, given by Binet.

152. An interesting theorem of Binet's relates to the momental ellipsoid at any point of a principal plane: taking the fixed point in the plane of xy , then $h = 0$, and we have

$$\frac{f^2}{O_x - A'} + \frac{g^2}{O_x - B'} = \frac{1}{M} - \frac{0}{O_x - C'} = \frac{1}{M},$$

which may be written

$$\frac{f^2}{\frac{1}{M}O_x - \frac{A'}{M}} + \frac{g^2}{\frac{1}{M}O_x - \frac{B'}{M}} = 1,$$

and thus shows that the section by the plane of xy of the system of quadric surfaces is a system of confocal conics; the foci thereof are the principal axes in the plane of xy , and the squares of the semiaxes thereof are $\frac{A'}{M}$ and $\frac{B'}{M}$, that is, the squares of the principal radii of gyration in the plane of xy .

153. Another theorem relates to the section of the ellipsoid of gyration by a principal plane. Taking this to be the plane of xy , and writing $A = Ma^2$, $B = Mb^2$, $C = Mc^2$, the equation of the ellipsoid of gyration is $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = k^2$; the section by the plane of xy is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = k^2$, an ellipse the squares of whose semiaxes are a^2 , b^2 , that is, the principal radii of gyration in the plane of xy . It may be added, that at any point of the plane of xy the section of the ellipsoid of gyration is an ellipse having its foci at the principal axes (or centres of gyration) in that plane; and hence also the theorem that the difference of the squares of the radii of gyration in any two directions at right angles to each other in a principal plane is constant, being equal to the difference of the squares of the principal radii of gyration in that plane.

154. To complete the theory, I give the following proof of the theorem for the principal axes through any point of a solid body. Starting from the ellipsoid of gyration for the centre of gravity, the axes at the point in question are (ante No. 145) the rectangular directions at the point in regard to the ellipsoid. Now the rectangular directions at a point in regard to an ellipsoid coincide with the normals to the three confocal quadric surfaces through the point. Hence the directions of the principal axes at the point in question are those of the normals to the three confocal quadric surfaces through that point; and if we determine, as above, the axes of the confocal ellipsoid through the point, these will be the principal axes in question. It is to be observed that the axes of the confocal ellipsoid are not the same thing as

the rectangular directions, and that the theorem does not apply to the axes of the confocal ellipsoid, but only to the normals thereof. The rectangular directions at any point in regard to an ellipsoid, and the normals to the confocal quadric surfaces through the point, are identical; and they coincide also with the axes of the cone circumscribed about the ellipsoid, having its vertex at the point. The notion of the rectangular directions is, I think, the most convenient one for the statement of the theorem.

155. The analytical expression of the theorem is as follows:—The equations of the confocal surfaces being

$$\frac{x^2}{a^2 + \theta} + \frac{y^2}{b^2 + \theta} + \frac{z^2}{c^2 + \theta} = k^2,$$

these determine for a given point (f, g, h) three values $\theta_1, \theta_2, \theta_3$; and if (l, m, n) are the direction cosines of an axis, then the equations for the determination of $l : m : n$ are

$$\begin{aligned} \left(\frac{f^2}{(a^2 + \theta_1)^2} + \frac{g^2}{(b^2 + \theta_1)^2} + \frac{h^2}{(c^2 + \theta_1)^2} \right) l &= \frac{f}{a^2 + \theta_1}, \\ \left(\frac{f^2}{(a^2 + \theta_1)^2} + \frac{g^2}{(b^2 + \theta_1)^2} + \frac{h^2}{(c^2 + \theta_1)^2} \right) m &= \frac{g}{b^2 + \theta_1}, \\ \left(\frac{f^2}{(a^2 + \theta_1)^2} + \frac{g^2}{(b^2 + \theta_1)^2} + \frac{h^2}{(c^2 + \theta_1)^2} \right) n &= \frac{h}{c^2 + \theta_1}, \end{aligned}$$

for $\theta = \theta_1$, and two other similar systems for $\theta = \theta_2$ and $\theta = \theta_3$; these give the directions of the three principal axes at the point (f, g, h) .

156. I may remark that the foregoing investigations belong to a purely geometrical theory which I call the “*Théorie des Directions Rectangulaires*”; the theory might, I think, be advantageously introduced into the ordinary treatises on solid geometry.

157. Another theorem relating to the ellipsoid of gyration is the following:—If through any point (f, g, h) we draw a line meeting the ellipsoid of gyration in the points P, Q , and if R be the radius vector parallel to this line drawn from the centre to the ellipsoid, then the product $OP \cdot OQ$ (where O is the point (f, g, h)) varies inversely as R^2 ; or, what is the same thing, the product $OP \cdot OQ \cdot R^2$ is constant for all directions through the point. This theorem is due to MacCullagh.

158. I do not stop to consider the various other theorems relating to moments of inertia and principal axes which are to be found in the works of Poinsot, Chasles, and other writers.

159. The theorem of the principal axes at any point, viz. that the directions thereof are the rectangular directions at the point in regard to the ellipsoid of gyration for the centre of gravity, has been already referred to, No. 145. It may be remarked that the theorem includes as a particular case the theorem for the principal axes through the centre of gravity itself; for at the centre of gravity the ellipsoid of gyration becomes the central ellipsoid of gyration, and the rectangular directions at the centre of gravity in regard to this ellipsoid are of course the axes thereof.

160. The ellipsoid of gyration at any point is connected with the central ellipsoid of gyration by a simple relation: taking the central ellipsoid to be $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = k^2$, then the ellipsoid of gyration at the point (f, g, h) is at once expressible in terms of f, g, h , and the coefficients a^2, b^2, c^2 ; the equation may be written

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = k^2 + \frac{f^2}{a^2} + \frac{g^2}{b^2} + \frac{h^2}{c^2} - \left(\frac{fx}{a^2} + \frac{gy}{b^2} + \frac{hz}{c^2} \right)^2 \bigg/ \left(\frac{f^2}{a^2} + \frac{g^2}{b^2} + \frac{h^2}{c^2} \right),$$

and from this the axes of the ellipsoid, that is, the principal radii of gyration at the point, may be found. But as regards the directions only, these are, as has been seen, the rectangular directions at the point in regard to the central ellipsoid of gyration.

161. The investigation of the principal axes by means of the ellipsoid of gyration, although in some respects the most complete and elegant method, is not the one originally employed. The first direct analytical solution of the problem is that of Segner (1755), already referred to, No. 146; and it is worthy of remark that the method employed by Segner is in principle identical with that which is now in common use, viz. the problem is made to depend on the solution of a cubic equation.

162. The modern form of the solution is as follows:—If A', B', C', F', G', H' are the integrals $\int x^2 dm$, &c., referred to any system of rectangular axes through the point, then the determination of the principal axes depends on the solution of the cubic equation

$$\begin{vmatrix} A' - \theta & H' & G' \\ H' & B' - \theta & F' \\ G' & F' & C' - \theta \end{vmatrix} = 0,$$

or, expanded,

$$(A' - \theta)(B' - \theta)(C' - \theta) - F'^2(A' - \theta) - G'^2(B' - \theta) - H'^2(C' - \theta) + 2F'G'H' = 0.$$

If the roots are $\theta_1, \theta_2, \theta_3$, then the direction cosines (l, m, n) of the corresponding principal axis are given by

$$(A' - \theta_1)l + H'm + G'n = 0, \quad H'l + (B' - \theta_1)m + F'n = 0, \quad G'l + F'm + (C' - \theta_1)n = 0,$$

with two other similar systems for θ_2 and θ_3 .

163. The foregoing cubic equation has three real roots, and the corresponding axes are at right angles to each other. That the roots are real may be shown in various ways; the most direct proof is that the cubic is the discriminant of the quadratic form $(A', B', C', F', G', H' \wedge l, m, n)^2$, which being a positive definite form, its discriminant has three real roots. The theory of principal axes is thus made to depend on the theory of quadratic forms, and the whole investigation is conducted by purely algebraical methods.

Rotation of a Solid Body

ARTICLE NOS. 164 TO 205.

164. The problem of the rotation of a solid body about a fixed point is one of the most interesting and important in dynamics. The first complete solution is due to Euler, who in the memoir "Du Mouvement de Rotation des Corps Solides autour d'un Point Variable" (1758) established the equations of motion and obtained the integrals of Vis Viva and of areas. But the complete solution, expressing the nine direction cosines in terms of the time, was not effected until the discoveries of Jacobi (1839) relating to elliptic functions.

165. The equations of motion, as given by Euler, are

$$A \frac{dp}{dt} + (C - B)qr = L, \quad B \frac{dq}{dt} + (A - C)rp = M, \quad C \frac{dr}{dt} + (B - A)pq = N,$$

where A, B, C are the principal moments of inertia, p, q, r the angular velocities about the principal axes, and L, M, N the moments of the external forces about the same axes. In the case where there are no external forces, these become

$$A \frac{dp}{dt} + (C - B)qr = 0, \quad B \frac{dq}{dt} + (A - C)rp = 0, \quad C \frac{dr}{dt} + (B - A)pq = 0.$$

166. The two integrals in this case are those of Vis Viva and of areas:

$$Ap^2 + Bq^2 + Cr^2 = 2h, \quad A^2p^2 + B^2q^2 + C^2r^2 = k^2,$$

where h and k are constants. From these equations, together with the third equation of motion, the values of p , q , r may be expressed in terms of elliptic functions of the time. The actual expressions were first obtained by Jacobi in 1839.

167. Euler's investigations on the rotation of a solid body extend over a long series of years. The memoir of 1758, already referred to, contains the complete establishment of the equations of motion and the proof of the existence of the two integrals; but the further integration was not effected. In the memoir "Problema Algebraicum propositum publice" (1770), Euler considered the transformation of coordinates, which is intimately connected with the problem; and in the memoir "Nova Methodus Motum Corporum rigidorum determinandi" (1775), he returned to the problem of rotation, obtaining results which, although interesting, do not constitute a complete solution.

168. The first step towards the complete solution was made by Lagrange in the *Mécanique Analytique* (1788), who showed that the equations of motion could be expressed in terms of the angles which determine the position of the body, and who obtained the integrals in a form suitable for further development. But the actual integration in terms of elliptic functions required the discoveries of Abel and Jacobi.

169. Poinso't's geometrical representation of the motion (1834) is one of the most beautiful results in the whole theory. The motion is represented by the rolling of the ellipsoid of gyration on a fixed plane, the distance of the plane from the centre being proportional to the angular velocity. This representation, although it does not give the analytical expressions for the motion, affords a complete and elegant geometrical description thereof.

170. The analytical solution was completed by Jacobi in the memoir "Sur la Rotation d'un Corps" (1839), and in the fundamental memoirs on elliptic functions in Crelle's Journal. The solution is expressed by means of the elliptic functions sn , cn , dn , and involves the introduction of the functions Θ and H , which Jacobi invented for this purpose. The expressions for the direction cosines are of the form

$$\alpha = \frac{H(u + ia)}{\Theta(u)}, \quad \&c.,$$

and the time t is proportional to the argument u .

171. Jacobi's solution was considerably simplified by the introduction of the notation of elliptic functions with the modulus k , and

by the use of the addition theorems and other properties of these functions. The solution has been presented in various forms by different writers; among the most important are those of Rosenhain, Richelot, and others.

172. The case of two equal moments of inertia, say $A = B$, is of special interest, as it includes the case of a body of revolution. In this case the elliptic functions degenerate into circular functions, and the solution takes a much simpler form. The motion is then represented by the rolling of a spheroid on a fixed plane, and the precession and nutation of the axis of the body are easily determined.

173. The case of three equal moments of inertia, $A = B = C$, is the simplest of all; the body moves as if it were a sphere, and the axis of rotation remains fixed in direction in space. This is the case of the free rotation of a sphere or cube about its centre.

174. Another particular case of interest is that in which the angular momentum is the least or greatest possible for the given value of the Vis Viva; in this case the motion is uniform, and the axis of rotation is a principal axis. This is the case of steady rotation about a principal axis, which is stable for the axes of greatest and least moment, and unstable for the intermediate axis.

175. The stability of the rotation about a principal axis was first investigated by Euler, who showed that the rotation about the axes of greatest and least moment is stable, while that about the intermediate axis is unstable. The analytical proof depends on the consideration of small oscillations about the state of steady motion; if the axis be that of greatest or least moment, the oscillations remain small, but if it be the intermediate axis, they increase indefinitely.

176. The motion of a solid body subject to external forces is in general much more complicated than that of the free body. The most important case is that in which the body is acted on by gravity, and rotates about a fixed point not coinciding with the centre of gravity; this is the case of the heavy rigid body, or the "top." The complete integration of this problem has not been effected, except in certain particular cases.

177. The particular cases in which the problem of the heavy top admits of complete solution are:—1. The case of Euler and Poinsot, where the fixed point coincides with the centre of gravity; 2. The case of Lagrange, where two of the principal moments are equal, and the centre of gravity lies on the axis of symmetry; 3. The case of Kowalevski, where two of the principal moments are equal to one-fourth of

the third, and the centre of gravity lies in the plane perpendicular to the axis of the unequal moment.

178. The case of Lagrange (the symmetrical top) is the one of greatest physical interest, as it includes the motion of the spinning top and the precession and nutation of the earth. The solution is effected by means of elliptic functions, and the motion consists of a steady precession combined with a periodic nutation.

179. The case of Kowalevski (1888) was a remarkable discovery, showing that there exists a third case in which the problem admits of complete integration. The solution is more complicated than that of Lagrange, and involves hyperelliptic functions; but the existence of this case is of great theoretical importance.

180. The investigations of Kowalevski were continued by various writers, among whom may be mentioned Tchapliguine, Husson, and others. The problem of the complete integration of the equations of motion of a heavy rigid body remains, however, one of the most difficult in dynamics, and it is not known whether there are any other cases besides those already discovered.

181. The equations of motion of a heavy rigid body may be written in various forms, according to the variables employed. The most usual form is that of Euler, in terms of the angular velocities p, q, r about the principal axes; but they may also be expressed in terms of the Eulerian angles θ, ϕ, ψ , or in terms of the parameters of Rodrigues, or by means of quaternions.

182. The use of quaternions in the theory of rotations, as developed by Hamilton and Tait, affords a very elegant method of treating the problem. The equation of motion may be written in the form

$$\frac{d\rho}{dt} = \frac{1}{2}\rho\omega,$$

where ρ is the quaternion representing the position of the body, and ω is the vector quaternion representing the angular velocity. This form of the equation is particularly adapted to the treatment of the composition of rotations.

183. The theory of the composition of finite rotations, as developed by Rodrigues, Hamilton, and Cayley, is intimately connected with the problem of the rotation of a solid body. The composition of two finite rotations about different axes is equivalent to a single rotation about a certain axis; and the determination of the resultant

rotation may be effected by various methods, among which the quaternion method is perhaps the most elegant.

184. The kinematical representation of the motion of a solid body, as given by Poinso, has been already referred to, No. 169. Analytical representations of a similar character have been given by various writers; among the most important are the “moving axes” of Thomson and Tait, and the “screws” of Ball.

185. The method of moving axes, as developed by Thomson and Tait, affords a very powerful method of treating the kinematics and dynamics of a solid body. The axes are supposed to move with the body, and the components of velocity and angular velocity are referred to them; the equations of motion then take a form which is particularly adapted to the treatment of problems involving the rotation of a solid body.

186. Rueb’s memoir (1834) on the rotation of a solid body is of great historical importance, as it contains the first successful attempt to express the direction cosines in terms of elliptic functions. The memoir is entitled “De motu gyatorio corporis rigidi” and was published in the *Commentationes Societatis Regiae Scientiarum Gottingensis*. The solution, although incomplete in some respects, prepared the way for the complete solution of Jacobi.

187. In Rueb’s solution, the direction cosines are expressed by means of certain integrals involving the square root of a quartic function; these integrals are in fact elliptic integrals of the first kind, but Rueb did not reduce them to the standard form. The reduction was effected by Jacobi, who thus completed the solution.

188. The connection between the problem of rotation and the theory of elliptic functions is one of the most interesting in the whole range of mathematics. The elliptic functions arise in the most natural manner from the attempt to express the direction cosines in terms of the time; and the theory of these functions receives its most beautiful applications in the solution of this problem.

189. The further development of the theory of elliptic functions, as given by Jacobi in the *Fundamenta Nova* (1829), and in his later memoirs, was largely influenced by the requirements of the problem of rotation. The functions Θ and H , which are fundamental in the theory, were invented for the purpose of expressing the direction cosines in the most elegant form.

190. The solution of the problem of rotation has been presented in various forms by different writers. Among the most important of

these forms are those of Jacobi (1839), Rosenhain (1846), Richelot (1850), and others. Each of these writers has contributed something to the simplification or extension of the solution.

191. Rosenhain's memoir (1846) contains a very elegant treatment of the problem, based on the theory of the hyperelliptic functions. The solution is expressed in terms of the quotients of the theta-functions, and the expressions are symmetrical and elegant. But the solution is less directly applicable to the physical problem than that of Jacobi.

192. Richelot's memoir (1850) contains an elaborate investigation of the problem, with especial reference to the case of unequal moments of inertia. The solution is expressed in terms of the elliptic functions, and the author has paid particular attention to the reduction of the integrals to the standard form. The memoir also contains various interesting particular cases.

193. The solution by Jacobi, § 27 of the memoir "Theoria Novi Multiplicatoris" (1845), is given as an application of the general theory, the author remarking that, as well in this question as in the problem of the two fixed centres, he purposely employed a somewhat inartificial analysis, in order to show that the principle (of the Ultimate Multiplier) would lead to the result without any special artifices. The principal axes are referred to the axes fixed in space by the ordinary three angles (here called q_1, q_2, q_3), and the solution is carried so far as to give the integral equations, without any reduction of the integrals contained in them to elliptic integrals. The solution is, however, in so far remarkable that the integrations are effected without the aid of the invariable plane.

194. Cayley, "On the Rotation of a Solid Body &c." (1846).—It appeared desirable to obtain the solution by means of the quantities λ, μ, ν , without the assistance of the invariable plane, and Jacobi's discovery of the theorem of the Ultimate Multiplier induced me to resume the problem, and at least attempt to bring it so far as to obtain a differential equation of the first order between two variables only, the multiplier of which could be obtained theoretically by Jacobi's discovery. The choice of two new variables to which the equations of the problem led me, enabled me to effect this in a simple manner; and the differential equation which I finally obtained turned out to be integrable *per se*, so that the laborious process of finding the multiplier became unnecessary.

195. The new variables Ω, v have the following geometrical si-

gnifications: $\Omega = I \tan \frac{1}{2}\theta \cos J$, where I is the principal moment ($A^2p^2 + B^2q^2 + C^2r^2 = I^2$), θ (as before) the angle of resultant rotation, and J is the inclination of the resultant axis to the invariable axis; and $v = I^2 \cos^2 \frac{1}{2}\theta \cdot J$, where if we imagine a line AQ having the same position relatively to the axes fixed in space that the invariable axis has to the principal axes of the body, then J is the inclination of this line to the invariable axis. It is found that p, q, r are functions of v only, whereas λ, μ, ν contain besides the variable Ω . In obtaining these relations, there occurs a singular relation $\Omega^2 = kv - 1$, which may also be written $1 + \tan^2 \frac{1}{2}\theta \cos^2 J = \text{const.}$, where the geometrical significations of the quantities I, J have just been explained.

The final results are that the time t , and the arc $\tan^{-1} \frac{\Omega}{\sqrt{kv-1}}$ are each of them expressible as the integrals of certain algebraical functions of v . There might be some interest in comparing the results with those of Euler's memoir of 1758, where the principal axes are also referred to an arbitrary system of axes fixed in space; but I was not then acquainted with Euler's memoir.

The concluding part of the paper relates to the determination of the variations of the constants in the disturbed problem.

196. Cayley, "Note on the Rotation of a Solid of Revolution" (1849), shows the simplification produced in the formulæ of the last-mentioned memoir in the case where two of the moments of inertia are equal, say $A = B$.

197. Jacobi's final solution of the problem of Rotation was given without demonstration in the letter to the Academy of Sciences at Paris; the demonstration is added in the memoir, Crelle, t. XX-XIX. (1849). The fundamental idea consists in taking in the invariable plane, instead of the fixed axes xy , a set of axes xy revolving with uniform velocity, such that the angular distance of the axis of x from its initial position is precisely $= -n'u$; and consequently if θ' be the longitude of the node of the equator on the invariable plane, measured from the moveable axis of x as the origin of longitude, we have

$$\theta' = \theta - n'u = \frac{1}{\sqrt{-1}} \log \frac{A - B}{A + B},$$

($\sqrt{-1} = i$); and in consequence of this form of the expression for θ' ($= \frac{1}{i} \log$ into a logarithmic function), in passing to the trigonometrical functions $\sin \theta, \cos \theta$ the logarithm disappears altogether; and we have in a simple form the expressions for the actual functions $\sin \theta, \cos \theta$, through which θ enters into the formulæ, and thus, Jacobi remarks,

the barrier is cleared which stands in the way when the expression of an angle is reduced to an elliptic integral of the third class.

198. For the better expression of the results, Jacobi joins to the functions H , Θ , considered in the “*Fundamenta Nova*,” the functions $\Theta_1 u = \Theta(K - u)$, $H_1 u = H(K - u)$, so that

$$\sqrt{k} \sin \operatorname{am} u = \frac{Hu}{i\sqrt{k}\Theta u}, \quad \sqrt{k} \cos \operatorname{am} u = \frac{H_1 u}{\sqrt{k}\Theta u}, \quad \Delta \operatorname{am} u = \frac{\Theta_1 u}{\Theta u};$$

and then considering the cosine inclinations of the principal axes to the invariable axis and the revolving axes in the invariable plane, these are all fractions which, neglecting constant factors, have the common denominator Θu ; α'' , β'' , γ'' (as shown by Rueb’s formula) have the numerators $H_1 u$, Hu , and $\Theta_1 u$ respectively; and α , α' have the numerators $H(u + ia) \pm H(u - ia)$, β , β' the numerators $H_1(u - ia) \pm H_1(u + ia)$, γ , γ' the numerators $\Theta(u + ia) + \Theta(u - ia)$ respectively: there are also expressions of a similar form for the angular velocities about the axis of x and y ; that about the axis of z (the invariable axis) having, as was known, the constant value n . The memoir is also very valuable analytically, as completing the systems of formulæ given in the *Fundamenta Nova* in reference to elliptic integrals of the third class.

199. It is worth noticing how the results connect themselves with Poincot’s theorem of the rolling and sliding cone: the velocity of the rolling motion depends only upon the position, on the cone, of the line of contact, so that the same series of velocities recur after any number of complete revolutions of the cone; that is, the total angle described by the line of contact in consequence of the rolling motion, consists of a part varying directly with the time (or say varying as u) and a periodic part; the former part combines with the similar term arising from the sliding motion; and the two together give Jacobi’s term $n'u$.

200. Somoff’s memoir (1851), written after Jacobi’s Note in the *Comptes Rendus*, but before the appearance of the memoir in Crelle, gives the demonstration of the greater part of Jacobi’s results.

201. Booth’s *Theory of Elliptic Integrals &c.* (1851) (contemporaneous with the publication of Poincot’s memoir of 1834) contains various interesting analytical developments, and, as an interpretation of them, the author obtains (p. 93) the theorem of the rolling and sliding cone. The investigations involve the elliptic integrals, but not the elliptic or Jacobian functions.

202. Richelot's two Notes (Crelle, tt. XLII. and XLIV.) relate to the solution of the problem of rotation given in his memoir "Eine neue Lösung &c." (1851). This is an application of Jacobi's theorem for the integration of a system of dynamical equations by means of the principal function V (see my "Report" of 1857, art. 34). Retaining Richelot's letters ϕ , ψ , θ , which signify

ϕ , the longitude of the node,

θ , the inclination,

ψ , the hour-angle,

the question is to find a complete solution of the partial differential equation

$$\frac{dV}{dt} + \frac{1}{2A} \left(\frac{dV}{d\phi} \right)^2 + \frac{1}{2} \left(\frac{1}{A \sin^2 \theta} + \frac{\cos^2 \theta}{C} \right) \left(\frac{dV}{d\psi} \right)^2 + \frac{1}{2C} \left(\frac{dV}{d\theta} \right)^2 - \frac{\cos \theta}{A \sin^2 \theta} \frac{dV}{d\phi} \frac{dV}{d\psi} = 0$$

that is, a solution involving (besides the constant attached to V by a mere addition) three arbitrary constants; these are a , b , p . Writing in the first place $V = W + at + b\phi + p\psi$, the resulting equation for W may be satisfied by taking W , a function of ϕ and θ , without ψ or t ; and it is sufficient to have a solution involving only a single arbitrary constant.

This leads to a solution which is as follows:

$$\begin{aligned} V = & at + b\phi + p\psi \\ & - \int \frac{d\theta}{\sqrt{\Theta}} \sqrt{(b - p \cos \theta)^2 - 2A(a - p^2/A - p^2/C)(\sin^2 \theta)} \\ & + \int \frac{d\theta}{\sqrt{\Theta}} \sqrt{(b - p \cos \theta)^2 - 2A(a - p^2/A - p^2/C)(\sin^2 \theta) + \dots} \end{aligned}$$

where Θ and the radicals are certain given functions of a , b , p , and of θ and ϕ . The solution of the dynamical problem is then obtained by putting the differential coefficients $\frac{dV}{da}$, $\frac{dV}{db}$, $\frac{dV}{dp}$ equal to arbitrary constants L , G , H respectively; the results are somewhat more simple than might be expected from the very complicated form of the function V . The foregoing results (although not by themselves very intelligible) will give an idea of the form in which the analytical solution in the first instance presents itself.

203. Richelot proceeds to remark that the solution in question, and the resulting integral equations of the problem, may be simplified in a peculiar manner by the method which he calls "the integration by the spherical triangle." For this purpose he introduces a spherical

triangle, the sides and angles whereof are $V, \lambda, \mu; N, \Lambda, M$, and then assuming N constant, $M = \pi - \psi$,

$$(B - C) \sin^2(\phi + \psi) \sin^2 \Lambda + (A - B) \cos^2(\phi + \psi) \sin^2 \Lambda = A - C,$$

where p and q are constant, the solution is

$$V = at - p(\psi - \lambda) \cos N - pM + p \int \cos \Lambda d(\phi + \psi);$$

and that this expression leads to all the results almost without calculation.

204. I have quoted the foregoing results from the Note (Crelle, t. XLII.), having seen, but without having studied, the Memoir itself: the results appear very interesting and valuable ones; but they seem to require a more complete geometrical development than they have received in the Memoir; and I am not able to bring them into connexion with the other researches on the subject.

205. The solution, § 3 of Donkin's memoir "On a Class of Differential Equations &c." (part I. 1854), is given as an illustration of the general theory to which the memoir relates; it contains, however, some interesting geometrical developments in regard to the case ($A = B$) of two equal moments of inertia. I have not compared the results with those in my Note of 1849.

[298]

CERTAIN SPECIAL PROBLEMS OF DYNAMICS.

(Report on the Progress of the Solution of)

206. The solution of the rotation problem, § 66 of Jacobi's memoir "Nova Methodus &c." (1862), has for its object to show the complete analogy which exists between this problem and the problem of a body attracted to a fixed centre. The section is in fact headed "Solutio simultanea problematis de motu puncti versus centrum attracti atque problematis de rotatione &c."; and Jacobi, after noticing that Poisson, in his memoir of 1816 (*Mém. de l'Inst. t. i.*), had shown that the expressions for the variations of the elements in the two problems could be investigated by a common analysis, remarks, "Sed ipsa problemata duo imperturbata hic primum, quantum credo, amplexus sum." The solution is in fact as follows:—Suppose that in the one problem the position of the point in space, and in the other problem the position of the body in regard to the fixed axes, is determined in any manner by the quantities q_1, q_2, q_3 . Let

$$\dot{q}_1 = \frac{dq_1}{dt}, \quad \dot{q}_2 = \frac{dq_2}{dt}, \quad \dot{q}_3 = \frac{dq_3}{dt},$$

and expressing the *Vis Viva* function T in terms of $q_1, q_2, q_3, \dot{q}_1, \dot{q}_2, \dot{q}_3$, let

$$\frac{\partial T}{\partial \dot{q}_1} = p_1, \quad \frac{\partial T}{\partial \dot{q}_2} = p_2, \quad \frac{\partial T}{\partial \dot{q}_3} = p_3,$$

and let H be the value of T expressed in terms of $q_1, q_2, q_3, p_1, p_2, p_3$, so that $H = a$ is the integral of *Vis Viva* (this is merely the transformation to the Hamiltonian form). And let $H_1 = a_1, \Phi = a'_2, \Psi = a''_2$ be the three integrals of areas (H, H_1, Φ, Ψ are functions of the variables only, not containing the arbitrary constants a, a_1, a'_2, a''_2). Then, expressing

$$p_1 dq_1 + p_2 dq_2 + p_3 dq_3$$

in terms of $p_1, p_2, p_3, q_1, q_2, q_3$, and by means of the equations

$$H = a, \quad H_1 = a_1, \quad H_2 = a_2$$

(where $a_2 = \sqrt{a_2'^2 + a_2''^2}$) expressing p_1, p_2, p_3 in terms of q_1, q_2, q_3 , we have $p_1 dq_1 + p_2 dq_2 + p_3 dq_3$ a complete differential; and putting

$$\int (p_1 dq_1 + p_2 dq_2 + p_3 dq_3) = V,$$

then ($a, a_1, a_2, b_1, b_2, b_3$ being arbitrary constants) we have

$$\begin{array}{lll} H = a, & H_1 = a_1, & H_2 = a_2, \\ \frac{\partial V}{\partial a_1} = b_1, & \frac{\partial V}{\partial a_2} = b_2, & \frac{\partial V}{\partial a} = t + b_3, \end{array}$$

which is the complete integrals of either problem, the last three of them being the final integrals.

And it is added that if in either problem we have $H + \Omega$ instead of H , the expressions for the variations of the elements are at once obtained by the general theorem for the variation of the elements.

207. It is part of the general theorem that any number of rotations about axes passing through one and the same point may be compounded into a single rotation about an axis through that point; this is, in fact, Euler's theorem. Poinsot, in the memoir "Théorie nouvelle de la rotation des corps," Liouville, t. xvi. (1851), pp. 9—130, gives the theorem a more complete form by showing that the axes are the principal axes of an ellipsoid which is derived from the momental ellipsoid by altering the lengths of its principal axes respectively in given ratios. This is equivalent to the theorem that any number of couples may be compounded into a single couple, the axis of which is the principal axis of a certain ellipsoid derived from the momental ellipsoid.

208. The general theory of the composition of rotations is further developed by Sylvester in the memoir "On an extension of a theorem of Euler," Quart. Math. Journ. t. iii. (1860), pp. 187—188. The theorem is that any number of rotations about axes meeting in a point may be compounded into a single rotation about an axis through the point, and a translation along the axis. The translation is what Sylvester calls the "pitch" of the resultant screw.

209. The theory of screws has been further developed by Ball in the memoir "The Theory of Screws," Phil. Trans. t. clxiv. (1874), pp. 187—233. The memoir contains a complete account of the theory,

with applications to the theory of rigid bodies and to the theory of mechanisms.

210. The theory of the deformation of surfaces is discussed in the memoir "On the deformation of surfaces" by Bour, Liouville, t. xxii. (1857), pp. 329—355. The memoir contains a general investigation of the conditions under which a surface can be deformed without stretching or tearing. The theory is applied to the particular case of the deformation of a surface of revolution into another surface of revolution.

A skew surface is said to be "deformed" if, considering the elements between consecutive generating lines as rigid, these elements be made in any manner to turn round and slide along the successive generating lines:—and that two skew surfaces can be made to roll and slide one upon the other, only if the one is a deformation of the other—and that then the rolling and sliding can be effected in an infinity of ways.

211. The theory of the deformation of surfaces is further discussed by Bonnet in the memoir "Mémoire sur la théorie des surfaces applicables sur une surface donnée," Liouville, t. xxv. (1860), pp. 1—63. The memoir contains a complete investigation of the conditions under which a surface can be deformed into another surface, with applications to the theory of minimal surfaces and to the theory of surfaces of constant curvature.

212. The theory of the variation of the elements is discussed in the memoir "Sur la variation des constantes arbitraires dans les problèmes de mécanique" by Poisson (1809). The variations are expressed in terms of the differential coefficients of the disturbing function in regard to the elements, and, as the author remarks, they are very similar in their form to, and can be rendered identical with, those for the problem of rotation.

213. The theory of the variation of the elements is further discussed by Lagrange in the memoir "Théorie des variations des éléments des planètes," Mém. de l'Inst. 1808. The memoir contains a complete investigation of the theory, with applications to the planetary theory.

214. In conclusion, the author expresses his regret that the Report, so far from being as complete as he should have wished to make it. To have reproduced, to any much greater extent than has been done, the

various mathematical investigations, would not have been proper, nor indeed practicable; at the same time, more especially as regards the subjects treated of in the second part of this Report, or those relating to the general theories of Dynamics, he fears that the account is in many respects very imperfect.

REFERENCES

Bonnet. Note sur un théorème de Mécanique. Liouv. t. ix. p. 113 (1844), and Note IV. of t. II. of the last edition of the Méc. Anal., pp. 329—331 (1855).

Booth. Theory of Elliptic Integrals. 8vo. Lond. 1851.

Bour. Mémoire sur le problème des trois corps. Journ. École Polyt. t. xxi. (cah. 36), pp. 35—58 (1856).

Bravais. Mémoire sur l'influence qu'exerce la rotation de la terre sur le mouvement d'un pendule à oscillations coniques. Liouv. t. XIX. pp. 1—50 (1854).

Bravais. Note sur une formule de Lagrange relative au mouvement pendulaire. Note VII. of t. II. of the last edition of the Méc. Anal., pp. 352—355 (1855).

Briot. Thèse sur le mouvement d'un corps solide autour d'un point fixe. Liouv. t. vii. pp. 70—84 (1842).

Cauchy. Sur les momens d'inertie. Exer. de Math. t. i. pp. 93—103 (1827).

Cauchy. Résumé d'un mémoire sur la Mécanique Céleste et sur un nouveau calcul appelé des limites. (Read at Turin Oct. 1831.) Exer. d'Anal. &c. t. ii. pp. 41—109 (1841).

Cayley. On certain expansions in multiple sines and cosines. Camb. Math. Journ. t. III. pp. 162—167 (1842), [4].

Cayley. On the motion of rotation of a solid body. Camb. Math. Journ. t. iii. pp. 224—232 (1842), [5].

Cayley. On certain results relating to quaternions. Phil. Mag. t. xxvi. (1845), p. 141, [20].

Cayley. On the geometrical representation of the motion of a solid body. Camb. and Dubl. Math. Journ. t. i. pp. 164—167 (1846), [36].

Cayley. On the rotation of a solid body round a fixed point. Camb. and Dubl. Math. Journ. t. I. pp. 167—173 and 264—274 (1846), [37].

Cayley. Note on a geometrical theorem in Prof Thomson's memoir on the principal axes of a solid body. Camb. and Dubl. Math. Journ. t. i. pp. 207, 208 (1846), [38].

Cayley. On the application of quaternions to the theory of Rotation. *Phil. Mag.* t. xxxiii. (1848), p. 196, [68].

Cayley. Note on the motion of rotation of a solid of revolution. *Camb. and Dubl. Math. Journ.* t. iv. pp. 268—271 (1849), [78].

Cayley. Sur les déterminants gauches. *Crelle*, t. xxxviii. (1849), pp. 93—96, [69].

Cayley. Note on the theory of Elliptic Motion. *Phil. Mag.* t. xi. (1856), pp. 425—428, [199].

Cayley. A demonstration of Sir W. R. Hamilton's theorem of the Isochronism of the Circular Hodograph. *Phil. Mag.* t. xiii. (1857), p. 427, [209].

Cayley. Report on the recent progress of Theoretical Dynamics. *Rep. Brit. Assoc. for 1857*, pp. 1—42, [195].

Cayley. On Lagrange's solution of the problem of two fixed Centres. *Quart. Math. Journ.* t. II. pp. 76—82 (1858), [182].

Cayley. Note on the expansion of the true anomaly. *Quart. Math. Journ.* t. II. pp. 229—232 (1858), [191].

Cayley. Tables in the theory of Elliptic Motion. *Mem. R. Astr. Soc.* t. XXIX. (1860), pp. 191—306, [216].

Cayley. A Memoir on the problem of the rotation of a solid body. *Mem. R. Astr. Soc.* t. XXIX. (1860), pp. 307—342, [217].

Cayley. On Lambert's theorem for Elliptic Motion. *Monthly Not. R. Astr. Soc.* t. xxii. pp. 238—242, (1861), [222].

Cayley. Note on a theorem of Jacobi's in relation to the problem of three bodies. *Monthly Not. R. Astr. Soc.* t. xi. pp. 76—79 (1861), [220].

Chasles. Note sur les propriétés générales du système de deux corps semblables entr'eux et placés d'une manière quelconque dans l'espace, et sur le déplacement fini ou infiniment petit d'un corps solide libre. (Read Feb. 1831.) *Bull. Univ. des Sciences (Férussac)*, t. XIV. pp. 321—326.

Chasles. Théorèmes généraux sur les systèmes de forces et leurs moments. *Liouv.* t. XII. pp. 213—224 (1847).

Clairaut. Théorie de la Lune déduite du seul principe de l'attraction

réciroquement proportionnelle aux carrés des distances. 4to. St Pét. 1752, and Paris, 1765.

Cohen. On the Differential Coefficients and Determinants of Lines, and their Application to Theoretical Mechanics. Phil. Trans. t. 152 (1862), pp. 469—510.

Cotes. Harmonia mensurarum sive analysis et synthesis per rationum et angulorum mensuras promotae; accedunt alia opuscula mathematica. 4to. Camb. 1722.

Cournot. Mémoire sur le mouvement d'un corps rigide soutenu par un plan fixe. Crelle, t. V. pp. 133—162 and 223—249 (1830); Suite, t. VIII. pp. 1—12 (1832).

Creedy. General and practical solution of Kepler's Problem. Quart. Math. Journ. t. I. pp. 259—271 (1855).

D'Alembert. Traité de Dynamique. Paris, 1743.

D'Alembert. Recherches sur la précession des équinoxes et sur la nutation de l'axe de la terre. Mém. de Berl. (1749).

Deaboves. Thèse sur le mouvement d'un point materiel attiré en raison inverse du carré des distances vers deux centres fixes. Liouv. t. XIII. pp. 369—396 (1848).

Donkin. On an application of the calculus of operations to the transformation of trigonometric series. Quart. Math. Journ. t. ii. pp. 1—15 (1858).

Donkin. On a class of Differential Equations, including those which occur in Dynamical Problems. Part I. Phil. Trans. t. cxliv. (1854), pp. 71—113; Part II. t. cxlv. (1855), pp. 299—358.

Droop. On the Isochronism of the Circular Hodograph. Quart. Math. Journ. t. I. (1856), pp. 374—378.

Dumas. Ueber die Bewegung des Raumpendels mit Rücksicht auf die Rotation der Erde. Crelle, t. L. pp. 52—78 and 126—185 (1855).

Durege. Theorie der elliptischen Functionen. 8vo. Leipzig, 1861. (§ xx. reproduces some results on the spherical pendulum obtained in an unpublished memoir of 1849.)

Euler. Determinatio Orbitae Cometæ anni 1742. Misc. Berl. t. vii. (1743), p. 1.

Euler. *Theoria motuum planetarum et cometarum.* 4to. Berl. 1744.

Euler. De motu corporis ad duo virium centra attracti. Nov. Comm. Petrop. t. X. for 1764, pub. 1766, pp. 207—242.

Euler. Problème: un corps étant attiré en raison réciproque carrée des distances vers deux points fixes donnés, trouver les cas où la courbe décrite par ce corps sera algébrique. Mém. de Berl. for 1760, pub. 1767, pp. 228—249.

Euler. De motu corporis ad duo centra virium fixa attracti. Nov. Comm. Petrop. t. XI. for 1765, pub. 1767, pp. 152—184.

Euler. Considerationes de motu corporum coelestium. Nov. Comm. Petrop. t. X. for 1764, pub. 1766, pp. 544—558.

Euler. De motu rectilineo trium corporum se mutuo attrahentium. Nov. Comm. Petrop. t. XI. for 1765, pub. 1767, pp. 144—151.

Euler. De motu trium corporum se mutuo attrahentium super eadem linea recta. Nov. Acta Petrop. t. iii. (1776), pp. 126—141.

Euler. Problema algebraicum ob affectiones prorsus singulares memorabile. Nov. Comm. Petrop. t. XV. (1770), p. 75; Comm. Arith. Coll. t. i. pp. 427—443.

Euler. Formulae generales pro translatione quacunque corporum rigidorum. Nov. Comm. Petrop. t. XX. 1775, pp. 189—207.

Euler. Nova methodus motum corporum rigidorum determinandi. Nov. Comm. Petrop. t. XX. (1775), p. 208.

Euler. Recherches sur la précession des Equinoxes et sur la nutation de l'axe de la terre. Mém. de Berl. t. v. for 1749, pub. 1751, pp. 326—338. (Euler mentions, t. VI., that this was written after he had seen D'Alembert's memoir.)

Euler. Découverte d'un nouveau principe de Mécanique. Mém. de Berl. t. VI. for 1750, pub. 1752, pp. 185—217.

Euler. Recherches sur la connaissance mécanique des corps. Mém. de Berl. for 1758, pub. 1767, pp. 132—153.

Euler. Du mouvement de rotation des corps solides autour d'une axe variable. Mém. de Berl. for 1758, pub. 1765, pp. 154—193.

Euler. Du mouvement d'un corps solide lorsqu'il tourne autour d'une axe mobile. Mém. de Berl. for 1760, pub. 1767, pp. 176—227.

Euler. *Theoria motus corporum solidorum.* 4to. Rostock, 1765.

Foucault. *Démonstration physique du mouvement de rotation de la terre au moyen du pendule.* Comptes Rendus, t. xxxii. (1851), pp. 135—138.

Gauss. *Fundamental-Gleichungen für die Bewegung schwerer Körper auf der rotirenden Erde,* 1804.

Gauss. *Theoria motus corporum coelestium.* 4to. Hamb. 1809. [Werke, t. vii. Gotha, 1871.]

Greatheed. Investigation of the general term of the expansion of the true anomaly in terms of the mean. *Camb. Math. Journ.* t. i. pp. 228—232 (1838).

Gudermann. De pendulis sphaericis et de curvis quae ab ipsis describuntur sphaericis. *Crelle*, t. xxxviii. pp. 185—215 (1849).

Hamilton, Sir W. R. A theorem of anthodographic isochronism. *Proc. R. Irish Acad.* 1847, t. iii. pp. 465—466.

Hamilton, Sir W. R. *Lectures on Quaternions.* 8vo. Dublin, &c. (1853).

Hansen. *Fundamenta Nova investigationis orbitae verae quam Luna perlustrat.* 4to. Gotha, 1838.

Hansen. Ermittlung der absoluten Storungen in Ellipsen von beliebiger Excentricität und Neigung. Gotha, 1843, pp. 1—167.

Hansen. Entwicklung des Products einer Potenz des Radius-Vectoris mit dem Sinus oder Cosinus eines Vielfachen der wahren Anomalie in Reihen die nach den Sinussen oder Cosinussen der Vielfachen der wahren excentrischen oder mittleren Anomalie fortschreiten. *Abh. d. K. Sachs. Ges. zu Leipzig*, t. ii. pp. 183—281 (1853).

Hansen. Entwicklung der negativen und ungeraden Potenzen der Quadratwurzel der Function $r^2 + r'^2 - 2rr'(\cos U \cos U' + \sin U \sin U' \cos J)$. *Abh. d. K. Sachs. Ges. zu Leipzig*, pp. 286—376 (1854).

Hansen. *Theorie der Pendelbewegung.* 4to. Dantzig, 1856.

Haton de la Goupillière. *Mémoire sur une théorie nouvelle de la géométrie des masses.* *Journ. École Polyt.* t. xxi. (cah. 37), 1858, 1er Mémoire, pp. 35—72; 2e Mémoire, pp. 73—96.

Haughton. On the rotation of a solid body round a fixed point, being an account of the late Professor MacCullagh's lectures on that subject, Hilary Term, 1844, in Trinity College, Dublin; compiled by the Rev. S. Haughton. Trans. R. Irish Acad. t. XXII. (1849), pp. 1—18.

Jacobi. Euleri formulae de transformatione coordinatarum. Crelle, t. ii. pp. 188, 189 (1827).

Jacobi. Zur Theorie der Variations-Rechnung und der Differential-Gleichungen. Crelle, t. XVII. (1837), pp. 68—82.

Jacobi. Formulae transformationis integralium definitorum. Crelle, t. xv. pp. 1—26 (1836).

Jacobi. De motu puncti singularis. Crelle, t. xxiv. pp. 5—27 (1842).

Jacobi. Elimination des noeuds dans le problème des trois corps. Crelle, t. xxvi. (1843), pp. 115—131.

Jacobi. Theoria novi multiplicatoris systemati aequationum differentialium vulgarium applicandi (§ 26, two centres). Crelle, t. xxix. pp. 333—337 (1845).

Jacobi. Sur la rotation d'un corps. Extrait d'une lettre adressée à l'Académie des Sciences. Comptes Rendus, t. xxix. p. 97; and Liouv. t. xiv. pp. 337—344 (1849).

Jacobi. Sur la rotation d'un corps. (With addition containing the demonstration of the formulas.) Crelle, t. XXXIX. pp. 293—350 (1850).

Jacobi. Nova methodus aequationes differentiales partiales primi ordinis inter numerum variabilium quemcunque propositas integrandi (posthumous, edited by A. Clebsch). Crelle, t. LX. pp. 1—181 (1862).

Lagrange. Mécanique Analytique. 1st ed. 1788; 2nd ed. t. i. 1811; t. ii. 1815; 3rd ed. 1855.

Lagrange. Sur une manière particulière d'exprimer le temps dans les sections coniques décrites par des forces tendantes au foyer et réciproquement proportionnelles aux carrés des distances. Mém. de Berlin for 1778; and Note V. of t. ii. of the 3rd edition of the Méc. Anal. pp. 332—349.

Lagrange. Recherches sur le mouvement d'un corps qui est attiré vers deux centres fixes. Premier Mémoire, où l'on suppose que l'attraction est en raison inverse des carrés des distances.

Anc. Mém. de Turin, t. iv. (1766—1769), pp. 118—215.

Lagrange. Recherches sur le mouvement d'un corps qui est attiré vers deux centres fixes. Second Mémoire, où l'on applique la méthode précédente à différentes hypothèses d'attraction. Anc. Mém. de Turin, t. iv. (1766—1769), pp. 215—243.

Lagrange. Nouvelle solution du problème du mouvement de rotation d'un corps. Mém. de Berl. for 1773.

Lambert. Insigniores Orbitae Cometarum Proprietates. 8vo. Aug. 1765.

Laplace. Mécanique Céleste, t. i. 1799; t. ii. 1799; t. iii. 1802; t. iv. 1805; t. v. 1823.

Laplace. Mémoire sur le développement de l'anomalie vraie et du rayon vecteur elliptique en series ordonnées suivant les puissances de l'excentricité. Mém. de l'Inst. t. VI. (1823), pp. 63—80.

Lefort. Expressions numériques des intégrales définies qui se présentent quand on cherche les termes généraux des développements des coordonnées d'une planète dans son mouvement elliptique. Liouv. t. xi. pp. 142—152 (1846).

Legendre. Exercices de Calcul Intégrale, t. II. (containing the dynamical applications) 1817.

Legendre. Traité des Fonctions Elliptiques, t. i. (1825) (but the dynamical applications are for the most part reproduced from the Exercises).

Leverrier. Annales de l'Observatoire de Paris, t. i. (1855).

Lexell. Theoremata nonnulla generalia de translatione corporum rigidorum. Nov. Comm. Petrop. t. xx. (1775), pp. 239—270.

Liouville. Sur l'intégrale $\int_0^\pi \cos i(u - x \sin u) du$. Liouv. t. vi. pp. 36, 37 (1841).

Liouville. Extrait d'un mémoire sur un cas particulier du problème de trois corps. Liouv. t. VII. pp. 110—113 (1842).

Liouville. Sur quelques cas particuliers où les Equations du mouvement d'un point materiel peuvent s'intégrer. 3 Memoirs. Liouv. t. xi. pp. 345—379 (1846), t. xii. pp. 410—444 (1847), and t. xiv. pp. 257—300 (1849).

Liouville. Mémoire sur un cas particulier du problème de trois corps. Conn. des Temps for 1845, and Liouv. t. i. (2 sér.), pp. 248—256 (1842).

Liouville. Note ajoutée au mémoire de M. Serret (two revolving centres). Liouv. t. XIII. pp. 34—37 (1848).

Lottner. Reduction der Bewegung eines schweren um einen festen Punct rotirenden Revolutions-Körpers auf die elliptischen Transcendenten. Crelle, t. L. pp. 111—125 (1855).

Lottner. Zur Theorie des Foucaultschen Pendelversuchs. Crelle, t. LII. pp. 52—58 (1856).

MacCullagh, see Haughton.

Möbius. Ueber die Zusammensetzung unendlich kleiner Drehungen. Crelle, t. XVIII. pp. 189—212 (1838).

Möbius. Der barycentrische Calcul. 8vo. Leipzig, 1827.

Möbius. Lehrbuch der Statik. 8vo. Leipzig, 1837.

Möbius. Die Elemente der Mechanik des Himmels. 8vo. Leipzig, 1843.

Newton (Sir Isaac). Philosophiae Naturalis Principia Mathematica. 4to. London, 1687.

Pagani. Démonstration d'un théorème de Lambert. Crelle, t. xv. pp. 350—352 (1834).

Pagani. Mémoire sur l'équilibre d'un corps solide suspendu à un cordon flexible. Crelle, t. xxix. pp. 185—204 (1839).

Painvin. Recherche du dernier multiplicateur pour deux formes spéciales et remarquables des Equations différentielles du problème de trois corps. Liouv. t. xix. pp. 88—111 (1854).

Poinsot. Mémoire sur la rotation. Extrait, 8vo. Paris, 1834.

Poinsot. Théorie nouvelle de la rotation des corps. Liouv. t. xvi. pp. 9—130 and 289—336 (1851).

Poinsot. Théorie des cônes circulaires roulants. Liouv. t. xviii. pp. 41—70 (1853).

Poinsot. Questions dynamiques sur la percussion des corps. Liouv. t. ii. (2 sér.), pp. 281—329 (1859).

Poisson. *Traité de Mécanique.* 1st ed. Paris, 1811; 2nd ed. Paris, 1833.

Poisson. Mémoire sur un cas particulier du mouvement de rotation des corps graves. *Journ. École Polyt.* t. ix. (cah. 16), pp. 247—262 (1813).

Poisson. Sur une nouvelle manière d'exprimer les coordonnées des planètes dans l'orbite elliptique. *Conn. des Temps* for 1809.

Poisson. Mémoire sur la variation des constantes arbitraires dans les problèmes de mécanique. *Journ. École Polyt.* t. viii. (cah. 15), pp. 266—344 (1809).

Poisson. Addition au mémoire précédent. *Journ. École Polyt.* t. viii. (cah. 16), pp. 1—69 (1809).

Poisson. Mémoire sur un cas particulier du problème des trois corps. *Journ. École Polyt.* t. xvi. (cah. 26), pp. 124—146 (1850).

Quet. Des mouvements relatifs en général, et spécialement des mouvements relatifs sur la terre. *Liouv.* t. XVIII. pp. 213—298 (1853).

Richelot. Bemerkung über einen Fall der Bewegung eines systems von materiellen Puncten. *Crelle*, t. XL. pp. 178—182 (1850).

Richelot. Auszug eines Schreibens an Herrn Prof. Jacobi. *Crelle*, t. XLII. pp. 32—34 (1851).

Richelot. Bemerkungen zur Theorie des Raumpendels. *Crelle*, t. XLII. pp. 233—238 (1853).

Richelot. Eine neue Lösung des Problems der Rotation eines festen Körpers um einen Punkt. (Auszug einer zu Berlin, 1851, erschienenen Schrift.) *Crelle*, t. XLIV. pp. 50—65.

Rodrigues. Des lois géométriques qui régissent les déplacements d'un système solide dans l'espace, et de la variation des coordonnées provenant de ces déplacements considérés indépendamment des causes qui peuvent les produire. *Liouv.* t. iii. pp. 380—440 (1840).

Rueb. Specimen inaugurale de motu gyratorio corporis rigidi nulla vi acceleratrice sollicitati. Utrecht, 1834.

Schellbach. Mathematische Miszellen. No. I.—IV. Ueber die Bewegung eines Puncts der von einem festen Puncte angezogen wird. *Crelle*, t. XLV. pp. 255—266 (1853).

Schubert. *Astronomie Théorique*. 4to. St Pet. (1822).

Segner. *Specimen theorise turbinum*. 4to. Halae, 1755.

Serret. Thèse sur le mouvement d'un point attiré par deux centres fixes en raison inverse du carré des distances. *Liouv. t. xiii.* pp. 17—33 (1848).

Serret. Sur la solution particulière que peut admettre le problème du mouvement d'un corps attiré vers deux centres fixes par des forces réciproquement proportionnelles aux carrés des distances. *Note III. to t. ii. of the 3rd ed. of the Méc. Anal., pp. 327—331* (1855).

Sohncke. *Motus corporum coelestium in medio resistente*. Crelle, t. x. pp. 23—40 (1833).

Somoff. Démonstration des formules de M. Jacobi relatives à la théorie de la rotation d'un corps solide. Crelle, t. XLII. pp. 95—116 (1851).

Stader. De orbitis et motibus puncti cujusdam corporei circa centrum attractivum aliis quam Neutoniana attractionis legibus sollicitati. Crelle, t. XLVI. pp. 262—327 (1852).

Steichen. De la propriété fondamentale du mouvement cyclor"dal, et de sa liaison avec le principe de la composition des mouvements de rotation autour des axes parallèles et des axes qui se coupent. Crelle, t. XLVI. pp. 24—42 (1853).

Steichen. Mémoire sur la question réciproque du centre de percussion. Crelle, t. XLVIII. pp. 1—68 (1854).

Steichen. Mémoire de Mécanique relatif au mouvement de rotation et au mouvement naissant des corps solides. Crelle, t. XLIII. pp. 161—244 (1852), and Addition t. XLIV. pp. 43—46 (1853).

Sylvester. Sur l'involution des lignes droites dans l'espace considérées comme des axes de rotation. *Comptes Rendus*, t. LII. (1861), p. 741.

Sylvester. Note sur l'involution de six lignes dans l'espace. *Comptes Rendus*, t. LII. (1861), p. 815.

Thomson. On the principal axes of a solid body. *Camb. and Du-bl. Math. Journ.* t. I. pp. 127—133 and 195—206 (1846).

Tissot. Thèse de Mécanique. *Liouv. t. xvII.* pp. 88—116 (1852).

Townsend. On the principal axes of a solid body, their moments of

inertia and distribution in space. *Camb. and Dubl. Math. Journ.* t. i. pp. 209—227, and t. ii. pp. 19—42, 140—171 and 241—251 (1846—47).

Walton. Instantaneous lines and planes in a body revolving round a fixed point. *Camb. and Dubl. Math. Journ.* t. vii. pp. 111—113 (1852).

Whewell. *A Treatise on Dynamics.* 1st ed. Cambridge, 1823.

Worms. *The Earth and its Mechanism*—being an account of the various proofs of the rotation of the Earth, with a description of the instruments used in the experimental demonstrations: to which is added the theory of Foucault's Pendulum and Gyroscope. 8vo. London, 1862.

[299]

MATHEMATICS, RECENT TERMINOLOGY IN.

[From the *English Cyclopædia*, vol. v. (1860), pp. 534—542.]

The terms intended to be explained in the present article relate to subjects distinct indeed, but intimately connected together, as well logically as historically. Determinants were devised as a means to the solution of a system of simple equations, but the principle of their construction is contained in the rule of signs; and the theory of determinants is the foundation of the theory of matrices, and of the theory of invariants or covariants. The theory of groups is closely connected with the theory of substitutions, and the theory of substitutions is the foundation of the theory of determinants.

The following list of terms may be convenient:

- Rule of signs.
- Group.
- Determinant.
- Minor.
- Symmetric, skew, skew symmetric.
- Commutant.
- Pfaffian.
- Permutant.
- Intermutant.
- Cumulant.
- Matrix.
- Resultant.
- Discriminant.

- Plexus.
- Rational and integral functions (notation and nomenclature of).
- Quantic, quadric, cubic, &c., binary, ternary, &c., facient, multipartite, lineo-linear.
- Emanant.
- Linear transformations.
- Covariant, invariant, contravariant.
- Canonisant, catalecticant.
- Bezoutic matrix.
- Concomitance, congruency, contragredieny.

Rule of Signs.—In any arrangement of a set of symbols, if the order of any two symbols is said to be “wrong” when it is opposite to their natural order, then the arrangement is positive or negative according as the number of wrong pairs is even or odd. For instance, in the case of three symbols, the arrangements 123, 231, 312 are respectively positive arrangements, and 132, 213, 321 are negative arrangements: in the case of four terms, the primitive arrangement being 1234, the arrangements 1234, 2341, 3412, 4123 are respectively positive, negative, positive, negative; there are in all twelve positive and twelve negative arrangements.

Group.—The term was originally used as applied to substitutions only, but the more general use of the term is as follows: A set of symbols, say x, y, z, w , is said to form a group when, if x operating upon y gives z , then x operating upon z gives w , and so on; that is, when the result of the operation of any one of the symbols upon any other of the symbols is always a symbol of the set. The number of symbols in the group is called the *order* of the group. The theory of groups is of great importance in algebra and in the theory of equations.

Determinant.—A determinant is a function of n^2 quantities arranged in a square, and is defined as follows: The quantities being

$$\begin{array}{cccc} a, & b, & c, & d \\ a', & b', & c', & d' \\ a'', & b'', & c'', & d'' \\ a''', & b''', & c''', & d''' \end{array}$$

the determinant is the sum of all the products of n quantities, one from each row and one from each column, each product being taken with the sign which belongs to the arrangement of the columns (or lines). The algebraical sum of these products is a determinant, and such determinant is or may be represented as above, by enclosing the terms within two vertical lines. Thus the developed value of the determinant of the fourth order is

$$\begin{vmatrix} a & b & c & d \\ a' & b' & c' & d' \\ a'' & b'' & c'' & d'' \\ a''' & b''' & c''' & d''' \end{vmatrix} = \sum \pm a \cdot b' \cdot c'' \cdot d'''.$$

Minor.—If in a determinant we omit any number of rows and the same number of columns, the determinant formed by the remaining rows and columns is called a *minor* of the original determinant. If only one row and one column are omitted, the resulting minor is called a *first minor*; if two rows and two columns are omitted, the resulting minor is called a *second minor*; and so on. The minors of a determinant are also called the determinants of the system of quantities which remain.

Symmetric, Skew, Skew Symmetric.—A determinant

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix} \quad \text{or} \quad \begin{vmatrix} 11, & 12, & 13 \\ 21, & 22, & 23 \\ 31, & 32, & 33 \end{vmatrix}$$

where the corresponding terms on opposite sides of the dexter diagonal are equal to each other (say $rs = sr$) is said to be *symmetrical*.

But if the terms are equal in magnitude only, but have opposite signs (say $rs = -sr$, this relation not extending to the terms in the diagonal, for which $s = r$), the determinant is said to be *skew*.

If the terms in the diagonal are each of them $= 0$, the determinant is said to be *skew symmetric*.

Commutant.—If instead of two we have three or more columns, the resulting function is a *commutant*. But a distinction is to be made according as the number of columns is even or odd. In the former case we may permute all but one of the columns, and it is indifferent which column is left unpermuted; and if all the columns are permuted, the effect is merely to introduce the numerical factor $(-1)^{n-1}$. In the latter case we may permute all but one of the columns, but the column left unpermuted is a determinate one; and if all the columns are permuted, the effect is to introduce the numerical factor $(-1)^n$.

Pfaffian.—A skew symmetric determinant of an even order is a perfect square; the square root of such determinant is called a *Pfaffian*. The name is derived from Pfaff, who first investigated the properties of these functions.

Permutant.—A permutant is a function of a set of quantities arranged in a square, analogous to a determinant, but involving all the permutations of the quantities, instead of only those which give one quantity from each row and each column.

Intermutant.—An intermutant is a function of two or more sets of quantities, involving the permutations of the quantities in each set.

Cumulant.—The cumulant is a particular form of determinant which naturally presents itself in the form of a determinant, thus the cumulant $(abcd)$ or numerator of the fractions

$$a + \frac{1}{b + \frac{1}{c + \frac{1}{d}}}$$

is

$$\begin{vmatrix} a, & 1, & \cdot, & \cdot \\ -1, & b, & 1, & \cdot \\ \cdot, & -1, & c, & 1 \\ \cdot, & \cdot, & -1, & d \end{vmatrix}$$

and so for a greater number of terms. The developed expression is

$abcd + ab + ad + cd + 1$, which is formed from the product $abcd$ by successively omitting each product of two consecutive letters.

Matrix.—The square arrangement of terms out of which a determinant is formed, and generally any square or rectangular arrangement of terms, is called a *matrix*. Consider a determinant

$$\begin{array}{cccc} a, & b, & c, & d \\ a', & b', & c', & d' \\ a'', & b'', & c'', & d'' \\ a''', & b''', & c''', & d''' \end{array}$$

and partitioning the lines in any manner, form with them the matrices

$$\begin{array}{cccc} a, & b, & c, & d \\ a', & b', & c', & d' \end{array} \quad \text{and} \quad \begin{array}{cccc} a'', & b'', & c'', & d'' \\ a''', & b''', & c''', & d''' \end{array}$$

and partition the columns in any manner, form with them the matrices

$$\begin{array}{cc} a, & b \\ a', & b' \\ a'', & b'' \\ a''', & b''' \end{array} \quad \text{and} \quad \begin{array}{cc} c, & d \\ c', & d' \\ c'', & d'' \\ c''', & d''' \end{array}$$

A matrix may be *compounded* with another matrix: thus if the lines of the first matrix are compounded with the columns of the second matrix, the result is a new matrix called the *compound* matrix. If the first matrix is of m lines and n columns, and the second is of n lines and p columns, the compound matrix is of m lines and p columns.

If the matrices are square, and of the same order n , the determinant of the compound matrix is equal to the product of the determinants of the component matrices, each raised to the power $\binom{n-1}{m-1}$.

Resultant.—The *resultant* of a system of equations is the function which, equated to zero, gives the condition for the existence of a common root. In the particular case of a system of linear equations between as many unknown quantities, the resultant is the determinant formed with the coefficients of the equations. In the case of two equations of degrees m and n respectively, the resultant is of the degree n in the coefficients of the first equation, and of the degree m in the coefficients of the second equation.

Discriminant.—If in a system of equations the functions equated to zero contain parameters, the condition that the equations may have a

common root is expressed by equating to zero the discriminant of the system. In the particular case of a single equation, the discriminant is the function which, equated to zero, expresses the condition that the equation has equal roots. For a binary quantic $(a, b, c, \dots)(x, y)^n$, the discriminant is a function of the coefficients $a, b, c, \dots$ which, equated to zero, gives the condition that the quantic has two equal linear factors.

Plexus.—The plexus of a system of equations is the condition that the equations may have a common root of a given multiplicity. The discriminant is the simplest case of a plexus, corresponding to a common root of multiplicity two.

Rational and Integral Functions.—A rational and integral function of any number of variables is represented by the notation

$$(a, b, c, \dots)(x, y, \dots)^n$$

where n is the degree of the function. Thus $(a, b, c)(x, y)^2$ denotes $ax^2 + 2bxy + cy^2$; and in like manner $(a, b, c, d)(x, y)^3$ denotes $ax^3 + 3bx^2y + 3cxy^2 + dy^3$.

A homogeneous function may be represented as a non-homogeneous function by putting one of the variables thereof equal to unity, and is represented accordingly: thus $ax^2 + 2bx + c$ is denoted by $(a, b, c)(x, 1)^2$.

But it is often proper to express the coefficients, and in regard to this the following distinction is made, namely

$$(a, b, c)(x, y)^2$$

denotes $ax^2 + 2bxy + cy^2$; and in like manner $(a, b, c, d)(x, y)^3$ denotes $ax^3 + 3bx^2y + 3cxy^2 + dy^3$.

Quantic, Quadric, Cubic, &c.—A homogeneous function of any number of variables is called a *quantic*. The particular cases have special names: a quantic of the second degree is called a *quadric*; of the third degree, a *cubic*; of the fourth degree, a *quartic*; of the fifth degree, a *quintic*; and so on.

A quantic of two variables is called *binary*; of three variables, *ternary*; of four variables, *quaternary*; and so on.

A quantic which is linear in each of its variables is called *lineo-linear*.

A quantic which is composed of factors of the form $\alpha x + \beta y + \gamma z + \dots$ is called *multipartite*.

Emanant.—If in a quantic $(a, b, c, \dots)(x, y)^n$ we replace x by $x + \xi$, y by $y + \eta$, and develop in powers of ξ, η , the coefficients of the several powers of ξ, η are called the *emanants* of the original quantic. The emanants are themselves quantics, and they possess important properties in relation to the theory of invariants and covariants.

Linear Transformations.—In this theory the variables of a function are supposed to be respectively linear functions of a new set of variables, so that the function is transformed into a similar function of these new variables, with of course altered values of the coefficients, and the question was to find the relations which existed between the original and new coefficients and the coefficients of the linear transformations.

Covariant.—A *covariant* of a given quantic is a function of the coefficients and variables of the quantic which, when the quantic is transformed by a linear transformation, undergoes the same transformation as the quantic itself. An *invariant* is a covariant which does not contain the variables, that is, a function of the coefficients only which is equal to the same function of the new coefficients. A *contravariant* is a function which, when the quantic is transformed, undergoes a transformation which is the inverse of that undergone by the quantic.

The foregoing definition gives the characteristic property of a covariant, but it does not directly show how the covariants of a given quantic are to be investigated. This is supplied as follows:—For any quantic with arbitrary coefficients, for example $(a, b, c, d)(x, y)^3$, there exist operators involving differentiations in respect to the coefficients, tantamount to the operators $x \frac{\partial}{\partial y}$ and $y \frac{\partial}{\partial x}$ in respect to the variables, and the covariants are the functions which are annihilated by these operators.

Canonisant, Catalecticant.—The *canonisant* is a function which presents itself in the problem of the reduction of a binary quantic to the canonical form. The canonical form of a binary function of an odd order is

$$\xi_1^n + \xi_2^n + \dots + \xi_n^n,$$

and there is in like manner a form (not admitting, however, of a simple definition) which is taken as the canonical form of a binary quantic of an even order. The *catalecticant* and the *canonisant* present themselves in the problem of the reduction of a binary quantic to the canonical form.

Bezoutic Matrix.—If U, V are any two binary quantics of the same order, the Bezoutic matrix is the matrix formed with the coefficients of the “Bezoutic secondary” derived functions, which present themselves in the expression for the resultant of U and V in the form given by Bezout.

Concomitance.—If we have a quantic in two sets of variables, and if by any linear transformations of the two sets respectively, the quantic is transformed into a function of the same form, then the quantic is said to be a *concomitant* of the two sets of variables. The two most important kinds of concomitance are, 1st. *Congrediency*, that is, when the substitution on the second set of variables is identical with that on the first set; 2nd. *Contragrediency*, that is, when the substitution on the second set is the inverse of that on the first set.

Notes and References

235. Contains the demonstration and I think the first publication (1858) of Hermite's formula alluded to 135, for the reduction of an elliptic differential to the form

$$\frac{dx}{\sqrt{4x^3 - g_2x - g_3}},$$

which is in fact the Weierstrassian form, the theory of which has been of late so extensively developed.

241, 242. Figures of Poinso't's stellated Polyhedra are given, Fouché et Combérusse, *Traité de Géométrie Élémentaire*, 8vo. Paris, 1866, and Dostor, "Propriétés générales des polyèdres réguliers étoilés," *Liouv. t. v.* (1879), pp. 209—226.

246. In connexion with this paper, On Contour and Slope Lines (1859), I refer to the memoir by Maxwell, "On Hills and Dales," *Phil. Mag.* 1860.

248. See Professor Mc Clintock's paper "On the resolution of the sextic equation" (1879), where the resolvent equation in ψ is in fact the equation in ϕ of 237; the sinister resolvent equation deduced from it by reversing the order of the coefficients (a, b, c, d, e, f) , this is in fact the equation in χ obtained from my covariant equation, p. 323, for $\psi = \frac{1}{10}(\Phi - \chi U)$ by writing therein $\chi = 0$ (viz. ψ , Mc Clintock = χ , Cayley), but he afterwards modifies the form of this sinister resolvent; and a central resolvent equation for t ($= T$, Mc Clintock = $\frac{\psi}{\psi'}$, Cayley). We obtain this equation by writing in the equation for ψ , $x = \psi$, $y = \phi$, whence $\phi = 0$, that is the equation becomes simply $A\psi - 2\psi y = 0$, where in the covariants A, ψ we have

$$t = r + a^{-1}\chi, \quad v = \frac{-a^{-1}\chi}{10\sqrt{t}}.$$

The two equations are

$$\begin{aligned} awT^2 + ce(w - 100ae)T + wf &= 0, \\ T(w - 20ae) + 20af &= 0, \end{aligned}$$

or, if for greater convenience we write $20ae = \theta$, then

$$\begin{aligned} awT^2 - b\theta(w - \theta)T + wf &= 0, \\ T(w - \theta) + 20af &= 0, \end{aligned}$$

we have

$$T = -\frac{20af}{w-\theta},$$

and thence

$$\frac{aw \cdot 400a^2f^2}{(w-\theta)^2} + b\theta(w-\theta)\frac{20af}{w-\theta} + wf = 0,$$

that is

$$w(w-\theta)^2 - 50(w-\theta)(w-\theta) - 3200000a^4f^2 = 0,$$

which should be identical with the before mentioned equation in w , that is with

$$(w^2 - 5\theta w + 10\theta^2)w + 50\theta^3 - (3200000 + 256 \cdot 81ae^4f^2)w = 0,$$

and it is in fact at once seen that each of these equations is

$$\begin{aligned} 0 = & w^4 \\ & + w^3 \cdot -10\theta \\ & + w^2 \cdot 55\theta^2 \\ & + w \cdot -140\theta^3 \\ & + 175\theta^4 \\ & - 10\theta^5 - 3200000a^4f^2. \end{aligned}$$

249. We have

$$t_1 = r + a^{-1}\chi, \quad v_1 = \frac{-a^{-1}\chi}{10\sqrt{\iota}},$$

viz. from the values just written down, we obtain by permutation of the roots the six values of each of the functions t and ϕ .

By means of these data, or say of t , $r + a^{-1}\chi$, and v , $= \frac{-a^{-1}\chi}{10\sqrt{\iota}}$ (I write v instead of his V_i), Mc Clintock completes in a very elegant manner the determination of the roots x_1, x_2, x_3, x_4, x_5 of the quintic equation: the solution contains also the coefficients $\gamma, \delta, \epsilon, \zeta$ which belong to the equation deprived of its second term, viz. for the definition of these, we have

$$(a, b, c, d, e, f)(x, 1)^5 = a(1, 0, \gamma, \delta, \epsilon, \zeta)(x + a^{-1}b, 1)^5 = 0.$$

I reproduce this solution: writing as usual

$$\omega u_1 = x_1 + \omega x_2 + \omega^2 x_3 + \omega^3 x_4 + \omega^4 x_5,$$

if as above $v = \frac{-a^{-1}\chi}{10\sqrt{\iota}}$; that is

$$u_1 - u_4 = 2v,$$

and combining herewith the equation

$$u_1 u_4 + u_2 u_3 = -2\gamma,$$

we have the first pair of equations. The second pair of equations is obtained by a similar process, but the work is longer. We have

$$\begin{aligned} 125u_1^3 u_3 &= F + A\omega + B\omega^2 + C\omega^3 + D\omega^4, \\ 125u_4^3 u_2 &= F + A\omega^2 + B\omega^4 + C\omega + D\omega^3, \\ 125u_2^3 u_1 &= F + A\omega^3 + B\omega + C\omega^4 + D\omega^2, \\ 125u_3^3 u_4 &= F + A\omega^4 + B\omega^3 + C\omega^2 + D\omega, \end{aligned}$$

where

$$\begin{aligned} F &= \sum x_1^3 x_2 + 2 \sum x_1 x_2 x_3, \\ A &= \{24135\} + 2\{12345\} + 2(24135), \\ B &= \{54321\} + 2\{24135\} + 2(54321), \\ C &= \{12345\} + 2\{53142\} + 2(12345), \\ D &= \{53142\} + 2\{54321\} + 2(53142), \end{aligned}$$

where

$$\{24135\} = x_1^3 x_2 + x_2^3 x_3 + x_3^3 x_4 + x_4^3 x_5 + x_5^3 x_1,$$

and as before

$$u_1 u_4 + u_2 u_3 = -2\gamma.$$

Hence

$$A + D - B - C = -5a^{-1}b\phi - \psi = -5a^{-1}b\phi - \phi\tau = -5\phi(a^{-1}b + \tau),$$

or substituting for ϕ and $a^{-1}b + \tau$ their values $10\sqrt{\iota}v$ and τ , this is $= -50\sqrt{\iota}\tau v$.

Hence the value of $125(u_1^3u_3 + u_4^3u_2 - u_2^3u_1 - u_3^3u_4)$ is $= -250\omega$, that is we have

$$u_1^3u_3 + u_4^3u_2 - u_2^3u_1 - u_3^3u_4 = -2\omega,$$

and combining herewith

$$u_1u_3 + u_2u_4 + u_1u_4 + u_4u_1 = -2\delta,$$

we have the second pair of equations.

We next write

$$2v\sqrt{\iota_3} = -\delta - tv + 2\eta_1,$$

$$2v\sqrt{\iota_4} = -\delta - tv - 2\eta_1,$$

$$2v\sqrt{\iota_1} = -\delta + tv + 2\eta_2,$$

$$2v\sqrt{\iota_2} = -\delta + tv - 2\eta_2,$$

whence

$$4\eta_1^2 = (\delta + tv)^2 + 4(\gamma^2 - v^2)(\gamma + v),$$

$$4\eta_2^2 = (\delta - tv)^2 + 4(\gamma^2 - v^2)(\gamma - v),$$

where

$$\eta_1 = \frac{-\delta + 2tv\gamma}{v^2 - v'^2} + \{3\delta\epsilon + 2\gamma\zeta + (-27\gamma^2 + \delta^2 + 4\epsilon^2)t + 7\delta\epsilon t^2\}v^2 \\ + \{\gamma^2\delta\epsilon - 7\gamma\delta^2\zeta\},$$

$$\eta_2 = tv^3\iota_1 + \frac{-\delta + 2tv\gamma}{v^2 - v'^2} + 3\gamma + 10\gamma\epsilon^2 + 5\gamma\delta^2,$$

and

$$\iota_1 = \frac{1}{v^2 - v'^2} + 3\gamma + 10\gamma\epsilon^2 + 5\gamma\delta^2, \quad \iota_2 = tv^3\iota_1 - 5\gamma^2\epsilon - 10\gamma^2\delta - \gamma\zeta.$$

To prove these results write for shortness

$$2m_1 = -\delta - tv, \quad 2m_2 = -\delta + tv,$$

and (as above)

$$2n_1 = u_1v_4 - u_2u_3, \quad 2n_2 = u_1v_3 - u_2v_4,$$

then

$$u_1^3 u_3 = m_1 + \eta_1, \quad u_4^3 u_2 = m_1 - \eta_1, \quad u_2^3 u_1 = m_2 + \eta_2, \quad u_3^3 u_4 = m_2 - \eta_2;$$

we have $u_1 v_4 \cdot u_2 u_3 = m_1^2 - \eta_1^2 = (\gamma + v)(-\gamma - v) = \gamma^2 - v^2$, and the third Eulerian equation thus becomes

$$\epsilon = \gamma + 3\eta^2 + (\eta^2 - \gamma)^{-1} \{u_1 v_4 \cdot u_3 u_4^2 + u_2 u_3^2 \cdot u_1 u_2 + u_2^2 u_3 \cdot u_1 u_2 + u_4^2 u_1 \cdot u_3 u_4\}.$$

250. And thence, instead of $4v\sqrt{\iota_3}\eta_1$, which occurs on the right-hand side writing its value $= p$, we find

$$\begin{aligned} (v^2 - v'^2)^2 V_1 &= (m_1 + m_2)^{-1} \{p(v^2 + v'^2) - 4m_1 m_2 \eta\} \\ &\quad + 2(m_1 - m_2) \{m_1 \eta^2 (v^2 + v'^2) - \psi\} \\ &\quad + 2(m_1^2 - m_2^2)(v^2 + v'^2) \\ &\quad - 4(m_1 \xi_1 + m_2 \xi_2) \eta v. \end{aligned}$$

We have

$$4m_1^2 = (\delta + tv)^2 + 4(\gamma^2 - v^2)(\gamma + v), \quad 4m_2^2 = (\delta - tv)^2 + 4(\gamma^2 - v^2)(\gamma - v),$$

p is given as above, and moreover $m_1 + m_2 = -\delta$, $m_1 - m_2 = -tv$, $m_1 m_2 = \frac{1}{4}(\delta^2 - t^2 v^2)$.

Substituting these values and dividing out by $(\gamma^2 - v^2)$, we find after some reductions the value given above for V_1 .

Finally we have

$$4(\gamma - \iota_1) = -4(u_1 u_4)^2 = (\iota_3 \iota_4 + \iota'_3 \iota'_4 - 4(u_1 u_4)(u_2 u_3)),$$

and in the second case

$$4(\gamma - \iota_2) = -4(u_2 u_3)^2 = (\iota_1 \iota_2 + \iota'_1 \iota'_2 - 4(u_2 u_3)(u_1 u_4)),$$

values which satisfy $ps + qt + ru = 0$. And I accordingly say that the equation of a line in space is

$$Ap + Bq + Cr + Fs + Gt + Hu = 0,$$

viz. this is the equation of the cone of arbitrary vertex $(\alpha, \beta, \gamma, \delta)$ (that is, of the plane through the point $(\alpha, \beta, \gamma, \delta)$) which passes through the line in question. But I go on to say that if $(\alpha', \beta', \gamma', \delta')$,

$(\alpha'', \beta'', \gamma'', \delta'')$ are the coordinates of any two points on the given line, or if the line be given as the intersection of the two planes

$$\alpha x + \beta y + \gamma z + \delta w = 0, \quad \alpha' x + \beta' y + \gamma' z + \delta' w = 0,$$

then in the first case

$$A = \alpha' \beta'' - \alpha'' \beta', \quad F = \gamma' \delta'' - \gamma'' \delta',$$

etc., and in the second case the six coordinates are the determinants

$$\begin{vmatrix} \alpha & \beta \\ \alpha' & \beta' \end{vmatrix}, \quad \begin{vmatrix} \alpha & \gamma \\ \alpha' & \gamma' \end{vmatrix}, \quad \begin{vmatrix} \alpha & \delta \\ \alpha' & \delta' \end{vmatrix}, \quad \begin{vmatrix} \beta & \gamma \\ \beta' & \gamma' \end{vmatrix}, \quad \begin{vmatrix} \beta & \delta \\ \beta' & \delta' \end{vmatrix}, \quad \begin{vmatrix} \gamma & \delta \\ \gamma' & \delta' \end{vmatrix}.$$

251. The leading idea of Plücker's memoir appears in the first words thereof, "I. On linear Complexes of Right lines." He works at first with the four coordinates of a line; as long as these are arbitrary the line is any line whatever; but considering them as connected by a single equation, then there is he says "a Complex," considering them as connected by two equations "a Congruency", and considering them as connected by three equations "a Configuration" or ruled surface. The establishment of these notions of a Complex and a Congruency, and the general idea of regarding the line as an element in the Geometry of Space are absolutely Plücker's. The theory of linear complexes was extensively developed by him in the memoir "Neue Geometrie des Raumes, gegründet auf die Betrachtung der geraden Linie als Raumelement" (1868—69).

Select Scientific Publications of the Cambridge University Press

SELECT SCIENTIFIC PUBLICATIONS OF THE CAMBRIDGE UNIVERSITY PRESS.

The *Collected Mathematical Papers* of Arthur Cayley, Sc.D., F.R.S., Sadlerian Professor of Pure Mathematics in the University of Cambridge.

Vol. I. demy 4to. £1. 5s.

Vol. II. demy 4to. £1. 5s.

Vol. III. demy 4to. £1. 5s.

Vol. IV. demy 4to. £1. 5s.

A Treatise on Natural Philosophy. By Sir W. Thomson, LL.D., D.C.L., F.R.S., and P. G. Tait, M.A.

Part I demy 8vo. 16s.

Part II demy 8vo. 18s.

A Treatise on the Motion of Vortex Rings. By W. M. Hicks, M.A., Fellow of St John's College, Cambridge. 4to. 6s.

Hydrodynamics. A Treatise on the Mathematical Theory of Fluid Motion. By Horace Lamb, M.A., Professor of Mathematics in the University of Adelaide. demy 8vo. 12s.

The Mathematical Works of Isaac Barrow, D.D. Edited by W. Whewell, D.D. 8vo. 9s.

The Mathematical Papers of the late William Kingdon Clifford. Edited by Robert Tucker, M.A. With an Introduction by H. J. Stephen Smith, M.A. 4to. 15s.

A Treatise on the Theory of Determinants and their applications in Analysis and Geometry. By Robert Forsyth Scott, M.A. Crown 8vo. 8s. 6d.

The Elements of Coordinate Geometry in three dimensions. By George Bruce Halsted. Crown 8vo.

End of Volume IV.